The Final Exam Textbook

General Psychology, Psychological Assessment, Social Psychology, Personality, Physiological Psychology, Developmental Psychology, Clinical Psychology, Psychiatric Disorders and Research Methodology, in one book

Written for Sihan, for the JSS AHER MA Clinical Psychology entrance, 27 October 2026. Built from every paper and mock on the site.

BLACK AND WHITE EDITION · A4, DOUBLE SIDED · 22 OCTOBER 2026

How this book works

This is the whole JSS syllabus in one book: one chapter for each of the nine subjects JSS names, in the order JSS lists them, cut so that all nine fit in under 1000 A4 pages. It is built from every past paper, question bank and practice mock on the site. Every question was sorted into the section that should teach it, and every option in it, wrong ones included, is explained somewhere in the book, because next year's paper can make a wrong option the right answer. Chapter 1 is the full-length General Psychology book (821 pages) cut down to its exam core; the full version stays in the Library for looking things up.

How to read it. Each chapter opens with a real case, lists its sections with a learning goal for each, and then teaches them in order, the way Pinel does: the question, the idea, the study behind it and who did it, then what it means. Read the High yield sections first if time is short (the tags are explained below), and do every Scan Your Brain before moving on. Each chapter ends with the case revisited, a summary of every section, a Who's who, the key terms, real past-paper questions with the answers explained, and a one-page High-yield sheet for your last read before the exam.

Key terms are in bold colour where they are defined, and the index at the back points to those pages. People are in bold where they are introduced, with their full name and years.

The boxes

! EXAM TRAP

The confusable pair, the bait option, the famous misattribution. Read every one.

KEY STUDY

A classic experiment told as a story: the question, the method, the result, the criticism.

CLINICAL LINK

Where the idea matters to a clinical psychologist, and to an interview panel.

REMEMBER

The one idea in the section you must not leave without.

INDIAN CONTEXT

Indian research, Indian psychology and Indian applications.

CASE

A person whose story makes the concept concrete.

SCAN YOUR BRAIN

Fill-in-the-blank checks after every two or three sections, answers underneath. Cover the answers first.

The colour code: how often each topic is asked

Every numbered section carries a tag, and every subsection that has been asked carries dots. Both are counted from the past papers and practice mocks on the site: each question was matched to the section that teaches it, a real past-paper question scores 2 points and a mock or question-bank item 1 point.

HIGH YIELD · 23 QS ●●●	MEDIUM YIELD · 8 QS ●●	LOWER YIELD · 1 Q ○
The most-asked quarter of the book's sections (at least 62 points). Read these first and know every trap.	Asked regularly (at least 33 points). Read them fully.	Asked rarely or not yet. Still in the syllabus, so still taught in full.

"23 Qs" is the number of questions on the site that this section answers. The same tags sit next to each section in the Contents, and the page after this one shows where the marks are across the book.

Printing

Print A4, double sided, flip on the long edge. Every chapter opens on a right-hand page, the inner margin is wider for binding, and page numbers sit on the outer corner. The colour and the black and white editions have identical pages, page numbers and index, so either can be printed.

Where the marks are

3180 questions from the past papers, question banks and practice mocks on the site are taught in this book. This is how they spread across the chapters.

Chapter	Questions	From real papers	Share of the book	High-yield sections
1 General Psychology	815	402	25.6%	9 of 11
2 Psychological Assessment	336	136	10.6%	2 of 11
3 Social Psychology	136	93	4.3%	0 of 11
4 Personality	202	103	6.4%	1 of 10
5 Physiological Psychology	364	151	11.4%	2 of 11
6 Developmental Psychology	230	101	7.2%	1 of 11
7 Clinical Psychology	366	191	11.5%	4 of 10
8 Psychiatric Disorders	411	205	12.9%	4 of 11
9 Research Methodology	320	175	10.1%	2 of 11

The 12 most-asked sections

143 QS **1.6** Learning: Conditioning, Cognition and Observation 49

112 QS **1.2** Schools and Systems, the Cognitive Revolution and Indian Psychology 18

111 QS **1.7** Memory: Encoding, Storage, Retrieval and Forgetting 60

88 QS **7.4** Behaviour Therapy 659

79 QS **8.4** Schizophrenia Spectrum and Other Psychotic Disorders 766

74 QS **1.11** Emotion and Stress 94

74 QS **1.10** Motivation 86

69 QS **7.5** Cognitive and Cognitive-Behavioural Therapies, and the Third Wave 669

64 QS **2.3** Reliability: Consistency and the Standard Error of Measurement 145

63 QS **1.9** Intelligence: Theories, Measurement, Heredity and Environment 78

61 QS **1.5** Attention and Consciousness 40

58 QS **9.8** Measurement Scales, Organising Data and Descriptive Statistics 893

Contents

1 General Psychology **9**

1.1 Psychology as a Science: Roots, Founders and the First Schools — MEDIUM · 46 QS — 11

1.2 Schools and Systems, the Cognitive Revolution and Indian Psychology — HIGH · 112 QS — 18

1.3 Sensation and Psychophysics — MEDIUM · 40 QS — 27

1.4 Perception: Organising and Interpreting the World — HIGH · 38 QS — 33

1.5 Attention and Consciousness — HIGH · 61 QS — 40

1.6 Learning: Conditioning, Cognition and Observation — HIGH · 143 QS — 49

1.7 Memory: Encoding, Storage, Retrieval and Forgetting — HIGH · 111 QS — 60

1.8 Thinking, Problem Solving, Reasoning, Decision Making and Language — HIGH · 53 QS — 71

1.9 Intelligence: Theories, Measurement, Heredity and Environment — HIGH · 63 QS — 78

1.10 Motivation — HIGH · 74 QS — 86

1.11 Emotion and Stress — HIGH · 74 QS — 94

High-yield sheet — 127

2 Psychological Assessment **131**

2.1 Psychological Testing: Nature, History, Types and Ethics — LOWER · 11 QS — 133

2.2 Building a Test: Items, Item Analysis, IRT and Response Bias — LOWER · 11 QS — 139

2.3 Reliability: Consistency and the Standard Error of Measurement — HIGH · 64 QS — 145

2.4 Validity and the Accuracy of Test Decisions — MEDIUM · 32 QS — 153

2.5 Norms, Standard Scores and Interpreting a Score — MEDIUM · 32 QS — 160

2.6 Intelligence Tests: Binet, Wechsler, Non-verbal and Indian Scales — MEDIUM · 35 QS — 166

2.7 Assessing Development, Adaptive Behaviour and Disability — LOWER · 21 QS — 173

2.8 Aptitude, Achievement, Creativity, Interest and Attitude Measures — LOWER · 8 QS — 180

2.9 Objective Personality Inventories — MEDIUM · 38 QS — 185

2.10 Projective Techniques — HIGH · 44 QS — 193

2.11 Neuropsychological Assessment and Clinical Rating Scales — MEDIUM · 40 QS — 201

High-yield sheet — 226

3 Social Psychology **229**

3.1 Social Psychology: History, Theories, Concepts and Methods — LOWER · 6 QS — 231

3.2 Social Cognition and First Impressions — LOWER · 7 QS — 238

3.3 Communication, Nonverbal Cues and Attribution — LOWER · 12 QS — 244

3.4 Attitudes, Persuasion and Cognitive Dissonance — LOWER · 15 QS — 251

3.5 Stereotypes, Prejudice and Intergroup Relations — LOWER · 13 QS — 258

3.6 Affiliation, Attraction and Close Relationships — LOWER · 7 QS — 266

3.7 Social Influence: Conformity, Compliance, Obedience and Power — LOWER · 18 QS — 272

3.8 Prosocial Behaviour: Why and When People Help — LOWER · 15 QS — 280

3.9 Aggression: Its Roots, Causes and Control LOWER · 3 QS 287
3.10 Groups: Structure, Performance, Decisions and Leadership MEDIUM · 25 QS 293
3.11 Applied Social Psychology: Law, Culture and Social Work LOWER · 15 QS 302
High-yield sheet 332

4 Personality 335

4.1 What Personality Is: Definitions, Approaches and Early Type Theories LOWER · 4 QS 337
4.2 Freud's Psychoanalysis: Instincts, Structure and Psychosexual Stages MEDIUM · 42 QS 343
4.3 Defence Mechanisms: How the Ego Protects Itself HIGH · 38 QS 351
4.4 Jung's Analytical Psychology and Adler's Individual Psychology MEDIUM · 25 QS 361
4.5 Horney, Fromm, Sullivan and the Object-Relations Theorists LOWER · 10 QS 369
4.6 Traits I: Allport, Cattell and the Lexical Search LOWER · 17 QS 375
4.7 Traits II: Eysenck, the Big Five and the Biology of Personality LOWER · 18 QS 383
4.8 Humanistic and Existential Theories: Rogers, Maslow and May LOWER · 16 QS 391
4.9 Behavioural, Social-Cognitive and Cognitive Theories LOWER · 16 QS 397
4.10 The Self, Positive Traits and Indian Approaches to Personality LOWER · 16 QS 404
High-yield sheet 426

5 Physiological Psychology 429

5.1 Biopsychology and the Organisation of the Nervous System LOWER · 21 QS 431
5.2 The Brain: Brainstem, Diencephalon and Cerebral Cortex MEDIUM · 27 QS 438
5.3 Neurons, Neural Conduction, Synapses and Neurotransmitters MEDIUM · 45 QS 445
5.4 Drugs, Reward and Psychopharmacology MEDIUM · 40 QS 453
5.5 Research Methods, Behaviour Genetics and the Developing Brain MEDIUM · 25 QS 461
5.6 Sensory Pathways and the Motor System LOWER · 22 QS 468
5.7 The Endocrine System, Stress and the Emotional Brain HIGH · 46 QS 475
5.8 Sleep, Biological Rhythms, Hunger, Thirst and Sex MEDIUM · 31 QS 484
5.9 The Biology of Learning, Memory and Amnesia HIGH · 47 QS 491
5.10 Lateralisation, the Split Brain and Language MEDIUM · 28 QS 498
5.11 Brain Damage, Neurodegeneration and Neuropsychological Syndromes MEDIUM · 32 QS 505
High-yield sheet 529

6 Developmental Psychology 533

6.1 Foundations: The Lifespan View, Key Issues and Ecological Contexts LOWER · 13 QS 535
6.2 Heredity, Prenatal Development, Birth and the Newborn LOWER · 7 QS 542
6.3 Infancy: Brain, Body, Motor Skills and Perception LOWER · 12 QS 548
6.4 Piaget's Theory of Cognitive Development HIGH · 43 QS 554
6.5 Vygotsky, Bruner, Information Processing and Theory of Mind MEDIUM · 25 QS 563
6.6 Emotional Development, Temperament and Self-Regulation LOWER · 19 QS 570
6.7 Attachment: Imprinting, Bowlby, Ainsworth and Beyond MEDIUM · 37 QS 577
6.8 Parenting, Family, Play and Gender in Childhood LOWER · 12 QS 584

6.9 Moral Development: Piaget, Kohlberg, Gilligan and Beyond LOWER · 20 QS 591
6.10 Erikson's Psychosocial Stages, Identity and Adolescence MEDIUM · 40 QS 597
6.11 Adulthood, Ageing, Death and Dying LOWER · 2 QS 606
High-yield sheet 629

7 Clinical Psychology 633
7.1 The Profession: History, Models, Ethics and Indian Law MEDIUM · 39 QS 635
7.2 The Clinical Interview, Mental Status Examination and Formulation LOWER · 11 QS 644
7.3 Psychodynamic Therapies and Their Descendants HIGH · 39 QS 651
7.4 Behaviour Therapy HIGH · 88 QS 659
7.5 Cognitive and Cognitive-Behavioural Therapies, and the Third Wave HIGH · 69 QS 669
7.6 Humanistic, Existential, Experiential and Brief Therapies HIGH · 49 QS 679
7.7 Family, Couples and Group Therapies MEDIUM · 28 QS 688
7.8 Does Psychotherapy Work? Evidence, Common Factors and the Alliance LOWER · 12 QS 695
7.9 Health, Community, Crisis and Rehabilitation Psychology LOWER · 15 QS 703
7.10 Working with Children, Older Adults and Special Settings LOWER · 16 QS 710
High-yield sheet 739

8 Psychiatric Disorders 741
8.1 Abnormality, Causes and the Classification of Mental Disorders MEDIUM · 30 QS 743
8.2 Descriptive Psychopathology: Signs and Symptoms HIGH · 45 QS 751
8.3 Neurodevelopmental Disorders MEDIUM · 29 QS 758
8.4 Schizophrenia Spectrum and Other Psychotic Disorders HIGH · 79 QS 766
8.5 Depressive and Bipolar Disorders, and Suicide HIGH · 49 QS 775
8.6 Anxiety and Obsessive-Compulsive and Related Disorders MEDIUM · 32 QS 783
8.7 Trauma, Dissociative and Somatic Symptom Disorders HIGH · 50 QS 790
8.8 Feeding, Eating, Elimination and Sleep-Wake Disorders LOWER · 20 QS 797
8.9 Sexual Dysfunctions, Paraphilic Disorders and Gender Dysphoria LOWER · 9 QS 803
8.10 Substance-Related, Addictive and Neurocognitive Disorders MEDIUM · 36 QS 809
8.11 Personality Disorders, Conduct and Impulse-Control Disorders MEDIUM · 32 QS 816
High-yield sheet 840

9 Research Methodology 843
9.1 Science and the Research Process: Problems and Hypotheses MEDIUM · 21 QS 845
9.2 Variables, Experimental Control and the Validity of Research MEDIUM · 22 QS 852
9.3 Sampling: Populations, Probability and Non-Probability Methods MEDIUM · 24 QS 859
9.4 Research Designs: Experimental, Single-Case and Non-Experimental MEDIUM · 21 QS 866
9.5 Clinical Epidemiology, Randomised Trials and Evidence Synthesis MEDIUM · 27 QS 873
9.6 Methods of Data Collection and Qualitative Research LOWER · 15 QS 879
9.7 Research Ethics and Writing the Research Report LOWER · 10 QS 886

9.8 Measurement Scales, Organising Data and Descriptive Statistics HIGH · 58 QS 893
9.9 Sampling Distributions and the Logic of Hypothesis Testing MEDIUM · 37 QS 900
9.10 Comparing Groups: t-Tests, ANOVA and Non-Parametric Tests HIGH · 57 QS 907
9.11 Correlation, Regression and Factor Analysis MEDIUM · 28 QS 915
High-yield sheet 937

Index **941**

CHAPTER 1

General Psychology

IN THIS CHAPTER

1.1 ●● Psychology as a Science: Roots, Founders and the First Schools: Explain how psychology grew out of philosophy and physiology into a laboratory science, and name the founders, methods and central ideas of Wundt's voluntarism, Titchener's structuralism and the functionalism of James and the American pioneers.

1.2 ●●● Schools and Systems, the Cognitive Revolution and Indian Psychology: Trace psychology's rival schools (behaviourism, Gestalt, psychoanalysis, humanistic and existential psychology) through the cognitive revolution to today's perspectives, and explain the core ideas of Indian psychology: its pioneers, Samkhya, the three gunas, Patanjali's yoga and the Vedantic self.

1.3 ●● Sensation and Psychophysics: Explain how physical energy becomes sensation, how thresholds, the psychophysical laws and signal detection theory measure it, and how vision, hearing, the chemical senses, the skin senses and the body senses work.

1.4 ●●● Perception: Organising and Interpreting the World: Explain how the brain organises and interprets sensory input, from the bottom-up and top-down theories, Gestalt laws, depth cues and constancies to illusions, motion, pattern recognition, perceptual set and perception without awareness.

1.5 ●●● Attention and Consciousness: Explain how attention selects, divides and sustains processing (the filter and attenuation models, automaticity, the Stroop effect and vigilance), and describe consciousness and its states: biological rhythms, sleep stages and disorders, dreams, hypnosis, meditation and psychoactive drugs.

1.6 ●●● Learning: Conditioning, Cognition and Observation: Define learning and explain habituation, classical and operant conditioning, the conditioning therapies, schedules of reinforcement, and cognitive and observational learning, naming who showed what and when.

1.7 ●●● Memory: Encoding, Storage, Retrieval and Forgetting: Explain how memories are encoded, stored, organised, retrieved, distorted and forgotten, using the multi-store and working memory models, the systems of long-term memory, the theories of forgetting and the classic amnesic cases.

1.8 ●●● Thinking, Problem Solving, Reasoning, Decision Making and Language: Explain how people form concepts, solve problems, reason, judge, decide and create, and how language is structured, acquired and related to thought, naming the classic studies and their traps.

1.9 ●●● Intelligence: Theories, Measurement, Heredity and Environment: Define intelligence, calculate ratio and deviation IQs, describe the Wechsler, Stanford-Binet, culture-fair and Indian tests, compare the factor, multiple, triarchic and emotional intelligence theories, interpret heritability correctly and classify intellectual disability and giftedness.

1.10 ●●● Motivation: Explain why people and animals act, from instinct, drive, arousal and incentive theories through Maslow's hierarchy, achievement motivation and self-determination theory, and apply these ideas to conflict, hunger, sex and work.

1.11 ●●● Emotion and Stress: Explain what an emotion is, compare the classic theories and brain systems of emotion, and describe how stress arises, how the body responds to it, and how coping, support and personality shape its effects on health.

Weightage. 815 of the 3180 questions mapped in this book (25.6%) test this chapter, 402 of them from real past papers. 9 of its 11 sections are High yield.

Leipzig, in the winter of 1879. In a small room of an old university building known as the Konvikt, a doctoral student named Max Friedrich sits in front of a Hipp chronoscope, a clockwork timer that measures intervals in thousandths of a second. A stimulus is presented; Friedrich presses a telegraph key. Then the rule changes: now he must press not the instant he senses the stimulus, but only once he has clearly grasped what it is. The second reaction takes longer, and the extra time, Friedrich and his supervisor reasoned, is the time the mind needs to *apperceive*, to bring an impression into the clear focus of attention. Trial after trial, the numbers go into a notebook.

His supervisor was a 47-year-old professor of philosophy, Wilhelm Wundt. When Wundt arrived in Leipzig four years earlier, the university had given him this room to store the apparatus he used in lecture demonstrations. By the winter of 1879 it had become something new: a place where students carried out original experiments on the mind and published them as research. Friedrich's study of the duration of apperception became the first dissertation to come out of that room, and historians who later combed the Leipzig archives dated the start of the work to the last weeks of 1879. That is why textbooks, and nearly every entrance paper in India, treat Leipzig in 1879 as the birth of psychology as a science.

Look closely and that one small act holds this whole chapter. The stimulus had to be sensed and then perceived as something; attention had to seize it; practice changed the times; the instruction had to be remembered; a decision released the press; and the readiness and strain of waiting were motivation and feeling in miniature. Remember Max Friedrich pressing his key. Each section that follows takes one step of that act and asks how it works.

1.1 Psychology as a Science: Roots, Founders and the First Schools

MEDIUM YIELD · 46 QS

"Psychology has a long past, but only a short history," wrote Hermann Ebbinghaus in 1908. The long past is philosophy and physiology; the short history, set out in Figure 1.1, begins in Wilhelm Wundt's Leipzig laboratory in 1879.

What psychology is ●●

The word joins the Greek *psyche* (soul) and *logos* (study); **Aristotle** (384 to 322 BCE) wrote the first treatise on the soul, *De Anima*. Later schools redefined the subject: mind for the philosophers, conscious experience for Wundt ("the Science of Mental Life" for William James in 1890), then behaviour for **William McDougall** (1905, 1908) and John B. Watson (1913). Woodworth's definition was the individual's activities in relation to the environment; the one Indian textbooks quote, "the scientific study of behaviour", is **Norman L. Munn**'s. Today psychology is *the scientific study of behaviour and mental processes*: behaviour is overt (speaking, blushing), **mental processes** internal and covert (perceiving, remembering).

Its goals are to describe, explain, predict and control (or influence); basic research seeks understanding for its own sake, applied research solves practical problems. Among its fields, **clinical psychology** assesses and treats psychological disorders, counselling psychology

helps people in the normal range with adjustment problems, and abnormal psychology studies the causes of disorders. Animal behaviour across species is the field of **comparative psychology**; **evolutionary psychology** explains behaviour as inherited adaptations shaped by natural selection, and differential psychology studies individual differences. Health, developmental, social, industrial-organisational, educational, forensic, experimental and neuropsychology study illness, the lifespan, people among others, work, schooling, the law, basic processes and the brain.

> **! EXAM TRAP**
>
> **Mental processes** are internal and covert, not outward behaviour, overt actions or animal behaviour. The bank credits **Munn** with "the scientific study of behaviour" ("J. B. William" is no known psychologist). Animal behaviour is **comparative** psychology, not social, abnormal or differential. Clinical psychology is the **assessment and treatment of mental illness**, not industry or organisations. "Behaviour in relation to genetic inheritance" is keyed **evolutionary psychology**.

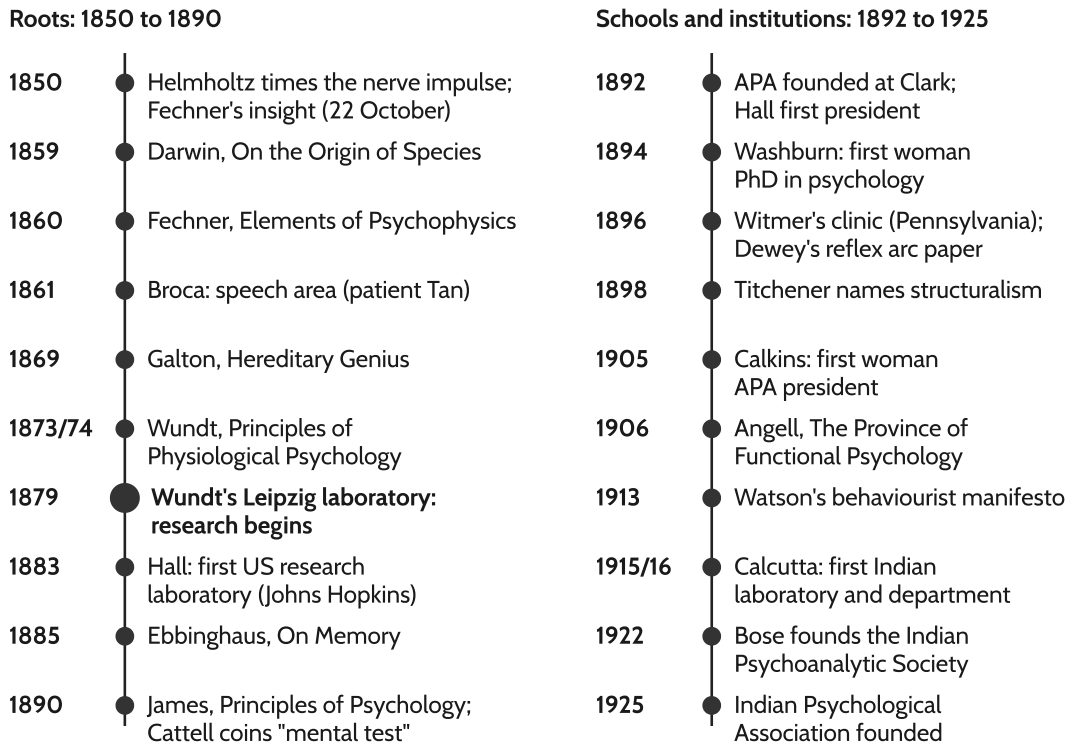


Figure 1.1 The short history of psychology at a glance. Notice how the roots (left) cluster in German physiology and British biology, and how quickly the institutions (right) followed once Wundt's laboratory opened in 1879.

Philosophical roots ●●

Philosophy asked where knowledge comes from: some is inborn, says **nativism**; all comes through the senses, says **empiricism**; reason is the chief route, says **rationalism**. **Plato** (about 427 to 347 BCE) saw learning as the recollection of inborn knowledge. Aristotle, an empiricist, gave the **laws of association**, similarity, contrast and contiguity: "Sachin" calls up "Tendulkar" by contiguity.

Rene Descartes (1596 to 1650) saw the body as a machine run by the reflex, yet gave humans a separate thinking mind that interacts with the body through the pineal gland: **dualism**. He also held **innate ideas** (God, self, infinity). His reflex led to behaviourism, and his nativism, through Kant, to Gestalt psychology.

John Locke (1632 to 1704) answered in 1690 that the mind at birth is "white paper", a **tabula rasa**, filled by sensation and reflection. Simple ideas combine into complex ones; primary qualities (size, shape, motion) lie in objects, secondary qualities (colour, taste, warmth) in the perceiver. Building all mental life from such links became the doctrine of **associationism**, founded by **David Hartley** (1749). **James Mill** (1773 to 1836) held a complex idea to be the sum of its parts, mental mechanics; his son **John Stuart Mill** (1806 to 1873) held that parts fuse into something new, as hydrogen and oxygen make water: **mental chemistry**. Before Locke, **Thomas Hobbes** (1588 to 1679) was a materialist; after him, **George Berkeley** (1685 to 1753) held that to be is to be perceived, **David Hume** (1711 to 1776) gave the laws of resemblance, contiguity and cause and effect, and **Alexander Bain** founded the journal *Mind* (1876).

German rationalists made the mind active. **Gottfried Wilhelm Leibniz** (1646 to 1716), who built reality from active units called monads, held that unnoticed *petites perceptions* (each wave in the sea's roar) add up to clear awareness, which he called apperception. **Johann Friedrich Herbart** (1776 to 1841) added a threshold of consciousness and the **apperceptive mass**, the existing ideas that must absorb a new idea for it to be understood. **Immanuel Kant** (1724 to 1804) gave the mind innate categories and denied that psychology could be a science.

On mind and body, interactionism (Descartes) lets each act on the other; **psychophysical parallelism** (Leibniz, Wundt) has them run in step without causation; epiphenomenalism (**Thomas Henry Huxley**, 1825 to 1895) makes mind a by-product of the brain; and monism admits one substance, or, in the double-aspect theory of **Baruch Spinoza** (1632 to 1677), one reality with two faces. Ontology asks what exists; epistemology, how we know.

> **! EXAM TRAP**
>
> Tabula rasa is **Locke**; Plato, Descartes and Kant were nativists. Empiricists draw knowledge from **sensory experience**; logical reasoning and contemplation are rationalist routes, and theories are products of knowledge. A practice key says Descartes' dualism influenced **behaviourism and Gestalt psychology**, not empiricism and rationalism. Post-positivism treats knowledge as fallible; scientific realism holds unobservables real. The apperceptive mass (Herbart's term, credited to Wundt by IHBAS 2025) is **pure perception and pre-existing ideas**, not associated elements (associationism), "sensation and pure perception" or Helmholtz's unconscious inference.

Physiological and biological roots ○

The **Bell-Magendie law** (**Charles Bell**, 1811; **Francois Magendie**, 1822) split the spinal nerve roots into sensory (dorsal) and motor (ventral). By the doctrine of specific nerve energies of **Johannes Muller** (1801 to 1858; *Handbook of Human Physiology*, 1833 to 1840), a sensation's quality depends on the nerve excited: press on the eye and you see light. **Hermann von Helmholtz** (1821 to 1894) timed the nerve impulse at about 27 metres per second (1850), called perception unconscious inference, and built the 1802 idea of **Thomas Young** into the trichromatic theory of colour vision. Localisation ran from the phrenology of **Franz Joseph Gall** and **Johann Gaspar Spurzheim** (character read from skull bumps, about 1800) through the

ablation studies of **Pierre Flourens** (1820s) to **Paul Broca**'s speech area (1861), **Carl Wernicke**'s comprehension area (1874) and the motor cortex of **Gustav Fritsch** and **Eduard Hitzig** (1870).

Charles Darwin (1809 to 1882; *On the Origin of Species*, 1859; *The Expression of the Emotions*, 1872) gave psychology continuity between species, the question of function and an interest in variation. **Francis Galton** (1822 to 1911) measured that variation, founding the study of individual differences: *Hereditary Genius* (1869), twin studies, an anthropometric laboratory (1884), correlation, the questionnaire, word association and eugenics. **George John Romanes** founded comparative psychology on anecdotes (1882), and **Conwy Lloyd Morgan**'s canon (1894) demanded parsimony: never a higher faculty where a lower one will do.

> ! EXAM TRAP
>
> Helmholtz is keyed for the **trichromatic theory**; the JND is Weber's, the law of effect Thorndike's, the unconscious mind Freud's.

Measuring the mind ●●

Kant was answered at Leipzig. **Ernst Heinrich Weber** (1795 to 1878) found that the just noticeable difference is a constant fraction of the standard stimulus. **Gustav Theodor Fechner** (1801 to 1887), after an insight on 22 October 1850, made sensation grow with the logarithm of intensity ($S = k \log I$) and founded **psychophysics** in *Elements of Psychophysics* (1860). (Seeing a contour needs a minimum intensity and exposure time, not a minimum reaction time; Section 1.3.) In 1868 **Franciscus Cornelis Donders** (1818 to 1889) subtracted the time of a simple reaction from that of one with an added stage, such as identifying the stimulus, founding **mental chronometry**. **Hermann Ebbinghaus** (1850 to 1909), working alone, measured memory with nonsense syllables (1885; Section 1.7).

> ! EXAM TRAP
>
> The founder of psychophysics is **Fechner**, not Weber, Wundt or Ebbinghaus. Ebbinghaus, Galton and Freud each worked independently as pioneers and each devised new methods (nonsense syllables; the questionnaire and word association; free association), but Freud founded no laboratory. The Panwar bank keys (d), "(a) and (c)"; RML 2020 keyed (a), "pioneers", alone. Both statements are true, so prefer (d).

Wundt and the Leipzig laboratory ●●●

Wilhelm Maximilian Wundt (1832 to 1920), a physician who had assisted Helmholtz, announced a new science in *Principles of Physiological Psychology* (1873 to 1874), "physiological" meaning borrowed experimental method. At Leipzig his students began original research in 1879, the conventional birth of psychology, and the journal *Philosophische Studien* followed in 1881. Weber, Fechner and Helmholtz were precursors; Wundt founded the discipline by giving it a laboratory, a journal and students.

He studied **immediate experience**, sensations and simple feelings that the mind actively organises through attention and will; hence **voluntarism**. Contents in the broad field of consciousness are perceived, those in its focus apperceived: the ceiling fan is perceived, the question paper apperceived. By **creative synthesis** a whole gains new properties, as a melody ex-

ceeds its notes. His **tridimensional theory of feeling** (1896) has three axes: pleasure-displeasure, tension-relaxation and excitement-depression. His method was **introspection** under experimental control (*innere Wahrnehmung*), short judgements of signalled stimuli, not armchair self-observation (*Selbstbeobachtung*). Language, myth and custom he studied in the ten-volume **Volkerpsychologie** (folk psychology, 1900 to 1920). His students included Hall, Cattell, Titchener, Kulpe, Witmer, Munsterberg, **Emil Kraepelin** (1856 to 1926; psychiatric classification), **Charles Spearman** (1863 to 1945; factor analysis) and **Vladimir Bekhterev** (1857 to 1927; reflexology).

> ! EXAM TRAP
>
> The first laboratory is **Wundt's**, at **Leipzig**, not Vienna, Paris or Cambridge; Galton's (1884) was anthropometric, Weber founded none, Jung used word association clinically, and James's 1875 Harvard room was for demonstration. *The Principles of Psychology* (1890) is **James's**. Introspection belongs to **Wundt** (Watson attacked it; Wertheimer founded Gestalt) and to **structuralism** (functionalism added tests, behaviourism rejected it, psychoanalysis used free association). Spearman is to factor analysis as Wundt is to **introspection** (humanism: Maslow, Rogers). Folk psychology (RML 2024) is **Wundt's**; Pavlov found conditioning, Roentgen X-rays.

Structuralism and its rivals ●●

Edward Bradford Titchener (1867 to 1927) took a stricter version to Cornell and named it **structuralism** (1898): consciousness analysed into elements, aiming at a "periodic table" of the mind. Its elements were sensations, images and affections, varying in quality, intensity, duration, clearness and extensity; of Wundt's three feeling dimensions he kept only pleasantness-unpleasantness. Introspection was "a highly specialised form of self-observation"; its cardinal sin, the **stimulus error**, was naming the object ("an apple") instead of describing the experience (a glossy red patch). By his **context theory of meaning**, perception is a complex of sensory and imaginal processes. His society, the Experimentalists (1904), barred women.

At Wurzburg **Oswald Kulpe** (1862 to 1915) found **imageless thought**, thought without images, and showed that the task (*Aufgabe*) steers thought through what Narziss Ach called a determining tendency, creating a set (*Einstellung*): "Mysuru" brings "palace" if you want a building, "Karnataka" if a state. The act psychology of **Franz Brentano** (1874) studied mental acts and their intentionality, their pointing at objects; **Carl Stumpf** (1848 to 1936) taught the Gestalt founders; **Georg Elias Muller** (1850 to 1934) extended memory research at Gottingen. Structuralism died with Titchener in 1927: introspection could not settle disputes, and the school excluded animals, children and applied work. Its topics returned in cognitive psychology.

> ! EXAM TRAP
>
> Introspection is **self-observation**, not self-perception (a later social-psychology theory), self-thinking or attention. "Sensory and imaginal" perception is **Titchener**; Bruner and Schafer were "New Look" perception researchers, Kimble a learning theorist. Structuralism aimed at a **periodic table of elements**; "understanding consciousness" is too general and "behaviour, not mind" is behaviourism. Its heir is **cognitive psychology**, not neuroscience, behaviourism or functionalism.

Functionalism and the American pioneers ●●●

Americans asked what the mind is *for*. Their answer, **functionalism**, studies the mind as it functions in adapting the organism to its environment; it drew on Darwin, on pragmatism (ideas judged by their consequences) and on a demand for useful science.

William James (1842 to 1910), father of American psychology and inspiration of functionalism, wrote *The Principles of Psychology* (1890). It described the **stream of consciousness** ("It is nothing jointed; it flows"), personal, changing, continuous and selective; habit, "the enormous fly-wheel of society"; the self, split into the material, social and spiritual "Me" and the knowing "I" or pure ego; and, with **Carl Lange** (1885), the James-Lange theory of emotion (Section 1.11).

At Chicago, **John Dewey** (1859 to 1952) attacked the **reflex arc** in 1896: stimulus and response are not separate links, since a child's burn changes how the next flame is seen. **James Rowland Angell** (1869 to 1949) defined functional psychology (1906) and taught Watson; **Harvey A. Carr** (1873 to 1954) took the adaptive act as its unit. At Columbia, **James McKeen Cattell** (1860 to 1944) measured reaction time and individual differences and coined **mental test** (1890), **Robert Sessions Woodworth** (1869 to 1962) wrote *Dynamic Psychology* (1918) and added the organism to stimulus and response (S-O-R), and **Edward Lee Thorndike** (1874 to 1949) studied animal learning (Section 1.6).

Granville Stanley Hall (1844 to 1924) earned the first American doctorate in psychology (1878), opened the first American research laboratory (Johns Hopkins, 1883), founded the *American Journal of Psychology* (1887) and was first president of the American Psychological Association (APA, 1892); he called adolescence "storm and stress", explained by recapitulation (each child relives human evolution). **Lightner Witmer** (1867 to 1956) opened the first **psychological clinic** (Pennsylvania, 1896) and first printed "clinical psychology" (1907). **Hugo Munsterberg** (1863 to 1916) founded forensic and industrial psychology. **Mary Whiton Calkins** (1863 to 1930), refused a Harvard doctorate as a woman, became the first woman APA president (1905); **Margaret Floy Washburn** (1871 to 1939) was the first woman PhD in psychology (1894) and wrote *The Animal Mind*; **Leta Stetter Hollingworth** (1886 to 1939) disproved claims of female inferiority.

KEY STUDY: THE FIRST CASE AT THE FIRST CLINIC (1896)

A teacher brought Witmer a fourteen-year-old "chronic bad speller". Testing his vision, memory and reading rather than labelling him, Witmer found poor eyesight behind part of the problem; corrected vision and individual teaching helped. Assess first; seek the simplest cause.

Functionalism dissolved into the mainstream: Angell's student Watson dropped consciousness to found behaviourism (Section 1.2), and the functional question passed to evolutionary psychology.

> **! EXAM TRAP**
>
> Functionalism ("how people work, play and adapt") is **William James**, not Titchener, Watson, Hall or "William Wundt"; Darwin inspired it but led no school, and Dewey or Angell is right only for the formal Chicago school. It studies the **mind** adapting the **organism**, grew from **Darwin's evolution** (not Freud, Pavlov or Gestalt laws) and gave rise to the **evolutionary perspective**. Dewey's arc is the **reflex arc**; "efferent" is only the outgoing half; "association" and "dynamic" arcs are invented. "Clinical psychology" in print: **Witmer**, not Freud, Erikson or Cattell. APA: **1892**, not 1896. First woman APA president: **Calkins**, not Washburn, Hollingworth or the psychoanalyst Karen Horney.

Zeitgeist, great persons and "fathers"

Why credit one man and one year? The **great-person theory** (**Thomas Carlyle**, 1841) credits exceptional individuals; the **Zeitgeist** ("spirit of the times") view, championed by **Edwin Garrigues Boring** (1886 to 1968), credits ripe times, pointing to multiple discovery, as when Darwin and **Alfred Russel Wallace** reached natural selection independently (1858). "Father" may mean founder, namer, chief influence or first in a country, so read the qualifier (Table 1.1). Psychology's arrival in India, through N. N. Sengupta at Calcutta (1915, 1916), is told in Section 1.2.

Table 1.1 Who is the "father" of what.

Father of	Usual answer	Common bait
Scientific, experimental psychology	Wundt	Aristotle; Fechner
Psychophysics	Fechner	Weber
Associationism	Hartley	Aristotle
Structuralism	Titchener	Wundt (in Indian texts)
American psychology; functionalism	James	Hall; Titchener (named it)
Behaviourism	Watson	Pavlov; Skinner
Clinical psychology	Witmer	Freud; Kraepelin
Individual differences	Galton	Cattell
Intelligence testing	**Alfred Binet** (1905)	Galton; Cattell
Modern psychology in India	N. N. Sengupta	G. Bose; M. V. Gopalaswamy

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Thinking and remembering are internal, covert _____ processes.
2. Locke called the mind at birth a tabula _____.
3. Herbart's store of existing ideas is the _____ mass.
4. Wundt's system, stressing will and attention, is _____.
5. Naming "an apple" instead of describing the experience is the _____ error.
6. Dewey's 1896 paper attacked the _____ concept.
7. The first psychological clinic (1896) was opened by Lightner _____.

Answers: (1) mental, (2) rasa, (3) apperceptive, (4) voluntarism, (5) stimulus, (6) reflex arc, (7) Witmer.

1.2 Schools and Systems, the Cognitive Revolution and Indian Psychology

HIGH YIELD · 112 QS

Section 1.1 ended with structuralism and functionalism. This section meets the rival schools that followed (about 1900 to 1960), the mind's return in the cognitive revolution, and the Indian traditions, which asked the same questions two thousand years earlier.

Schools, paradigms and the four forces ●●

A school of thought is a group of psychologists sharing assumptions about subject matter, method and explanation. **Thomas Samuel Kuhn** (1922 to 1996), in *The Structure of Scientific Revolutions* (1962), argued that a mature science works within one paradigm, a shared framework of problems and methods, until anomalies force a paradigm shift; a field of competing schools, as psychology was, is a **preparadigmatic science**. The humanists later numbered the schools as "forces": psychoanalysis first, behaviourism second, humanistic psychology third, transpersonal psychology fourth.

The schools divided on enduring issues, which the historian **Robert Irving Watson** (1909 to 1980) called **prescriptions** (1967): mind-body, nature-nurture, free will-determinism, mechanism-vitalism, molar-molecular, nomothetic-idiographic (general laws or the unique individual), objectivism-subjectivism and empiricism-rationalism. Rene Descartes fed two schools: his reflexes led to behaviourism, his innate ideas to Gestalt nativism.

! EXAM TRAP

The **fourth force** is **transpersonal** psychology; "experiential" is not a force. **Two Watsons**: John B. founded behaviourism (1913); Robert I. proposed the prescriptions (1967).

Behaviourism: Watson, the neobehaviourists and Skinner ●●●

Behaviourism threw consciousness out of psychology altogether. Animal psychology (Thorndike's puzzle-box cats, 1898) and Russian objective psychology had prepared the ground: Pavlov's conditioned reflexes and the reflexology of **Vladimir Mikhailovich Bekhterev** (1857 to 1927), who conditioned movements.

KEY STUDY: WATSON'S MANIFESTO (1913)

John Broadus Watson (1878 to 1958) launched **behaviourism** with "Psychology as the Behaviorist Views It": psychology is "a purely objective experimental branch of natural science", its subject matter behaviour, not consciousness, and "introspection forms no essential part of its methods". Its goal is prediction and control: given the stimulus, to predict the response, and given the response, the stimulus. The same laws hold for humans and animals.

Watson treated thinking as subvocal speech, tiny movements of the larynx; this **peripheralism** puts mental life in the muscles rather than the brain. Such implicit (covert) responses cannot be seen but can be measured with instruments, so covert behaviour is "measurable but not observable", and the "mental processes" in today's definition of psychology are internal, covert processes. Infants, he held, have only three unlearned emotions, fear, rage and love; the rest are conditioned, as he and **Rosalie Rayner** tried to show with Little Albert (1920). His boast "give me a dozen healthy infants" (1925) is environmentalism at its purest.

The neobehaviourists (about 1930 to 1960) drew on logical positivism (only statements that observation can verify are meaningful) and on **operationism**, proposed by the physicist **Percy Williams Bridgman** (1882 to 1961) in 1927: a concept means the operations used to measure it, hence the operational definition. **Edward Chace Tolman** (1886 to 1959) studied molar, purposive acts and inferred intervening variables between stimulus and response, above all the **cognitive map**, a mental representation of the environment. His evidence was **latent learning**: rats that explored a maze unrewarded for ten days matched rewarded rats the day after food appeared (Tolman and Honzik, 1930). **Clark Leonard Hull** (1884 to 1952) built a hypothetico-deductive system of drive reduction in which reaction potential is habit strength multiplied by drive; **John Dollard** and **Neal Elgar Miller** (1950) recast Freud in his terms. **Edwin Ray Guthrie** (1886 to 1959) needed only contiguity and one-trial learning. **Burrhus Frederic Skinner** (1904 to 1990; *Walden Two*, 1948; *Verbal Behavior*, 1957; *Beyond Freedom and Dignity*, 1971), who studied operant behaviour, was a radical ("thoroughgoing") behaviourist: thoughts are real behaviour but not causes, whereas methodological behaviourism simply sets them aside. **Zing-Yang Kuo** (1898 to 1970) rejected instinct outright.

From the late 1950s Chomsky's attack (1959, below), instinctive drift (1961) and Kohler's insight weakened behaviourism; it left operational definitions and behaviour therapy, such as **Joseph Wolpe**'s systematic desensitisation by reciprocal inhibition (1958).

! EXAM TRAP

Watson founded behaviourism (not rationalism) and defined psychology as the science of **behaviour** (not soul, mind or consciousness). One Panwar item keys **N. L. Munn**, whose textbooks popularised "the scientific study of behaviour": prefer Watson if offered, else Munn (not Woodworth, of the S-O-R formula, or Wundt). **Cognitive map and latent learning** are Tolman's, not Kohler's (insight), Harlow's (learning sets), Skinner's, Gould's (he only tested the idea in honeybees, 1986) or Bandura's. **Dollard and Miller** = early social learning; **Bandura and Walters** (1963) = later.

Gestalt psychology and Kurt Lewin ●●●

A second rebellion, in Germany, kept consciousness but denied that experience is built from elements.

KEY STUDY: WERTHEIMER'S PHI PHENOMENON (1912)

At Frankfurt, **Max Wertheimer** (1880 to 1943) flashed two lines in quick succession. About 30 milliseconds apart they looked simultaneous, about 200 apart successive, but at about 60 a single line seemed to move: the **phi phenomenon**. The motion lay in neither stimulus, so no sum of sensations could produce it. The paper founded Gestalt psychology.

Gestalt means an organised whole, and the school's slogan is that the whole is different from (popularly, greater than) the sum of its parts: a square has closedness and an inside that none of its four lines has. Field theory holds that perception, thought and behaviour are set by the whole field of forces present at the moment. Against the elements of structuralism and the reflexes of behaviourism, Gestalt studied wholes and relations and leaned nativist; its laws of organisation (Pragnanz, figure and ground) are in Section 1.4.

Wolfgang Kohler (1887 to 1967), stranded on Tenerife from 1913 to 1920, saw the chimpanzee Sultan join two sticks to reach a banana: **insight**, a sudden reorganisation of the perceptual field (German 1917; *The Mentality of Apes*, 1925). His hens learned the relation "darker" rather than a shade (**transposition**), and his isomorphism matched the structure of experience to brain fields. **Kurt Koffka** (1886 to 1941; *Growth of the Mind*, 1921; *Principles of Gestalt Psychology*, 1935) separated the geographical environment from the behavioural one, the world as perceived. Wertheimer's *Productive Thinking* (1945) set understanding against rote. Wertheimer, Kohler and Lewin fled the Nazis to America, where Koffka had settled in 1927.

Kurt Lewin (1890 to 1947) carried field theory into motivation and social life. In B = f(P, E), behaviour is a function of the person and the environment, together the **life space**: the totality of facts determining behaviour at a moment, the environment being psychological, as it exists for that person. Topological psychology maps its regions and boundaries, hodological space its paths; valences attract or repel, and vectors push. His students found that interrupted tasks are better recalled (**Bluma Zeigarnik**, 1927) and spontaneously resumed (**Maria Ovsiankina**, 1928). He coined group dynamics, compared autocratic, democratic and laissez-faire leaders (Social chapter), and founded **action research**, a cycle of planning, acting, observing and reflecting with the people affected; he is the father of modern social psychology.

Gestalt therapy is different: an experiential, here-and-now therapy of **Frederick Salomon Perls** (Fritz Perls, 1893 to 1970), Laura Perls and Paul Goodman, famous for the empty chair, in which the client addresses an imagined person or a disowned part of the self.

! EXAM TRAP

Founders: Wertheimer, Kohler, Koffka (not Watson, James, Pavlov or Lewin); the school most important for perception. **Empty chair** = Gestalt therapy, not psychoanalysis, cognitive-behavioural or dialectical behaviour therapy, or person-centred therapy. Loose keys stretch "Gestalt" to cover the therapy: "here-and-now" is keyed to Gestalt psychology, and "Perls was a proponent of the Gestalt perspective" is accepted because "Gestalt focuses on wholeness in the present moment". A management stem adding that the parts relate to each other and to the whole means **systems theory**. **Life space** = person plus psychological environment, not groups. Lewin founded **action research**, not psychoanalysis, psychophysics or functionalism.

Psychoanalysis: the first force ●●

Sigmund Freud (1856 to 1939) began as a neurologist treating hysteria (paralysis or blindness without nerve damage). In Paris in 1885 **Jean-Martin Charcot** (1825 to 1893) showed him hysterical symptoms appearing and vanishing under hypnosis; **Pierre Janet** (1859 to 1947) explained hysteria as dissociation, and **Hippolyte Bernheim** (1840 to 1919) of Nancy saw hypnosis as suggestion. **Josef Breuer** (1842 to 1925) found that his patient Anna O. lost symptoms by talking about their origins, the "talking cure" or cathartic method, and their *Studies on Hysteria* (1895) began psychoanalysis. Freud turned to free association, called dreams "the royal road" to the unconscious (*The Interpretation of Dreams*, 1900) and lectured at Clark University in 1909.

His topographic model divides the mind into conscious, preconscious and unconscious; his **structural model** (1923) into the id (impulsive, irrational, seeking immediate gratification), the ego (reality) and the superego (conscience), the ego guarding against anxiety with defence mechanisms such as repression. (Freud's neurotic anxiety is the fear that id impulses will break through.) Dissenters and neo-Freudians (Personality chapter) stressed relationships and culture: Adler (individual psychology), Jung (analytical psychology, the collective unconscious), Karen Horney (**basic anxiety**, a pervasive loneliness and helplessness), Fromm (humanistic psychoanalysis), Sullivan (interpersonal psychiatry), Erikson (psychosocial stages), Anna Freud (ego defences) and Melanie Klein (object relations). From psychoanalysis, the first force, descends the psychodynamic perspective, which explains behaviour, and many disorders, by unconscious urges and the hidden struggle over their expression.

> **! EXAM TRAP**
>
> Freud began as a **neurologist**. **Father of psychoanalysis**: Freud, not Bernheim, Adler or Jung. Psychoanalytic theory is Freud's, **psychosocial** theory Erikson's. Gratification, irrational, impulsive = **id**. Horney's **basic anxiety** is not Freud's neurotic anxiety, existential anxiety or "innate anxiety" (no such technical term).

Humanistic, existential and positive psychology ●●●

Against the determinism of psychoanalysis (instinct) and behaviourism (reward), **Abraham Harold Maslow** (1908 to 1970) launched a third force and studied self-actualisation, the fulfilment of one's potential (Section 1.10); **Carl Ransom Rogers** (1902 to 1987) described an innate actualising tendency that thrives on unconditional positive regard. **Charlotte Buhler** (1893 to 1974) added a lifespan view and **Rollo May** (1909 to 1994) existentialism. The Association for Humanistic Psychology formed in 1961 or 1962 (sources differ), and the Old Saybrook conference (1964) marked the movement's formal birth. Humanistic psychology emphasises free will, the whole person, an innate tendency to grow and reaching maximum potential, understood from inside the person's experience: **phenomenology**, after **Edmund Husserl** (1859 to 1938), describes experience by bracketing assumptions about its causes. Disorder means blocked growth; healing comes from acceptance, respect and belief in the person's potential, not from reinforcement or uncovering unconscious conflict. Critics call humanistic and phenomenological approaches vague and blind to the wider social and cultural contexts of experience.

Existential psychology starts from the givens no one escapes, which **Irvin David Yalom** (1980) named the four ultimate concerns: death, freedom, isolation and meaninglessness.

Soren Kierkegaard (1813 to 1855) called anxiety the dizziness of freedom; **Martin Heidegger** (1889 to 1976) named human existence **Dasein**, being-in-the-world (1927); **Jean-Paul Sartre** (1905 to 1980) held that existence precedes essence, so each person must keep recreating himself and meaning is never fixed. **Ludwig Binswanger** (1881 to 1966) and **Medard Boss** (1903 to 1990) built Daseinsanalysis on Heidegger. **Viktor Emil Frankl** (1905 to 1997) founded **logotherapy**: the will to meaning is the primary human motive, and its frustration brings an existential vacuum or a **noogenic neurosis**. May (*The Meaning of Anxiety*, 1950) separated normal, usable anxiety, existential anxiety included, from neurotic anxiety.

Maslow later foresaw a fourth force (1968), **transpersonal psychology**, the study of peak experiences, meditation and self-transcendence. Positive psychology, the science of positive experiences, traits and institutions, was launched by **Martin Elias Peter Seligman** (born 1942) as President of the American Psychological Association in 1998, with **Mihaly Csikszentmihalyi** (1934 to 2021), known for flow. Its themes include the PERMA model of well-being (Seligman, 2011; positive emotion, engagement, relationships, meaning and accomplishment), C. R. Snyder's hope (the will and the ways to reach a goal), which its interventions promote, and **savouring**, deliberately enhancing the enjoyment of positive experiences.

> ! EXAM TRAP
>
> **Third force** = humanism (Maslow, Rogers), not cognitivism or the biopsychosocial model. "Uncovering unconscious fears" is psychoanalytic, not existential. **Dasein** and **noogenic anxiety** belong to the existential school. **Will to meaning** = logotherapy, not Ellis's rational emotive behaviour therapy, acceptance and commitment therapy or the broader existential-humanistic therapy. Positive psychology was founded by **Seligman**, not Gardner (multiple intelligences), Pinker (language and evolutionary psychology) or Bandura; its interventions promote **hope**; savouring is not avoidance, relaxation or trauma work.

The cognitive revolution and today's perspectives ●●●

Other critics charged behaviourism with leaving out the mind. Tolman's maps, Kohler's insight and **Karl Spencer Lashley**'s (1890 to 1958) argument of 1948 that rapid sequences such as speech need central plans were anomalies, and the computer made inner states respectable. The symbolic birth year of the **cognitive revolution** is 1956: **Allen Newell** and **Herbert Alexander Simon**'s Logic Theorist modelled problem solving, **Noam Chomsky** (born 1928) presented formal models of language, **George Armitage Miller** (1920 to 2012) gave "the magical number seven, plus or minus two", and the Dartmouth project launched artificial intelligence. **Donald Eric Broadbent** (1926 to 1993) produced the first major information-processing theory (1958); Chomsky's 1959 review of Skinner's *Verbal Behavior* argued that children produce sentences never reinforced; and **Ulric Neisser** (1928 to 2012), the father of cognitive psychology, named the field in *Cognitive Psychology* (1967).

Cognitive psychology studies internal mental processes such as attention, memory and problem solving through the **information-processing approach**, which treats the mind as a computer passing information through stages of limited capacity. Its forerunners were **Frederic Charles Bartlett** (schemas, 1932) and **Jean Piaget** (1896 to 1980), who studied children's thinking from the 1920s. Cognitive neuroscience asks how the brain does it, and **connectionism**, or parallel distributed processing (David Rumelhart and James McClelland,

1986), stores knowledge as weights across many units working at once. **Albert Bandura**'s (1925 to 2021) self-efficacy (1977), belief in one's capacity to act, and the cognitive therapies of Albert Ellis and Aaron Beck, which target maladaptive thoughts and cognitive distortions, fed cognitive-behavioural therapy (CBT; Clinical chapter).

By the 1970s schools had given way to coexisting perspectives (Table 1.2, applied to Ravi, a Mysuru student anxious before every examination). Some lists add a developmental perspective, the study of change as people mature. Cross-cultural psychology compares processes across cultures, cultural psychology studies how culture and mind make each other up, indigenous psychology builds theory from within one culture, and **eclecticism** selects concepts and techniques from several schools to fit the problem.

Table 1.2 The contemporary perspectives, applied to one case of examination anxiety.

Perspective	Explains behaviour through	Ravi's anxiety comes from
Biological (neurobiological)	Brain, nervous system, genes	An over-reactive threat system
Evolutionary	Natural selection	An ancient alarm misfiring
Psychodynamic	Unconscious conflict, early childhood	Fear of parental disapproval
Behavioural	Stimulus-response conditioning	Failure paired with humiliation
Cognitive	Internal mental processes	Catastrophic thoughts
Humanistic	Free will and growth	Worth conditional on marks
Sociocultural	Culture and social context	Competitive family pressure
Biopsychosocial (George Engel, 1977)	All three levels together	All of these, interacting

> **! EXAM TRAP**
>
> **First major information-processing theory**: Broadbent, 1958 (not Treisman, about 1960; Chomsky, 1957; or Watson, 1920). **Pioneer of cognitive psychology** against Kohler, Chomsky and Kohlberg (moral development): **Piaget**. "Behaviour is governed by the brain and nervous system" = **neurobiological**, not psychophysical (Fechner's stimulus-sensation measurement, Section 1.3) or neo-behaviouristic. Cultural differences are studied by **cross-cultural psychologists** (a Panwar key says social psychologists, the vaguer answer).

Indian psychology: the idea and its pioneers ●●●

Psychology in India is the discipline as practised here, long mostly Western psychology on Indian soil. **Indian psychology** is an approach that integrates the classical Indian traditions with psychological concepts. It is consciousness-centred; holistic, treating body, mind, spirit, family and community as one system; transformative; open to disciplined first-person methods such as meditation; and value-laden. In mental health it combines traditional practices such as yoga with modern therapy.

Modern psychology entered India at Calcutta. **Narendra Nath Sengupta** (1889 to 1944) set up the first laboratory there in 1915 and the first Department of Psychology in 1916, and founded the Indian Psychological Association (1925), whose *Indian Journal of Psychology* began in 1926; he is the father of modern psychology in India. **Girindrasekhar Bose** (1887 to 1953) founded the Indian Psychoanalytical Society (1922) and corresponded with Freud; the father of psychoanalysis in India, he proposed the **theory of opposite wishes**, in which every wish is

paired with its opposite, one expressed and one repressed. **M. V. Gopalaswamy** began psychology at Mysore in 1924; Madras (1943) and Pune (1950) followed.

Durganand Sinha (1922 to 1998) urged **indigenisation**, making psychology's concepts and methods fit Indian culture, either from without, by adapting Western tools (as in **Uma Chowdhury**'s Indian Thematic Apperception Test, about 1960), or from within, from the culture's own traditions. **Jai B. P. Sinha** proposed the **nurturant-task leader** (1980), whose warm care depends on hard work; **Udai Pareek** (1925 to 2010) studied role stress; and **Sudhir Kakar** (born 1938) applied psychoanalysis to Indian childhood and society. The Indian Psychology movement dates from the Pondicherry Manifesto (2002). **Matthijs Cornelissen** founded the Indian Psychology Institute; **Koneru Ramakrishna Rao**, **Anand C. Paranjpe** and **Ajit K. Dalal** edited the *Handbook of Indian Psychology* (2008); and Cornelissen, **Girishwar Misra** and **Suneet Varma** edited *Foundations of Indian Psychology* (2011).

> **! EXAM TRAP**
>
> Indian psychology is not Western theory applied exclusively, folklore, or meditation without psychological context. Keys naming Bose (IHBAS) or Uma Chaudhuri (Panwar) for the **first laboratory** are wrong: it was Sengupta's; without Sengupta, choose Bose (Mangal and A. K. Singh are textbook authors; neither they nor Rathi founded a laboratory). **Sequence**: Calcutta (1916), Mysore (1924), journal (1926), Madras, Pune; a key putting the journal before Mysore is wrong. Bose's theory is **opposite wishes** (not repression, dharma-sankata or "guilt theory"). "Amitosh Mukherjee" garbles Asutosh Mookerjee, a Calcutta vice-chancellor. Kakar is keyed for Horney's basic anxiety only as the sole psychoanalyst offered ("Kamala Chaudhuri" there probably garbles Uma Chowdhury). "G. R. S. Rao" as father of Indian psychology (RML 2025) probably means K. Ramakrishna Rao.

Samkhya, the three gunas and the inner instrument ●●●

Indian philosophies (darshanas) are **astika** if they accept the Vedas (Nyaya, logic; Vaisheshika; Samkhya; Yoga; Mimamsa, ritual; Vedanta) and nastika if not (Buddhism, Jainism, the materialist Charvaka); Sufism, Islamic mysticism, lies outside the scheme.

Samkhya, attributed to the sage **Kapila** and set out in Ishvarakrishna's *Samkhya Karika*, is a dualism of **purusha**, pure consciousness, the passive, unchanging witness, and **prakriti**, primordial matter, unconscious, active and ever-changing, from which all forms of life emerge. In purusha's presence prakriti evolves, down to the five great elements (pancha mahabhutas), as Figure 1.2 shows. We suffer by mistaking prakriti's activity for the self ("I am angry"); discriminating the seer from the seen brings liberation, kaivalya.

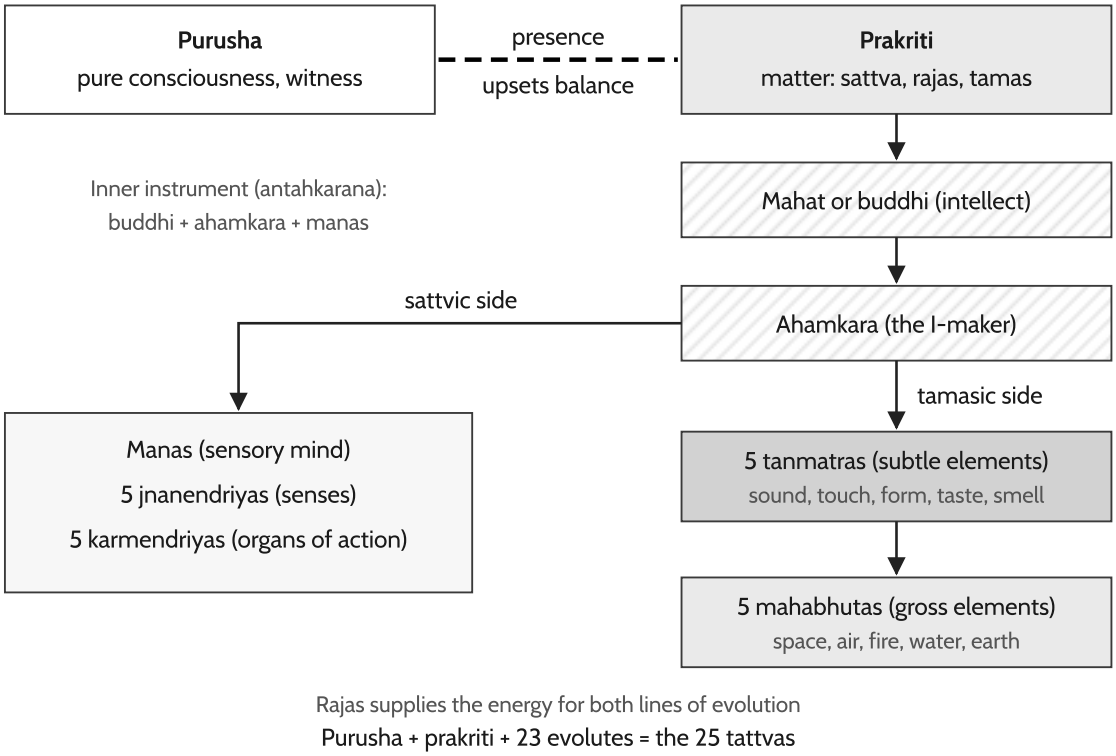


Figure 1.2 Samkhya's map of mind and matter. In the presence of purusha, prakriti evolves into intellect, the I-maker, the sensory mind, the ten organs and the elements. Notice that everything below the top row, including thinking, feeling and the sense of "I", is a product of prakriti; only purusha is conscious.

Prakriti is woven of three strands, the **gunas**, always mixed in varying proportions: sattva (clarity, calm, harmony, balance), rajas (activity, passion, desire, restlessness) and tamas (inertia, lethargy, darkness, dullness, ignorance). The *Bhagavad Gita* (chapters 14, 17 and 18) made them a psychology of personality. Before the same examination, Ananya keeps her plan and sleeps on time (sattva), Rohan flits between six guidebooks and lies awake racing (rajas), and Karthik sleeps till noon, telling himself there is no point (tamas). Nobody is pure; diet, routine and practice shift the balance.

The mind's working part is the **antahkarana**, the inner instrument: manas, the sensory mind that attends and doubts; ahamkara, the I-maker or ego; buddhi, the intellect that decides; and, in later Vedanta, chitta, the store of memory impressions.

> **! EXAM TRAP**
>
> Purusha is **not** dynamic or ever-changing; prakriti is not pure consciousness, unchanging or the intelligent self. The gunas belong to **prakriti** (not purusha, atman, Brahman, samadhi or samskara) and come from **Samkhya** with the *Gita* (not Nyaya, Charvaka or Mimamsa). Bavisha, who has stopped eating and cannot act, shows **tamas alone** (yama is a yoga limb, not a state). Not gunas: vata (a dosha), ahimsa, prakriti, ahamkara, or "karma, dharma, moksha". The ego is **ahamkara**; the *Gita*'s three elements of mind are manas, buddhi and ahamkara, not "prakriti, purusha, kshetra" (kshetra, the field, is the body that the self knows) or "dharma, artha, kama".

Yoga, the self and other Indian paths ●●●

Samkhya says what the mind is made of; yoga says how to change it. **Patanjali**'s *Yoga Sutras* adopt Samkhya's map, and sutra 1.2 defines yoga as **chitta-vritti-nirodha**, the stilling of the mind's fluctuations; the goal is liberation (kaivalya). The five fluctuations (vrittis) are valid cognition, error, verbal imagination, sleep and memory. They are stilled by practice (abhyasa) and non-attachment (vairagya), and driven by five afflictions (kleshas): ignorance (avidya), egoism, attachment, aversion and clinging to life. The method is the eight limbs, in fixed order (Table 1.3).

Table 1.3 The eight limbs of Patanjali's yoga, in order.

Limb	Meaning
1. Yama	Ethical restraints, such as non-violence (ahimsa)
2. Niyama	Personal observances, such as purity
3. Asana	Steady, comfortable posture
4. Pranayama	Breath regulation
5. Pratyahara	Withdrawal of the senses inward
6. Dharana	Concentration on one point
7. Dhyana	Meditation: an unbroken flow of attention
8. Samadhi	Absorption in the object

A student meditating on a candle flame turns her senses away from the hostel's noise and smells (pratyahara), fixes on the flame and brings attention back whenever it drifts (dharana), finds it flowing unbroken (dhyana), and for a moment loses even "I am meditating" (samadhi).

The *Gita*'s principle of disinterestedness is **nishkama karma**, acting without attachment to the fruits. It drives karma yoga, beside the paths of knowledge (jnana) and devotion (bhakti), and answers Arjuna's dharma-sankata, a dilemma in which every action breaks some duty. The Upanishads call the true self **atman**, eternal consciousness distinct from the body and from passing states of mind; **Adi Shankaracharya**'s Advaita Vedanta (eighth century CE) makes it one with Brahman, ultimate reality, and calls the world of many things maya, appearance. The *Mandukya Upanishad*'s four states are waking (jagrat), dreaming (svapna), deep sleep (sushupti) and **turiya**, "the fourth", the consciousness that witnesses the other three. The *Taittiriya Upanishad*'s five sheaths (**panchakosha**), outer to inner, are annamaya (the food body), pranamaya (breath), manomaya (mind and senses, the sheath of perception), vijnanamaya (intellect) and anandamaya (bliss); the middle three form the subtle body (linga sharira).

Buddhism (**Siddhartha Gautama**, the Buddha, traditionally 563 to 483 BCE) teaches four Noble Truths: suffering, its cause in craving, its cessation and the Noble Eightfold Path. Its anatta denies a permanent self, its Abhidhamma catalogues states of mind, and vipassana (insight meditation) trains sati, mindfulness. Sufi psychology refines the nafs, the lower self, from the commanding ammara to the tranquil mutma'inna and the ego's passing away (fana). **Sri Aurobindo** (1872 to 1950) put the **psychic being**, the evolving soul, at the centre of integral yoga, behind an outer being of body, vital (life-energy, desire and emotion) and

mind. Ayurveda's three doshas (vata, pitta, kapha) govern the body. Karma's impressions (samskaras) work like habits, and dharma, the order of duties, heads the four aims of life (purusharthas): dharma, artha (wealth), kama (pleasure) and moksha (liberation).

CLINICAL LINK

Buddhist mindfulness underlies third-wave CBT, such as **Marsha Linehan**'s dialectical behaviour therapy, which blends CBT with Zen mindfulness and dialectical philosophy (Clinical chapter). Psychoanalysis is no wave of CBT.

! EXAM TRAP

Goal of yoga: control of the mind and kaivalya, not fitness, cognitive enhancement or emotion regulation only. **Last four limbs**: pratyahara, dharana, dhyana, samadhi. One point = **dharana**; senses turned inward = **pratyahara** (not pranayama or niyama); karma and jnana are not limbs. Nishkama karma is the *principle*, karma yoga the path. **Atman** is not the body, a passing state or Jung's collective unconscious; turiya is not deep sleep. **Annamaya** is outermost, **manomaya** perceptual; the atman is never a sheath, so a key adding it to the subtle body is wrong.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Kuhn called a field of competing schools a _____ science.
2. Tolman's unrewarded rats showed _____ learning.
3. Lewin's life space is the person plus the _____ environment.
4. Frankl called neurosis from frustrated meaning _____ neurosis.
5. India's first psychology laboratory (1915) was set up by _____.
6. The guna of inertia and ignorance is _____.
7. The yoga limb that withdraws the senses inward is _____.

Answers: (1) preparadigmatic, (2) latent, (3) psychological, (4) noogenic, (5) Narendra Nath Sengupta, (6) tamas, (7) pratyahara.

1.3 Sensation and Psychophysics MEDIUM YIELD · 40 QS

Everything you know about the world arrives as physical energy. This section follows that energy into experience, shows how sensation is measured, and then takes each sense in turn.

From energy to experience ●●

In **sensation** the sense organs detect physical energy and convert it into neural signals; in **perception** the brain organises and interprets those signals. Optical illusions therefore distort perception, not sensation: in the Muller-Lyer figure the receptors register the lines correctly (Section 1.4). The structuralist **Edward Bradford Titchener** (1867 to 1927) defined a perception as a complex of sensory and imaginal elements.

A receptor is a cell specialised for one form of energy, and its conversion of that energy into neural signals is **transduction**. Because all impulses look alike, **Johannes Peter Muller** (1801 to 1858) proposed the **doctrine of specific nerve energies** (today, labelled-line coding): the quality of a sensation depends on which nerve is stimulated, not on what stimulates it, so pressing a closed eyelid makes you see light. Intensity is coded differently. By the **all-or-none law** an action potential fires at full size once threshold is reached; a stronger stimulus never makes a bigger impulse, and a sub-threshold one makes none, not a miniature or graded one. Intensity is therefore carried by rate coding (faster firing) and by recruiting more neurons. Every sense except smell relays to the cortex through the thalamus (vision via its lateral geniculate nucleus, hearing via the medial). Finally, **sensory adaptation** is the fading of response to an unchanging stimulus, which is why you soon stop feeling your wristwatch.

> **! EXAM TRAP**
>
> Illusions distort *perception*; "translation" and "adaptation" are bait. "Sensory and imaginal" is Tit*chener*, not Bruner (Section 1.4). Specific nerve energies is *Muller*, not Helmholtz. The relay for every sense but smell is the *thalamus*, not the hypothalamus (drives), medulla (breathing), corpus callosum (joins the hemispheres), basal ganglia (movement) or hippocampus (memory).

Thresholds and the psychophysical laws ●●

Gustav Theodor Fechner (1801 to 1887) founded **psychophysics**, the measurement of the relation between stimuli and sensations, with his *Elements of Psychophysics* (1860). The **absolute threshold** (or limen) is the minimum intensity a person can detect. It is not a switch: as a faint tone grows stronger, "yes" answers climb gradually along an S-shaped psychometric function, since neurons fire spontaneously. The threshold is therefore defined as the intensity detected on *50 percent* of trials; weaker stimuli are subliminal. **Eugene Galanter** (1962) gave the classic examples: a candle flame 48 kilometres away on a dark night, a watch ticking at 6 metres. The difference threshold, or **just noticeable difference** (JND), is the smallest change detected 50 percent of the time.

Ernst Heinrich Weber (1795 to 1878), comparing lifted weights in 1834, found the JND to be a constant fraction of the standard: **Weber's law**, $\Delta I/I = k$. If k for weights is 0.02, 500 grams needs 10 grams more to feel heavier, but 5 kilograms needs 100. The fraction is finest for pitch and among the coarsest for taste and smell. Fechner then assumed that every JND is an equal unit of sensation; counting JNDs gives **Fechner's law**, $S = k \log I$, so each doubling of light adds about one equal step of brightness. **Stanley Smith Stevens** (1906 to 1973) instead had people give each stimulus a number proportional to its strength (**magnitude estimation**) and found **Stevens' power law** (1957), $S = kI^n$: exponents below 1 compress (brightness, about 0.33), exponents above 1 expand (electric shock, about 3.5). Cross-modality matching (squeezing a hand grip as hard as a sound is loud) confirmed them (Table 1.4).

Table 1.4 The three psychophysical laws compared.

Law	Formula	Main limitation
Weber (1834)	$\Delta I/I = k$	Fails near threshold
Fechner (1860)	$S = k \log I$	Assumes equal JNDs; no expansion

Law	Formula	Main limitation
Stevens (1957)	$S = kI^n$	Swayed by range and number habits

> **! EXAM TRAP**
>
> Minimum detectable stimulus: *absolute threshold*; smallest detectable change: *difference threshold*. "Sensory adaptation point", "perceptual boundary" and "signal detection of response" are invented; response bias is a signal detection term, and the phi phenomenon is apparent movement (Section 1.4). Weber's JND is a *constant fraction*, not a fixed amount. The equal-JND assumption is *Fechner's*, not Weber's. The father of psychophysics is Fechner, not Ernst Weber (whose JND he built on), Wilhelm Wundt (experimental psychology), Hermann Ebbinghaus (memory, inspired by Fechner) or Edward Tolman (learning).

Measuring thresholds and signal detection ●●

Fechner's **method of limits** changes the stimulus in steps, in alternating ascending and descending series. Its errors of *habituation* (keeping an answer too long) and *anticipation* (switching too early) are cancelled by counterbalancing. The **method of constant stimuli** presents fixed values many times in random order: most accurate, but slow. In the method of adjustment (average error) the observer sets the stimulus: fastest, but least precise. Modern staircases weaken the stimulus after each "yes" and strengthen it after each "no".

Yet a threshold mixes what an observer can sense with how readily the observer says "yes", as wartime radar operators showed; **signal detection theory** (SDT), set out by **David M. Green** and **John A. Swets** in 1966, separates the two. Noise alone and signal plus noise produce overlapping distributions of internal evidence, and the observer says "yes" above a cut-off, the criterion. "Yes" to a signal is a *hit*, "no" a *miss*; "yes" to noise alone is a *false alarm*, "no" a *correct rejection*. Sensitivity, or **d prime** (d′), is the distance between the distributions, *z*(hit rate) minus *z*(false alarm rate): zero when hits equal false alarms, larger as hits outnumber them. The criterion (β or *c*), or **response bias**, follows expectations and payoffs. A *liberal* observer (a smoke alarm) gives many hits and false alarms; a *conservative* one (a court) few false alarms but more misses. Plotting hit rate against false alarm rate as the criterion moves gives the **receiver operating characteristic** (ROC) curve. It is bowed, not straight as a fixed threshold predicts, which is why SDT replaced the threshold. Lie detection and claims of subliminal perception are judged the same way.

> **! EXAM TRAP**
>
> Telling a signal from background noise is *sensitivity* (*d′*); willingness to say "yes" is the *criterion* (β); the ROC curve plots both; the JND and Weber's constant are classical psychophysics. SDT added the *response criterion*. "Receiver-operant characteristics in communication studies" (IHBAS 2025) means *audience factors influencing persuasion* (Social Psychology chapter), not the ROC curve.

Vision: light, the eye, rods and cones ●●●

The eye responds to wavelengths of about 380 to 760 *nanometres*; shorter waves are ultraviolet, longer ones infrared. Since speed equals frequency times wavelength, frequency and wavelength are *inversely* related, for light and sound alike. Wavelength is seen as hue, inten-

sity as brightness and purity as saturation. Light passes through the cornea, the pupil (sized by the iris) and the lens, which changes shape to focus (**accommodation**), to the retina. In the retina's macula lies the **fovea**, packed only with cones and giving the sharpest vision. The optic disc, where the optic nerve leaves, has no receptors and forms the **blind spot**.

About 120 million rods, with one pigment, rhodopsin, serve **scotopic vision**, vision in dim light: a rod responds to a single quantum of light, but many rods converge on each ganglion cell, so rod vision is blurred (and, with one pigment, colourless). Rods dominate the periphery and are absent from the fovea. About 6 to 7 million cones, with three pigments, serve photopic vision: colour and fine detail in moderate to bright light. Walk into a dark cinema and **dark adaptation** slowly raises your sensitivity to dim light, first in the cones and then, after the rod-cone break, in the rods as bleached rhodopsin regenerates (Figure 1.3). Light adaptation takes seconds to a minute; both are visual adaptation, adjustment to different light intensities. At dusk, as peak sensitivity shifts from cones (about 555 nanometres) to rods (500), reds darken before blues: the **Purkinje shift**, described by **Jan Evangelista Purkyne** (Purkinje) in 1825.

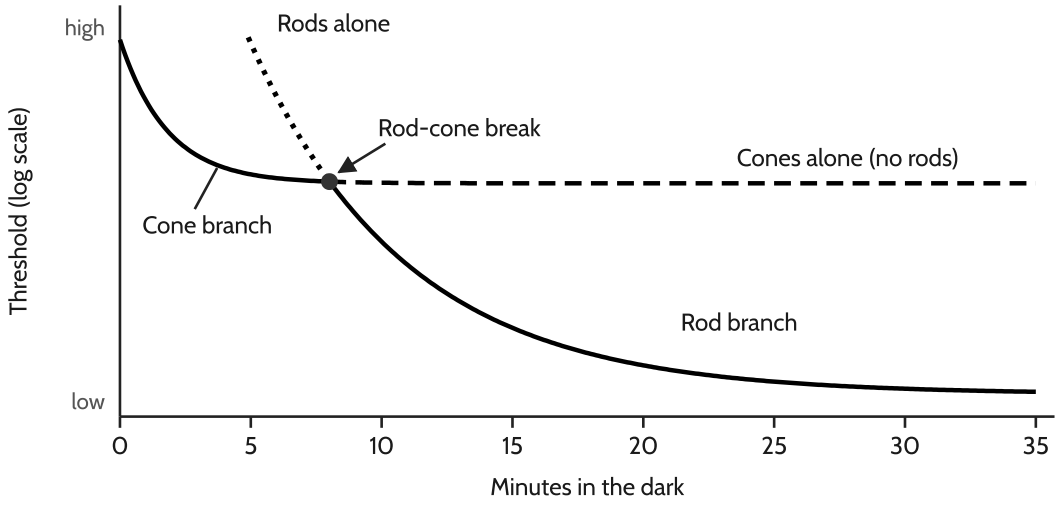


Figure 1.3 The dark adaptation curve. The threshold for detecting light falls rapidly as the cones adapt and levels off after a few minutes (cone branch). At the rod-cone break, about 7 to 10 minutes into darkness, the more sensitive rods take over and the threshold continues to fall for 20 minutes or more (rod branch). A person with no functioning rods would follow the dashed line; one with no functioning cones would show only a rod curve.

Visual acuity, the ability to see fine detail, peaks at the fovea. Rapid jumps between fixations, **saccades**, and tiny microsaccades keep the image moving on the retina; a stabilised image fades within seconds, so these movements keep perception constant. Illusory light and dark stripes at a light-dark edge, **Mach bands** (**Ernst Mach**, 1865), come from **lateral inhibition**: retinal cells suppress their neighbours, so a cell on the light side of the edge, having dark neighbours, is less inhibited and signals "brighter", exaggerating contrast.

! EXAM TRAP

Scotopic vision is *rods*; the retina and iris are structures, not receptors. Cones peak in the *fovea centralis*, not the optic disc, the periphery, "the macula but not the fovea" (the fovea lies inside the macula) or the ora serrata (its front edge). "Cones give only monochromatic information" is false: rods do. Dark adaptation raises sensitivity to *low* illumination, not bright or coloured light; adjusting to light levels is *visual adaptation*, not accommodation or Piaget's assimilation. *Saccades* ensure constant perception; fixations are pauses. Mach bands come from *lateral inhibition*, not the opponent process (colour afterimages), the rod-cone dichotomy (day versus night vision) or temporal summation (summing weak signals over time).

Colour vision ○

Lights mix additively (red, green and blue light make white, as on a screen); paints mix subtractively (each absorbs wavelengths, so blue and yellow leave green). **Thomas Young** (1802) and **Hermann von Helmholtz** (1821 to 1894, in 1852) proposed the **trichromatic theory**: three receptor types whose ratio of activity codes colour, confirmed in 1964 as short-, medium- and long-wavelength cones. **Ewald Hering** (1834 to 1918) proposed the **opponent-process theory** (1878): opposed pairs of red versus green, blue versus yellow and black versus white. It explains why nobody sees a reddish green and why staring at red leaves a green negative afterimage. The dual-process theory unites them: trichromatic in the cones, opponent in later cells. In colour vision deficiency, anomalous trichromacy (one cone type shifted) is commonest; in dichromacy one cone type is missing (protanopia, long, and deuteranopia, medium, confuse red with green; tritanopia, short, blue with yellow); in monochromacy at most one works. Red-green defects are X-linked, so far commoner in men; the plates of **Shinobu Ishihara** (1917) screen for them.

! EXAM TRAP

Helmholtz gave the trichromatic theory (with Young), the place theory of hearing and unconscious inference, not the JND (Weber and Fechner), the unconscious (Sigmund Freud) or the law of effect (Edward Thorndike).

Hearing ●●

A sound's frequency, in hertz (Hz), is heard as pitch (humans hear about 20 to 20,000 Hz); its amplitude as loudness, in decibels (dB), a logarithmic scale on which each 10 dB is a tenfold rise in intensity; its complexity as timbre. Sound passes from the pinna to the tympanic membrane, across the ossicles (malleus, incus, stapes) to the oval window and into the cochlea, where waves along the basilar membrane bend the hair cells of the organ of Corti. The bending opens channels mechanically, so potassium flows in and the cell depolarises.

Helmholtz's **place theory** (1863), confirmed by the travelling wave of **Georg von Bekesy** (Nobel Prize 1961), holds that each frequency peaks at its own place: high near the narrow base, low near the floppy apex. It best explains high pitches. In the **frequency theory** of **William Rutherford** (1886) the nerve fires in step with the sound, which suits low pitches, but no neuron fires much faster than 1,000 times a second. **Ernest Glen Wever** and **Charles William Bray** (1930) answered with the **volley principle**: fibres fire in staggered turns, like soldiers in rotation, following up to about 4,000 to 5,000 Hz. Today, timing codes low pitches and place

high ones. Sounds are located by interaural differences in time (low frequencies) and intensity (high). Outer or middle ear damage (wax, otosclerosis) causes conductive hearing loss; damage to hair cells or the auditory nerve (noise, ageing) causes permanent sensorineural loss.

> **! EXAM TRAP**
>
> Frequencies flexing different parts of the basilar membrane is *place coding*; frequency coding is firing in time with the wave, rate coding signals loudness, and "pitch coding" is not a term. Place theory does not explain low pitches, localisation (comparing the ears) or loudness (firing rate). Hair cells transduce by *mechanical bending*; closing sodium channels happens in photoreceptors, and GABA and chemical signalling are wrong. The frequency theory is William, not Ernest, Rutherford's.

Taste and smell ○

Taste receptors sit in taste buds inside the **papillae** of the tongue (fungiform, foliate and circumvallate; filiform papillae have none and sense texture). The five basic tastes are sweet, salty, sour, bitter and **umami**, the savoury taste of glutamate named by **Kikunae Ikeda** (1908); the "tongue map" of taste zones is a myth. Smell receptors lie in the olfactory epithelium high in the nose. Pheromones, chemical signals between members of a species, are clear in many animals but doubtful in humans. Flavour is taste plus smell, which is why food tastes flat with a cold. **Hans Henning** (1916) arranged six primary odours on a smell prism and four tastes on a taste tetrahedron, but the hundreds of smell receptor types found since 1991 rule out a small set of primary odours.

> **! EXAM TRAP**
>
> Papillae serve *taste*. The olfactory nerve (I) is the only sensory nerve whose pathway bypasses the thalamus; the optic (II), trigeminal (V) and vestibulocochlear (VIII) relay there.

Skin, pain and the body senses ●●

The skin senses are touch, pressure, temperature and pain. Pain is detected by free nerve endings, nociceptors, and ascends in the anterolateral system; fine touch, vibration and limb position take the dorsal columns. In the **gate control theory** of **Ronald Melzack** and **Patrick David Wall** (1965), a gate in the spinal cord's substantia gelatinosa is opened by small pain fibres, closed by large touch fibres (so rubbing a bump helps) and set from above by attention and emotion; endorphins, the body's own opiates, help to close it.

> **KEY STUDY: RAMACHANDRAN FINDS A HAND ON THE FACE (1990S)**
>
> Most amputees still feel the missing limb, a **phantom limb** (the term of **Silas Weir Mitchell**, 1871). **Vilayanur Subramanian Ramachandran** (born 1951) touched an arm amputee's face with a cotton bud, and the man felt it on specific fingers of the phantom hand. The face area of the somatosensory cortex borders the hand area, so phantoms reflect *cortical reorganisation*: neighbours invade the lost limb's territory.

Kinesthesis is the sense of limb position and movement, from proprioceptors in muscles, tendons and joints. The vestibular sense (balance) comes from the inner ear. The three **semicir-**

cular canals detect rotational acceleration, so after spinning the still-moving fluid makes you feel you are turning; the otolith organs (utricle and saccule), whose calcium carbonate crystals are pulled by gravity, detect tilt and linear acceleration.

> **! EXAM TRAP**
>
> The anterolateral system carries *nociceptive* signals, not proprioception, vibration or touch. Phantoms are *cortical remapping*, not nerve regrowth, denial or cerebellar damage. Dizziness after spinning is the *semicircular canals*, which detect rotation in three planes, not static position or linear motion (otolith organs and their crystals), limb position (proprioceptors) or pitch (cochlea).

Senses in concert and senses deprived ●●

Sensory interaction is one sense shaping another: in the McGurk effect (Harry McGurk and John MacDonald, 1976), "ba" heard with lips saying "ga" is perceived as "da". Combining senses aids learning, so teaching for learning difficulties uses multisensory input with regular repetition. In **synaesthesia** a minority automatically and consistently experience a second sensation, such as coloured letters. Remove input and the brain makes its own: in the McGill laboratory of **Donald Olding Hebb** (Bexton, Heron and Scott, 1954), students in goggles and gloves quit within days, thought poorly and hallucinated, the classic sensory deprivation study. Likewise, **Charles Bonnet syndrome** (**Charles Bonnet**, 1760) is complex visual hallucinations in people with sight loss who are cognitively intact and keep insight.

> **! EXAM TRAP**
>
> Charles Bonnet syndrome is not auditory hallucinations in dementia or deafness, tactile ones in alcohol withdrawal, multimodal ones in delirium or olfactory auras before seizures.

> **SCAN YOUR BRAIN**
>
> Fill in the blanks. The answers are below; cover them first.
>
> 1. The absolute threshold is detected on _____ percent of trials.
> 2. SDT indexes sensitivity by _____.
> 3. Scotopic vision depends on the _____.
> 4. Mach bands are explained by lateral _____.
> 5. High frequencies peak near the _____ of the basilar membrane.
> 6. Rotational acceleration of the head is detected by the _____ canals.
>
> **Answers:** (1) 50, (2) d′ (d prime), (3) rods, (4) inhibition, (5) base, (6) semicircular.

1.4 Perception: Organising and Interpreting the World HIGH YIELD · 38 QS

Section 1.3 followed light as far as the retina. This section asks what the brain does with it next, and why seeing, as its illusions show, is far less direct than it feels.

Inference or direct pickup: the theories ●●●

Perception is the selection, organisation and interpretation of sensory signals into meaningful objects and events: sensation gives you a pattern of light, perception your mother's face. **Bottom-up processing** (data-driven) builds the percept from features of the stimulus; **top-down processing** (conceptually driven) lets knowledge, expectation and context shape how those features are read. The theories differ in the weight they give each.

For **Hermann von Helmholtz** (1821 to 1894), perception was **unconscious inference**: rapid, automatic conclusions from sensory evidence plus past experience. **Richard Langton Gregory** (1923 to 2010) built this into the **constructivist approach**, the indirect or top-down theory (*Eye and Brain*, 1966), in which a percept is a best-guess hypothesis; that is why a hollow mask looks like a normal, convex face. The **Bayesian approach** puts the idea in probabilities: the percept combines prior beliefs with sensory evidence, and the weaker the evidence, the more the priors decide.

The classic bottom-up theory is the **direct perception** (ecological) approach of **James Jerome Gibson** (1904 to 1979). The light reaching a moving observer, he argued, specifies the world without inference: optic flow streams outwards from the point you are heading for, and objects offer **affordances**, possibilities for action perceived directly (a chair affords sitting). **Ulric Neisser** (1928 to 2012) joined the two directions in his perceptual cycle (1976): a schema directs exploration, and the information sampled modifies the schema.

> **! EXAM TRAP**
>
> Unconscious inference and probabilistic reasoning define the **constructivist** approach. The computational approach (Marr, below) asks what each stage of vision computes, and the information-processing approach describes a chain of stages; neither centres on inference. Bayesian perception rests on **prior beliefs and sensory evidence**, not genes, culture or emotion. The bottom-up theorist, and the coiner of affordances, is **James J. Gibson**: Helmholtz and Gregory are top-down, Wertheimer is a Gestalt psychologist, and Gibson's wife Eleanor built the visual cliff. The apperceptive mass (Herbart's term, often credited to Wundt; Section 1.1) is not unconscious inference.

Gestalt organisation: figure, ground and grouping ●●●

No school has contributed more to perception than Gestalt psychology (Section 1.2), launched in 1912 by **Max Wertheimer** (1880 to 1943) with **Kurt Koffka** (1886 to 1941) and **Wolfgang Köhler** (1887 to 1967). Its claim is that the whole is different from the sum of its parts.

First the field splits into a **figure**, which stands out, and a ground behind it. As **Edgar Rubin** (1886 to 1951) described in 1915, the figure is thing-like (high in object character), clearly outlined, usually smaller and in front; the ground is shapeless, diffuse and seems farther away. Rubin's vase is a reversible figure: it flips into two faces although the stimulus never changes, so the perceiver does the organising.

Elements are then grouped (Figure 1.4). By the **law of proximity**, elements close together in space or time belong together. By **similarity**, alike elements group; by the **law of closure**, gaps are filled, so a broken outline is seen as a circle; by **good continuation** (continuity), elements on a smooth path group; by **common fate**, elements moving together group, like a

flock of birds. Symmetry, common region (a shared enclosure) and connectedness (joining lines) group elements too. The master law is **Prägnanz** (good figure, simplicity): every pattern is seen in the simplest, most complete organisation the conditions allow, so two overlapping rings look like two circles.

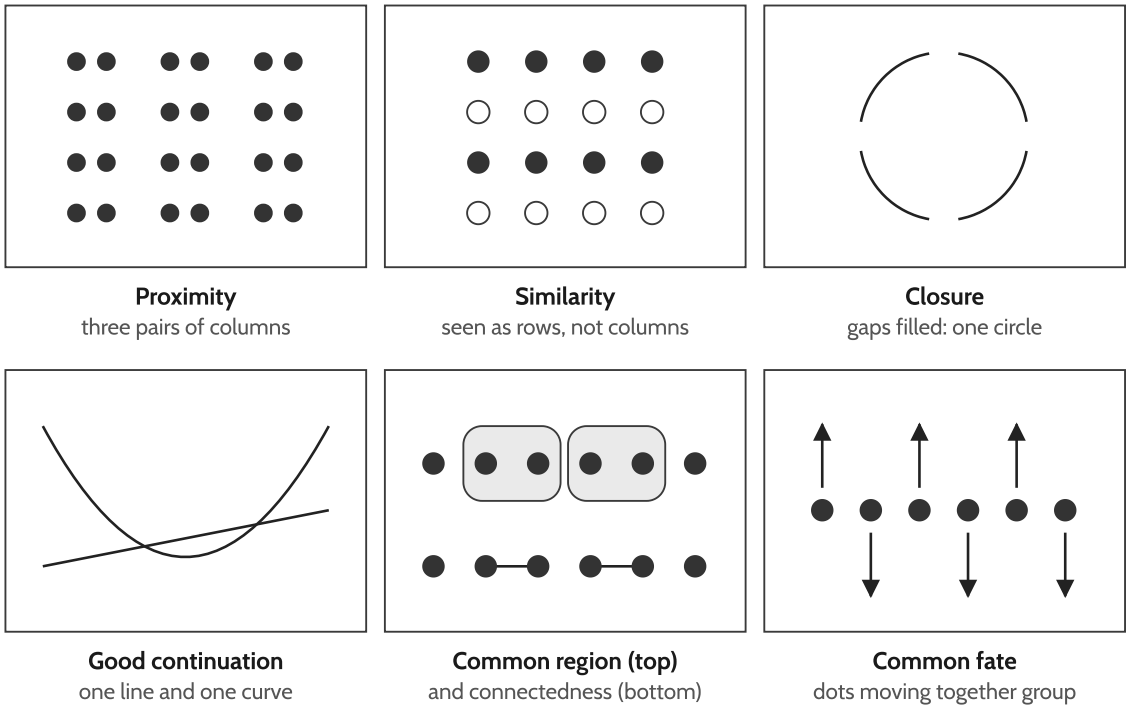


Figure 1.4 Six grouping laws you can watch in your own perception. In each panel the elements are physically identical or evenly spaced in some respect, yet you see definite groups; the grouping is supplied by your visual system.

> **! EXAM TRAP**
>
> Close in **space and time** is **proximity**. The tendency to **complete** a figure is **closure**; continuation and continuity both name the smooth-path law. Organised "in the best possible whole form" is **Prägnanz**. The school that contributed most to perception is **Gestalt**, not structuralism, behaviourism or psychoanalysis. Interposition is a depth cue, interpolation fills in between known points, extrapolation extends beyond them; one paper keyed **structuring** for working out a rule from incomplete evidence and picking the items that fit it.

Depth cues and the visual cliff ●●●

How does a flat retinal image give you three dimensions? The brain combines depth cues, and only two are **binocular cues**, needing both eyes. In **retinal disparity**, the eyes, about 6 centimetres apart, receive slightly different views, and the nearer the object, the greater the difference; fusing the two gives the solid depth of stereopsis. In **convergence**, the eyes turn inwards to fixate a near object, and the brain registers how far they have turned.

All other cues are **monocular cues**, available to one eye (Table 1.5). Most are static pictorial cues, the ones a painter uses to put depth on a flat canvas. In the **texture gradient**, pebbles look smaller and more densely packed with distance; engineers exaggerate it by painting stripes ever closer together before a toll plaza or on an exit ramp, so that drivers feel their

speed and slow down. **Motion parallax** needs movement: from a train, nearby poles flash past while distant hills crawl.

Table 1.5 Depth cues classified.

Cue	Type	Depth signal
Retinal disparity	Binocular	Images differ more when near
Convergence	Binocular, oculomotor	Eyes turn in more when near
Accommodation	Monocular, oculomotor	Lens thickens for near objects
Motion parallax	Monocular, movement	Near objects sweep past faster
Interposition (overlap)	Monocular, pictorial	A partly hidden object is farther
Relative and familiar size	Monocular, pictorial	Smaller image, farther object
Linear perspective	Monocular, pictorial	Parallel lines converge
Texture gradient	Monocular, pictorial	Texture finer and denser
Aerial perspective	Monocular, pictorial	Distant objects hazy, bluish
Height in the field	Monocular, pictorial	Nearer the horizon, farther
Shading	Monocular, pictorial	Shadows reveal shape

KEY STUDY: THE VISUAL CLIFF (GIBSON AND WALK, 1960)

Question. Can infants perceive depth before learning it from falls? **Method. Eleanor Jack Gibson** (1910 to 2002) and **Richard Walk** (1920 to 1999) built a **visual cliff**: glass over a checkerboard lying just beneath one half (the shallow side) and far below the other (the deep side). Mothers called 36 infants aged 6 to 14 months across it. **Results.** Of the 27 who left the centre board, only three crept onto the deep side; newborn animals avoided it too. **Conclusion.** Depth perception is present by crawling age. **Criticism.** Crawlers have already had months of visual experience, so innateness is not proved.

! EXAM TRAP

The binocular cues are only **retinal disparity** and **convergence**. Interposition, accommodation, motion parallax and aerial perspective are all monocular. Assimilation (Piaget's term) is no depth cue. The cliff tested **depth perception**, not object permanence, colour or imprinting, and showed it present **by crawling age**, not fully formed at birth or only after age two. One paper keyed **180 degrees** for each eye's visual field; the better figure is about 160 degrees, with about 200 for both eyes together.

Perceptual constancy ●●

Perceptual constancy is the perception of objects as stable despite changing sensory input. Hold one hand at arm's length and the other halfway to your face: the far hand's retinal image is half the size, yet both look alike. This **size constancy** follows **size-distance invariance**: perceived size is retinal size scaled by perceived distance, a rule that **Emmert's law** (1881) shows with afterimages, which look larger on a farther wall. Shape constancy keeps an opening door rectangular; brightness constancy keeps coal black in sunlight, because lightness depends on the ratio of an object's light to its surroundings'; colour constancy keeps a

banana yellow under any lamp; location constancy keeps the world still as the eyes jump in saccades. Tiny saccades also refresh the retinal image, which would otherwise fade, and so ensure constant perception.

> **! EXAM TRAP**
>
> The outstretched hand is **size constancy**, not shape constancy, motion parallax or depth perception. Lateral inhibition (Section 1.3) **enhances** edges; it does not dull them. A fixation is the pause between saccades, and sensory adaptation the fading they prevent.

Illusions and what they reveal ●●

An **illusion** is a misinterpretation of a stimulus that really exists. The error lies in perception, not sensation: the retinal image is accurate. Indian textbooks contrast **personal illusions**, which vary with observer and occasion (a rope seen as a snake at dusk), with permanent or universal illusions, which everyone sees every time (railway tracks converging).

In the **Müller-Lyer illusion**, published in 1889 by **Franz Carl Müller-Lyer** (1857 to 1916), a shaft ending in outward fins looks longer than an equal shaft ending in arrowheads (Figure 1.5). Gregory (1963) blamed **inappropriate constancy scaling**: arrowheads suggest the near outside corner of a building, fins the far inside corner of a room, and size constancy enlarges the "farther" shaft on a flat page. **Marshall Segall**, **Donald Thomas Campbell** (1916 to 1996) and **Melville Jean Herskovits** (1895 to 1963) showed that experience matters (1963, 1966): by their **carpentered-world hypothesis**, people raised among rectangular buildings are more susceptible than people raised among round huts.

The Ponzo illusion, the Ames room (a distorted room that looks rectangular through a peephole, shrinking a person in its far corner) and the moon illusion (terrain makes the horizon moon seem farther) are misapplied size scaling too: whatever is registered as farther is seen as bigger. The Ebbinghaus illusion (Titchener circles) is size contrast, and the Zöllner, Poggendorff and Hering figures distort lines. The Necker cube is an ambiguous figure that flips in depth; the Penrose triangle is an impossible figure.

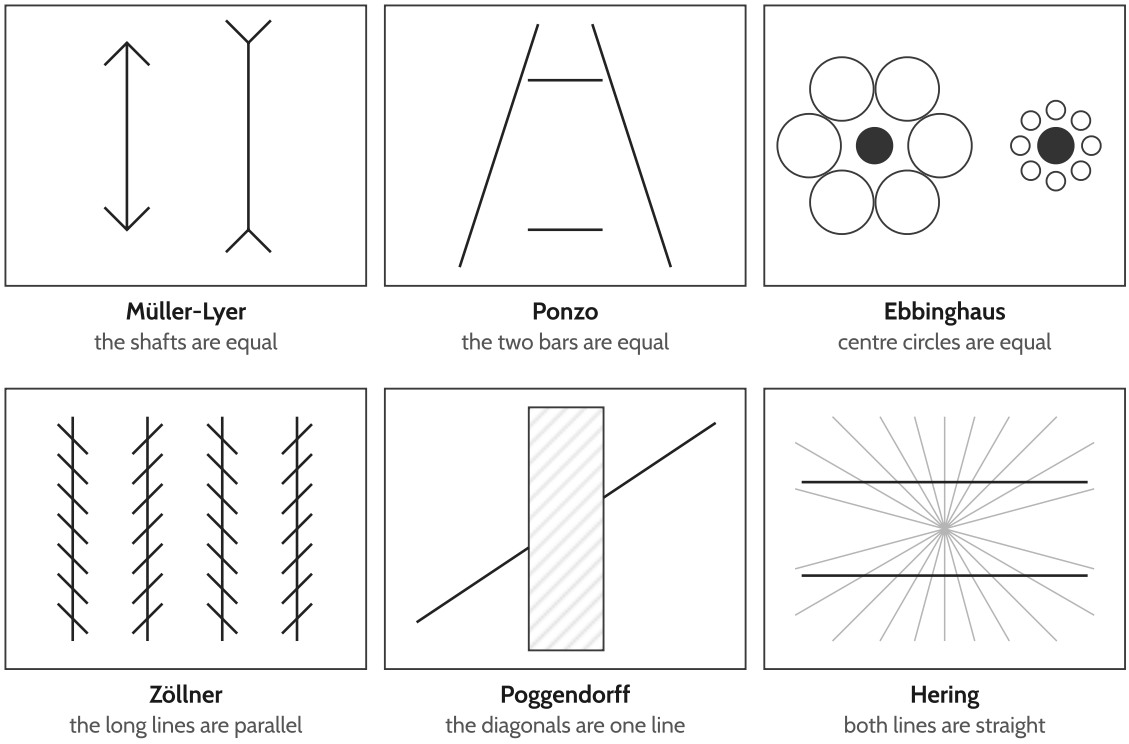


Figure 1.5 Six classic geometric illusions. In every panel the caption underneath is physically true, yet you see something else: the finned shaft looks longer, the upper bar looks longer, the circle among small circles looks larger, the long lines look tilted, the diagonal looks broken and the straight lines look bowed. Check them with a ruler.

> **! EXAM TRAP**
>
> Rails converging for everyone is a **permanent** illusion (visual too, but the item tests the classification), not a personal one or the Müller-Lyer. Illusions distort **perception**, not sensation, adaptation or "translation" (transduction). The Müller-Lyer is strongest with **buildings with corners** and best explained by **misapplied size constancy**, not disparity (it works with one eye), lateral inhibition or figure-ground reversal.

Illusion, hallucination and delusion ••

An illusion misinterprets a real external stimulus. A **hallucination** is a perception without an adequate external stimulus, such as a voice in an empty room. A delusion is not a perception at all but a false belief held with unshakeable conviction. Psychopathology adds the sensory distortion, a real object perceived but altered in size, intensity or quality (micropsia), and imagery, a picture in the mind's eye under voluntary control. Déjà vu is the illusion of having seen something before; déjà entendu, of having heard it before.

! EXAM TRAP

Illusions are a **misinterpretation of stimuli arising from an external object**, or **distortions of existing stimuli**. "Without an adequate external stimulus" and "not based on external reality" are **hallucinations**; "a real object perceived in a distorted way" is a sensory distortion; "vivid mental images" is imagery; "innate mechanisms" fails because learning shapes illusions. Clear memory of the past is hypermnesia; involuntary repeated speech is palilalia (Psychiatric Disorders chapter).

Motion perception

Real motion is signalled by direction-selective neurons, concentrated in area MT (V5). Apparent motion is seen where nothing moves: in Wertheimer's phi phenomenon (1912) two lines flashed in quick succession look like one line moving (strictly, smooth object motion is beta movement), and films give stroboscopic motion. In induced motion a still object seems to move because its frame moves, like the moon behind racing clouds. In the **autokinetic effect** a still light in darkness seems to wander for lack of a frame of reference; **Muzafer Sherif** (1906 to 1988) used it in 1935 to study how group norms form. The motion aftereffect (waterfall illusion) is a still scene drifting opposite to a motion just watched.

Pattern and object recognition ○

Recognising an A in any handwriting defeats **template matching**, comparison with exact stored copies, so feature analysis uses parts instead. In the **Pandemonium** model of **Oliver Selfridge** (1959), image demons record the pattern, feature demons shout when they find their line or curve, cognitive demons (one per letter) shout in proportion to their matching features, and a decision demon picks the loudest. **David Hubel** and **Torsten Wiesel** found such feature detectors, cells tuned to lines of one orientation, in cat visual cortex from 1959. A **prototype**, in the work of **Eleanor Rosch** (born 1938), is an idealised average member of a category. The **recognition-by-components** theory of **Irving Biederman** (1987) builds objects from about 36 three-dimensional geons, and **David Marr** (1945 to 1980) traced vision from a primal sketch through a viewer-centred 2.5-D sketch to an object-centred 3-D model. Prosopagnosia (face blindness) is a selective loss of face recognition. In the word superiority effect (**Gerald Reicher**, 1969), a letter is identified better in a word than alone: knowledge helps from the top.

Set, culture, subliminal perception and ESP ●●

A **perceptual set** is a readiness to perceive in a particular way. **Jerome Seymour Bruner** (1915 to 2016) and **Leigh Minturn** (1955) found one shape read as B among letters and as 13 among numbers, a context effect. Bruner's New Look stressed needs: with **Cecile Goodman** in 1947 he found that poor ten-year-olds overestimated coin sizes more than rich ones, and **Elliott McGinnies** (1949) found that taboo words needed longer exposures (perceptual defence). Experience reshapes perception. Perceptual learning sharpens discrimination; **George Malcolm Stratton** (1865 to 1957) showed perceptual adaptation by adjusting to lenses that turned his world upside down (1896, 1897); and S.B., given sight at 52 and studied by Gregory and **Jean Wallace** (1963), could not at first judge distance. Culture matters too, as the carpentered world showed. **Herman Witkin** (1916 to 1979) found a perceptual style, **field de-**

pendence-independence (rod-and-frame and embedded figures tests): field-dependent people are swayed by the surrounding field, field-independent people separate items from it.

Subliminal perception is the processing of stimuli too faint or brief to be noticed. **James Vicary**'s 1957 "Eat Popcorn" cinema study was a hoax, admitted in 1962, but in masked semantic priming (**Anthony Marcel**, 1983) an unseen word speeds responses to a related one, and in blindsight (**Lawrence Weiskrantz**, 1974) patients with damaged primary visual cortex deny seeing yet locate targets above chance. **Extrasensory perception** claims perception with no sensory input: telepathy (mind to mind), clairvoyance (hidden objects) and precognition (the future), studied with psychokinesis (mind over matter) by parapsychology. **Joseph Banks Rhine** (1895 to 1980) used Zener cards at Duke (chance is 5 hits in 25), and later researchers the Ganzfeld procedure, guessing a target picture in a uniform sensory field. The results do not replicate, so most scientists remain sceptical.

> **! EXAM TRAP**
>
> Semantic priming studies **subliminal perception**, not field dependence-independence (Witkin), locus of control (a personality variable) or signal detection (Section 1.3). Failing to notice obvious changes is change blindness, an attention topic (Section 1.5), not a failure of constancy. Deprivation and coin size is **Bruner and Goodman**, not Segall, Herskovits, or **K. D. Broota** and **H. C. Ganguli** (Indian work on cultural differences in perceptual selectivity, 1970s).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Processing driven by knowledge, expectation and context is called _____ processing.
2. Gibson's term for the possibilities for action an object offers is _____.
3. The Gestalt tendency to fill the gaps in a figure is the law of _____.
4. The two binocular depth cues are retinal disparity and _____.
5. Gregory explained the Müller-Lyer illusion as inappropriate constancy _____.
6. A perception without an adequate external stimulus is a _____.
7. Biederman's recognition-by-components theory builds objects from about 36 _____.
8. Bruner and Goodman found that poor children overestimated the size of _____ more than rich children did.

Answers: (1) top-down, (2) affordances, (3) closure, (4) convergence, (5) scaling, (6) hallucination, (7) geons, (8) coins.

1.5 Attention and Consciousness HIGH YIELD · 61 QS

Attention selects the small part of the world that reaches awareness, and consciousness, the awareness itself, changes state many times a day and night: in sleep and dreams, under hypnosis, in meditation and on drugs.

What attention is: span and determinants ○

William James (1842 to 1910) defined attention in 1890 as the mind "taking possession … in clear and vivid form, of one out of what seem several simultaneously possible objects or trains of thought": selection plus concentration, needed because the brain's capacity is limited. Classical texts give it five characteristics: *selectivity*, *shifting* to a new object, a limited *span*, *fluctuation* (attention to an unchanging object waxes and wanes, as a faint ticking watch seems to fade and return) and *division* between tasks.

The **span of attention** is the number of objects apprehended in one brief glance. **William Stanley Jevons** (1871), glancing at handfuls of beans, was never wrong with three or four; **tachistoscope** studies, using displays flashed for a fraction of a second, put the span at about six to eight unrelated items. **External determinants** of attention are properties of the stimulus (intensity, size, contrast, change, movement, novelty, repetition); **internal determinants** are properties of the person (motives, interests, attitudes and **mental set**, a temporary readiness to perceive something).

> **! EXAM TRAP**
>
> Number plates carry about four digits plus a few letters because of the **span of attention**, not selective, divided or sustained attention. Miller's memory span of seven (Section 1.7) is a different process.

Selective attention: filters, attenuators and spotlights ●●●

The British engineer **Colin Cherry** (1953) named the cocktail party problem: how do we follow one conversation among many? Hearing your own name from a conversation you were ignoring is the **cocktail party phenomenon**.

KEY STUDY: CHERRY'S SHADOWING EXPERIMENTS (1953)

In **dichotic listening** a different message goes to each ear, and the listener repeats one aloud as it arrives (**shadowing**). People recalled almost nothing of the unattended message's meaning, missing even a switch from English to German, yet noticed physical changes such as a male voice turning female. **Neville Moray** (1959) then found that a word list repeated 35 times in that ear left no memory, yet a third of listeners noticed their own name there.

Donald Broadbent's **filter model** (1958), the first major information-processing theory, passes one message on a physical feature (ear, pitch) and blocks the rest before meaning is analysed: **early selection**, with the filter as a **bottleneck**. **Anne Treisman**'s **attenuation model** (1964) turns unattended messages down, not off. Each word in a mental dictionary has a threshold, and important words such as your name have permanently low thresholds, so they fire even from the weakened channel. **J. Anthony Deutsch** and **Diana Deutsch** (1963) instead proposed **late selection**, at the response, after every message is analysed for meaning (Table 1.6).

Table 1.6 Theories of selective auditory attention.

Theory	Theorist and year	Where selection happens
Filter (switch)	Broadbent, 1958	Early; the rest is blocked

Theory	Theorist and year	Where selection happens
Attenuation	Treisman, 1964	Early but leaky
Late selection	Deutsch and Deutsch, 1963	At the response, after meaning
Pertinence	**Donald Norman**, 1968	Late; signal strength plus relevance
Multimode	**William Johnston** and **Steven Heinz**, 1978	Flexible; later costs more effort
Perceptual load	**Nilli Lavie**, 1994 to 1995	Early if the task is demanding, late if easy

Vision has the same limits. **Michael Posner** likened covert attention, moved without moving the eyes, to a **spotlight** (others prefer a **zoom lens**, wider but less detailed). In his cueing task (1980) a central arrow directs voluntary **endogenous** attention, a peripheral flash captures reflexive **exogenous** attention, and after about 300 milliseconds the flashed spot is answered more slowly (**inhibition of return**). In Treisman and **Garry Gelade**'s **feature integration theory** (1980), features register in parallel at a preattentive stage, so a red letter pops out among green ones, but binding them into objects needs serial focused attention, so conjunction search slows with every added item; unattended features can combine wrongly (illusory conjunctions). Missing a person in a gorilla suit while counting basketball passes (**Daniel Simons** and **Christopher Chabris**, 1999) is **inattentional blindness**, missing an unexpected object in plain view; missing a large change made across a blink or flicker is **change blindness**; missing a second target 200 to 500 milliseconds after a first is the **attentional blink**.

> **! EXAM TRAP**
>
> "Filter-attenuation theory" is **Treisman** (1964), not Broadbent, Johnston or Heinz, and her attenuator model, not Broadbent's switch model, best explains the cocktail party effect. A bottleneck is a narrow processing stage, not the cocktail party phenomenon. Failing to notice obvious changes is **change blindness**, not inattentional blindness or perceptual constancy (Section 1.4).

Divided attention, automaticity, Stroop and vigilance ●●●

Divided attention asks how far we can do two things at once. **Daniel Kahneman**'s **capacity model** (*Attention and Effort*, 1973) treats attention as one limited pool of mental effort; **Christopher Wickens**'s **multiple resource theory** (1984) adds separate pools by modality, code and stage, so dissimilar tasks interfere less: driving combines better with the radio than with texting, though even a hands-free phone conversation impairs it.

Practice helps because some processes stop needing capacity. An **automatic process** runs without intention, awareness or interference with other activity (Posner and Charles Snyder, 1975). **Richard Shiffrin** and **Walter Schneider** (1977) contrasted fast, parallel, effortless and inflexible automatic processing, learned through consistent practice, with slow, serial, effortful and flexible **controlled processing**. In **Patricia Devine**'s model (1989), stereotypes are activated automatically in everyone, but low-prejudiced people use controlled processing to refute them (Chapter 3).

KEY STUDY: STROOP'S COLOUR-WORD EXPERIMENTS (1935)

John Ridley Stroop (1897 to 1973) asked people to read colour words printed in a conflicting ink (RED printed in green) or to *name the ink colour* of such words. Reading was barely slowed, but naming the inks took about 74 per cent longer than naming plain colour patches, and errors rose.

The **Stroop effect** is this slowing of colour naming when word and ink conflict. Reading is automatic, so the word cannot be shut out, and its name competes with the controlled naming of the ink at the response stage. The effect thus shows a limit of **selective attention**: interference in reaction time from conflicting information, one attribute of the stimulus (the word) dominating the other (the ink).

Watching radar for hours for a rare blip demands **sustained attention**, or **vigilance**. **Norman Mackworth** (1948) had observers watch a clock pointer for occasional double jumps; detection fell across the two-hour watch, the **vigilance decrement**. In an oddball task, where a rare target appears among frequent non-targets, the brain produces a positive event-related potential about 300 milliseconds later, the **P300**, which indexes the allocation of attention to the target and is reduced in schizophrenia and Alzheimer's disease. Attention fails in ADHD, in hemispatial neglect (ignoring the left side of space after right parietal damage) and in delirium (Chapter 8).

! EXAM TRAP

The Stroop effect is an **attention** phenomenon (selective attention; interference in reaction time from conflicting information), not long-term memory retrieval, working memory, procedural memory, perception or language development. RML 2024 asks what it is "an account of" and keys **stimulus dominance**: the printed word dominates its ink colour. "Semantic dominance" tempts because the word is meaningful; "instructional" and "conceptual" dominance are not terms used for it. Air traffic controllers and radar readers exemplify **sustained attention**, not selective, divided or "perceptual" attention.

Consciousness: James's stream and Freud's levels ●●●

Consciousness is our awareness of ourselves and our environment at a given moment. Its level, from coma to alert waking, depends on the brainstem reticular activating system and is scored on the **Glasgow Coma Scale** (eye, verbal and motor responses; 3 to 15). The structuralists analysed its contents into elements by introspection (Section 1.1); James objected that consciousness "does not appear to itself chopped up in bits … it flows": it is a **stream of consciousness** (1890), personal, ever changing, continuous and selective. Its most researched feature is **daydreaming**, which **Jerome L. Singer** showed to be normal and nearly universal.

Sigmund Freud (1856 to 1939) held that most mental life never enters the stream. His **topographic model** (1900) has three levels: the conscious, the small tip of present awareness; the **preconscious**, accessible memories recallable at will (your mobile number); and the much larger **unconscious**, repressed wishes known only indirectly, through dreams, symptoms and **Freudian slips**, errors of speech that betray unconscious impulses. By **psychic determinism**, no mental event is accidental. In his structural model (1923) the id works on the pleasure principle and the ego on the reality principle (Chapter 4). The **psychodynamic perspective** holds that behaviour is driven by unconscious inner forces largely beyond our con-

trol and shaped in early childhood. Research has not supported the Oedipus complex, but it does support the concept of an unconscious mind.

Modern psychology adds the **cognitive unconscious**, processing outside awareness that is automatic, not repressed, such as the unconscious stereotyping measured by the Implicit Association Test (Anthony Greenwald and colleagues, 1998). Processes never open to awareness, such as blood-pressure control, are **nonconscious**. **David Chalmers** (1995) called the question of why brain processing is accompanied by subjective experience at all the **hard problem of consciousness**. An **altered state of consciousness**, a term made standard by **Charles Tart** (1969), is a temporary, qualitative change in the overall pattern of mental functioning, arising naturally (sleep), psychologically (hypnosis, meditation) or chemically (drugs).

> **! EXAM TRAP**
>
> "Stream of consciousness" is **William James** (1890), not Carl Jung, Wilhelm Wundt or B. F. Skinner; its most researched feature is **daydreaming**, not gender differences, brain-wave patterns or eye movements. Memories recallable at will are **preconscious**, not unconscious. The three levels form the **topographic** model (1900); id, ego and superego the **structural** model (1923).

> **INDIAN CONTEXT**
>
> The *Mandukya Upanishad* names four states: jagrat (waking), svapna (dreaming), sushupti (dreamless sleep) and **turiya**, "the fourth", pure consciousness witnessing all three, identified with the **Atman**, the true self or soul, eternal consciousness (Section 1.2).

Biological rhythms and the suprachiasmatic nucleus ●●

A **circadian rhythm** is a biological cycle of about 24 hours, such as sleep and waking or body temperature. Without time cues, in caves, human rhythms persist but drift slightly beyond 24 hours, so they come from an internal clock, reset daily by **zeitgebers** ("time givers"), above all light. The clock is the **suprachiasmatic nucleus** (SCN) of the hypothalamus, just above the optic chiasm: animals without it sleep the usual total amount but at random times. Light reaches it directly along the **retinohypothalamic tract**, and it drives the pineal gland to secrete **melatonin** in darkness. **Jet lag** and shift work put the clock out of step with the day. **Ultradian rhythms** are shorter than a day (the 90-minute sleep cycle), **infradian rhythms** longer (the menstrual cycle).

> **! EXAM TRAP**
>
> The SCN is the **master circadian pacemaker**; its lesions disrupt circadian rhythms, not total sleep. It is not the hunger and satiety centre (lateral and ventromedial hypothalamus), the thermoregulation centre (preoptic area), the paraventricular stress nucleus, the controller of anterior pituitary hormones, or a seat of memory (hippocampus, mammillary bodies), language or pain perception.

The stages and cycles of sleep ●●●

Sleep opened to study after **Hans Berger** published the first recordings of the human electroencephalogram (EEG) in 1929: neurons firing independently give small, fast, **desynchronised** waves, neurons firing together large, slow, **synchronised** ones. In 1953 **Eugene Aserinsky** and **Nathaniel Kleitman** found recurring bursts of rapid eye movement with a waking-like EEG, from which woken sleepers usually reported vivid dreams; **William Charles Dement** and Kleitman (1957) then described the cycle of stages. Sleep thus divides into **REM sleep** and NREM sleep, whose old stages 1 to 4 became N1, N2 and N3 in 2007 (Table 1.7).

Table 1.7 EEG waves and the stages of sleep.

State	EEG	What marks it
Alert waking	Beta, 13 to 30 Hz	Active thinking
Relaxed, eyes closed	Alpha, 8 to 12 Hz	Rest before sleep
N1	Theta, 4 to 7 Hz	Drowsy transition
N2 (about half the night)	Sleep spindles, K-complexes	Light but definite sleep
N3 (old stages 3 and 4)	Delta, 0.5 to 4 Hz, high amplitude	Deepest, **slow-wave sleep**; growth hormone peak
REM (20 to 25 per cent)	Fast, low voltage, desynchronised; sawtooth waves	Rapid eye movements, atonia, dreams

REM is also called **paradoxical sleep**, because the brain looks awake while the body is limp: the eyes dart, the core muscles are almost paralysed (**REM atonia**), breathing turns irregular and genital arousal accompanies nearly every REM period. One run of NREM stages followed by REM is a **sleep cycle** of about 90 minutes, so seven to eight hours hold **four to six cycles**; N3 dominates the early cycles, and REM periods lengthen toward morning (Figure 1.6). A newborn sleeps about 16 hours, half of it REM, and older adults about six, with little N3 (**Howard Roffwarg** and colleagues, 1966). Slow-wave sleep is switched on by GABA neurons of the **ventrolateral preoptic area** (VLPO), which silence the arousal systems; REM is generated in the pons.

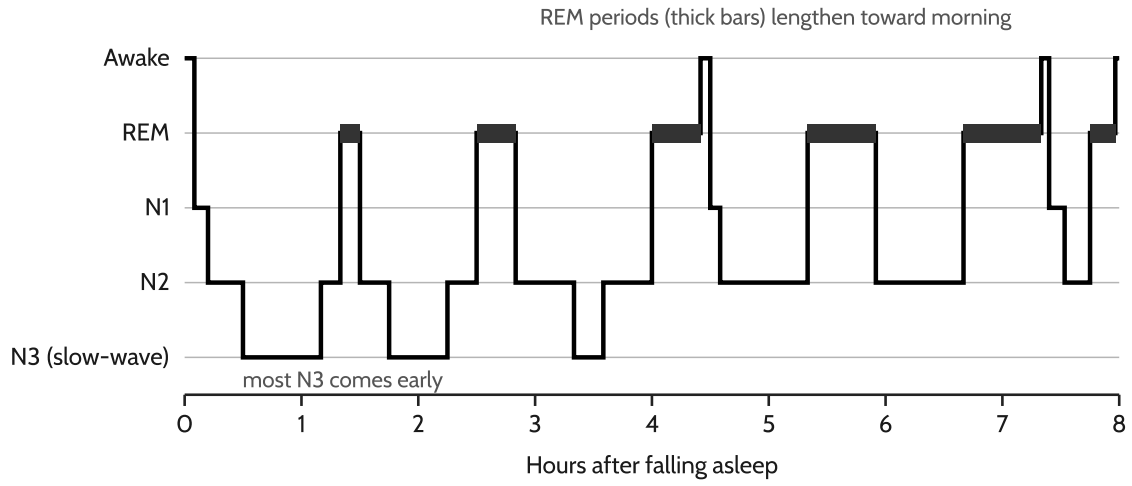


Figure 1.6 A typical night of young adult sleep (a hypnogram). Cycles of about 90 minutes repeat four to six times; N3 is concentrated early in the night, and REM periods become longer toward morning.

> **! EXAM TRAP**
>
> REM sleep is **not** marked by EEG synchrony, delta waves or a growth hormone surge (all N3). Cycles per night: **four to six** (RML 2024); RML 2022, offering no 4 to 6, keys **4 to 7**, the option containing it. Slow-wave sleep begins when the **VLPO** becomes active; the SCN is the clock, and the sublaterodorsal nucleus (REM-on) and ventrolateral periaqueductal grey (REM-off) govern REM.

Why we sleep, and sleep disorders ●●

Recuperation theories say sleep restores what waking runs down; **adaptation theories** say evolution timed sleep to keep animals safe and save energy; **memory consolidation** theories note that sleep after learning reduces forgetting (Section 1.7). Sleep loss brings sleepiness, low mood and lapses of sustained attention, including **microsleeps** of a few seconds. Volunteers woken at every REM onset tried to enter REM ever more often, then showed **REM rebound**, extra REM (Dement, 1960). In **cataplexy**, part of narcolepsy, strong emotion triggers a sudden loss of muscle tone in full wakefulness: REM atonia while awake. Sleep disorders fall into insomnia, **hypersomnia** (excessive sleepiness, as in narcolepsy), breathing-related disorders and the **parasomnias**, abnormal behaviours during sleep: NREM parasomnias (sleepwalking and sleep terrors from N3, and bedwetting) and REM parasomnias (nightmares, REM sleep behaviour disorder). Table 1.8 lists the main ones (Chapter 8).

Table 1.8 The main sleep disorders in brief.

Disorder	Key features	Stage or cause
Insomnia	Trouble falling or staying asleep	Stress, anxiety, poor habits
Narcolepsy	Daytime sleep attacks, cataplexy, sleep paralysis	REM intrudes into waking
Sleep apnoea	Breathing stops repeatedly; snoring	Airway collapse
Sleepwalking (somnambulism)	Sitting, walking or complex acts while asleep	N3, early in the night
Sleep terrors	Screaming fear, inconsolable, little recall	N3, early in the night

Disorder	Key features	Stage or cause
Nightmares	Frightening dream, vivid recall	REM, late in the night
REM sleep behaviour disorder	Acting out dreams	Loss of REM atonia
Restless legs syndrome	Leg discomfort, urge to move	Delays sleep onset

! EXAM TRAP

Somnambulism is sitting, walking or complex behaviour while asleep; screaming fear is sleep terrors, acting out nightmares is REM sleep behaviour disorder, and leg discomfort is restless legs syndrome. Sleepwalking arises from **N3**, not REM: a sleeper in REM is paralysed. Treatments for insomnia, such as stimulus control and Viktor Frankl's paradoxical intention (trying to stay awake), are in Chapter 7.

Dreams: Freud, Jung and the brain ●●●

Freud's *The Interpretation of Dreams* (1900) called dreams "the royal road to a knowledge of the unconscious". Each dream is a **wish fulfilment**, the disguised fulfilment of a repressed wish. The **manifest content** is the dream as remembered, the **latent content** its hidden meaning, and the **dream work** disguises one as the other by condensation (several ideas in one image), displacement (feeling shifted to a trivial object), symbolisation (thoughts made into images) and secondary revision (tidying it into a story).

Carl Gustav Jung (1875 to 1961) saw dreams as undisguised symbolic messages that compensate for a one-sided conscious attitude. He held that the unconscious has two levels: a personal (individual) unconscious of forgotten and repressed experience and, beneath it, the **collective unconscious**, an inherited layer shared by all humanity, which he explained by evolution, through the similar brain structure of all human races. It holds **archetypes**, universal images recurring in myths: the persona (social mask), the shadow (disowned dark side), the **anima** (the feminine side of a man) and the **animus** (the masculine side of a woman). Jung also devised the word association test, in which hesitations reveal unconscious complexes (Chapter 4).

In the **activation-synthesis hypothesis** of **J. Allan Hobson** and **Robert McCarley** (1977), the pons fires randomly during REM and the cortex synthesises a story from the activation, which is why dreams are visual, bizarre and soon forgotten. Information-processing views link dreams to the overnight consolidation of recent experience, and **Antti Revonsuo**'s **threat-simulation theory** (2000) says dreams evolved to rehearse responses to danger. In a **lucid dream** the dreamer knows he is dreaming, confirmed by pre-arranged eye-movement signals from REM sleep (**Stephen LaBerge**). **Hypnagogic imagery** comes while falling asleep, **hypnopompic imagery** while waking.

! EXAM TRAP

The Interpretation of Dreams: **1900** (printed November 1899). The collective unconscious is from **Carl Jung**, not Sigmund Freud, Alfred Adler or Karen Horney; his two levels are **collective and individual**, not anima and animus or extraversion and introversion.

Hypnosis, meditation and psychoactive drugs ●●●

Hypnosis is a social interaction in which one person's suggestions of changes in perception, memory or behaviour are experienced by the other as real and involuntary. Its crux is **suggestibility**, and it is not sleep. **Franz Anton Mesmer**'s **animal magnetism** (1770s) was put down to imagination by a royal commission in 1784. The Scottish surgeon **James Braid** coined "hypnotism" (*Neurypnology*, 1843). **Jean-Martin Charcot** took hypnotisability for a sign of hysteria; the **Nancy school** of **Ambroise-Auguste Liébeault** and **Hippolyte Bernheim** showed it was normal suggestion. Freud learned from both, published the mechanism of hysteria with **Josef Breuer** (1893; *Studies on Hysteria*, 1895), then replaced hypnosis with **free association**, the patient on the couch reporting whatever comes to mind.

Susceptibility is measured by the **Stanford Hypnotic Susceptibility Scales** (**André Weitzenhoffer** and **Ernest Ropiequet Hilgard**, 1959). Hypnosis can relieve pain and produce posthypnotic suggestions (acted on at a cue after hypnosis ends) and posthypnotic amnesia, but it cannot give superhuman strength or recover memories reliably, since it raises false recall and confidence together. Hilgard's **neodissociation theory** holds that hypnosis splits consciousness, a **hidden observer** registering pain that the hypnotised person denies; the **social-cognitive theory** of **Theodore Xenophon Barber** and **Nicholas Spanos** says hypnotised people enact an expected role.

Meditation trains attention. **Concentrative meditation** holds it on one object, such as the repeated mantra of **transcendental meditation**, introduced by **Maharishi Mahesh Yogi** in the 1950s; **mindfulness meditation** (opening-up) observes whatever arises without judging it, from Buddhist vipassana. **Herbert Benson** showed that meditation's slowed heart rate and breathing, the **relaxation response** (1975), follow from any similar technique. Surveys showing that regular meditators are less stressed are correlational, however: without random assignment, calmer people may simply meditate more, and only an experiment can show cause (Chapter 9). In McGill University's **sensory deprivation** studies (1954), students lying in a cubicle with goggles and gloves mostly quit within two or three days, and many hallucinated. A **psychoactive drug** alters perception, mood or consciousness by acting on the nervous system (Table 1.9). With repeated use, **tolerance** means a larger dose is needed for the same effect, **dependence** means the drug is needed to function, and **withdrawal** is the unpleasant reaction, usually opposite to the drug's effects, on stopping (Chapter 8).

Table 1.9 The main classes of psychoactive drugs.

Class	Examples	Main effects
Depressants	Alcohol, barbiturates, benzodiazepines	Enhance GABA; sedation, sleep
Stimulants	Caffeine, nicotine, amphetamines, cocaine	Arousal, euphoria, then a crash
Opiates	Morphine, heroin, codeine	Pain relief; mimic endorphins
Hallucinogens	LSD, psilocybin, mescaline	Distorted perception
Cannabis	Ganja, charas, bhang (THC)	Relaxation, slowed time sense

! EXAM TRAP

The crux of hypnosis is **suggestibility**, not the analytical mind or the logical mind. Freud and Breuer, not Freud and Charcot, published the mechanism of hysteria; Freud, not Bernheim, fathered psychoanalysis. Valium, Xanax, Rohypnol and Halcion are **benzodiazepines**: "sedatives" and "hypnotics" name uses, not the class, and they are not barbiturates.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The number of objects apprehended in one brief glance is the _____ of attention.
2. In the Stroop task, automatic _____ slows naming of the ink colour.
3. In Freud's model, memories recallable at will are _____.
4. The master circadian clock is the _____ nucleus.
5. Delta waves mark stage _____.
6. Jung's inherited layer of the psyche is the _____ unconscious.

Answers: (1) span, (2) reading, (3) preconscious, (4) suprachiasmatic, (5) N3, (6) collective.

1.6 Learning: Conditioning, Cognition and Observation HIGH YIELD · 143 QS

Almost everything that makes you a particular person was learned, from your language to the way your stomach tightens in an exam hall. This section runs from the simplest learning, through conditioning, to the maps, insights and imitation that conditioning alone could not explain.

What learning is ●●

The standard definition descends from **Gregory Kimble** (1961): **learning** is a relatively permanent change in behaviour, or in the potential for behaviour, that results from experience. "Potential" matters because learning is inferred from *performance*, what an organism actually does, and the two can come apart. "Experience" rules out *maturation*, change on a biological timetable (a baby walks when her body is ready), as well as fatigue, drugs, motivation and instinct. **Nikolaas Tinbergen** (1907 to 1988) showed that a male stickleback attacks any crude model with a red belly: a *fixed action pattern*, an unlearned, stereotyped sequence triggered by a sign stimulus and run to completion.

The simplest learning involves one stimulus and is called non-associative. In **habituation** the response to a repeated, harmless stimulus decreases: trains outside a new hostel room wake you the first night but not by the end of the week. It is stimulus-specific and shows dishabituation (a novel strong stimulus restores the response), proof that it is learning, not fatigue. Its near opposite, **sensitisation**, is an increased response to many stimuli after a strong or noxious one. **Eric Kandel** (born 1929) traced both to synapses of the sea slug *Aplysia* (Physiological Psychology chapter). In infants, habituation reveals discrimination before speech (**Robert Fantz**, 1964): a baby bored with one picture who looks longer at a new one can tell them apart.

Between instinct and ordinary learning lies **imprinting**, the rapid, unrewarded and largely irreversible attachment to the first suitable moving object, possible only within a *critical period*: in 1935 **Konrad Lorenz** (1903 to 1989) reported incubator-hatched greylag goslings following him as their mother. Everything else here is associative learning: learning that two events go together.

> **! EXAM TRAP**
>
> Fixed action patterns are **innate**, not products of classical or operant conditioning. The near opposite of habituation is **sensitisation**; "habitation" and "inhabitation" are sound-alikes.

Pavlov and classical conditioning ●●●

Ivan Petrovich Pavlov (1849 to 1936), a physiologist who won the 1904 Nobel Prize for work on digestion, collected his dogs' saliva through a tube in the cheek. In the late 1890s he noticed that they salivated before food arrived, at the sight of the dish or the attendant's footsteps. From about 1897 he studied these "psychic secretions": a metronome was sounded, meat powder followed, and after several pairings the metronome alone produced salivation. This is **classical conditioning** (Pavlovian or respondent conditioning): learning in which a neutral stimulus, repeatedly paired with a stimulus that already produces a response, comes to produce a similar response itself.

Five terms describe it. The **unconditioned stimulus** (UCS) produces a response without learning (food in the mouth), and that unlearned response is the unconditioned response (UCR: salivation to food). A neutral stimulus (NS), the metronome on day one, does not produce it. After pairing, the metronome is a **conditioned stimulus** (CS), and its learned response is the conditioned response (CR: salivation to the metronome). So the metronome is an NS before conditioning and a CS after it, and salivation is a UCR to food but a CR to the metronome. The fundamental element is association, the pairing of two stimuli close in time. Table 1.10 adds three cases; each response is involuntary.

Table 1.10 Three classical conditioning situations broken into their parts.

Situation	UCS	UCR	CS	CR
Little Albert	Loud clang	Startle and fear	White rat	Fear of the rat
Student who lost a match	Defeat and humiliation	Distress	The sports ground	Anxiety at the ground
Cancer patient	Chemotherapy drug	Nausea	Hospital smell	Nausea on arrival

Timing matters. In delay conditioning the CS begins first and is still on when the UCS arrives; this is the most effective arrangement, because the CS works as a warning. Trace conditioning, the other forward arrangement, leaves a gap between CS and UCS and is weaker; simultaneous conditioning gives no warning and is weak; backward conditioning (UCS first) is the weakest of all.

! EXAM TRAP

The bell is a **neutral stimulus before** pairing and a **conditioned stimulus after**; never a UCS, never an "operant stimulus". It becomes a CS by **repeated pairing with a UCS**, not by reward, punishment or random occurrence. "Delay" does not mean slow: delay conditioning is the most effective; trace has the gap.

The CR grows during acquisition, quickly at first and then more slowly. If the CS is then presented repeatedly without the UCS, the CR fades: **extinction**. (Pavlov called the UCS the "reinforcement", so withdrawing reinforcement means omitting the UCS.) Extinction suppresses rather than erases learning (Pavlov saw it as active inhibition): after a rest the CR returns with no new pairings (**spontaneous recovery**); a novel distracting stimulus restores it briefly (disinhibition); so does the UCS alone (reinstatement) or a return to the original context (renewal). Hence fears can relapse after therapy.

Responding to stimuli similar to the CS, the more so the more similar they are (a generalisation gradient), is **stimulus generalisation**; responding only to the CS, learned when a similar stimulus is never followed by the UCS, is **stimulus discrimination**. A dog whose circle-versus-ellipse discrimination **Shenger-Krestovnikova** (1921) made impossible became agitated and lost even easy discriminations: *experimental neurosis*, the first laboratory model of anxiety. In *higher-order conditioning* an established CS acts as the UCS for a new stimulus (a black square paired only with the metronome comes to produce salivation), which is how the word "exam" alone can make a heart race. In *sensory preconditioning* (**W. J. Brogden**, 1939) two neutral stimuli are paired first; when one is later conditioned, the other also produces the CR.

! EXAM TRAP

Omitting the UCS produces **extinction**. Spontaneous recovery needs only time, disinhibition a novel stimulus, reinstatement the UCS alone and reconditioning new pairings; sensitisation is no comeback. Avoiding a black cat but playing with a tabby is **discrimination**, not extinction, successive approximation (shaping) or negative transfer.

Conditioned fear and its therapies ●●●

John Broadus Watson (1878 to 1958), the founder of behaviourism, set out with **Rosalie Rayner** to show that emotions are conditioned. In 1920 they paired a white rat seven times with the clang of a steel bar behind the head of an eleven-month-old infant, "Albert B.". Afterwards the rat alone made him cry and crawl away, and the fear generalised to a rabbit, a dog and a fur coat but not to his wooden blocks: a conditioned emotional response. With one infant and no control, the study would be unethical today.

Can fear be unconditioned? In 1924 **Mary Cover Jones** (1897 to 1987) treated Peter, a boy who feared rabbits: while he ate a favourite food, a caged rabbit was brought a little closer each day until he let it nibble his fingers. This is **counterconditioning**: replacing a conditioned response with an incompatible one, so that the client learns the stimulus is no longer threatening. **Joseph Wolpe** (1915 to 1997) borrowed a physiologist's term for the principle, **reciprocal inhibition**: a response incompatible with anxiety, made to occur in the presence of the feared stimulus, weakens the bond between them. His **systematic desensitisation**

(*Psychotherapy by Reciprocal Inhibition*, 1958) has three steps: relaxation training, usually the progressive muscle relaxation of **Edmund Jacobson** (1888 to 1983); an anxiety hierarchy, its items rated on Wolpe's 0 to 100 Subjective Units of Disturbance Scale (SUDS); and desensitisation proper, in which the relaxed client imagines each item in turn, moving up only when it no longer arouses anxiety. A relaxed student who pictures himself on the stage, then introducing himself, then giving his feared talk, is being desensitised.

In **flooding**, another conditioning therapy, the client stays in the most feared situation at full intensity, with no steps or relaxation, until anxiety subsides: extinction with escape blocked. Implosive therapy (**Thomas Stampfl**) is flooding in imagination. Aversion therapy pairs a cue such as the taste of alcohol with a nausea-producing drug, and in covert sensitisation (**Joseph Cautela**, 1967) the behaviour and an unpleasant outcome are both imagined. Exposure and response prevention (**Victor Meyer**, 1966) exposes a person with obsessive-compulsive disorder (OCD) to the feared cue and blocks the compulsion, so that anxiety falls by habituation and extinction (Clinical Psychology chapter).

> **! EXAM TRAP**
>
> Systematic desensitisation is **Wolpe's** and rests on **reciprocal inhibition**, not operant extinction, shaping, habituation, the law of effect, catharsis or vicarious reinforcement. Relaxation is **Jacobson's**; hierarchy ratings use **SUDS**, not the Fear Survey Schedule, State-Trait Anxiety Inventory or Hamilton Anxiety Rating Scale. Graded steps with relaxation is desensitisation, full intensity is flooding, and imagined behaviour plus imagined unpleasantness is covert sensitisation. Exposure and response prevention rests on **habituation and extinction**; "graded exposure" is a procedure, not a principle. **Symptom substitution** is not a technique but the psychoanalytic prediction that a removed symptom will be replaced.

Prediction and preparedness ●●

Conditioning was long thought to need only contiguity, the CS and UCS close together in time. In 1968 **Robert Rescorla** (1940 to 2020) showed that rats did not fear a tone if shocks were as likely without it as with it: conditioning depends on contingency, how well the CS predicts the UCS. In blocking (**Leon Kamin**, 1969), rats already afraid of a noise learned almost nothing about a redundant light added to it. In overshadowing the most salient part of a compound CS takes most of the learning; in latent inhibition (**Robert Lubow** and **A. U. Moore**, 1959) a stimulus often met alone beforehand conditions slowly.

The Rescorla-Wagner model (Rescorla and **Allan Wagner**, 1972), $\Delta V = \alpha\beta(\lambda - \Sigma V)$, says that an animal learns only when surprised. A CS gains associative strength (ΔV) in proportion to the prediction error, the gap between the most the UCS can support (λ) and what all stimuli present already predict (ΣV), scaled by salience and learning rate ($\alpha\beta$). The error shrinks as the UCS becomes expected, so each trial adds less; in blocking it is zero from the start. In the 1990s **Wolfram Schultz** found dopamine neurons of the ventral tegmental area signalling exactly this error. The CS is thus a signal that predicts the UCS, not, as Pavlov's stimulus substitution theory held, a substitute for it.

KEY STUDY: GARCIA AND KOELLING'S "BRIGHT, NOISY, TASTY WATER" (1966)

Question. Are all stimuli equally easy to associate with all consequences? **Method. John Garcia** (1917 to 2012) and **Robert Koelling** let rats drink saccharin-flavoured water whose every lick also produced a flash and a click; then some were made ill, others shocked. **Result.** Ill rats avoided the taste but not the light and click; shocked rats avoided the light and click but not the taste. **Conclusion.** Taste "belongs" with illness, and sights and sounds with pain; the finding so contradicted theory that Garcia struggled to publish it.

Such **conditioned taste aversion** forms in a single trial, even when illness follows the taste by hours, and refutes equipotentiality, the assumption that any stimulus can be associated with any consequence equally easily. **Martin Seligman** (born 1942) called the principle **preparedness** (1970): each species is biologically prepared to learn some associations easily, which is why phobias of snakes and heights are common and phobias of electric sockets rare. Chemotherapy patients are given a novel "scapegoat" food to protect their usual diet, and advertisers pair a brand with a beloved film star so that the liking transfers.

! EXAM TRAP

Taste aversion is **classical conditioning** (taste = CS, illness = UCS), not operant, second-order conditioning or a "spurious association" (a statistical term). It illustrates **biological preparedness**, defying equipotentiality and contiguity. Prediction error is signalled by **dopamine neurons in the ventral tegmental area**, not serotonin, GABA or acetylcholine. Latent *inhibition* is not Tolman's latent *learning*.

Thorndike, Skinner and consequences ●●●

Most behaviour, however, is emitted rather than elicited, and its consequences decide whether it recurs. **Edward Lee Thorndike** (1874 to 1949) shut hungry cats in puzzle boxes (*Animal Intelligence*, 1898). Each cat scrambled until it hit the release by accident, and over trials its escape time fell gradually, not suddenly: trial-and-error learning. His **law of effect** (1911) states that responses followed by satisfaction are strengthened and those followed by discomfort weakened. His connectionism, which treats learning as forming stimulus-response bonds, added the law of exercise (use strengthens a bond), later abandoned, and the law of readiness (acting when ready is satisfying).

The radical behaviourist **Burrhus Frederic Skinner** (1904 to 1990), in *The Behavior of Organisms* (1938), separated respondent behaviour, elicited by a stimulus, from operant behaviour, emitted by the organism and operating on the environment. In **operant conditioning** the probability of a response is changed by its consequences (Thorndike's trial-by-trial version is often called instrumental conditioning). In the Skinner box a rat pressed a lever freely while a cumulative recorder plotted its rate. A discriminative stimulus, such as a chai stall's "Open" board, signals that a response will be reinforced, bringing it under stimulus control. Applied behaviour analysis and functional analysis, which ask what a problem behaviour gets the person or lets him escape, rest on these principles; merely recording a behaviour often changes it (reactivity).

Two questions classify every consequence (Table 1.11): does the behaviour increase (reinforcement) or decrease (punishment), and is a stimulus added (positive) or removed (negative)? Positive and negative mean plus and minus, not good and bad. A friend's warm

agreement with an opinion you voice is positive reinforcement, so the attitude strengthens. Pressing a lever to switch off a shock already on is **negative reinforcement**. A fine for jumping a red light is negative punishment. A parent who buys a sweet to end a tantrum positively reinforces the tantrum and is negatively reinforced for giving in.

Table 1.11 The four kinds of operant consequence.

Effect on behaviour	Stimulus added (positive)	Stimulus removed (negative)
Increases (reinforcement)	Positive reinforcement: something wanted is given (praise)	Negative reinforcement: something unpleasant is taken away (a headache ends)
Decreases (punishment)	Positive punishment: something unpleasant is given (a scolding)	Negative punishment: something wanted is taken away (time-out, response cost)

Food is a primary reinforcer, effective without learning. Money and praise are **secondary reinforcers**, neutral stimuli that became reinforcing through association with primary ones; one linked with many primary reinforcers, as money is, is a generalised reinforcer. (Harry Harlow's infant monkeys preferred a cloth mother to a wire one that fed them: attachment rests on contact comfort, not secondary reinforcement.) In the 1960s **Teodoro Ayllon** and **Nathan Azrin** built the **token economy** at Anna State Hospital, Illinois: long-stay psychiatric patients earned tokens for self-care and ward work and exchanged them for privileges (*The Token Economy*, 1968).

Escape learning terminates an aversive stimulus already present; avoidance learning prevents its arrival. **Orval Hobart Mowrer** (1907 to 1982) explained avoidance with his **two-factor theory** (1947): a warning signal comes to elicit fear by classical conditioning, and avoiding it is negatively reinforced by fear reduction, an operant process that also keeps the fear from extinguishing. Compulsive washing in OCD is such an escape or safety behaviour. Time-out (brief removal from a setting where reinforcement is available) and **response cost** (loss of a reinforcer already held) are negative punishment. As **William Kaye Estes** (1919 to 2011) showed in 1944, punishment suppresses a response without erasing it; it also breeds fear of the punisher and models aggression.

New behaviour is built by **shaping**, reinforcing successive approximations to a target: facing the lever, approaching it, touching it, finally pressing it. Shaping works on a response's speed and force and needs no words. Chaining links responses into a sequence; a prompt adds a cue and fading withdraws it. Operant extinction, the fading of a response no longer reinforced, often begins with an extinction burst of harder, more varied responding (jabbing at a lift button), after which responding falls back to its operant level, the rate before reinforcement.

> **! EXAM TRAP**
>
> The law of effect is **Thorndike's** (not Skinner's, Pavlov's or Watson's), not modelling, insight or unconscious drive. Operant conditioning works through **reinforcement and punishment**, not modelling or cognitive dissonance. **Negative reinforcement is not punishment**, and it too increases behaviour, though one real item keyed only positive reinforcement. Removing a desirable stimulus to reduce behaviour is **response cost**, not negative reinforcement, positive punishment or extinction; time-out is **negative punishment**. OCD compulsions are **escape (avoidance) behaviours**, not conditioned stimuli, unconditioned responses or positively reinforced acts.

> **! EXAM TRAP**
>
> The token economy is **operant** (secondary reinforcement), the therapy **not** derived from classical conditioning, and **Ayllon and Azrin's** (Amity 2023 keyed Skinner, who supplied only the principles). Stickers that buy a homework exemption are **tokens**, not primary or generalised reinforcers or "prepotent" (dominant) responses. Ignoring tantrums is **extinction**, not "operant escape" (another wrong Amity 2023 key). Shaping is not chaining, prompting, fading or the law of effect; RML 2024 keyed its principles as speed, power and non-verbal.

Schedules and other operant effects ●●●

A **schedule of reinforcement** is the rule deciding which responses are reinforced. Continuous reinforcement, after every correct response, teaches fastest but extinguishes fastest; partial reinforcement makes behaviour persist far longer once reward stops (the partial reinforcement extinction effect). Reward may depend on the number of responses (ratio) or on time elapsed (interval), fixed or variable, giving the four schedules that **Charles Ferster** and Skinner mapped in 1957 (Figure 1.7).

- Fixed ratio (FR), as in piece-rate pay: high, steady responding with a brief post-reinforcement pause.
- Variable ratio (VR), as in slot machines: the highest, steadiest rate and the greatest resistance to extinction.
- Fixed interval (FI), as in checking a dish due out of the oven: a pause, then faster and faster responding, the scallop.
- Variable interval (VI), as in checking for a reply to a message: moderate, very steady responding.

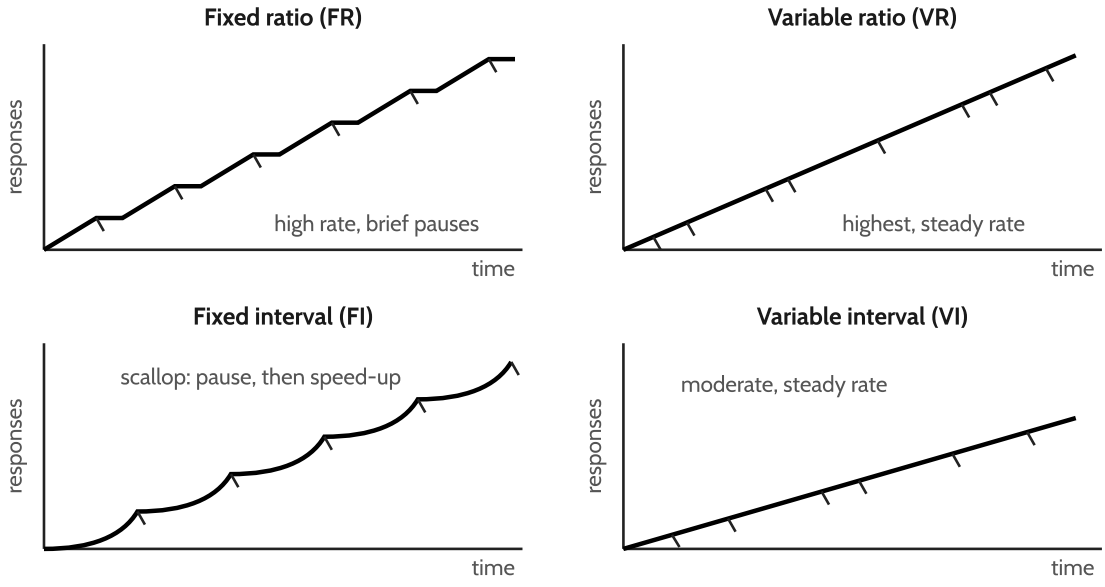


Figure 1.7 Stylised cumulative records for the four basic schedules. The steeper the line, the faster the responding; short diagonal ticks mark reinforcers. Notice the flat steps after each reinforcer on FR, the unbroken steep line on VR, the scallops on FI and the steady, shallower line on VI.

> **! EXAM TRAP**
>
> "Number" means ratio, "time" interval, "unpredictable" variable. Reinforcement after every response is **continuous**, the least resistant to extinction. Sheena's chocolate "for every 45 minutes" of study is keyed **fixed interval**.

Differential reinforcement extinguishes a problem behaviour while reinforcing other behaviour, that is, its absence (DRO), an incompatible (DRI) or alternative (DRA) behaviour, or a low rate of it (DRL). The **Premack principle** of **David Premack** (1959) holds that a more probable activity can reinforce a less probable one: homework first, then video games. Skinner's pigeons, fed every 15 seconds whatever they did (1948), developed odd rituals: superstitious behaviour, strengthened by accidental reinforcement.

KEY STUDY: LEARNED HELPLESSNESS (SELIGMAN AND MAIER, 1967)

Martin Seligman and **Steven Maier** gave dogs shocks they could switch off, identical shocks they could not control (the yoked group), or none. Later, in a shuttle box, most yoked dogs passively took shocks they could have escaped by jumping a barrier. Lack of control, not shock, produced **learned helplessness**: passive acceptance of aversive events, even avoidable ones, after repeated unavoidable ones. It became a model of depression, later recast in attributional terms (Psychiatric Disorders chapter).

Biology limits operant learning too: **Keller Breland** and **Marian Breland** (1961) found raccoons trained to bank coins rubbing them together as if washing food (instinctive drift). **Richard Herrnstein**'s matching law (1961) states that responses are divided among options in proportion to the reinforcement each yields. Rats that **Leo Crespi** (1942) shifted to a medium reward ran slower if shifted down, and faster if shifted up, than rats always given it: contrast effects, too abrupt to be learning, showing that reward size changes motivation. In **biofeedback** a bodily process is measured and displayed so that changes in the desired direction are reinforced, following **Neal Miller**'s (1909 to 2002) claim that autonomic responses can be operantly conditioned. Electromyographic (EMG) biofeedback suits tension headache, and **Arnold Kegel**'s perineometer (1948) was an early form.

> **! EXAM TRAP**
>
> Premack: **high-probability behaviour reinforces low-probability behaviour**, never the reverse. Learned helplessness is **Seligman and Maier's**, not Beck's, Sullivan's or Kahneman's. The Crespi effect is a **change in motivation when reward size changes**, not retroactive inhibition, extinction or insight. Autonomic control by operant conditioning is **biofeedback**, not hypnosis, sublimation or desensitisation.

Table 1.12 compares the two conditionings. They are not "entirely different" but interact constantly: Albert's fear was classical, but had he crawled away and felt relief, his avoidance would have been operant. The hallmark of behaviour therapy is the use of both to eliminate unwanted behaviour, not a search for underlying causes, insight, cognitive restructuring, dream analysis or free association.

Table 1.12 Classical versus operant conditioning.

Feature	Classical conditioning	Operant conditioning
Key figures	Pavlov, Watson	Thorndike, Skinner
What is associated	Two stimuli (CS and UCS)	A response and its consequence
Behaviour	Elicited, involuntary, emotional	Emitted, voluntary, goal-directed
Extinction	CS presented without the UCS	Response no longer reinforced

> **! EXAM TRAP**
>
> The main difference: **classical pairs two stimuli; operant relies on reinforcement and punishment**, not "classical relies on reinforcement" (backwards) or "only operant involves cognition".

Latent learning, insight and learning sets ●●●

Against the view that learning is only stimulus-response connections stamped in by reward, **Edward Chace Tolman** (1886 to 1959), a purposive behaviourist, argued that animals learn knowledge.

> **KEY STUDY: TOLMAN AND HONZIK'S LATENT LEARNING EXPERIMENT (1930)**
>
> Tolman and **Charles Honzik** ran three groups of hungry rats through a maze daily for seventeen days: one always found food in the goal box, one never did, and one found food only from day 11. The third group made as many errors as the unrewarded rats for ten days, then at once matched the always-rewarded rats. It had learned the maze all along; reward merely gave it a reason to show it, so learning and performance differ.

This is **latent learning**: learning without obvious reinforcement that stays hidden until there is an incentive to show it. Tolman (1948) said the rats had formed a **cognitive map**, a mental representation of the layout; rats learn places and expectancies of what leads to what, not only turns. **Wolfgang Köhler** (1887 to 1967), a founder of Gestalt psychology (Section 1.2), studied chimpanzees on Tenerife (1913 to 1920). Sultan, unable to reach a banana with either of two short sticks, later fitted one into the other and at once raked it in; others stacked boxes. Such **insight learning**, the sudden grasp of a solution by perceiving the relations among a problem's elements, is abrupt, smooth once found, well retained and readily transferred; its flash of understanding is the aha experience. **Harry Harlow** (1905 to 1981) found that monkeys given hundreds of discrimination problems came to solve each new one in a single trial: a learning set, or learning to learn (1949), suggesting that apparent insight rests on earlier, gradual learning. Concept formation (Section 1.8) runs from observing instances, through discriminating them from non-instances and generalising to all that share the essential features, to abstracting the defining properties.

> **! EXAM TRAP**
>
> Latent learning and the cognitive map are **Tolman's**, not Bandura's, Skinner's, Köhler's or Harlow's (Gould studied honeybee navigation). Sultan means **insight**, and the realisation is the **"aha" experience**, not the "light bulb"; insight graded in a mental status examination is psychiatric. Concept formation runs **observation, discrimination, generalisation, abstraction**; assimilation is Piaget's term.

Observational learning ●●●

Most complex skills, from rolling a chapati to a cover drive, are learned largely by watching. This is **observational learning** (social learning, modelling): learning by watching others, called models, and the consequences of their behaviour. **John Dollard** (1900 to 1980) and Neal Miller (1941) held that imitation must itself be reinforced, the early social learning theory; **Albert Bandura** (1925 to 2021) showed that it need not be.

KEY STUDY: BANDURA, ROSS AND ROSS'S BOBO DOLL EXPERIMENT (1961)

Bandura, **Dorothea Ross** and **Sheila Ross** let 72 nursery-school children watch an adult punch, kick and hammer an inflatable Bobo doll. Left later with the doll, these children copied the model's specific acts and words far more than children who saw a quiet model or none, with no reinforcement at all. In 1965 children who saw the model punished imitated less, until offered a reward for showing what the model had done, when all groups did equally well. **Criticism.** Bobo is made to be hit, and only short-term effects were measured.

The model's consequences, then, affect performance, not learning: seeing a model rewarded increases imitation (vicarious reinforcement), and seeing one punished reduces it. Bandura's four processes are attention (noticing the model), retention (storing the act in images and words), motor reproduction (turning the memory into action) and motivation (a reason to perform). The first three govern learning and the fourth performance, so reinforcement is not necessary, and the model must actually demonstrate the target behaviour. Modelling is also a therapy (Clinical Psychology chapter).

With **Richard Walters** (1963) Bandura developed later social learning theory, renamed social cognitive theory in 1986. Its **reciprocal determinism** (1978) holds that the person (beliefs, expectations), behaviour and environment influence one another in both directions, against Skinner's view that environment alone shapes behaviour. Self-regulation means that most behaviour is guided by self-set standards and self-reward rather than external rewards. The theory's central motivational construct is **self-efficacy** (1977), a person's belief in his capability to carry out the actions a specific task requires. (**Julian Rotter**'s separate social learning theory of 1954 belongs to the Personality chapter: behaviour potential, the likelihood of a behaviour in a given situation in relation to a reinforcement, depends on expectancy and reinforcement value. Social learning frameworks in general explain behaviour by expectancies, situational demands and incentives rather than by reinforcement history alone.)

! EXAM TRAP

The four processes are **attention, retention, motor reproduction and motivation**, not sensory-perceptual or memory stages; the one NOT necessary is **reinforcement**, a view that is **Bandura's**, not Skinner's, Tolman's or Miller and Dollard's. Reciprocal causation belongs to **social cognitive theory**, not two-factor, drive-reduction or constructivist theory, and is neither reciprocal inhibition nor reciprocal altruism. Self-efficacy (Bandura, not Beck, Ellis or Skinner) is not self-esteem, external locus of control or Adlerian striving; it, not goal difficulty, rewards or drives, is the motivational core of social cognitive theory. Most behaviour stems from **self-regulation**, not "reciprocal dualism" or "behavioural determinism". Match: Dollard and Miller, early social learning; Bandura and Walters, later; Skinner, radical behaviourism; Maslow, self-actualisation.

Other theorists and learning in practice ●●

The grand theories divided on two questions: is what is learned a stimulus-response connection, as the stimulus-response (S-R) behaviouristic approach holds, or knowledge (Tolman, Köhler); and does learning need only contiguity, or reinforcement too? **Edwin Ray Guthrie** (1886 to 1959) chose contiguity: a stimulus combination that accompanied a movement will tend to be followed by it again, at full strength after one trial. A habit is broken by attaching a new response to its cue: present the cue too weakly to trigger the habit and strengthen it gradually, as with a light blanket before a saddle (threshold method); repeat the response to exhaustion (fatigue method); or present the cue while a competing response occurs (incompatible response method). **Clark Leonard Hull** (1884 to 1952) chose reinforcement, defined as drive reduction: reaction potential equals habit strength multiplied by drive (${}_{S}E_{R} = {}_{S}H_{R} \times D$). Table 1.13 adds Estes and **Robert Mills Gagné** (1916 to 2002). Mowrer and his wife devised the bell-and-pad for bed-wetting (1938), and constructivists (Piaget, Vygotsky, Bruner) hold that learners actively build knowledge.

Table 1.13 Other learning theorists at a glance.

Theorist	Family	Central idea	Key work
Guthrie	S-R contiguity	One-trial learning; three ways to break habits	*The Psychology of Learning* (1935)
Hull	S-R reinforcement	Drive reduction; habit strength × drive	*Principles of Behavior* (1943)
Estes	Mathematical	Stimulus sampling: each trial conditions a fraction (θ) of stimulus elements	Stimulus sampling theory (1950)
Gagné	Instructional	Eight types in a hierarchy: signal, S-R, chaining, verbal association, multiple discrimination, concept, principle, problem solving	*The Conditions of Learning* (1965)

A learning curve plots performance against practice: usually negatively accelerated (fast early gains, then diminishing returns), sometimes positively accelerated (a slow start on a hard task) or S-shaped. **William Lowe Bryan** and **Noble Harter** (1897, 1899) found telegraphers stalling on a plateau, a temporary flat stretch while letter habits became automatic, before word habits lifted them again; it is not the final physiological limit. Overlearning, practice beyond the first errorless recall, aids retention (**William C. F. Krueger**, 1929). Distributed (spaced) practice usually beats massed practice; whole learning suits short, meaningful material and part learning long, difficult material. Knowledge of results (feedback) is essential,

and a specific goal makes practice purposeful (**Edwin Locke** and **Gary Latham**). Learners with learning difficulties need multisensory input and regular repetition.

Transfer of training, the effect of one task on learning another, is positive (cycling helps with a motorcycle), negative (an old habit interferes) or zero. **William James** (1890) and Thorndike refuted formal discipline, the belief that Latin trains the mind; Thorndike's identical elements theory ties transfer to elements the tasks share. **Sidney Leavitt Pressey** built the first teaching machine (1920s); Skinner's programmed instruction (1954) used small steps, active responses, immediate feedback and self-pacing.

> **! EXAM TRAP**
>
> Learning through S-R associations is the **stimulus-response behaviouristic approach**, not the cognitive, psychoanalytic or existential approach. "Learning always leads to efficacy" given **a specific goal**, not ability, aptitude or knowledge. A plateau is not the physiological limit or extinction.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The near opposite of habituation is _____.
2. Pavlov's bell was a neutral stimulus before pairing and a _____ stimulus after it.
3. Wolpe's systematic desensitisation rests on _____.
4. Removing an unpleasant stimulus to increase a behaviour is _____ reinforcement.
5. The schedule with the highest rate and greatest resistance to extinction is _____.
6. Bandura's four processes are attention, retention, _____ and motivation.

Answers: (1) sensitisation, (2) conditioned, (3) reciprocal inhibition, (4) negative, (5) variable ratio and (6) motor reproduction.

1.7 Memory: Encoding, Storage, Retrieval and Forgetting HIGH YIELD · 111 QS

Memory, the capacity to retain information and use it later, is not one thing. After encephalitis destroyed both his hippocampi in 1985, the British musician **Clive Wearing** could follow a conversation only while it lasted and had lost almost every episode of his past, yet he could still sight-read and conduct. To see how, begin with how memory was first measured.

Ebbinghaus and the measurement of memory ●●

Cognitive psychology treats the mind as an information processor (Section 1.2) and divides remembering into *encoding* (turning information into a code the system can hold), *storage* and *retrieval*. Encoding needs attention: fewer than half of the adults tested by **Raymond Nickerson** and **Marilyn Adams** (1979) could pick out the correct drawing of a coin they had handled thousands of times. Details never attended to are never encoded (*encoding failure*) and so cannot be retrieved.

The first memory experiments were made by **Hermann Ebbinghaus** (1850 to 1909), his own sole participant. To escape the associations of real words he invented the *nonsense syllable* (DAX, BUP), learned lists to one perfect recitation and relearned them later. Relearning was faster even when nothing could be recalled. The **savings method** expresses this as original learning time minus relearning time, divided by original learning time, multiplied by 100: a list learned in 20 readings and relearned in 12 shows (20 minus 12) divided by 20, or 40 per cent, savings. His *forgetting curve* (*Über das Gedächtnis*, 1885) falls steeply, losing more than half within an hour and about two thirds within a day, then levels off without reaching zero. He also first noted the spacing effect and the advantage of a list's beginning and end.

Retention is measured by *recall*, which is free (any order), serial (original order) or cued (from a hint); *paired-associate learning*, introduced by **Mary Whiton Calkins** in 1894, is cued recall of pairs, such as a Kannada word and its English meaning. *Recognition*, picking old items from new ones as in a multiple-choice question, is easier than recall. Relearning is the most sensitive measure. The *reconstruction method* hands back the presented items shuffled for the person to put in order.

> **! EXAM TRAP**
>
> Items presented, scrambled and put back in order (RML 2024) is the reconstruction method, a measuring technique, not Bartlett's reconstructive memory. Ebbinghaus, **Francis Galton** and Sigmund Freud were independent pioneers, not laboratory founders, and each devised a new method (nonsense syllables; questionnaires and word association; free association). RML 2020 keys "independent pioneers" alone, as Ebbinghaus's method was experimental, not observational; a bank version keys "(a) and (c)", adding "novel observational methods".

The multi-store model ●●●

In 1968 **Richard C. Atkinson** and **Richard M. Shiffrin** proposed the **multi-store model**, also called the modal or stage model: attended information passes from sensory memory to short-term memory, and rehearsed information passes on to long-term memory, from which it is retrieved when needed. Attention and rehearsal are *control processes* that the person chooses.

The first store, **sensory memory** (the sensory register), briefly holds a large amount of raw, pre-categorical information, one store per sense. **Ulric Neisser** named the visual store *iconic memory* and the auditory store *echoic memory* in 1967; a touch store, haptic memory, is little studied.

> **KEY STUDY: SPERLING'S PARTIAL REPORT (1960)**
>
> Shown twelve letters for 50 milliseconds, people report only four or five but insist they saw more. **George Sperling** sounded a tone just after the flash to signal which of the three rows to report. People gave about three of the four cued letters, so about nine of the twelve had been briefly available. Delaying the tone by about one second removed the advantage: the icon fades within roughly 250 to 500 milliseconds.

Echoic memory lasts two to four seconds, as the "three-eared man" study of **Christopher Darwin**, **Michael Turvey** and **Robert Crowder** (1972) showed.

Short-term memory (STM) holds a little information for seconds unless it is rehearsed. **George Armitage Miller** put its span at "the magical number seven, plus or minus two" (1956) meaningful units, or chunks. Grouping items into larger meaningful units is **chunking**: I N D I A I I T I S R O is twelve letters but three chunks (INDIA, IIT, ISRO). Without chunking and rehearsal, capacity is nearer four chunks (**Nelson Cowan**, 2001). In the Brown-Peterson technique (**John Brown**, 1958; **Lloyd Peterson** and **Margaret Peterson**, 1959), counting backwards blocked rehearsal, and recall of a consonant trigram fell from about 80 per cent at 3 seconds to about 10 per cent at 18, hence a duration of 18 to 30 seconds; **Geoffrey Keppel** and **Benton Underwood** (1962) traced the loss to interference, not decay. STM codes mainly by sound: **Reuben Conrad** (1964) found people writing V for B, and **Alan Baddeley** showed in 1966 that sound-alike words confuse immediate recall and meaning-alike words delayed recall. *Long-term memory* (LTM) codes mainly by meaning (Table 1.14).

Table 1.14 The three stores of the multi-store model compared.

Feature	Sensory memory	Short-term memory	Long-term memory
Capacity	Large	7 ± 2 items (Miller); about 4 chunks (Cowan)	No known limit
Duration	Icon 0.25 to 0.5 s; echo 2 to 4 s	About 18 to 30 s unrehearsed	Minutes to a lifetime
Main code	Raw sensory	Acoustic	Semantic
Main cause of loss	Decay and masking	Displacement and interference	Interference and retrieval failure

> **! EXAM TRAP**
>
> "Largest capacity but shortest duration" (JEMAS 2025) is sensory memory; "largest capacity" alone in the Atkinson and Shiffrin model is long-term memory. "Icon and echo" are always sensory memory, and the store that "lasts only a few seconds" is sensory memory, not eidetic imagery.

Free recall gives the U-shaped **serial position effect**, or primacy-recency effect (**Bennet Murdock**, 1962; Figure 1.8). The *primacy effect* reflects extra rehearsal of early items into LTM; the *recency effect* reflects the last items still sitting in STM. **Murray Glanzer** and **Anita Cunitz** (1966) found that 30 seconds of counting before recall abolished recency but spared primacy.

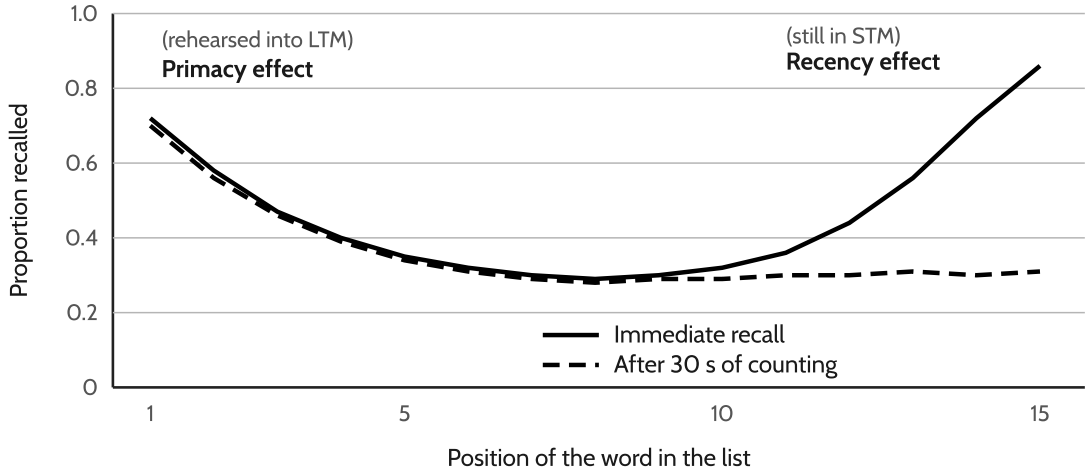


Figure 1.8 The serial position curve in free recall (schematic, with the pattern reported by Glanzer and Cunitz, 1966). Immediate recall (solid line) shows both primacy and recency. A 30-second filled delay (dashed line) removes the recency effect but leaves primacy intact.

Amnesic patients such as H.M., met below, have a normal span and recency effect but no new long-term learning, while patient K.F. (**Tim Shallice** and **Elizabeth Warrington**, 1970) had a two-item span and normal long-term learning: a double dissociation that supports separate stores. Yet neither store proved unitary, and mere repetition does little for LTM.

> **! EXAM TRAP**
>
> The stage model has three stores; working memory is NOT one of them (RML 2025), and information must pass through short-term (sometimes "short-term (working)") memory to reach LTM. RML 2024 keys "McCray and Hunter" (**J. W. McCrary** and **W. S. Hunter**, 1953) for first observing the primacy-recency effect, although Ebbinghaus noted it in 1885. Better recall of the last items is recency, not the Flynn effect (rising IQ scores) or halo effect (a rating bias). As order effects in a psychophysics experiment, primacy and recency are controlled by counterbalancing (Section 1.3).

Working memory ●●●

Multiply 24 by 7 in your head and you must hold numbers, calculate and store a partial result at once. Alan Baddeley and **Graham Hitch** (1974) therefore recast STM as **working memory**, a limited-capacity system for holding and manipulating information during reasoning and comprehension. Using the dual-task method, they found that people holding six digits still reasoned accurately, so STM must have several parts (Figure 1.9).

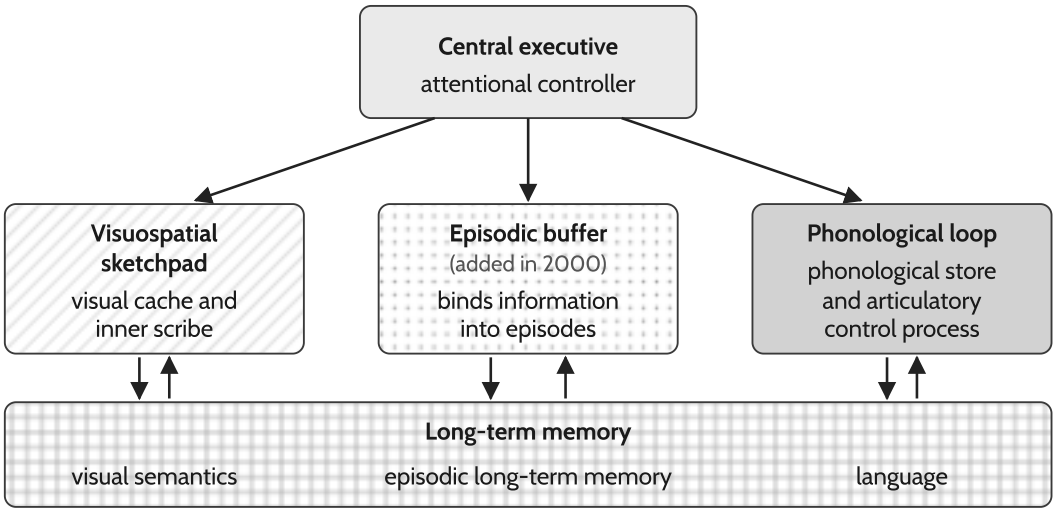


Figure 1.9 Baddeley's working memory model (Baddeley and Hitch, 1974; episodic buffer added by Baddeley, 2000). The central executive directs attention but stores nothing; the three subsidiary systems hold information and exchange it with long-term memory.

The **phonological loop** stores and rehearses auditory-verbal information: a phonological store (inner ear) holds speech sounds for about two seconds, and an articulatory control process (inner voice) refreshes them. Similar-sounding items are confused, spans are larger for short words (the word-length effect), and *articulatory suppression* (saying "the, the, the") abolishes the word-length effect. The **visuospatial sketchpad** holds visual and spatial information, such as a mentally rotated object; **Robert Logie** (1995) split it into a visual cache and an inner scribe.

The **central executive**, an attentional controller that stores nothing, focuses, divides and switches attention and inhibits unwanted responses. Random number generation loads it, and dorsolateral prefrontal damage causes a dysexecutive syndrome. In the Stroop task (**John Ridley Stroop**, 1935), naming the ink colour of RED printed in blue is slow because automatic reading competes with controlled colour naming, an interference in selective attention (Section 1.5). The **episodic buffer**, added by Baddeley in 2000, holds about four chunks and binds information from the loop, the sketchpad and LTM into single, coherent episodes. STM is storage alone (forward span); working memory is storage plus processing, tested by complex span and by backward digit span.

> **! EXAM TRAP**
> Random number generation loads the central executive (RML 2022); repeating a word loads the loop, a spatial configuration the sketchpad, irrelevant speech the phonological store. A "visual executive loop" (Amity 2023) does not exist, and the Stroop effect is selective attention, not memory.

Levels of processing ●●

Fergus Craik and **Robert Lockhart** (1972) argued that memory is a by-product of how deeply information is processed. Their **levels of processing** run from shallow structural processing (is the word in capitals?) through phonemic processing (does it rhyme with WEIGHT?) to

deep semantic processing (does it fit the sentence?), and deeper processing leaves more durable traces, as Craik and Endel Tulving (1975) confirmed. Maintenance rehearsal, mere repetition, adds little to LTM; elaborative rehearsal, linking an item to meaning, builds it.

Words judged for "Does this describe you?" are recalled about twice as well as words judged for meaning: the **self-reference effect** (**T. B. Rogers**, **Nicholas Kuiper** and **W. S. Kirker**, 1977). Words you generate also beat words you merely read (the generation effect). Critics called depth circular, since it is defined by the memory it explains, and **Donald Morris**, **John Bransford** and **Jeffery Franks** (1977) found that rhyme processing wins on a rhyming test: memory depends on the match between study and test (transfer-appropriate processing).

> **! EXAM TRAP**
>
> The self-reference effect is Rogers, Kuiper and Kirker's (keyed "Rogers and Kuiper", RML 2024), not Craik and Lockhart's, **Allan Paivio**'s (dual coding theory, 1971: verbal plus imagery codes) or **Michael Posner**'s (attention). It means better recall of self-related information (IHBAS 2020), not self-concept, self-schema or self-awareness (Personality chapter).

The systems of long-term memory ●●●

Clive Wearing kept skills and facts but lost his past, so long-term memory is not one store. In 1972 **Endel Tulving** (1927 to 2023) separated **episodic memory**, personal events with their what, where and when (your first day of college), from **semantic memory**, the network of facts and concepts that make up our knowledge of the world, also called generic memory. Episodic recall carries autonoetic consciousness, a sense of travelling back in time; semantic memory carries only noetic knowing. Knowing how (riding a bicycle) is **procedural memory**. Tulving's patient K.C. kept his general knowledge but lost every episode of his life.

Explicit memory involves conscious recollection, tested by recall and recognition; **implicit memory** is an influence of past experience on behaviour, feelings or thought without conscious recollection (**Peter Graf** and **Daniel Schacter**, 1985), roughly **Larry Squire**'s declarative and nondeclarative memory. Declarative memory needs the hippocampus; nondeclarative memory covers procedural skills (striatum), priming, classical conditioning (amygdala, cerebellum) and habituation. In *priming*, earlier exposure eases processing of the same item (repetition priming) or a related one: BUTTER is recognised faster after BREAD (semantic priming; **David Meyer** and **Roger Schvaneveldt**, 1971), even with a masked, unseen prime, so semantic priming is a tool for studying subliminal perception.

> **! EXAM TRAP**
>
> Implicit memory does NOT involve conscious recollection (RML 2020); it "affects performance without conscious awareness" (RML 2025), and limited capacity describes STM. Procedural memory is one kind of implicit memory.

Autobiographical memory holds the specific episodes of one's own life; older adults recall a surplus from ages 10 to 30, the *reminiscence bump* (**David Rubin** and colleagues, 1986). *Prospective memory*, remembering to act in future (a tablet at 8 p.m.), contrasts with retrospective memory. **Roger Brown** and **James Kulik** (1977) named the **flashbulb memory**, a vivid memory of the circumstances in which one learned of a shocking event, and credited it to a "Now Print!" mechanism.

KEY STUDY: NEISSER AND HARSCH, THE CHALLENGER DISASTER (1992)

The morning after the Challenger shuttle exploded in January 1986, Ulric Neisser and **Nicole Harsch** had students write down how they heard the news. Two and a half years later, a quarter of 44 were wrong on every major detail, yet confident. After 11 September 2001, **Jennifer Talarico** and David Rubin (2003) found that flashbulb and everyday memories lost consistency equally, but flashbulb memories stayed more vivid and more confidently believed.

Flashbulb memories are thus special in vividness and confidence, not accuracy, and arise from emotional arousal at encoding plus rehearsal. *Eidetic imagery*, a visual image that lingers half a minute or more, occurs in a few children and rarely in adults. *Metamemory* (**John Flavell**, 1971), knowledge and control of one's own memory, is part of metacognition and includes the feeling of knowing, a sense that one could recognise an answer one cannot recall. In the **tip-of-the-tongue state**, which Roger Brown and **David McNeill** (1966) produced with definitions of rare words (sextant), people knew the first letter and syllable count: the word was stored but not retrievable.

! EXAM TRAP

A vivid recollection of an important event is a flashbulb memory (RML 2025), not an engram, hindsight bias (the "knew it all along" distortion of memory) or equipotentiality. Its confidence stays very high as accuracy falls (NIMHANS 2026 recall), and arousal plus rehearsal explains it, not Now Print!, lack of interference or eidetic imagery. Tip-of-the-tongue is a retrieval failure (RML 2024), not storage, repression or aphasia (Wernicke's or Broca's, where language itself is damaged). Planning, monitoring and controlling memory is metacognition (IHBAS 2020), not attention, inhibition or a knowledge base.

Networks, schemas and reconstruction ●●●

In the hierarchical network of **Allan Collins** and **Ross Quillian** (1969), concepts are nodes (ANIMAL, BIRD, CANARY) and, by cognitive economy, each property is stored once at the highest node it fits, so "A canary has skin" takes longer to verify than "A canary can sing". In the spreading activation model of Collins and **Elizabeth Loftus** in 1975, activation spreads to related concepts and primes them. In **connectionism**, or parallel distributed processing (**David Rumelhart**, **James McClelland** and colleagues, 1986), a concept is a pattern of simultaneous activation across many interconnected units, and knowledge lies in the connection weights. A proposition is the smallest unit of knowledge that can be true or false.

! EXAM TRAP

"A network of interconnected nodes" (IHBAS 2024) keys the connectionist model, although Collins and Quillian's network also has nodes; "associative" and "linkage" describe any network rather than name a model. Parallel distributed processing means associative networks and simultaneous activation (IHBAS 2025), not serial processing.

A **schema** is organised knowledge, built from experience, that guides perception and memory; the schema approach studies how such knowledge is used to understand behaviour. A script, a schema for a routine (the restaurant script of **Roger Schank** and **Robert Abelson**,

1977: enter, order, eat, pay, leave), belongs to generic memory. The schema idea came from **Frederic Charles Bartlett** (1886 to 1969). In *Remembering* (1932), English students retelling the Native American tale "The War of the Ghosts" shortened it and emphasised a few details (levelling and sharpening, in **Gordon Allport** and **Leo Postman**'s later terms), explained away puzzling events (rationalisation) and turned canoes into boats. Remembering is an "effort after meaning": *reconstructive memory* rebuilds the past from fragments and schemas.

Asked how fast filmed cars were going when they "smashed" rather than "hit" each other, students gave higher speeds and a week later more often recalled broken glass that was never shown (32 against 14 per cent; Elizabeth Loftus and **John Palmer**, 1974). This is the **misinformation effect**: memory distorted by leading questions or misleading later information. Whole false events can be implanted (being lost in a shopping mall as a child), and people who study bed, rest, tired and dream recall the absent "sleep" (the Deese-Roediger-McDermott paradigm). In **source monitoring** (**Marcia Johnson** and colleagues, 1993) we attribute a memory to its origin from its contextual details; reality monitoring, perceived versus imagined, is one kind. Source errors cause source amnesia (a fact remembered without its source), cryptomnesia (another's idea recalled as one's own) and false eyewitness identifications. *Déjà vu* is the illusion of recognition, jamais vu its reverse, and pseudologia fantastica pathological lying. In the recovered memory controversy Loftus warned that suggestive therapy can implant false memories of abuse, though real abuse can also be forgotten and recalled.

> **! EXAM TRAP**
>
> Failing to recall the relevant elements of an autobiographical experience is a source-monitoring deficit (RML 2022); "personal-" and "social-monitoring" are invented. Defence lawyers fear source errors (Amity 2023), not forgetting or poor eyesight; confidence does not mean accuracy. Schema theory is Bartlett's, not Wundt's, Titchener's or Watson's, and his rationalisation is not the defence mechanism. Scripts come from generic memory (RML 2025).

Retrieval and forgetting ●●●

Much forgetting is a failure to find what is still stored: given category names as cues, people recall far more words (Endel Tulving and **Zena Pearlstone**, 1966). By the **encoding specificity principle** (Tulving and **Donald Thomson**, 1973), a cue works to the extent that it was encoded with the memory; recognition beats recall because the test supplies the item. In **context-dependent memory** the physical setting must match: divers recalled words best where they had learned them, underwater or on the beach (**Duncan Godden** and Alan Baddeley, 1975). In **state-dependent memory** the internal state must match: material learned while drunk is recalled better when drunk again (**Donald Goodwin** and colleagues, 1969). **Gordon Bower** (1981) added mood-dependent memory and mood-congruent memory, the easier recall of material matching one's current mood.

> **! EXAM TRAP**
>
> Rahul, who expects to do better in his usual classroom (IHBAS 2024), shows encoding specificity, keyed because context-dependent memory is not offered; "the principle of congruity" is a distractor. Offered both (JEMAS 2025), choose context-dependent for surroundings and state-dependent for internal states. Alcohol-induced amnesia is state-dependent memory (RML 2022).

Forgetting is useful: it clears out unneeded detail and reduces interference, and the more is retained, the less is forgotten. First, **decay theory**, credited to **Edward Lee Thorndike**'s (1874 to 1949) law of disuse, says unused traces fade with time. But **John Jenkins** and **Karl Dallenbach** (1924) found that after eight hours of sleep students recalled over half of their nonsense syllables, and after eight waking hours about a tenth: events, not time, caused the loss. Table 1.15 lists other named effects on retention.

Interference theory blames other memories. In **proactive interference** earlier learning disrupts newer learning; in **retroactive interference** newer learning disrupts older memories (PRO: old disrupts new; RETRO: new disrupts old). Dialling your old phone number instead of the new one is proactive, as is earlier psychology intruding on a later sociology test; losing the old number to the new one is retroactive. After lists A-B and then A-C, recalling A-B measures retroactive and recalling A-C proactive interference. **Arthur Melton** and **J. M. Irwin** (1940) blamed response competition and unlearning.

> **! EXAM TRAP**
>
> RML 2024 keys an English speaker applying English grammar to Spanish as "retroactive". The key is wrong: old disrupts new, so it is proactive. "Interference from new information" (IHBAS 2024) is retroactive. Decay theory is keyed to Thorndike (RML 2024), not Ebbinghaus, Tulving or Melton and Irwin.

In retrieval failure, or cue-dependent forgetting (Tulving, 1974), the trace survives but the cues do not match. Retrieval can even cause forgetting: practising some items of a category (FRUIT-orange) makes the unpractised ones (FRUIT-banana) harder to recall than items of unpractised categories (**Michael Anderson**, **Robert Bjork** and **Elizabeth Bjork**, 1994), through inhibition of competitors. **Sigmund Freud** (1856 to 1939) proposed motivated forgetting through **repression**, the unconscious exclusion of threatening memories from awareness; suppression is conscious, and evidence for repression is weak. Last, consolidation (**Georg Elias Müller** and **Alfons Pilzecker**, 1900) can fail: a head injury or electroconvulsive therapy erases recent memories, and by Ribot's law (**Théodule Ribot**, 1881) the new perishes before the old.

> **! EXAM TRAP**
>
> Retrieval-induced forgetting (RML 2020) is keyed "practised information remembered better than unpractised", but its defining half is that related unpractised items suffer. Repression is unconscious, suppression conscious (RML 2025); motivated forgetting covers both, and inhibition weakens competitors. "Less retention leads to less forgetting" is the incorrect statement (RML 2025).

Table 1.15 Classic named effects on retention.

Effect	Who and when	What happens
Zeigarnik effect	**Bluma Zeigarnik**, 1927	Interrupted tasks are recalled better
Ovsiankina effect	**Maria Ovsiankina**, 1928	Interrupted tasks tend to be resumed
Reminiscence	**Philip Ballard**, 1913	Recall rises after a delay without study
Von Restorff effect	**Hedwig von Restorff**, 1933	A distinctive list item is remembered better
Spacing effect	Ebbinghaus, 1885	Distributed practice beats massed practice
Testing effect	**Arthur Gates**, 1917	Retrieval practice beats restudy

Amnesia and the biology of memory ●●●

Amnesia is a pathological loss of memory. In **anterograde amnesia** new memories cannot be formed after an injury; in **retrograde amnesia** memories from before it are lost, usually sparing the remote past. Of the other "a-" disorders, aphasia (with alexia and acalculia) affects language, apraxia skilled movement and agnosia recognition; dementia is a progressive decline in memory, judgement and abstraction.

In 1953 the neurosurgeon **William Beecher Scoville** removed both medial temporal lobes, with most of the hippocampus and amygdala, from **Henry Gustav Molaison** (1926 to 2008), known as H.M., to relieve his epilepsy. Scoville and **Brenda Milner** reported in 1957 a profound anterograde amnesia: he reread magazines with fresh interest and never learned his new address. Yet his intelligence, digit span and childhood memories were intact, and over three days he improved at tracing a star in a mirror while denying that he had ever done it.

> **! EXAM TRAP**
> H.M.'s main deficit was anterograde amnesia for declarative memory (Real PYQ 2024); procedural, short-term and old memories were spared. Both medial temporal lobes were removed, not frontal, parietal or occipital cortex. The hippocampus is the structure damaged in anterograde amnesia and critical for consolidation (JEMAS 2025, RML 2024, RML 2023), not the hypothalamus, pituitary, cortex, amygdala (emotional memory), prefrontal cortex (working memory) or cerebellum (motor learning).

Korsakoff's syndrome (Korsakoff's psychosis; **Sergei Korsakoff**, late 1880s) results mainly from thiamine (vitamin B1) deficiency in chronic alcoholism, often after Wernicke's encephalopathy (**Carl Wernicke**, 1881: confusion, ataxia, nystagmus); together they form the Wernicke-Korsakoff syndrome, which damages the mammillary bodies and mediodorsal (dorsomedial) thalamus. It brings dense anterograde amnesia, extensive retrograde amnesia and **confabulation**, sincere false stories that fill memory gaps, unlike malingering (deliberate faking), delirium (acute, fluctuating consciousness) or confusion. Alzheimer's disease, by contrast, first erodes recent episodic memory, and Huntington's and Parkinson's diseases impair procedural learning.

> **! EXAM TRAP**
>
> Memory deficits, confabulation and nystagmus in a chronic drinker (Real PYQ 2024) is Wernicke-Korsakoff syndrome, not Alzheimer's, vascular or frontotemporal dementia. The alcohol-related illness with profound retrograde amnesia (RML 2024) is Korsakoff's psychosis; the vitamin is B1, not B12, D or folate.

Infantile amnesia, the lack of memories from the first three or four years, reflects an immature hippocampus and the lack of language. Transient global amnesia is a sudden anterograde amnesia lasting hours. Dissociative amnesia, a loss of personal memories with no medical cause (generalised when it covers the whole life), belongs to the Psychiatric Disorders chapter.

Richard Semon (1904) called the physical memory trace the **engram**. **Karl Spencer Lashley** (1890 to 1958) cut away rats' cortex after maze learning and found that memory loss depended on how much was removed (mass action), not where, with areas substituting for one another (equipotentiality): memories are distributed across the cortex ("In search of the engram", 1950). **Donald Olding Hebb** (1904 to 1985) proposed in 1949 that neurons active together strengthen their links, and **Timothy Bliss** and **Terje Lømo** (1973) found such **long-term potentiation** (LTP), a lasting strengthening of hippocampal synapses after high-frequency stimulation and the leading cellular candidate for memory. The hippocampus consolidates new memories into the neocortex, partly during sleep, so remote memories survive its damage.

Improving memory ○

A mnemonic adds organisation, meaning or imagery. The method of loci, credited to **Simonides of Ceos**, places vivid images along a familiar route; the peg-word system links items to rhyming number pegs (one is a bun); the keyword method (Richard Atkinson, 1975) links a foreign word to a sound-alike image (Spanish "pato", duck: a duck in a pot). An acronym is a word of first letters (VIBGYOR); an acrostic is a sentence of first letters ("Every Good Boy Deserves Fruit"). SQ3R (**Francis P. Robinson**, 1946) is Survey, Question, Read, Recite, Review; PQRST is Preview, Question, Read, Self-recitation, Test. Testing and spacing beat rereading, overlearning gives diminishing returns, and sleep consolidates.

> **! EXAM TRAP**
>
> A first-letter sentence is an acrostic, not an acronym, peg-word or keyword. The "S" of PQRST is self-recitation (RML 2024), not storing, sequencing or semantics.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Learning in 30 trials and relearning in 18 gives savings of _____ per cent.
2. A 30-second filled delay before recall abolishes the _____ effect.
3. Baddeley added the _____ to working memory in 2000.
4. Flashbulb memories keep their vividness and _____ as accuracy falls.
5. Distortion of a memory by a leading question asked afterwards is the _____ effect.
6. Dialling your old phone number instead of the new one is _____ interference.
7. Korsakoff's syndrome follows deficiency of _____ (vitamin B1).

Answers: (1) 40, (2) recency, (3) episodic buffer, (4) confidence, (5) misinformation, (6) proactive, (7) thiamine.

1.8 Thinking, Problem Solving, Reasoning, Decision Making and Language

HIGH YIELD · 53 QS

Thinking is the mental manipulation of internal representations to form new ones, as in solving, reasoning, deciding or creating. Seeing the chai stall is not thinking; working out whether you can afford a second cup is. Banished by Watson, thought returned with the cognitive revolution (Section 1.2): cognitive psychology studies the structuralists' topics (consciousness, imagery, thought) by objective measures.

Elements, Imagery and Types of Thought ○

Thought uses five elements: *images*, *concepts*, *symbols*, *propositions* (the smallest units of knowledge that can be true or false) and *language*. **Roger Newland Shepard** and Jacqueline Metzler (1971) found that the time to judge two rotated block figures identical rose linearly with the angle between them, as if the image were turned at a steady speed: **mental rotation**. **Stephen Michael Kosslyn**'s image-scanning studies (1978) suggested that images depict space, against **Zenon Walter Pylyshyn**'s view (1973) that knowledge is stored as propositions. **Allan Paivio**'s **dual-coding theory** (1971) gives pictures and concrete words both a verbal and an imagery code, so they are remembered better than abstract words. Table 1.16 lists types of thinking.

Table 1.16 Types of thinking.

Type	Associated with	What it is
Directed versus undirected	Würzburg school; Freud	Goal-guided and reality-checked versus drifting by association (daydreaming)
Autistic (dereistic)	**Eugen Bleuler**, 1911	Wish-driven fantasy that ignores reality
Convergent versus divergent	J. P. Guilford	One correct answer versus many varied answers
Critical versus creative	Robert Ennis; Guilford	Evaluating claims versus generating novel, useful ideas
Reflective	**John Dewey**, 1910	Careful weighing of a belief and its grounds

Type	Associated with	What it is
Productive versus reproductive	**Max Wertheimer**, 1945	Restructuring a problem versus reapplying a method

> **! EXAM TRAP**
>
> Dual coding favours **visual** information, not verbal, auditory or semantic. Autistic thinking is fantasy, not autism. Loosening of associations, circumstantiality and word salad are disorders of the *form* of thought (Psychiatric Disorders chapter).

Concepts: Rules and Prototypes ●●

A **concept** is a mental category of things sharing properties, formed through observation, discrimination, generalisation and abstraction. The classical view defines it by necessary and sufficient features, which suits *artificial concepts* set by a rule: *conjunctive* (all attributes: a voter is a citizen, adult *and* enrolled), *disjunctive* (any one: "out" in cricket) or *relational* ("taller than"). **Jerome Seymour Bruner**, Jacqueline Goodnow and George Austin (1956) found that people learn them by testing hypotheses, by *scanning* or *focusing*; *conservative focusing*, changing one attribute at a time, works best, and *focus gambling* changes several.

Natural concepts are fuzzier. **Ludwig Wittgenstein** (1953) saw that games share only a **family resemblance**, and **Eleanor Rosch** showed in the 1970s that typicality is graded: "a robin is a bird" is verified faster than "a penguin is a bird". We classify by similarity to a **prototype**, the abstract, averaged best example, and learn the *basic level* (chair, not furniture) first; *exemplar theory* stores many instances instead. How concepts are networked in memory is told in Section 1.7.

> **! EXAM TRAP**
>
> Artificial concepts by hypothesis testing = **Bruner**, not Rosch (natural concepts), **Clark Leonard Hull** (gradual associative abstraction, 1920) or Watson. A prototype is **abstract and averaged**, not a template (exact copy), a feature (a part) or a geon (a simple solid).

Problem Solving: Strategies and Obstacles ●●●

A problem is a gap between an initial state and a goal state, well defined (a sudoku) or ill defined ("choose a career"). **Allen Newell** and **Herbert Alexander Simon** (*Human Problem Solving*, 1972; the General Problem Solver) treated solving as search through a **problem space** of states linked by operators (legal moves). An **algorithm** guarantees the solution but may be slow (trying every padlock code); a **heuristic** is a quick rule of thumb with no guarantee. **Means-end analysis** sets subgoals that reduce the biggest difference from the goal (to reach Bengaluru, first reach the station); *working backward* starts from the goal; **hill climbing** always takes the step that gets closest, so it stalls when a backward step is needed; *analogy* reuses an earlier solution, though people rarely notice one unprompted (**Mary Gick** and **Keith Holyoak**'s radiation problem, 1980); *trial and error* is the most primitive. Stage models range from Dewey (1910) to Sternberg's problem-solving cycle.

The Gestaltists stressed **insight**, a sudden restructuring of the problem: **Wolfgang Köhler**'s chimpanzee Sultan suddenly joined two sticks to reach a banana (1917), and **Norman Maier**'s nine-dot problem (1930) defeats people until they let the lines leave the square. Janet

Metcalfe and David Wiebe (1987) found "warmth" ratings flat until just before an insight solution, but rising steadily on algebra. *Incubation*, setting a problem aside, often helps. Experience can also block a solution.

KEY STUDY: DUNCKER'S CANDLE PROBLEM (1945)

Karl Duncker asked people to fix a candle to a wall with matches and a box of drawing pins. Few thought of pinning up the emptied box as a shelf, but many more did when the box came empty. Knowing an object's usual use (top-down processing) blinded them to its other uses: **functional fixedness**.

Functional fixedness has a strategic cousin, **mental set** (Einstellung): in **Abraham Luchins**'s water jars (1942), people drilled on B minus A minus 2C kept using it when A minus C was far simpler. Confirmation bias and irrelevant information also obstruct.

! EXAM TRAP

A heuristic is "a general rule that is usually right": efficient, **not guaranteed**, not more accurate than an algorithm. Functional fixedness concerns **objects** (a matchbox seen only as a container, not a platform) and reflects overactive top-down processing; mental set concerns **strategies**. As fixedness is a special case of set, one paper keyed **mental set** "the most basic and common obstacle"; *fixation* is the umbrella word, "mental barrier" no term, trial and error a strategy. In a mental status examination, insight is awareness of illness, graded 1 to 6; **intellectual insight** is grade 5.

Reasoning: Deduction and Induction ●●●

Deductive reasoning moves from general premises to a conclusion that must be true if they are ("All Mysuru autos must use the meter; this is one; so it must"). **Inductive reasoning** moves from specific observations to a probable generalisation that the next case may overturn ("every auto I took overcharged, so all do"). An argument is *valid* if its conclusion follows from its form, *sound* if its premises are also true. **Charles Sanders Peirce** (1839 to 1914) called inference to the best explanation, as in diagnosis, **abductive reasoning**. In research, deduction tests theory against data (quantitative work), induction builds theory from data (qualitative work), and the hypothetico-deductive method of **Karl Raimund Popper** (1902 to 1994) is the full cycle of hypothesis, deduced prediction and test.

! EXAM TRAP

"Starts with a theory and tests it with observations" = **deductive**. "Gathers data, looks for patterns, then generates a theory", "specific observations to broader generalisations", "theory development" and "qualitative research" = **inductive**. Keys prefer "deductive reasoning" to "hypothetico-deductive method" and rest deductive validity on **the soundness of the initial theory**. **Heuristics** was keyed as NOT reflecting induction; probabilistic conclusions, decision making and concept formation do.

In syllogisms (two premises, one conclusion), **belief bias** makes people accept invalid arguments with believable conclusions, and the *atmosphere effect* (Robert Woodworth and Saul Sells, 1935) lets a "some" or "no" premise suggest a matching conclusion. For "if p, then q",

modus ponens (p, so q) and *modus tollens* (not q, so not p) are valid; *affirming the consequent* (q, so p) and *denying the antecedent* (not p, so not q) are fallacies. In **Peter Cathcart Wason**'s selection task (1966), under one person in ten turns the right cards (Figure 1.10), yet a permission rule ("if drinking beer, must be over 19") makes the same logic easy.

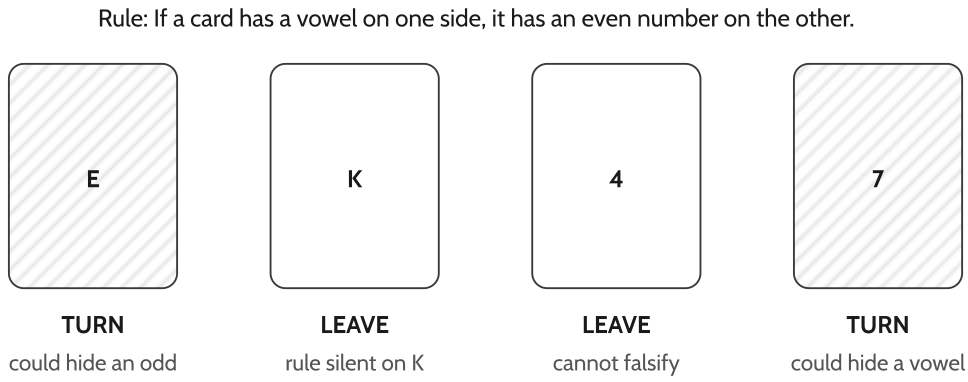


Figure 1.10 The abstract Wason selection task. The two cards with heavy outlines (E and 7) are the only ones that could reveal a counterexample. The popular choice of the 4 card tests for confirmation, which can never prove a general rule.

Philip Nicholas Johnson-Laird (1983) argued that we reason with *mental models* of possible situations, not formal rules. In Wason's 2-4-6 task (1960), people guessed "rising by two" and tested only fitting sequences (the rule was any increasing numbers): **confirmation bias**, seeking and favouring evidence for what one already believes.

> **! EXAM TRAP**
>
> Confirmation bias was keyed as "closely related to" **representativeness**, a debatable key (recognition is least related). Acting on an expectation until it comes true is a **self-fulfilling prophecy**, not confirmation bias, the base-rate fallacy or "motivational reaction" (no term). Piaget's formal operations bring hypothetico-deductive (a capacity, not a stage), counterfactual and metacognitive reasoning; **dialectical** reasoning, integrating contradictions, is postformal (Developmental Psychology chapter).

Judgement and Decision Making ●●

Normative models say how a rational agent should decide, maximising expected utility; *descriptive* models say how people actually decide. Simon's **bounded rationality** holds that information, mental capacity and time limit choice, so people practise **satisficing**, taking the first good-enough option (Nobel Prize, 1978). **Gerd Gigerenzer** (born 1947) adds that fast and frugal heuristics, such as the *recognition heuristic* (choosing the option you recognise), are often ecologically rational.

Amos Tversky and **Daniel Kahneman** (Kahneman's Nobel Prize, 2002) catalogued the heuristics of judgement and their biases (Table 1.17). Representativeness invites *base-rate neglect*: a description that "sounds like" an engineer is judged an engineer whether engineers are 30 or 70 in 100. Most people judged Linda, a socially concerned philosophy graduate, likelier to be "a feminist bank teller" than "a bank teller", although a conjunction cannot be likelier than its parts: the **conjunction fallacy**.

Table 1.17 The classic heuristics and their typical biases.

Heuristic	Proposed	We judge by	Typical error
Availability	1973	Ease of recall	More words seem to begin with K than have K third
Representativeness	1972	Similarity to the typical case	Base-rate neglect; conjunction fallacy; gambler's fallacy (after many reds, black seems "due")
Anchoring and adjustment	1974	Adjusting from a starting value	Estimates stay near an arbitrary anchor

In the Asian disease problem (1981), most people preferred saving 200 of 600 lives for sure to a gamble, but chose the gamble when the sure option read "400 people will die": risk aversion for gains, risk seeking for losses. **Prospect theory** (1979) explains this *framing effect*, the same choice decided differently when worded as gains or as losses: outcomes are gains or losses from a reference point, and losses loom about twice as large as equal gains (**loss aversion**), which also drives the endowment effect (overvaluing what we own), status quo bias (preferring no change) and the sunk-cost fallacy (persisting because of past investment). **Hindsight bias**, the "I knew it all along" effect (**Baruch Fischhoff**, 1975), and overconfidence (confidence exceeding accuracy) distort self-knowledge. Dual-process theory contrasts fast, intuitive *System* 1, home of the heuristics, with slow, deliberate *System* 2.

> **! EXAM TRAP**
>
> Bounded rationality = **Herbert Simon**, not Kahneman, Tversky or Howard Gardner. Representativeness errs when people **ignore base rates and focus solely on similarity**; the other options describe availability, anchoring and confirmation bias. Overrating how well we would have predicted events = **hindsight**, not the just-world hypothesis or "decisiveness".

Creativity ●●

Creativity is producing work that is both novel and appropriate. **Joy Paul Guilford**, whose 1950 American Psychological Association address launched its study, equated it with divergent production, scored by *fluency* (number of ideas), *flexibility* (number of categories), *originality* (statistical rarity) and *elaboration* (detail). **Graham Wallas** (*The Art of Thought*, 1926) described four stages: *preparation* (defining and working on the problem), *incubation* (setting it aside while work continues outside awareness), *illumination* (the sudden insight) and *verification* (testing and refining it).

> **! EXAM TRAP**
>
> Order: **preparation, incubation, illumination, verification** ("Please Incubate Ideas Very carefully"). Sensitivity, fluency, flexibility and originality are Guilford's abilities; analytical, creative and practical are Sternberg's intelligences; "problem-finding, brainstorming, selection, implementation" is not Wallas.

Creativity is measured by the *Torrance Tests of Creative Thinking* (**Ellis Paul Torrance**, 1966) and **Sarnoff Mednick**'s *Remote Associates Test* (1962): which word links cottage, Swiss and cake? (Cheese.) **Edward de Bono**'s *lateral thinking* (1967) challenges assumptions; **Alex Faickney Osborn**'s brainstorming (*Applied Imagination*, 1953) defers judgement and seeks

quantity. By the threshold hypothesis, creativity and IQ correlate only up to an IQ of about 120. In **Robert Jeffrey Sternberg** and Todd Lubart's investment theory (1991), creative people "buy low and sell high". The creative personality is open to experience, independent and tolerant of ambiguity.

The Structure of Language ●●

Language combines arbitrary symbols by rules into unlimited messages. **Charles Francis Hockett** (1960) listed its design features, notably semanticity (meaning), arbitrariness (nothing in "dog" is doglike), displacement (talking about what is absent), generativity (endless new sentences) and duality of patterning (meaningless sounds build meaningful units). The **phoneme** is the smallest unit of sound that distinguishes words (/p/ and /b/ in pat and bat) and has no meaning; the **morpheme** is the smallest unit of meaning, free (cat) or bound (-s), so "cats" has two. Syntax governs how words combine, *semantics* is meaning, **pragmatics** is use in social context ("Can you pass the salt?" is a request) and **prosody** is rhythm and intonation. **Noam Chomsky**'s transformational grammar (*Syntactic Structures*, 1957; *Aspects*, 1965) separates *surface structure*, the words as spoken, from **deep structure**, who did what to whom: "Visiting relatives can be boring" has one surface and two deep structures. *Competence* is tacit knowledge of the rules, *performance* their imperfect use.

> **! EXAM TRAP**
>
> Smallest unit of meaning = **morpheme**; basic unit of sound = **phoneme**; a syllable is a unit of pronunciation, a consonant one kind of phoneme. Emotional prosody depends on **right-hemisphere regions homotopic to the language areas**, not Broca's area, motor cortex or occipital lobe. **Recursive thought** ("I think that you think...") is perspective taking, not reversibility (Piaget's mental undoing of an action), referential style (below) or reconstruction (rebuilding a memory).

Acquiring Language ●●●

Burrhus Frederic Skinner's *Verbal Behavior* (1957) explained language by reinforcement of verbal operants such as the *mand* (a request) and the *tact* (naming). Chomsky's 1959 review replied that children produce sentences they never heard, make errors no adult models ("I goed") and are rarely corrected for grammar. He proposed an innate **language acquisition device** (LAD) holding a *universal grammar*, argued from the **poverty of the stimulus**: children's input is too sparse to explain the grammar they reach. Bruner's *language acquisition support system* (LASS, 1983) stresses adult scaffolding and child-directed speech (motherese): higher and wider pitch, slower rate, more pauses, shorter utterances. **Dan Isaac Slobin**'s *language-making capacity* (1985) is a set of cognitive and perceptual operating principles specialised for language. **Eric Heinz Lenneberg** (1967) proposed a **critical period** from about two to puberty, now usually called a sensitive period; Genie, isolated until 13, learned words but never grammar.

Infants coo at about 2 months and babble at about 6; at about 12 months a single word, a **holophrase**, does a sentence's work; at 18 to 24 months comes what **Roger Brown** (1925 to 1997) called **telegraphic speech**, two words without function words ("more milk"); and at 2 to 3 years **overregularisation** ("two foots"). Children overextend words ("dog" for every animal), underextend them ("dog" only for their own) and learn new words from one exposure

(fast mapping); **Katherine Nelson** (1973) found some children's first words mostly object names, a *referential* style. Language is most often compromised in autism spectrum disorder.

> ! EXAM TRAP
>
> Chomsky's innate capacity is the **LAD** (an inborn processor activated by verbal input, holding universal grammar, the rules common to all languages), not the phonological loop (working memory), the zone of proximal development (Vygotsky) or a schema; LASS = **Bruner**, LMC = **Slobin**. Irreversible deficits after early deprivation support **sensitive periods**, not semantics or syntax (levels of language) or nativism (the broader theory that language is innate).

Language and Thought ●●●

Benjamin Lee Whorf (1897 to 1941), a student of **Edward Sapir** (1884 to 1939), argued that language shapes thought. The strong version, linguistic determinism, has failed: **Brent Berlin** and **Paul Kay** (1969) found colour terms entering languages in a near-universal order, and Rosch found that the Dani, with two colour terms, remember focal colours as English speakers do. The weak version, **linguistic relativity** (language influences thought), survives: Hindi kinship terms (*chacha*, *mama*, *phupha*) make speakers track which side an uncle is on. **Steven Pinker** (1994) holds that we think in "mentalese".

For **Jean Piaget** (1896 to 1980), thought comes first and language expresses it; preschoolers' egocentric speech simply fades. For **Lev Semyonovich Vygotsky** (*Thinking and Speech*, 1934), thought and speech have separate roots that merge at about two, after which thought becomes verbal; social speech becomes self-guiding private speech and then **inner speech**. He traced concepts from heaps through complexes to true concepts in adolescence, tied to society's tasks, social institutions and self-reflection.

> ! EXAM TRAP
>
> **Vygotsky: thought depends on language; Piaget: language depends on thought.** "One studied early, the other later language" is wrong: both studied the same years. Private speech **guides behaviour and is internalised**; it is not regression, imitation, disorder or fading egocentrism. Gender plays no part in Vygotsky's account. "Language and thought involve..." expects **simultaneous** processes; "independent" fits only their early roots. Linguistic relativity = **Whorf**, not Chomsky, Piaget or Pinker.

Bilingualism is the use of two languages, and a *polyglot* speaks many (a linguist studies language scientifically); bilingual children reach milestones on time. Apes such as Washoe, Nim Chimpsky and Kanzi learn meaningful symbols but little grammar. Speech perception is *categorical* (/ba/ switches abruptly to /pa/ as voice onset time rises) and audiovisual: in the McGurk effect, a heard "ba" dubbed onto lips saying "ga" is heard as "da". Reading builds most directly on spoken language, above all awareness of its sounds, not on theory of mind, imagination or mental representation. **Aphasia** is an acquired disorder of language from brain damage, not of intelligence, memory or abstract thinking (Broca's aphasia impairs production, Wernicke's comprehension: Physiological Psychology chapter).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Seeing a box only as a container shows _____.
2. Reasoning from specific observations to a probable generalisation is _____.
3. Wallas's second stage is _____.
4. The smallest unit of meaning is the _____.
5. "More milk" at 18 to 24 months is _____ speech.
6. For Vygotsky, private speech becomes _____ speech.

Answers: (1) functional fixedness, (2) induction, (3) incubation, (4) morpheme, (5) telegraphic, (6) inner.

1.9 Intelligence: Theories, Measurement, Heredity and Environment

HIGH YIELD · 63 QS

Everyone thinks they recognise intelligence, yet psychologists still argue over what it is, whether it is one thing or many, and how much of it a test can capture.

Defining Intelligence ●●

Alfred Binet (1857 to 1911) put judgement at the centre: "to judge well, to comprehend well, to reason well." **Edwin Garrigues Boring** (1886 to 1968) retorted in 1923 that intelligence is "what the tests test", an **operational definition** (one framed by its measuring procedure). **David Wechsler** (1896 to 1981) gave the most quoted definition (1944): "the aggregate or global capacity of the individual to act purposefully, to think rationally and to deal effectively with his environment", aggregate because it comprises many abilities, global because they work as one. **Robert Jeffrey Sternberg** (born 1949), who re-polled the experts with Douglas Detterman in 1986, stresses adapting to, selecting and shaping real-world environments.

Achievement tests look back at what a specific course taught (a board examination); **aptitude** tests look ahead to the capacity to learn a specific skill with training (the Differential Aptitude Tests), so they anchor guidance; intelligence tests predict learning in general. Among lay (implicit) theories, **Carol Susan Dweck** (born 1946) contrasted an entity theory, or fixed mindset, with an incremental theory, or **growth mindset** (*Mindset*, 2006), the belief that effort develops intelligence. Fixed-mindset children avoid challenges; children praised for effort, not cleverness, persist after failure.

! EXAM TRAP

The WAIS is an *intelligence* test, not an aptitude, achievement or attitude (evaluative stance) test. RML 2022 keyed the aptitude test as "the heart of the guidance programme", over the case study (a life history), the interview and the cumulative record (a running school file); textbooks give the phrase to counselling, so choose it if offered.

From Galton to Binet: The First Tests and the IQ ●●

Francis Galton (1822 to 1911), author of *Hereditary Genius* (1869) and father of differential psychology, measured reaction time and sensory acuity in his London anthropometric laborato-

ry (1884). **James McKeen Cattell** (1860 to 1944) coined the term **mental test** (1890) for a similar battery, which failed to predict college grades (Clark Wissler, 1901). Asked by the French ministry to pick out children needing special schooling, Binet and the physician **Théodore Simon** (1873 to 1961) measured judgement and reasoning instead, keeping items that older children passed more often than younger ones. Their 1905 scale had 30 items of rising difficulty; the 1908 revision grouped them by age, a child's **mental age** being the age level of the items passed; the 1911 revision reached adults. (**Jean Piaget** standardised tests in Simon's laboratory around 1920 but did not build the first test.)

Henry Herbert Goddard (1866 to 1957) translated the scale into English in 1908. In 1912 **William Stern** (1871 to 1938) divided mental age by chronological age, coining the **intelligence quotient (IQ)**; **Lewis Madison Terman** (1877 to 1956) multiplied it by 100 in his 1916 **Stanford-Binet**. So the **ratio IQ** = (MA ÷ CA) × 100: a 10-year-old with a mental age of 8 scores 80. In 1917 **Robert Mearns Yerkes** (1876 to 1956) led the first group tests, for army recruits: the verbal Army Alpha and, for the illiterate, the pictorial Army Beta.

> **! EXAM TRAP**
>
> Stern coined IQ; Terman popularised it. Inverting the ratio (10 ÷ 8 × 100 = 125) is the usual slip: a mental age below chronological age always gives an IQ below 100. James McKeen Cattell coined "mental test"; Raymond Cattell, unrelated, gave us fluid and crystallised intelligence and the 16PF personality questionnaire.

Deviation IQ and What a Score Means ●●●

The ratio IQ fails for adults: mental age stops growing in the late teens, so an ordinary 40-year-old with a mental age of 16 would score (16 ÷ 40) × 100 = 40. The Wechsler-Bellevue scale (1939) dropped mental age and compared each person with norms from a large representative sample of the same age. In this **deviation IQ** (the "standard IQ"; loosely, the actual score relative to the expected mean score for one's age) every age group has a mean of 100 and a standard deviation (SD), the typical distance of scores from the mean, of 15: IQ = 100 + 15z (z: distance from the mean in SDs).

Because IQs are roughly normally distributed, any IQ converts to a percentile, the percentage of people scoring below it (Figure 1.11): 100 is the 50th percentile and 85, one SD below, the 16th. Stanford-Binet deviation IQs used an SD of 16 from 1960 until the fifth edition (2003), so there an IQ of 136 gives z = 36 ÷ 16 = 2.25, about the 98.7th percentile. WAIS-IV bands: 130 or above very superior, 90 to 109 average, 80 to 89 low average, 70 to 79 borderline ("very low" in newer editions), 69 or below extremely low. DSM-IV-TR's borderline intellectual functioning (71 to 84) straddles two bands, so 70 to 84 is "low average or borderline, depending on the edition".

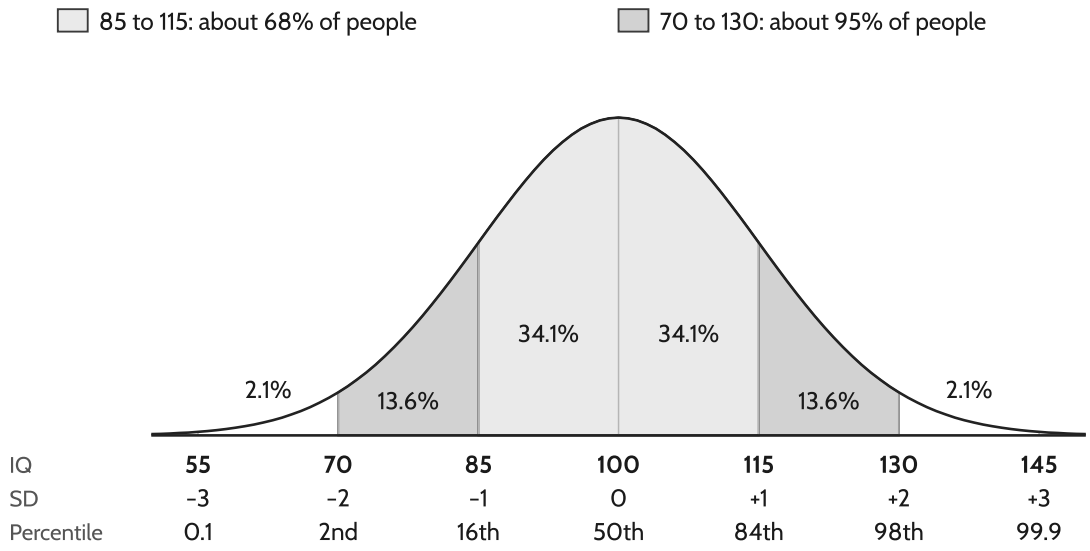


Figure 1.11 The distribution of deviation IQ (mean 100, SD 15). Read across the bottom rows to convert an IQ into standard deviations and percentiles: 85 is the 16th percentile, 115 the 84th, 130 about the 98th, and 70 about the 2nd.

> **! EXAM TRAP**
>
> The original Stanford-Binet used the ratio IQ. The deviation IQ is not "only for adults", not built from unstandardised raw scores, and no less reliable than the ratio IQ, whose meaning shifts with age.

The Wechsler Scales, the SB5 and Other Tests ●●●

Wechsler, at New York's Bellevue Hospital, set out to fix the Stanford-Binet's faults: little appeal for adults, too much word manipulation, too much weight on speed and reliance on mental age. His tests add non-verbal performance subtests and give a profile as well as a total. The family is the WAIS (1955; WAIS-IV 2008, ages 16 to 90), the WISC (1949; WISC-V 2014, 6 years to 16 years 11 months) and the WPPSI (from 2 years 6 months). In the WAIS-IV, four index scores (Table 1.18) replaced the Verbal and Performance IQs and combine into the Full Scale IQ (FSIQ), with an optional General Ability Index; the WISC-V and WAIS-5 (2024) have five.

Table 1.18 The WAIS-IV (2008): 10 core subtests in four indices.

Index	Core subtests	Supplemental
Verbal Comprehension (VCI)	Similarities, Vocabulary, Information (3)	Comprehension
Perceptual Reasoning (PRI)	Block Design, Matrix Reasoning, Visual Puzzles (3)	Picture Completion, Figure Weights
Working Memory (WMI)	Digit Span, Arithmetic	Letter-Number Sequencing
Processing Speed (PSI)	Symbol Search, Coding	Cancellation

> **! EXAM TRAP**
>
> The Working Memory Index reflects attention, concentration and mental control. "VIQ, PIQ, FSIQ and memory quotient" mixes the old WAIS with the old Wechsler Memory Scale (1945), a memory test whose single memory quotient became five WMS-IV (2009) indices: Auditory Memory, Visual Memory, Visual Working Memory, Immediate Memory and Delayed Memory. Letter-Number Sequencing, keyed as core in the WAIS-IV but not the WISC-V, is strictly supplemental in one and secondary in the other; it is still the best answer, as the other options are core in both.

The Stanford-Binet Fifth Edition (SB5; Gale Roid, 2003; ages 2 to 85 and over; mean 100, SD 15) measures five factors drawn from Carroll's model (below) and Alan Baddeley's working memory: Fluid Reasoning, Knowledge, Quantitative Reasoning, Visual-Spatial Processing and Working Memory, each verbal and non-verbal. The Kaufman Brief Intelligence Test (KBIT-2) has a Verbal (crystallised) and a Nonverbal (fluid) scale. For infants, the Bayley Scales (cognitive, language and motor development up to 42 months) yield a developmental quotient (DQ); the McCarthy Scales cover about 2½ to 8½ years.

The WAIS-IV FSIQ is highly reliable (about .98). Content validity asks whether items sample the domain, **concurrent validity** whether scores agree with a criterion measured at the same time (an established test given alongside), predictive validity whether they forecast a later one, and construct validity whether the test behaves as theory predicts. IQ correlates about .5 with school grades and job performance (Frank Schmidt and John Hunter, 1998). The WAIS documents intellectual disability or decline, as for a disability allowance; children's special educational needs call for the WISC.

Culture-Fair and Indian Tests ●●●

Because vocabulary and information depend on schooling, psychologists sought a culture-free test; that proved impossible, so a **culture-fair test** instead minimises language, reading and specific knowledge. The classic is **Raven's Progressive Matrices** (Standard form 1938) by Spearman's student **John Carlyle Raven** (1902 to 1970): each of its 60 progressively harder items is a pattern with a piece missing, solved by inferring the rule. It is the classic test of fluid intelligence and the best single marker of g, widely used with Indian norms. Raymond Cattell's figure-only Culture Fair Intelligence Test works the same way.

In India, **V. V. Kamat**'s Binet-Kamat Test (1930s) adapted the Stanford-Binet and is heavily verbal; **Arthur J. Malin**'s Intelligence Scale for Indian Children (MISIC, 1969) adapts the WISC and is half verbal; **C. M. Bhatia**'s Battery of Performance Tests of Intelligence (1955) has five performance subtests (Kohs block design, Alexander's pass-along, pattern drawing, immediate memory, picture construction), needs no reading and has rural and urban norms; normed on 11- to 16-year-olds, it is also used with illiterate adults. So for an illiterate rural 10-year-old, where the WISC-IV and SB5 rest on schooling and foreign norms, Bhatia's Battery best minimises cultural and educational bias.

> **! EXAM TRAP**
>
> Raven's measures *fluid* intelligence: prefer it to crystallised, verbal, social or even "general intelligence". It is culture-fair, not culture-free. The Battery is C. M. Bhatia's, not that of Girishwar Misra, Durganand Sinha or J. B. P. Sinha (cultural, indigenous and organisational psychologists).

Factor Theories: From g to CHC ●●●

Factor analysis asks how few underlying factors explain a table of correlations among tests. Its inventor, **Charles Edward Spearman** (1863 to 1945), found in 1904 that schoolchildren's marks and sensory scores all correlated positively, the **positive manifold**, and explained it by a **general factor (g)** common to every test plus a specific factor (s) peculiar to each: the **two-factor theory**. **Louis Leon Thurstone** (1887 to 1955) found seven **primary mental abilities** (1938): verbal comprehension, word fluency, number, space, associative memory, perceptual speed and reasoning. Because these correlate, g returned at the top of hierarchies such as **Philip Ewart Vernon**'s (1950) and **John Bissell Carroll**'s three strata (1993); Kevin McGrew merged Carroll's model with Cattell and Horn's into the CHC theory behind most modern tests (Table 1.19).

Joy Paul Guilford (1897 to 1987) went the other way. His **structure of intellect** (1967) crosses five operations (cognition, memory, divergent production, convergent production, evaluation), four contents (figural, symbolic, semantic, behavioural) and six products, giving 5 × 4 × 6 = 120 abilities, later 150 and then 180.

Raymond Bernard Cattell (1905 to 1998), later with **John Leonard Horn** (1928 to 2006), split g in two (1963). **Fluid intelligence (Gf)** is the ability to reason and solve novel problems independently of acquired knowledge; measured by matrices, it peaks in the early twenties and then declines. **Crystallised intelligence (Gc)** is the repertoire of information, cognitive skills and strategies acquired through education and experience; measured by vocabulary and information, it holds or rises into the sixties. In Cattell's investment theory, Gf invested in learning crystallises as Gc; Horn later extended the list to about nine broad abilities.

Table 1.19 Other factor theories and the hierarchies.

Theorist	Year	Theory	Key structure
Edward Thorndike	1920s	Multifactor ("anarchic")	Abstract, mechanical, social; CAVD test; no g
Godfrey Thomson	1916	Sampling theory	Tests sample overlapping sets of many small bonds
P. E. Vernon	1950	Hierarchical	g; v:ed and k:m; minor; specific
Arthur Jensen	1969	Level I and II	Associative (rote) versus conceptual
J. B. Carroll	1993	Three strata	g; 8 broad; about 70 narrow

> **! EXAM TRAP**
>
> g is general and s specific; neither means group factors (Thurstone, Vernon) or a genetic predisposition. Guilford's cube has three dimensions but 120 abilities; RML 2021 keyed "120 dimensions", so give 120 there. "Ability to deal with new problems" is Gf; block design, matrix reasoning and coding do not measure Gc, and processing speed and working memory are correlates, not definitions, of it.

Gardner, Sternberg, PASS and Emotional Intelligence ●●

Howard Gardner (born 1943) built his theory from brain damage and prodigies, not factor analysis. *Frames of Mind* (1983) proposed seven relatively independent **multiple intelligences**: linguistic, logical-mathematical, spatial, musical, bodily-kinaesthetic, interpersonal and intrapersonal; naturalist was added in 1999 and existential only considered. Critics note that most correlate with one another and with g.

Sternberg's **triarchic theory** (*Beyond IQ*, 1985) has three parts. The componential or analytical part is the mental machinery: metacomponents that plan and monitor, performance components that execute, and knowledge-acquisition components that learn. The experiential or creative part copes with novelty and automatises the familiar. The contextual or practical part adapts to the environment, shapes it or selects a better one: a clerk who learns her office's unwritten rules adapts, one who changes a pointless procedure shapes, one who leaves for a better job selects. In the late 1990s this grew into **successful intelligence**, the ability to achieve success in life by one's own standards and within one's culture through a balance of analytical, creative and practical abilities; it rests partly on tacit knowledge, the unspoken know-how a setting demands.

> **! EXAM TRAP**
>
> Componential = analytical, experiential (not "experimental", and not Dweck's "incremental") = creative, contextual = practical. "Verbal, performance, full scale" is Wechsler; "fluid, crystallised" is Cattell; "linguistic, logical-mathematical, spatial" is Gardner.

The **PASS model** of the Odisha-born **Jagannath Prasad Das** (born 1931), **Jack A. Naglieri** and **John R. Kirby** (1994), built on **Alexander Romanovich Luria**'s (1902 to 1977) functional units of the brain, has four processes: Planning, Attention, Simultaneous (parts into a whole) and Successive (serial order), measured by the Cognitive Assessment System. **Stephen J. Ceci**'s bioecological view (1990) ties each cognitive potential to its context.

Edward Lee Thorndike's (1874 to 1949) social intelligence (1920), the ability to understand and manage people, foreshadowed **emotional intelligence**, defined by **Peter Salovey** and **John D. Mayer** (1990) as the ability to monitor and use one's own and others' emotions; their four-branch ability model (perceiving, using, understanding and managing emotions) is measured by the MSCEIT. **Daniel Goleman** (born 1946) popularised a mixed model (1995): knowing one's emotions, managing them, motivating oneself, recognising others' emotions and handling relationships (relabelled in 1998 as self-awareness, self-regulation, motivation, empathy and social skills). **Reuven Bar-On**'s EQ-i (emotional quotient, 1997) and trait EI are self-report versions; **Danah Zohar** proposed spiritual intelligence (2000).

> **! EXAM TRAP**
>
> Salovey and Mayer coined emotional intelligence; Goleman popularised it. His 1998 labels rename the same five parts, so match the question's wording, and if both lists appear choose the original 1995 one ("knowing one's emotions" to "handling relationships"). "IQ, EQ, SQ, CQ" is no one's model of EI.

Heredity, Environment and the Brain ○

Higher IQ goes with faster reaction and inspection times, larger working memory, slightly larger brains and, in Richard Haier's 1988 study, lower brain glucose use (neural efficiency); intelligence is polygenic. Family studies compare relatives of differing closeness, adoption studies compare children with their biological and adoptive parents, and **twin studies** compare monozygotic (MZ) twins, who share virtually all genes, with dizygotic (DZ) twins, who share half, most powerfully when reared apart. In **Thomas J. Bouchard Jr.** and Matt McGue's 1981 review, MZ twins correlated .86 on IQ when reared together and .72 when reared apart (about .70 in Bouchard's Minnesota Study of Twins Reared Apart, from 1979), DZ twins .60.

Heritability is the proportion of *variation* in a trait, within one population at one time, associated with genetic variation; it says nothing about one individual. It rises with age, and **Eric Turkheimer** (2003) found it near .10 in the poorest families, a gene-environment interaction. **Richard Lewontin**'s seeds (1970) show why it cannot explain group gaps: varied seed differs genetically within each tray, yet the gap between a rich-soil and a poor-soil tray is wholly environmental. In **Irving Gottesman**'s **reaction range** (1963), genes set limits and environment decides where within them a person lands.

Iodine, nutrition and schooling raise IQ; early intervention (Head Start, 1965; the Milwaukee Project; the Abecedarian Project, 1972, from infancy) brings IQ gains that often fade but lasting school and life benefits. **James Robert Flynn** (1934 to 2020) showed in 1984 that average scores rose about 3 points a decade, the **Flynn effect**, too fast for genes. Neither **Arthur Robert Jensen** (1923 to 2012), who argued in 1969 that group IQ gaps are partly genetic, nor *The Bell Curve* (**Richard J. Herrnstein** and **Charles Murray**, 1994) has direct support, and **Claude Steele** and **Joshua Aronson** (1995) showed that **stereotype threat**, fear of confirming a group stereotype, depresses scores.

> **! EXAM TRAP**
>
> Individual differences come from many genetic and environmental factors together, not from genes, environment or income alone. Linkage studies, which track DNA markers through families, are neither twin nor adoption designs.

Intellectual Disability and Giftedness ●●●

Intellectual disability (DSM-IV-TR's mental retardation) requires deficits in intellectual functions, an IQ about two SDs below the mean (about 70; 85 is merely low average); deficits in **adaptive functioning**, the conceptual, social and practical skills of independent living; and onset in the developmental period. DSM-IV-TR graded severity by IQ (Table 1.20); DSM-5 grades it by adaptive functioning. **Global developmental delay** is a provisional DSM-5 label for children under 5 who cannot yet be fully assessed, so an assessed 7-year-old with an IQ of 65 and severe adaptive deficits has intellectual disability itself. Causes include Down syndrome, fragile X and phenylketonuria.

Table 1.20 Levels of intellectual disability (DSM-IV-TR).

Level	IQ range	Historical label
Mild (about 85% of cases)	50 to 55 to about 70	Moron (Goddard, 1910)

Level	IQ range	Historical label
Moderate	35 to 40 to 50 to 55	Imbecile
Severe	20 to 25 to 35 to 40	Imbecile or idiot
Profound	Below 20 or 25	Idiot

Adaptive behaviour is measured by **Edgar Arnold Doll**'s (1889 to 1968) **Vineland Social Maturity Scale** (1935), which yields a social age and a social quotient = (SA ÷ CA) × 100, and by the Vineland Adaptive Behavior Scales (communication, daily living skills, socialisation, motor). India's National Trust Act (1999) covers autism, cerebral palsy, mental retardation and multiple disabilities, but not specific learning disability.

At the other tail, an IQ of 130 or above is the usual cut-off for giftedness, though **Joseph Renzulli** (born 1936) required three rings (1978): above-average ability, task commitment and creativity.

KEY STUDY: TERMAN'S GIFTED CHILDREN (FROM 1921)

Question. Are bright children frail and odd? **Method.** Terman followed 1,528 California children with IQs of about 135 or more (mean about 151), the "Termites", for life. **Result.** They were healthier, better adjusted and more successful than average; social and behavioural problems clustered among those with childhood IQs of about 180 or more. **Criticism.** The mostly white, middle-class sample missed two future Nobel physicists.

Savant syndrome pairs a developmental disability, often autism, with an island of exceptional skill such as calendar calculation; a prodigy excels without disability. Creativity is treated with thinking (Section 1.8).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The term intelligence quotient was coined in 1912 by _____.
2. On a Wechsler scale, an IQ of 85 lies at the _____ percentile.
3. For an illiterate rural Indian child, the best choice is _____ Battery of Performance Tests.
4. Cattell called crystallised intelligence a repertoire of information, cognitive skills and _____.
5. Adapting to, shaping and selecting environments make up Sternberg's _____ subtheory.
6. Heritability is the proportion of _____ in a trait, within a population, associated with genetic variation.

Answers: (1) William Stern, (2) 16th, (3) Bhatia's, (4) strategies, (5) contextual, (6) variation.

1.10 Motivation HIGH YIELD · 74 QS

Why does a student study until 2 a.m., or a monkey dismantle a puzzle that hides no food? Each of psychology's answers below grew from the failures of the last.

What motivation is ●●●

In psychology, **motivation** (Latin *movere*, to move) is the set of processes that start, direct, energise and sustain behaviour.

Take thirst. A **need** is a state of deficiency (the body lacks water, felt or not). A **drive** is the aroused, tense state the need produces, which energises behaviour. A **motive** is any internal state directing behaviour towards a goal, usually a learned one such as achievement. An **incentive** is an external object that attracts or repels; the **goal** is the end state pursued. The **motivational cycle** runs need, drive, instrumental behaviour, goal, relief, until the need returns.

The **primary motives** (biological, unlearned, mostly homeostatic) include hunger, thirst, sleep, pain avoidance, sex and the maternal drive; the **secondary motives** (psychosocial, learned), such as achievement, affiliation, power and gregariousness, grow from experience. Henry Murray's **viscerogenic** (bodily) and **psychogenic** (mental) needs draw the same line.

Doing something for its own sake is **intrinsic motivation**; doing it for a separable outcome such as marks or praise is **extrinsic motivation**. A toddler who struggles to open a box she knows is empty is intrinsically motivated: no bodily imbalance (homeostasis), payoff (extrinsic) or learned social motive (secondary) is involved. Malingering (feigning for an external reward) and factitious disorder (feigning without one) differ on this line.

> ! EXAM TRAP
>
> **Gregariousness** is a **social motive**, though McDougall called it an instinct ("psychological" and "personal" motives are not standard classes). Among the perspectives (Section 1.2), inner forces beyond individual control are **psychodynamic**; free will and an innate tendency to grow to one's maximum potential are **humanistic** (Maslow, Rogers).

Instinct theories ●●

An **instinct** is a complex, unlearned, stereotyped pattern of behaviour shared by a whole species. **William James** (1842 to 1910) held in 1890 that humans have more instincts than animals, not fewer. **William McDougall** (1871 to 1938), in *An Introduction to Social Psychology* (1908), made instincts, each paired with an emotion (flight with fear, pugnacity with anger), the prime movers of all behaviour; he later listed eighteen. His **hormic psychology** (Greek *horme*, urge) holds that behaviour is purposive.

Sigmund Freud (1856 to 1939) divided the instincts (*Trieb*) in 1920 into the **life instincts** (Eros), whose energy is the **libido**, and the **death instinct** (Thanatos), which, turned outwards, appears as aggression and war. Investing psychic energy in an object is **cathexis**; the **id** is the reservoir of the instincts.

The ethologists **Konrad Lorenz** and **Nikolaas Tinbergen** (Nobel Prize 1973) described **fixed action patterns** released by a **sign stimulus**: a male stickleback attacks anything with a red belly. Yet as an account of human motives, instinct theory had already collapsed by the late

1920s: naming is not explaining (a "gregarious instinct" is circular), the lists ran into thousands, and **Zing-Yang Kuo** (1930) found that kittens reared with rats seldom killed them.

> **! EXAM TRAP**
>
> The instinct theory of motivation is **McDougall's**, not James's, Freud's or Maslow's. Freud's "innate forces" are **instincts**, not needs (Maslow, Murray), drives (Hull) or the id (their reservoir). War and violence express the **death instinct**.

Drives, arousal and incentives ●●●

Drive theory grew from physiology. **Walter Bradford Cannon** (1871 to 1945) named **homeostasis** (1926; *The Wisdom of the Body*, 1932), the body's thermostat-like correction around a set point, building on the *milieu intérieur* of **Claude Bernard** (1813 to 1878). On it **Clark Leonard Hull** (1884 to 1952) built **drive-reduction theory** (1943): a need creates a drive, a state of tension, and any response that reduces the drive is reinforced. In Hull's equation (${}_{S}E_{R} = {}_{S}H_{R} \times D \times K \times V$), K is **incentive motivation** (reward size) and V **stimulus-intensity dynamism** (stronger stimuli evoke the learned response more readily). **Primary drives** arise from tissue needs; **secondary drives**, such as learned fear, are acquired.

> **! EXAM TRAP**
>
> "Greater stimulus intensity, greater probability of the learned response" is **stimulus-intensity dynamism**, not incentive motivation, response amplitude (vigour) or reaction threshold (the minimum for any response). A deviation from homeostasis creates a **drive**, not equilibrium (the restored state), regulation (the correction) or dissonance (Festinger). Drive theory is about **reducing physiological needs**.

Yet **Harry Frederick Harlow**'s (1905 to 1981) monkeys took apart puzzles that hid no food (1950), rats worked for non-nutritive saccharin, and students paid to lie in sensory-deprivation cubicles (1954) quit within days. **Donald Olding Hebb**'s (1904 to 1985) **optimal arousal theory** (1955) says organisms seek a moderate level of arousal. In 1908 **Robert Mearns Yerkes** and **John Dillingham Dodson** had found that mice learned a brightness discrimination fastest under intermediate shock, and the harder the discrimination, the weaker the best shock. The **Yerkes-Dodson law**, as later taught (Figure 1.12), makes performance an **inverted-U** function of arousal or anxiety, with a lower optimum for complex tasks than for simple ones.

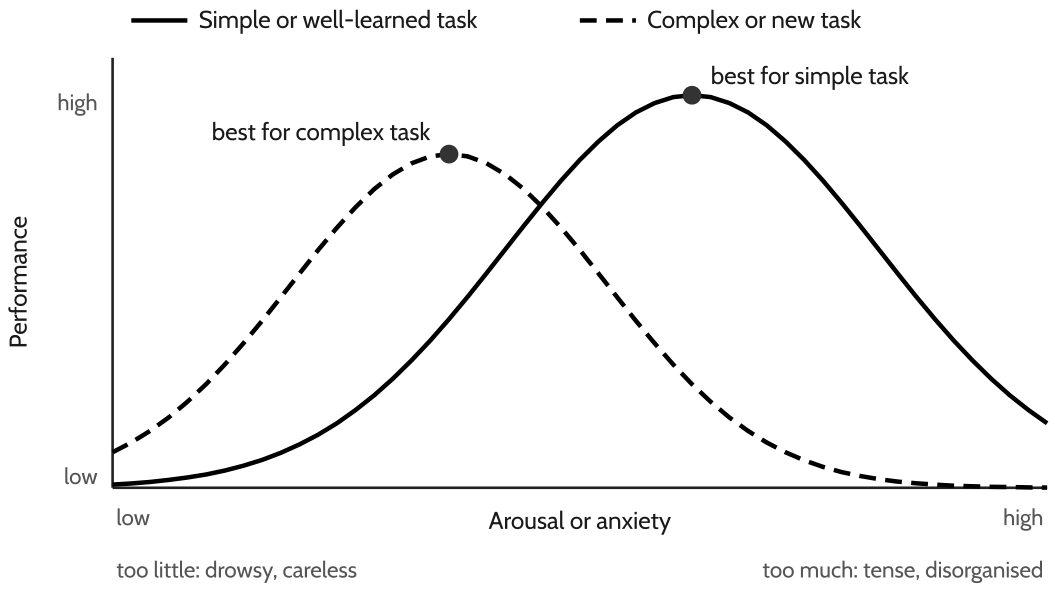


Figure 1.12 The Yerkes-Dodson law. Performance is an inverted-U function of arousal, and the peak for a complex or new task (dashed) lies at a lower arousal level than the peak for a simple or well-learned task (solid).

Robert Zajonc (1965) explained social facilitation the same way: an audience raises arousal, which strengthens the dominant response, helping easy, well-learned tasks and impairing novel or difficult ones. People differ too: **Marvin Zuckerman**'s (1928 to 2018) **sensation seeking** is the urge for varied, novel, intense experience despite risk.

> **! EXAM TRAP**
>
> Yerkes-Dodson is an **inverted-U** relationship between performance and anxiety: more precise than "curvilinear", and never linear or M-shaped. A complex task is done best at **moderate arousal, with a lower optimum**, not at very high arousal or at a level independent of difficulty.

Drives push; incentives pull. The **incentive theory** holds that behaviour is driven mainly by external rewards: **Otto Tinklepaugh**'s monkeys (1928) refused lettuce after seeing a banana hidden. The **expectancy-value theory** multiplies the expectancy of success by the value of the outcome, so either at zero kills motivation. Its roots are the learned expectancies of **Edward Chace Tolman** (1886 to 1959) and the **social learning** theory of **Julian Bernard Rotter** (1954), in which behaviour depends on expectancy and reinforcement value; organisational texts give the social learning name to the framework built on expectancy, demand and incentive, which combines the **cognitive** framework (Tolman's expectancies) with the **behaviouristic** one (reinforcement); "supportive" is not a framework but a model of organisational behaviour (autocratic, custodial, supportive, collegial). **Richard Lester Solomon** and John Corbit's **opponent-process theory** (1974) explains acquired motives such as addiction: an opposite after-reaction grows with repetition, bringing tolerance and withdrawal.

Maslow's hierarchy and other lists of needs ●●●

Abraham Harold Maslow (1908 to 1970), with Carl Rogers a founder of humanistic psychology (Section 1.2), set out in 1943 ("A Theory of Human Motivation") and 1954 (*Motivation and Personality*) a **hierarchy of needs** (Figure 1.13): physiological, safety, belongingness and love (social), esteem, and self-actualisation, becoming all that one can be. By **prepotency**, unmet lower needs are strongest and must be largely satisfied before higher ones take charge: a starving man dreams of food, not fame. Physiological and safety needs are thus the **lower needs**. The four lower levels are **deficiency needs** (D-needs), which stop motivating once met; self-actualisation is a **growth** or being (B) need, which strengthens as it is satisfied.

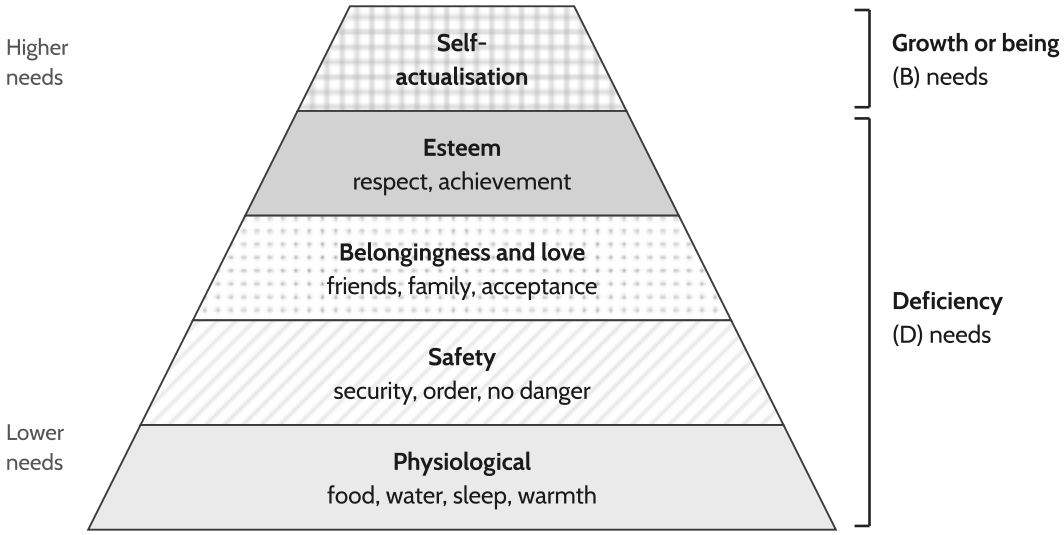


Figure 1.13 Maslow's hierarchy of needs (1943, 1954). The four lower levels are deficiency (D) needs; self-actualisation is a growth or being (B) need. Lower needs are prepotent: when unmet they dominate behaviour, and they must be largely met before higher needs take charge.

> **! EXAM TRAP**
>
> Bottom to top: **physiological, safety, belongingness, esteem, self-actualisation**. "Which is NOT a level?" **Ego** (Freud). Physiological and safety needs are **lower needs**: "greatest potency" is half-true (they are prepotent), and "transformational" is a leadership term. The hierarchical theory is **A. H. Maslow's**, not McClelland's, Atkinson's or Mowrer's. RML 2025 keyed **physiological** as the need mostly met before esteem; the level immediately below esteem is **belongingness (social belonging)**, so choose that when offered.

Self-actualisers, by Maslow's estimate 1 per cent of people or fewer, show **metamotivation** towards metaneeds or B-values such as truth, beauty and justice; frustrated, these breed **metapathologies** such as cynicism and apathy. They report **peak experiences**, moments of ecstatic awe. Maslow later added cognitive and aesthetic needs and, beyond self-actualisation, **self-transcendence**. The term self-actualisation was coined by **Kurt Goldstein** (1878 to 1965) in 1939. Critics fault Maslow's hand-picked sample, weak evidence for the ranking and Western bias.

> **! EXAM TRAP**
>
> "Striving for fulfilment of one's potential" is **self-actualisation**, not self-concept, ideal self (Rogers) or self-efficacy (Bandura). Self-actualisers show **metamotivation**, not D-motivation; "E-" and "F-motivation" are invented.

Clayton Paul Alderfer's **ERG theory** (1969) shrinks the ladder to existence, relatedness and growth, lets several needs act at once and adds **frustration-regression:** blocked at a higher need, a person falls back on a lower one (Table 1.21).

Table 1.21 Who proposed which list of needs.

Theorist	Year	The needs
Erich Fromm	1955	Relatedness, transcendence, rootedness, identity, frame of orientation
David McClelland	1953, 1961	Achievement, affiliation, power (learned)
Clayton Alderfer	1969	Existence, relatedness, growth
Deci and Ryan	1985	Autonomy, competence, relatedness (innate)
William Glasser	1965, 1998	Survival, love and belonging, power (achievement), freedom, fun

> **! EXAM TRAP**
>
> One of **William Glasser**'s five basic genetic needs in reality therapy is **power or achievement**; self-actualisation (Maslow), the collective unconscious (Jung) and unconditional positive regard (Rogers) are baits. Viktor Frankl's **will to meaning** belongs to **logotherapy** (Section 1.2), not REBT, existential-humanistic therapy or ACT.

Murray, McClelland and achievement ●●●

Henry Alexander Murray (1893 to 1988), in *Explorations in Personality* (1938), listed about twenty psychogenic needs, among them n Achievement, n Affiliation (companionship), n Dominance (control of others) and n Exhibition (being seen and heard). A **press** is an environmental force that helps or hinders a need. **Christiana Morgan** and Murray's **Thematic Apperception Test** (TAT, 1935) reads needs and presses from stories told about ambiguous pictures (Assessment chapter).

> **! EXAM TRAP**
>
> Murray's **need-press theory** underlies the **TAT**, a projective test whose stories reveal unconscious needs, conflicts and motives; not the Rorschach, Jung's word association test or sentence completion, and not a test of IQ or memory.

David Clarence McClelland (1917 to 1998) scored TAT stories for the **need for achievement** (nAch, 1953), a learned desire to meet standards of excellence, and later for **power** (nPow) and **affiliation** (nAff). High scorers prefer moderate risk, personal responsibility and quick feedback; businessmen he trained in Kakinada, Andhra Pradesh, in the 1960s went on to start more businesses. **John William Atkinson** (1923 to 2003) explained why in 1957: the tendency to approach success = motive to succeed × probability of success (Ps) × incentive (1 − Ps), and Ps × (1 − Ps) is 0.25 at Ps = 0.5 but only 0.09 at 0.1 or 0.9, so success-oriented people choose in-

termediate tasks, while those ruled by the **motive to avoid failure** pick very easy or very hard ones. **Matina Souretis Horner** (1972) proposed a **fear of success** in women, later read as cultural stereotyping.

Explanations matter too. Rotter's **locus of control** (1966) is internal when people credit their own effort, external when they credit luck or others (not low self-efficacy or Adler's inferiority complex). **Bernard Weiner** (born 1935) classified the causes of success and failure by **locus**, **stability** and **controllability**: ability is internal, stable and uncontrollable; effort internal, unstable and controllable. "I am useless at maths" breeds hopelessness; "I did not practise enough" breeds effort.

Intrinsic motivation and the self ●●●

Daniel Ellis Berlyne (1960) traced curiosity to novelty, complexity and incongruity, and **Robert White** (1959) united Harlow's manipulation drive, exploration and play as **effectance motivation**, the urge to deal effectively with the environment. That competence motive has two modern heirs.

The first is **self-efficacy**, introduced by **Albert Bandura** (1925 to 2021) in 1977: the belief that one can carry out the actions a particular task needs. It governs which tasks people attempt, how hard they try and how long they persist, and it grows mainly from mastery experiences, then from watching others, persuasion and bodily states. It is the core of his social cognitive theory (1986), built on **triadic reciprocal causation**: person, behaviour and environment influence one another. (Observational learning belongs to Section 1.6.)

> **! EXAM TRAP**
>
> The reciprocal causation model belongs to **social cognitive theory**, not drive reduction or constructivism (learners building their own knowledge, after Piaget and Vygotsky), and that theory highlights **self-efficacy**, not external rewards only, biological drives or goal difficulty. "Two-factor theories" are Herzberg's (work), Mowrer's (avoidance), Schachter and Singer's (emotion) and Spearman's (intelligence).

KEY STUDY: LEPPER, GREENE AND NISBETT'S MAGIC MARKERS (1973)

Mark Lepper, **David Greene** and **Richard Nisbett** let nursery-school children who liked drawing with magic markers draw for an expected certificate, an unexpected one, or nothing. One to two weeks later, the expected-award children spent about half as much free time drawing as the others; surprise awards did no harm. **Criticism.** Verbal or informative rewards do not undermine interest either, so the effect is narrower than often claimed.

This **overjustification effect** occurs because the reward moves the reason for an enjoyed activity outside the person. **Edward L. Deci** and **Richard M. Ryan** built such findings into the second heir, **self-determination theory** (SDT, 1985), with three innate needs: **autonomy**, **competence** and **relatedness**. Its **cognitive evaluation theory** explains the magic markers: rewards felt as controlling lower intrinsic motivation, while informational ones, such as unexpected praise, raise it. Motivation runs from amotivation through increasingly internalised extrinsic regulation to intrinsic motivation, driven by internal reward and growth.

> ! EXAM TRAP
>
> SDT's needs are **autonomy, competence and relatedness**, not **affiliation** or achievement (McClelland), independence, superiority or excellence. Its motivation driven by internal reward and growth is **intrinsic**; amotivation is its absence, external regulation the least autonomous form.

Mihaly Csikszentmihalyi (1934 to 2021) called intrinsic motivation at its most absorbing **flow** (1975, 1990): action and awareness merge and time vanishes when **challenge and skill are balanced** (too hard brings anxiety, too easy boredom). It is not Maslow's peak experience, Rogers's fully functioning person or Frankl's will to meaning. Finally, **Leon Festinger**'s (1957) **cognitive dissonance**, the tension of acting against one's beliefs (Arvind, who finds smoking disgusting, smokes at a party), motivates its reduction by a change of attitude or behaviour; effort justification is one form of that reduction, not the state (Social Psychology chapter).

Conflict and frustration ●●

The blocking of goal-directed behaviour by obstacles, personal limits or conflicting motives is **frustration**. **Pressure** is the demand to perform or conform: Ramesh, made to work hard, fast and a lot in a call centre, is under pressure, not blocked (frustration), torn (conflict) or facing unpredictability.

Kurt Lewin (1890 to 1947) described the conflicts in 1935. **Approach-approach** (two attractive goals) is easiest to resolve; **avoidance-avoidance** (two evils) produces vacillation and attempts to leave the field; in an **approach-avoidance conflict** one goal both attracts and repels; and **double approach-avoidance**, several such goals, is the commonest in real life. **Neal Elgar Miller**'s (1909 to 2002) gradients (1944) explain the vacillation: both tendencies strengthen near the goal, but avoidance is steeper, so the person hovers where they cross.

Frustration may lead to **aggression**, direct or displaced; **regression**, as in Roger Barker, Tamara Dembo and Lewin's children (1941), who played like toddlers once attractive toys were screened off; **fixation**, as in **Norman Maier**'s (1949) rats, which kept jumping one way after the problem became soluble; withdrawal; or defence mechanisms (Personality chapter). The **frustration-aggression hypothesis** of **John Dollard**, Leonard Doob, Miller, **Orval Hobart Mowrer** and Robert Sears (1939) held that frustration always leads to aggression; **Leonard Berkowitz** (1989) recast it in terms of negative affect.

> ! EXAM TRAP
>
> An after-the-fact, socially acceptable reason for behaviour really driven by other motives is **rationalisation**; re-channelling sexual or aggressive impulses into valued outlets is **sublimation**; a safer target is **displacement**; attributing one's own impulse to others is **projection**; turning it into its exaggerated opposite is **reaction formation**; pushing a threatening thought out of awareness is **repression**; refusing to acknowledge an unpleasant reality is **denial**; compensation pushed to excess is **overcompensation** (the full list: Personality chapter).

Hunger, thirst and sex ●●●

In 1912 Walter Cannon and **A. L. Washburn** showed that stomach contractions coincide with hunger pangs, yet people without stomachs still feel hunger. **Glucostatic** (blood sugar) and **lipostatic** (body fat) set-point theories followed; John Pinel and Steven Barnes argue instead that we eat mainly for anticipated pleasure, with weight drifting to a **settling point**.

In 1940 Albert Hetherington and Stephen Ranson found that lesions of the **ventromedial hypothalamus** (VMH) made rats overeat and grow obese; in 1951 the Indian physiologist **Bal Krishan Anand** and John Brobeck found that lesions of the **lateral hypothalamus** (LH) caused aphagia (no eating) and adipsia (no drinking). Eliot Stellar's **dual-centre theory** (1954) made the LH the feeding centre and the VMH the satiety centre. Since revised (VMH lesions raise insulin; other nuclei matter), it now shares the stage with leptin from fat, insulin from the pancreas and ghrelin from the empty stomach.

> **! EXAM TRAP**
>
> Satiety centre: **VMH**, lesion gives hyperphagia and obesity. Feeding centre: **LH**, lesion gives aphagia and weight loss (**L**ateral, **L**ose). "Ventromedial hippocampus" and "lateral thalamus" are invented; the cerebrum and basal ganglia are not hunger centres. The hypothalamus as a whole is the master homeostatic regulator (hunger, thirst, temperature; its suprachiasmatic nucleus is the circadian clock), not the cerebellum (movement) or pineal gland (melatonin).

Stanley Schachter's externality hypothesis (1968 to 1971) held that obese people respond more to external food cues, such as the clock, than to internal ones (eating disorders: Psychiatric Disorders chapter). Thirst is **osmotic** (salty body fluid draws water from cells) or **hypovolaemic** (blood volume falls, signalled through angiotensin II). **Alfred Charles Kinsey** surveyed the sexual behaviour of men (1948) and women (1953); **William Howell Masters** and **Virginia Eshelman Johnson** (1966) observed a four-phase **sexual response cycle** of excitement, plateau, orgasm and resolution; **Helen Singer Kaplan** (1979) put **desire** first (desire, excitement, orgasm). Hormones such as testosterone matter less in human sexuality than in other mammals. In Carl Warden's obstruction box (1931) the maternal drive outran thirst and hunger; pain avoidance (aroused by tissue damage) and sleep (Section 1.5) complete the primary motives.

> **! EXAM TRAP**
>
> Masters and Johnson: **excitement, plateau, orgasm, resolution**. Sequences that insert desire (Kaplan), start with plateau or end in "refractory" (the male recovery period) are scrambles. The reports on sexual behaviour are **Kinsey's**, not Freud's, Anna Freud's or Horney's.

Motivation at work, school and clinic ●●●

Frederick Herzberg (1923 to 2000) asked about two hundred engineers and accountants when they had felt exceptionally good or bad at work. His **two-factor theory** (1959) separates **motivators** (Table 1.22), in the work itself, which create satisfaction, from **hygiene factors**, in its context, which cause dissatisfaction when poor but, when improved, merely remove it.

Table 1.22 Herzberg's two factors.

Motivators (job content)	Hygiene factors (job context)
Achievement, recognition	Company policy, supervision
The work itself, responsibility	Relationships, working conditions
Advancement, growth	Salary, status, job security

Victor Harold Vroom's **expectancy theory** (1964) makes motivational force = expectancy (effort leads to performance) × instrumentality (performance leads to outcomes) × valence (value of the outcomes). If Priya, a sales executive, has E = 0.8, I = 0.9 and V = 0.7, her force is about 0.50; if her firm is known not to pay bonuses, I drops to 0.2 and her force to about 0.11. **John Stacy Adams**'s **equity theory** (1963) says people compare their ratio of outcomes to inputs with others' and act to remove inequity. In the **goal-setting theory** of **Edwin Locke** and **Gary Latham** (from 1968), specific, difficult but attainable goals, with commitment and feedback, beat vague "do your best" goals.

> **! EXAM TRAP**
>
> Hygiene factors **prevent dissatisfaction but do not by themselves motivate** (they are not self-actualisation needs, nor for high achievers only); job satisfaction depends on **work motivation and job environment**, not "or" or "only". Vroom's outcomes must be **valuable**; guaranteed rewards, effort unrelated to performance and unimportant rewards kill motivation. Goals should be **specific and challenging**, with feedback, not vague, easy or value-free.

In school, informational feedback, praise for effort and attribution retraining (Carol Dweck, 1975) sustain effort. In the clinic, **motivational interviewing** (William Richard Miller and Stephen Rollnick, 1991) treats resistance to change as ambivalence, an approach-avoidance conflict, and draws out the client's own reasons to change.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. A physiological deficiency is a _____; the tense state it produces is a _____.
2. For a complex task the optimal arousal is _____ than for a simple one.
3. The level immediately below esteem in Maslow's hierarchy is _____.
4. Murray's environmental force that helps or hinders a need is a _____.
5. Self-determination theory's three needs are autonomy, competence and _____.
6. In Herzberg's theory, salary is a _____ factor.

Answers: (1) need, drive, (2) lower, (3) belongingness and love, (4) press, (5) relatedness, (6) hygiene.

1.11 Emotion and Stress HIGH YIELD · 74 QS

Your heart pounds as the question paper is handed out. Is that pounding the fear itself, or something your mind has still to interpret? This section takes emotion apart, then turns to stress and coping.

What an emotion is ●●

An **emotion** is a brief, whole-person response to an event that matters to you, with three components: physiological arousal (a pounding heart, sweating palms), expressive behaviour (face, voice, posture, action) and conscious experience ("I am afraid"). The theories that follow quarrel mainly about the order of these three. A feeling is the subjective part of an emotion. An emotion lasts seconds to minutes and has an object (you are angry *at* someone); a **mood** lasts hours or days, is weaker and often has no object. **Affect** is the umbrella term for your present good or bad feeling state and, clinically, its observable expression, rated for quality, range and reactivity; persistence is what defines mood. Temperament is the inborn style of emotional reactivity (Alexander Thomas and Stella Chess's easy, difficult and slow-to-warm-up types are in the Developmental chapter).

Are there basic emotions, innate and universal, each with its own face? **Paul Ekman** (born 1934) and **Wallace V. Friesen** found six with universally recognised expressions: happiness, sadness, fear, anger, surprise and disgust, with contempt added in 1986. **Silvan Tomkins** (1962) listed nine innate affects; **Carroll Izard**'s differential emotions theory (1977) listed ten, including shame and guilt; and **Robert Plutchik**'s wheel (1980) set eight primaries opposite one another (joy and sadness, trust and disgust, fear and anger, surprise and anticipation), with neighbours blending into dyads such as love (joy plus trust). Dimensional models place emotions in a space instead: Wundt's three-dimensional theory of feeling (Section 1.1) and the circumplex of **James Russell** (1980), a circle of valence (pleasant to unpleasant) by arousal.

> ! EXAM TRAP
>
> **Jealousy, shame, guilt, pride and love are not among Ekman's six.** Shame and guilt are Izard's, trust and anticipation Plutchik's. "Our current mood and feelings" is **affect**, not mood.

Body first or label first ●●●

Common sense says you see a bear, feel afraid, and therefore tremble. **William James** (1842 to 1910) in 1884 and the Danish physician **Carl Georg Lange** (1834 to 1900) in 1885 reversed the order. The **James-Lange theory** holds that the stimulus first triggers bodily changes and that the emotion *is* our perception of them: "we feel sorry because we cry, angry because we strike, afraid because we tremble." The textbook sequence is stimulus, arousal, awareness ("interpretation") of the arousal, then emotion (Figure 1.14, top row), where the "interpretation" is just the brain noticing what the body is doing.

Walter Bradford Cannon (1871 to 1945) objected in 1927 that emotion survives when the viscera are cut off from the brain; that the same visceral changes accompany fear, anger, fever and exercise; that the viscera are too insensitive and slow for quick, fine distinctions; and that injected adrenaline produces symptoms without real emotion. In the **Cannon-Bard theory**, developed with **Philip Bard** (1898 to 1977), the thalamus signals the cortex and the body at once, so feeling and arousal occur simultaneously and independently (second row). If arousal is so similar across emotions, what tells fear from joy? In 1962 **Stanley Schachter** and **Jerome Singer** answered with the **two-factor theory**: emotion is physiological arousal, which sets its intensity, plus a cognitive label drawn from the situation, which names it.

KEY STUDY: SCHACHTER AND SINGER'S ADRENALINE EXPERIMENT (1962)

Students given adrenaline (epinephrine), described as a vitamin, were told its true effects, told nothing or misinformed; a placebo group got saline. Each then waited with a stooge acting euphoric or angry. The aroused but uninformed caught the stooge's mood most; the informed, who could blame the injection, least. **Criticism.** The effects were small, and in replications unexplained arousal simply felt anxious.

A label can be wrong. In **misattribution of arousal** people mistake the source of their arousal: in 1974 **Donald Dutton** and **Arthur Aron** found that men approached by an attractive woman on the swaying Capilano suspension bridge telephoned her far more often than men on a low, solid bridge. Fear had been read as attraction. **Dolf Zillmann**'s excitation transfer theory (1970s) adds that leftover arousal from one event intensifies the next emotion.

! EXAM TRAP

Match theory to sequence. Common sense: emotion, then body (the view James rejected). James-Lange: body first, then emotion. Cannon-Bard: emotion and visceral reaction *simultaneous*, via the thalamus. Schachter-Singer: *cognition plus arousal*, and the only one that explains misattribution. No classic theory makes the somatic nervous system the seat of emotion, and "emotions are learned through reinforcement" is behaviourism.

Appraisal, faces and opposing processes

Richard S. Lazarus (1922 to 2002) put thought first. In his cognitive appraisal (cognitive-mediational) theory, **primary appraisal** asks "Does this matter to me?" (harm-loss, threat or challenge), **secondary appraisal** asks "What can I do about it?", and reappraisal revises both (Figure 1.14, bottom row). **Robert Zajonc** (1923 to 2008) argued for affective primacy ("preferences need no inferences", 1980; debated with Lazarus in 1984): by the mere exposure effect, repeated exposure raises liking even for shapes seen too briefly to be recognised, so affect can come first. The **facial feedback hypothesis**, rooted in Tomkins's work, says that our own expressions modulate our feelings; in the pen study of **Fritz Strack**, **Leonard Martin** and **Sabine Stepper** (1988), holding a pen in the teeth, which forces a smile, made cartoons seem funnier, an effect now judged real but small. **Charles Darwin**'s *The Expression of the Emotions in Man and Animals* (1872) founded the evolutionary view of expressions as inherited adaptations. **Richard Solomon**'s opponent-process theory (1974) holds that every strong emotion triggers a slow, opposite after-reaction that grows with repetition: a drug's high fades while withdrawal deepens.

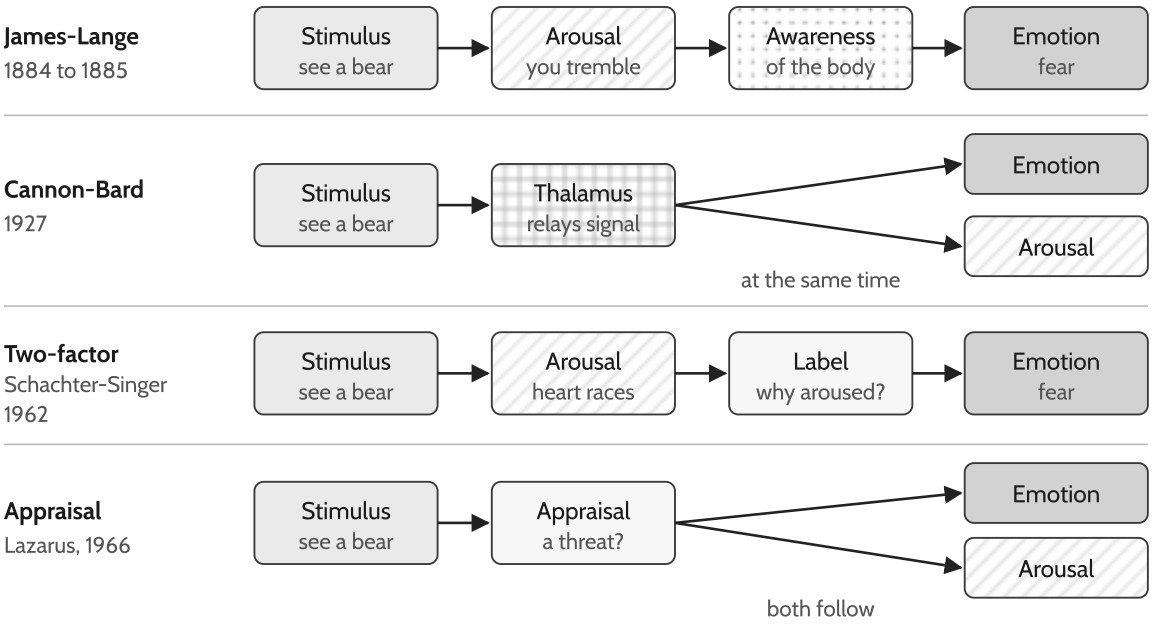


Figure 1.14 Four classic theories of emotion compared. Follow each row from the stimulus to the emotion. James-Lange: the body reacts first and the felt emotion is awareness of that reaction. Cannon-Bard: the thalamus sends signals that produce feeling and arousal at the same time. Two-factor: arousal plus a cognitive label. Appraisal (cognitive-mediational): an evaluation of the stimulus comes first, and both emotion and arousal follow from it.

The emotional brain and body ●●●

The body's side of emotion is run by the autonomic nervous system. Its sympathetic division prepares for fight or flight: it dilates the pupils, speeds the heart and breathing, dries the mouth, halts digestion and releases adrenaline from the adrenal medulla, and its sweating lowers skin resistance (the galvanic skin response). The parasympathetic division ("rest and digest") reverses each effect, constricting the pupils. Heartbeat and breathing themselves are paced by the medulla of the hindbrain; the hypothalamus only orchestrates them.

Bard's cats, with the cortex removed, showed **sham rage**, an undirected hissing fury that vanished only when the hypothalamus was removed too: the hypothalamus organises rage and the cortex restrains it. In 1937 **James Papez** (1883 to 1958) proposed an emotional loop, the **Papez circuit**: hippocampus, fornix, mammillary bodies, anterior thalamus, cingulate gyrus, entorhinal cortex and back to the hippocampus. **Paul MacLean** (1913 to 2007) added the amygdala and other structures in 1952 and named the whole the **limbic system**, the classic "seat of emotion", which regulates emotion, motivation and memory (Table 1.23).

Its key member is the **amygdala**, deep in each temporal lobe. When **Heinrich Klüver** and **Paul Bucy** removed both anterior temporal lobes of rhesus monkeys (1937, 1939), wild animals became tame. **Klüver-Bucy syndrome** is placidity (loss of fear, even of snakes), psychic blindness or visual agnosia, hyperorality, hyperphagia and hypersexuality; human cases often add aphasia and amnesia, because nearby temporal cortex is damaged too. The amygdala detects threat, drives fear conditioning and gives emotional events their extra memorability, an advantage that disappears when noradrenaline is blocked with propranolol (Cahill and McGaugh, 1994). **Joseph LeDoux** (born 1949) found a fast, crude low road from the thalamus straight to the amygdala and a slower, detailed high road through the cortex. Such subcorti-

cal routes probably explain affective blindsight, the recognition of emotional faces shown in a cortically blind field.

The prefrontal cortex regulates emotion (**Phineas Gage**'s rod injury of 1848; **Antonio Damasio**'s somatic markers, 1994). Damage to its control of the brainstem, after stroke, multiple sclerosis, ALS or brain injury, can cause **pseudobulbar affect**: sudden, involuntary laughing or crying out of proportion to the actual mood, which is not depression. The polygraph records arousal, not lies, so frightened innocent suspects can fail it; **David T. Lykken**'s guilty knowledge test (1959), which asks about details only the culprit knows, is more accurate.

Table 1.23 Brain structures in emotion.

Structure	Role	Classic evidence
Hypothalamus	Rage, stress hormones, hunger, thirst	Sham rage (Bard)
Limbic system	Emotion, motivation, memory	Papez (1937); MacLean (1952)
Amygdala	Fear, aggression, emotional memory	Klüver and Bucy; LeDoux
Hippocampus	Memory for facts	Amnesia, as in H.M.
Distractors	Reticular formation (arousal), pituitary, corpus callosum, cerebellum (coordination), caudate (movement), suprachiasmatic nucleus (body clock)	Not emotion centres

> **! EXAM TRAP**
>
> **Limbic system**, not one of its parts or the broad forebrain, for "seat of emotion"; its function is emotion, while hunger and thirst are the hypothalamus's. **Amygdala**, not hippocampus, for fear, aggression and the emotional enhancement of memory. Amygdala removal gives tameness, not attack (sham rage), obesity (ventromedial hypothalamus) or amnesia (hippocampus); options with "hyperalertness" or "avolition, alogia" are wrong. Pupil constriction is the odd one out among sympathetic effects.

Expression and four emotions ●●

The Fore of Papua New Guinea, who had seen no films or magazines, matched Ekman and Friesen's faces to stories far above chance (1971), confusing only fear with surprise. Cultures differ instead in **display rules**, learned norms for when an emotion may be shown. The Facial Action Coding System (1978) codes faces as muscle action units, and a genuine Duchenne smile engages the muscles around the eyes. Posture, gesture and voice carry emotion too. Women are, on average, more facially expressive, largely through socialisation. Fear answers a present danger; anxiety is diffuse and future-oriented; a phobia is a persistent, irrational fear of a specific object or situation. The catharsis hypothesis, that venting anger drains it, fails: hitting a punching bag while thinking of an insulter increased aggression (**Brad Bushman**, 2002). Disgust guards against contamination (**Paul Rozin** and **April Fallon**, 1987).

Ed Diener (1946 to 2021) defined **subjective well-being** (1984) by three components: life satisfaction (cognitive), frequent positive affect and infrequent negative affect. Good fortune fades: by **Harry Helson**'s adaptation-level theory (1964) and the hedonic treadmill of **Philip**

Brickman and **Donald Campbell** (1971), lottery winners are no happier than others a year or so later, and relative deprivation (**Samuel Stouffer**, 1949) makes us judge ourselves against others. **Barbara Fredrickson**'s **broaden-and-build theory** (1998) says positive emotions widen thought and build lasting resources. **Savoring** (**Fred Bryant** and **Joseph Veroff**, 2007) is enhancing and prolonging the enjoyment of positive experiences.

> ! EXAM TRAP
>
> **Diener:** life satisfaction, not physical, spiritual or social wellness. **Fredrickson:** broaden-and-build; flow is Csikszentmihalyi, cognitive distortions Beck, psychodynamic theory Freud. **Savoring** is not avoiding negatives, relaxing or reflecting on trauma.

How emotions develop and are regulated ●●●

Katharine Banham Bridges (1932) saw the newborn's general excitement split first into *distress* and *delight*, and only then into anger, fear, elation and the rest. Primary emotions are present by about six months. The **self-conscious emotions** (embarrassment, pride, shame, guilt) need a sense of self, so they appear after 12 months, from about 18 months. **Edward Tronick**'s still-face procedure (1978), in which a mother suddenly freezes her face, shows that even young infants expect emotional give-and-take.

In the process model of **James Gross** (1998), emotion regulation has five families of strategies, in the order an emotion unfolds: situation selection, situation modification, attentional deployment (distraction, rumination), cognitive change, chiefly **cognitive reappraisal** ("my racing heart is my body getting ready"), and response modulation, which includes expressive suppression. **Alexithymia** (**Peter Sifneos**, 1973) is difficulty identifying and describing one's own feelings, not anhedonia (loss of pleasure). **Daniel Goleman**'s emotional intelligence (1995; Section 1.9) has five domains: knowing one's emotions, managing them, motivating oneself, recognising others' emotions and handling relationships (later relabelled self-awareness, self-regulation, motivation, empathy and social skills); empathy has emotional, cognitive and empathic-concern components.

> ! EXAM TRAP
>
> **Reappraisal is cognitive change**; suppression is response modulation. **First infant emotions:** distress and delight (Bridges); fear, rage and love are Watson's. **Goleman:** if the 1995 list (knowing and managing one's emotions, motivating oneself, recognising others' emotions, handling relationships) and the later self-awareness relabelling both appear, choose the 1995 wording.

Stress and the body's emergency systems ●●●

Stress has been defined as a stimulus (the stressor), as a response (Selye's "nonspecific response of the body to any demand") and as a transaction in which demands are appraised as exceeding one's resources. Cannon's **fight-or-flight response** (1915) is the fast route: within seconds the sympathetic nerves act and the adrenal medulla releases adrenaline.

What if the emergency never ends? **Hans Selye** (1907 to 1982) found that every noxious agent he gave rats, from tissue extracts to cold, produced the same triad: enlarged adrenal glands, shrunken immune tissue and stomach ulcers. In 1936 he proposed the **general adaptation**

syndrome (*The Stress of Life*, 1956), with three stages in fixed order. In the *alarm reaction* the body mobilises, as in fight or flight. In the *stage of resistance* cortisol stays high and the body adapts, at a cost. In the *stage of exhaustion* adaptive capacity runs out and diseases of adaptation such as ulcers and hypertension appear. He later (1974) separated **eustress**, beneficial stress, from distress.

Resistance runs on the slower **hypothalamic-pituitary-adrenal axis** (HPA axis): the hypothalamus releases corticotropin-releasing hormone (CRH), the anterior pituitary adrenocorticotropic hormone (ACTH), and the adrenal cortex cortisol, which feeds back on the hypothalamus and anterior pituitary to switch the system off. **Shelley E. Taylor** (2000) added tend-and-befriend: females especially meet stress by protecting offspring and seeking allies, a pattern linked to oxytocin.

> ! EXAM TRAP
>
> **Alarm, resistance, exhaustion.** Illness is not a stage, and the stages are Selye's, not life events theory, the health belief model or protection motivation theory. **The adrenal glands** are the stress glands; the pituitary only releases ACTH, and the posterior pituitary is not in the HPA axis. Insulin (pancreas), thyroxine (thyroid) and melatonin (pineal) are distractors.

Sources of stress and the transactional model ●●●

Thomas Holmes and **Richard Rahe**'s **Social Readjustment Rating Scale** (1967) scores 43 life events in life change units, from death of a spouse (100) through marriage (50) to minor violations of the law (11); because change itself is the stressor, even holidays count. **Allen Kanner**, Lazarus and colleagues (1981) showed that **daily hassles**, the irritations of everyday life, predict symptoms better than major events, while uplifts, the small pleasures that offset them, matter less. An acute stressor is brief (a pop quiz); a chronic one lasts (HIV disease, cancer, marital discord); a cataclysmic one, such as a flood or war, strikes many people at once. Frustration and conflict are covered under motivation (Section 1.10). At work, organisational stressors are task, role and interpersonal demands, structure and leadership, as distinct from environmental (political uncertainty) and individual (family, money) factors. Role conflict (incompatible expectations) and **Robert Karasek**'s high-demand, low-control jobs (1979) can end in **burnout** (**Herbert Freudenberger**, 1974), which **Christina Maslach**'s inventory (1981) measures as exhaustion, depersonalisation and reduced accomplishment.

In Lazarus and **Susan Folkman**'s **transactional model of stress** (1984), stress is a person-environment relationship appraised as taxing or exceeding one's resources: primary appraisal judges harm-loss, threat or challenge, secondary appraisal judges the resources for coping, and coping and reappraisal follow. In one exam hall, Ananya appraises the paper as a threat and her preparation as thin, and shakes; Kiran sees a challenge, trusts his practice, and feels keyed up but focused. The **diathesis-stress model** (**Paul Meehl**, 1962; **Joseph Zubin** and **Bonnie Spring**, 1977) explains why stress makes one person ill and spares another: a disorder appears when a pre-existing vulnerability, often biological, is triggered by stress; usually neither is enough alone.

! EXAM TRAP

The SRRS measures life events, not hassles, physiological reactions or cognitive change. **Secondary appraisal** is the resources question; "coping resource assessment" is not the model's term. **Diathesis-stress** is not "genes only", "environment only", "unconscious conflicts only", "learned experiences alone" or supernatural; humanistic, sociocultural and cognitive models blame blocked growth, social forces and faulty thinking.

Coping, personality and health ●●●

Problem-focused coping acts on the stressor (a revision plan); **emotion-focused coping** acts on the feelings (reappraisal, support, acceptance, but also denial). The best choice fits controllability: plan while the exam can still be prepared for, reappraise once the result is out. Approach coping beats avoidance in the long run. Defence mechanisms (Personality chapter) differ in being unconscious: repression involuntarily blocks painful feelings (suppression is deliberate); displacement redirects anger to a safer target (anger at a husband shouted at a child); sublimation channels it into a valued outlet (kickboxing); projection attributes one's own feelings to others; isolation of affect strips the feeling from an idea (a gruesome fantasy told calmly), whereas denial rejects the reality; rationalisation offers a plausible excuse; and reaction formation turns a feeling into its opposite.

Social support protects both directly and as a buffer when stress is high (**Sheldon Cohen** and **Thomas Wills**, 1985). **Robert S. Weiss** (1973) separated **emotional isolation**, lacking a deep attachment to one specific person, from social isolation, lacking a circle of friends. **Suzanne Kobasa**'s **hardiness** (1979) is commitment, control and challenge; **Aaron Antonovsky**'s sense of coherence (1979) is comprehensibility, manageability and meaningfulness; resilience is bouncing back from adversity. Dispositional optimism (**Michael Scheier** and **Charles Carver**, 1985), a general expectancy of good outcomes, aids coping and recovery. **Martin Seligman** and **Steven Maier** (1967) found that dogs given inescapable shock stopped trying to escape, learned helplessness (Section 1.6), which Seligman later reversed into learned optimism.

In 1959 the cardiologists **Meyer Friedman** and **Ray Rosenman** described the **Type A behaviour pattern**: competitive striving, time urgency and hostility. In their Western Collaborative Group Study, Type A men developed coronary heart disease about twice as often as relaxed Type B men; later work found hostility to be the toxic component. A cancer-prone Type C, unassertive and suppressing negative emotion, has weak support. **Johan Denollet**'s **Type D personality** (mid-1990s) is negative affectivity plus social inhibition, and predicts a poor outlook in heart disease.

George Engel's **biopsychosocial model** (1977) explains illness through biological, psychological and social factors together. **Psychoneuroimmunology** began when **Robert Ader** and **Nicholas Cohen** (1975) conditioned immune suppression in rats by pairing saccharin with an immunosuppressive drug; chronic stress suppresses immunity and slows healing. Three models predict health behaviour: the health belief model (**Irwin Rosenstock** and colleagues, 1950s: perceived susceptibility, severity, benefits, barriers and cues to action), **Ronald Rogers**'s protection motivation theory (1975: threat appraisal and coping appraisal) and **Icek Ajzen**'s theory of planned behaviour (1985: attitude, subjective norm and perceived behavioural control, close to Albert Bandura's self-efficacy, belief in one's capability to act).

Stress management (Clinical chapter) includes **Edmund Jacobson**'s progressive muscle relaxation (1929), tensing and releasing muscle groups in turn; **Herbert Benson**'s relaxation response (1975); biofeedback, learning to control a bodily signal displayed on a monitor; meditation, as in **Jon Kabat-Zinn**'s mindfulness-based stress reduction (1979); regular aerobic exercise; and cognitive restructuring, replacing distorted thoughts with realistic ones, which **Donald Meichenbaum** combined with coping skills in stress inoculation training (1970s).

> **! EXAM TRAP**
>
> **Type A** is competitiveness, impatience and hostility; options with "non-competitiveness" are wrong, and introversion or extraversion is no risk factor. Offered both Type A and hostility, RML 2024 keyed Type A; for the toxic *component*, answer hostility. **Type D** is not hostility, optimism or perfectionism. **Learned helplessness** is Seligman and Maier, not Beck, Sullivan or Kahneman. **Biopsychosocial** means social, not cultural. Emotional isolation is not incompetence or low self-esteem.

> **SCAN YOUR BRAIN**
>
> Fill in the blanks. The answers are below; cover them first.
>
> 1. "We feel afraid because we tremble" summarises the _____ theory.
> 2. MacLean's name for the classic "seat of emotion" is the _____ system.
> 3. Tameness, hyperorality and hypersexuality after bilateral amygdala damage make up _____ syndrome.
> 4. In Gross's process model, reappraisal belongs to _____ change.
> 5. Selye's general adaptation syndrome runs alarm, _____, exhaustion.
> 6. Denollet's Type D combines negative affectivity with social _____.
>
> **Answers:** (1) James-Lange, (2) limbic, (3) Klüver-Bucy, (4) cognitive, (5) resistance, (6) inhibition.

The Case of Max Friedrich's Key, Revisited

Go back to the winter of 1879 and the student pressing his key. You can now take his act apart with the whole chapter. The stimulus had to cross a threshold (Section 1.3) and be organised into an object (Section 1.4); apperception was attention bringing it into focus (Section 1.5); practice shortened the times, which is learning (Section 1.6); the instruction was held in memory (Section 1.7); choosing when to press was a small decision (Section 1.8); observers differed in speed, one root of the study of individual differences and intelligence (Section 1.9); and the readiness and strain of waiting were motivation and emotion (Sections 1.10 and 1.11). Friedrich's numbers are forgotten, but his question, how long a mental act takes and what it is made of, runs through every section of this chapter.

Chapter summary

1.1 Psychology as a Science: Roots, Founders and the First Schools. Psychology, once the study of the soul, is now the scientific study of behaviour and covert mental processes. Its long past lies in philosophy (Plato's nativism, Aristotle's laws of association, Descartes' dualism, Locke's *tabula*

rasa, the associationists, Leibniz and Herbart on apperception, Kant) and in physiology (Muller, Helmholtz, localisation), with Darwin and Galton adding function and individual differences. Fechner's psychophysics (1860) and Donders' chronometry measured the mind. **Wundt** founded the first laboratory at Leipzig in 1879 and taught voluntarism through controlled introspection. **Titchener**'s structuralism sought the elements of consciousness; **James**'s functionalism asked what the mind is for, and Witmer opened the first clinic (1896).

1.2 Schools and Systems, the Cognitive Revolution and Indian Psychology. From about 1900 to 1960 psychology was a *preparadigmatic* field of rival schools. **Watson** founded behaviourism in 1913; Tolman, Hull, Guthrie and Skinner refined it. **Gestalt psychology** (Wertheimer's phi phenomenon, 1912; Kohler; Koffka) held that the whole differs from the sum of its parts, and Lewin defined the life space. **Psychoanalysis** was the first force, **humanism** (Maslow, Rogers) the third and transpersonal psychology the fourth, beside existential psychology and Seligman's positive psychology. The **cognitive revolution** (1956 onwards) restored the mind. Sengupta and Bose pioneered modern psychology at Calcutta; **Indian psychology** rests on Samkhya's purusha, prakriti and three gunas, Patanjali's eight limbs, the four states and the five koshas.

1.3 Sensation and Psychophysics. Sensation is the detection and transduction of physical energy; perception is its interpretation, so illusions distort perception. Muller's specific nerve energies and the all-or-none law explain how quality and intensity are coded, and every sense but smell relays through the thalamus. Fechner's psychophysics (1860) defines the absolute threshold as 50 percent detection and the JND as the smallest detectable change; Weber's law, Fechner's logarithmic law and Stevens' power law link stimulus to sensation, and signal detection theory separates sensitivity (d′) from response bias. Rods serve scotopic vision and foveal cones colour and acuity; colour is trichromatic then opponent; pitch is coded by place and timing; gate control explains pain.

1.4 Perception: Organising and Interpreting the World. Perception is the selection, organisation and interpretation of sensory signals. **Constructivist** theories (Helmholtz's unconscious inference, Gregory's hypotheses, the Bayesian union of priors and evidence) treat it as top-down inference; **Gibson's** direct, ecological theory, with its affordances, is the classic bottom-up account, and Neisser's perceptual cycle joins the two. The Gestalt laws, crowned by Prägnanz, organise the field into figure, ground and groups. Depth comes from two binocular cues (retinal disparity, convergence) and many monocular ones; the visual cliff (Gibson and Walk, 1960) showed depth perception by crawling age. Constancies keep objects stable, while illusions such as the Müller-Lyer reveal misapplied size scaling and differ from hallucinations and delusions.

1.5 Attention and Consciousness. Attention selects the little that reaches awareness; its span, studied by Jevons (1871) and with the tachistoscope, is about six to eight items. Cherry's dichotic listening (1953) led to Broadbent's early filter (1958), Treisman's attenuation model (1964) and late selection (Deutsch and Deutsch, 1963). In the **Stroop effect** (1935) automatic reading defeats selective attention, and vigilance declines over a watch (Mackworth). James described a stream of consciousness, Freud conscious, preconscious and unconscious levels. The suprachiasmatic nucleus times sleep, whose NREM and REM stages cycle every 90 minutes, four to six times a night. Dreams are explained by Freud, Jung and activation-synthesis; hypnosis rests on suggestibility.

1.6 Learning: Conditioning, Cognition and Observation. Learning is a relatively permanent change in behaviour, or its potential, resulting from experience (Kimble). In **classical conditioning** (Pavlov)

a neutral stimulus paired with a UCS becomes a CS; extinction, spontaneous recovery, generalisation and discrimination follow. Watson's Little Albert showed conditioned fear, and Jones and Wolpe reversed it through counterconditioning and systematic desensitisation (reciprocal inhibition). Rescorla, Kamin and Garcia showed that conditioning depends on prediction and preparedness. In **operant conditioning** (Thorndike's law of effect, Skinner) consequences change behaviour: reinforcement increases it, punishment decreases it, and schedules shape its pattern, variable ratio most strongly. Tolman's latent learning, Köhler's insight and Bandura's observational learning showed much learning without direct reinforcement.

1.7 Memory: Encoding, Storage, Retrieval and Forgetting. Memory is studied as **encoding**, **storage** and **retrieval**; Ebbinghaus (1885) measured forgetting with nonsense syllables and savings. Atkinson and Shiffrin's **multi-store model** (1968) separates sensory, short-term (seven plus or minus two chunks, mainly acoustic) and long-term memory, and Baddeley and Hitch (1974) recast short-term memory as **working memory**: central executive, phonological loop, visuospatial sketchpad and, from 2000, the episodic buffer. Depth of processing, Tulving's episodic and semantic memory, implicit memory, schemas and Bartlett's reconstruction, encoding specificity, decay, interference and repression explain remembering and forgetting, while H.M., Korsakoff's syndrome and Lashley's search for the engram reveal its biology.

1.8 Thinking, Problem Solving, Reasoning, Decision Making and Language. Thinking manipulates internal representations: images (Shepard and Metzler's mental rotation; Paivio's dual coding), concepts (Bruner's hypothesis testing for artificial concepts, Rosch's prototypes for natural ones) and propositions. Problems yield to algorithms, heuristics or Gestalt insight, and are blocked by **functional fixedness** (Duncker, 1945) and **mental set** (Luchins, 1942). Deduction runs from theory to data, induction from data to theory. Simon's bounded rationality and satisficing, and Tversky and Kahneman's heuristics and prospect theory, describe real decisions. Creativity is divergent production passing through Wallas's four stages. Language is built from phonemes and morphemes; Chomsky's LAD, the critical period, Whorf's linguistic relativity and Vygotsky's inner speech complete the picture.

1.9 Intelligence: Theories, Measurement, Heredity and Environment. Intelligence has been defined as judgement (Binet), *what the tests test* (Boring) and Wechsler's aggregate or global capacity (1944). Binet and Simon's 1905 scale introduced mental age; Stern coined the IQ (1912), Terman's Stanford-Binet (1916) made the ratio IQ famous, and Wechsler's deviation IQ (mean 100, SD 15) replaced it. The WAIS-IV has four indices; Raven's matrices test fluid intelligence, and Bhatia's Battery suits illiterate Indian children. Spearman's g, Thurstone's primary abilities, Guilford's cube, Cattell's fluid and crystallised intelligence, Gardner's multiple intelligences and Sternberg's triarchic theory offer rival structures. Heritability is a population statistic, and intellectual disability requires adaptive deficits as well as low IQ.

1.10 Motivation. Motivation energises and directs behaviour: a **need** produces a **drive** that pushes, while incentives pull. McDougall's instincts gave way to Hull's **drive reduction** (1943), which could not explain curiosity; Hebb proposed optimal arousal, and the **Yerkes-Dodson law** (1908) makes performance an inverted U of arousal with a lower optimum for complex tasks. Maslow's **hierarchy of needs** runs physiological, safety, belongingness, esteem, self-actualisation; Alderfer reduced it to ERG. Murray's needs and press underlie the TAT; McClelland studied achievement, power and affiliation. Self-efficacy, **self-determination theory** and flow explain intrinsic motivation. Lewin classified conflicts; the VMH is the satiety centre, the LH the feeding centre; Herzberg, Vroom and Locke explain work.

1.11 Emotion and Stress. An **emotion** joins arousal, expression and conscious experience; a mood is longer and diffuse, and affect is the current feeling state. Ekman named six universal basic emotions. **James-Lange** puts bodily change before feeling, **Cannon-Bard** makes them simultaneous via the thalamus, **Schachter and Singer** add a cognitive label (explaining misattribution), and Lazarus puts appraisal first. The **limbic system**, above all the **amygdala**, organises fear. Stress runs through Selye's alarm, resistance and exhaustion and the HPA axis; life events, hassles and appraisal shape it, and coping, support, hardiness and Type A or D personality link it to illness.

Who's who

Table 1.24 The people in this chapter.

Name	When	Examined for
Hermann Ebbinghaus	1850 to 1909	Memory measured with nonsense syllables and savings (On Memory, 1885); "Psychology has a long past, but only a short history" (1908)
Wilhelm Maximilian Wundt	1832 to 1920	First psychology laboratory, Leipzig (1879); Principles of Physiological Psychology (1873 to 1874); voluntarism; experimental introspection; Volkerpsychologie (folk psychology)
Aristotle	384 to 322 BCE	De Anima, the first treatise on the soul; empiricist; laws of association (similarity, contrast, contiguity)
William James	1842 to 1910	Father of American psychology; The Principles of Psychology (1890); stream of consciousness; habit; the self; inspiration of functionalism
William McDougall	1871 to 1938	Defined psychology as the science of conduct or behaviour (1905, 1908)
John Broadus Watson	1878 to 1958	Founder of behaviourism (1913); Angell's student at Chicago
Robert Sessions Woodworth	1869 to 1962	S-O-R formula; Dynamic Psychology (1918); psychology as the activities of the individual in relation to the environment
Norman L. Munn	Twentieth century	"The scientific study of behaviour", the definition credited to him in Indian textbooks
Plato	About 427 to 347 BCE	Nativist and rationalist; learning as recollection of inborn knowledge
Rene Descartes	1596 to 1650	Dualism; interaction through the pineal gland; innate ideas; the reflex
John Locke	1632 to 1704	Tabula rasa (1690); sensation and reflection; simple and complex ideas; primary and secondary qualities
David Hartley	1705 to 1757	Founder of associationism as a doctrine (1749)
James Mill	1773 to 1836	Mental mechanics: a complex idea is the sum of its parts
John Stuart Mill	1806 to 1873	Mental chemistry: ideas fuse into a new whole
Thomas Hobbes	1588 to 1679	Materialism; forerunner of British empiricism
George Berkeley	1685 to 1753	To be is to be perceived (esse est percipi)
David Hume	1711 to 1776	Laws of association: resemblance, contiguity, cause and effect
Alexander Bain	1818 to 1903	Associationism tied to the nervous system; founded the journal Mind (1876)
Gottfried Wilhelm Leibniz	1646 to 1716	Monads; petites perceptions; coined apperception; psychophysical parallelism
Johann Friedrich Herbart	1776 to 1841	Threshold of consciousness; apperceptive mass
Immanuel Kant	1724 to 1804	Innate categories; held that psychology could not be a science
Thomas Henry Huxley	1825 to 1895	Epiphenomenalism: mind as a by-product of the brain
Baruch Spinoza	1632 to 1677	Double-aspect theory of mind and body
Charles Bell	1774 to 1842	Bell-Magendie law (1811)
Francois Magendie	1783 to 1855	Bell-Magendie law (1822): dorsal roots sensory, ventral roots motor
Johannes Muller	1801 to 1858	Doctrine of specific nerve energies; Handbook of Human Physiology (1833 to 1840)

Name	When	Examined for
Hermann von Helmholtz	1821 to 1894	Speed of the nerve impulse (1850); unconscious inference; trichromatic theory of colour vision
Thomas Young	1773 to 1829	Suggested three colour receptors (1802): the Young-Helmholtz theory
Franz Joseph Gall	1758 to 1828	Founder of phrenology
Johann Gaspar Spurzheim	1776 to 1832	Popularised phrenology under that name
Pierre Flourens	1794 to 1867	Ablation studies of the brain (1820s)
Paul Broca	1824 to 1880	Speech production area in the left frontal lobe; patient Tan (1861)
Leborgne ("Tan")	1861	Broca's patient who understood speech but could say little more than "tan"
Carl Wernicke	1848 to 1905	Area for understanding speech in the left temporal lobe (1874)
Gustav Fritsch	1838 to 1927	Motor cortex mapped by electrical stimulation, with Hitzig (1870)
Eduard Hitzig	1838 to 1907	Motor cortex mapped by electrical stimulation, with Fritsch (1870)
Charles Darwin	1809 to 1882	Natural selection (1859); The Expression of the Emotions (1872); continuity, function and variation
Francis Galton	1822 to 1911	Individual differences; Hereditary Genius (1869); anthropometric laboratory (1884); correlation; questionnaire; word association; twin studies; eugenics
George John Romanes	1848 to 1894	Animal Intelligence (1882); anecdotal comparative psychology
Conwy Lloyd Morgan	1852 to 1936	Morgan's canon of parsimony (1894)
Ernst Heinrich Weber	1795 to 1878	Just noticeable difference as a constant fraction of the standard (Weber's law)
Gustav Theodor Fechner	1801 to 1887	Founder of psychophysics; insight of 22 October 1850; Elements of Psychophysics (1860); S = k log I
Franciscus Cornelis Donders	1818 to 1889	Subtraction method (1868); mental chronometry
Emil Kraepelin	1856 to 1926	Worked in Wundt's laboratory; classification of psychiatric disorders
Charles Spearman	1863 to 1945	Leipzig doctorate; factor analysis
Vladimir Mikhailovich Bekhterev	1857 to 1927	Studied with Wundt; reflexology
Edward Bradford Titchener	1867 to 1927	Structuralism (1898) at Cornell; elements and attributes; stimulus error; context theory of meaning; the Experimentalists (1904)
Oswald Kulpe	1862 to 1915	Wurzburg school; imageless thought; Aufgabe and set
Narziss Ach	1871 to 1946	Wurzburg: the determining tendency and set (1905)
Franz Brentano	1838 to 1917	Act psychology; intentionality (1874)
Carl Stumpf	1848 to 1936	Tone psychology; teacher of the Gestalt founders
Georg Elias Muller	1850 to 1934	Gottingen laboratory; memory research
Carl Lange	1834 to 1900	Theory of emotion (1885), independently of James: the James-Lange theory
John Dewey	1859 to 1952	"The Reflex Arc Concept in Psychology" (1896); Chicago functionalism
James Rowland Angell	1869 to 1949	"The Province of Functional Psychology" (1906); taught Watson
Harvey A. Carr	1873 to 1954	The adaptive act; mature Chicago functionalism
James McKeen Cattell	1860 to 1944	Coined "mental test" (1890); reaction time and individual differences; Columbia
Edward Lee Thorndike	1874 to 1949	Animal learning at Columbia; law of effect
Granville Stanley Hall	1844 to 1924	First American PhD in psychology (1878); first American research laboratory (1883); American Journal of Psychology (1887); first APA president (1892); adolescence as storm and stress; recapitulation
Lightner Witmer	1867 to 1956	First psychological clinic (1896); first used "clinical psychology" in print (1907); father of clinical psychology
Hugo Munsterberg	1863 to 1916	Founder of forensic and industrial psychology; Wundt's student
Mary Whiton Calkins	1863 to 1930	First woman APA president (1905); refused a Harvard doctorate
Margaret Floy Washburn	1871 to 1939	First woman PhD in psychology (1894); The Animal Mind (1908)
Leta Stetter Hollingworth	1886 to 1939	Rejected claims of female inferiority with experimental evidence
Thomas Carlyle	1795 to 1881	Great-man (great-person) theory of history (1841)
Edwin Garrigues Boring	1886 to 1968	Historian of experimental psychology; champion of the Zeitgeist view
Alfred Russel Wallace	1823 to 1913	Natural selection, independently of Darwin (1858): a multiple discovery

Name	When	Examined for
Alfred Binet	1857 to 1911	First practical intelligence test (1905); father of intelligence testing
Narendra Nath Sengupta	1889 to 1944	First Indian laboratory (1915) and department (1916) at Calcutta; father of modern psychology in India
Girindrasekhar Bose	1887 to 1953	Indian Psychoanalytic Society (1922); father of psychoanalysis in India
M. V. Gopalaswamy	1924	Founded the psychology department at the University of Mysore
Sigmund Freud	1856 to 1939	Founder of psychoanalysis; free association; the unconscious mind
Carl Jung	1875 to 1961	Used word association as a clinical technique
Max Wertheimer	1880 to 1943	Founder of Gestalt psychology (1912)
Ivan Pavlov	1849 to 1936	Classical conditioning
Wilhelm Roentgen	1895	Physicist who discovered X-rays (1895); a distractor for folk psychology
Abraham Maslow	1908 to 1970	Humanistic psychology
Carl Rogers	1902 to 1987	Humanistic psychology
Burrhus Frederic Skinner	1904 to 1990	Operant conditioning; radical behaviourism
Jerome Bruner	1915 to 2016	"New Look" research on perception
Roy Schafer	1940s	"New Look" research on perception; a distractor for Titchener's definition of perception
Gregory Kimble	1961	Learning theorist (the standard definition of learning, 1961); a distractor for Titchener's definition of perception
Erik Erikson	1902 to 1994	Psychosocial stages; a distractor for "clinical psychology" in print
Karen Horney	1885 to 1952	Psychoanalyst; never APA president, a distractor for Calkins
Thomas Samuel Kuhn	1922 to 1996	Paradigm, paradigm shift and preparadigmatic science (The Structure of Scientific Revolutions, 1962)
Robert Irving Watson	1909 to 1980	Historian who named the prescriptions, the enduring issues of psychology (1967)
Ivan Petrovich Pavlov	1849 to 1936	Conditioned reflexes, a forerunner of behaviourism
Rosalie Rayner	1898 to 1935	Little Albert study with Watson (1920)
Percy Williams Bridgman	1882 to 1961	Operationism (1927), the basis of the operational definition
Edward Chace Tolman	1886 to 1959	Purposive behaviourism; molar behaviour; intervening variables; cognitive map; latent learning
Charles Honzik	1930 (study)	Latent learning experiment with Tolman
Clark Leonard Hull	1884 to 1952	Hypothetico-deductive system; drive reduction; reaction potential = habit strength × drive
John Dollard	1950 (with Miller)	Early social learning theory: Freud recast in Hull's terms
Neal Elgar Miller	1950 (with Dollard)	Early social learning theory with Dollard
Edwin Ray Guthrie	1886 to 1959	Contiguity; one-trial learning
Zing-Yang Kuo	1898 to 1970	Behaviourist who rejected the concept of instinct
Harry Harlow	1905 to 1981	Learning sets (a wrong option for the cognitive map)
James L. Gould	1986 (honeybee study)	Tested Tolman's cognitive map idea in honeybees; did not propose it
Joseph Wolpe	1915 to 1997	Systematic desensitisation (1958)
Norman Leslie Munn	Modern	Textbook author who popularised "the scientific study of behaviour"; keyed in one Panwar item as first to establish it (prefer Watson if offered)
Albert Bandura	1925 to 2021	Self-efficacy (1977); later social learning theory with Walters (1963)
Richard Walters	1963 (with Bandura)	Later social learning theory
Wolfgang Kohler	1887 to 1967	Insight in Sultan the chimpanzee on Tenerife (1913 to 1920); transposition; isomorphism
Kurt Koffka	1886 to 1941	Gestalt co-founder; geographical versus behavioural environment; Principles of Gestalt Psychology (1935)
Kurt Lewin	1890 to 1947	Field theory; life space; B = f(P, E); group dynamics; action research; father of modern social psychology
Bluma Zeigarnik	1900 to 1988	Better recall of interrupted tasks (1927)
Maria Ovsiankina	1898 to 1993	Spontaneous resumption of interrupted tasks (1928)

Name	When	Examined for
Frederick Salomon Perls	1893 to 1970	Fritz Perls: Gestalt therapy, here-and-now awareness, the empty chair
Laura Perls	1940s and 1950s	Co-developer of Gestalt therapy
Paul Goodman	1951 (Gestalt Therapy)	Co-developer of Gestalt therapy
Jean-Martin Charcot	1825 to 1893	Showed Freud hysterical symptoms produced and removed under hypnosis (Paris, 1885)
Pierre Janet	1859 to 1947	Explained hysteria as dissociation
Hippolyte Bernheim	1840 to 1919	Nancy school: hypnosis as suggestion
Josef Breuer	1842 to 1925	Anna O. and the cathartic method; Studies on Hysteria with Freud (1895)
Anna O. (Bertha Pappenheim)	Treated 1880 to 1882	Breuer's patient; the "talking cure"
Alfred Adler	1870 to 1937	Individual psychology
Carl Gustav Jung	1875 to 1961	Analytical psychology; the collective unconscious
Erich Fromm	1900 to 1980	Humanistic psychoanalysis
Harry Stack Sullivan	1892 to 1949	Interpersonal psychiatry
Erik Homburger Erikson	1902 to 1994	Psychosocial stages
Anna Freud	1895 to 1982	Ego psychology; defence mechanisms
Melanie Klein	1882 to 1960	Object relations
Abraham Harold Maslow	1908 to 1970	Named the third force; self-actualisation; foresaw the fourth force (1968)
Carl Ransom Rogers	1902 to 1987	Actualising tendency; unconditional positive regard
Charlotte Buhler	1893 to 1974	Humanistic founder; lifespan view
Rollo May	1909 to 1994	Brought existentialism to American psychology; normal versus neurotic anxiety (1950)
Edmund Husserl	1859 to 1938	Phenomenology; bracketing
Irvin David Yalom	Born 1931	Four ultimate concerns: death, freedom, isolation, meaninglessness (1980)
Soren Kierkegaard	1813 to 1855	Anxiety as the dizziness of freedom
Martin Heidegger	1889 to 1976	Dasein, being-in-the-world (1927)
Jean-Paul Sartre	1905 to 1980	Existence precedes essence
Ludwig Binswanger	1881 to 1966	Daseinsanalysis
Medard Boss	1903 to 1990	Daseinsanalysis
Viktor Emil Frankl	1905 to 1997	Logotherapy; will to meaning; existential vacuum; noogenic neurosis
Martin Elias Peter Seligman	Born 1942	Founded positive psychology as President of the American Psychological Association (1998)
Mihaly Csikszentmihalyi	1934 to 2021	Co-founder of positive psychology; flow
C. R. Snyder	Modern	Hope theory: the will and the ways to reach a goal
Howard Gardner	Born 1943	Multiple intelligences (a wrong option for the founder of positive psychology)
Steven Pinker	Born 1954	Language and evolutionary psychology (The Language Instinct, 1994); a wrong option for the founder of positive psychology
Karl Spencer Lashley	1890 to 1958	Serial order in behaviour needs central plans (1948)
Allen Newell	1927 to 1992	Logic Theorist with Simon (1956)
Herbert Alexander Simon	1916 to 2001	Logic Theorist with Newell (1956)
Noam Chomsky	Born 1928	Formal models of language (1956); review of Skinner's Verbal Behavior (1959)
George Armitage Miller	1920 to 2012	"The magical number seven, plus or minus two" (1956)
Donald Eric Broadbent	1926 to 1993	First major information-processing theory (1958)
Ulric Neisser	1928 to 2012	Father of cognitive psychology; Cognitive Psychology (1967)
Frederic Charles Bartlett	1886 to 1969	Schemas (1932); forerunner of cognitive psychology
Jean Piaget	1896 to 1980	Studied children's thinking from the 1920s; pioneer of cognitive psychology
David Rumelhart	1942 to 2011	Parallel distributed processing with McClelland (1986)
James McClelland	Born 1948	Parallel distributed processing with Rumelhart (1986)

Name	When	Examined for
Albert Ellis	1913 to 2007	Rational emotive behaviour therapy
Aaron Temkin Beck	1921 to 2021	Cognitive therapy; cognitive distortions
Anne Treisman	1935 to 2018	Attenuation work from about 1960 (a wrong option for the first information-processing theory)
Lawrence Kohlberg	1981 (The Philosophy of Moral Development)	Moral development (a wrong option for the pioneer of cognitive psychology)
George Libman Engel	1913 to 1999	Biopsychosocial model (1977)
Asutosh Mookerjee	1864 to 1924	Calcutta vice-chancellor (garbled as "Amitosh Mukherjee" in one item); not a founder of psychology
Durganand Sinha	1922 to 1998	Indigenisation of psychology in India
Uma Chowdhury	About 1960	Indian adaptation of the Thematic Apperception Test (also spelled Chaudhuri; probably the "Kamala Chaudhuri" of one item)
Jai B. P. Sinha	1980	Nurturant-task leader
Udai Pareek	1925 to 2010	Role stress
Sudhir Kakar	Born 1938	Psychoanalysis of Indian childhood and society
S. K. Mangal	Modern	Indian textbook author; did not found a laboratory
A. K. Singh	Modern	Indian textbook author; did not found a laboratory
G. S. Rathi	Modern	A wrong option for India's first laboratory
Matthijs Cornelissen	Early 2000s	Led the Indian Psychology movement; founded the Indian Psychology Institute
Koneru Ramakrishna Rao	Born 1932	Co-editor of the Handbook of Indian Psychology (2008); probably the "G. R. S. Rao" of one item
Anand C. Paranjpe	2008 (Handbook)	Co-editor of the Handbook of Indian Psychology
Ajit K. Dalal	2008 (Handbook)	Co-editor of the Handbook of Indian Psychology
Girishwar Misra	2011	Co-editor of Foundations of Indian Psychology
Suneet Varma	2011	Co-editor of Foundations of Indian Psychology
Kapila	Ancient (traditional)	Traditional founder of Samkhya
Ishvarakrishna	About fourth or fifth century CE	Samkhya Karika
Patanjali	Between about the second century BCE and the fourth century CE	Yoga Sutras; chitta-vritti-nirodha; the eight limbs
Adi Shankaracharya	Eighth century CE	Advaita Vedanta: atman is Brahman; the world is maya
Siddhartha Gautama (the Buddha)	Traditionally 563 to 483 BCE	Four Noble Truths; Eightfold Path; anatta
Sri Aurobindo	1872 to 1950	Integral yoga; the psychic being
Marsha Linehan	Born 1943	Dialectical behaviour therapy, with Zen mindfulness
Jerome Seymour Bruner	1915 to 2016	New Look in perception (Section 1.4); distractor for Titchener's definition
Johannes Peter Muller	1801 to 1858	Doctrine of specific nerve energies
Eugene Galanter	1962	Classic absolute threshold examples (a candle flame 48 km away, a watch tick at 6 m)
Stanley Smith Stevens	1906 to 1973	Power law of sensation (1957); magnitude estimation; cross-modality matching
Wilhelm Wundt	1832 to 1920	Father of experimental psychology; distractor for the founder of psychophysics
David M. Green	1966	Signal detection theory, with Swets (1966)
John A. Swets	1966	Signal detection theory, with Green (1966)
Jan Evangelista Purkyne (Purkinje)	1787 to 1869	Purkinje shift: reds darken before blues at dusk (1825)
Ernst Mach	1838 to 1916	Mach bands (1865), explained by lateral inhibition
Ewald Hering	1834 to 1918	Opponent-process theory of colour vision (1878)

Name	When	Examined for
Shinobu Ishihara	1879 to 1963	Ishihara colour vision plates (1917)
Georg von Bekesy	1899 to 1972	Travelling wave on the basilar membrane, place theory of pitch; Nobel Prize 1961
William Rutherford	1839 to 1899	Frequency (telephone) theory of pitch (1886)
Ernest Rutherford	1871 to 1937	Physicist wrongly credited with the frequency theory of pitch in some answer keys
Ernest Glen Wever	1902 to 1991	Volley principle of pitch (1930, with Bray)
Charles William Bray	born 1904	Volley principle of pitch (1930, with Wever)
Kikunae Ikeda	1864 to 1936	Named umami, the fifth basic taste (1908)
Hans Henning	1916	Smell prism (six primary odours) and taste tetrahedron (four tastes)
Ronald Melzack	1929 to 2019	Gate control theory of pain (1965, with Wall)
Patrick David Wall	1925 to 2001	Gate control theory of pain (1965, with Melzack)
Silas Weir Mitchell	1829 to 1914	Coined the term phantom limb (1871)
Vilayanur Subramanian Ramachandran	born 1951	Phantom limbs explained by cortical reorganisation (1990s)
Harry McGurk	1976	McGurk effect, with MacDonald (1976)
John MacDonald	1976	McGurk effect, with McGurk (1976)
Donald Olding Hebb	1904 to 1985	McGill sensory deprivation studies in his laboratory
William Bexton	1954	McGill sensory deprivation study (Bexton, Heron and Scott, 1954)
Woodburn Heron	1954	McGill sensory deprivation study (Bexton, Heron and Scott, 1954)
Thomas Scott	1954	McGill sensory deprivation study (Bexton, Heron and Scott, 1954)
Charles Bonnet	1720 to 1793	Charles Bonnet syndrome: visual hallucinations with insight in people with sight loss (1760)
Richard Langton Gregory	1923 to 2010	Constructivist approach and perceptual hypotheses (Eye and Brain, 1966); inappropriate constancy scaling (1963); studied S.B. (1963)
James Jerome Gibson	1904 to 1979	Direct (ecological) perception, the classic bottom-up theory; affordances; optic flow
Wolfgang Köhler	1887 to 1967	Gestalt co-founder
Edgar Rubin	1886 to 1951	Figure and ground (1915); Rubin's vase
Eleanor Jack Gibson	1910 to 2002	Visual cliff (with Walk, 1960)
Richard Walk	1920 to 1999	Visual cliff (with Eleanor Gibson, 1960)
Emil Emmert	1844 to 1911	Emmert's law of afterimage size (1881)
Franz Carl Müller-Lyer	1857 to 1916	Müller-Lyer illusion (1889)
Marshall Segall	1963, 1966	Carpentered-world hypothesis (with Campbell and Herskovits)
Donald Thomas Campbell	1916 to 1996	Carpentered-world hypothesis (with Segall and Herskovits)
Melville Jean Herskovits	1895 to 1963	Carpentered-world hypothesis (with Segall and Campbell)
Mario Ponzo	1882 to 1960	Ponzo illusion (1911)
Adelbert Ames Jr.	1880 to 1955	Ames room
Johann Karl Friedrich Zöllner	1834 to 1882	Zöllner illusion (1860)
Johann Christian Poggendorff	1796 to 1877	Poggendorff illusion (1860)
Louis Albert Necker	1786 to 1861	Necker cube, an ambiguous figure (1832)
Lionel Penrose	1898 to 1972	Penrose triangle, an impossible figure (1958, with his son Roger)
Roger Penrose	born 1931	Penrose triangle (1958, with his father Lionel)
Muzafer Sherif	1906 to 1988	Used the autokinetic effect to study norm formation (1935)
Oliver Selfridge	1926 to 2008	Pandemonium model of feature analysis (1959)
David Hubel	1926 to 2013	Feature detectors in cat visual cortex (with Wiesel, from 1959)
Torsten Wiesel	born 1924	Feature detectors in cat visual cortex (with Hubel, from 1959)
Eleanor Rosch	born 1938	Prototypes (1970s)
Irving Biederman	1987	Recognition-by-components theory; about 36 geons
David Marr	1945 to 1980	Computational theory: primal sketch, 2.5-D sketch, 3-D model
Gerald Reicher	1969	Word superiority effect
Leigh Minturn	1955	B or 13 context effect (with Bruner)

Name	When	Examined for
Cecile Goodman	1947	Coin study: economic deprivation affects perception (with Bruner)
Elliott McGinnies	1949	Perceptual defence for taboo words
George Malcolm Stratton	1865 to 1957	Inverting lenses (1896, 1897): perceptual adaptation
Jean Wallace	1963	Studied the restored-sight patient S.B. with Gregory
Herman Witkin	1916 to 1979	Field dependence-independence; rod-and-frame and embedded figures tests
James Vicary	1915 to 1977	Subliminal advertising hoax (1957, admitted 1962)
Anthony Marcel	1983	Masked semantic priming without awareness
Lawrence Weiskrantz	1926 to 2018	Blindsight (1974)
Joseph Banks Rhine	1895 to 1980	Parapsychology at Duke; Zener-card studies
Karl Zener	1903 to 1964	Zener cards
K. D. Broota	1970s	Indian work on cultural differences in perceptual selectivity (with Ganguli)
H. C. Ganguli	1970s	Indian work on cultural differences in perceptual selectivity (with Broota)
William Stanley Jevons	1835 to 1882; 1871	Bean experiment on the span of attention
George Miller	1956	Magical number seven, plus or minus two: memory span, not span of attention (Section 1.7)
Colin Cherry	1914 to 1979; 1953	Cocktail party problem; dichotic listening and shadowing
Neville Moray	1959	Own name heard in the unattended ear; no memory for words repeated 35 times
Donald Broadbent	1926 to 1993; 1958	Filter model (early selection); first major information-processing theory
J. Anthony Deutsch and Diana Deutsch	1963	Late selection model
Donald Norman	born 1935; 1968	Pertinence model of late selection
William Johnston and Steven Heinz	1978	Multimode (flexible) theory of selection
Nilli Lavie	1994 to 1995	Perceptual load theory
Michael Posner	born 1936	Spotlight; cueing task (1980); endogenous and exogenous attention; automaticity criteria (with Charles Snyder, 1975)
Charles Snyder	1975	Criteria for automatic processing (with Posner)
Garry Gelade	1980	Feature integration theory (with Treisman)
Daniel Simons and Christopher Chabris	1999	The invisible gorilla (inattentional blindness)
Daniel Kahneman	1934 to 2024; 1973	Capacity model (Attention and Effort)
Christopher Wickens	1984	Multiple resource theory
Richard Shiffrin and Walter Schneider	1977	Automatic versus controlled processing
Patricia Devine	1989	Stereotypes activated automatically; low-prejudiced people refute them by controlled processing
John Ridley Stroop	1897 to 1973; 1935	The Stroop effect
Norman Mackworth	1948	Clock test; vigilance decrement
Jerome L. Singer	1924 to 2010	Research on daydreaming
B. F. Skinner	1904 to 1990	Operant conditioning (Section 1.6); not the coiner of stream of consciousness
Anthony Greenwald and colleagues	1998	Implicit Association Test (unconscious stereotyping)
David Chalmers	born 1966; 1995	The hard problem of consciousness
Charles Tart	born 1937; 1969	Altered states of consciousness
Hans Berger	1873 to 1941; 1929	First human EEG
Eugene Aserinsky	1921 to 1998; 1953	Discovered REM sleep with Kleitman
Nathaniel Kleitman	1895 to 1999	REM discovery (1953); sleep cycle with Dement (1957)

Name	When	Examined for
William Charles Dement	1928 to 2020	Sleep cycle (1957, with Kleitman); REM deprivation and REM rebound (1960)
Howard Roffwarg and colleagues	1966	Sleep and REM across the lifespan
Viktor Frankl	1905 to 1997	Paradoxical intention (logotherapy), used for insomnia (Chapter 7)
J. Allan Hobson and Robert McCarley	1977	Activation-synthesis hypothesis of dreams
Antti Revonsuo	2000	Threat-simulation theory of dreams
Stephen LaBerge	early 1980s	Verified lucid dreaming with eye-movement signals
Franz Anton Mesmer	1734 to 1815	Animal magnetism (1770s)
James Braid	1795 to 1860	Coined hypnotism; Neurypnology (1843)
Ambroise-Auguste Liébeault	1823 to 1904	Founder of the Nancy school
André Weitzenhoffer	1959	Stanford Hypnotic Susceptibility Scales (with Hilgard)
Ernest Ropiequet Hilgard	1904 to 2001	Stanford scales (1959); neodissociation theory and the hidden observer
Theodore Xenophon Barber	1927 to 2005	Social-cognitive view of hypnosis
Nicholas Spanos	1942 to 1994	Social-cognitive theory of hypnosis
Maharishi Mahesh Yogi	1918 to 2008	Transcendental meditation (1950s)
Herbert Benson	1935 to 2022; 1975	The relaxation response
Nikolaas Tinbergen	1907 to 1988	Fixed action patterns triggered by sign stimuli (stickleback)
Eric Kandel	Born 1929	Habituation and sensitisation traced to synapses in Aplysia
Robert Fantz	1964	Habituation method for studying infant discrimination
Konrad Lorenz	1903 to 1989	Imprinting in greylag goslings (1935); critical period
Shenger-Krestovnikova	1921	Experimental neurosis (circle and ellipse), in Pavlov's laboratory
W. J. Brogden	1939	Sensory preconditioning
Mary Cover Jones	1897 to 1987	Peter and the rabbit (1924); counterconditioning
Edmund Jacobson	1888 to 1983	Progressive muscle relaxation
Thomas Stampfl	1960s	Implosive therapy
Joseph Cautela	1967	Covert sensitisation
Victor Meyer	1966	Exposure and response prevention for obsessive-compulsive disorder
Robert Rescorla	1940 to 2020	Contingency (1968); Rescorla-Wagner model (1972)
Leon Kamin	1927 to 2017	Blocking (1969)
Robert Lubow	1959	Latent inhibition (with Moore)
A. U. Moore	1959	Latent inhibition (with Lubow)
Allan Wagner	1934 to 2018	Rescorla-Wagner model (1972)
Wolfram Schultz	1990s	Dopamine neurons of the ventral tegmental area signal reward prediction error
John Garcia	1917 to 2012	Taste aversion; Garcia and Koelling (1966); equipotentiality refuted
Robert Koelling	1966	Bright, noisy, tasty water (with Garcia)
Martin Seligman	Born 1942	Preparedness (1970); learned helplessness (1967, with Maier)
Teodoro Ayllon	1960s; book 1968	Token economy at Anna State Hospital (with Azrin)
Nathan Azrin	1960s; book 1968	Token economy (with Ayllon)
Orval Hobart Mowrer	1907 to 1982	Two-factor theory of avoidance (1947); bell-and-pad for bed-wetting (1938)
William Kaye Estes	1919 to 2011	Punishment suppresses rather than erases a response (1944); stimulus sampling theory (1950)
Charles Ferster	1957	Schedules of reinforcement (with Skinner)
David Premack	1925 to 2015	Premack principle (1959)
Steven Maier	1967	Learned helplessness (with Seligman)
Keller Breland	1961	Instinctive drift (with Marian Breland)
Marian Breland	1961	Instinctive drift (with Keller Breland)
Richard Herrnstein	1930 to 1994	Matching law (1961)
Leo Crespi	1942	Contrast effects (the Crespi effect)

Name	When	Examined for
Neal Miller	1909 to 2002	Operant conditioning of autonomic responses and biofeedback; imitation (with Dollard, 1941)
Arnold Kegel	1948	Perineometer, an early form of biofeedback
Dorothea Ross	1961	Bobo doll studies (with Bandura and Sheila Ross)
Sheila Ross	1961	Bobo doll studies (with Bandura and Dorothea Ross)
Julian Rotter	1916 to 2014	Social learning theory (1954): behaviour potential, expectancy, reinforcement value
Robert Mills Gagné	1916 to 2002	Eight types of learning; The Conditions of Learning (1965)
William Lowe Bryan	1860 to 1955	Telegraphy learning curves and the plateau (with Harter, 1897, 1899)
Noble Harter	1897, 1899	The plateau in telegraphy learning (with Bryan)
William C. F. Krueger	1929	Overlearning improves retention
Edwin Locke	Born 1938	Goal setting: specific goals improve performance
Gary Latham	1970s onwards	Goal setting (with Locke)
Sidney Leavitt Pressey	1888 to 1979	The first teaching machine (1920s)
Clive Wearing	1985	British musician whose encephalitis (1985) left dense amnesia with spared musical skill
Raymond Nickerson	1979	Coin-drawing study showing encoding failure (with Marilyn Adams)
Marilyn Adams	1979	Coin-drawing study showing encoding failure (with Raymond Nickerson)
Richard C. Atkinson	1968	Multi-store (modal) model with Shiffrin; keyword method (1975)
Richard M. Shiffrin	1968	Multi-store (modal) model with Atkinson
George Sperling	born 1934	Partial-report experiments on iconic memory (1960)
Christopher Darwin	1972	Three-eared man study of echoic memory (with Turvey and Crowder)
Michael Turvey	1972	Three-eared man study of echoic memory (with Darwin and Crowder)
Robert Crowder	1972	Three-eared man study of echoic memory (with Darwin and Turvey)
Nelson Cowan	2001	Pure short-term memory capacity of about four chunks
John Brown	1958	Brown-Peterson technique for short-term memory duration
Lloyd Peterson	1959	Brown-Peterson technique: unrehearsed short-term memory lasts about 18 to 30 seconds
Margaret Peterson	1959	Brown-Peterson technique: unrehearsed short-term memory lasts about 18 to 30 seconds
Geoffrey Keppel	1962	Showed Brown-Peterson forgetting was interference, not decay (with Underwood)
Benton Underwood	1962	Showed Brown-Peterson forgetting was interference, not decay (with Keppel)
Reuben Conrad	1964	Acoustic confusions show acoustic coding in short-term memory
Alan Baddeley	born 1934	Acoustic versus semantic coding (1966); working memory model with Hitch (1974); episodic buffer (2000); divers study with Godden (1975)
Bennet Murdock	1962	Reliability of the serial position curve in free recall
Murray Glanzer	1966	A filled delay abolishes recency but spares primacy (with Cunitz)
Anita Cunitz	1966	A filled delay abolishes recency but spares primacy (with Glanzer)
Tim Shallice	1970	Patient K.F.: impaired short-term memory with normal long-term learning (with Warrington)
Elizabeth Warrington	1970	Patient K.F.: impaired short-term memory with normal long-term learning (with Shallice)
J. W. McCrary	1953	Keyed in RML 2024 (as McCray) for first observing the primacy-recency effect, with Hunter
W. S. Hunter	1953	Keyed in RML 2024 for first observing the primacy-recency effect, with McCrary
Graham Hitch	1974	Working memory model with Baddeley
Robert Logie	1995	Visual cache and inner scribe of the visuospatial sketchpad
Fergus Craik	1972	Levels of processing with Lockhart; Craik and Tulving (1975)
Robert Lockhart	1972	Levels of processing with Craik
T. B. Rogers	1977	Self-reference effect (with Kuiper and Kirker)
Nicholas Kuiper	1977	Self-reference effect (with Rogers and Kirker)

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W. S. Kirker	1977	Self-reference effect (with Rogers and Kuiper)
Donald Morris	1977	Transfer-appropriate processing (with Bransford and Franks)
John Bransford	1977	Transfer-appropriate processing (with Morris and Franks)
Jeffery Franks	1977	Transfer-appropriate processing (with Morris and Bransford)
Allan Paivio	1925 to 2016	Dual coding theory (1971): verbal and imagery codes
Endel Tulving	1927 to 2023	Episodic versus semantic memory (1972); encoding specificity with Thomson (1973); cue-dependent forgetting (1974); patient K.C.
Peter Graf	1985	Named explicit and implicit memory (with Schacter)
Daniel Schacter	1985	Named explicit and implicit memory (with Graf)
Larry Squire	1980	Declarative versus nondeclarative memory and their taxonomy
David Meyer	1971	Semantic priming in lexical decision, BREAD and BUTTER (with Schvaneveldt)
Roger Schvaneveldt	1971	Semantic priming in lexical decision, BREAD and BUTTER (with Meyer)
David Rubin	1986	Reminiscence bump (with colleagues); flashbulb versus everyday memories with Talarico (2003)
Roger Brown	1925 to 1997	Flashbulb memory with Kulik (1977); tip-of-the-tongue state with McNeill (1966)
James Kulik	1977	Flashbulb memory and the Now Print! mechanism (with Brown)
Nicole Harsch	1992	Challenger flashbulb memory study (with Neisser)
Jennifer Talarico	2003	9/11 flashbulb memories: confidence, not consistency (with Rubin)
John Flavell	1971	Coined metamemory (1971); metacognition
David McNeill	1966	Tip-of-the-tongue state (with Brown)
Allan Collins	1969	Hierarchical network model with Quillian (1969); spreading activation with Loftus (1975)
Ross Quillian	1969	Hierarchical network model and cognitive economy (with Collins)
Elizabeth Loftus	born 1944	Misinformation effect (Loftus and Palmer, 1974); spreading activation with Collins (1975); false memory implantation; critic in the recovered memory controversy
Roger Schank	1977	Scripts, such as the restaurant script (with Abelson)
Robert Abelson	1977	Scripts, such as the restaurant script (with Schank)
Gordon Allport	1947	Levelling and sharpening, from the study of rumour (with Postman)
Leo Postman	1947	Levelling and sharpening, from the study of rumour (with Allport)
John Palmer	1974	Smashed versus hit study of the misinformation effect (with Loftus)
Marcia Johnson	1993	Source monitoring framework (with colleagues)
Zena Pearlstone	1966	Category cues show available but inaccessible memories (with Tulving)
Donald Thomson	1973	Encoding specificity principle (with Tulving)
Duncan Godden	1975	Context-dependent memory in divers (with Baddeley)
Donald Goodwin	1969	State-dependent memory with alcohol
Gordon Bower	1932 to 2020	Mood-dependent and mood-congruent memory (1981)
John Jenkins	1924	Less forgetting after sleep than after waking (with Dallenbach)
Karl Dallenbach	1924	Less forgetting after sleep than after waking (with Jenkins)
Arthur Melton	1940	Two-factor theory of interference: response competition and unlearning (with Irwin)
J. M. Irwin	1940	Two-factor theory of interference (with Melton)
Michael Anderson	1994	Retrieval-induced forgetting (with Robert and Elizabeth Bjork)
Robert Bjork	1994	Retrieval-induced forgetting (with Anderson and Elizabeth Bjork)
Elizabeth Bjork	1994	Retrieval-induced forgetting (with Anderson and Robert Bjork)
Georg Elias Müller	1900	Consolidation (with Pilzecker)
Alfons Pilzecker	1900	Consolidation (with Müller)
Théodule Ribot	1881	Ribot's law: the new perishes before the old
Philip Ballard	1913	Reminiscence: recall rises after a delay without study
Hedwig von Restorff	1933	Von Restorff (isolation) effect: a distinctive item is remembered better
Arthur Gates	1917	Testing effect: recitation beats rereading

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William Beecher Scoville	1953	Removed H.M.'s medial temporal lobes; reported the case with Milner (1957)
Henry Gustav Molaison	1926 to 2008	Patient H.M.: profound anterograde amnesia after bilateral medial temporal lobe removal, with spared procedural learning
Brenda Milner	1957	Studied H.M. (Scoville and Milner, 1957); mirror-tracing learning without awareness
Sergei Korsakoff	late 1880s	Described Korsakoff's syndrome
Richard Semon	1904	Coined the term engram
Timothy Bliss	1973	Discovered long-term potentiation (with Lømo)
Terje Lømo	1973	Discovered long-term potentiation (with Bliss)
Simonides of Ceos	about 556 to 468 BCE	Credited with the method of loci
Francis P. Robinson	1946	SQ3R study method
Roger Newland Shepard	1929 to 2022	Mental rotation (1971, with Jacqueline Metzler)
Jacqueline Metzler	1971	Mental rotation, with Shepard
Stephen Michael Kosslyn	born 1948	Image scanning (1978): images as depictive representations
Zenon Walter Pylyshyn	1937 to 2022	Propositional theory and the imagery debate (1973)
Eugen Bleuler	1857 to 1939	Autistic (dereistic) thinking, 1911
Robert Ennis	20th century	A widely used definition of critical thinking
Jacqueline Goodnow	1956	A Study of Thinking, with Bruner and Austin
George Austin	1956	A Study of Thinking, with Bruner and Goodnow
Ludwig Wittgenstein	1889 to 1951	Family resemblance (1953)
Mary Gick	1980	Analogical problem solving: the radiation problem, with Holyoak
Keith Holyoak	1980	Analogical problem solving: the radiation problem, with Gick
Robert Jeffrey Sternberg	born 1949	Problem-solving cycle; investment theory of creativity (1991, with Lubart); analytical, creative and practical intelligence
Norman Maier	1900 to 1977	Nine-dot problem (1930)
Janet Metcalfe	1987	Warmth ratings show insight solutions arrive suddenly (with Wiebe)
David Wiebe	1987	Warmth ratings, with Metcalfe
Karl Duncker	1903 to 1940	Candle problem and functional fixedness (1945)
Abraham Luchins	1914 to 2005	Water jars; mental set (Einstellung), 1942
Charles Sanders Peirce	1839 to 1914	Abductive reasoning, inference to the best explanation
Karl Raimund Popper	1902 to 1994	Hypothetico-deductive method
Robert Woodworth	1935	Atmosphere effect in syllogisms, with Sells
Saul Sells	1935	Atmosphere effect in syllogisms, with Woodworth
Peter Cathcart Wason	1924 to 2003	2-4-6 task (1960), selection task (1966), confirmation bias
Philip Nicholas Johnson-Laird	born 1936	Mental models theory of reasoning (1983)
Gerd Gigerenzer	born 1947	Fast and frugal heuristics; recognition heuristic; ecological rationality
Amos Tversky	1937 to 1996	Heuristics and biases (1972 to 1974), prospect theory (1979), framing (1981), with Kahneman
Baruch Fischhoff	born 1946	Hindsight bias, the knew-it-all-along effect (1975)
Joy Paul Guilford	1897 to 1987	Convergent and divergent thinking; 1950 APA address; fluency, flexibility, originality, elaboration
Graham Wallas	1858 to 1932	The Art of Thought (1926): preparation, incubation, illumination, verification
Ellis Paul Torrance	1915 to 2003	Torrance Tests of Creative Thinking (1966)
Sarnoff Mednick	1928 to 2015	Remote Associates Test (1962)
Edward de Bono	1933 to 2021	Lateral thinking (1967)
Alex Faickney Osborn	1888 to 1966	Brainstorming; Applied Imagination (1953)
Todd Lubart	1991	Investment theory of creativity, with Sternberg
Charles Francis Hockett	1916 to 2000	Design features of language (1960)
Dan Isaac Slobin	born 1939	Language-making capacity (1985)
Eric Heinz Lenneberg	1921 to 1975	Critical period for language (1967)

Name	When	Examined for
Genie	found 1970	Child isolated until 13: learned words but not grammar
Katherine Nelson	1973	Referential style of early word learning
Benjamin Lee Whorf	1897 to 1941	Linguistic relativity (the Sapir-Whorf hypothesis)
Edward Sapir	1884 to 1939	Whorf's teacher; the Sapir-Whorf hypothesis
Brent Berlin	1969	Basic colour terms in a near-universal order, with Kay
Paul Kay	1969	Basic colour terms, with Berlin
Lev Semyonovich Vygotsky	1896 to 1934	Thinking and Speech (1934): thought depends on language; private speech becomes inner speech; heaps, complexes, true concepts
David Wechsler	1896 to 1981	Aggregate or global capacity definition (1944); Wechsler-Bellevue (1939); deviation IQ; WAIS, WISC and WPPSI
Douglas Detterman	1986	With Sternberg, the 1986 expert symposium on defining intelligence
Carol Susan Dweck	Born 1946	Entity versus incremental theories of intelligence; fixed and growth mind-sets (Mindset, 2006); praise effort, not ability
Clark Wissler	1901	Showed that Cattell's mental tests did not predict college grades
Théodore Simon	1873 to 1961	Binet's collaborator on the 1905, 1908 and 1911 scales
Henry Herbert Goddard	1866 to 1957	Translated the Binet-Simon scale into English (1908); coined moron (1910)
William Stern	1871 to 1938	Coined the intelligence quotient (1912): mental age divided by chronological age
Lewis Madison Terman	1877 to 1956	Stanford-Binet (1916) and ratio IQ = MA ÷ CA × 100; gifted Termites study (from 1921)
Robert Mearns Yerkes	1876 to 1956	Army Alpha (verbal) and Army Beta (non-verbal) group tests (1917)
Gale Roid	2003	Stanford-Binet Fifth Edition (SB5): five factors, verbal and non-verbal, mean 100 and SD 15
Frank Schmidt and John Hunter	1998	General mental ability predicts job performance with a validity of about .5
John Carlyle Raven	1902 to 1970	Standard Progressive Matrices (1938): culture-fair test of fluid intelligence and the best single marker of g
Raymond Bernard Cattell	1905 to 1998	Fluid versus crystallised intelligence (1963); investment theory; Culture Fair Intelligence Test; the 16PF (personality)
V. V. Kamat	1930s	Binet-Kamat Test, a heavily verbal Indian adaptation of the Stanford-Binet
Arthur J. Malin	1969	Malin's Intelligence Scale for Indian Children (MISIC), a half-verbal adaptation of the WISC
C. M. Bhatia	1955	Bhatia's Battery of Performance Tests of Intelligence: five performance subtests, no reading, rural and urban norms
J. B. P. Sinha	1980	Nurturant-task leader (organisational psychology); a distractor, not the author of Bhatia's Battery
Charles Edward Spearman	1863 to 1945	Two-factor theory (1904): general factor g and specific factors s; positive manifold; first factor analysis
Louis Leon Thurstone	1887 to 1955	Seven primary mental abilities (1938)
Philip Ewart Vernon	1905 to 1987	Hierarchical model (1950): g, then the major group factors v:ed and k:m
John Bissell Carroll	1916 to 2003	Three-stratum theory (1993): g, eight broad and about 70 narrow abilities
Kevin McGrew	Late 1990s	Merged the Cattell-Horn and Carroll models into the CHC theory
John Leonard Horn	1928 to 2006	Developed the Gf-Gc theory with Cattell; extended it to about nine broad abilities
Godfrey Hilton Thomson	1881 to 1955	Sampling theory (1916): test correlations arise from overlapping samples of many small bonds
Arthur Robert Jensen	1923 to 2012	Level I (associative) and Level II (conceptual) abilities; 1969 claim that group IQ gaps are partly genetic
Jagannath Prasad Das	Born 1931	PASS model with Naglieri and Kirby (1994); Cognitive Assessment System
Jack A. Naglieri	1994	Co-author of the PASS model and the Cognitive Assessment System
John R. Kirby	1994	Co-author of the PASS model
Alexander Romanovich Luria	1902 to 1977	Functional units of the brain, the basis of the PASS model

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Stephen J. Ceci	1990	Bioecological view: cognitive potentials develop only in supporting contexts
Peter Salovey	1990	With Mayer, emotional intelligence (1990); four-branch ability model; MSCEIT
John D. Mayer	1990	With Salovey, emotional intelligence (1990); four-branch ability model; MSCEIT
Daniel Goleman	Born 1946	Popularised emotional intelligence (1995); five-part mixed model
Reuven Bar-On	1997	Emotional Quotient Inventory (EQ-i), a self-report mixed model; the term EQ
Danah Zohar	2000	Spiritual intelligence
Richard Haier	1988	PET study: high scorers used less brain glucose (neural efficiency)
Thomas J. Bouchard Jr.	1981; from 1979	Review of IQ correlations among relatives with McGue (1981); Minnesota Study of Twins Reared Apart
Matt McGue	1981	With Bouchard, the 1981 review of IQ correlations among relatives
Eric Turkheimer	2003	Heritability of IQ near .10 in the poorest families: gene-environment interaction
Richard Lewontin	1929 to 2021	Seed analogy (1970): heritability within groups says nothing about differences between groups
Irving Gottesman	1930 to 2016	Reaction range (1963): genes set limits, environment decides where within them a person lands
James Robert Flynn	1934 to 2020	The Flynn effect: average IQ scores rose about 3 points per decade (1984)
Richard J. Herrnstein	1930 to 1994	The Bell Curve (1994) with Murray
Charles Murray	Born 1943	The Bell Curve (1994) with Herrnstein
Claude Steele	1995	Stereotype threat, with Aronson (1995)
Joshua Aronson	1995	Stereotype threat, with Steele (1995)
Edgar Arnold Doll	1889 to 1968	Vineland Social Maturity Scale (1935): social age and social quotient
Joseph Renzulli	Born 1936	Three-ring conception of giftedness (1978): above-average ability, task commitment, creativity
Henry Alexander Murray	1893 to 1988	Needs and press (Explorations in Personality, 1938); viscerogenic and psychogenic needs; TAT with Morgan (1935)
Walter Bradford Cannon	1871 to 1945	Coined homeostasis (1926; The Wisdom of the Body, 1932); stomach contractions and hunger with Washburn (1912)
Claude Bernard	1813 to 1878	The milieu intérieur, the constant internal environment
Leon Festinger	1919 to 1989	Cognitive dissonance (1957)
Harry Frederick Harlow	1905 to 1981	Puzzle-solving monkeys and the manipulation drive (1950); Maslow's doctoral adviser
John Dillingham Dodson	1879 to 1955	Yerkes-Dodson law (1908)
Robert Zajonc	1923 to 2008	Drive theory of social facilitation (1965): arousal strengthens the dominant response
Marvin Zuckerman	1928 to 2018	Sensation seeking
Otto Tinklepaugh	1928	Monkeys expecting banana refused lettuce: incentive theory
Julian Bernard Rotter	1916 to 2014	Social learning theory: expectancy and reinforcement value (1954); locus of control (1966)
Richard Lester Solomon	1918 to 1995	Opponent-process theory of acquired motivation (with Corbit, 1974)
John Corbit	1974	Opponent-process theory with Solomon
Kurt Goldstein	1878 to 1965	Coined the term self-actualisation (1939)
Clayton Paul Alderfer	1940 to 2015	ERG theory (1969): existence, relatedness, growth; frustration-regression
David Clarence McClelland	1917 to 1998	Need for achievement (1953), power and affiliation; the Kakinada training studies
Edward L. Deci	born 1942	Self-determination theory with Ryan (1985); cognitive evaluation theory
Richard M. Ryan	born 1953	Self-determination theory with Deci (1985): autonomy, competence, relatedness
William Glasser	1925 to 2013	Reality therapy: five basic needs, including power or achievement
Christiana Morgan	1897 to 1967	Thematic Apperception Test with Murray (1935)

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John William Atkinson	1923 to 2003	Achievement motivation theory (1957): motive to succeed versus motive to avoid failure; preference for intermediate risk
Matina Souretis Horner	born 1939	Fear of success (1972)
Bernard Weiner	born 1935	Attribution theory of achievement: locus, stability, controllability
Daniel Ellis Berlyne	1924 to 1976	Curiosity: novelty, complexity, incongruity (1960)
Robert White	1904 to 2001	Effectance or competence motivation (1959)
Lev Vygotsky	1896 to 1934	Constructivism (a wrong option for reciprocal causation)
Frederick Herzberg	1923 to 2000	Two-factor (motivator-hygiene) theory (1959)
Stanley Schachter	1922 to 1997	Externality hypothesis of obesity (1968 to 1971); two-factor theory of emotion with Singer
Jerome Singer	1962	Two-factor theory of emotion with Schachter
Mark Lepper	1973	Overjustification effect: the magic-marker study with Greene and Nisbett
David Greene	1973	Magic-marker study with Lepper and Nisbett
Richard Nisbett	born 1941	Magic-marker study with Lepper and Greene (1973)
Roger Barker	1903 to 1990	Regression after frustration: toys behind a screen (with Dembo and Lewin, 1941)
Tamara Dembo	1941	Regression study with Barker and Lewin
Leonard Doob	1909 to 2000	Co-author of Frustration and Aggression (1939)
Robert Sears	1908 to 1989	Co-author of Frustration and Aggression (1939)
Leonard Berkowitz	1926 to 2016	Reformulated frustration-aggression hypothesis (1989): negative affect
A. L. Washburn	1912	Swallowed the balloon in Cannon's hunger study
John P. J. Pinel	2021 (Biopsychology, with Barnes)	Positive-incentive view of eating; settling point
Steven J. Barnes	2021 (Biopsychology, with Pinel)	Positive-incentive view of eating; settling point
Albert Hetherington	1940 (with Ranson)	VMH lesions cause hyperphagia and obesity
Stephen Walter Ranson	1880 to 1942	VMH lesions cause hyperphagia and obesity (with Hetherington, 1940)
Bal Krishan Anand	1951 (with Brobeck)	Lateral hypothalamic lesions cause aphagia: the feeding centre
John Brobeck	1951 (with Anand)	Lateral hypothalamic lesions cause aphagia
Eliot Stellar	1919 to 1993	Dual-centre theory of hunger (1954)
Alfred Charles Kinsey	1894 to 1956	Reports on male (1948) and female (1953) sexual behaviour
William Howell Masters	1915 to 2001	Sexual response cycle: excitement, plateau, orgasm, resolution (1966)
Virginia Eshelman Johnson	1925 to 2013	Sexual response cycle with Masters (1966)
Helen Singer Kaplan	1929 to 1995	Triphasic model: desire, excitement, orgasm (1979)
Carl Warden	1931	Columbia obstruction box: maternal drive strongest
Victor Harold Vroom	born 1932	Expectancy (VIE) theory of work motivation (1964)
John Stacy Adams	1963	Equity theory
Carol Dweck	born 1946	Attribution retraining of helpless children (1975)
William Richard Miller	born 1947	Motivational interviewing (with Rollnick, 1991)
Stephen Rollnick	1991 (with Miller)	Motivational interviewing
Paul Ekman	born 1934	Six universal basic emotions (contempt added 1986); Fore study (1971); Facial Action Coding System (1978)
Wallace V. Friesen	1971; 1978	With Ekman: Fore universality study and the Facial Action Coding System
Silvan Tomkins	1962	Affect theory: nine innate affects; the face as the seat of affect, root of the facial feedback hypothesis
Carroll Izard	1977	Differential emotions theory: ten basic emotions, including shame and guilt
Robert Plutchik	1980	Wheel of eight primary emotions and their dyads
James Russell	1980	Circumplex model: emotions on valence and arousal
Alexander Thomas		With Chess: easy, difficult and slow-to-warm-up temperaments (pointer)

Name	When	Examined for
Stella Chess		With Thomas: easy, difficult and slow-to-warm-up temperaments (pointer)
Carl Georg Lange	1834 to 1900	James-Lange theory of emotion (1885)
Philip Bard	1898 to 1977	Cannon-Bard theory; sham rage in decorticate cats
Donald Dutton	1974	Capilano bridge study: misattribution of arousal
Arthur Aron	1974	Capilano bridge study: misattribution of arousal
Dolf Zillmann	1970s	Excitation transfer theory
Richard S. Lazarus	1922 to 2002	Cognitive appraisal theory (primary and secondary appraisal); daily hassles (1981); transactional model of stress (1984)
Fritz Strack	1988	Pen study supporting the facial feedback hypothesis
Leonard Martin	1988	Pen study of facial feedback, with Strack and Stepper
Sabine Stepper	1988	Pen study of facial feedback, with Strack and Martin
Richard Solomon	1918 to 1995	Opponent-process theory of emotion (1974)
James Papez	1883 to 1958	Papez circuit of emotion (1937)
Paul MacLean	1913 to 2007	Named the limbic system (1952), the classic seat of emotion
Heinrich Klüver	1897 to 1979	Klüver-Bucy syndrome after bilateral temporal lobe removal in monkeys (1937, 1939)
Paul Bucy	1904 to 1992	Klüver-Bucy syndrome (1937, 1939)
Joseph LeDoux	born 1949	Low road and high road to the amygdala in fear
Larry Cahill	1994	With McGaugh: propranolol abolishes the emotional enhancement of memory
James McGaugh	1994	With Cahill: the amygdala and noradrenaline in emotional memory
H.M. (Henry Molaison)	operated on 1953; died 2008	Amnesic patient: the hippocampus, not the amygdala, stores facts
Phineas Gage	1823 to 1860	Prefrontal injury (1848) that changed his emotional and social conduct
Antonio Damasio	born 1944	Somatic marker hypothesis (1994)
David T. Lykken	1959	Guilty knowledge test, more accurate than the polygraph
Brad Bushman	2002	Venting experiment against the catharsis hypothesis
Paul Rozin	1987	With Fallon: disgust as protection against contamination
April Fallon	1987	With Rozin: disgust
Ed Diener	1946 to 2021	Subjective well-being, tripartite model (1984): life satisfaction, positive affect, negative affect
Harry Helson	1898 to 1977	Adaptation-level theory (1964)
Philip Brickman	1971	With Campbell: the hedonic treadmill
Donald Campbell	1971	With Brickman: the hedonic treadmill
Samuel Stouffer	1949	Relative deprivation (The American Soldier)
Barbara Fredrickson	1998	Broaden-and-build theory of positive emotions
Fred Bryant	2007	With Veroff: savoring
Joseph Veroff	2007	With Bryant: savoring
Aaron Beck	1967	Cognitive model of depression (1967) and cognitive distortions (a distractor in questions on Fredrickson and learned helplessness)
John B. Watson	1878 to 1958	Three innate emotions: fear, rage and love
Katharine Banham Bridges	1932	Differentiation of infant emotions: distress and delight first
Edward Tronick	1978	Still-face procedure: infants expect emotional give-and-take
James Gross	1998	Process model of emotion regulation: reappraisal is cognitive change, suppression is response modulation
Peter Sifneos	1920 to 2008	Coined alexithymia (1973)
Hans Selye	1907 to 1982	General adaptation syndrome (1936; The Stress of Life, 1956): alarm, resistance, exhaustion; eustress (1974)
Shelley E. Taylor	born 1946	Tend-and-befriend (2000)
Thomas Holmes	1967	With Rahe: Social Readjustment Rating Scale (life events)
Richard Rahe	1967	With Holmes: Social Readjustment Rating Scale (life events)
Allen Kanner	1981	With Lazarus and colleagues: daily hassles and uplifts

Name	When	Examined for
Robert Karasek	1979	Job demand-control model
Herbert Freudenberger	1926 to 1999	Introduced the term burnout (1974)
Christina Maslach	born 1946	Maslach Burnout Inventory (1981): exhaustion, depersonalisation, reduced accomplishment
Susan Folkman	1984	With Lazarus: transactional model of stress
Paul Meehl	1920 to 2003	Diathesis-stress model of schizophrenia (1962)
Joseph Zubin	1977	With Spring: vulnerability-stress model
Bonnie Spring	1977	With Zubin: vulnerability-stress model
Sheldon Cohen	1985	With Wills: direct and stress-buffering effects of social support
Thomas Wills	1985	With Cohen: stress-buffering effect of social support
Robert S. Weiss	1973	Emotional isolation versus social isolation
Suzanne Kobasa	1979	Hardiness: commitment, control, challenge
Aaron Antonovsky	1923 to 1994	Sense of coherence (1979): comprehensibility, manageability, meaningfulness
Michael Scheier	1985	With Carver: dispositional optimism
Charles Carver	1985	With Scheier: dispositional optimism
Meyer Friedman	1910 to 2001	With Rosenman: Type A behaviour pattern (1959); Western Collaborative Group Study
Ray Rosenman	1920 to 2013	With Friedman: Type A behaviour pattern (1959)
Johan Denollet	mid-1990s	Type D personality: negative affectivity plus social inhibition
George Engel	1913 to 1999	Biopsychosocial model (1977)
Robert Ader	1932 to 2011	With Nicholas Cohen: conditioned immunosuppression (1975), the founding study of psychoneuroimmunology
Nicholas Cohen	1975	With Ader: conditioned immunosuppression
Irwin Rosenstock	1950s	Health belief model
Ronald Rogers	1975	Protection motivation theory
Icek Ajzen	1985	Theory of planned behaviour: attitude, subjective norm, perceived behavioural control
Jon Kabat-Zinn	born 1944	Mindfulness-based stress reduction (1979)
Donald Meichenbaum	born 1940	Stress inoculation training (1970s)

Key terms

mental processes, clinical psychology, comparative psychology, evolutionary psychology, nativism, empiricism, rationalism, laws of association, dualism, innate ideas, tabula rasa, associationism, mental chemistry, apperceptive mass, psychophysical parallelism, Bell-Magendie law, psychophysics, mental chronometry, immediate experience, voluntarism, creative synthesis, tridimensional theory of feeling, introspection, Volkerpsychologie, structuralism, stimulus error, context theory of meaning, imageless thought, functionalism, stream of consciousness, reflex arc, mental test, psychological clinic, great-person theory, Zeitgeist, preparadigmatic science, prescriptions, behaviourism, peripheralism, operationism, cognitive map, latent learning, phi phenomenon, insight, transposition, life space, action research, Gestalt therapy, structural model, basic anxiety, phenomenology, Dasein, logotherapy, noogenic neurosis, transpersonal psychology, savouring, cognitive revolution, information-processing approach, connectionism, eclecticism, Indian psychology, theory of opposite wishes, indigenisation, nurturant-task leader, astika, purusha, prakriti, gunas, antahkarana, chitta-vritti-nirodha, nishkama karma, atman, turiya, panchakosha, psychic being, sensation, perception, transduction, doctrine of specific nerve energies, all-or-none law, sensory adaptation, absolute threshold, just noticeable difference, Weber's law, Fechner's law, magnitude estimation, Stevens' power law, method of limits, method of constant

stimuli, signal detection theory, d prime, response bias, receiver operating characteristic, accommodation, fovea, blind spot, scotopic vision, dark adaptation, Purkinje shift, saccades, Mach bands, lateral inhibition, trichromatic theory, opponent-process theory, place theory, frequency theory, volley principle, papillae, umami, gate control theory, phantom limb, semicircular canals, synaesthesia, Charles Bonnet syndrome, Bottom-up processing, top-down processing, unconscious inference, constructivist approach, Bayesian approach, direct perception, affordances, law of proximity, law of closure, Prägnanz, binocular cues, retinal disparity, convergence, monocular cues, texture gradient, Motion parallax, visual cliff, Perceptual constancy, size constancy, size-distance invariance, Emmert's law, illusion, Müller-Lyer illusion, inappropriate constancy scaling, carpentered-world hypothesis, hallucination, autokinetic effect, Pandemonium, prototype, recognition-by-components, perceptual set, field dependence-independence, Subliminal perception, Extrasensory perception, span of attention, cocktail party phenomenon, dichotic listening, filter model, early selection, attenuation model, late selection, feature integration theory, inattentional blindness, change blindness, automatic process, controlled processing, Stroop effect, sustained attention, Consciousness, topographic model, preconscious, unconscious, psychodynamic perspective, hard problem of consciousness, altered state of consciousness, circadian rhythm, suprachiasmatic nucleus, REM sleep, slow-wave sleep, sleep cycle, REM rebound, cataplexy, manifest content, latent content, dream work, collective unconscious, activation-synthesis hypothesis, Hypnosis, suggestibility, neodissociation theory, Concentrative meditation, mindfulness meditation, tolerance, learning, habituation, sensitisation, imprinting, classical conditioning, unconditioned stimulus, conditioned stimulus, extinction, spontaneous recovery, stimulus generalisation, stimulus discrimination, counterconditioning, reciprocal inhibition, systematic desensitisation, flooding, conditioned taste aversion, preparedness, law of effect, operant conditioning, negative reinforcement, secondary reinforcers, token economy, two-factor theory, response cost, shaping, schedule of reinforcement, Premack principle, learned helplessness, biofeedback, insight learning, observational learning, reciprocal determinism, self-efficacy, savings method, multi-store model, sensory memory, chunking, serial position effect, working memory, phonological loop, visuospatial sketchpad, central executive, episodic buffer, levels of processing, self-reference effect, episodic memory, semantic memory, procedural memory, implicit memory, flashbulb memory, tip-of-the-tongue state, schema, misinformation effect, source monitoring, encoding specificity principle, context-dependent memory, state-dependent memory, decay theory, proactive interference, retroactive interference, repression, anterograde amnesia, retrograde amnesia, Korsakoff's syndrome, confabulation, engram, long-term potentiation, Thinking, mental rotation, dual-coding theory, concept, family resemblance, problem space, algorithm, heuristic, Means-end analysis, hill climbing, functional fixedness, mental set, Deductive reasoning, Inductive reasoning, abductive reasoning, belief bias, confirmation bias, bounded rationality, satisficing, conjunction fallacy, Prospect theory, loss aversion, Hindsight bias, Creativity, phoneme, morpheme, pragmatics, prosody, deep structure, language acquisition device, poverty of the stimulus, critical period, holophrase, telegraphic speech, overregularisation, linguistic relativity, inner speech, Aphasia, operational definition, Achievement, aptitude, growth mindset, mental age, intelligence quotient (IQ), Stanford-Binet, ratio IQ, deviation IQ, concurrent validity, culture-fair test, Raven's Progressive Matrices, Factor analysis, positive manifold, general factor (g), primary mental abilities, structure of intellect, Fluid intelligence (Gf), Crystallised intelligence (Gc), multiple intelligences, triarchic theory, successful intelligence, PASS model, emotional intelligence, twin studies, Heritability, reaction range, Flynn effect, stereotype threat, Intellectual disability, adaptive functioning, Global developmental delay, Vineland Social Maturity Scale, Savant syndrome, motivation, need, drive, primary motives, secondary motives, intrinsic motivation, extrinsic motivation, instinct, hormic psychology, homeostasis, drive-reduction theory, optimal arousal theory, Yerkes-

Dodson law, incentive theory, expectancy-value theory, hierarchy of needs, prepotency, deficiency needs, metamotivation, ERG theory, press, need for achievement, triadic reciprocal causation, overjustification effect, self-determination theory, cognitive evaluation theory, flow, frustration, ventromedial hypothalamus, lateral hypothalamus, sexual response cycle, expectancy theory, goal-setting theory, emotion, mood, Affect, James-Lange theory, Cannon-Bard theory, misattribution of arousal, primary appraisal, secondary appraisal, facial feedback hypothesis, sham rage, Papez circuit, limbic system, amygdala, Klüver-Bucy syndrome, pseudobulbar affect, display rules, subjective well-being, broaden-and-build theory, Savoring, self-conscious emotions, cognitive reappraisal, Alexithymia, fight-or-flight response, general adaptation syndrome, eustress, hypothalamic-pituitary-adrenal axis, Social Readjustment Rating Scale, daily hassles, burnout, transactional model of stress, diathesis-stress model, Problem-focused coping, emotion-focused coping, emotional isolation, hardiness, Type A behaviour pattern, Type D personality, biopsychosocial model, Psychoneuroimmunology

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From the past papers

1. Functionalism, an early school of thought in psychology, was most closely associated with which psychologist? (IHBAS 2024)

   (a) Edward Titchener (b) John Watson (c) William James (d) G. Stanley Hall

   Answer: (c) William James. James's *Principles of Psychology* (1890) inspired the functionalist question of what the mind is for, and Dewey and Angell made it a formal school at Chicago. Titchener founded the rival structuralism, Watson founded behaviourism, and Hall, though functionalist in spirit, is examined for institutions such as the first American research laboratory (1883) and the APA (1892).

2. The first person to use the term clinical psychology in print was (RML 2020)

   (a) Erik Erikson (b) Sigmund Freud (c) Lightner Witmer (d) Cattell

   Answer: (c) Lightner Witmer. Witmer opened the first psychological clinic at the University of Pennsylvania in 1896 and used the term in print in his journal *The Psychological Clinic* (1907). Erikson is known for the psychosocial stages, Freud for psychoanalysis, and James McKeen Cattell for coining "mental test".

3. Match List I with List II LIST I (Psychologist) LIST 2 (Theory) A. Dollard and Miller 1. Radical Behaviorism B. Skinner 2. Self-Actualization C. Bandura and Walters 3. Early Social Learning D. Maslow 4. Later Social Learning (IHBAS 2024)

   (a) A-1, B-2, C-3, D-4 (b) A-3, B-1, C-4, D-2 (c) A-2, B-1, C-3, D-4 (d) A-3, B-2, C-4, D-1

   Answer: (b) A-3, B-1, C-4, D-2. Dollard and Miller recast Freud in Hull's terms in 1950 (early social learning); Skinner's system is radical behaviourism; Bandura and Walters gave the later social learning theory (1963); Maslow is self-actualisation. Option (d) tempts only if you forget that Skinner, not Maslow, is the radical behaviourist.

4. Which is the correct order of the last four of Patanjali's eight limbs? (NIMHANS 2026 recall)

   (a) Dharana, pratyahara, dhyana, samadhi (b) Pratyahara, dharana, dhyana, samadhi (c) Pratyahara, dhyana, dharana, samadhi (d) Dhyana, dharana, pratyahara, samadhi

   Answer: (b) Pratyahara, dharana, dhyana, samadhi. The senses are first withdrawn inward (pratyahara), the mind is then fixed on one point (dharana), the fixing becomes an unbroken flow (dhyana) and finally absorption (samadhi). Each wrong option swaps two of these steps.

5. Which statement is INCORRECT? (Amity 2023)

   (a) Cones are most prevalent in the central retina (b) Cones are sensitive to moderate to high levels of light (c) Cones provide only monochromatic information (d) Cones provide excellent acuity

   Answer: (c) Cones provide only monochromatic information. Cones, with three pigments, give colour vision; it is the rods, with one pigment (rhodopsin), that give colourless scotopic vision. The other three statements are true: cones are packed in the fovea, work in moderate to bright light and give excellent acuity.

6. In signal detection theory, the ability to distinguish a stimulus from background noise is: (IHBAS 2025)

   (a) Sensitivity (d′) (b) Criterion (β) (c) ROC curve (d) JND

   Answer: (a) Sensitivity (d'). d′ is the separation between the noise and signal-plus-noise distributions and does not depend on the observer's willingness to say "yes". The criterion (β) is that willingness (response bias), the ROC curve plots hit rate against false alarm rate, and the JND belongs to classical psychophysics.

7. Among monocular cues for depth perception is (JEMAS 2025)

   (a) Convergence (b) Retinal disparity (c) Assimilation (d) Interposition

   Answer: (d) Interposition. An object that partly hides another is seen as nearer, and one eye is enough to see it. Convergence and retinal disparity are the only two binocular cues, and assimilation is not a depth cue at all (it is Piaget's term).

8. Illusions are (JEMAS 2025)

   (a) misinterpretation of stimuli arising from an external object (b) real perceptual object perceived in a distorted way (c) vivid mental images (d) perception without an adequate external stimulus

   Answer: (a) Misinterpretation of stimuli arising from an external object. In an illusion a real stimulus is misread. Option (b) is a sensory distortion such as micropsia, (c) is imagery, and (d) is the definition of a hallucination.

9. The Stroop effect demonstrates interference in (JEMAS 2025)

   (a) Long-term memory retrieval (b) Working memory (c) Selective attention (d) Procedural memory

   Answer: (c) Selective attention. Automatic word reading cannot be filtered out while naming the ink colour, so the task shows a failure of selective attention. The memory options are distractors; no stored memory is being retrieved or rehearsed.

10. According to Freud, the ___ mind includes accessible memories, which can be recalled at will (RML 2020)

    (a) Conscious (b) Preconscious (c) Unconscious (d) Psychic determinism

    Answer: (b) Preconscious. The conscious holds only what is in awareness now, and the unconscious cannot be recalled at will; psychic determinism is a principle (no mental event is accidental), not a level of mind.

11. Systematic desensitization technique is based upon the idea of: (IHBAS 2025)

    (a) Reciprocal inhibition (b) Law of effect (c) Habituation (d) Adaptation to stress

    Answer: (a) Reciprocal inhibition. Wolpe paired each step of an anxiety hierarchy with relaxation, a response incompatible with anxiety, so the anxiety bond weakens. The law of effect is Thorndike's operant principle, habituation underlies exposure and response prevention, and adaptation to stress belongs to stress theory.

12. Among the following schedules of reinforcement, which typically produces the highest rate of response and is most resistant to extinction? (Real PYQ 2024)

    (a) Fixed Ratio (b) Variable Ratio (c) Fixed Interval (d) Variable Interval

    Answer: (b) Variable Ratio. Ratio schedules reward fast responding, and unpredictability removes any reason to pause and makes it hard to tell that reinforcement has stopped, as with slot machines. Fixed ratio shows a pause after each reinforcer, fixed interval a scallop and variable interval only a moderate, steady rate.

13. The famous patient H.M. (Henry Molaison) underwent surgical removal of the medial temporal lobes, including the hippocampus, to treat severe epilepsy. While he retained most of his intellectual abilities and working memory, his primary and most profound deficit was in: (Real PYQ 2024)

    (a) Procedural memory (b) Semantic memory (c) Anterograde amnesia (d) Retrograde amnesia

    Answer: (c) Anterograde amnesia. After the 1953 operation H.M. could not form new declarative memories, although his intelligence, digit span and childhood memories survived. Procedural memory was spared (he learned mirror tracing), old semantic knowledge was intact, and his retrograde amnesia covered only the years before surgery.

14. Which of the following terms refers to difficulty recalling information due to interference from new information? (IHBAS 2024)

    (a) Retroactive interference (b) Proactive interference (c) Decay theory (d) Retrieval failure

    Answer: (a) Retroactive interference. New learning reaching back to disrupt older memories is retroactive (RETRO: new disrupts old); proactive interference is the reverse, old disrupting new. Decay blames the passage of time and retrieval failure a mismatch of cues, not competition from new learning.

15. In which method does the researcher gather data, look for patterns, and then generate a theory or hypothesis? (RML 2023)

    (a) Deductive reasoning (b) Hypothetico-deductive method (c) Inductive reasoning (d) Abductive reasoning

    Answer: (c) Inductive reasoning. Induction runs from specific observations to a probable generalisation, so data come first and theory last, as in most qualitative research. Deduction and the hypothetico-deductive method run the other way, from a theory to a tested prediction; abduction infers the best explanation of a surprising fact.

16. What is the primary difference between Piaget and Vygotsky's theories of language development? (RML 2021)

    (a) Piaget argued thought depends on language while Vygotsky argued language depends on thought (b) Vygotsky argued thought depends on language while Piaget argued language depends on thought (c) Vygotsky was concerned with early language development while Piaget was concerned with later language development. (d) None of the above

    Answer: (b). For Piaget, language expresses concepts already built through action; for Vygotsky, thought and speech merge at about two, after which thought becomes verbal and depends on language. Option (a) swaps the names, and (c) is false because both studied the same preschool years.

17. The term IQ was invented by (JEMAS 2025)

    (a) Lewis Termon (sic) (b) Gordon Allport (c) William Stern (d) John Watson

    Answer: (c) William Stern. Stern proposed the intelligence quotient, mental age divided by chronological age, in 1912. Terman is the tempting wrong answer: he multiplied the quotient by 100 and popularised it in the 1916 Stanford-Binet, but did not coin it; Allport (traits) and Watson (behaviourism) never worked on the IQ.

18. A 10-year-old child from a rural area, with no formal schooling, is brought for an intellectual assessment due to academic struggles. The child is illiterate and has had minimal exposure to formal testing materials. Which of the following tests would be the MOST appropriate to use to minimize cultural and educational bias? (Real PYQ 2024)

    (a) Bhatia's Battery of Performance Tests of Intelligence (b) Wechsler Intelligence Scale for Children (WISC-IV) (c) Stanford-Binet Intelligence Scale (5th ed.) (d) Malin's Intelligence Scale for Indian Children (MISIC)

    Answer: (a) Bhatia's Battery of Performance Tests of Intelligence. Its five performance subtests need no reading, and it has Indian rural norms. The WISC-IV and SB5 rest on school-built verbal knowledge and foreign norms; MISIC is the tempting Indian option, but half of it is verbal, adapted from the WISC.

19. The 'satiety centre' of the brain is located in the: (RML 2024)

    (a) Lateral hypothalamus (b) Ventromedial hippocampus (c) Lateral thalamus (d) Ventromedial hypothalamus

    Answer: (d) Ventromedial hypothalamus. Bilateral VMH lesions make rats overeat and grow obese (Hetherington and Ranson, 1940), so Stellar's dual-centre theory (1954) called the VMH the satiety centre. The lateral hypothalamus is the opposite, the feeding centre, whose destruction causes aphagia (Anand and Brobeck, 1951); "ventromedial hippocampus" and "lateral thalamus" are invented by changing one word.

20. According to Herzberg's Two-Factor Theory, which of the following factors are considered motivators that lead to job satisfaction? (RML 2023)

    (a) Salary and benefits (b) Job security and work conditions (c) Recognition and opportunities for advancement (d) Company policies and interpersonal relationships

    Answer: (c) Recognition and opportunities for advancement. Herzberg's motivators lie in the content of the work: achievement, recognition, the work itself, responsibility, advancement and growth. Salary, benefits, job security, working conditions, company policies and relationships are hygiene factors, which prevent dissatisfaction when good but do not by themselves motivate.

21. The seat of emotion is to be found in _____. (RML 2024)

    (a) Reticular formation (b) Hind brain (c) Limbic system (d) Forebrain

    Answer: (c) Limbic system. MacLean's limbic system (1952), built on Papez's circuit and including the amygdala, is the classic seat of emotion. The reticular formation governs arousal and wakefulness, the hindbrain paces heartbeat and breathing, and "forebrain" is too broad, although the limbic system lies within it.

22. 'Psychological disorders develop when a biological predisposition is set off by a stressful situation', is proposed by (JEMAS 2025)

    (a) Humanistics existential models. (b) Socio-cultural model. (c) Diathesis-stress model. (d) Cognitive model.

    Answer: (c) Diathesis-stress model. A pre-existing vulnerability (diathesis) set off by a stressful trigger is the model's definition (Meehl, 1962; Zubin and Spring, 1977). Humanistic-existential models blame blocked growth, sociocultural models social forces and cognitive models faulty thinking; none is defined by predisposition plus stress.

High-yield sheet

The facts this chapter's questions ask most, most-asked first. Last read before the exam.

1 **Systematic desensitisation** (Joseph Wolpe, *Psychotherapy by Reciprocal Inhibition*, 1958) rests on **reciprocal inhibition**: relaxation, a response incompatible with anxiety, is paired step by step with a graded anxiety hierarchy. ASKED 17 TIMES

2 **Working memory** (Alan Baddeley and Graham Hitch, 1974) has four components: the central executive (an attentional controller that stores nothing, loaded by random number generation), the phonological loop (auditory-verbal information), the visuospatial sketchpad (visual and spatial information, including mental rotation) and the episodic buffer, added in 2000, which binds information into coherent episodes. ASKED 11 TIMES

3 The **token economy** is operant conditioning with tokens as secondary reinforcers, developed by Teodoro Ayllon and Nathan Azrin (*The Token Economy*, 1968); it is the listed therapy not derived from classical conditioning. ASKED 10 TIMES

4 **Abraham Maslow**'s hierarchy of needs (1943, 1954), bottom to top: physiological, safety, belongingness and love (social), esteem, self-actualisation. Physiological and safety needs are the lower needs; ego is not a level. ASKED 10 TIMES

5 **Korsakoff's syndrome** follows thiamine (vitamin B1) deficiency in chronic alcoholism, often after Wernicke's encephalopathy (confusion, ataxia, nystagmus); it damages the mammillary bodies and mediodorsal thalamus and causes dense anterograde amnesia, extensive retrograde amnesia and confabulation, sincere false stories that fill memory gaps. ASKED 8 TIMES

6 The **empty chair** technique belongs to **Gestalt therapy** (Fritz Perls, with Laura Perls and Paul Goodman), an experiential here-and-now therapy that is not the Gestalt psychology of Wertheimer, Kohler and Koffka. ASKED 7 TIMES

7 **Variable ratio** reinforcement gives the highest, steadiest rate and the greatest resistance to extinction; fixed ratio gives high rates with a post-reinforcement pause, fixed interval the scallop, variable interval moderate steady rates, and continuous reinforcement the fastest learning and fastest extinction. ASKED 7 TIMES

8 **Extinction** is the fading of a conditioned response when the CS is repeatedly presented without the UCS, or of an operant response that is no longer reinforced; a parent who ignores tantrums is using extinction, not operant escape. ASKED 7 TIMES

9 Reinforcement always increases behaviour and punishment decreases it; positive adds and negative removes a stimulus, so switching off a shock is **negative reinforcement**, while time-out and response cost are **negative punishment**. ASKED 7 TIMES

10 **H.M.** (Henry Molaison) had both medial temporal lobes, including the hippocampus and amygdala, removed by William Beecher Scoville in 1953; Scoville and Brenda Milner (1957) reported a profound anterograde amnesia for declarative memory, with normal short-term memory, limited retrograde amnesia and spared procedural learning (mirror tracing). ASKED 7 TIMES

11 **Raymond Cattell** (with John Horn) split g into **fluid intelligence** (Gf), reasoning with novel problems independently of acquired knowledge and measured best by Raven's Progressive Matrices, and **crystallised intelligence** (Gc), the repertoire of information, cognitive skills and strategies, measured by vocabulary and information. ASKED 7 TIMES

12 **The limbic system** is the classic "seat of emotion": Paul MacLean named it in 1952, enlarging James Papez's 1937 circuit, and it regulates emotion, motivation and memory; hunger and thirst belong to the hypothalamus. ASKED 7 TIMES

13 **The amygdala** is the structure for fear and aggression: it detects threat, drives fear conditioning and gives emotional events their extra memorability, while the hippocampus stores the facts. ASKED 7 TIMES

14 The **hippocampus** is the structure damaged in anterograde amnesia and is critical for consolidating new declarative memories; it is not the store of remote memories. ASKED 6 TIMES

15 **Narendra Nath Sengupta** set up India's first psychology laboratory (1915) and first Department of Psychology (1916) at the University of Calcutta and is the father of modern psychology in India; keys naming Bose or Uma Chaudhuri for the first laboratory are wrong. ASKED 5 TIMES

16 **Humanistic psychology** (Maslow, Rogers) is the **third force**, after psychoanalysis (first) and behaviourism (second); **transpersonal psychology** is the fourth force. ASKED 5 TIMES

17 The **Stroop effect** (John Ridley Stroop, 1935): naming the ink colour of a conflicting colour word (RED printed in green) is slowed because automatic word reading interferes; it shows a limit of selective attention (interference in reaction time), not a memory effect. ASKED 5 TIMES

18 **Proactive interference**: earlier learning disrupts newer learning (dialling your old phone number); **retroactive interference**: newer learning disrupts older memories. RML 2024 keys English grammar intruding on newly learned Spanish as retroactive, but it is proactive. ASKED 5 TIMES

19 **Inductive reasoning** moves from specific observations to probable generalisations (gather data, find patterns, build theory: the logic of theory development and qualitative research); **deductive reasoning** starts from a theory and tests its predictions against observations. ASKED 5 TIMES

20 **Graham Wallas** (*The Art of Thought*, 1926): preparation, incubation, illumination, verification; incubation (setting the problem aside) is second and illumination (the insight) third. ASKED 5 TIMES

21 **Functional fixedness** (Karl Duncker's candle problem, 1945) is seeing an object only in its usual use; **mental set** (Abraham Luchins's water jars, 1942) is reusing a strategy that worked before, and was keyed as the most basic and common obstacle to problem solving. ASKED 5 TIMES

22 **Henry Murray**'s need-press theory (1938) underlies the **Thematic Apperception Test** (Morgan and Murray, 1935), a projective test whose stories reveal the teller's needs, conflicts and motives. ASKED 5 TIMES

23 **Wilhelm Wundt** opened the first psychology laboratory at **Leipzig in 1879** (not Vienna, Paris or Cambridge); Galton's London laboratory (1884) was anthropometric. ASKED 4 TIMES

24 Patanjali's **eight limbs**, in order: yama, niyama, asana, pranayama, pratyahara (withdrawal of the senses), dharana (concentration on one point), dhyana (unbroken meditation), samadhi (absorption); karma and jnana are not limbs. ASKED 4 TIMES

25 **Tamas** is the guna of inertia, lethargy, darkness, dullness and ignorance; a person who has stopped eating and cannot get herself to act shows tamas alone. ASKED 4 TIMES

26 **Every sense except smell relays through the thalamus** on its way to the cortex (vision via the lateral geniculate nucleus, hearing via the medial geniculate nucleus); the olfactory nerve (cranial nerve I) bypasses it. ASKED 4 TIMES

27 The **suprachiasmatic nucleus** (SCN) of the hypothalamus is the master circadian pacemaker, entrained by light through the retinohypothalamic tract; SCN lesions abolish circadian rhythms, not total sleep. ASKED 4 TIMES

28 **REM (paradoxical) sleep**: a desynchronised, fast, low-voltage EEG like waking, rapid eye movements, atonia of the core muscles, genital arousal and vivid dreams; EEG synchrony and delta waves belong to N3, spindles and K-complexes to N2. ASKED 4 TIMES

29 **Carl Jung** split the unconscious into a personal (individual) unconscious and the **collective unconscious**, an inherited layer of archetypes shared by all humanity, which he explained by the similar brain structure of all human races. ASKED 4 TIMES

30 **William Stern** coined the intelligence quotient in 1912; **Lewis Terman**'s Stanford-Binet (1916) multiplied it by 100, giving the ratio IQ = (MA ÷ CA) × 100, so a mental age of 8 at age 10 gives an IQ of 80. ASKED 4 TIMES

CHAPTER 2

Psychological Assessment

IN THIS CHAPTER

2.1 ○ Psychological Testing: Nature, History, Types and Ethics: Define a psychological test and assessment, trace testing from Chinese civil-service examinations to the MMPI, classify tests (objective versus subjective, speed versus power, maximum versus typical performance), list the characteristics of a good test, recognise the errors of clinical judgement and apply the ethics of test use.

2.2 ○ Building a Test: Items, Item Analysis, IRT and Response Bias: Write and format test items, run a classical item analysis (difficulty, discrimination, item-total correlations, distractors, ceiling and floor effects), explain item response theory and the item characteristic curve, and recognise response sets and rater errors such as acquiescence and halo.

2.3 ●●● Reliability: Consistency and the Standard Error of Measurement: Explain how classical test theory defines reliability, choose and compute the right estimate (test-retest, parallel forms, split-half with Spearman-Brown, Rulon, Flanagan, KR-20, KR-21, alpha, kappa) and put a standard-error band around one person's score.

2.4 ●● Validity and the Accuracy of Test Decisions: Define validity in its classical and unified senses, tell face, content, criterion-related (concurrent and predictive) and construct validity apart by their evidence, link validity to reliability through the validity ceiling, and judge test decisions with sensitivity, specificity, predictive values, base rates and ROC curves.

2.5 ●● Norms, Standard Scores and Interpreting a Score: Interpret any test score by placing it against norms: tell norm- from criterion-referenced scores, judge a norm sample, and convert between percentiles, z, T, stanine, sten, scaled scores and deviation IQs on the normal curve.

2.6 ●● Intelligence Tests: Binet, Wechsler, Non-verbal and Indian Scales: Identify and compare the major intelligence tests (the Binet scales, the Wechsler family, the Kaufman and Das-Naglieri batteries, non-verbal and culture-fair tests and the Indian scales), say what an IQ test can and cannot tell you, and choose the right test for a given person.

2.7 ○ Assessing Development, Adaptive Behaviour and Disability: Explain how development, social maturity and adaptive behaviour are measured (DQ, SQ, Gesell, Bayley, Denver, VSMS, Vineland and the Indian scales), how intellectual disability and global developmental delay are assessed, and how intellectual disability, specific learning disability and autism are certified in India.

2.8 ○ Aptitude, Achievement, Creativity, Interest and Attitude Measures: Tell achievement, aptitude and intelligence tests apart, name the main aptitude batteries, creativity tests and interest and value inventories, and explain how Thurstone, Likert, Guttman, Bogardus, semantic differential and implicit attitude measures are built and scored.

2.9 ●● Objective Personality Inventories: Distinguish objective inventories from projective tests, name Cattell's L-, Q- and T-data, match inventories to their construction strategy, interpret the MMPI's validity and clinical scales and its revisions, and identify the authors, items, scales and scores of the 16PF, EPQ, NEO-PI-R, MBTI, EPPS, CPI, MCMI, STAI, Rep test and Q-sort.

2.10 ●●● Projective Techniques: Explain the projective hypothesis, sort projective techniques by Lindzey's five categories, describe how the Rorschach and TAT are given, scored and interpreted, recognise the completion, expressive and choice techniques with their authors, and weigh the evidence on their reliability and validity.

2.11 ●● Neuropsychological Assessment and Clinical Rating Scales: Explain what neuropsychological assessment measures and how, recognise the main batteries and the memory, visual-motor, executive, attention and language tests and bedside screens, and match each clinical rating scale to its author, format, rater and purpose.

Weightage. 336 of the 3180 questions mapped in this book (10.6%) test this chapter, 136 of them from real past papers. 2 of its 11 sections are High yield.

London, 1884. Among the displays on diet, dress and sanitation at the International Health Exhibition in South Kensington stood a long, narrow room unlike anything else in the show. For threepence a visitor could walk along a counter of instruments while attendants measured height, arm span, weight, breathing power, strength of squeeze, keenness of sight and hearing, colour sense and speed of reaction. The figures went onto a card: the visitor kept one copy and the laboratory kept the other. By the time the exhibition closed, 9,337 people had passed through **Francis Galton**'s Anthropometric Laboratory.

Galton was not running a sideshow. He had argued in *Hereditary Genius* (1869) that ability is inherited, and he believed that an able mind showed itself in keen senses and quick reactions. Measure the senses accurately enough, in enough people, and you would be measuring intelligence. His data were superb: taken the same way for every visitor and from thousands of people, they showed how human traits spread around an average, and to analyse them Galton worked out the idea of correlation. At the University of Pennsylvania, **James McKeen Cattell** built the same kinds of measure into a battery and in 1890 gave it a name: the mental test.

Then, in 1901, Cattell's student **Clark Wissler** asked the obvious question. Did the tests predict anything that mattered? At Columbia University they barely related to students' grades. Careful, precise measurement had captured the wrong thing. That gap, between measuring something consistently and measuring what matters, runs through this whole chapter, and at its end we return to Galton's counter of instruments.

2.1 Psychological Testing: Nature, History, Types and Ethics

LOWER YIELD · 11 QS

Later sections each study one kind of test or one property of tests. This one builds their frame: what a test is, where testing came from, how tests are sorted and judged, how clinicians reason from scores, and testing ethics.

What a test is ○

Anne Anastasi (1908 to 2001), author of the standard text *Psychological Testing* (1954), defined a **psychological test** as "essentially an objective and standardized measure of a sample of behavior". *Objective*: the score does not depend on who gives or scores it. *Standardised*: everyone meets the same items, instructions, time and scoring. *Sample*: the test observes a small, chosen part of behaviour and infers the rest, as a doctor judges your blood from one drop. **Lee J. Cronbach** (1916 to 2001) stressed comparison: a test is "a systematic procedure for comparing the behavior of two or more persons".

Measurement assigns numbers to attributes by rules (Ravi scored 42); **evaluation** judges the number against a standard (42 is a pass); **testing** obtains a behaviour sample with a standardised instrument; and **assessment**, the widest, answers a referral question by integrating tests with interview, observation, history and records. A test yields a score; an assessment yields a reasoned conclusion about a person.

The **psychometric approach** measures individual differences with standardised tests, and its logic of fixed items, norms, reliability and validity serves intelligence, symptoms of psychopathology and neurological or cognitive deficits alike. Levels of measurement (nominal, ordinal, interval, ratio) are taught in Chapter 9.

From Chinese examinations to the MMPI ○

Testing is older than psychology: China examined candidates for public office under the Han dynasty (206 BC to AD 220), and under the T'ang dynasty written civil-service examinations became the rule. Modern testing has two nineteenth-century roots, one clinical: the French psychiatrist **Jean-Étienne Esquirol** (1772 to 1840) separated mental retardation from mental illness, grading retardation by the person's use of language, and **Édouard Séguin** (1812 to 1880), who trained retarded children through sensory and motor exercises, left us the **form board**, a performance task of fitting shapes into matching holes.

The other root is individual differences. **Francis Galton** (1822 to 1911) argued in *Hereditary Genius* (1869) that ability is inherited and, believing that all knowledge enters through the senses, measured sensory discrimination and reaction time in his London **Anthropometric Laboratory** of 1884 (the scene that opened this chapter). His idea of correlation became, in the hands of **Karl Pearson** (1857 to 1936), the product-moment correlation coefficient. Galton is called the father of mental testing and of psychometrics. **James McKeen Cattell** (1860 to 1944), who had worked with Wundt and Galton, coined the term **mental test** in his 1890 paper "Mental Tests and Measurements"; his tests, too, were sensory and motor. In 1901 **Clark Wissler** (1870 to 1947) found that they barely correlated with Columbia students' grades. Measuring the senses precisely was not measuring intelligence.

The breakthrough came in Paris, where **Alfred Binet** (1857 to 1911) and **Théodore Simon** tested judgement, comprehension and reasoning instead (Table 2.1). From the 1960s, the period of accountability, American courts heard challenges to tests as biased against minorities and women. Indian tests (Kamat 1934, Bhatia 1955, Malin 1969, the NIMHANS and PGIMER batteries) appear in Sections 2.6, 2.7 and 2.11.

Table 2.1 Milestones of the testing century.

Year	Who	Milestone
1905	Binet and Simon	Binet-Simon scale, the first practical intelligence test
1912	**William Stern**	Intelligence quotient: mental age divided by chronological age
1916	**Lewis Terman**	Stanford-Binet; IQ = MA/CA × 100
1917	**Robert Yerkes**, using **Arthur Otis**'s group items	Army Alpha (literate recruits) and Army Beta (non-verbal) group tests
1917	**Robert S. Woodworth**	Personal Data Sheet, the first personality inventory
1921	**Hermann Rorschach**	Inkblot test
1927	**Edward K. Strong**	Strong Vocational Interest Blank
1928 to 1929	**L. L. Thurstone**	Attitude scaling; later multiple factor analysis
1935	**Christiana Morgan** and **Henry Murray**	Thematic Apperception Test (TAT)
1938	**Oscar Buros**	First *Mental Measurements Yearbook* of test reviews

Year	Who	Milestone
1939	**David Wechsler**	Wechsler-Bellevue scale for adults
1943	**Starke Hathaway** and **J. Charnley McKinley**	Minnesota Multiphasic Personality Inventory (MMPI)

! EXAM TRAP

"Mental tests" was coined by Cattell (1890), not Binet (first practical intelligence test, 1905), Terman (Stanford-Binet, 1916) or Wechsler (1939). Galton is the father of mental testing; Stern proposed the intelligence quotient and Terman multiplied it by 100. Woodworth's data sheet, built in 1917 to 1918, is dated 1919 or 1920 in some books.

Kinds of tests ●●

Four ways of sorting tests matter most; Table 2.2 gives the rest.

Scoring. On an **objective test** a fixed key leaves no room for the scorer's judgement, so any two scorers agree: multiple-choice items, rating scales and self-report inventories such as the MMPI are objective. On a **subjective test** (an essay, an interview rating, a projective test such as the Rorschach) the scorer judges each free response. "Projective" names a technique of ambiguous stimuli (Section 2.10); "introjective" is no kind of test (introjection is a defence mechanism, Chapter 4).

Time. A **speed test** has easy items of similar difficulty under a time limit so strict that few finish; the score is how many are done, as in a clerical checking test. A **power test** gives generous time, or none, but its items climb in difficulty until some are too hard for anyone; the score reflects the hardest level mastered (Figure 2.1). Most real tests are **speeded tests**, a blend of both.

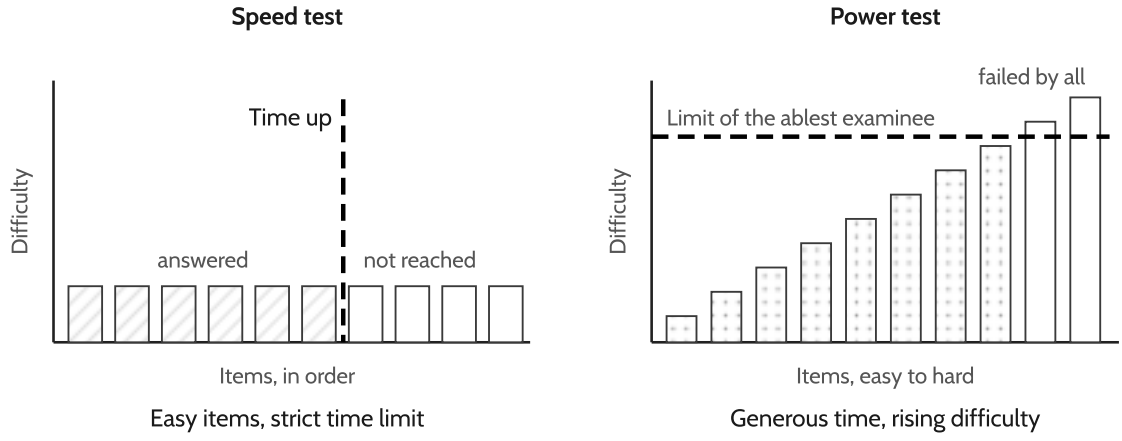


Figure 2.1 Speed versus power. Left: easy items, and time runs out first (white items are never reached). Right: time for everything, but difficulty rises until the white items defeat everyone.

What is asked. Cronbach, in *Essentials of Psychological Testing* (1949), separated tests of **maximum performance**, which ask you to do your best under standard conditions (intelligence, aptitude and achievement tests: the **ability tests**), from tests of **typical performance**, which

ask how you usually feel or act (personality, interest and attitude measures). An IQ test is an ability test.

Purpose. An **intelligence test** measures general mental ability; an **aptitude test**, the potential to learn a specific skill; an **achievement test**, what has already been learned; a **personality test**, traits, emotions, attitudes and motives; an interest inventory, liking for activities and occupations; a **neuropsychological test**, brain-behaviour functions in suspected brain damage; and a diagnostic test, specific weaknesses such as reading errors.

Table 2.2 Ways of classifying tests.

Basis	Types	The difference
Administration	Individual; group	One person, trained examiner (WAIS); many at once (Army Alpha)
Item content	Verbal; non-verbal; non-language	Words in items; pictures, words only in instructions (Raven's); gesture only (Army Beta)
Medium	Paper-and-pencil; performance	Written answers; handling blocks or form boards
Construction	Standardised; teacher-made	Fixed procedures and norms; classroom use, no norms
Reference	Norm-referenced; criterion-referenced	Against other people; against a mastery standard (Section 2.5)
Culture	Culture-fair	Minimal language and schooling (Section 2.6)
Delivery	Computerised; computer-adaptive	On screen; items chosen by earlier answers (Section 2.2)

> **! EXAM TRAP**
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> Speed versus power concerns time limit and item difficulty only, not reaction time and not individual versus group administration; and it is power items that rise in difficulty. "Maximum performance" means an **ability test**: speed and power tests are ability tests differing in timing, and a personality inventory measures typical performance.

Characteristics of a good test ●●

A good test has five characteristics:

- **Objectivity**: freedom from the examiner's judgement in administration, scoring and interpretation, so that different examiners agree.
- **Reliability**: consistency of scores over occasions, forms and scorers (Section 2.3). Consistency is a criterion of good measurement; high cost, low reliability and subjectivity mark poor measurement.
- **Validity**: measuring what the test claims to measure (Section 2.4). Reliability is necessary but not sufficient: head size can be measured consistently, yet it is a worthless intelligence test.
- **Norms**: a representative sample's performance, against which a raw score is read (Section 2.5).
- **Practicability** (usability): reasonable time, length, cost and ease of scoring.

Standardisation means fixing uniform procedures for administration and scoring (the same instructions, materials, time limits and key for everyone) and establishing norms, so that

scores are comparable across people and occasions. It is a precondition of validity, not a guarantee of it; it does not remove cultural bias; and it has nothing to do with computer administration.

> **! EXAM TRAP**
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> RML 2024 keyed **objectivity** as the most important characteristic of a psychological test, following Indian texts such as A. K. Singh's that list it first: without it the score reflects the examiner. Most textbooks, and the 2014 *Standards* ("the most fundamental consideration"), name **validity**, since an objective test of the wrong thing is useless. On RML papers expect objectivity; elsewhere choose validity. Subjectivity and rationality are not test characteristics.

Building, using and misusing tests

A test is built in the eight steps of Figure 2.2, which Sections 2.2 to 2.5 take in turn (A. K. Singh counts seven, folding item analysis into the try-out).

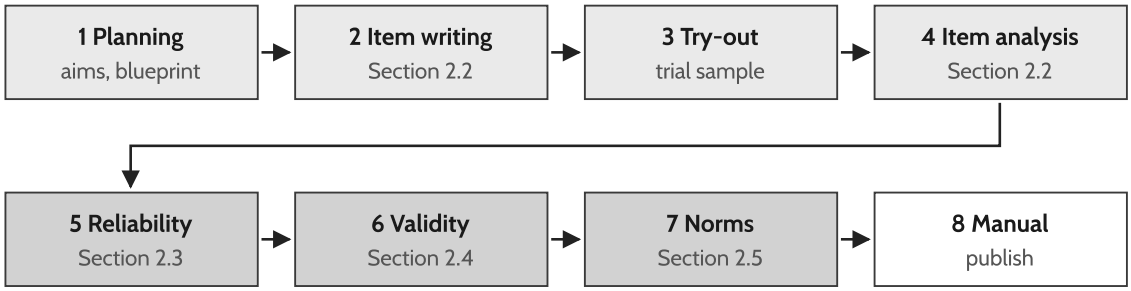


Figure 2.2 The steps of test construction. The top row builds and selects the items; the bottom row checks the scores, and the manual reports it all.

Tests serve selection (choosing applicants), classification and placement (assigning people to categories or programmes), diagnosis and treatment planning, self-knowledge and guidance, programme evaluation and research. Their limitations: they can invade privacy; a label such as "retarded" or "gifted" may stick after the person has changed; they sample only limited aspects of behaviour; test anxiety depresses some scores; and fixed keys penalise creative examinees whose unusual answers earn no credit.

Clinical judgement and its errors ○

Clinicians treat assessment information as samples, correlates or signs. A **sample** directly resembles the behaviour of interest: how a client tackles a frustrating puzzle samples how she meets frustration. A **correlate** has a statistically demonstrated link with something outside the test, as a high depression-scale score goes with clinicians' ratings of depression. A **sign** is read as an indicator of a hidden state, as when a Rorschach response is taken to signal disordered thinking.

Interpretation can be **nomothetic**, placing the person against norms for everyone (her anxiety is at the 95th percentile), or **idiographic**, tracing what is unique to her (what this worry means in her life); good assessment uses both. In *Clinical versus Statistical Prediction* (1954)

Paul E. Meehl (1920 to 2003) compared **clinical prediction**, combining the data in the clinician's head, with **actuarial prediction**, combining them by a formula derived from past cases. The formula usually equalled or beat the clinician, and later reviews agree.

Judgement also errs systematically. **Confirmatory bias** covers ignoring information that does not support one's hypothesis, ignoring information that does not fit one's stereotypes, and reading ambiguous information as support (confirmation bias in general: Chapter 1). **Illusory correlation** is perceiving a relation that is not there (Key study). **Overconfidence** grows with more data even when accuracy does not, and **neglect of base rates** over-diagnoses rare conditions (Section 2.4). The **Barnum effect** is accepting vague, universally true descriptions as uniquely accurate: in 1949 **Bertram Forer** (1914 to 2000) tested his students, then gave each the same "personal" sketch, largely from an astrology book ("You have a great need for other people to like and admire you"); they rated it 4.26 out of 5 for accuracy. Meehl named the effect in 1956 after the showman P. T. Barnum.

KEY STUDY: ILLUSORY CORRELATION (CHAPMAN AND CHAPMAN, 1967 AND 1969)

Clinicians believed Draw-a-Person signs, such as large eyes meaning suspiciousness, that research did not support. **Loren Chapman** and **Jean Chapman** showed naive students drawings, each paired at random with a symptom statement. The students "found" the very relations clinicians reported, though none existed, because the pairs were already linked in their minds (eyes go with suspicion). In 1969 they found the same for Rorschach signs: popular invalid signs were seen and valid ones missed.

The safeguards are **multimethod assessment** (tests, interview, observation, records), multiple informants (client, family, teachers), actuarial aids and a deliberate search for disconfirming evidence. Assessment ends in a written report (referral question, background, tests, behaviour, results, interpretation, recommendations) and feedback the client can understand. Interviewing, the mental status examination and behavioural assessment are in Chapter 7.

! EXAM TRAP

All three statements describe confirmatory bias: answer "All of the above". Assessment information is conceived as **samples, correlates or signs**; the other triads offered (factual, hypothetical or imaginary; predictions, estimates or suggestions; examples, prototypes or categories) are invented.

Ethics and responsible test use

Tests can harm as well as help. The rules are the *Standards for Educational and Psychological Testing* of the American Educational Research Association, American Psychological Association and National Council on Measurement in Education (AERA, APA and NCME; 2014 edition), Section 9 (Assessment) of the APA Ethics Code and, in India, the code of the Rehabilitation Council of India (RCI), where clinical psychologists register (ethics and law in full: Chapter 7). The main duties:

- **Competence.** Use only tests you are trained for. Publishers sell by **test-user qualification levels**: Level A (achievement and proficiency tests) needs only the manual; Level B (group intelligence, aptitude and interest tests) needs training in test construction and

statistics; Level C (individual intelligence tests, clinical personality inventories, projective techniques) needs an advanced degree and supervised experience.

- **Informed consent.** Explain purpose, nature, fees, who sees results and the limits of confidentiality, unless law or a court orders the testing.
- **Confidentiality and release of test data.** Results go only to the client, those the client authorises, or as law requires; raw data may be withheld to prevent harm or misuse.
- **Test security.** Items, manuals and protocols are kept from the public; a leaked test measures nothing.
- **Current editions and relevant norms.** Obsolete tests and outdated or foreign norms mislead.
- **Fairness.** A test is **biased** when its scores mean different things for different groups (Section 2.4).
- **Disability. Accommodations** (Braille or large print, a sign-language interpreter, extra time, a scribe) remove barriers unrelated to what is measured; the report records them.
- **Translation.** A translated test is **back-translated** into the original language by an independent translator and the versions compared; reliability, validity and norms must then be re-established.
- **Explaining results and labelling.** Clients are owed an understandable explanation; labels are used sparingly, because they stigmatise and can fulfil themselves.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Anastasi called a test an objective, standardised measure of a _____ of behaviour.
2. The term "mental tests" was coined in 1890 by _____.
3. Easy items under a time limit so strict that few finish make a _____ test.
4. Intelligence, aptitude and achievement tests measure _____ performance.
5. Uniform administration and scoring procedures plus norms define _____.
6. Clinicians treat assessment information as samples, correlates or _____.
7. Accepting a vague, universal description as uniquely one's own is the _____ effect.

Answers: (1) sample, (2) James McKeen Cattell, (3) speed, (4) maximum, (5) standardisation, (6) signs, (7) Barnum.

2.2 Building a Test: Items, Item Analysis, IRT and Response Bias

LOWER YIELD · 11 QS

A test begins as a pile of draft items, and most will not survive. The constructor writes far more than needed, tries them out and keeps those of the right difficulty that separate people high and low on the trait. Then come the biases that test-takers and raters add.

Writing items

An **item** is the smallest scorable unit of a test, a question or task with a scoring rule. A good item measures only its trait, is unambiguous, gives no clues, is of suitable difficulty and discriminates.

Essay items (free-response items) tap organisation and evaluation but are scored slowly and subjectively. **Objective items**, once called *new-type* items, are scored alike by anyone. In **supply type** items the examinee produces the answer (short answer, completion); in **selection type** items he chooses it: true-false or alternative response (a 50 percent guessing chance), matching (more responses than stems, to block elimination), rearrangement (put steps in order) and the **multiple-choice item**, with a stem (the question), a key (the correct option) and **distractors**, wrong options that attract those who do not know.

The writing rules make the trait alone decide the answer: one idea per item ("I enjoy parties and sports" defeats someone who likes only one), clear wording, no double negatives, no clues such as a longer key or **specific determiners** (*always* and *never* suggest false), plausible distractors and, on personality scales, **balanced keying** (half the items keyed "agree", half "disagree").

Item formats, scoring and guessing

Psychological tests use the wider formats of Table 2.3.

Table 2.3 Item formats in psychological tests.

Format	The respondent
Dichotomous	Chooses between two answers (true or false)
Polytomous	Chooses among three or more
Likert format	Rates agreement, usually on five points (Section 2.8)
Category format	Rates on numbered categories, often 1 to 10
Visual analogue scale	Marks a point on an unbroken line
Adjective checklist	Ticks the adjectives that apply
Forced choice	Picks between options matched for desirability
Q-sort	Sorts statements from most to least like me (Section 2.9)

Answers are scored with a **stencil key** (overlay key: a card with holes at the correct answers, laid over the sheet) or a **strip key** (the answers in a column laid beside the test's). A **correction for guessing** removes lucky hits: S = R - W / (n - 1), with R right, W wrong (omissions excluded) and n options. A blind guesser on four options gets one right per three wrong, so 60 right and 30 wrong give S = 60 - 10 = 50. The formula assumes, wrongly, that every error is a blind guess.

Item difficulty, ceiling and floor ○

Item analysis uses tryout data to decide which items to keep, revise or discard, judging each by its difficulty, its discrimination and its distractors.

The **item difficulty index** (p) is the proportion passing the item (120 of 200 gives 0.60), so a high p means an easy item. An item everyone passes or everyone fails tells no one apart. The

variance of a pass-fail item, p × q, peaks at p = 0.50, which gives maximum variance and discrimination. Guessing raises the target: for multiple choice the optimum lies midway between chance and 1.00, about 0.62 for four options, 0.75 for true-false (texts aiming at maximum reliability give 0.74 and 0.85). Most tests keep p between 0.30 and 0.70.

A **ceiling effect** occurs when a test is too easy: most respondents score near the maximum, so it cannot discriminate among high performers or register further gains; scores are negatively skewed (Figure 2.3). A **floor effect** is the reverse: the test is too difficult, scores pile up near the minimum (positively skewed) and low performers cannot be told apart. Both shrink the spread of scores, lowering reliability and validity.

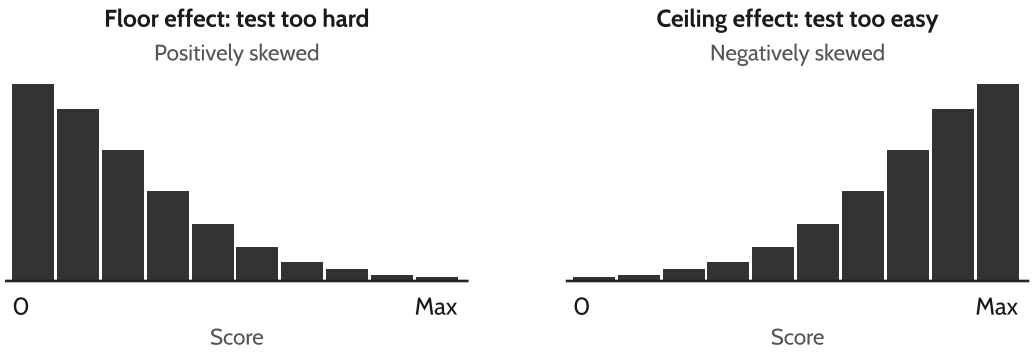


Figure 2.3 Floor and ceiling effects: scores pile up at the minimum (test too hard) or the maximum (too easy), and the piled-up people cannot be told apart.

> **! EXAM TRAP**
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> "The test is too difficult for everyone" is a **floor** effect. A ceiling effect does not mean "scores are normally distributed" (they are negatively skewed) or "perfect reliability": no reliability coefficient exceeds 1, and bunched scores lower it.

Item discrimination and distractors ○

The **item discrimination index** (D) asks whether high scorers on the whole test did better on this item. Take the top and bottom 27 percent by total score, the split that **Truman Lee Kelley** (1884 to 1961) showed in 1939 best balances extremeness against group size; then $D = p_{upper} - p_{lower}$, ranging from -1 to +1. If 40 of 50 upper and 15 of 50 lower examinees pass, D = 0.80 - 0.30 = 0.50 (and p = 0.55).

Robert L. Ebel's guide: 0.40 and above, very good; 0.30 to 0.39, reasonably good; 0.20 to 0.29, marginal; below 0.20, poor. A **negative D** means the lower group did better: the item is miskeyed, ambiguous (able examinees see a second defensible answer) or otherwise faulty, and must be revised or discarded.

> **! EXAM TRAP**
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> D = -0.20 does not mean "well-functioning", "too easy" or "at ideal difficulty": difficulty is p, not D. A negative D says the item works backwards; check the key first.

Discrimination can also be an **item-total correlation**: point-biserial r (pass-fail item against a continuous total), biserial r (the pass-fail split assumed to cut a normal ability), phi (two true dichotomies) or tetrachoric r (two artificial dichotomies). Since the item is part of the total, the correlation is inflated, most in short tests; the corrected form uses the other items. The item's correlation with an outside criterion, times its standard deviation, is the **item validity index**.

Distractor analysis checks each option: a working distractor draws more of the lower group; a non-functioning one draws almost nobody; one drawing the upper group signals ambiguity or miskeying. Plausible distractors make an item harder and usually more discriminating. Difficulty caps discrimination: at p = 1.00 or 0, D must be 0, and D can approach 1 only near p = 0.50, though an ambiguous item can sit at p = 0.50 with D near zero. These indices assume a **power test**; in a **speed test** the last items are reached only by fast workers, so their p and D reflect speed, not content.

Item response theory ●●

All this is **classical test theory** (CTT; X = T + E, Section 2.3), whose p and D depend on the sample: an item with p = 0.90 among postgraduates may have p = 0.30 in Class 8. **Item response theory** (IRT), or latent trait theory, models each item instead. It assumes an unobservable **latent trait**, θ (theta), and expresses the probability of a correct (or keyed) response as a function of the person's θ and the item's parameters.

Graphed, this is the **item characteristic curve** (ICC): the probability of a correct response plotted against the examinee's level on the latent trait, not against time taken, number of examinees or reliability. It is an S-shaped ogive (Figure 2.4). Its location on θ is the difficulty, **b** (where the probability reaches 0.50, or halfway between c and 1); its slope there is the discrimination, **a**; its lower asymptote is the pseudo-guessing parameter, **c**, the chance that a very low-θ examinee still succeeds.

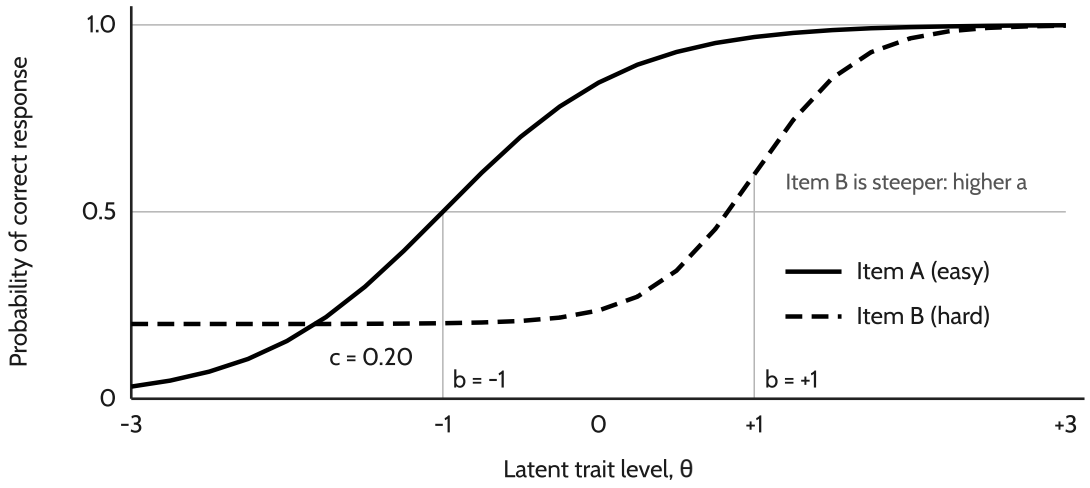


Figure 2.4 Item A (solid) lies left, so it is easy (b = -1). Item B (dashed) lies right (b = +1), rises more steeply (higher a) and never drops below its guessing floor, c = 0.20.

The **Rasch model**, or one-parameter logistic model (1PL), published in 1960 by the Danish mathematician **Georg Rasch** (1901 to 1980), estimates difficulty only. The two-parameter model adds discrimination and the three-parameter model adds guessing; **Allan Birnbaum** set out these logistic models in 1968, building on the work of **Frederic M. Lord** (1912 to 2000) from 1952. IRT assumes **unidimensionality** (one trait) and **local independence** (at a fixed θ, responses to different items are unrelated). An **item information function** shows how precisely an item measures at each θ, peaking near b; summed, they give the test information function.

When the model fits, item parameters are sample-invariant and θ does not depend on which items were given, since persons and items share one scale. This allows **computerised adaptive testing** (CAT), which picks each next item to suit the current estimate of θ and so needs far fewer items; item banking; test equating across forms; and detection of **differential item functioning** (DIF), or item bias, when equally able people from two groups have different ICCs for an item (Table 2.4).

Table 2.4 Classical test theory and item response theory.

Feature	Classical test theory	Item response theory
Model	X = T + E	Probability of a response given θ and item parameters
Level	Whole test	Each item
Item statistics	p and D, sample-dependent	b, a, c, sample-invariant
Person score	Number right, tied to the items given	θ, on the item difficulty scale
Precision	One standard error of measurement for all	Varies with θ

> **! EXAM TRAP**
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> IRT does not treat item statistics as sample-dependent (CTT does), is not limited to one parameter (only the Rasch model is), estimates item difficulty and a person parameter (θ), and makes adaptive testing possible. Group means and variances, test-retest stability, internal consistency and inter-rater agreement belong to classical reliability (Section 2.3); validity coefficients pooled across studies are validity generalisation (Section 2.4).

Response sets and response styles ○

Even a good item can be answered for the wrong reason. **Lee Joseph Cronbach** (1916 to 2001) showed in 1946 and 1950 that habitual ways of answering, **response sets**, shift scores regardless of content. A response set now usually means a temporary, situational tendency and a **response style** a stable, trait-like one; **Douglas N. Jackson** (1929 to 2004) and **Samuel Messick** (1931 to 1998) argued in 1958 that much variance on inventories reflects style, not content (Table 2.5).

Table 2.5 Response sets and their controls.

Set or style	The respondent	Control
Acquiescence (yea-saying), nay-saying	Agrees, or disagrees, regardless of content	Balanced keying
Social desirability, faking good	Gives answers that look good	Forced choice; L and K scales
Faking bad, malingering	Exaggerates problems	F scale

Set or style	The respondent	Control
Deviation	Gives atypical answers	Read as a sign in itself
Extreme or central-tendency responding	Uses only the ends, or only the middle	Clear anchors; no midpoint
Evasiveness, cautiousness	Marks "?" or omits items	Fewer "?" options; instructions
Guessing, working speedily	Guesses; rushes long tests	Correction formula; ample time

Acquiescence is agreeing regardless of content, so a yea-sayer scores high on an all-"agree" scale without having the trait; nay-saying is the opposite. **Social desirability** is choosing the answer that presents oneself well, often without intent; **Allen L. Edwards** showed, in work summed up in his 1957 book, that it strongly predicts which items are endorsed. Its deliberate forms are faking good and faking bad, and faking bad for external gain is malingering. **Irwin A. Berg**'s **deviation hypothesis** holds that atypical responding is general: people who answer deviantly even where content does not matter also deviate in behaviour.

A consistent style can raise reliability yet lower validity. Reverse-scored items make a yea-sayer's agreements cancel out; the Edwards Personal Preference Schedule pairs statements matched for desirability; the MMPI's L (lie), F (infrequency) and K (correction) scales are covered in Section 2.9; subtle items, rapport and honest-answering instructions also help.

> **! EXAM TRAP**
>
> Agreeing *regardless of content* is **acquiescence**. Social desirability depends on content; central tendency is choosing the middle; halo is a rater's error, not a test-taker's.

Rating scales and rater errors ●●

When another person judges the trait, bias moves to the rater. Rating scales may be numerical (numbered points), graphic (a line to mark), descriptive (a phrase per level), percentage (a percentile band), standard or man-to-man (known people as anchors), cumulated points or checklist (ticked statements summed) or forced-choice (equally favourable statements, curbing leniency). The **behaviourally anchored rating scale** (BARS) of **Patricia Cain Smith** and **Lorne M. Kendall** (1963) anchors each point with a concrete behaviour. Ranking orders people; paired comparison compares every pair.

The **halo effect** (halo error), named by **Edward Lee Thorndike** (1874 to 1949) in 1920 from officers' ratings of soldiers, is judging a person on the basis of one attribute or a general impression that colours every other rating: a clinician who sees anxiety improve rates depression improved too. Table 2.6 lists the other errors, reduced by rater training, behavioural anchors, rating one trait across all persons at a time and pooling raters.

Table 2.6 Rater errors.

Error	The rater
Leniency (generosity)	Rates everyone too favourably
Severity	Rates everyone too harshly
Central tendency	Avoids both ends; rates everyone average
Logical error	Rates traits alike because he believes they go together

Error	The rater
Contrast error	Rates others as opposite to himself, or against the previous ratee
Proximity error	Rates traits placed close together on the form alike
Similarity ("similar to me")	Favours people who resemble himself

! EXAM TRAP

Anxiety improvement spuriously improving depression ratings is **halo**, not a **ceiling effect** (a test too easy to show gains), the **Hawthorne effect** (people change because they are studied; Chapter 9) or a **therapeutic effect** (real change from treatment): the rating improved, not the depression.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The proportion of examinees passing an item is its _____ index.
2. Kelley's extreme groups are the top and bottom _____ percent.
3. When the lower group outscores the upper group, D is _____.
4. A test so easy that most score near the maximum shows a _____ effect.
5. The ICC plots the probability of a correct response against the _____.
6. The one-parameter logistic model is the _____ model.
7. Agreeing with items regardless of content is _____.
8. Judging a person on one attribute is the _____ effect.

Answers: (1) difficulty, (2) 27, (3) negative, (4) ceiling, (5) latent trait (θ), (6) Rasch, (7) acquiescence, (8) halo.

2.3 Reliability: Consistency and the Standard Error of Measurement

HIGH YIELD · 64 QS

Step on a bathroom scale three times in a minute, read 61, 64 and 58 kilograms, and you stop trusting it. A psychological test is a scale for something invisible, so the first question about any score is the same: measured again, would it come out the same? What follows separates score from error, estimates the error four ways, and puts an honest band around one person's score.

True scores, error and the meaning of reliability ●●●

In 1904 **Charles Spearman** proposed the model on which **classical test theory** rests: every observed score is a true score plus error, **X = T + E**. The **true score** is the average a person would earn over endless independent testings. The model assumes that errors are random, averaging zero; that they are uncorrelated with true scores; and that errors on one test are uncorrelated with errors on another. The variances therefore add: observed variance = true variance + error variance.

The **reliability coefficient** (r_{xx}) is the share of observed-score variance that is true-score variance: $r_{xx} = \sigma^2_T / \sigma^2_X = 1 - \sigma^2_E / \sigma^2_X$. If scores vary with variance 100, of which 80 is true and 20 error, reliability is .80. It runs from 0 (all error) to 1 (no error). The **domain-sampling model** sees a test's items as a sample from a large domain, such as all the words a spelling test could use; the score on the whole domain is the universe score. The more items sampled, the closer the test score comes to it, so longer tests are more reliable.

Only **random error** (a lapse of attention, a lucky guess, a noisy hall), which moves scores unpredictably, lowers reliability. A **systematic error** moves every score the same way, like a scale that always reads two kilograms heavy; being consistent, it leaves reliability intact but biases the scores and so threatens validity. Reliability therefore means **consistency**, **dependability** and **freedom from measurement error**. It is distinct from validity (measuring what the test claims) and sensitivity (detecting a condition or a change), both in Section 2.4, from generalizability (findings extending to other people and settings), and from fairness to groups (Section 2.1). A **standardised** test has uniform administration and scoring (Section 2.1); a **normed** test is read against a reference group (Section 2.5).

> ! EXAM TRAP
>
> Reliability is "freedom from measurement error, consistency and dependability": **all of the above**. A study is reliable when **its measures are stable on different occasions**. "Conducted by a reputable researcher" is trust in a person, "findings can be generalised to other settings" is external validity, and "methods stated clearly enough to be repeated" is replicability (Chapter 9).

Stability and equivalence: test-retest and parallel forms ●●●

In **test-retest reliability** the same test is given to the same people on two occasions and the scores are correlated, giving the **coefficient of stability**. Giving a new anxiety scale to students twice, two weeks apart, and correlating the scores is the classic case. It detects time-sampling error: changes in health, mood and conditions between sittings. Too short an interval lets memory and practice inflate the coefficient; too long lets real change lower it, so a fortnight to a few weeks is usual. The method suits traits expected to stay stable over the interval, such as intelligence and personality, not rapidly changing states such as current mood, where a low coefficient reflects true change.

> ! EXAM TRAP
>
> Test-retest reliability means the test **gives consistent results over time**. Correlating two halves is split-half reliability; two raters scoring one protocol is inter-rater reliability; correlating with a future criterion, or "measuring what it claims", is validity (Section 2.4); the client doing better the second time is a **practice effect**. It fits stable traits, not mood states, single-item measures or open-ended interviews.

A second method, **alternate-forms reliability** (**parallel-forms reliability**), uses two versions built to one blueprint: the same number, kind, content and format of items, equal difficulty, and equal means and variances. Given in one session, their correlation is a **coefficient of equivalence**, which detects content-sampling error; given weeks apart, it is a **coefficient of stability and equivalence**, the most demanding estimate. Parallel forms avoid remem-

bered answers and suit retesting and speed tests, but they are costly to write, rarely truly equivalent, and practice on the type of task still carries over.

Split-half reliability and the Spearman-Brown formula ●●●

Can reliability come from one sitting? Methods that do so estimate **internal consistency reliability**: how well the parts of a test agree with each other within one administration. In **split-half reliability** one test is scored as two halves, which are then correlated. First half against second half is a poor split, since warming up, fatigue and rising difficulty make the halves unequal; an odd-even split, or random halves matched for content and difficulty, is better. The major challenge is dividing the test into two nearly equivalent halves, and different splits give different answers.

The half correlation describes a test half as long, so it underestimates the whole. Spearman and **William Brown** published the correction independently in 1910. The **Spearman-Brown prophecy formula** steps the half correlation up: $r_{full} = 2r_{half} / (1 + r_{half})$, so halves correlating .60 give 1.20 / 1.60 = **.75**. Its general form predicts reliability when length changes by a factor k (new length over old): $\mathbf{r_{new} = kr / [1 + (k - 1)r]}$. Adding parallel items (k above 1) raises reliability, because true-score parts add up while random errors partly cancel; shortening (k below 1) lowers it. Rearranged, the length needed for a target is $k = r_{target}(1 - r) / [r(1 - r_{target})]$. Table 2.7 works four cases, and Figure 2.5 shows the diminishing returns.

Table 2.7 The Spearman-Brown formula, worked four ways.

Question	k	Working	Answer
Halves correlate .60; whole test?	2	1.20 / 1.60	.75
r = .50; test made three times as long?	3	1.50 / (1 + 2 × .50)	.75
r = .80; test cut to half length?	0.5	.40 / (1 - .5 × .80)	.67
r = .60; how much longer to reach .90?	?	(.90 × .40) / (.60 × .10)	6 times

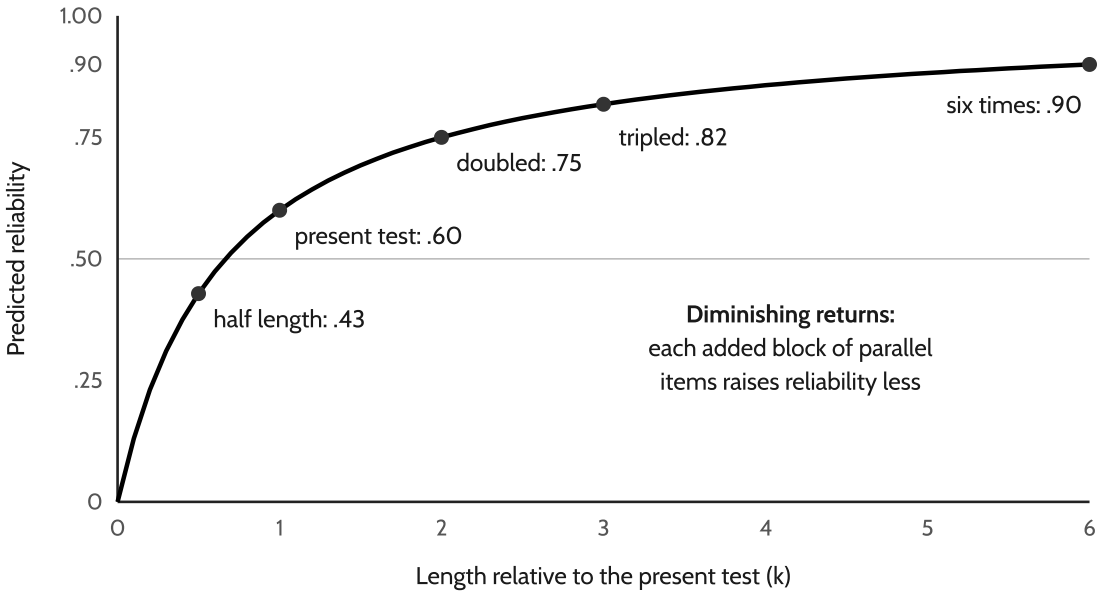


Figure 2.5 Spearman-Brown predictions for a test whose present reliability is .60. Doubling gains .15, tripling only .07 more, and .90 needs six times the items.

Spearman-Brown assumes parallel halves, with equal means and variances; when the half variances differ it overestimates. Two formulas need no step-up. **Phillip Rulon** (1939) treated the difference between a person's two half-scores as error: **Rulon's formula** is $r = 1 - \sigma^2_d / \sigma^2_X$, the variance of the differences over the variance of totals. **John Flanagan** (1937) used the half variances: $r = 2[1 - (\sigma^2_a + \sigma^2_b) / \sigma^2_X]$ (**Louis Guttman** gave the same formula in 1945). If each half has variance 6, totals 20 and differences 4, Rulon gives 1 - 4 / 20 = .80 and Flanagan 2(1 - 12 / 20) = .80; the halves correlate .67, which Spearman-Brown also steps up to .80.

> **! EXAM TRAP**
>
> Spearman-Brown **estimates full-test reliability from a split-half correlation, or predicts reliability when length changes**; it raises reliability by adding parallel items, not by removing items, making them ambiguous or shrinking the sample. It does not estimate rater agreement, correct validity for unreliability, estimate a speed test's reliability from one sitting, or predict validity, effect sizes or factor structure. Rulon needs **variances**: SDs of 3 (differences) and 6 (totals) give 1 - 9 / 36 = .75.

Odd-even splitting suits a **power test**: graded items, generous time. On a pure **speed test** of easy items the score is how far a person gets, so whoever reaches item 60 has about 30 odd and 30 even right. The halves match by construction and the correlation nears 1.00 spuriously, as do KR-20 and alpha from one timed sitting. Speed tests need test-retest, parallel forms or **separately timed halves**, each given with half the time and stepped up with Spearman-Brown. Projective and many neuropsychological tests lack comparable, objectively scored items to split. Every internal consistency method shares one limitation: a single administration says nothing about stability over time.

> **! EXAM TRAP**
>
> The limitation of internal consistency methods: they **measure consistency within a single administration only**. Needing several items and falling with mixed content are conditions of use, so a mock offering "all of the above" still keys the single administration.

Kuder-Richardson formulas and coefficient alpha ●●●

Why trust one arbitrary split? In 1937 **G. Frederic Kuder** and **Marion Richardson** derived an internal consistency coefficient from item statistics. **KR-20** suits items scored dichotomously (right or wrong, yes or no) that may vary in difficulty: **KR-20 = [k / (k - 1)] × [1 - Σpq / σ^2_X]**, where k is the number of items, p the proportion passing an item, q = 1 - p, and σ^2_X the variance of total scores.

The product pq is the item's variance: an item passed by 30 percent has pq = .30 × .70 = .21, and the maximum is .25 at p = .50. Items measuring one thing covary, so total variance far exceeds the summed item variances. For 20 items with Σpq = 3.2, mean 12 and variance 16: 3.2 / 16 = .20, 1 - .20 = .80, and × 20 / 19 gives **.84**. **KR-21** assumes equal item difficulty, replacing Σpq with M(k - M) / k, so it needs only k, mean and variance: KR-21 = [k / (k - 1)] × [1 - M(k - M) / (kσ^2)]. For the same test, 12 × 8 = 96, 96 / (20 × 16) = .30, 1 - .30 = .70, and × 20 / 19 gives **.74**. When item difficulties differ, KR-21 is lower than KR-20: a quick, conservative estimate.

Lee Cronbach (1951) generalised KR-20 to items with several score points, such as Likert ratings. The resulting **coefficient alpha** is α = [k / (k - 1)] × [1 - Σσ^2_i / σ^2_X]: five Likert items with variances summing to 6 and total variance 20 give 1.25 × .70 = **.875**. For 0 or 1 items σ^2_i = pq, so alpha equals KR-20. Cronbach showed that alpha is the mean of all possible split-half coefficients (each by Rulon's or Flanagan's formula); averaging over every split increases the accuracy of the estimate. **Cyril Hoyt** (1941) reached the same value by analysis of variance of persons by items: r = ($MS_{persons}$ - $MS_{residual}$) / $MS_{persons}$.

For alpha, as for any reliability coefficient, about .70 is acceptable for research and group comparisons, decisions about an individual patient need .80 to .90 or more, and .40 is poor: the items do not hang together. Alpha rises with test length, very high values can mean redundant items, and a high alpha does not prove that a test measures a single trait.

> **! EXAM TRAP**
>
> Cronbach's alpha measures **internal consistency reliability**: how well a scale's items hang together. It is not validity of any kind (content, construct, concurrent, predictive or criterion), not test-retest, parallel-forms or inter-rater reliability, and not a variance or regression statistic. KR-20 differs from alpha only in being limited to dichotomous items, whatever their content.

Agreement between scorers ●●

Scorer or **inter-rater reliability** is the degree to which two or more clinicians agree when independently scoring or interpreting the same test responses. It matters wherever judgement enters, since scorer disagreement becomes error in the score: projective protocols, essays, behaviour observation and psychiatric diagnosis. Agreement of two tests on one construct is convergent validity, predicting future behaviour is predictive validity, and items relating to each other is internal consistency.

Simple percentage agreement flatters raters, since some agreement is chance. **Jacob Cohen** (1960) corrected for it with **Cohen's kappa** for categorical judgements: $\kappa = (P_o - P_e) / (1 - P_e)$. Two clinicians classify 100 patients: both say depressed for 40 and not depressed for 30, so observed agreement $P_o = .70$. Clinician A calls 50 depressed and B calls 60, so chance agreement $P_e = (.50 \times .60) + (.50 \times .40) = .50$, and $\kappa = .20 / .50 =$ **.40**. Seventy percent raw agreement is only fair beyond chance. **J. Richard Landis** and **Gary Koch** (1977) labelled kappa below 0 poor, .00 to .20 slight, .21 to .40 fair, .41 to .60 moderate, .61 to .80 substantial and .81 to 1.00 almost perfect, so .85 is almost perfect. Table 2.8 lists related statistics.

Table 2.8 Statistics of agreement between raters.

Statistic	Ratings	Raters	Note
Percentage agreement	Categories	Two	Inflated by chance
Cohen's kappa (1960)	Categories	Two	Corrected for chance
Weighted kappa (Cohen, 1968)	Ordered categories	Two	Near-misses earn partial credit
Fleiss' kappa (1971)	Categories	Three or more	After **Joseph Fleiss**
Intraclass correlation	Continuous scores	Two or more	Share of variance from true differences
Kendall's W	Rankings	Several	**Maurice Kendall**'s concordance, 0 to 1

> **! EXAM TRAP**
>
> Cohen's kappa measures **inter-rater agreement on categorical data, corrected for chance**. It is not an effect size for ANOVA (that is eta squared, Chapter 9), and not internal consistency, test-retest reliability, or convergent, criterion or predictive validity.

The standard error of measurement ●●●

A reliability coefficient describes a group. A psychologist facing Ravi, whose IQ came out at 110, needs to know how far his true score might lie from 110. The **standard error of measurement** (SEM), the standard deviation of the errors one person's scores would show over many retestings, answers that: $\mathbf{SEM = SD \times \sqrt{(1 - r_{xx})}}$.

With SD 15 and reliability .91, SEM = 15 × √.09 = 15 × .30 = **4.5**; with reliability .80, SEM = 15 × √.20 = about **6.7**. At r = 1 the SEM is 0; at r = 0 it equals the SD. The true score lies within ±1 SEM of the observed score about 68 percent of the time and within ±1.96 SEM 95 percent of the time, so Ravi's true IQ is probably 105.5 to 114.5 (68 percent) or 101.2 to 118.8 (95 percent), as Figure 2.6 shows.

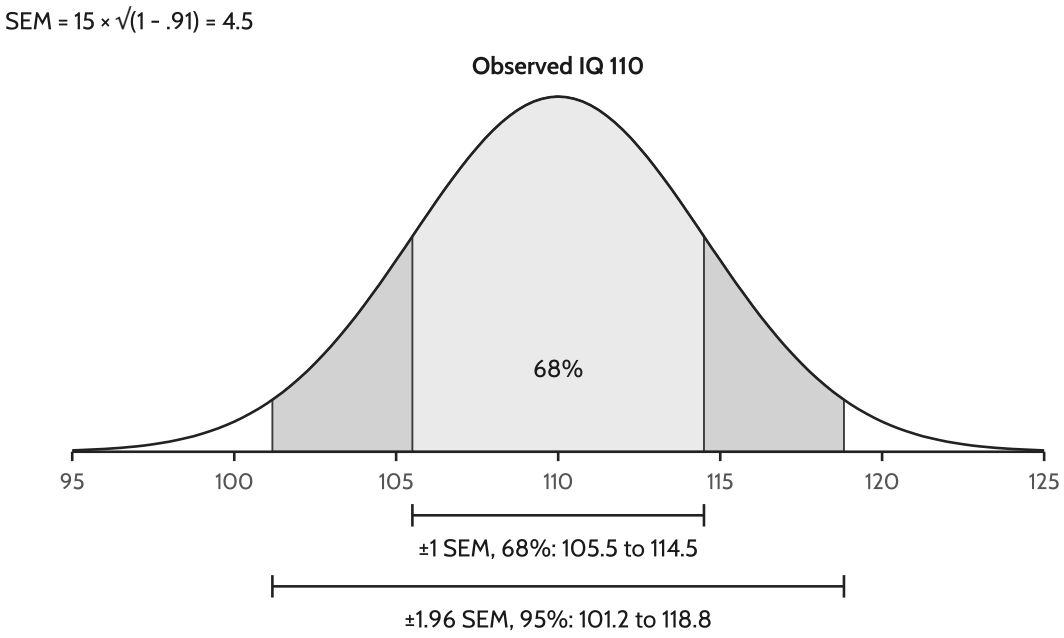


Figure 2.6 Confidence bands around an observed IQ of 110 (SD 15, reliability .91). The narrow band holds the true score about two times in three; the wide band, 19 times in 20.

> **! EXAM TRAP**
>
> SEM = **SD × √(1 - r)**, used to **estimate the range within which a person's true score falls.** Wrong options: SD × √r, SD × r (15 × .91 = 13.65), SD × (1 - r) without the root (1.35), √(1 - r) alone (.30), SD / r, SD × (1 + r), reliability × validity, mean minus SD. It does not correlate two tests, index item difficulty (Section 2.2) or predict performance.

> **CLINICAL LINK**
>
> DSM-5 places the IQ criterion for intellectual disability at about 70 ± 5 (65 to 75), a margin for measurement error. Report bands, never a single number treated as exact.

Comparing two scores needs the **standard error of the difference**, $\sqrt{(SEM_1^2 + SEM_2^2)}$. For two index scores (SD 15) with reliabilities .91 and .84, the SEMs are 4.5 and 6.0 and the standard error of the difference is √56.25 = 7.5, so the scores must differ by about 1.96 × 7.5 ≈ 15 points to be trusted at 95 percent. A **difference score** is unreliable when its two tests correlate highly: $r_{dd} = [(r_{xx} + r_{yy}) / 2 - r_{xy}] / (1 - r_{xy})$, so reliabilities .90 and .80 with a correlation of .70 give .15 / .30 = .50, because the shared true score cancels and the errors remain. A **composite score** from several positively correlated tests is usually more reliable than its parts.

The **index of reliability**, $\sqrt{r_{xx}}$, is the correlation between observed and true scores (√.81 = .90) and caps validity (Section 2.4). The **estimated true score**, $M + r_{xx}(X - M)$, regresses towards the mean: an IQ of 130 with reliability .90 gives 100 + .90 × 30 = 127, because extreme scores contain some luck. Spearman's (1904) **correction for attenuation**, $r_{xy} / \sqrt{(r_{xx}\, r_{yy})}$, estimates the correlation between perfectly reliable measures: a predictor and criterion correlating .36, with reliabilities .81 and .64, give .36 / (.90 × .80) = .36 / .72 = **.50** (Section 2.4). Divide by the root; multiplying (.26) or dropping the root (.69) are the usual slips.

What raises or lowers reliability

Reliability belongs to scores from a particular group under particular conditions, not to a test for all time.

- **Length.** More parallel items raise it, as Spearman-Brown predicts.
- **Group heterogeneity.** A wide range of ability raises r; restriction of range (testing only admitted students) lowers it.
- **Item difficulty and discrimination.** Items near p = .50 with high item-total correlations add most; items everyone passes or fails add nothing (Section 2.2).
- **Guessing.** It adds random error, most on true-false items.
- **Speededness and interval.** Speed inflates single-session estimates; a longer retest interval lowers test-retest coefficients.
- **Scoring and conditions.** Objective keys raise reliability; subjective scoring, ambiguous items, noise, fatigue and low motivation lower it.

Cronbach and **Goldine Gleser** (1965) applied the **bandwidth-fidelity dilemma** from information theory to testing: a test covering many traits briefly (wide bandwidth) measures each with low precision (fidelity), while a narrow test measures one trait precisely but says little else.

Beyond classical theory ○

Classical theory lumps all error into one E; **generalizability theory**, from Cronbach, Gleser, **Harinder Nanda** and **Nageswari Rajaratnam** (1972), uses analysis of variance to split observed-score variance into persons and facets such as raters, items and occasions. A **G study** estimates these variance components; a **D study** (decision study) forecasts dependability under other designs, such as two raters instead of one. The result is a **generalizability coefficient**, of which classical reliability is a special case. Item response theory, which models each item's response curve, replaces one reliability for everyone with **test information**, which varies along the trait (Section 2.2). Table 2.9 gathers the methods.

Table 2.9 The methods of estimating reliability.

Method	Error it detects	Statistic	Best use
Test-retest	Time sampling	r (stability)	Stable traits; speed tests
Parallel forms	Content; time if delayed	r (equivalence)	Retesting; speed tests
Split-half	Content sampling	Spearman-Brown; Rulon; Flanagan	Power tests
KR-20, KR-21, alpha	Content sampling, heterogeneity	Internal consistency	Homogeneous scales
Inter-rater	Scorer differences	Kappa, ICC, W	Projectives, diagnosis
Generalizability	Several facets at once	G coefficient	Designing measurement

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Reliability is the ratio of _____ variance to observed-score variance.
2. The same test given twice to the same people yields the coefficient of _____.
3. Halves correlating .60 give a full-test reliability of _____ by Spearman-Brown.
4. An odd-even split from one timed sitting is least appropriate for a _____ test.
5. Cronbach's alpha measures _____ reliability.
6. KR-21 assumes all items are of equal _____.
7. Rater agreement on categories, corrected for chance, is Cohen's _____.
8. With SD 15 and reliability .91, the SEM is _____.

Answers: (1) true-score, (2) stability, (3) .75, (4) speed, (5) internal consistency, (6) difficulty, (7) kappa, (8) 4.5.

2.4 Validity and the Accuracy of Test Decisions MEDIUM YIELD · 32 QS

A bathroom scale that always reads two kilograms light is perfectly consistent and consistently wrong. Section 2.3 asked whether a test measures consistently; this section asks whether it measures the right thing, and then follows its scores into the decisions they drive.

What validity means

The classic definition, associated with **Henry Edward Garrett** (1894 to 1973), is that **validity** is the extent to which a test measures what it claims (purports) to measure. Validity is specific to a purpose and a population (a test valid in one culture need not be valid in another), it is a matter of degree, and every new use needs new evidence.

The American Psychological Association's *Technical Recommendations* of 1954 named content, predictive, concurrent and construct validity, and the 1966 *Standards* merged predictive and concurrent into criterion-related validity, giving the classical trinity of content, criterion-related and construct validity. In 1989 **Samuel Messick** replaced the trinity with a **unified view of validity**: one integrated judgement of how well evidence and theory support the interpretations and uses of scores, so validity belongs to an inference drawn from scores, not to the test itself. The 2014 *Standards for Educational and Psychological Testing* list five sources of evidence: test content, response processes, internal structure, relations with other variables, and the consequences of testing.

Face validity and content validity ●●

Face validity is the appearance of validity: on superficial inspection a test seems to measure what it claims, because its questions intuitively seem relevant, as when a depression questionnaire asks "I feel sad most of the day". It is judged by test takers and lay people, not by experts or by any statistic, so it is not validity in the technical sense. It still earns rapport and acceptance, but an obvious purpose makes faking good or faking bad easy, which is why some inventories use subtle items (Section 2.9).

Content validity is how adequately the items sample the whole domain: a depression scale should cover mood, interest, sleep, appetite, energy, guilt, concentration and suicidal think-

ing, not sadness alone. The developer drafts a **table of specifications** (test blueprint), a grid of content areas against objectives that fixes the number of items in each cell, and experts then judge every item. **C. H. Lawshe** quantified their judgement in 1975 with the **content validity ratio**, CVR = (n_e − N/2) / (N/2), where n_e of N experts rate an item essential; it runs from −1 to +1. Item validity means each item belongs to the domain; sampling validity means the items together cover all of it. The two threats are **construct under-representation** (part of the domain left out) and **construct-irrelevant variance** (scores also reflecting something else, as when a mathematics test set in English partly measures reading). Content validity is the main evidence for achievement tests.

> **! EXAM TRAP**
>
> Face validity has nothing to do with faces: "facial expressions used for diagnosis" and "a scale of emotional responding" are invented distractors, and "a hypothetical attribute that may not be directly observable" defines a construct (2.4.5). Face validity is judged by lay people; content validity by experts against a blueprint.

Criterion-related validity: concurrent and predictive ●●●

A criterion is an independent, external measure of the attribute, such as a psychiatrist's diagnosis, a gold-standard scale, grades or job ratings, and **criterion-related validity** is the correlation between test and criterion. Its two types differ only in timing.

In **concurrent validity** test and criterion are measured at about the same time. The classic design gives a new intelligence test together with an established one; likewise, a new 20-item depression questionnaire that correlates r = 0.85 with a gold-standard scale completed the same day shows concurrent validity. A high correlation with an established measure of the same attribute shows that the test reflects present status. In **predictive validity** the criterion is measured later, as when entrance scores are correlated with first-year marks. Selection tests need it, and it usually runs lower than concurrent validity because people change in the interval. A criterion recorded before the test gives postdictive validity.

The correlation is the validity coefficient; its square, the coefficient of determination, is the share of criterion variance the test explains (r = 0.60 explains 36 percent). The standard error of estimate, $SE_{est} = SD_y \sqrt{(1 - r^2)}$, is the error of an individual prediction: with r = 0.60 and a criterion SD of 10 marks it is 8 marks. It is not the standard error of measurement (Section 2.3).

> **! EXAM TRAP**
>
> Concurrent and predictive are the two criterion-related types; face, content, factorial and construct validity are not criterion based. RML 2024 keyed "predictive validity" as the one NOT criterion based: that key is wrong. Of its options, face validity (appearance only) and factorial validity (construct evidence) are not criterion based; face validity is the safer mark. RML 2021 defined concurrent validity as "scores correlate highly with scores from tests that measure the same attribute"; "two tests done at the same time" is only the timing, "two clinicians agree" is inter-rater reliability, and split-half or retest correlations are reliability (Section 2.3).

Criteria, cut-offs and cross-validation ○

A coefficient is no better than its criterion, and finding a good one is the criterion problem. A good criterion is relevant, reliable (an unreliable one caps the coefficient, 2.4.6), free from bias and available at reasonable cost. In **criterion contamination** the criterion is influenced by knowledge of the predictor scores, or includes predictor-related content, so the correlation is inflated artificially: supervisors who know employees' selection-test scores tend to rate high scorers highly. The cure is blind ratings. Criterion deficiency is the opposite flaw, a criterion that omits important parts of performance, and plain random error in the criterion lowers rather than inflates the coefficient.

Restriction of range also lowers validity: grades exist only for students the entrance test admitted, and their narrow band of scores underestimates the test. Items chosen because they correlated with the criterion in one sample capitalise on chance, so the coefficient falls in a fresh sample. Re-checking it there is **cross-validation**, and the fall is shrinkage, largest with small samples and many items tried for few kept.

An expectancy table shows the chance of criterion success at each band of test scores. A cut-off score divides those selected or diagnosed from the rest, and since no test is perfectly valid every cut-off makes some errors (2.4.8). **Incremental validity** (**Lee Sechrest**, 1963) is the prediction a test adds beyond measures already available, such as school marks. Synthetic validity, a term of Lawshe's, infers validity for a job too small to study directly from known validities for its components.

Construct validity ●●

What is the criterion for anxiety? Many attributes have none. A **construct** is a hypothetical or inferred attribute, such as intelligence, anxiety or extraversion, that cannot be directly observed but is used to explain behaviour. **Construct validity**, introduced in the 1954 recommendations and set out by **Lee Joseph Cronbach** (1916 to 2001) and **Paul Everett Meehl** (1920 to 2003) in 1955, is the degree to which a test measures the theoretical construct it claims to measure. They placed each construct in a **nomological network**, the web of lawful relations the theory predicts with other constructs and with behaviour. Validation then tests whether scores behave as the network predicts.

Convergent validity is a high correlation with other measures of the same construct; **discriminant validity** (divergent validity) is a low correlation with measures of theoretically unrelated constructs. A new resilience scale should correlate highly with an established resilience measure and weakly with an unrelated trait. **Donald Thomas Campbell** (1916 to 1996) and **Donald W. Fiske** showed in 1959 how to test both at once.

KEY STUDY: CAMPBELL AND FISKE'S MULTITRAIT-MULTIMETHOD MATRIX (1959)

Does a score reflect the trait, or only the way it was measured? Campbell and Fiske measured two or more traits, each by two or more methods (self-report, peer rating, observation), and intercorrelated everything in a **multitrait-multimethod matrix**. The monotrait-heteromethod values (same trait, different methods) should be high, showing convergence, and higher than the heterotrait-monomethod values (different traits, same method). When different traits correlate highly merely because both were self-reported, the matrix has exposed method variance, which proved common in the published matrices they examined. Criticism: the criteria are judged by eye, without a single statistical test, so confirmatory factor analysis is now used to model the matrix.

Factorial validity is shown when items load on the expected factor in exploratory or confirmatory factor analysis, evidence of internal structure (the technique is in Chapter 9). In the **known-groups method** (contrasted groups) the test separates groups known to differ, such as depressed inpatients and community volunteers. Scores should also change with development as the theory expects and move with experimental interventions (anxiety scores falling after therapy). Internal consistency (Section 2.3) supports a construct claim but cannot establish it alone. **Consequential validity**, Messick's addition, weighs the social consequences of using the scores; **ecological validity** is how well test performance represents everyday functioning, a key question in neuropsychology. The validity of research designs (internal, external, statistical conclusion and the construct validity of a manipulation) belongs to Chapter 9.

! EXAM TRAP

Construct validity was keyed as "the degree to which a test measures what it claims to measure" (RML 2025) and as "the level to which the theoretical construct is related to other similar measures" (RML 2021); in the 2021 paper "scores correlate highly with tests of the same attribute" defined concurrent validity. The 2025 wording is Garrett's general definition, which fits because the unified view treats all validity as construct validity. The distractors are reliability (consistency across administrations or among items), content validity (the whole domain) and inter-rater reliability (rater agreement). For discriminant validity choose the *weak* correlation with an unrelated construct; a high correlation with an established measure of the same construct is convergent, "covers all facets" is content and "predicts future coping" is predictive.

Reliability and validity together ●●

A test can be reliable without being valid: a scale with high internal consistency but low criterion validity consistently measures something, but not the intended construct. It cannot be valid without being reliable, because scores that are mostly random error correlate with nothing. Reliability is therefore necessary but not sufficient for validity; consistency does not make a test valid, and validity is more than consistency.

A validity coefficient cannot exceed $\sqrt{(r_{xx} \times r_{yy})}$, the square root of the product of the test and criterion reliabilities. With a perfectly reliable criterion this **validity ceiling** is $\sqrt{r_{xx}}$, the index of reliability: $r_{xx} = 0.81$ allows at most 0.90. Run in reverse, the logic gives the **correction for attenuation** of **Charles Edward Spearman** (1863 to 1945), published in 1904, which estimates the correlation if both measures were perfectly reliable: $r_c = r_{xy} / \sqrt{(r_{xx} \times r_{yy})}$. An observed 0.36 with reliabilities of 0.81 and 0.64 becomes 0.36 / 0.72 = 0.50.

One apparent conflict remains. Very high homogeneity maximises internal consistency but narrows coverage, so a broad criterion is often predicted better by a heterogeneous test with moderate inter-item correlations: the bandwidth-fidelity trade-off of Section 2.3 seen from the validity side.

> **! EXAM TRAP**
>
> The TRUE statement (RML 2024) is "a test can be reliable without being valid". "If a test measures what it is intended to measure, then by definition it is reliable" is the near-miss: a valid test must be reliable, but not by definition. Validity need not come "first chronologically". The ceiling is the *square root* of reliability, not reliability itself, its square or reliability minus the SEM.

Sensitivity, specificity and predictive values ●●

Once a test is used to decide (depressed or not, select or reject), validity shows up as right and wrong calls. Crossing the test result with the true state, set by a gold-standard diagnosis, puts every person in one of four cells (Figure 2.7): a true positive, a **false positive** (positive without the condition), a false negative (the condition missed) or a true negative.

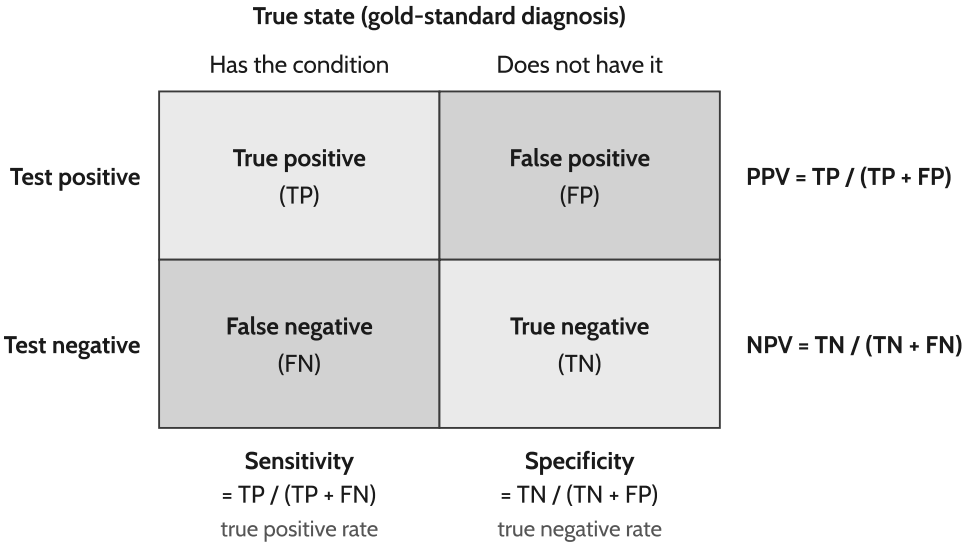


Figure 2.7 The four outcomes of a diagnostic decision. Sensitivity and specificity read down the columns; the predictive values read across the rows. Only the diagonal cells are correct decisions.

Sensitivity = TP / (TP + FN) is the true positive rate: the proportion of people who have the condition whom the test correctly identifies. **Specificity** = TN / (TN + FP) is the proportion of people without the condition whom the test correctly clears. Both describe the test itself at a given cut-off. A screen with sensitivity 90 percent and specificity 70 percent calls 90 percent of depressed people depressed and 70 percent of healthy people healthy. A high false positive rate means normal people are diagnosed as disordered; false negatives leave disordered people undetected. The proportion of all decisions that are correct is the hit rate.

The clinician, though, holds a result, not a diagnosis, and needs the rows. The **positive predictive value**, PPV = TP / (TP + FP), is the probability that a positive result is correct, and the negative predictive value, NPV = TN / (TN + FN), the probability that a negative one is. Both

depend heavily on the **base rate** (prevalence) of the condition in the people tested, as Meehl and **Albert Rosen** showed in 1955 (Table 2.10).

Table 2.10 One screen (sensitivity 90 percent, specificity 70 percent) given to 1000 people in two settings.

Outcome	Clinic: base rate 50 percent	Community: base rate 5 percent
Depressed: caught (TP) / missed (FN)	450 / 50	45 / 5
Healthy: cleared (TN) / flagged (FP)	350 / 150	665 / 285
PPV: positives truly depressed	450 / 600 = 75 percent	45 / 330 = 14 percent
NPV: negatives truly healthy	350 / 400 = 88 percent	665 / 670 = 99 percent
Hit rate: all correct calls	800 / 1000 = 80 percent	710 / 1000 = 71 percent

The test is unchanged, yet in the community six positive results in seven are false alarms, and a clinician who simply called everyone healthy would be right 95 percent of the time, beating the test's 71 percent. A sign is worth using only if it improves on the base rate.

> **! EXAM TRAP**
>
> TP / (TP + FP) is the PPV, not sensitivity. "Probability that a positive test reflects true disease" is the PPV; "probability of disease prior to testing" is the base rate. Sensitivity of 90 percent does not mean 90 percent of positive results are correct. The PPV is most affected by the base rate, not by sensitivity alone, specificity alone or the number of items. An overstrict diagnosis produces false negatives, not false positives.

Cut-offs, ROC curves and selection ○

Sensitivity and specificity hold only for one cutting score. Lower the cut-off and sensitivity rises while specificity falls; raise it and the reverse happens. The choice weighs the costs of the two errors: missing a suicidal patient is worse than a false alarm, so screens use lenient cut-offs. The **receiver operating characteristic curve** (ROC curve), borrowed from signal detection theory, plots sensitivity (true positive rate) against 1 – specificity (false positive rate), one point for each possible cut-off (Figure 2.8). A useless test follows the diagonal; a better one bows toward the top-left corner. The area under the curve (AUC) summarises discrimination: 0.5 is chance, 1.0 is perfect, and it equals the probability that a randomly chosen case scores higher than a randomly chosen non-case.

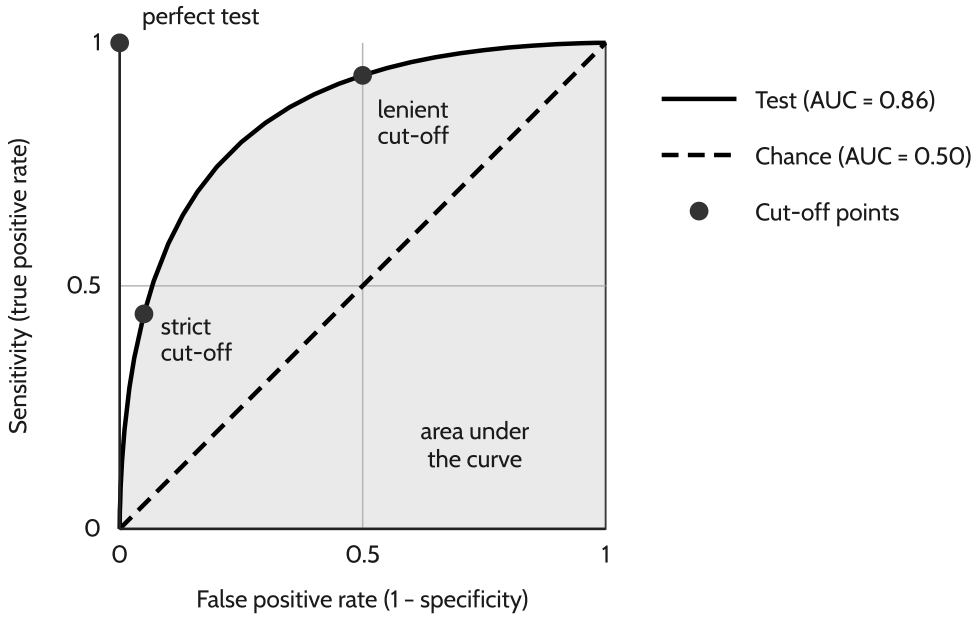


Figure 2.8 A receiver operating characteristic (ROC) curve. The strict cut-off flags few healthy people but misses cases; the lenient one catches nearly every case at the price of many false alarms. The shaded area is the AUC.

In selection, the **selection ratio** is the proportion of applicants who can be accepted (50 posts for 500 applicants gives 0.10). **H. C. Taylor** and **J. T. Russell** published the **Taylor-Russell tables** in 1939, giving the percentage of selectees who will succeed for a given validity coefficient, selection ratio and base rate. Even a modest validity helps a great deal when the selection ratio is low, and little when almost everyone must be accepted or the base rate is extreme. Utility analysis, developed by Cronbach and **Goldine Gleser** (1957), adds costs and gains: a test is worth using when better decisions repay the cost of testing.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. A test that merely looks relevant to the people taking it has _____ validity.
2. Correlating a new intelligence test with an established one given at the same time tests _____ validity.
3. Supervisors who know applicants' test scores and then rate their work cause criterion _____.
4. A weak correlation with a measure of an unrelated construct is evidence of _____ validity.
5. A test with reliability 0.64 can have a validity coefficient no higher than _____.
6. Sensitivity equals TP divided by _____.
7. As the base rate of a disorder falls, the positive predictive value of a screen _____.

Answers: (1) face, (2) concurrent, (3) contamination, (4) discriminant (divergent), (5) 0.80, (6) TP + FN, (7) falls.

2.5 Norms, Standard Scores and Interpreting a Score MEDIUM YIELD · 32 QS

A ten-year-old answers 34 reasoning items correctly. Is that good? Until you know how hard the items were and how other ten-year-olds do, the **raw score** (items passed or points earned) means nothing; it needs a frame of reference. There are two such frames, and the commoner one, comparison with other people, rests on norms, on derived scores and on the normal curve that ties them together.

Norm-referenced and criterion-referenced scores ○

In **norm-referenced interpretation** a score is compared with those of a **norm group** or **standardisation sample**, a representative sample of the people the test is meant for. "Higher than 90 per cent of her peers", "5th in his class" and "1 standard deviation (SD) above the mean" are norm-referenced: they rank a person among others, as selection requires. In **criterion-referenced interpretation** the score is compared with a fixed standard, whatever others do: "85 per cent of the addition problems correct, meeting the mastery cut-off". **Robert Glaser** drew the distinction in 1963. A criterion-referenced test asks what a person can do, so a whole class can pass it, or no one.

A **domain-referenced test** samples items from a defined domain (all two-digit additions, say), so the percentage correct estimates how much of the domain the person commands; a **mastery test** adds a cut-off and reports pass or fail, like a driving test. **Norms are not standards**: they describe how people do perform, not how they should, so matching the norm of a poorly taught school is not being good enough. An **ipsative score** compares a person with no one else: it expresses one trait relative to the same person's other traits, as in forced-choice inventories (Section 2.9), and cannot be compared across people.

> **! EXAM TRAP**
> Of the usual options, only "85 per cent correct, meeting the mastery cut-off" is criterion-referenced; percentile standing, class rank and "1 SD above the mean" are norm-referenced. Norm-referencing compares with a representative standardisation sample, not with a cut-off, an absolute criterion or the person's own earlier scores.

Standardisation and the norm sample ○

The process of **standardisation** means giving and scoring a test identically for everyone (same instructions, materials, time limits and scoring rules) and giving it so to a large sample whose scores become the **norms**. The test maker defines the **target population**, the people the test is meant for, and samples it: simple **random sampling** is ideal but rarely practical, so **stratified sampling** (quotas matching the census on age, sex, region, residence and socio-economic status) or **cluster sampling** (whole schools chosen at random) is used (Section 9.3). Good norms are large, **representative**, **recent** (the Flynn effect, a rise of about 3 intelligence quotient (IQ) points a decade, makes old norms too easy; Section 1.9), **relevant** to the person tested and fully described in the manual. Hence **national norms**, **local norms** (one region or school) and **subgroup norms** (by age, sex, grade, language or occupation). American norms compare an Indian child with American children, so Indian versions of Western tests were renormed (Section 2.6).

Standardisation lets the clinician do all three of the following: see where a client's score falls on the test's distribution of scores; estimate from it whether the client might meet the diagnostic criteria for a disorder (an IQ about 2 SD below the mean is one criterion for intellectual disability, Section 2.7); and estimate how probable that diagnosis is. Computers now score tests against the norm tables and write narrative reports (as for the Minnesota Multiphasic Personality Inventory-2, MMPI-2), saving time and errors, though not replacing clinical judgement.

Age and grade norms; ratio and deviation IQ ○

The oldest derived scores express performance as the age or school grade at which it is typical. An **age-equivalent norm** is the average performance at each age; **Alfred Binet**'s 1908 scale grouped items by age, and the age level a child reached was the **mental age** (MA). A **grade-equivalent norm** does the same for school grades, interpolating in-between values. Both mislead. Growth is uneven (faster at 2 than at 14), so a year is not a fixed unit; traits such as personality do not grow with age; grade equivalents in different subjects are not comparable; and a Class 4 child with an arithmetic grade equivalent of 7.2 has not mastered Class 7 work, only scored on a Class 4 test as an average Class 7 pupil is estimated to.

William Stern divided MA by chronological age (CA) in 1912, and **Lewis Madison Terman** multiplied by 100 in the 1916 Stanford-Binet: the **ratio IQ** = (MA ÷ CA) × 100, so MA 8 and CA 10 give 80. It fails for adults, because mental age stops rising in the late teens (a 40-year-old with an MA of 16 would score 40). **David Wechsler**'s Wechsler-Bellevue scale (1939) introduced the **deviation IQ**, a standard score comparing a person with same-age peers, with a mean of 100 and an SD of 15 at every age (Sections 1.9 and 2.6).

> **! EXAM TRAP**
>
> MA 8 and CA 10 give 80; 125 inverts the ratio (CA ÷ MA). The deviation IQ is not "only for adults", is not less reliable and is not built from unstandardised raw scores: it is a standard score against age peers.

Percentile ranks

The **percentile rank** (PR) is the percentage of the norm group scoring below a given raw score (some books say "at or below"). If 70 per cent of the sample scored below Meera's 65, her PR is 70 and her score is the 70th percentile (P_{70}). It is not the **percentage correct**, a raw score: 70 per cent correct could be the 30th percentile on an easy test and the 99th on a hard one. The **quartiles** cut the distribution into four ($Q_1 = P_{25}$; $Q_2 = P_{50}$, the median; $Q_3 = P_{75}$) and **deciles** into ten ($D_5 = P_{50}$).

Percentiles are **ordinal**, with unequal units. Scores crowd around the middle of a normal distribution, so a small raw difference there jumps many percentile points while a large one in the tails moves few: the 50th to the 60th percentile is a quarter of an SD (IQ 100 to 104), the 90th to the 99th more than a whole SD (IQ 119 to 135). So percentiles exaggerate differences near the middle, hide them in the tails and should not be averaged; equal-unit standard scores solve this.

The z score ●●

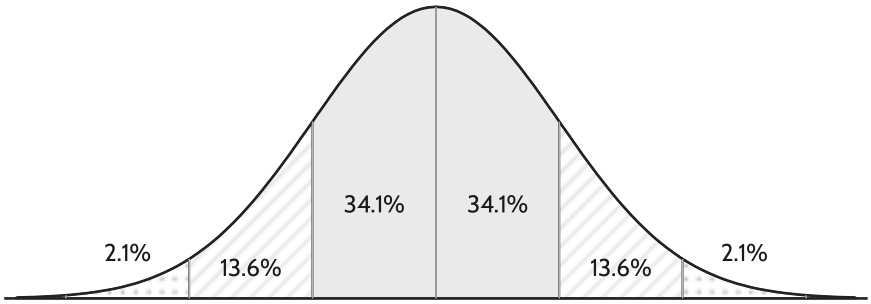
A **standard score** states how far a score lies from the mean in SD units. The basic one is the **z score**: $z = (X - M) \div SD$, with X the raw score and M and SD those of the norm group. z scores have a mean of 0 and an SD of 1; z = +1.0 is exactly one SD above the mean, and negative values lie below it. Their great use is comparing results from different distributions. Arjun scores 60 in psychology (mean 50, SD 5) and 70 in statistics (mean 60, SD 10). The raw scores favour statistics, but psychology is z = +2.0 and statistics z = +1.0, so psychology was his better paper.

Look-alikes answer other questions (Sections 9.8 and 9.9). The **variance** (SD squared) and the **semi-interquartile range**, $(Q_3 - Q_1) \div 2$, measure the spread of a whole distribution; the **coefficient of variation**, $(SD \div M) \times 100$, compares spread across distributions in different units; the **standard error** is the SD of a sampling distribution, such as that of sample means (the standard error of measurement is in Section 2.3); the median, not z, describes the centre of a skewed distribution. Only z places one person within a distribution.

> **! EXAM TRAP**
>
> The measure of how far a data point lies from the mean in SD units, which lets scores from different distributions be compared, is the z score. It is not a measure of variability or of the central tendency of a skewed distribution.

The normal curve: from z to percentile ●●●

Why does z = +1 always mean the 84th percentile? Because test scales are built on the **normal curve**, the symmetrical bell in which mean, median and mode coincide, and its areas are fixed: about 34.1 per cent of cases lie between the mean and 1 SD on each side, 13.6 per cent between 1 and 2 SD and 2.1 per cent between 2 and 3 SD. So about 68 per cent fall within ±1 SD, 95 per cent within ±2 SD and 99.7 per cent within ±3 SD (Figure 2.9). The **z table** of areas under the curve is derived from the normal distribution and gives the proportion of cases below any z.



z score	−3	−2	−1	0	+1	+2	+3
Percentile	0.1	2	16	50	84	98	99.9
T score	20	30	40	50	60	70	80
IQ (SD 15)	55	70	85	100	115	130	145
Scaled score	1	4	7	10	13	16	19

Stanine	1	2	3	4	5	6	7	8	9
% of cases	4	7	12	17	20	17	12	7	4

Figure 2.9 The normal curve and the scales built on it. Read straight down from any point to convert between z, percentile, T score, IQ (SD 15), scaled score and stanine; stanine bands are half an SD wide except the open-ended 1 and 9.

The landmarks: z = 0 is the 50th percentile; +1 the 84th and −1 the 16th; +0.5 about the 69th; +1.5 about the 93rd; +2 about the 98th (97.7) and −2 the 2nd; +3 the 99.9th. With IQ (mean 100, SD 15), 100 is the 50th percentile, 85 the 16th, and 130, two SDs above the mean, about the 98th, so only about 2 in 100 score higher. On a Binet-type scale with an SD of 16, an IQ of 136 is z = 36 ÷ 16 = 2.25; the z table gives .9878 below it, so about 98.8 per cent score lower (98.7 if the entry is truncated).

> **! EXAM TRAP**
>
> ±2 SD holds about 95 per cent (exactly 95 within ±1.96, so 97.5 per cent lies below +1.96); ±1 SD holds 68 and ±3 SD 99.7. The z table comes from the normal distribution, not from the SD, t scores or random sampling. The 84th percentile is z = +1, not +1.5 (93rd) or +2 (98th).

Linear and normalised scores; the T score ●●

z scores carry signs and decimals, so other standard scores rescale them: new score = new mean + (new SD × z). This is a **linear transformation** (new = a + b × old), as are z itself and the **deviation score** X − M; it keeps the raw distribution's shape, so skewed raw scores stay skewed. A **normalised standard score** (an area, or non-linear, transformation) instead turns each raw score into its percentile rank, looks up the z that cuts off that percentage of the normal curve and rescales it, forcing a normal shape. Stanines, stens, the C scale, normalised T scores and most modern intelligence-test scales are normalised, which is justified only with large samples and when the skew lies in the test, not the trait.

The **T score** of **William A. McCall** (1922) has a mean of 50 and an SD of 10: T = 50 + 10z, usually 20 to 80 (±3 SD). Computed from z it is linear; McCall's own scale was normalised. T scores

are the metric of the MMPI and most personality inventories, including the Revised NEO Personality Inventory (NEO-PI-R; Section 2.9). T 70, 2 SD above the mean (about the 98th percentile), was the original MMPI's clinical cut-off; the MMPI-2 uses T 65, 1.5 SD above (about the 93rd).

> **! EXAM TRAP**
>
> T scores have a mean of 50 and an SD of 10, not 100 and 15 (IQ), 0 and 1 (z) or 500 and 100 (CEEB). T 70 = +2 SD, T 65 = +1.5 SD, T 60 = +1 SD.

Stanines, stens and the C scale ●●

The **stanine** ("standard nine"), developed by the United States Army Air Forces in World War II, runs from 1 to 9 with a mean of 5 and an SD of about 2. Stanines are assigned by rank: the lowest 4 per cent of the norm group get 1, the next 7 per cent 2, and so on in the proportions 4, 7, 12, 17, 20, 17, 12, 7 and 4 per cent. Each band but the open-ended 1 and 9 is half an SD wide. Stanine 5 spans z = −0.25 to +0.25, the 40th to 60th percentile, and holds 20 per cent of cases; stanine 4 covers about the 23rd to 40th percentile, and 9 begins at the 96th (Figure 2.9). Built from percentile ranks, stanines are a normalised, non-linear transformation. Stanines 1 to 3 are below average, 4 to 6 average and 7 to 9 above average; single digits are easy to record but coarse.

The **sten** ("standard ten") runs from 1 to 10 with a mean of 5.5 and an SD of 2. It has no middle band: stens 5 and 6 lie just below and above the mean, and 4 to 7 cover the middle 68 per cent. Stens are the scale of **Raymond Bernard Cattell**'s Sixteen Personality Factor Questionnaire (16PF, 1949), not of the Wechsler scales, the Stanford-Binet or the Rorschach. **J. P. Guilford**'s **C scale** (1954) extends the stanine to 11 units (0 to 10; mean 5, SD 2) by splitting off about the extreme 1 per cent at each end.

> **! EXAM TRAP**
>
> Stanine 5 holds 20 per cent, not 17 (stanines 4 and 6). Of z, 50 + 10z, X − M and the stanine, only the stanine is a non-linear transformation.

Scaled scores, deviation IQs and CEEB scores ●●

Each Wechsler subtest is reported as a **scaled score** (mean 10, SD 3, range 1 to 19), and the index scores and Full Scale IQ as deviation IQs (mean 100, SD 15). A scaled score of 10 is exactly average, the 50th percentile; 7 is 1 SD below (about the 16th percentile, like an IQ of 85), which the Wechsler Adult Intelligence Scale, fourth edition (WAIS-IV), calls low average; 13 is 1 SD above (84th); 4 is 2 SD below (2nd), the borderline region. The Stanford-Binet used an SD of 16 from Form L-M (1960) through the fourth edition (1986), returning to 15 in the fifth (2003), and Cattell's Culture Fair Intelligence Test also uses 16. So IQ 130 is z = +2.0 (97.7th percentile) on a Wechsler scale but +1.875 (97th) on an SD-16 scale: compare tests through z. A **CEEB score** (College Entrance Examination Board) has a mean of 500 and an SD of 100, as on the older SAT and Graduate Record Examinations (GRE). Tables 2.11 and 2.12 summarise them all.

Table 2.11 The standard-score systems.

Score	Mean	SD	Range	Type; main use

Score	Mean	SD	Range	Type; main use
z score	0	1	about −3 to +3	Linear; comparing distributions
T score	50	10	20 to 80	Linear or normalised; MMPI, NEO-PI-R
Deviation IQ	100	15	40 to 160	Normalised; Wechsler, SB5
Older Binet IQ	100	16	varies	Normalised; SB L-M to SB4, Culture Fair
Scaled score	10	3	1 to 19	Normalised; Wechsler subtests
Stanine	5	about 2	1 to 9	Normalised; school tests
Sten	5.5	2	1 to 10	Normalised; 16PF
C scale	5	2	0 to 10	Normalised; Guilford
CEEB score	500	100	200 to 800	Older SAT, GRE

Table 2.12 Converting between scores (IQ in brackets for an SD of 16).

z (percentile)	T score	IQ, SD 15 (16)	Scaled score	Stanine
−3.0 (0.1)	20	55 (52)	1	1
−2.0 (2)	30	70 (68)	4	1
−1.5 (7)	35	77.5 (76)	5.5	2
−1.0 (16)	40	85 (84)	7	3
−0.5 (31)	45	92.5 (92)	8.5	4
0 (50)	50	100 (100)	10	5
+0.5 (69)	55	107.5 (108)	11.5	6
+1.0 (84)	60	115 (116)	13	7
+1.5 (93)	65	122.5 (124)	14.5	8
+2.0 (98)	70	130 (132)	16	9
+3.0 (99.9)	80	145 (148)	19	9

! EXAM TRAP

Mean 10 and SD 3 identify the scaled score, and 10 is the 50th percentile. One mock keys a scaled score of 7 as "within average range (1 SD below mean)": the best of its options, since 7 is neither impaired nor borderline (4 is), but the WAIS-IV calls it low average.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. "85 per cent correct, meeting the mastery cut-off" is a _____-referenced interpretation.
2. A mental age of 8 and a chronological age of 10 give a ratio IQ of _____.
3. About _____ per cent of cases fall within ±2 SD of the mean.
4. A percentile rank of 84 corresponds to z = _____.
5. A T score of 65 lies _____ SD above the mean.
6. Stanine 5 holds _____ per cent of cases.
7. Stens (mean 5.5) are the scale of Cattell's _____.

Answers: (1) criterion, (2) 80, (3) 95, (4) +1, (5) 1.5, (6) 20, (7) 16PF.

2.6 Intelligence Tests: Binet, Wechsler, Non-verbal and Indian Scales

MEDIUM YIELD · 35 QS

In 1904 the French education ministry asked which Paris schoolchildren could not keep up without special help. The answer, a list of graded tasks, founded psychology's most used family of tests. This section follows it through the Binet, Wechsler and Kaufman scales, the non-verbal tests and the Indian adaptations; theories of intelligence and the IQ itself were introduced in Section 1.9.

What an intelligence test can and cannot do ●●

An **individual test** is given by one examiner to one person, who can be watched and questioned (the Stanford-Binet, the Wechsler scales); a **group test** is given to many people at once. A **verbal test** uses language in its items; a **non-verbal test** keeps language out of the items; a **performance test** asks the person to handle materials, such as blocks or a form board, usually against the clock.

What does a good IQ test offer? High test-retest reliability, high internal consistency (both above .90 for the major full-scale IQs) and good validity for predicting school achievement (it correlates about .5 with grades). It does not offer a good prediction of behaviour in general: honesty, persistence and social skill depend on personality, motivation and circumstance.

Intelligence tests diagnose intellectual disability and giftedness, guide educational placement, set a neuropsychological baseline and document eligibility for a disability allowance, but not for housing benefit (an income question), suicide risk (a clinical risk assessment, Chapter 7) or, with an adult scale, a child's special educational needs. The Wechsler Adult Intelligence Scale (WAIS) is an intelligence test, not a personality, aptitude (predicting future success), attitude or achievement (measuring past learning) test (Sections 2.8 and 2.9), and it does not look for neurological soft signs, subtle non-localising signs such as clumsiness.

> **! EXAM TRAP**
>
> "An IQ test does NOT provide ..." is answered by "a good predictor of behaviour". High test-retest reliability, high internal consistency and good validity (for school achievement) are exactly what it does provide.

The Binet scales ●●

Alfred Binet (1857 to 1911) and the physician **Theodore Simon** (1872 to 1961) published the first practical intelligence test in 1905: 30 items in order of difficulty. In 1908 they grouped the items by the age at which most children pass them, creating the first **age scale**; the highest level a child reaches is the **mental age**. The 1911 revision, the year Binet died, extended it to adults.

Lewis Madison Terman (1877 to 1956) standardised it at Stanford University as the Stanford-Binet (1916), using William Stern's ratio IQ (mental age ÷ chronological age × 100): the original Stanford-Binet used the ratio IQ, not a deviation IQ, percentile rank or T-score (Section 2.5). Later editions (Table 2.13) moved away from mental age: parallel Forms L and M with **Maud Merrill** (1937); Form L-M (1960), with a deviation IQ of SD 16; and the SB4 (1986) of **Robert L. Thorndike**, **Elizabeth Hagen** and **Jerome M. Sattler**, a **point scale** (each subtest scored by items passed, whatever their age level) reporting Standard Age Scores, SD 16.

The Stanford-Binet Fifth Edition (SB5, 2003) of **Gale H. Roid** covers ages 2 to 85 and over, with mean 100 and SD 15. Its five factors, Fluid Reasoning, Knowledge, Quantitative Reasoning, Visual-Spatial Processing and Working Memory, are each tested in a verbal and a non-verbal domain: 10 subtests yielding Full Scale, Verbal and Nonverbal IQs. Two **routing subtests** (Object Series/Matrices and Vocabulary) set the starting level for the rest. Within a subtest the examiner finds a **basal level**, below which all items are assumed passed, and a **ceiling level**, where failures stop testing.

Table 2.13 The Binet line, from mental age to deviation IQ.

Edition	Year	Authors	Scale and score
Binet-Simon	1905, 1908, 1911	Binet, Simon	30 items; age scale, mental age (1908)
Stanford-Binet	1916	Terman	Ratio IQ
Forms L and M	1937	Terman, Merrill	Parallel forms; ratio IQ
Form L-M	1960	Terman, Merrill	Deviation IQ, SD 16
SB4	1986	Thorndike, Hagen, Sattler	Point scale; Standard Age Scores, SD 16
SB5	2003	Roid	Ages 2 to 85+; 5 factors, 10 subtests; SD 15

> **! EXAM TRAP**
>
> The Stanford-Binet grew from Binet and Simon, revised by Terman at Stanford; not Wechsler and Terman, Spearman and Thurstone (factor theorists, Section 1.9) or Galton and Cattell (sensory tests, Section 2.1). The SB5's factors are not "sensory, motor, memory, attention, language" (a neuropsychological checklist) or "Verbal and Performance IQ only" (old Wechsler).

Wechsler, the deviation IQ and the WAIS-IV ●●●

David Wechsler (1896 to 1981), chief psychologist at Bellevue Hospital in New York, found the Stanford-Binet unsuited to the adults he tested. It had little appeal for adults; it put too much emphasis on word manipulation; it stressed speed over accuracy; and it relied on mental age, which stops growing in the late teens, so an adult's ratio IQ falls with every birthday. His Wechsler-Bellevue Intelligence Scale (1939) answered each flaw: a point scale with a Performance scale of non-verbal tasks beside the Verbal scale, scored by a new **deviation IQ**, also called the standard IQ because it is a standard score.

The deviation IQ compares the actual score with the expected mean score for the person's age group: IQ = 100 + 15 × (score minus age-group mean) ÷ age-group SD. Scoring at your age group's mean gives 100 at any age; one SD above gives 115. Wechsler defined intelligence (1944) as "the aggregate or global capacity of the individual to act purposefully, to think rationally, and to deal effectively with his environment".

> **! EXAM TRAP**
>
> Wechsler's new method was the deviation ("standard") IQ, not a mental-age, physical-age or age-relative formula IQ; mocks write it as actual score ÷ expected mean score for age. His target flaws were no appeal for adults, word manipulation, speed over accuracy and mental age, never "children", "physical age" or "accuracy over speed".

The adult line runs from the WAIS (1955; 6 verbal and 5 performance subtests) through the WAIS-R (1981) and WAIS-III (1997), which still gave Verbal, Performance and Full Scale IQs, to the WAIS-IV (2008; ages 16:0 to 90:11), which dropped the first two. It yields a **Full Scale IQ** (FSIQ) from 10 **core subtests** and four **index scores**, composites for narrower abilities: Verbal Comprehension (VCI), Perceptual Reasoning (PRI), Working Memory (WMI) and Processing Speed (PSI) (Figure 2.10). Five **supplemental subtests** can replace a spoiled core subtest. The VCI reflects verbal reasoning and acquired knowledge; the PRI, reasoning with visual, non-verbal material. The **Working Memory Index**, built on holding and manipulating digits and mental sums, reflects attention, concentration and mental control; the **Processing Speed Index**, how quickly simple visual material is scanned and acted on. The optional **General Ability Index** (GAI, VCI plus PRI) estimates reasoning with less weight on memory and speed; the **Cognitive Proficiency Index** (CPI) pairs WMI and PSI. The WAIS-5 (2024) splits perceptual reasoning into Visual Spatial and Fluid Reasoning: five primary indices.

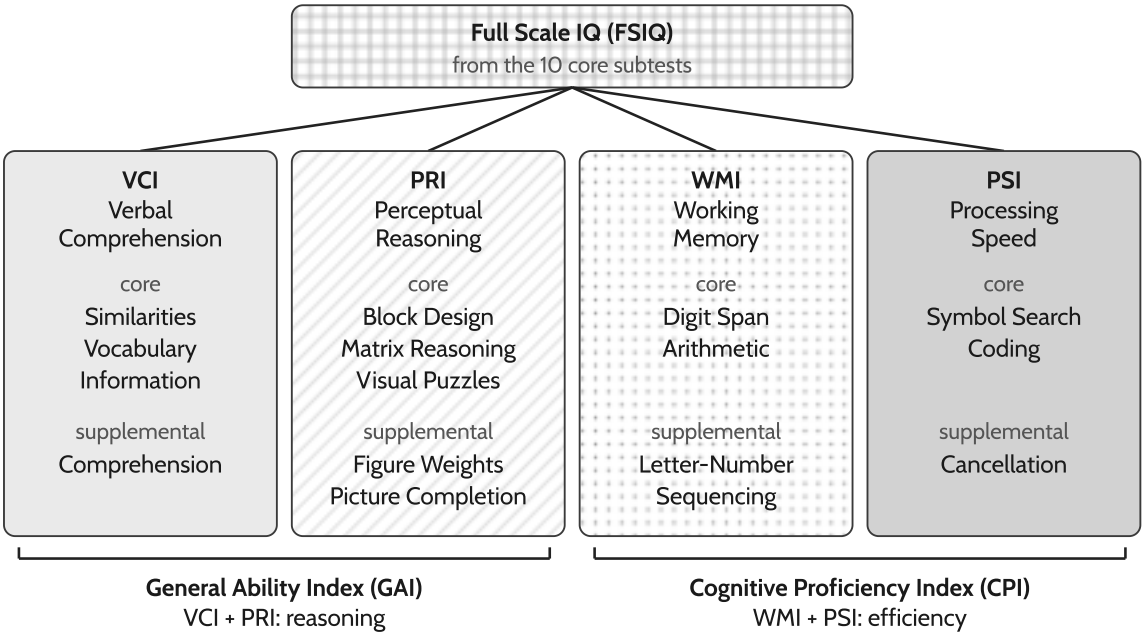


Figure 2.10 The WAIS-IV. Count the core subtests under each index: three verbal, three perceptual, two working memory, two processing speed. The optional GAI and CPI pair the indices.

> **! EXAM TRAP**
>
> The WAIS-IV gives four indices (VCI, PRI, WMI, PSI) beside the FSIQ, not two, three or five; the GAI is an optional summary. Core subtests: 3 verbal, 3 perceptual. "VIQ, PIQ, FSIQ and Memory Quotient" mixes the WAIS-III with the Wechsler Memory Scale (Section 2.11); "crystallised, fluid, visual, auditory (or quantitative)", "VIQ, PIQ, memory, attention" and "cognitive, behavioural, emotional, adaptive" are not WAIS-IV indices.

WISC, WPPSI and reading a Wechsler profile ●●

The Wechsler Intelligence Scale for Children (WISC, 1949) and the Wechsler Preschool and Primary Scale of Intelligence (WPPSI, 1967) extend the design downwards (Table 2.14). The WISC-V (2014; ages 6:0 to 16:11) has five primary indices, Verbal Comprehension, Visual Spatial, Fluid Reasoning, Working Memory and Processing Speed, from 10 primary subtests, and takes the FSIQ from seven; Information, Arithmetic and Letter-Number Sequencing are among its secondary subtests.

Table 2.14 The Wechsler family today.

Scale	Editions	Ages	Structure
WAIS-IV	1955, 1981, 1997, 2008	16:0 to 90:11	FSIQ + 4 indices; 10 core, 5 supplemental subtests
WAIS-5	2024	16:0 to 90:11	5 primary indices
WISC-V	1949, 1974, 1991, 2003, 2014	6:0 to 16:11	5 primary indices; FSIQ from 7 subtests
WPPSI-IV	1967 to 2012	2:6 to 7:7	Preschool and early school

> **! EXAM TRAP**
>
> Age ranges: WISC-V 6:0 to 16:11; WPPSI-IV 2:6 to 7:7; WAIS-IV 16:0 to 90:11; Bayley scales to 42 months (Section 2.7). A 2024 paper keys Letter-Number Sequencing as "core in the WAIS-IV but not the WISC-V". In fact it is supplemental in the WAIS-IV and secondary in the WISC-V (core in the WISC-IV). Digit Span, Symbol Search and Matrix Reasoning are core or primary in both, so mark Letter-Number Sequencing, but Information and Arithmetic truly fit the stem.

Which ability does each subtest tap? Vocabulary and Information measure crystallised intelligence, knowledge acquired through schooling and culture. Matrix Reasoning (completing a visual pattern) measures fluid reasoning, solving novel problems without learned knowledge; Block Design (copying a design with coloured cubes) measures visuospatial analysis and construction; Coding (formerly Digit Symbol: copying number-paired symbols at speed) measures processing speed.

The FSIQ is placed in a descriptive band (Table 2.15). Below 70 is extremely low, but intellectual disability also requires deficits in adaptive behaviour (Section 2.7). Scores of 70 to 84 straddle borderline (70 to 79) and low average (80 to 89); older systems called 71 to 84 borderline intellectual functioning. The examiner then studies **index discrepancies**, differences between index scores, and **scatter**, unevenness among subtest scores. A VCI of 115 beside a PSI of 82 is interpreted only if the gap is statistically significant and rare in the norm sample; the FSIQ then averages over a real split, and the separate indices or the GAI tell more.

Table 2.15 Wechsler descriptive classification of IQ (WAIS-IV labels).

IQ	Classification
130 and above	Very superior
120 to 129	Superior
110 to 119	High average
90 to 109	Average
80 to 89	Low average
70 to 79	Borderline
69 and below	Extremely low

Kaufman, Das-Naglieri and Woodcock-Johnson ○

Alan S. Kaufman and **Nadeen L. Kaufman** built their tests on theory. The Kaufman Assessment Battery for Children (K-ABC, 1983) used Alexander Luria's sequential processing (handling items in serial order) and simultaneous processing (integrating many at once), and scored achievement apart from ability. The KABC-II (2004, ages 3 to 18) lets the examiner choose a Luria model or a Cattell-Horn-Carroll (CHC) model. The Kaufman Brief Intelligence Test (K-BIT, 1990; KBIT-2, 2004, ages 4 to 90) is a quick screen with a Verbal or Crystallised scale and a Nonverbal or Fluid scale (matrices). Reading, mathematics, quantitative and analytical scales belong to achievement and scholastic aptitude tests, such as the ACT, SAT and GRE admission tests (Section 2.8); an "EQ scale" belongs to emotional intelligence (Section 1.9).

The Indian-born **J. P. Das** and **Jack A. Naglieri** turned Luria's three functional units of the brain into the **PASS theory** (Planning, Attention, Simultaneous and Successive processing), measured by their Cognitive Assessment System (1997). The Woodcock-Johnson Tests of Cognitive Abilities (**Richard W. Woodcock** and **Mary Bonner Johnson**, 1977) are, in current editions, built on CHC theory.

Non-verbal and culture-fair tests ●●

Vocabulary items reward schooling and language as much as intelligence. A **culture-free test** would leave culture out entirely; none exists, since even a pattern on paper assumes familiarity with pictures and timing. A **culture-fair test** merely minimises language, reading and specific knowledge, so that people from different backgrounds start more nearly level.

The classic is Raven's Progressive Matrices by **John Carlyle Raven** (1902 to 1970). Each item is a pattern with a piece missing. The Standard Progressive Matrices (1938) has 60 items in five sets, A to E, growing progressively harder; the Coloured version (1947) serves children of 5 to 11, elderly people and disabled people; the Advanced version serves able adults. Raven aimed at Spearman's g through **eductive ability**, seeing relations in confusion, as against recalling learned information. The test measures fluid, non-verbal reasoning by abstract pattern completion with minimal language, and is one of the strongest single markers of g.

> **! EXAM TRAP**
>
> Raven's measures fluid intelligence; it is culture-fair, not culture-free, and does not measure crystallised, verbal or social intelligence, reading comprehension or memory alone. Offered "general intelligence" beside "fluid", choose fluid: Raven's marks g, but its content is fluid and non-verbal. "Both" is wrong.

Raymond Bernard Cattell (1905 to 1998) named his test the Culture Free Intelligence Test (1940), then renamed it Culture Fair. Its three Scales are graded by age and ability; Scales 2 and 3 use four figural subtests: series, classification, matrices and conditions. Other tests remove speech from the task: the Leiter International Performance Scale of **Russell Graydon Leiter** (1929; Leiter-3, 2013) gives its instructions by pantomime, for deaf or autistic children; the Columbia Mental Maturity Scale needs only pointing to the drawing that does not belong, suiting children with cerebral palsy; and the Peabody Picture Vocabulary Test of **Lloyd M. Dunn** (1959) measures receptive vocabulary as the person points to the picture the examiner names.

The simplest non-verbal test is a pencil and the request "draw a man". **Florence Laura Goodenough** (1886 to 1959) noticed that children add details to drawings in a regular developmental order. Her **Draw-a-Man test** (1926) credits up to 51 details as a non-verbal estimate of intellectual maturity and visual-motor ability; **Dale B. Harris** revised it in 1963. Because it asks so little, it works with a client too disorganised for verbal or performance subtests. **Karen Machover**'s Draw-a-Person uses the same request projectively, to read personality (Section 2.10).

> **! EXAM TRAP**
>
> A disorganised client who cannot manage verbal or performance tasks is asked to draw a person: the domain targeted is visual-motor integration and cognitive estimation (Goodenough), not projective personality assessment (Machover), verbal comprehension or processing speed.

Group testing arrived with the First World War. In 1917 a committee led by **Robert Mearns Yerkes** (1876 to 1956) built the Army Alpha, a verbal test for literate recruits, and the Army Beta, a non-verbal one for illiterate and non-English-speaking recruits. **Arthur Otis**, Terman's student, devised the multiple-choice group format the Alpha borrowed (today's Otis-Lennon School Ability Test). The Primary Mental Abilities tests of **Louis Leon Thurstone** (1887 to 1955) score seven primary abilities separately instead of giving one IQ.

Performance tests and Indian scales ●●

India needed tests that did not lean on literacy or on one language, so its psychologists normed performance tests and adapted Western scales (Table 2.16). In the Kohs Block Design test (**Samuel C. Kohs**, 1920; book 1923) the person arranges coloured cubes to copy a printed design; Wechsler borrowed it as Block Design. In the Alexander Pass-Along test (**W. P. Alexander**, 1932) blocks are slid along a tray, never lifted, to reach a target pattern. The Seguin Form Board, a teaching device of **Edouard Seguin** (1812 to 1880) for children with intellectual disability, became a test in **Henry Herbert Goddard**'s version: a child replaces 10 shapes over three trials, and the shortest time converts to a mental age. In the Gesell Drawing Test of **Arnold Gesell** (1880 to 1961) a child copies geometric figures mastered at typical ages.

C. M. Bhatia assembled **Bhatia's Battery of Performance Tests of Intelligence** (1955), standardised on Indian children. Its five subtests are Kohs Block Design, Alexander Pass-Along, Pattern Drawing, Immediate Memory (digits, the one non-performance task) and Picture Construction; the Seguin Form Board is not among them. Needing no reading or writing, it is the test of choice for an illiterate child or adult from a rural area, because it minimises cultural and educational bias; the WISC-IV and SB5 carry Western verbal content, and MISIC is half verbal and assumes schooling. Being visual, it cannot serve a patient with cortical blindness.

> **! EXAM TRAP**
>
> For an illiterate rural 10-year-old, choose Bhatia's Battery over the WISC-IV, SB5 or MISIC. Its population is illiterate or rural Indians, adolescents and adults included, not urban English speakers or preschoolers only.

The adaptations follow their parents. The **Binet-Kamat Test** (**V. V. Kamat**, 1934; revised 1967) adapted the 1916 Stanford-Binet; it is a heavily verbal age scale giving a mental age and ratio IQ. **Malin's Intelligence Scale for Indian Children** (MISIC; **Arthur J. Malin**, Nagpur, 1969) adapted the WISC for ages 6 to 15, so it can be given from 6 years, and yields Verbal and Performance Quotients and a Full Scale IQ. The **Wechsler Adult Performance Intelligence Scale** (WAPIS; **P. Ramalingaswamy**, 1974) adapted the WAIS performance subtests for Indian adults. **Pramila Phatak** of Baroda produced Indian norms for the Draw-a-Man test, and

Raven's matrices also have Indian norms. Pearson's Indian editions of the WISC-IV and WAIS-IV are newer and poorly documented.

> **! EXAM TRAP**
>
> A mock keys "WAIS-IV India edition" as the Indian WAIS common in clinics. Learn WAPIS (Ramalingaswamy) as the classic Indian adult adaptation; the BKT and MISIC are children's tests, so "all of the above" fails.

Table 2.16 Performance and Indian tests compared.

Test	Author, year	Ages	Key fact
Raven's SPM	J. C. Raven, 1938	About 6 to adult	60 items; Indian norms exist
Draw-a-Man	Goodenough, 1926	About 3 to 15	51 points; Indian norms, Phatak
Seguin Form Board	Seguin; Goddard	Young children	10 shapes, 3 trials
Bhatia's Battery	C. M. Bhatia, 1955	Normed 11 to 16	5 subtests; no reading
Binet-Kamat	V. V. Kamat, 1934	3 to 22	From 1916 Stanford-Binet; ratio IQ
MISIC	A. J. Malin, 1969	6 to 15	From WISC; VQ, PQ, FSIQ
WAPIS	P. Ramalingaswamy, 1974	Adults	From WAIS performance scale

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. An IQ test is reliable and predicts school achievement, but it is not a good predictor of _____ in general.
2. The original 1916 Stanford-Binet reported the _____ IQ.
3. The WAIS-IV Working Memory Index has two core subtests, Digit Span and _____.
4. The WISC-V is normed for ages 6:0 to _____.
5. The SB5 subtests that set the starting level for the rest are the _____ subtests.
6. Raven's Progressive Matrices is culture-fair, not culture-_____.
7. Arthur J. Malin's MISIC is the Indian adaptation of the _____.

Answers: (1) behaviour, (2) ratio, (3) Arithmetic, (4) 16:11, (5) routing, (6) free, (7) WISC.

2.7 Assessing Development, Adaptive Behaviour and Disability

LOWER YIELD · 21 QS

An intelligence test asks how well a person reasons when tested. Two other questions matter as much. Has this infant reached the milestones expected for her age? And how well does a child or adult actually manage everyday life? Their answers decide who is diagnosed with intellectual disability and who in India is certified as disabled.

Beyond IQ: quotients, adaptive behaviour and informants ○

An infant cannot follow test instructions, so you measure how far she has come along the expected path. A **developmental scale** lists milestones at the ages most children reach them. The level reached is the **developmental age** (DA), and the **developmental quotient** is DQ = DA / CA × 100, CA being chronological age. A social maturity scale gives a **social age** (SA) and a **social quotient**, SQ = SA / CA × 100: if 6-year-old Ravi (72 months) manages daily life like a typical child of 54 months, his SQ is 75. Modern scales replace these crude ratios with deviation scores, mean 100 and SD 15 (Section 2.5).

Why measure everyday competence apart from IQ? Because one child scoring 65 dresses, shops and takes the bus alone, while another scoring 75 cannot look after herself. The American Association on Intellectual and Developmental Disabilities (AAIDD, formerly the AAMD) defines **adaptive behaviour** as the conceptual (language, money, number), social (relationships, rules, avoiding victimisation) and practical (self-care, safety, travel, work) skills that people learn and perform in everyday life. Since it is what a person habitually does, not what he can do at his best, it is usually rated by an **informant**, a parent, caregiver or teacher, which also suits infants and the severely disabled.

A **screening test** is short, is given to many children and only flags those needing a closer look; a **diagnostic test** is longer and individual, and yields a level on which a diagnosis can rest. And because symptoms vary across contexts (the boy restless in a noisy classroom may be calm at home), clinicians use **multi-informant assessment**, combining parents, teachers, the child and clinic observation.

KEY STUDY: ACHENBACH'S CROSS-INFORMANT CORRELATIONS (1987)

Thomas M. Achenbach and colleagues pooled studies in which several people rated the same children. Similar informants, such as two parents, agreed moderately (mean correlation about 0.60); different kinds, such as parent and teacher, only weakly (about 0.28). Disagreement, they argued, reflects real differences between settings, so each informant adds valid information: the logic of Achenbach's parallel parent Child Behavior Checklist, Teacher's Report Form and Youth Self-Report. Criticism: it does not say whose report should decide.

! EXAM TRAP

Multi-informant assessment is recommended because **symptoms may vary across contexts**, not because parents over-report, teachers add nothing or clinic observation alone is insufficient.

Infant and preschool developmental scales ○

Arnold Gesell (1880 to 1961) of the Yale Clinic of Child Development filmed thousands of infants and turned the timing of their behaviour into norms (1925, 1940). His **Gesell Developmental Schedules** place observed behaviour at an age level in four fields: motor, adaptive (solving problems with objects, not adaptive behaviour in the Vineland sense), language and personal-social, each giving a DQ.

Nancy Bayley (1899 to 1994) built the leading diagnostic infant test, the **Bayley Scales of Infant Development** (BSID, 1969). The BSID-II (1993) gave a Mental Development Index

(MDI) and a Psychomotor Development Index (PDI). The Bayley-III (2006), renamed Infant and Toddler Development, and the Bayley-4 (2019) have five scales: cognitive, language and motor, tested with the child, and social-emotional and adaptive behaviour, reported by the caregiver. Its range since 1993, the first month to 42 months, is its signature.

For screening, the **Denver Developmental Screening Test** of **William K. Frankenburg** and **Josiah B. Dodds** (1967; Denver II, 1990) covers birth to 6 years in four sectors: personal-social, fine motor-adaptive, language and gross motor. The result is normal, suspect or untestable, never a quotient. The **McCarthy Scales of Children's Abilities** (**Dorothea McCarthy**, 1972; 2½ to 8½ years) yield a General Cognitive Index. **Virginia Apgar**'s newborn score (1953) and **T. Berry Brazelton**'s Neonatal Behavioral Assessment Scale (1973) are taught in Chapter 6.

The Indian **Developmental Screening Test** (DST) of **J. Bharath Raj** (Mysore, 1977, 1983) places 88 milestones on Indian norms from birth to 15 years and gives a DQ. The **Developmental Assessment Scales for Indian Infants** (DASII) of **Pramila Phatak** and colleagues (Baroda) restandardised the Bayley on Indian infants from birth to 30 months, giving motor and mental DQs. The **Trivandrum Developmental Screening Chart** of **M. K. C. Nair** and colleagues (1991), built from the Baroda norms, screens children under 2 in minutes.

> **! EXAM TRAP**
>
> The scale for **birth to 42 months** across cognitive, language and motor domains is the **Bayley**, not the WPPSI (from 2½ years), the Stanford-Binet (from 2; both Section 2.6) or the McCarthy (from 2½). Do not confuse Bharath Raj's Indian DST with the American Denver, or the DASII (the Indian Bayley) with the TDSC (a Kerala screening chart).

Social maturity: the Vineland Social Maturity Scale ••

Intelligence tests missed something institution staff saw daily: residents of the same mental age differed hugely in self-care. **Edgar Arnold Doll** (1889 to 1968), director of research at the Vineland Training School in New Jersey, measured that difference with the **Vineland Social Maturity Scale** (VSMS, 1935), the first standardised measure of social competence, or adaptive behaviour.

Its 117 items run in the order in which they are usually achieved, grouped year by year through childhood and in wider bands up to "25 and above", so the scale spans birth to adulthood. They fall into eight categories: self-help general, self-help eating, self-help dressing, self-direction, occupation, communication, locomotion and socialisation. The scale is not given to the person: an informant who knows him well is interviewed, habitually performed items are credited under standardised scoring rules, and the total converts to a social age and an SQ.

The Indian version, **Malin's VSMS**, is by **Arthur J. Malin** of Nagpur, who also adapted the WISC as MISIC (Section 2.6). It keeps the eight categories but has 89 items for birth to 15 years on Indian norms, and its SQ is used with IQ in Indian disability certification.

> **! EXAM TRAP**
>
> The VSMS was devised by **Doll**, not Malin (the Indian adaptation), Bharath Raj (the DST), C. M. Bhatia (a performance battery), Terman, Wechsler or Binet. It yields **social age and social quotient**, not mental age and IQ, developmental age and DQ, or emotional age and EQ. Its items run past 25 years, so "applicable from 0 to 25 years only" is false; age-graded items set out in age-level groups (the "separate forms for different age groups" of the options) and standardised scoring are genuine features. Malin VSMS: 89 items, SQ; Bharath Raj DST: 88 items, DQ; both 0 to 15 years.

Adaptive behaviour scales and structured interviews ●●

Sara S. Sparrow, **David A. Balla** and **Domenic V. Cicchetti** rebuilt Doll's scale as the **Vineland Adaptive Behavior Scales** (1984; Vineland-II, 2005; Vineland-3, 2016); the current edition covers birth to 90 years. They measure adaptive functioning in daily life in four domains, communication, daily living skills, socialisation and motor skills (for young children), plus an optional maladaptive behaviour domain; the first three form the **Adaptive Behavior Composite** (mean 100, SD 15). Given as a caregiver interview or parent and teacher forms, the Vineland supplies the adaptive half of an intellectual disability diagnosis. The older **AAMD Adaptive Behavior Scale** (**Kazuo Nihira** and colleagues, 1969) rates adaptive skills in Part One and maladaptive behaviour in Part Two.

The Vineland belongs to a family of clinician-administered instruments with the **K-SADS** (Kiddie Schedule for Affective Disorders and Schizophrenia; **Joaquim Puig-Antich**, 1978), a semi-structured psychiatric interview of parent and child, and the ADI-R and ADOS-2 for autism (below). Table 2.17 compares this section's scales.

> **! EXAM TRAP**
>
> The Vineland measures **adaptive behaviour**, not IQ or verbal IQ, personality, psychopathology, psychotic symptoms, neurological soft signs, memory deficits or suicidal ideation. The Vineland-3, K-SADS, ADI-R and ADOS-2 are **interviews and observation schedules**, not EEG paradigms, projective techniques or self-report cognitive tests.

Table 2.17 Developmental, social maturity and adaptive behaviour scales compared.

Scale	Author, year	Items or parts	Ages	Score
Gesell Schedules	Arnold Gesell, 1925, 1940	4 fields	Infancy to 5 or 6 years	DQ
Bayley	Nancy Bayley, 1969 to 2019	5 scales from 2006	1 to 42 months	MDI, PDI (BSID-II); composites
Denver II	Frankenburg, Dodds, 1967, 1990	4 sectors	0 to 6 years	Normal, suspect, untestable
McCarthy Scales	Dorothea McCarthy, 1972	18 subtests	2½ to 8½ years	General Cognitive Index
DST (India)	J. Bharath Raj, 1977, 1983	88 items	0 to 15 years	DQ
DASII (India)	Pramila Phatak, Baroda	Motor, mental	0 to 30 months	Motor and mental DQ
TDSC (India)	M. K. C. Nair, 1991	Short chart	0 to 2 years	Delay or not

Scale	Author, year	Items or parts	Ages	Score
VSMS	Edgar A. Doll, 1935	117 items, 8 categories	Birth to 25 and above	SA, SQ
Malin's VSMS	A. J. Malin, Nagpur	89 items, 8 categories	0 to 15 years	SA, SQ
Vineland ABS	Sparrow, Balla, Cicchetti, 1984	4 domains	Birth to 90 (since 2005)	ABC, mean 100
AAMD ABS	Kazuo Nihira, 1969	2 parts	Children, adults	Domain scores

Assessing intellectual disability ●●

DSM-5 (2013) diagnoses **intellectual disability** (formerly mental retardation) on three criteria. (A) Deficits in intellectual functions such as reasoning and learning, confirmed by clinical assessment and individual standardised testing: about 2 SD below the mean, an IQ of about 65 to 75 allowing for measurement error (Section 2.3). (B) Deficits in adaptive functioning in conceptual, social and practical domains. (C) Onset in the developmental period. The disorder itself is in Chapter 8.

Severity (mild, moderate, severe, profound) is graded by **adaptive functioning**, not IQ, since that decides the support needed. Older systems used IQ bands (Table 2.18): ICD-10's mild 50 to 69, moderate 35 to 49, severe 20 to 34, profound below 20, and DSM-IV-TR's overlapping bands. ICD-11 (in force 2022) renamed the condition **disorders of intellectual development**, adding a provisional level.

Old educational labels set expectations: mild was **educable** (academics to about sixth grade by the late teens), moderate trainable (about second grade), severe and profound custodial. So an IQ of about 50, at the mild-moderate border, legitimately predicts learning to about fifth grade, not lifelong institutional care, constant supervision or high-school graduation. And **borderline intellectual functioning** (IQ about 71 to 84) is not intellectual disability, and an FSIQ of 70 without adaptive deficits is no diagnosis.

Table 2.18 Levels of intellectual disability by IQ band (DSM-5 grades by adaptive functioning instead). Indian percentages are those widely quoted from the earlier certification guidelines.

Level	ICD-10 IQ (adult mental age)	DSM-IV-TR IQ	Old label	Indian disability
Borderline (not ID)	Not a category	71 to 84	None	25 percent (IQ 70 to 79)
Mild	50 to 69 (9 to under 12 years)	50 to 55 up to about 70	Educable	50 percent
Moderate	35 to 49 (6 to under 9)	35 to 40 up to 50 to 55	Trainable	75 percent
Severe	20 to 34 (3 to under 6)	20 to 25 up to 35 to 40	Custodial	90 percent
Profound	Below 20 (under 3)	Below 20 or 25	Custodial	100 percent

> **! EXAM TRAP**
>
> An RML 2020 paper keys severe intellectual disability as IQ 21 to 34; the standard ICD-10 band is **20 to 34**, so pick 21 to 34 only when 20 to 34 is absent. An FSIQ of 70 is never "mild ID solely on score": without adaptive deficits the keyed answer is borderline intellectual functioning, because ID needs both criteria.

Global developmental delay or intellectual disability? ●●

What if the child is 3 and cannot sit through an IQ test? DSM-5 reserves **global developmental delay** for children under 5 years who miss milestones in several areas but are too young for reliable standardised testing; a DQ describes them until they are reassessed. From 5 years the diagnosis is intellectual disability. The key difference is the **age of assessment**. So the label is inappropriate for a 7-year-old with an FSIQ of 65 and severe adaptive impairment, who meets criteria for intellectual disability, but fits a child too young to test or one whose test or adaptive data are incomplete (Figure 2.11).

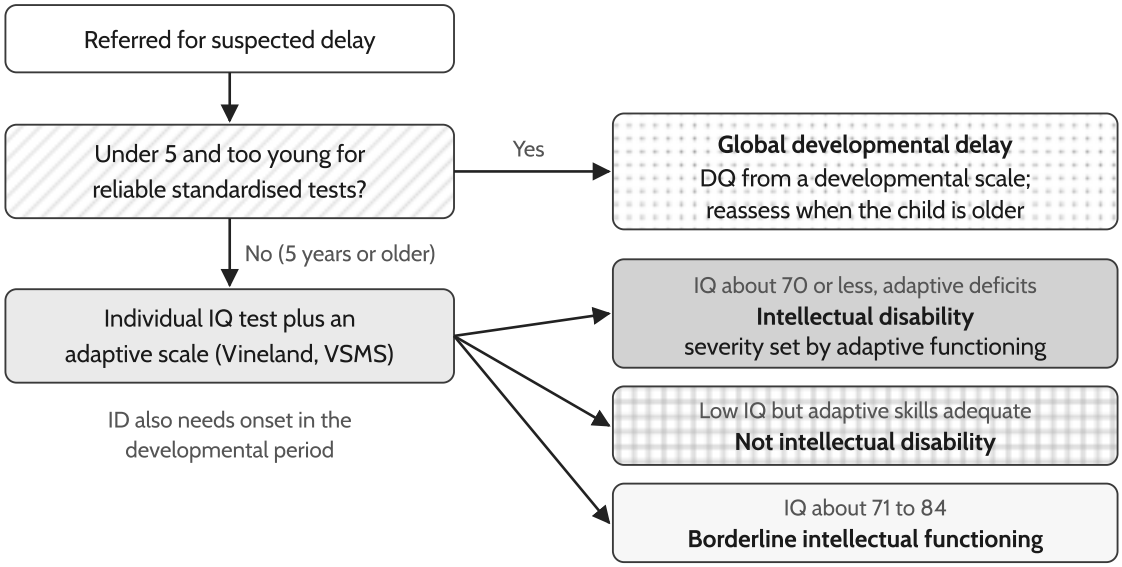


Figure 2.11 Age decides between global developmental delay and intellectual disability; after 5, both IQ and adaptive deficits are needed.

> **! EXAM TRAP**
>
> Global developmental delay versus intellectual disability turns on the **age of assessment** (under 5 versus 5 and above), not on social deficits, language delay alone or motor impairment.

Disability certification in India ○

The **Rights of Persons with Disabilities Act 2016** (RPWD Act) recognises 21 disabilities, up from 7 under the 1995 Act, among them intellectual disability, specific learning disabilities, autism spectrum disorder and mental illness. Forty percent or more of a specified disability is a **benchmark disability**, which opens reservations and other benefits. Guidelines notified under the Act in 2018 set out how each disability is assessed.

For intellectual disability, IQ comes from Indian or Indian-normed tests (Binet-Kamat, MISIC, WAPIS, Bhatia's battery, the Seguin Form Board, Raven's matrices: Section 2.6), the social quotient from the VSMS, and a DQ for children under 5, who are certified with global developmental delay. A medical board that includes a clinical psychologist certifies. The percentages widely quoted for each IQ band (Table 2.18) come from earlier guidelines; check them against the current notification.

A **specific learning disability** (dyslexia, dysgraphia, dyscalculia; Chapter 8) is assessed and certified only from 8 years of age, since before then ordinary variation in learning to read, write and calculate cannot be told apart from a disorder. An intelligence test shows that general ability is not the problem, and the **NIMHANS Index for Specific Learning Disabilities**, with curriculum-based tests, measures the deficits. Disability from mental illness is rated with IDEAS, the Indian Disability Evaluation and Assessment Scale (Section 2.11).

> **! EXAM TRAP**
>
> The minimum age for assessing specific learning disability under the RPWD guidelines is **8 years**, not 7, 10 or 14.

Assessing autism: ISAA, CARS and the diagnostic pair ●●

Two rating scales dominate Indian autism assessment (the disorder is in Chapter 8). In the **Childhood Autism Rating Scale** (CARS) of **Eric Schopler**, **Robert J. Reichler** and **Barbara Rochen Renner** (North Carolina; 1980, manual 1988; CARS-2, 2010), the clinician observes the child, hears the parents and rates 15 areas, such as relating to people and imitation, from 1 (normal for age) to 4 (severely abnormal). Totals run from 15 to 60; 30 and above indicates autism.

The **Indian Scale for Assessment of Autism** (ISAA) was developed in 2009 by the National Institute for the Mentally Handicapped, Secunderabad, partly from CARS items. Its 40 items cover six domains (social relationship and reciprocity, emotional responsiveness, speech-language and communication, behaviour patterns, sensory aspects, cognitive component), each rated 1 (rarely) to 5 (always) from observation and parent interview. Totals run from 40 to 200: below 70 is not autism, and higher scores grade mild, moderate or severe autism with a disability percentage. ISAA is the instrument prescribed for certifying autism in India.

Diagnosis pairs the **Autism Diagnostic Interview-Revised** (ADI-R; **Michael Rutter**, **Ann Le Couteur** and **Catherine Lord**, 2003), a 93-item parent interview, with the **Autism Diagnostic Observation Schedule** (ADOS-2; Lord and colleagues, 2012), structured play and conversation in one of five modules. The M-CHAT-R/F (**Diana Robins** and colleagues, 2009) screens toddlers of 16 to 30 months with 20 parent questions. Not autism tools: the **Children's Global Assessment Scale** (CGAS; **David Shaffer**, 1983), a single 1 to 100 rating of overall functioning, and the Child Behavior Checklist (Section 2.11).

> **! EXAM TRAP**
>
> IHBAS papers of 2020 and 2024 key **CARS** for severity and disability in autism, even for "the Indian tool" with ISAA offered. CARS is American; the Indian instrument prescribed for certifying autism disability is **ISAA**, which drew on CARS. Choose ISAA when offered. DAAS, PAAS, CGAS and the CBCL are distractors.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. DQ is developmental age divided by _____ age, multiplied by 100.
2. The VSMS was devised in 1935 by _____ at the Vineland Training School.
3. The infant scale for the first month to 42 months is the _____.
4. Malin's Indian VSMS has _____ items; Bharath Raj's DST has 88.
5. DSM-5 grades the severity of intellectual disability by _____ functioning.
6. Global developmental delay is diagnosed only under the age of _____ years.
7. In India, specific learning disability is certified only from _____ years, and autism with the _____.

Answers: (1) chronological, (2) Edgar A. Doll, (3) Bayley Scales, (4) 89, (5) adaptive, (6) 5, (7) 8; ISAA.

2.8 Aptitude, Achievement, Creativity, Interest and Attitude Measures

LOWER YIELD · 8 QS

Two Class 10 students in Mysuru both score 85 percent in mathematics. The marks show what each has learnt, but not what each could still learn, how original each is, what each enjoys or how each feels about a career in engineering. Each of those questions has its own family of tests.

Aptitude, achievement and intelligence ○

An **achievement test** measures what a person has already learnt: the knowledge and understanding gained in a specific academic area, usually through instruction. An **aptitude test** looks forward, estimating how well a person will learn or succeed in a specific area in future, such as mechanical, musical or clerical work. An intelligence test samples general ability and gives one broad score (Section 2.6). Achievement looks back and aptitude looks ahead, so achievement tests rest mainly on content validity and aptitude tests on predictive validity (Section 2.4).

Anne Anastasi (1908 to 2001) stressed that the difference is one of degree: all ability tests measure developed ability, and they lie on a continuum by how specific and how controlled the underlying learning is. Course-based achievement tests (a term of algebra) sit at one end; broad achievement tests, verbal aptitude and intelligence tests, and non-verbal tests follow; culture-fair tests, drawing on the least controlled everyday experience, sit at the other.

A teacher-made test serves one class and has no norms; a standardised achievement test has tried-out items, fixed administration and norms. By purpose, a **survey test** gives an overall level of attainment, a **diagnostic test** pinpoints specific weaknesses (exactly which kinds of subtraction a child gets wrong), and a **prognostic test** forecasts success in a later course.

Clinicians use individual batteries: the Wide Range Achievement Test (**Joseph Jastak**, 1946; reading, spelling and arithmetic), the Peabody Individual Achievement Test (**Lloyd Dunn** and **Frederick Markwardt**, 1970; many items answered by pointing), the Wechsler Individual Achievement Test (1992) and the Woodcock-Johnson Tests of Achievement (**Richard Woodcock** and **Mary Bonner Johnson**, 1977). The last two are co-normed with

ability tests to assess the **ability-achievement discrepancy**: achievement well below what measured intelligence predicts has traditionally signalled a specific learning disability (Section 2.7).

> **! EXAM TRAP**
>
> The primary purpose of an achievement test is to measure understanding and knowledge in specific academic areas. "Future success potential" is aptitude, "cognitive abilities related to brain function" is neuropsychological testing (Section 2.11), and "unconscious conflicts" belong to projective techniques (Section 2.10).

Aptitude batteries and special aptitude tests

One intelligence score told a careers counsellor little. After **Louis Leon Thurstone** (1887 to 1955) split intelligence into primary mental abilities (Section 1.9), the 1940s produced the **multiple aptitude battery**: separately normed subtests that give a profile instead of one number. The **Differential Aptitude Tests** (DAT) of **George K. Bennett**, **Harold G. Seashore** and **Alexander G. Wesman** (1947), built to guide secondary school students, has eight subtests: verbal reasoning, numerical ability, abstract reasoning, clerical speed and accuracy, mechanical reasoning, space relations, spelling and language usage. Verbal reasoning plus numerical ability indexes scholastic aptitude, and **J. M. Ojha** adapted the DAT for India. The **General Aptitude Test Battery** (GATB) of the United States Employment Service (1947) measures nine aptitudes, from general learning ability to finger and manual dexterity, to match people to occupations, and the Armed Services Vocational Aptitude Battery (1968) serves US military recruitment.

A special aptitude test measures one narrow talent; an early classic is the Measures of Musical Talents of **Carl Seashore** (1866 to 1949), published in 1919 (Table 2.19). Admission tests such as the SAT, ACT and GRE are scholastic aptitude tests predicting success in college or graduate study.

Table 2.19 Classic special aptitude tests.

Test	Aptitude	The examinee
Seashore Measures of Musical Talents (1919)	Musical	Compares tones for pitch, loudness and time; recalls tone sequences
Meier Art Judgment Test	Artistic	Picks the better of an original and an altered artwork
Minnesota Clerical Test	Clerical	Checks pairs of numbers and names, at speed
Bennett Mechanical Comprehension Test	Mechanical	Reasons about pictured gears, pulleys and levers
Minnesota Paper Form Board	Spatial	Picks the figure that loose pieces would form

> **! EXAM TRAP**
>
> Carl Seashore (music, 1919) is not Harold Seashore (DAT, 1947). The DAT is a multiple aptitude battery of eight subtests, not an intelligence test; the GATB has nine aptitudes.

Tests in the guidance programme ○

These batteries serve guidance, the organised help a school gives students in educational and vocational choices. A guidance programme's tools are the case study (an in-depth picture of one pupil from all sources), the interview (a face-to-face conversation to gather information and advise), the **cumulative record** (a running file of marks, test scores, health and activities kept over the years), observation, and tests of intelligence, aptitude, achievement and interest. Counselling then draws on them all.

> **! EXAM TRAP**
>
> Guidance textbooks call *counselling* "the heart of the guidance programme", so choose counselling whenever it is offered. RML 2022 offered only case study, interview, cumulative record and aptitude test, and keyed the aptitude test, which no standard text supports: an aptitude test only predicts trainability. With exactly those four options, choose the interview, counselling's chief instrument and itself called the heart of the counselling process (Section 7.10), and expect some keys to differ.

Creativity and divergent thinking ○

Intelligence tests reward **convergent thinking**, narrowing information down to one correct answer. In his presidential address to the American Psychological Association in 1950, **Joy Paul Guilford** (1897 to 1987) tied creativity to **divergent thinking**, producing many varied answers to an open problem, the operation he called divergent production in his Structure of Intellect model (Section 1.9). His Alternate Uses task asks for uses of an object such as a brick. If Kavya writes wall, house, doorstop, paperweight and red powder for rangoli, her **fluency** (number of ideas) is five; her **flexibility** (number of categories: building, weighting, pigment) is three; her **originality** is the statistical rarity of each idea (rangoli powder scores, a wall does not); and **elaboration** is the detail she adds.

Ellis Paul Torrance (1915 to 2003) standardised such tasks in the **Torrance Tests of Creative Thinking** (1966), with verbal and figural (picture-completing) forms. **Sarnoff Mednick** (1962), in his associative theory, held that creative people reach remote associations easily. His **Remote Associates Test** (RAT) is a timed paper-and-pencil test of about 40 minutes in which the examinee finds one word linking three cue words: cottage, Swiss, cake gives "cheese". **Michael Wallach** and **Nathan Kogan** (1965), testing children in relaxed, game-like conditions, found creativity largely independent of intelligence. The Indian tests are the verbal and non-verbal Tests of Creative Thinking of **Baqer Mehdi** (1973) and the six Passi Tests of Creativity (**B. K. Passi**, 1979).

> **! EXAM TRAP**
>
> "A timed, 40-minute paper-and-pencil test" is the RAT. Each item has one right answer, so its format is convergent, unlike the TTCT. The Alternate Uses task is Guilford's, not Torrance's.

Interest inventories and value scales

An **interest inventory** asks which activities, subjects and occupations a person likes. On the empirical route an item joins an occupational scale if people in that occupation answer it dif-

ferently from people in general, whatever its content; on the homogeneous route items are grouped by content into interest areas.

Edward Kellogg Strong Jr. (1884 to 1963) took the empirical route in the **Strong Vocational Interest Blank** (1927), revised by **David Campbell** as the Strong-Campbell Interest Inventory (1974) and now the Strong Interest Inventory. **G. Frederic Kuder** (1903 to 2000) took the homogeneous route in the Kuder Preference Record (1939): the person marks the most and least liked activity in each forced-choice triad, and the answers score ten areas (outdoor, mechanical, computational, scientific, persuasive, artistic, literary, musical, social service, clerical). As each choice raises one area by lowering another, the scores are **ipsative**: they show relative strengths within the person, not standing against others (Section 2.5). The later Kuder Occupational Interest Survey compares people with occupational groups.

John L. Holland (1919 to 2008), in a theory first published in 1959 and set out fully in 1973, sorted people and work environments into six **RIASEC** types (Realistic, Investigative, Artistic, Social, Enterprising, Conventional) arranged on a hexagon: adjacent types are alike, opposite ones (Realistic and Social) least alike. His self-scored **Self-Directed Search** (1971) gives a three-letter code matched to occupations.

For values, the Study of Values of **Gordon Allport** (1897 to 1967) and **Philip Vernon** (1931; with **Gardner Lindzey** from 1951; revised 1960) measures six values from the types of **Eduard Spranger** (1882 to 1963): theoretical, economic, aesthetic, social, political and religious. **Milton Rokeach** (1973) had people rank 18 terminal values (end states such as freedom) and 18 instrumental values (conduct such as honesty).

Thurstone and Likert scales ●●

An attitude is an evaluative disposition towards an object such as a group or an idea (its components and theories are in Chapter 3). An attitude scale infers it from responses to statements, combined into one score from unfavourable to favourable. The classic scales differ in how the statements are chosen and how the score is formed.

Thurstone and **Ernest J. Chave** (1929), measuring attitude towards the church, devised the **method of equal-appearing intervals**. Judges sorted over a hundred statements into 11 piles from most unfavourable to most favourable, without giving their own opinion. A statement's **scale value** is the median pile; its ambiguity is Q, the interquartile range of the placements, and high-Q statements are dropped. About 20 statements with evenly spread values form the scale. The respondent ticks those they agree with and scores the median (or mean) of their values: agreeing with statements valued 7.2, 8.1 and 8.4 gives 8.1, a favourable attitude.

This is laborious. **Rensis Likert** (1903 to 1981) offered a short cut in 1932, the **method of summated ratings**. A **Likert scale** is a set of closed-ended statements, each with a fixed number of ordered alternatives, classically five from strongly agree to strongly disagree, scored 5 to 1 (reversed for unfavourable statements). Respondents rate their own agreement on this symmetric continuum, and item scores are summed with equal weights. No judges are used: items are kept by item analysis if high and low total scorers answer them differently (Section 2.2). Easier to build and usually as reliable, Likert scales dominate questionnaires; strictly ordinal, their scores are usually treated as interval. Figure 2.12 contrasts the two methods.

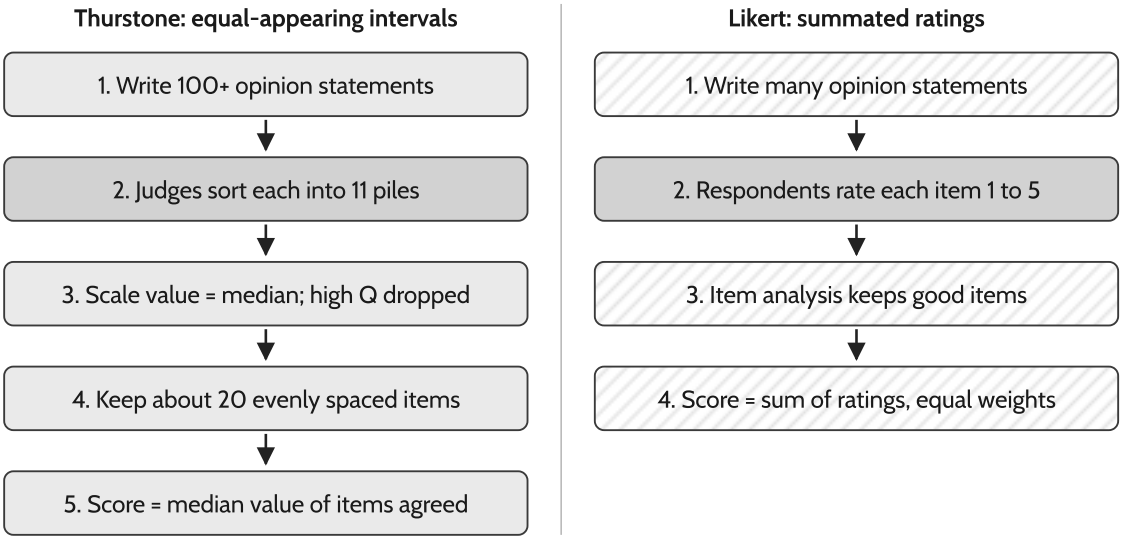


Figure 2.12 The step shaded alike is the key difference: Thurstone's judges pre-scale the statements, whereas Likert's respondents rate their own agreement and the ratings are summed.

> **! EXAM TRAP**
>
> A summated rating scale is a Likert scale, never a Thurstone, Guttman or equal-appearing interval scale. Likert items are closed-ended with a fixed number of ordered alternatives, neither open-ended nor dichotomous. Judges pre-scaling statements is Thurstone; respondents rating their agreement, summed with equal weights, is Likert; a sentence-completion stimulus belongs to projective techniques (Section 2.10). Likert is a type of attitude scale, whereas nominal, ordinal and interval are levels of measurement (Chapter 9).

Guttman, Bogardus and the semantic differential

Louis Guttman (1916 to 1987) built a one-dimensional **cumulative scale** by scalogram analysis (1944). Statements are ordered so that endorsing a stronger one implies endorsing every weaker one ("I would accept a de-addiction centre in my city", "in my locality", "on my street"), so the total score reveals the exact pattern of answers. Fit is judged by the **coefficient of reproducibility**, which should be 0.90 or more.

The **social distance scale** of **Emory Bogardus** (1882 to 1973), published in 1925, is generally regarded as the oldest attitude scale. For each group the respondent marks the closest relation they would accept, on seven steps: close kinship by marriage, club chums, neighbours, co-workers, fellow citizens, visitors only, and exclusion from the country. In the **semantic differential** of **Charles Osgood**, **George Suci** and **Percy Tannenbaum** (1957), a concept such as "the police" is rated on seven-point bipolar adjective pairs (good to bad, strong to weak, fast to slow), which factor into evaluation, potency and activity; evaluation is the attitude proper.

In paired comparisons each object is paired with every other and one of each pair is chosen; Thurstone's **law of comparative judgement** (1927) converts the choices into scale values, and rank ordering is a quicker variant. **Allen Edwards** and **Franklin Kilpatrick** (1948) combined Thurstone judging, Likert item analysis and Guttman scaling in the **scale-discrimination technique**. Table 2.20 compares the main scales.

Table 2.20 The classic attitude scales compared.

Scale	Author, year	Respondent	Score	Hallmark
Equal-appearing intervals	Thurstone and Chave, 1929	Ticks agreed statements	Median scale value	Judges, 11 piles, Q
Summated ratings	Likert, 1932	Rates every item, usually 5 points	Sum of ratings	Item analysis, no judges
Cumulative	Guttman, 1944	Agrees or not with graded items	Number endorsed	Reproducibility 0.90
Social distance	Bogardus, 1925	Marks closest relation accepted	Step, 1 to 7	Oldest attitude scale
Semantic differential	Osgood, Suci and Tannenbaum, 1957	Rates a concept on bipolar adjectives	Factor means	Evaluation, potency, activity

Implicit and indirect measures ○

Direct scales invite socially desirable answers, and some bias lies outside awareness. In the **Implicit Association Test** (IAT) of **Anthony Greenwald**, **Debbie McGhee** and **Jordan Schwartz** (1998), words or pictures are sorted on a computer with two keys into paired categories, such as "flower or pleasant" versus "insect or unpleasant", and then with the pairing reversed. Sorting is faster when concepts sharing a key are already linked in memory, so reaction times index implicit, often unconscious attitudes and stereotypes. In the **bogus pipeline** of **Edward Jones** and **Harold Sigall** (1971), people answer more honestly when wired to a machine they believe detects true feelings. Physiological indices (skin conductance, pupil size, facial muscle activity) show how strongly a person reacts, rarely in which direction.

> **! EXAM TRAP**
>
> The IAT measures implicit, unconscious attitudes and stereotypes through reaction times. It does not measure verbal ability, hidden mathematical ability or dream content.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. A test of knowledge already gained in a specific academic area is an _____ test.
2. The number of different categories of response in a divergent task is _____.
3. The RAT asks for one word linking _____ cue words.
4. Kuder's forced-choice triads yield _____ scores.
5. A summated rating scale is a _____ scale.
6. A Guttman scale needs a coefficient of reproducibility of at least _____.

Answers: (1) achievement, (2) flexibility, (3) three, (4) ipsative, (5) Likert, (6) 0.90.

2.9 Objective Personality Inventories MEDIUM YIELD · 38 QS

In 1917, as the United States entered World War I, its army needed a quick way to spot recruits likely to break down under fire, and there were far too few psychiatrists to interview them

all. **Robert Sessions Woodworth** (1869 to 1962) put the psychiatrist's screening questions on paper, and the personality inventory was born. Ever since, psychologists have argued over how to choose the questions and how far to trust the answers.

Objective tests and Cattell's three kinds of data ●●

An **objective personality test** fixes the items, the answer format (true or false, yes or no, a rating) and the scoring key, so any two scorers get the same result. Rating scales and **personality inventories** (long self-report questionnaires scored on several scales) are objective; projective tests (Section 2.10) use ambiguous stimuli, free responses and interpretive scoring.

Raymond Bernard Cattell (1905 to 1998), whose trait theory is taught in Chapter 4, accepted a trait only if it appeared in three kinds of data:

- **L-data** (life record data): actual behaviour in everyday life, from school performance, work and criminal records, and ratings by observers who know the person.
- **Q-data** (questionnaire data): self-report ratings of one's own characteristics, attributes and interests.
- **T-data** (objective test data): behaviour in a controlled test situation whose purpose the person cannot see, and so cannot easily fake, such as a laboratory task scored for a trait he does not know is being measured.

> **! EXAM TRAP**
>
> Cattell's scheme has exactly three letters, **L, Q and T**: there is no P-data (projective) or Z-data. School and criminal records are L-data; self-report questionnaires are Q-data; projective-test performance and Q-sort responses are not L-data. Q-data is not the Q-sort (2.9.7).

Four ways to build an inventory ○

The **logical-content** or **rational strategy** reasons out how people with a trait (anxiety, say) describe themselves and writes items whose content plainly reflects it, trusting face validity. Woodworth's **Personal Data Sheet** (built in 1917, published in 1919 and often dated 1920), the first personality inventory, asked 116 yes-no questions about symptoms and gave one score of emotional instability; **Hugh M. Bell**'s Adjustment Inventory (1934) scored home, health, social and emotional adjustment. Its weakness: respondents see what each item is after, and the method assumes insight and honesty.

The **theoretical strategy** writes items for the constructs of a personality theory: the Edwards Personal Preference Schedule (Henry Murray's needs), the Myers-Briggs Type Indicator (Carl Jung's types) and the Millon Clinical Multiaxial Inventory (Theodore Millon's theory).

Empirical criterion-group keying ignores content. A large item pool goes to a criterion group (say, patients diagnosed with depression) and a normal control group; only the items the two groups answer differently survive, whatever they say, and the scale is cross-validated on fresh groups. If depressed patients happened to answer "true" to "I like mechanics magazines" more often than controls, that item would join the depression scale. The MMPI, the California Psychological Inventory and the Strong Vocational Interest Blank (Section 2.8) were built this way.

The **factor-analytic strategy** factor-analyses the correlations among many items, keeps the items that load on each factor and names the factor afterwards: the 16PF, the Eysenck inventories, the NEO-PI-R, the **Guilford-Zimmerman Temperament Survey** (**J. P. Guilford** and Wayne Zimmerman, 1949; 300 items, ten traits such as sociability and restraint) and the **Comrey Personality Scales** (**Andrew L. Comrey**, 1970; 180 items, eight factors such as trust and orderliness). Modern inventories, such as the NEO-PI-R and the MCMI, use **combined strategies**.

Against response sets (Section 2.2) and deliberate faking, test makers use rapport and confidentiality, **forced-choice** items pairing statements of equal social desirability, disguised items, and **validity scales** and correction keys.

> ! EXAM TRAP
>
> The MMPI was derived by finding items that discriminate between criterion and control groups, **not** by the rational method or by assuming traits and writing questions for them. Comrey's 180-item test is **factor-analytic**. "Correlational" and "experimental" are research designs (Chapter 9), and criterion-referencing is a way of interpreting scores (Section 2.5), not of choosing items.

The MMPI and its validity scales ●●●

Around 1940, at the University of Minnesota Hospitals, the psychologist **Starke R. Hathaway** (1903 to 1984) and the neuropsychiatrist **J. Charnley McKinley** (1891 to 1950) set out to aid psychiatric diagnosis. They gave over a thousand statements to patient groups with known diagnoses and to about 700 "Minnesota normals", mostly hospital visitors, and kept for each scale only the items that separated that patient group from the normals. The **Minnesota Multiphasic Personality Inventory** (MMPI, 1943) had 550 different statements (566 printed, 16 repeated), answered true, false or cannot say. It is "multiphasic" because its items cover physical health, mood, family, sexual, religious and social attitudes, fears and unusual experiences.

The MMPI is therefore an **objective personality inventory** of **personality and psychopathology**, used mainly for psychiatric diagnosis. It is not a projective test, intelligence test, aptitude battery, interest inventory or neuropsychological screening battery, nor a tool for educational, career or physical-health assessment, and it does not measure unconscious drives (projective tests), the Big Five (the NEO) or leadership potential (the CPI). Because self-reports can mislead, it carries validity scales:

- **Cannot Say (?)**: items left blank or marked both ways; 30 or more make the record doubtful.
- **L (Lie) scale**: small faults nearly everyone admits, of the kind "I get angry sometimes". Answering "false" to them asserts a degree of personal virtue rarely seen in our culture, so a high L marks a *naive* attempt to look good by denying common minor faults.
- **F (Infrequency) scale**: items fewer than one normal person in ten endorses. A high F flags random or careless responding, over-reporting of symptoms (faking bad) or severe psychopathology; on the MMPI-2 an F above T 100 suggests random responding or gross exaggeration.

- **K (Correction) scale**: added by **Paul E. Meehl** (1920 to 2003) and Hathaway in 1946 to catch *subtle* defensiveness. A fraction of K is added to clinical scales 1, 4, 7, 8 and 9 (the **K-correction**) to compensate for defensive responding.

The MMPI-2 added **VRIN** (Variable Response Inconsistency: contradictory answers to similar item pairs, a sign of random responding), **TRIN** (True Response Inconsistency: answering "true", or "false", regardless of content), **F(B)** (Back F, for later items), **F(p)** (Infrequency-Psychopathology, items rare even among psychiatric inpatients, which separates faking bad from real disorder) and **S** (Superlative Self-Presentation).

> **! EXAM TRAP**
>
> The three classic validity scales are **L, F and K** (four if Cannot Say is counted, as some texts do); VRIN, TRIN and S came later. L is naive faking good, K is subtle defensiveness plus a correction, and F is rare responding (random answers, faking bad or severe pathology). A very high F never means faking good, defensiveness or an excellent profile.

Clinical scales, profiles and the MMPI family ●●

The MMPI has **ten clinical scales** (Table 2.21): not 4, 13 or 16 (the 13 "basic scales" are the ten plus L, F and K). Eight were keyed against psychiatric diagnostic groups; scales 5 and 0 were not. Scales 1, 2 and 3 form the **neurotic triad**; 6, 7, 8 and 9 the **psychotic tetrad**.

Table 2.21 The ten MMPI clinical scales.

Scale	Name	High scores suggest
1 Hs	Hypochondriasis	Bodily preoccupation, vague physical complaints
2 D	Depression	Low mood, hopelessness, slowed activity
3 Hy	Hysteria	Physical symptoms under stress, denial of problems
4 Pd	Psychopathic Deviate	Antisocial behaviour, conflict with family and authority, impulsivity, alienation
5 Mf	Masculinity-Femininity	Interests atypical for one's sex (not pathology)
6 Pa	Paranoia	Suspiciousness, ideas of persecution
7 Pt	Psychasthenia	Anxiety, worry, obsessions and compulsions
8 Sc	Schizophrenia	Odd thinking, alienation, unusual experiences
9 Ma	Hypomania	Overactivity, excitement, flight of ideas
0 Si	Social Introversion	Shyness, social discomfort

Raw scores become T scores (mean 50, SD 10; Section 2.5): T 70 or more was elevated on the MMPI, and the MMPI-2 uses uniform T scores with a cut-off of 65. Clinicians read the **code type**, the two highest scales: **2-7** suggests anxious, worried depression; **4-9** impulsive, antisocial acting out; and the **conversion V** (1 and 3 high, 2 clearly lower) a person who turns stress into physical complaints without feeling much distress (Figure 2.13).

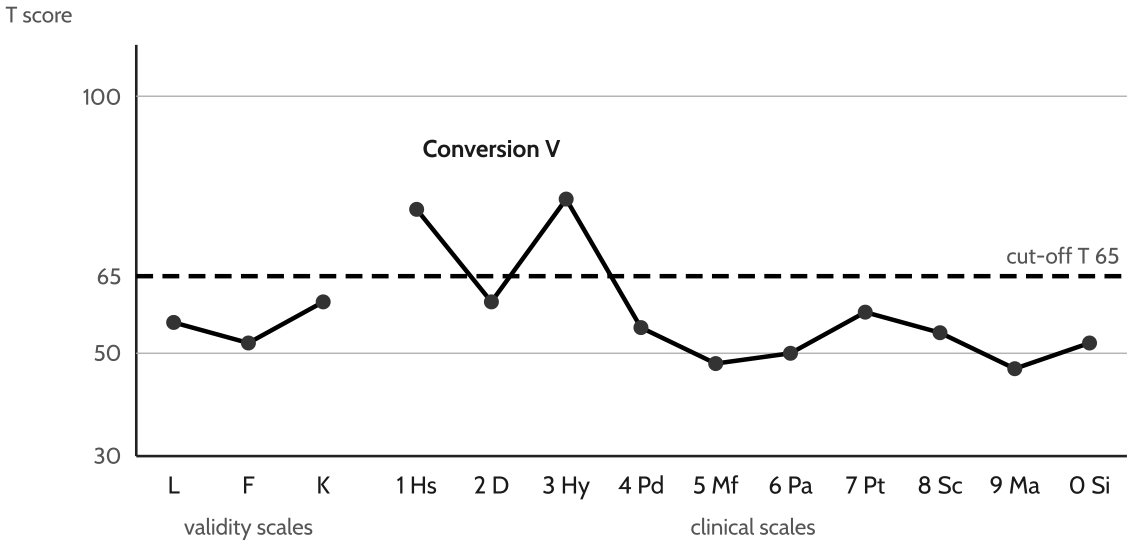


Figure 2.13 A valid MMPI-2 profile (L, F and K near 50) with a conversion V: scales 1 and 3 rise above the dashed cut-off while scale 2 stays lower.

The **MMPI-2** (**James N. Butcher** and colleagues, 1989) has 567 items, rewrote outdated ones, was normed on 2,600 US adults and added content and supplementary scales (such as health concerns and the MacAndrew alcoholism scale). The old clinical scales intercorrelate because all share a general distress factor, **demoralisation**. **Auke Tellegen** and colleagues (2003) extracted it first as **RCd** and rebuilt the **Restructured Clinical (RC) scales** (such as RC2 low positive emotions and RC4 antisocial behaviour) so that each measures a distinctive dimension. RC scales are the core of the **MMPI-2-RF** (**Yossef S. Ben-Porath** and Tellegen, 2008; 338 items) and the **MMPI-3** (2020; 335 items). The **MMPI-A** (1992; 478 items) serves adolescents aged 14 to 18.

> INDIAN CONTEXT
>
> **H. N. Murthy**'s **Multiphasic Questionnaire** (MPQ; 1964) is a short (100-item) Indian adult inventory built from MMPI-type items, with a Lie scale and seven clinical scales such as hysteria, depression, anxiety, paranoia and schizophrenia. An elevated MPQ Lie scale indicates **all** of these: a tendency to fake and manipulate, social desirability, and mistrust (a guarded person who admits no flaw to the examiner).

Factor-analytic inventories: 16PF, Eysenck and NEO ●●●

Cattell turned his source traits into the **Sixteen Personality Factor Questionnaire** (16PF), first published in 1949; the fifth edition (1993) has 185 three-choice items for ages 16 and over. It scores **16 primary factors** found by factor analysis and labelled by letter (A warmth, B reasoning, C emotional stability, E dominance, and so on to Q4 tension), plus **five global, second-order factors** (extraversion, anxiety, tough-mindedness, independence, self-control) that approximate the Big Five. Scores are **stens** (standard tens: 1 to 10, mean 5.5, SD 2). Factor B is a brief reasoning scale, but the 16PF measures personality traits, not fluid, crystallised or general intelligence (Section 2.6).

> **! EXAM TRAP**
>
> The 16PF is **Raymond B. Cattell's**: Kelly is to the Rep test as Cattell is to the 16PF. Hans Eysenck built the EPQ, Costa and McCrae the NEO-PI-R, Henry Murray the TAT, Gordon Allport the Study of Values, William Sheldon a body-type scheme, and Freud no inventory.

Hans Jurgen Eysenck (1916 to 1997), at the Maudsley Hospital, London, built three inventories. The **Maudsley Personality Inventory** (1959; 48 items) scored extraversion (E) and neuroticism (N); the **Eysenck Personality Inventory** (1964, with **Sybil B. G. Eysenck**; 57 items) added a Lie scale; and the **Eysenck Personality Questionnaire** (EPQ, 1975; 90 items) added psychoticism (P), giving PEN plus L. The EPQ-R followed in 1985.

Paul T. Costa Jr. and **Robert R. McCrae** built the **NEO Personality Inventory** on the Five-Factor Model (Chapter 4). The **NEO-PI-R** (1992) has 240 items on a five-point agree-disagree scale and scores five domains, **Openness, Conscientiousness, Extraversion, Agreeableness and Neuroticism** (OCEAN), each with **six facets** (30 in all), as T scores. The 60-item **NEO-FFI** scores domains only; the **NEO-PI-3** (2005) simplified the wording. The NEO is not based on Cattell's 16 factors, Eysenck's PEN, Murray's needs or MBTI types (personality-disorder scales belong to the MCMI), and OCEAN is neither the four humours (sanguine, choleric, melancholic, phlegmatic; Chapter 4) nor achievement, affiliation and power motives (Chapter 1). Shorter Big Five measures include the 44-item **Big Five Inventory** (Oliver John and colleagues, 1991); the **HEXACO** inventory of Kibeom Lee and Michael Ashton adds a sixth factor, Honesty-Humility.

Theory-based, clinical and Indian inventories ○

The **Myers-Briggs Type Indicator** (MBTI) of **Katharine Cook Briggs** (1875 to 1968) and her daughter **Isabel Briggs Myers** (1897 to 1980), first issued in 1943, is directly based on **Carl Gustav Jung**'s theory of psychological types. Forced-choice items place people on four dichotomies, extraversion-introversion, sensing-intuition, thinking-feeling and judging-perceiving, giving 16 types such as INTP. It is popular in counselling and team building but psychometrically weak: continuous scores that bunch near the middle are forced into types, and many people get a different type on retest weeks later.

The **Edwards Personal Preference Schedule** (EPPS) of **Allen L. Edwards** (1953, manual 1954; revised 1959) measures 15 needs from **Henry A. Murray**'s list, such as achievement, order, affiliation, dominance and aggression. To control social desirability, every item is a **forced choice** between two statements matched for desirability (225 pairs: 210 different and 15 repeated for a consistency score). The scores are therefore **ipsative**: each need is strong only relative to the person's other needs, so people cannot be compared directly.

The **California Psychological Inventory** (CPI) of **Harrison G. Gough** (1921 to 2010), published in 1956 (manual 1957), is the normal person's MMPI: largely criterion-keyed, with many MMPI items, but scoring **folk concepts**, everyday trait ideas found in all cultures, such as dominance, sociability, responsibility and self-control. It began with 480 items (now 434) and predicts achievement, leadership and delinquency. The **Millon Clinical Multiaxial Inventory** (MCMI) of **Theodore Millon** (1928 to 2014), first published in 1977, follows Millon's theory and the DSM, measuring personality disorders and clinical syndromes in patients (175 items; 195 in the MCMI-IV of 2015). It reports **base rate scores**, anchored to how common each disorder is in clinical samples rather than to normal T scores.

The **State-Trait Anxiety Inventory** (STAI) of **Charles D. Spielberger** (1927 to 2013) and colleagues (1970; Form Y 1983) has 40 items: 20 ask how you feel right now (state anxiety, passing) and 20 how you generally feel (trait anxiety, a stable proneness). Its purpose is to differentiate state from trait anxiety. Measuring psychoticism, extraversion and neuroticism is the EPQ's job; "normal and abnormal personality" fits broad inventories such as the MMPI and CPI; checking response validity is the job of validity scales. The older **Taylor Manifest Anxiety Scale** of **Janet A. Taylor** (1953) used 50 MMPI items judged to express overt anxiety.

> **INDIAN CONTEXT**
>
> Common Indian tools include **Sinha's Comprehensive Anxiety Test** (A. K. P. Sinha and L. N. K. Sinha), Hindi adaptations of Bell's Adjustment Inventory (Saxena, 1959; Mohsin and Hussain, 1981), the **PGI Health Questionnaire N-1** (N. N. Wig and S. K. Verma, PGIMER Chandigarh), a neuroticism screen derived from the Cornell Medical Index, and Hindi versions of the MPI and EPQ.

Idiographic and situational methods ○

Some methods study how one person construes the world. In the **Role Construct Repertory Test** (Rep test, 1955) of **George A. Kelly** (1905 to 1967), the person names people who fill roles in his life (mother, a liked teacher, someone disliked). Shown three at a time, he says how two are alike and differ from the third; each answer ("generous versus selfish") is a bipolar **personal construct**, and a **repertory grid** of figures by constructs maps his construct system (theory in Chapter 4).

In the **Q-sort**, set out by **William Stephenson** (1902 to 1989) in 1953, a person sorts statements, such as self-descriptions, into piles from least to most characteristic, with a fixed number per pile forming a forced, quasi-normal distribution (Figure 2.14). Since every sort has the same mean and spread, it shows which statements are more characteristic *of this person*: within-person, not between-person, variation. **Carl Rogers** (1902 to 1987) had clients sort for the real self and the ideal self; Butler and Haigh (1954) found the mean real-ideal correlation rose from about zero before client-centred therapy to about .3 after it (.34 at the end, .31 at follow-up). **Jack Block** (1924 to 2010) built the 100-item **California Q-set** for observers. The Q-sort complements inventories; it never replaced the MMPI.

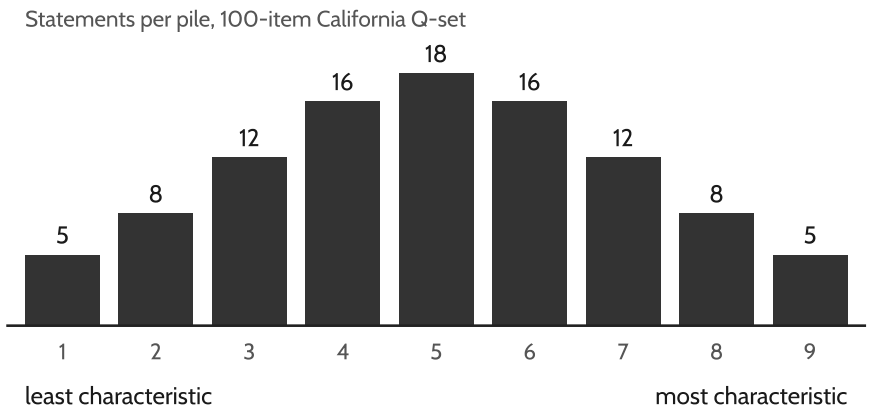


Figure 2.14 A Q-sort forces statements into a fixed quasi-normal shape: only five may go in each extreme pile, eighteen in the neutral middle pile.

On an **adjective checklist**, such as Gough's Adjective Check List of 300 adjectives, the person ticks every word that applies. **Situational tests** watch behaviour in staged real-life tasks. In the Character Education Inquiry, **Hugh Hartshorne** and **Mark A. May** (1928) gave schoolchildren natural-seeming chances to cheat, lie and steal; honesty in one situation predicted honesty in another only weakly. The US Office of Strategic Services (OSS) selected World War II agents through stressful staged tasks; in the **leaderless group discussion** candidates tackle a problem with no appointed leader while assessors rate who leads, as India's Services Selection Board still does.

Judging inventories

Inventories offer objective scoring, large norms, good reliability and economy (one psychologist can test a hundred people at once). Their weaknesses are those of self-report: response sets and faking, limited self-insight, the reading level they demand, and cultural bias when items and norms from one country (rural Minnesota in 1940) travel to another. **Walter Mischel** (1968) added that trait scores predict behaviour in any one situation only weakly, a debate for Chapter 4. Table 2.22 compares the main inventories.

Table 2.22 The main objective personality inventories compared.

Inventory	Author, year	Items	Strategy	Scales and scores
MMPI	Hathaway and McKinley, 1943	550	Empirical keying	?, L, F, K; 10 clinical; T
MMPI-2	Butcher and colleagues, 1989	567	Empirical keying	Adds content scales; uniform T
MMPI-2-RF	Ben-Porath and Tellegen, 2008	338	Restructured	RC scales; T
CPI	Gough, 1956 (manual 1957)	480, now 434	Empirical keying	Folk concepts; T
16PF	Cattell, 1949 (5th edition 1993)	185	Factor-analytic	16 primary, 5 global; stens
EPQ	H. J. and S. B. G. Eysenck, 1975	90	Factor-analytic	P, E, N, L
NEO-PI-R	Costa and McCrae, 1992	240	Factor-analytic, combined	5 domains, 30 facets; T
GZTS	Guilford and Zimmerman, 1949	300	Factor-analytic	10 traits
Comrey Scales	Comrey, 1970	180	Factor-analytic	8 factors
EPPS	Edwards, 1953	225 pairs	Theoretical (Murray)	15 needs; ipsative
MBTI	Briggs and Myers, 1943	Forced choice	Theoretical (Jung)	4 dichotomies, 16 types
MCMI	Millon, 1977	175 to 195	Theoretical, combined	Disorders, syndromes; base rates

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. School performance and criminal records are Cattell's _____-data.
2. The MMPI kept items that discriminated between _____ groups and control groups.
3. Denying common minor faults raises the MMPI _____ scale.
4. A fraction of the _____ scale is added to clinical scales 1, 4, 7, 8 and 9.
5. The RC scales first remove the common factor called _____.
6. The NEO-PI-R scores five domains with _____ facets each.
7. Forced-choice pairs make EPPS scores _____.

Answers: (1) L, (2) criterion, (3) L (Lie), (4) K (Correction), (5) demoralisation (RCd), (6) six, (7) ipsative.

2.10 Projective Techniques HIGH YIELD · 44 QS

Hand someone a symmetrical smudge of ink and ask, "What might this be?" One person sees two dancers, another a crushed bat, a third "blood, because it is red". The blot is the same, so the differences must come from the viewers. That idea produced clinical psychology's most famous and most disputed tests. This section explains it, sorts the tests into five families, works through the Rorschach and the Thematic Apperception Test, then the smaller techniques and the evidence on their worth.

The projective hypothesis •••

The objective inventories of Section 2.9 ask clear questions and score fixed answers. A **projective technique** reverses that logic. It presents an ambiguous, unstructured stimulus (an inkblot, a vague picture, half a sentence, a blank page), gives minimal instructions and accepts any answer. Because the stimulus supplies so little structure, whatever structure appears in the response is assumed to come from the person. This is the **projective hypothesis**: faced with an ambiguous stimulus, a person projects needs, conflicts, unconscious processes and a characteristic perceptual style into the response, usually without knowing it.

The word comes from **Sigmund Freud** (1856 to 1939), for whom **projection** was a defence that attributes one's own unacceptable impulse to someone else (Section 4.3). Testing borrowed a broader sense: everyone reads their inner world into an unclear outer one. **Lawrence K. Frank** coined the term **projective methods** in 1939, likening them to an X-ray that shows inner structure without opening the body.

Projective tests share four features: an **ambiguous stimulus**; an **open response**, with almost unlimited possible answers; a **disguised purpose**, so the person cannot tell how answers will be read; and **global interpretation** of the whole personality rather than one trait. They therefore claim to reveal **unconscious processes**, not intellectual attainment, parenting style or career aptitude. Their main advantage follows from the disguise: a person who cannot tell what a "good" answer is cannot easily fake one, so the tests avoid the self-report biases of inventories (social desirability, faking, defensiveness).

> **! EXAM TRAP**
>
> The advantage is **avoiding self-report bias**. "They let people use the defence of projection" twists the theory; superior retest reliability and numerical data are strengths of objective inventories. Beside the EPQ, 16PF, MMPI-2, NEO-PI-R, MBTI or MCMI-III, only the **Rorschach and TAT** are projective: the rest are objective self-report inventories (Section 2.9). Intelligence tests and direct observation are not projective either.

Lindzey's five families ●●

Gardner Lindzey (1959) sorted the many techniques by the kind of response they demand (Figure 2.15). In **association techniques** the person reports the first word, image or percept the stimulus calls up. In **construction techniques** the person builds a product, usually a story with a past, present and future. In **completion techniques** the person finishes an incomplete stimulus. In **choice or ordering techniques** the person picks or ranks items by preference, a step towards objective testing. In **expression techniques** the person produces something freely (a drawing, a painting, play), and the manner of production counts as much as the product.

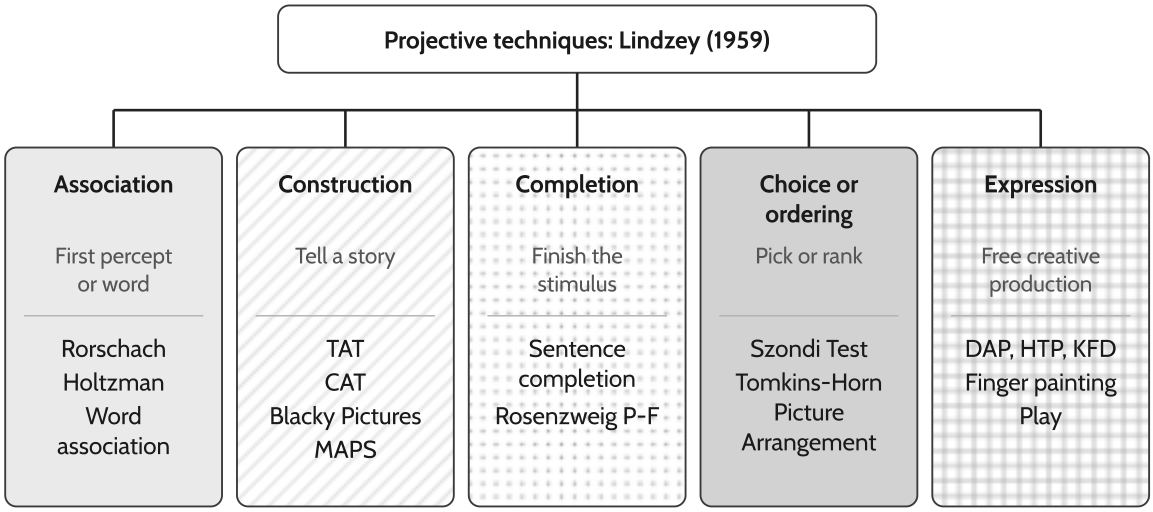


Figure 2.15 Lindzey's five families, named by the response each demands. Anchor one test per family: Rorschach, TAT, sentence completion, Szondi, Draw-A-Person.

Frank (1939) had grouped methods by what the person does with the material: **constitutive** (structuring formless material, as in finger painting), **constructive** (building a directed product), **interpretive** (giving material personal meaning, as in the TAT), **cathartic** (releasing emotion, as in play) and **refractive** (personality refracted through a conventional medium, as in handwriting). The categories overlapped, and Lindzey's replaced them.

! EXAM TRAP

The TAT is **construction**, the Rorschach **association**, sentence completion completion and Draw-A-Person expression. RML 2021 keyed the **word association test** as "not a projective test". That key is wrong: Lindzey lists word association as an association technique, and Galton's word association (1879) is usually called the first projective technique. By the standard classification all four options (word association, Rorschach, TAT, sentence completion) are projective, so no option is correct. Learn word association as projective; if the identical item returns, the paper's key was word association.

The Rorschach: ten cards and how they are given ●●●

Hermann Rorschach (1884 to 1922), a Swiss psychiatrist, published the test in *Psychodiagnostik* (1921). From many blots he kept the ten that best separated psychiatric groups, then died the next year, aged 37. The **Rorschach Inkblot Test** has **ten** bilaterally symmetrical cards (not 12, 15 or 20), always shown in the same order. Five are achromatic, black and grey (I, IV, V, VI, VII); two are black and red (II, III); three are multicoloured (VIII, IX, X), as Figure 2.16 shows. Hence the textbook phrase "five black-and-white and five coloured".

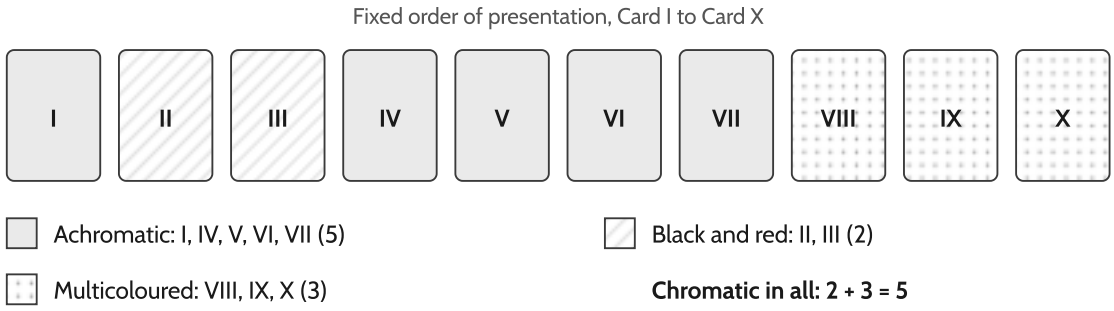


Figure 2.16 The ten Rorschach cards. Five are achromatic; the other five carry colour, two only as red patches on black and three as full pastel colour.

Administration has three phases. In the **response phase** (free association, or performance proper) the examiner asks "What might this be?", records every response verbatim with its reaction time, and does not prompt. In the **inquiry phase**, after all ten cards, the examiner goes back through each response to clarify where on the blot it was seen (location) and what made it look that way (determinants), using non-leading questions ("What made it look like that?"). Optional **testing the limits** follows: the examiner asks directly, for example, whether the person can see a common percept they missed. A longer reaction time, rejection or confusion on the coloured cards than on the achromatic ones is **colour shock**, read as a sign of emotional disturbance.

The Rorschach assesses personality, cognition and perception: how a person organises an ambiguous field, how they think, and whether thinking is disordered. Rorschach built it as a diagnostic aid, so it is used to detect thought disorder, help identify mental illness and infer unconscious intentions. It does not interpret dreams, record conscious desires, measure IQ, memory or brain size, or diagnose physical illness.

> **! EXAM TRAP**
>
> The inquiry clarifies each response **without leading questions**; cards are never shuffled, responses are recorded from the first card, and inquiry is not "scoring only". Ten cards, Hermann Rorschach, 1921: not 20 by Henry Murray, 15 by John Exner or 5 by Carl Jung, and no blank card (that is TAT card 16). Five chromatic and five achromatic, not three and seven. RML 2025 keyed "five black and white and five multi-colored cards": the five-five split is right but the wording is loose, since the achromatic cards are black and grey and only VIII, IX and X are fully multicoloured (II and III are black and red). It is still the only correct option; "only black and white", "only coloured" and "ten multicoloured" are wrong.

Scoring and reading a Rorschach record ●●●

Each response is coded first for its **location**, the part of the blot used: **W** the whole blot, **D** a common detail, **Dd** an unusual detail, and **S** white space (always coded with another location, as WS or DS). In Exner's system many W responses, especially well-organised ones, reflect **organisational and integrative effort**, the drive to pull a complex field into one meaningful whole. **Developmental quality** grades that integration: synthesised (+), ordinary (o) or vague (v). The **determinants** record what feature of the blot shaped the percept (Table 2.23); the colour determinants index emotional reactivity and impulse control.

Table 2.23 Rorschach determinants, with an example response and the usual reading.

Code	Determinant	Example	Usually read as
F	Pure form: shape alone	"A bat" (outline only)	Control, reasoning
M	Human movement	"Two people dancing"	Imagination, inner life
FM	Animal movement	"Two bears fighting"	Basic urges, need states
m	Inanimate movement	"A rocket exploding"	Situational stress
FC	Form first, colour adds	"A red butterfly"	Controlled emotion
CF	Colour first, vague form	"Fire"	Weaker emotional control
C	Pure chromatic colour	"Blood, because it is red"	Impulsive affect
Cn	Colour naming	"That is pink"	Serious disorganisation
C'	Achromatic colour	"A black cloud"	Held-in, painful feeling
T, V, Y	Shading: texture, vista (depth), diffuse	"Fur"; "a deep well"; "smoke"	Need for closeness; self-criticism; helplessness
FD	Form dimension: depth from size	"A giant seen from below"	Introspection
Fr, (2)	Reflection; pair	"A girl in a mirror"; "two dogs"	Self-focus

The **form quality** rating shows how well the percept fits the blot's actual shape, from good or ordinary to minus (distorted). It is the main index of **reality testing**: many minus responses mean perception driven by inner preoccupation rather than by the stimulus, as in psychosis. **Content** codes what was seen: H (whole human), Hd (human detail), A (animal), An (anatomy), Bl (blood) and others. A **popular response** (P) is one given very often, such as a bat or butterfly on Card V, and shows conventional perception; an **original response** is rare, a sign of creativity when well formed and of bizarre thinking when not.

Two further signs matter. An **absence of human responses** suggests **poor interpersonal orientation**, little interest in or understanding of people (not apathy or lack of empathy as such). A **confabulatory response** (DW) generalises from one detail to the whole blot ("a crab, because of these claws, so the whole thing is a crab"), and indicates **impaired reality testing**, not a regressed ego, cognitive immaturity or lack of empathy.

> ! EXAM TRAP
>
> Shape alone is **F**; "blood because it is red", with no form, is pure **C**. FC and CF mix form and colour; Cn only names the colour. **M** is human movement only (FM animal, m inanimate); C' is achromatic.

Scoring systems: Exner, R-PAS and Holtzman ●●

Rorschach left no full scoring system, and five American ones grew up in parallel: those of **Samuel Beck**, **Bruno Klopfer**, **Marguerite Hertz**, **Zygmunt Piotrowski**, and **David Rapaport** with **Roy Schafer**. Their codes differed, so research could not be compared. **John E. Exner** (1928 to 2006) combined their best-supported elements in his **Comprehensive System** (1974). It standardised administration, required at least 14 responses for a valid record, and tallied the codes into a **Structural Summary** of ratios and indices compared with norms. **Lambda** compares pure F responses with all others; a high Lambda suggests a simplifying, avoidant style. **Experience Balance** (EB) weighs human movement against colour: an **introversive** person (more M) works things out internally, an **extratensive** person (more colour) through feeling and action. Constellations flag specific risks: PTI (Perceptual-Thinking Index, formerly the Schizophrenia Index, SCZI), DEPI (depression), S-CON (suicide), CDI (coping deficit) and HVI (hypervigilance). The system gives **no IQ score**.

After Exner's death, **Gregory Meyer** and colleagues published the **Rorschach Performance Assessment System** (R-PAS, 2011). It keeps only research-supported variables, asks for two or three responses per card ("prompt and pull") so records are comparable in length, and reports standard scores against international norms. **Wayne Holtzman** had earlier solved the length problem differently: the **Holtzman Inkblot Technique** (1961) has two parallel forms of 45 cards each and allows one response per card, so every record is the same length and alternate-form reliability can be computed.

> ! EXAM TRAP
>
> There is **no "Murray system"** of Rorschach scoring: Henry Murray gave us the TAT. The Comprehensive System is Exner's (not Beck's, Klopfer's or Piotrowski's) and scores the Rorschach, not the MMPI, TAT or Wechsler scales.

The Thematic Apperception Test ●●●

The **Thematic Apperception Test** (TAT) was introduced in 1935 by **Christiana Morgan** (1897 to 1967) and **Henry Alexander Murray** (1893 to 1988) at the Harvard Psychological Clinic; Murray's manual followed in 1943. To apperceive is to perceive in the light of past experience. The theory is Murray's **need-press theory** of psychogenic needs (*Explorations in Personality*, 1938; Section 4.6), not personal constructs (George Kelly), operant conditioning (B. F. Skinner) or cognitive distortions (Aaron Beck). For each picture of people in an ambiguous

social scene, the person tells a dramatic story: what led up to it, what is happening, what the characters think and feel, and how it ends. The full set has **31 cards**: 30 pictures plus **blank card 16**, on which the person imagines a scene. Classically 20 cards were given over two sessions; in practice a clinician selects about **5 to 12** per session.

The interpreter assumes the storyteller identifies with the **hero** and lends the hero their own **needs**, such as need for achievement (n Ach), affiliation (n Aff) and aggression (n Agg). The **press** is the force of the environment on the hero: **alpha press** as it really is, **beta press** as the person perceives it. A recurring need-press combination is a **thema**, and the **outcome** (success, failure, rescue) shows the person's expectations. **Leopold Bellak** later systematised this scoring, and **David McClelland** scored stories for the achievement motive (1953). The TAT therefore reveals unconscious thoughts, conflicts and motives, and a client's characteristic ways of thinking, feeling and behaving in close relationships; when stories about ambiguous pictures are wanted, it is the test of choice. It is a projective personality test, not an intelligence, achievement or neuropsychological test, and it does not measure vocabulary or memory.

Unlike the Rorschach, the TAT uses ambiguous social scenes with characters, not symmetrical inkblots, and asks for stories, not percepts; it shows the content of motives and relationships more than the structure of thought. Its descendants adapt the pictures to the teller. The **Children's Apperception Test** (CAT; Leopold and **Sonya Sorel Bellak**, 1949) has 10 cards of animals in human situations for children of 3 to 10, who identify readily with animals; the CAT-H (1965) uses human figures and the CAT-S is a supplement. The Bellaks' Senior Apperception Technique (1973) serves older adults, and the Roberts Apperception Test for Children (1982) adds normed scoring. **Gerald Blum**'s Blacky Pictures (1950) show a dog family in scenes built around Freud's psychosexual stages, and **Edwin Shneidman**'s Make-A-Picture-Story test (MAPS, 1947) lets the person place cut-out figures on backgrounds before telling the story.

> **! EXAM TRAP**
>
> TAT: **Murray and Morgan, 1930s**, a **construction** technique. Not Freud, Jung, Skinner, Rorschach, Exner or Bellak (Bellak made the CAT). The CAT differs mainly in its **animal figures designed for children**; it is not objective, not an intelligence test and uses no inkblots.

> **INDIAN CONTEXT**
>
> **Uma Chowdhury** adapted the TAT (1960) and the CAT for India, with Indian figures, dress and settings. **Udai Pareek** adapted the Rosenzweig Picture-Frustration Study (below); do not swap them.

Completion and verbal techniques ○

The oldest projective method is the **word association test**. **Francis Galton** tried it on himself in 1879. **Carl Gustav Jung** made it clinical (1910) with 100 stimulus words, timing each reply and watching for **complex indicators** (a delayed reaction, no reply, repeating the stimulus word, an odd answer) that betray an emotionally charged complex (Section 4.4). **Grace Kent** and **Aaron Rosanoff** published a 100-word list in 1910 with frequency norms from 1,000 normal adults, so a reply could be scored as common or idiosyncratic.

The **sentence completion** method grew from **Hermann Ebbinghaus**'s completion method: the person finishes stems such as "My father..." or "I regret...". The **Rotter Incomplete Sentences Blank** (**Julian Rotter** and **Janet Rafferty**, 1950) has 40 stems, each rated for conflict or positive adjustment and summed into one adjustment score. The Sacks Sentence Completion Test (**Joseph Sacks** and **Sidney Levy**, 1950) has 60 items covering 15 attitude areas in four groups: family, sex, interpersonal relationships and self-concept.

The **Rosenzweig Picture-Frustration Study** (**Saul Rosenzweig**, 1945; some Indian texts give later dates) shows 24 cartoon-style frames of everyday frustrations; the respondent writes what the frustrated character says in an empty speech balloon. Replies are scored for **direction of aggression**: **extrapunitive** (blame turned outward), **intropunitive** (blame turned on oneself) or **impunitive** (blame avoided, the frustration glossed over); and for **type of reaction**: obstacle-dominance (the obstacle stands out), ego-defence (protecting the self) or need-persistence (seeking a solution). The **Group Conformity Rating** shows how often the person gives the commonest reply.

Expressive and choice techniques ○

In **Karen Machover**'s **Draw-A-Person** test (1949) the person draws a person, then one of the other sex. Her **body-image hypothesis** holds that the drawing reflects the drawer's self-concept and unconscious attitudes, so size, placement, erasures and omitted parts are read as signs; it is not a test of fine motor skill alone, academic achievement or brain lesions. Contrast **Florence Goodenough**'s Draw-a-Man, which counts details as an index of intellectual maturity (Section 2.6). **John Buck**'s **House-Tree-Person** test (1948) asks for a house (home and family), a tree (deeper feelings about the self) and a person (self-concept and relationships), first in pencil (achromatic phase), then in crayon (chromatic phase), each followed by questioning. In the **Kinetic Family Drawing** (**Robert Burns** and **S. Harvard Kaufman**, 1970) the child draws everyone in the family doing something, revealing perceived closeness and conflict. **Elizabeth Koppitz** listed emotional indicators in children's figure drawings, such as a tiny figure or gross asymmetry. **Max Hutt** read the copying of the Bender designs projectively (Section 2.11).

The **Szondi Test** (**Leopold Szondi**, 1947) is the classic choice technique: from 48 photographs of psychiatric patients, in six sets of eight, the person picks the faces they like and dislike most; its genetic theory is unsupported. **Margaret Lowenfeld**'s World Technique (a world built in a sand tray) and Mosaic Test (patterns from coloured pieces), the **Hand Test** (**Edward Wagner**, 1962: "What might this hand be doing?", read for acting-out tendencies), play, finger painting and **graphology** (handwriting analysis, with little validity) complete the expressive family.

> **! EXAM TRAP**
>
> House-Tree-Person is **John Buck**'s, not Aaron T. Beck's (cognitive therapy), David D. Burns's (author of *Feeling Good*) or Karen Machover's (Draw-A-Person). Kinetic Family Drawing is Robert Burns's, a different Burns.

How good are projective tests?

The case against them is psychometric. Many techniques have competing scoring systems, so two clinicians may not be giving the same test. Inter-scorer reliability is acceptable only with

structured systems such as Exner's or Rotter's; retest reliability is weak, because stories and percepts change between occasions. Global, intuitive interpretation has poor validity, and even scored variables show doubtful **incremental validity**: they add little to an interview and an inventory. Defenders reply that some Rorschach indices, notably of thought disorder and perceptual accuracy, hold up, and that the tests generate rich clinical hypotheses; they remain in clinical training and practice in India. Table 2.24 gathers the main tests.

KEY STUDY: ILLUSORY CORRELATION (CHAPMAN AND CHAPMAN, 1967, 1969)

Loren Chapman and **Jean Chapman** asked why clinicians trusted signs that research did not support. They paired figure drawings with symptom statements at random and showed them to students, who "discovered" the links clinicians reported, such as odd eyes with suspiciousness, though the real correlation was zero. In 1969 the same happened with invalid Rorschach signs, while valid signs went unnoticed. This **illusory correlation** comes from expectation, not data. The task was artificial, but practising clinicians reported the same invalid signs.

Table 2.24 The main projective tests compared.

Test	Author, year	Stimulus	Lindzey	Key fact
Rorschach	Rorschach, 1921	10 inkblots	Association	5 achromatic, 5 chromatic
Holtzman (HIT)	Holtzman, 1961	2 forms of 45 inkblots	Association	One response per card
Word association	Galton 1879; Jung 1910	100 words (Jung)	Association	Complex indicators
TAT	Morgan and Murray, 1935	31 cards, card 16 blank	Construction	Need and press
CAT	Bellak and Bellak, 1949	10 animal pictures	Construction	Ages 3 to 10
Rotter ISB	Rotter and Rafferty, 1950	40 stems	Completion	Adjustment score
Rosenzweig P-F	Rosenzweig, 1945	24 cartoons	Completion	Extra-, intro-, impunitive
Szondi Test	Szondi, 1947	48 patient photographs	Choice	Liked and disliked faces
Draw-A-Person	Machover, 1949	Blank page	Expression	Body-image hypothesis
House-Tree-Person	Buck, 1948	Blank pages	Expression	Achromatic then chromatic

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. The term "projective methods" was coined by _____ in 1939.
2. Of the ten Rorschach cards, cards II and III are black and _____.
3. The Rorschach phase in which each response is clarified without leading questions is the _____ phase.
4. A Rorschach response based on the shape of the blot alone is scored _____.
5. The Rorschach Comprehensive System (1974) was developed by _____.
6. In Lindzey's classification the TAT is a _____ technique.
7. The TAT rests on Murray's theory of needs and _____.
8. The House-Tree-Person test was developed by _____ in 1948.

Answers: (1) Lawrence K. Frank, (2) red, (3) inquiry, (4) F (pure form), (5) John Exner, (6) construction, (7) press, (8) John Buck.

2.11 Neuropsychological Assessment and Clinical Rating Scales

MEDIUM YIELD · 40 QS

A retired engineer forgets appointments and loses words mid-sentence. Is it depression, early dementia or a small stroke, and which abilities are gone? Neuropsychological tests profile what the brain can still do; clinical rating scales measure how severe symptoms are and whether treatment helps.

What neuropsychological assessment measures ●●

Neuropsychological assessment infers how the brain is working from behaviour: each standardised task samples one function, and a characteristic pattern of failures points to the damaged system. Its domains are orientation and consciousness, attention and concentration, processing speed, memory, language, visuospatial and constructional skill, **executive functions** (planning, shifting, inhibiting and monitoring goal-directed behaviour) and motor and sensory function; mood is checked too, because it colours every score. Opinions, attitudes and reactions are not neuropsychological domains. The purposes are diagnosis, localisation, a baseline for measuring change, rehabilitation planning and judgments of capacity. Localisation itself is in Section 5.11.

A **fixed battery** gives every patient the same tests scored against norms (Halstead-Reitan, Luria-Nebraska); a **flexible battery** adds tests to a core to fit the referral question. The **process approach** of **Edith Kaplan** (1924 to 2009) in Boston holds that how a patient fails matters as much as the score (a piecemeal copy and an incomplete one may score alike). Its roots lie in the qualitative method of **Alexander Romanovich Luria** (1902 to 1977), who used no norms but watched exactly where a function broke down.

A score needs a standard. The **normative comparison** (nomothetic) standard compares the patient with healthy people of the same age and education. The **individual comparison** standard compares present performance with the person's own estimated **premorbid level**, which one Indian paper calls the 'ideometric' approach: an engineer who now scores average has declined. **David Wechsler** (1896 to 1981) noticed that some subtests 'hold' with age and ill-

ness (Vocabulary, Information, Picture Completion, Object Assembly) while others 'don't hold' (Digit Span, Similarities, Digit Symbol, Block Design). His **deterioration index** or quotient, (hold minus don't hold) divided by hold, estimates how far brain or psychiatric illness has eroded cognitive and intellectual ability. The **National Adult Reading Test** (**Hazel Nelson**, 1982) predicts premorbid IQ from reading aloud 50 irregular words such as 'debt', a skill that survives early dementia. Such profiles replaced the old screening question 'organic or functional?', once answered with a single test such as the Bender-Gestalt.

> **! EXAM TRAP**
>
> Estimating the impact of brain or psychiatric illness on intellectual ability means the **deterioration quotient**. Distractors: a mediation index belongs to statistics (Chapter 9), sense of coherence to **Aaron Antonovsky**'s salutogenesis (Section 7.9), theory of mind to social cognition (Chapter 6). The 'ideometric' approach weighs premorbid functioning against current status; comparison with population norms is the normative approach, and neither quantitative scoring nor emotional well-being is meant.

Neuropsychological batteries ••

The classic fixed battery began with **Ward Halstead**'s studies of frontal-lobe patients (1947) and was validated by his student **Ralph Reitan** (1922 to 2014). The **Halstead-Reitan Neuropsychological Battery** is used for comprehensive neuropsychological assessment: to detect and localise brain dysfunction and gauge its severity. Its tests are the **Category Test** (abstract reasoning and concept formation); the **Tactual Performance Test** (the blindfolded patient fits blocks into a form board, then draws it from memory, giving time, memory and localisation scores); the **Seashore Rhythm Test**; the **Speech-Sounds Perception Test**; the **Finger Tapping** (Finger Oscillation) **Test** of motor speed; and the **Trail Making Test**, Parts A and B. Reitan added the Reitan-Indiana Aphasia Screening Test, a sensory-perceptual examination, grip strength and lateral dominance. A **Halstead Impairment Index** (the proportion of key scores in the impaired range) of 0.5 or more suggests brain damage.

The **Luria-Nebraska Neuropsychological Battery** gave Luria's methods American norms: **Anne-Lise Christensen** compiled them (1975), and **Charles Golden**, **Thomas Hammeke** and **Arnold Purisch** standardised them (1978; manual 1980). Form I has 269 items scored 0 to 2 in 11 clinical scales, from motor and rhythm to memory and intellectual processes; high T scores mean impairment.

The Indian **NIMHANS Neuropsychological Battery** (Bangalore; adult manual by **Shobini Rao**, D. K. Subbakrishna and K. Gopukumar, 2004, plus a children's battery) is a set of individually normed tests with Indian norms by age, sex and education: colour trails, digit vigilance, N-back, WCST, Tower of London, Stroop, verbal fluency, auditory verbal learning and a complex figure. The **PGI Battery of Brain Dysfunction** (PGIMER, Chandigarh; **Dwarka Pershad** and colleagues) combines the PGI Memory Scale, verbal and performance intelligence tests (the latter built on **C. M. Bhatia**'s, Section 2.6), the Nahor-Benson perceptual test and the Bender-Gestalt. The **PGI Memory Scale** (Pershad and **N. N. Wig**, 1977) has 10 subtests: remote and recent memory, mental balance, attention and concentration, delayed and immediate recall, retention for similar and for dissimilar pairs, visual retention and recognition.

> **! EXAM TRAP**
>
> The Halstead-Reitan assesses brain function, not personality, vocational interest, attachment, self-reported depression or unconscious conflict, and it is not a projective test. The Rorschach is never part of it (the WAIS and MMPI are often given alongside). Luria did not write the Luria-Nebraska. PGI means Chandigarh; NIMHANS means Bangalore.

Memory and visual-motor tests ●●●

The **Wechsler Memory Scale** (Wechsler, 1945) began as seven brief subtests summed into a **memory quotient**. The WMS-IV (2009) gives five indices: **Auditory Memory, Visual Memory, Visual Working Memory, Immediate Memory and Delayed Memory**. Verbal and Performance IQ, like crystallised and fluid scores, belong to intelligence tests (Section 2.6); Mechanical Reasoning is a Differential Aptitude Tests subtest (Section 2.8).

In the **Rey Auditory Verbal Learning Test** (**André Rey**, Geneva) the examiner reads 15 words over five trials, then an interference list, then asks for the first list again immediately and after a delay; a flat learning curve or heavy delayed loss signals a memory disorder. The California Verbal Learning Test (1987) uses a 16-word 'shopping list' of four categories, showing whether the person groups words by meaning. Visual memory is tested by the **Rey-Osterrieth Complex Figure**, a detailed drawing first copied (planning and construction) and later drawn from memory, and by the **Benton Visual Retention Test** (**Arthur Benton**, 1946), in which 10 designs are each shown briefly and drawn from memory.

The classic visual-motor test is the **Bender Visual-Motor Gestalt Test** (1938) of **Lauretta Bender** (1897 to 1987). She adapted nine figures (A and 1 to 8) from designs the Gestalt psychologist **Max Wertheimer** (1880 to 1943) had used to study perceptual organisation. Copying them requires perceiving each design as a whole and reproducing it by hand, so the test assesses **visual-motor integration**. Its errors screen for organic brain damage and neurological dysfunction in adults and for developmental immaturity in children, scored by **Elizabeth Koppitz**'s developmental system (1963). The Bender-Gestalt II (2003) has 16 designs and a recall phase. **Max Hutt** also read the drawings projectively, a secondary use.

Simplest of all is **digit span**. Repeating digits in the order heard (forward span, about seven in adults) measures attention span. Repeating them in reverse (backward span) forces the person to hold and rearrange the digits at once, so it measures **working memory** (Section 1.7), not long-term, sensory or procedural memory.

> **! EXAM TRAP**
>
> The Bender does not test verbal intelligence, vocational interest, personality, unconscious conflict, depression, schizophrenia, ADHD, reading, auditory or working memory, verbal abstraction or premorbid intelligence (that is the NART). Bender (1938) means copying nine figures; Benton (1946) means drawing ten designs from memory.

Executive function, attention, thinking and language ●●●

The most widely used test of executive function is the **Wisconsin Card Sorting Test** (WCST) of **David Grant** and **Esta Berg** (1948), with a clinical manual by **Robert Heaton** (1981). The patient sorts 128 response cards under four stimulus cards by colour, form or number, told only 'right' or 'wrong', and after ten correct sorts the rule changes without warning (Figure 2.17). A

healthy person notices the run of 'wrong' and shifts; a frontal patient keeps sorting by the old rule. The scores are **perseverative errors**, categories achieved and failure to maintain set. The WCST measures **set-shifting** and cognitive flexibility and is most sensitive to dysfunction of the **dorsolateral prefrontal cortex**.

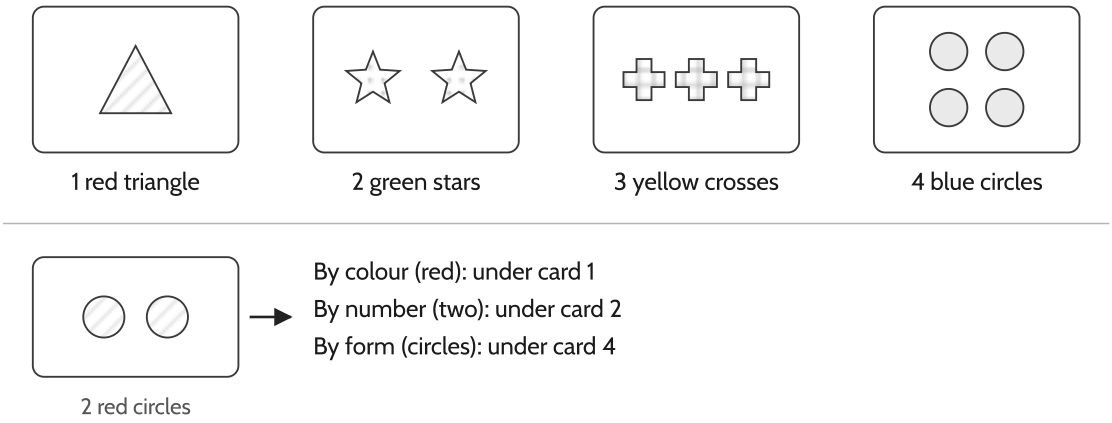


Figure 2.17 The Wisconsin Card Sorting Test. Each response card fits more than one rule; clinging to a rule after it has changed is perseveration.

KEY STUDY: MILNER AND THE FRONTAL LOBES (1963)

In Montreal **Brenda Milner** gave the WCST to patients having brain surgery for epilepsy. Those with dorsolateral frontal excisions achieved few categories and made many perseverative errors, some while saying the rule had changed; temporal and posterior cases did not. **Criticism:** perseveration also follows damage elsewhere and occurs in schizophrenia, so the test is sensitive to frontal damage but not specific to it.

! EXAM TRAP

The 'Minnesota Number Ordering Test' and 'Manhattan Card Sorting Test' do not exist. The WCST does not measure memory, language, reading, visual acuity, mood or manual dexterity, and its region is the dorsolateral prefrontal cortex, not the hippocampus, primary motor cortex or cerebellum.

The **Trail Making Test**, from the Army Individual Test Battery (1944), has two parts: Part A joins 25 numbered circles in order (attention and speed), and Part B alternates numbers and letters (1, A, 2, B and so on), adding set-shifting. In children with ADHD it assesses inattention; impulsivity, over-activity and social skills are better judged from continuous performance errors and rating scales. The **Stroop Colour-Word Test** (**John Ridley Stroop**, 1935) asks for the ink colour of conflicting colour words ('RED' printed in green), measuring **response inhibition**. The **Tower of London** (**Tim Shallice**, 1982), moving beads on three pegs to a goal in the fewest moves, measures planning, as does the Tower of Hanoi. **Verbal fluency** asks for words beginning with a letter in a minute (the Controlled Oral Word Association or FAS test) or from a category such as animals.

Sustained attention is tested by the **Continuous Performance Test** (**H. Enger Rosvold** and colleagues, 1956): in a long stream of letters the person responds only to a target, so omissions reflect inattention and commissions impulsivity. The Test of Everyday Attention measures sustained, selective and divided attention through everyday tasks such as map searching. Neither shows which side of the body is affected; Benton's **Finger Localization Test** does. The patient, without looking, identifies fingers the examiner touches, and consistently more errors on one hand point to the opposite hemisphere.

Concept formation, and with it **thought disorder**, is assessed by sorting. In the **Object Sorting Test** of **Kurt Goldstein** (1878 to 1965) and **Martin Scheerer** (1941) the patient groups everyday objects and explains why they belong together; concrete, bizarre or loose groupings expose disordered thinking directly. The Rorschach reveals it only indirectly (Section 2.10), the TAT taps needs and conflicts, and the Bender visual-motor skill. Proverbs and similarities test abstract against **concrete thinking** in the same way.

Language tests include the **Boston Naming Test** (60 line drawings), the **Token Test** of **Ennio De Renzi** and **Luigi Vignolo** (1962), which checks comprehension with commands such as 'touch the small red circle', and the Boston Diagnostic Aphasia Examination and Western Aphasia Battery. The aphasias are in Section 5.10. Table 2.25 gathers the tests by domain.

Table 2.25 Which neuropsychological test for which domain.

Domain	Classic tests
Consciousness, orientation	Glasgow Coma Scale; MMSE orientation
Attention and speed	Digit span forward, Trail Making A, CPT, Test of Everyday Attention
Working memory	Digit span backward, N-back
Verbal memory	WMS-IV, Rey AVLT, CVLT, PGI Memory Scale
Visual memory	Benton Visual Retention Test, Rey figure recall
Visuoconstruction	Bender-Gestalt, Rey figure copy, clock drawing
Executive function	WCST, Trail Making B, Stroop, Tower of London, fluency, Category Test
Concept formation	Object Sorting Test, proverbs, similarities
Language	Boston Naming Test, Token Test, BDAE, WAB
Motor and sensory	Finger Tapping, grip strength, Finger Localization

Bedside screening and staging ○

A short screen settles whether cognition is impaired at all. The **Mini-Mental State Examination** (MMSE) of **Marshal Folstein**, **Susan Folstein** and **Paul McHugh** (1975) takes about ten minutes and is scored out of 30: orientation (10), registration (3), attention by serial sevens (5), recall (3), and language and copying (9). A score of 23 or below (cut-off 23/24) suggests impairment. It screens for and tracks global cognitive impairment and dementia, not personality, depression or IQ. Because it is verbal and education-bound, **Mary Ganguli** built the Hindi Mental State Examination (1995) for rural, largely illiterate older Indians.

The **Montreal Cognitive Assessment** (MoCA) of **Ziad Nasreddine** (2005) is also out of 30 and takes about ten minutes, but adds executive tasks (a short trail, fluency, abstraction), cube copying and **clock drawing**; 26 and above is normal. It is therefore more sensitive than the

MMSE to mild cognitive impairment. Addenbrooke's Cognitive Examination III extends the idea to 100 points.

Staging scales grade severity. The **Glasgow Coma Scale** of **Graham Teasdale** and **Bryan Jennett** (1974) rates level of consciousness from the best eye opening (1 to 4), verbal response (1 to 5) and motor response (1 to 6), totalling 3 to 15; 8 or below means severe injury or coma. In dementia the Clinical Dementia Rating grades six domains of cognition and daily function from 0 to 3, the Global Deterioration Scale (**Barry Reisberg**, 1982) describes seven stages, and the Hachinski Ischemic Score separates vascular from Alzheimer-type dementia (Section 8.10).

> **! EXAM TRAP**
>
> The Glasgow Coma Scale measures consciousness, not attention, memory or intelligence, and its minimum is 3, not 0. The MoCA's advantage over the MMSE is sensitivity to mild impairment, not brevity, and neither is reserved for neurologists.

Rating scales for depression and anxiety ●●

Rating scales grade symptom severity and track change. What matters first is who completes them. In a **self-report inventory** the patient rates himself; in a **clinician-rated scale** a trained interviewer rates the patient from interview and observation. Structured diagnostic interviews (SCID, PSE) are in Section 7.2, and life-event scales such as Holmes and Rahe's in Section 1.11.

The **Beck Depression Inventory** of **Aaron Temkin Beck** (1921 to 2021), first published in 1961, is the most used self-report depression measure. The BDI-II (1996) has 21 items, each rated 0 to 3 for the past two weeks, so totals run from 0 to 63: **0 to 13 minimal, 14 to 19 mild, 20 to 28 moderate, 29 to 63 severe**. For a patient with two months of sadness, anhedonia and worthlessness, the BDI is the instrument for further evaluation, not the Rorschach (projective), the MMPI (broad personality, Section 2.9) or the WAIS (IQ). Clinician-rated alternatives are the **Hamilton Rating Scale for Depression** (**Max Hamilton**, 1960) and the Montgomery-Asberg scale, which is sensitive to treatment change; the Zung scale (1965) and PHQ-9 are brief self-report screens, and Beck's Hopelessness Scale and Scale for Suicide Ideation assess suicide risk. For anxiety, the clinician-rated **Hamilton Anxiety Rating Scale** pairs with the self-report Beck Anxiety Inventory and GAD-7. The Fear Survey Schedule of **Joseph Wolpe** and **Peter Lang** (1964) lists feared situations; the SUDS rating used in exposure is in Section 7.4.

> **! EXAM TRAP**
>
> RML 2025 keyed the BDI as 'typically used for neuropsychological assessment' against a Likert scale (an attitude-scaling format, Section 2.8), the 16PF (a personality inventory, Section 2.9) and the Peabody Individual Achievement Test (an achievement test, Section 2.8). The BDI is a self-report depression inventory, not a neuropsychological test; neuropsychologists add it to see whether low mood is depressing test scores. Choose it only because the others are further off.

Scales for OCD, psychosis, substances, children and forensic work ●●

The **Yale-Brown Obsessive Compulsive Scale** (Y-BOCS) of **Wayne Goodman** and colleagues (1989) is the standard clinician-rated measure of OCD severity. After a symptom checklist, ten core items rate obsessions (five) and compulsions (five) on the same five dimensions, time, interference, distress, resistance and control, each 0 to 4, for a total of 0 to 40. It does not rate depression, bipolar disorder, schizophrenia, PTSD or suicidal ideation.

For psychosis the **Brief Psychiatric Rating Scale** (**John Overall** and **Donald Gorham**, 1962) covers general psychopathology, and the **Positive and Negative Syndrome Scale** (PANSS) of **Stanley Kay**, **Abraham Fiszbein** and **Lewis Opler** (1987) measures symptom severity in schizophrenia with 30 items: 7 positive, 7 negative and 16 general psychopathology. **Nancy Andreasen**'s SANS and SAPS rate negative and positive symptoms separately; the Young Mania Rating Scale rates mania. General screens such as the **General Health Questionnaire** (**David Goldberg**, 1972) and **Leonard Derogatis**'s SCL-90-R detect distress without making a diagnosis. For substance use there are the WHO's Alcohol Use Disorders Identification Test (AUDIT), CAGE (Cut down, Annoyed, Guilty, Eye-opener), the Michigan Alcoholism Screening Test (MAST), the CIWA-Ar for alcohol withdrawal, the SADQ for dependence severity and the Fagerstrom Test for nicotine dependence.

The Clinical Global Impression rates severity and improvement from 1 to 7; the Global Assessment of Functioning, dropped in DSM-5, gave way to the WHO's WHODAS 2.0. In India, disability from mental illness is certified with **IDEAS**, the Indian Disability Evaluation and Assessment Scale (Indian Psychiatric Society, 2002). Children are rated with the Conners Rating Scales (ADHD), **Thomas Achenbach**'s **Child Behavior Checklist** and its Indian adaptation, **Savita Malhotra**'s Childhood Psychopathology Measurement Schedule (PGIMER); the Edinburgh Postnatal Depression Scale (1987) screens new mothers. In forensic work the **Psychopathy Checklist-Revised** of **Robert Hare** (1980, revised 1991) rates 20 items 0 to 2 from interview and case records. Hare built it on **Hervey Cleckley**'s portrait of the psychopath in *The Mask of Sanity* (1941), but Cleckley devised no checklist. Table 2.26 compares the main scales.

Table 2.26 Clinical rating scales compared.

Scale	Author, year	Items	Rater	Measures
BDI-II	Beck 1961; 1996	21 (0 to 3)	Self	Depression severity
HAM-D	Hamilton 1960	17	Clinician	Depression severity
MADRS	Montgomery, Asberg 1979	10	Clinician	Depression, change
HAM-A	Hamilton 1959	14	Clinician	Anxiety
BAI	Beck 1988	21	Self	Anxiety
Y-BOCS	Goodman 1989	10 (0 to 4)	Clinician	OCD severity
BPRS	Overall, Gorham 1962	18 (16 at first)	Clinician	General psychopathology
PANSS	Kay, Fiszbein, Opler 1987	30	Clinician	Schizophrenia severity
YMRS	Young 1978	11	Clinician	Mania
AUDIT / CAGE	WHO / Ewing	10 / 4	Self or interviewer	Problem drinking
EPDS	Cox 1987	10	Self	Postnatal depression

Scale	Author, year	Items	Rater	Measures
PCL-R	Hare 1980, 1991	20 (0 to 2)	Clinician	Psychopathy

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Comparing current performance with the person's own premorbid level is the _____ comparison standard.
2. Wechsler's index contrasting hold and don't-hold subtests is the _____ quotient.
3. A Halstead Impairment Index of _____ or more suggests brain damage.
4. The Bender-Gestalt assesses _____ integration.
5. Backward digit span measures _____ memory.
6. The WCST is most sensitive to dysfunction of the _____ prefrontal cortex.
7. A BDI-II score of 24 falls in the _____ band.

Answers: (1) individual, (2) deterioration, (3) 0.5, (4) visual-motor, (5) working, (6) dorsolateral, (7) moderate.

Galton's Anthropometric Laboratory, Revisited

Look again at Galton's counter with this chapter behind you. Every visitor met the same instruments, instructions and recording, so his procedures were standardised and objective (Section 2.1). Measures such as height and grip strength are highly reliable (Section 2.3), yet Wissler showed that sensory and motor tests of this kind had almost no criterion validity for academic success (Section 2.4): consistency is necessary, not sufficient. Galton's thousands of cards were among the first large norms, and his spread of scores around an average foreshadowed standard scores (Section 2.5). Binet's turn to judgement and reasoning founded the intelligence tests of Section 2.6, and every later test has had to prove what Galton assumed. His belief in inherited ability, which fed eugenics, is one reason testing now carries rules of fairness.

Chapter summary

2.1 Psychological Testing: Nature, History, Types and Ethics. A **psychological test** is an objective, standardised measure of a sample of behaviour (Anastasi); assessment integrates tests with interview, observation and records. Testing runs from Chinese examinations through Galton's laboratory (1884), Cattell's 'mental tests' (1890) and Wissler's disproof (1901) to Binet and Simon (1905), the Army Alpha and Beta, Woodworth's inventory and the MMPI (1943). Tests are objective or subjective, speed or power, of maximum or typical performance. A good test is objective, reliable, valid, normed and practicable; standardisation fixes procedures and norms. Clinicians read data as samples, correlates or signs; actuarial prediction usually beats clinical judgement, which suffers confirmatory bias, illusory correlation and the Barnum effect.

2.2 Building a Test: Items, Item Analysis, IRT and Response Bias. Items are **essay** or **objective** (supply or selection type), and guessing is corrected by S = R - W/(n - 1). **Item analysis** gives difficulty p (pro-

portion passing; ideal 0.50, about 0.62 for four options) and discrimination D = p upper - p lower on Kelley's 27 percent groups; a negative D flags a miskeyed item. **Ceiling** and **floor** effects stop a test discriminating at the top or bottom. In **IRT** the item characteristic curve plots success against the latent trait θ (b, a, c; Rasch 1PL to 3PL), and sample-invariant parameters allow adaptive testing. **Acquiescence**, social desirability and **halo** (Thorndike, 1920) bias scores.

2.3 Reliability: Consistency and the Standard Error of Measurement. Reliability is the consistency of a measure: dependability and freedom from measurement error. In Spearman's classical test theory (1904) every observed score is a true score plus random error, and reliability is the ratio of true-score to observed-score variance. **Test-retest** correlation gives the coefficient of stability, for stable traits; **parallel forms** give the coefficient of equivalence. Single-sitting **internal consistency** methods are split-half (stepped up by the **Spearman-Brown** formula, so .60 becomes .75), Rulon and Flanagan, **KR-20** and KR-21 for right-wrong items, and **Cronbach's alpha** (1951). **Cohen's kappa** measures rater agreement beyond chance. The **standard error of measurement**, SD × √(1 - r), bands one person's score.

2.4 Validity and the Accuracy of Test Decisions. Validity is the extent to which a test measures what it claims to measure; since **Messick** (1989) it is a unified judgement about the interpretations and uses of scores. **Face validity** is mere appearance; **content validity** is adequate sampling of the domain (blueprint, Lawshe's CVR). **Criterion-related validity** is **concurrent** (criterion measured now) or **predictive** (criterion measured later). **Construct validity** (Cronbach and Meehl, 1955) rests on convergent and discriminant evidence and the multitrait-multimethod matrix (Campbell and Fiske, 1959). A test can be reliable without being valid, and validity cannot exceed the square root of reliability. **Sensitivity** and **specificity** describe the test; predictive values depend on the base rate.

2.5 Norms, Standard Scores and Interpreting a Score. A raw score means nothing without a frame of reference. **Norm-referenced** interpretation compares a person with a representative standardisation sample; **criterion-referenced** interpretation (Glaser, 1963) compares with a fixed mastery standard. Good norms are large, representative, recent and relevant. The **ratio IQ** (MA ÷ CA × 100) gave way to Wechsler's **deviation IQ** (mean 100, SD 15). Percentile ranks are ordinal. The **z score** (mean 0, SD 1) and the normal curve (68, 95, 99.7 per cent within 1, 2, 3 SD) link T scores (50, 10), stanines (5, 2), stens (5.5, 2), scaled scores (10, 3) and CEEB scores (500, 100).

2.6 Intelligence Tests: Binet, Wechsler, Non-verbal and Indian Scales. Intelligence tests are reliable and predict school achievement but not behaviour in general. **Binet and Simon** (1905) built the first test; **Terman's** Stanford-Binet (1916) used the ratio IQ, and the SB5 (Roid, 2003) measures five factors in verbal and non-verbal domains. **Wechsler's** Bellevue scale (1939) introduced the deviation IQ; the WAIS-IV (2008; 16:0 to 90:11) gives an FSIQ and four indices (VCI, PRI, WMI, PSI) from 10 core subtests, and the WISC-V covers 6:0 to 16:11. Raven's Progressive Matrices measures fluid reasoning and is culture-fair. In India, Bhatia's performance battery suits illiterate people, while the Binet-Kamat, MISIC and WAPIS adapt the Stanford-Binet, WISC and WAIS.

2.7 Assessing Development, Adaptive Behaviour and Disability. Development and everyday competence are scored apart from IQ as **DQ** (developmental age / CA × 100) and **SQ** (social age / CA × 100), from several informants because symptoms vary across contexts. Infant scales include the **Bayley** (1 to 42 months), Gesell's schedules, Denver II and the Indian DST, DASII and TDSC.

Doll's **VSMS** (1935; 117 items; SQ) and Malin's Indian version (89 items) measure social maturity; the **Vineland Adaptive Behavior Scales**, adaptive functioning. DSM-5 **intellectual disability** needs IQ and adaptive deficits, with severity graded adaptively; **global developmental delay** is for under-5s. Under the RPWD Act 2016, learning disability is certified from 8 years, autism with **ISAA**.

2.8 Aptitude, Achievement, Creativity, Interest and Attitude Measures. An **achievement test** measures what has been learnt in a specific academic area; an **aptitude test** predicts future learning in a specific area, and Anne Anastasi placed both on one continuum. Multiple aptitude batteries (the DAT with eight subtests, 1947; the GATB with nine aptitudes) serve guidance. Guilford's **divergent thinking** is scored for fluency, flexibility, originality and elaboration (TTCT; Mednick's RAT). Strong keyed interests empirically, Kuder used ipsative forced-choice triads and Holland proposed RIASEC. Thurstone's **equal-appearing intervals** use judges and median scale values; Likert's **summated ratings** sum respondents' agreement ratings; Guttman's cumulative scale needs reproducibility of 0.90; Bogardus measured social distance; the IAT measures implicit attitudes by reaction time.

2.9 Objective Personality Inventories. An **objective personality test** fixes items, answer format and scoring. Cattell accepted three data sources: **L-data** (life records), **Q-data** (questionnaires) and **T-data** (objective tests). Inventories are built rationally (Woodworth's Personal Data Sheet, 1917), theoretically (EPPS, MBTI, MCMI), by **empirical criterion keying** (MMPI, CPI) or by factor analysis (16PF, EPQ, NEO-PI-R, GZTS, Comrey). The **MMPI** (Hathaway and McKinley, 1943) assesses personality and psychopathology with validity scales **L, F and K** and ten clinical scales; the MMPI-2, MMPI-2-RF (RC scales) and MMPI-3 followed. The Rep test and Q-sort study one person at a time.

2.10 Projective Techniques. **Projective techniques** rest on the **projective hypothesis** (term coined by Frank, 1939): faced with an ambiguous stimulus, people project needs, conflicts and unconscious processes into their answers, which resists self-report bias. **Lindzey** (1959) grouped them as association, construction, completion, choice or ordering and expression. The **Rorschach** (1921) has 10 cards (5 achromatic, 2 black-red, 3 multicoloured), a response and inquiry phase, and codes for location, determinants and form quality; Exner's Comprehensive System (1974) and R-PAS (2011) standardised it. The **TAT** (Morgan and Murray, 1935; 31 cards) is read for needs and press. Sentence completion, the Rosenzweig P-F, figure drawings and the Szondi complete the family, all psychometrically weak.

2.11 Neuropsychological Assessment and Clinical Rating Scales. **Neuropsychological assessment** infers brain function from behaviour across attention, memory, language, visuospatial, executive, motor and sensory domains, using fixed batteries (**Halstead-Reitan**, Luria-Nebraska), flexible batteries or Kaplan's process approach, and comparing scores with norms or with the person's own premorbid level (Wechsler's **deterioration quotient**, the NART). India has the NIMHANS and PGI batteries. Key tests are the WMS-IV, Rey AVLT, the **Bender-Gestalt** (visual-motor integration), backward digit span (working memory) and the **WCST** (set-shifting, dorsolateral prefrontal cortex). The MMSE, MoCA and Glasgow Coma Scale screen and stage. Rating scales are self-report (BDI) or clinician-rated (Hamilton scales, Y-BOCS, PANSS, PCL-R).

Who's who

Table 2.27 The people in this chapter.

Name	When	Examined for
Anne Anastasi	1908 to 2001	Defined a test as an objective and standardised measure of a sample of behaviour; Psychological Testing (1954)
Lee J. Cronbach	1916 to 2001	A test as a systematic procedure for comparing persons; maximum versus typical performance (1949)
A. K. Singh	Indian textbook author	Lists objectivity first among the five characteristics of a good test
Jean-Étienne Esquirol	1772 to 1840	Separated mental retardation from mental illness; graded it by use of language
Édouard Séguin	1812 to 1880	Sensory-motor training of retarded children; the Séguin form board
Francis Galton	1822 to 1911	Hereditary Genius (1869); Anthropometric Laboratory, London (1884); father of mental testing and psychometrics
Karl Pearson	1857 to 1936	Product-moment correlation coefficient
Wilhelm Wundt	1832 to 1920	Cattell trained in his Leipzig laboratory (Chapter 1)
James McKeen Cattell	1860 to 1944	Coined the term 'mental tests' (1890, Mental Tests and Measurements)
Clark Wissler	1870 to 1947	Showed in 1901 that Cattell's sensory tests did not predict college grades
Alfred Binet	1857 to 1911	With Simon, the first practical intelligence test (1905)
Théodore Simon	1873 to 1961	Co-author of the Binet-Simon scale
William Stern	1871 to 1938	Proposed the intelligence quotient, mental age divided by chronological age (1912)
Lewis Terman	1877 to 1956	Stanford-Binet (1916); IQ = MA/CA x 100
Arthur Otis	1917	Group test items used in the Army Alpha
Robert Yerkes	1876 to 1956	Led the Army Alpha (literate) and Army Beta (non-verbal) group tests (1917)
Robert S. Woodworth	1869 to 1962	Personal Data Sheet (1917 to 1918), the first personality inventory
Hermann Rorschach	1884 to 1922	Inkblot test (1921)
Edward K. Strong	1927	Strong Vocational Interest Blank
L. L. Thurstone	1887 to 1955	Attitude scaling (1928 to 1929); multiple factor analysis
Christiana Morgan	1897 to 1967	Thematic Apperception Test with Murray (1935)
Henry Murray	1893 to 1988	Thematic Apperception Test with Morgan (1935)
Oscar Buros	1905 to 1978	Mental Measurements Yearbook (from 1938)
David Wechsler	1896 to 1981	Wechsler-Bellevue scale (1939)
Starke Hathaway	1903 to 1984	MMPI with McKinley (1943)
J. Charnley McKinley	1891 to 1950	MMPI with Hathaway (1943)
V. V. Kamat	1934	Indian revision of the Binet scale (Section 2.6)
C. M. Bhatia	1955	Battery of performance tests of intelligence (Section 2.6)
A. J. Malin	1969	Malin's Intelligence Scale for Indian Children, an Indian WISC (Section 2.6)
Paul E. Meehl	1920 to 2003	Clinical versus Statistical Prediction (1954); named the Barnum effect (1956)
Bertram Forer	1914 to 2000	Demonstrated the Barnum (Forer) effect with a universal personality sketch (1949)
P. T. Barnum	1810 to 1891	Showman after whom Meehl named the Barnum effect
Loren Chapman	1967 and 1969	Illusory correlation in Draw-a-Person and Rorschach signs, with Jean Chapman
Jean Chapman	1967 and 1969	Illusory correlation in Draw-a-Person and Rorschach signs, with Loren Chapman
Truman Lee Kelley	1884 to 1961	Upper and lower 27 percent groups for item analysis (1939)
Robert L. Ebel	1965 (definition of item discrimination)	Rule of thumb for grading the discrimination index D (0.40 and above very good, below 0.20 poor)

Name	When	Examined for
Georg Rasch	1901 to 1980	One-parameter logistic (Rasch) model of item response theory (1960)
Allan Birnbaum	1968 (logistic IRT models)	Two- and three-parameter logistic models of item response theory
Frederic M. Lord	1912 to 2000	Laid the foundations of item response theory (from 1952)
Lee Joseph Cronbach	1916 to 2001	Response sets (1946, 1950)
Douglas N. Jackson	1929 to 2004	With Messick, content versus style in personality assessment (1958): response styles
Samuel Messick	1931 to 1998	With Jackson, content versus style in personality assessment (1958): response styles
Allen L. Edwards	1957 (The Social Desirability Variable)	Social desirability response set; forced-choice Edwards Personal Preference Schedule
Irwin A. Berg	1950s	Deviation hypothesis of response sets
Patricia Cain Smith	1963	Behaviourally anchored rating scales (BARS), with Lorne Kendall
Lorne M. Kendall	1963	Behaviourally anchored rating scales (BARS), with Patricia Cain Smith
Edward Lee Thorndike	1874 to 1949	Named the halo effect (1920), from officers' ratings of soldiers
Charles Spearman	1863 to 1945	True-score model X = T + E and correction for attenuation (1904); Spearman-Brown formula (1910)
William Brown	1881 to 1952	Spearman-Brown prophecy formula, published independently of Spearman in 1910
Phillip Rulon	1939	Rulon's split-half formula: 1 minus the variance of half-score differences over total-score variance
John Flanagan	1906 to 1996	Flanagan's split-half formula from the two half variances (1937)
Louis Guttman	1916 to 1987	Gave the same split-half formula as Flanagan (1945)
G. Frederic Kuder	1903 to 2000	KR-20 and KR-21 with Marion Richardson (1937)
Marion Richardson	1937	KR-20 and KR-21 with G. Frederic Kuder
Lee Cronbach	1916 to 2001	Coefficient alpha (1951); bandwidth-fidelity dilemma with Gleser (1965); generalizability theory (1972)
Cyril Hoyt	1941	Reliability by analysis of variance of persons by items, equal to alpha
Jacob Cohen	1923 to 1998	Cohen's kappa for chance-corrected rater agreement (1960); weighted kappa (1968)
J. Richard Landis	1977	Kappa benchmarks with Gary Koch: .81 to 1.00 almost perfect, .21 to .40 fair
Gary Koch	1977	Kappa benchmarks with J. Richard Landis
Joseph Fleiss	1937 to 2003	Fleiss' kappa for three or more raters (1971)
Maurice Kendall	1907 to 1983	Kendall's coefficient of concordance W for agreement among rankings
Goldine Gleser	1915 to 2004	Bandwidth-fidelity dilemma with Cronbach (1965); generalizability theory (1972)
Harinder Nanda	1972	Co-author of generalizability theory with Cronbach, Gleser and Rajaratnam
Nageswari Rajaratnam	1972	Co-author of generalizability theory with Cronbach, Gleser and Nanda
Henry Edward Garrett	1894 to 1973	Classic definition: validity is the extent to which a test measures what it claims (purports) to measure
C. H. Lawshe	1975	Content validity ratio (1975); the term synthetic validity
Lee Sechrest	1963	Incremental validity: the prediction a test adds beyond existing measures
Paul Everett Meehl	1920 to 2003	Construct validity with Cronbach (1955); base rates and predictive values with Rosen (1955)
Donald Thomas Campbell	1916 to 1996	Convergent and discriminant validity; multitrait-multimethod matrix with Fiske (1959)
Donald W. Fiske	1959	Multitrait-multimethod matrix with Campbell (1959)
Charles Edward Spearman	1863 to 1945	Correction for attenuation (1904)
Albert Rosen	1955	With Meehl, showed that the accuracy of a test sign or cutting score depends on the base rate (1955)
H. C. Taylor	1939	Taylor-Russell tables: success rate from validity, selection ratio and base rate (1939)

Name	When	Examined for
J. T. Russell	1939	Taylor-Russell tables, with Taylor (1939)
Robert Glaser	1921 to 2012	Distinguished criterion-referenced from norm-referenced measurement (1963)
Lewis Madison Terman	1877 to 1956	Stanford-Binet (1916); ratio IQ = (MA / CA) x 100
William A. McCall	1922	T score: mean 50, SD 10 (1922)
Raymond Bernard Cattell	1905 to 1998	16PF (1949), scored in stens; Culture Fair Intelligence Test (SD 16)
J. P. Guilford	1897 to 1987	C scale: 11 units, mean 5, SD 2 (1954)
James Robert Flynn	1934 to 2020	Flynn effect: IQ scores rising about 3 points a decade, so old norms grow too easy
Theodore Simon	1872 to 1961	Co-author of the Binet-Simon scale (1905, 1908, 1911)
Maud Merrill	1937	Co-author with Terman of Stanford-Binet Forms L and M (1937) and Form L-M (1960)
Robert L. Thorndike	1986	Senior author of the Stanford-Binet Fourth Edition (SB4), a point scale
Elizabeth Hagen	1986	Co-author of the SB4
Jerome M. Sattler	1986	Co-author of the SB4
Gale H. Roid	2003	Stanford-Binet Fifth Edition (SB5): five factors, verbal and non-verbal domains, SD 15; co-author of the Leiter-3
Louis Leon Thurstone	1887 to 1955	Primary Mental Abilities tests: seven abilities scored separately
Alan S. Kaufman	1983	K-ABC (1983), KABC-II (2004) and the Kaufman Brief Intelligence Test (K-BIT 1990, KBIT-2 2004)
Nadeen L. Kaufman	1983	Co-author of the K-ABC, KABC-II and K-BIT
Alexander Luria	1902 to 1977	Sequential and simultaneous processing and three functional units, the basis of the K-ABC and the PASS theory
J. P. Das	1997	Indian-born co-author of the PASS theory and the Das-Naglieri Cognitive Assessment System
Jack A. Naglieri	1997	Co-author of the Cognitive Assessment System (PASS)
Richard W. Woodcock	1977	Woodcock-Johnson Tests of Cognitive Abilities, now built on CHC theory
Mary Bonner Johnson	1977	Co-author of the Woodcock-Johnson tests
John Carlyle Raven	1902 to 1970	Raven's Progressive Matrices: Standard (1938), Coloured (1947), Advanced; fluid reasoning, culture-fair
Russell Graydon Leiter	1929	Leiter International Performance Scale, instructions by pantomime (Leiter-3, 2013)
Lloyd M. Dunn	1959	Peabody Picture Vocabulary Test: receptive vocabulary by pointing to pictures
Florence Laura Goodenough	1886 to 1959	Draw-a-Man test (1926), 51 scoring points, intellectual maturity
Dale B. Harris	1963	Revised the Draw-a-Man as the Goodenough-Harris Drawing Test
Karen Machover	1949	Projective Draw-a-Person test (Section 2.10)
Robert Mearns Yerkes	1876 to 1956	Army Alpha (verbal) and Army Beta (non-verbal) group tests, 1917
Samuel C. Kohs	1920 (book 1923)	Kohs Block Design test, borrowed by Wechsler and by Bhatia
W. P. Alexander	1932	Alexander Pass-Along test, part of Bhatia's Battery
Edouard Seguin	1812 to 1880	Seguin Form Board, a teaching device for children with intellectual disability
Henry Herbert Goddard	1866 to 1957	Goddard's version of the Seguin Form Board: 10 shapes, three trials, shortest time
Arnold Gesell	1880 to 1961	Gesell Drawing Test: copying geometric figures as a developmental measure
Arthur J. Malin	1969	Malin's Intelligence Scale for Indian Children (MISIC), Nagpur adaptation of the WISC, ages 6 to 15
P. Ramalingaswamy	1974	Wechsler Adult Performance Intelligence Scale (WAPIS), Indian adaptation of the WAIS performance scale
Pramila Phatak	Baroda	Indian norms for the Draw-a-Man test
Thomas M. Achenbach	1987	Cross-informant approach: parent, teacher and child ratings agree only modestly; Child Behavior Checklist, Teacher's Report Form, Youth Self-Report

Name	When	Examined for
Nancy Bayley	1899 to 1994	Bayley Scales of Infant Development (1969); BSID-II (1993) MDI and PDI; Bayley-III (2006), Bayley-4 (2019), 1 to 42 months
William K. Frankenburg	1967	Denver Developmental Screening Test with Josiah Dodds; Denver II (1990), 0 to 6 years
Josiah B. Dodds	1967	Co-author of the Denver Developmental Screening Test
Dorothea McCarthy	1972	McCarthy Scales of Children's Abilities (2½ to 8½ years); General Cognitive Index
Virginia Apgar	1953	Apgar score for newborns (Chapter 6)
T. Berry Brazelton	1973	Neonatal Behavioral Assessment Scale (Chapter 6)
J. Bharath Raj	1977	Developmental Screening Test, Mysore (88 items, 0 to 15 years, DQ)
M. K. C. Nair	1991	Trivandrum Developmental Screening Chart (0 to 2 years)
Edgar Arnold Doll	1889 to 1968	Vineland Social Maturity Scale (1935), Vineland Training School, New Jersey; social age and social quotient
Sara S. Sparrow	1984	Vineland Adaptive Behavior Scales with Balla and Cicchetti; Vineland-II (2005), Vineland-3 (2016)
David A. Balla	1984	Co-author of the Vineland Adaptive Behavior Scales
Domenic V. Cicchetti	1984	Co-author of the Vineland Adaptive Behavior Scales
Kazuo Nihira	1969	AAMD Adaptive Behavior Scale (Part One adaptive, Part Two maladaptive)
Joaquim Puig-Antich	1978	K-SADS, semi-structured child psychiatric interview
Eric Schopler	1980	Childhood Autism Rating Scale (CARS) with Reichler and Renner; TEACCH
Robert J. Reichler	1980	Co-author of the Childhood Autism Rating Scale
Barbara Rochen Renner	1988	Co-author of the CARS manual
Michael Rutter	2003	Autism Diagnostic Interview-Revised (ADI-R) with Le Couteur and Lord
Ann Le Couteur	2003	Co-author of the ADI-R
Catherine Lord	2012	ADOS-2 (autism observation schedule); co-author of the ADI-R
Diana Robins	2009	M-CHAT-R/F toddler autism screen (16 to 30 months)
David Shaffer	1983	Children's Global Assessment Scale (CGAS), a 1 to 100 rating of overall functioning
Joseph Jastak	1946	Wide Range Achievement Test (reading, spelling, arithmetic)
Lloyd Dunn	1970	Peabody Individual Achievement Test (with Frederick Markwardt)
Frederick Markwardt	1970	Peabody Individual Achievement Test (with Lloyd Dunn)
Richard Woodcock	1977	Woodcock-Johnson Tests of Achievement (with Mary Bonner Johnson)
George K. Bennett	1947	Differential Aptitude Tests (with Seashore and Wesman); Bennett Mechanical Comprehension Test
Harold G. Seashore	1947	Co-author of the Differential Aptitude Tests
Alexander G. Wesman	1947	Co-author of the Differential Aptitude Tests
J. M. Ojha	Indian adaptation	Indian adaptation of the Differential Aptitude Tests
Carl Seashore	1866 to 1949	Seashore Measures of Musical Talents (1919), an early special aptitude test
Joy Paul Guilford	1897 to 1987	Convergent versus divergent thinking (APA presidential address, 1950); Alternate Uses task; Structure of Intellect
Ellis Paul Torrance	1915 to 2003	Torrance Tests of Creative Thinking (1966), verbal and figural forms
Sarnoff Mednick	1962	Associative theory of creativity; Remote Associates Test (timed, about 40 minutes)
Michael Wallach	1965	With Kogan: creativity largely independent of intelligence when children are tested in game-like conditions
Nathan Kogan	1965	With Wallach: creativity largely independent of intelligence
Baqer Mehdi	1973	Verbal and Non-verbal Tests of Creative Thinking (India)
B. K. Passi	1979	Passi Tests of Creativity, six tests (India)
Edward Kellogg Strong Jr.	1884 to 1963	Strong Vocational Interest Blank (1927), empirical occupational keying
David Campbell	1974	Strong-Campbell Interest Inventory
John L. Holland	1919 to 2008	Theory of vocational choice (1959; set out fully 1973); RIASEC hexagon; Self-Directed Search (1971)

Name	When	Examined for
Gordon Allport	1897 to 1967	Study of Values (with Philip Vernon, 1931; Gardner Lindzey from 1951)
Philip Vernon	1931	Co-author of the Study of Values
Gardner Lindzey	1951	Joined the Study of Values (revised 1960)
Eduard Spranger	1882 to 1963	Six value types: theoretical, economic, aesthetic, social, political, religious
Milton Rokeach	1973	Rokeach Value Survey: 18 terminal and 18 instrumental values ranked
Ernest J. Chave	1929	Method of equal-appearing intervals with Thurstone (attitude towards the church)
Rensis Likert	1903 to 1981	Method of summated ratings (1932): the Likert scale
Emory Bogardus	1882 to 1973	Social distance scale (1925), seven steps; generally regarded as the oldest attitude scale
Charles Osgood	1916 to 1991	Semantic differential (1957, with Suci and Tannenbaum): evaluation, potency, activity
George Suci	1957	Co-author of the semantic differential
Percy Tannenbaum	1957	Co-author of the semantic differential
Allen Edwards	1948	Scale-discrimination technique (with Kilpatrick)
Franklin Kilpatrick	1948	Scale-discrimination technique (with Edwards)
Anthony Greenwald	1998	Implicit Association Test (with McGhee and Schwartz)
Debbie McGhee	1998	Co-author of the Implicit Association Test
Jordan Schwartz	1998	Co-author of the Implicit Association Test
Edward Jones	1971	Bogus pipeline (with Harold Sigall)
Harold Sigall	1971	Bogus pipeline (with Edward Jones)
Robert Sessions Woodworth	1869 to 1962	Personal Data Sheet (built 1917, published 1919, often dated 1920), the first personality inventory, rational strategy
Hugh M. Bell	1934	Bell Adjustment Inventory (home, health, social, emotional adjustment)
Wayne Zimmerman	1949	Co-author of the Guilford-Zimmerman Temperament Survey
Andrew L. Comrey	1970	Comrey Personality Scales: factor-analytic, 180 items, eight factors
Starke R. Hathaway	1903 to 1984	MMPI (1943) with McKinley; empirical criterion keying
James N. Butcher	1989	MMPI-2 (567 items) with colleagues
Auke Tellegen	2003	Restructured Clinical (RC) scales; MMPI-2-RF (2008) with Ben-Porath
Yossef S. Ben-Porath	2008	MMPI-2-RF (338 items) and MMPI-3 (2020) with Tellegen
H. N. Murthy	1964	Multiphasic Questionnaire (MPQ), 100-item Indian MMPI-type inventory with a Lie scale
Hans Jurgen Eysenck	1916 to 1997	MPI (1959), EPI (1964), EPQ (1975): psychoticism, extraversion, neuroticism
Sybil B. G. Eysenck	1964 and 1975	Co-author of the EPI and EPQ
Paul T. Costa Jr.	1992	NEO-PI-R with McCrae: Five-Factor Model, 240 items
Robert R. McCrae	1992	NEO-PI-R with Costa
Oliver John	1991	Big Five Inventory (44 items), with colleagues
Kibeom Lee	2004	HEXACO inventory with Ashton (adds Honesty-Humility)
Michael Ashton	2004	HEXACO inventory with Lee
Katharine Cook Briggs	1875 to 1968	Myers-Briggs Type Indicator (first form 1943), from Jung's types
Isabel Briggs Myers	1897 to 1980	Myers-Briggs Type Indicator with her mother
Carl Gustav Jung	1875 to 1961	Psychological types, the basis of the MBTI
Henry A. Murray	1893 to 1988	Theory of needs behind the EPPS; the TAT
Harrison G. Gough	1921 to 2010	California Psychological Inventory (1956, manual 1957; folk concepts); Adjective Check List
Theodore Millon	1928 to 2014	Millon Clinical Multiaxial Inventory (1977; MCMI-IV 2015); base rate scores
Charles D. Spielberger	1927 to 2013	State-Trait Anxiety Inventory (1970; Form Y 1983)
Janet A. Taylor	1953	Taylor Manifest Anxiety Scale (50 MMPI items)
A. K. P. Sinha and L. N. K. Sinha	Indian	Sinha's Comprehensive Anxiety Test
Saxena	1959	Hindi adaptation of the Bell Adjustment Inventory
Mohsin and Hussain	1981	Hindi adaptation of the Bell Adjustment Inventory (students' form)

Name	When	Examined for
N. N. Wig and S. K. Verma	PGIMER Chandigarh	PGI Health Questionnaire N-1, from the Cornell Medical Index
George A. Kelly	1905 to 1967	Role Construct Repertory Test (Rep test, 1955); repertory grid
William Stephenson	1902 to 1989	Q-sort (Q-methodology, 1953)
Carl Rogers	1902 to 1987	Real-self and ideal-self Q-sorts
Butler and Haigh	1954	Real-ideal Q-sort correlation rose from about zero to about .3 after client-centred therapy (.34 at the end, .31 at follow-up)
Jack Block	1924 to 2010	California Q-set (100 items)
Hugh Hartshorne	1928	Character Education Inquiry with May: honesty is situation-specific
Mark A. May	1928	Character Education Inquiry with Hartshorne
Walter Mischel	1968	Critique of trait consistency (taught in Chapter 4)
William Sheldon	1898 to 1977	Body types; distractor for the 16PF
Sigmund Freud	1856 to 1939	Psychoanalysis; built no inventory (distractor)
Lawrence K. Frank	1939	Coined 'projective methods'; constitutive, constructive, interpretive, cathartic and refractive categories
Samuel Beck	1944 (Rorschach's Test)	One of the five American Rorschach scoring systems
Bruno Klopfer	1942 (The Rorschach Technique)	One of the five American Rorschach scoring systems
Marguerite Hertz	Mid-twentieth century	One of the five American Rorschach scoring systems
Zygmunt Piotrowski	1957 (Perceptanalysis)	One of the five American Rorschach scoring systems
David Rapaport	1945 to 1946 (Diagnostic Psychological Testing)	Rorschach scoring system with Roy Schafer
Roy Schafer	1945 to 1946	Rorschach scoring system with David Rapaport
John E. Exner	1928 to 2006	Rorschach Comprehensive System (1974); Structural Summary, Lambda, EB, constellations
Gregory Meyer	2011	Rorschach Performance Assessment System (R-PAS), with colleagues
Wayne Holtzman	1961	Holtzman Inkblot Technique: two forms of 45 cards, one response per card
Henry Alexander Murray	1893 to 1988	TAT with Morgan (1935; manual 1943); need-press theory, Explorations in Personality (1938)
George Kelly	1955	Personal construct theory; not the basis of the TAT
B. F. Skinner	1904 to 1990	Operant conditioning; distractor for the TAT's theory and author
Aaron T. Beck	1921 to 2021	Cognitive therapy and cognitive distortions; not the author of House-Tree-Person
Leopold Bellak	1949	Children's Apperception Test with Sonya Sorel Bellak; TAT scoring; Senior Apperception Technique (1973)
Sonya Sorel Bellak	1949	Children's Apperception Test with Leopold Bellak
David McClelland	1953	Achievement motive (n Ach) scoring of TAT stories
Gerald Blum	1950	Blacky Pictures: dog family, psychosexual themes
Edwin Shneidman	1947	Make-A-Picture-Story test (MAPS)
Uma Chowdhury	1960	Indian adaptation of the TAT; Indian CAT
Udai Pareek	1925 to 2010	Indian adaptation of the Rosenzweig Picture-Frustration Study
Grace Kent	1910	Kent-Rosanoff word list with frequency norms
Aaron Rosanoff	1910	Kent-Rosanoff word list with frequency norms
Hermann Ebbinghaus	1850 to 1909	Completion method, forerunner of sentence completion tests
Julian Rotter	1950	Rotter Incomplete Sentences Blank (40 stems, adjustment score), with Janet Rafferty
Janet Rafferty	1950	Rotter Incomplete Sentences Blank, with Julian Rotter
Joseph Sacks	1950	Sacks Sentence Completion Test (60 items), with Sidney Levy

Name	When	Examined for
Sidney Levy	1950	Sacks Sentence Completion Test, with Joseph Sacks
Saul Rosenzweig	1945	Picture-Frustration Study: 24 cartoons; extrapunitive, intropunitive, impunitive
Florence Goodenough	1926	Draw-a-Man as a measure of intellectual maturity (contrast with Machover)
John Buck	1948	House-Tree-Person; achromatic then chromatic phases
Robert Burns	1970	Kinetic Family Drawing, with S. Harvard Kaufman
S. Harvard Kaufman	1970	Kinetic Family Drawing, with Robert Burns
David D. Burns	1980 (Feeling Good)	Cognitive therapy writer; distractor for House-Tree-Person
Elizabeth Koppitz	1968	Emotional indicators in children's human figure drawings
Max Hutt	From the 1940s	Projective use of the Bender-Gestalt designs
Leopold Szondi	1947	Szondi Test: 48 photographs of patients, liked and disliked faces
Margaret Lowenfeld	From 1929	World Technique (sand tray) and Mosaic Test
Edward Wagner	1962	Hand Test: 'What might this hand be doing?'; acting out
Edith Kaplan	1924 to 2009	Boston process approach to neuropsychological assessment; co-author of the CVLT and Boston Naming Test
Alexander Romanovich Luria	1902 to 1977	Qualitative, syndrome-analysis approach to neuropsychological assessment
Hazel Nelson	1982	National Adult Reading Test for estimating premorbid IQ
Aaron Antonovsky	1923 to 1994	Sense of coherence (salutogenesis), a distractor for the deterioration quotient
Ward Halstead	1947	Founded the Halstead-Reitan battery (Brain and Intelligence, 1947)
Ralph Reitan	1922 to 2014	Validated and extended the Halstead-Reitan Neuropsychological Battery
Anne-Lise Christensen	1975	Compiled Luria's methods, the basis of the Luria-Nebraska battery
Charles Golden	1978, 1980	Luria-Nebraska Neuropsychological Battery (with Hammeke and Purisch)
Thomas Hammeke	1978, 1980	Co-author of the Luria-Nebraska Neuropsychological Battery
Arnold Purisch	1978, 1980	Co-author of the Luria-Nebraska Neuropsychological Battery
Shobini Rao	2004	NIMHANS Neuropsychological Battery (with Subbakrishna and Gopukumar)
Dwarka Pershad	1977	PGI Memory Scale (with Wig) and the PGI Battery of Brain Dysfunction
N. N. Wig	1977	PGI Memory Scale (with Pershad)
Andre Rey	1941	Rey Auditory Verbal Learning Test and the Rey complex figure
Paul-Alexandre Osterrieth	1944	Standardised the Rey-Osterrieth Complex Figure
Arthur Benton	1909 to 2006	Benton Visual Retention Test (1946); Finger Localization Test
Lauretta Bender	1897 to 1987	Bender Visual-Motor Gestalt Test (1938)
Max Wertheimer	1880 to 1943	Gestalt designs adapted for the Bender-Gestalt
David Grant	1948	Wisconsin Card Sorting Test (with Esta Berg)
Esta Berg	1948	Wisconsin Card Sorting Test (with David Grant)
Robert Heaton	1981	Clinical manual of the Wisconsin Card Sorting Test
Brenda Milner	1963	Showed perseveration on the WCST after dorsolateral frontal excisions
John Ridley Stroop	1935	Stroop Colour-Word Test of response inhibition
Tim Shallice	1982	Tower of London test of planning
H. Enger Rosvold	1956	Continuous Performance Test of sustained attention
Kurt Goldstein	1878 to 1965	Object Sorting Test (with Scheerer, 1941); abstract versus concrete attitude
Martin Scheerer	1941	Object Sorting Test (with Goldstein)
Ennio De Renzi	1962	Token Test of language comprehension (with Vignolo)
Luigi Vignolo	1962	Token Test (with De Renzi)
Marshal Folstein	1975	Mini-Mental State Examination (with Susan Folstein and Paul McHugh)
Susan Folstein	1975	Mini-Mental State Examination
Paul McHugh	1975	Mini-Mental State Examination
Mary Ganguli	1995	Hindi Mental State Examination

Name	When	Examined for
Ziad Nasreddine	2005	Montreal Cognitive Assessment
Graham Teasdale	1974	Glasgow Coma Scale (with Bryan Jennett)
Bryan Jennett	1974	Glasgow Coma Scale (with Graham Teasdale)
Barry Reisberg	1982	Global Deterioration Scale (seven stages)
Vladimir Hachinski	1975	Hachinski Ischemic Score (vascular versus Alzheimer-type dementia)
Aaron Temkin Beck	1921 to 2021	Beck Depression Inventory (1961; BDI-II 1996), Beck Anxiety Inventory, Hopelessness Scale
Max Hamilton	1959, 1960	Clinician-rated Hamilton Anxiety (1959) and Depression (1960) scales
William Zung	1965	Self-Rating Depression Scale
Stuart Montgomery and Marie Asberg	1979	Montgomery-Asberg Depression Rating Scale (10 items)
Joseph Wolpe	1964	Fear Survey Schedule (with Peter Lang)
Peter Lang	1964	Fear Survey Schedule (with Joseph Wolpe)
Wayne Goodman	1989	Yale-Brown Obsessive Compulsive Scale
John Overall	1962	Brief Psychiatric Rating Scale (with Donald Gorham)
Donald Gorham	1962	Brief Psychiatric Rating Scale (with John Overall)
Stanley Kay	1987	Positive and Negative Syndrome Scale (with Fiszbein and Opler)
Abraham Fiszbein	1987	Positive and Negative Syndrome Scale
Lewis Opler	1987	Positive and Negative Syndrome Scale
Nancy Andreasen	1982, 1984	Scales for the Assessment of Negative (SANS) and Positive (SAPS) Symptoms
Robert Young	1978	Young Mania Rating Scale (11 items)
David Goldberg	1972	General Health Questionnaire
Leonard Derogatis	1970s	Symptom Checklist-90-Revised
John Ewing	1984	CAGE alcohol screening questionnaire
Thomas Achenbach	1983	Child Behavior Checklist
Savita Malhotra	1980s	Childhood Psychopathology Measurement Schedule (PGIMER), Indian adaptation of the CBCL
John Cox	1987	Edinburgh Postnatal Depression Scale
Robert Hare	1980, 1991	Psychopathy Checklist-Revised (20 items scored 0 to 2)
Hervey Cleckley	1903 to 1984	The Mask of Sanity (1941), the clinical basis of the PCL-R
Thomas Holmes and Richard Rahe	1967	Life-event (social readjustment) rating scale; see Section 1.11

Key terms

psychological test, Measurement, evaluation, assessment, psychometric approach, form board, mental test, objective test, subjective test, speed test, power test, maximum performance, ability tests, typical performance, aptitude test, achievement test, personality test, neuropsychological test, Objectivity, Reliability, Validity, Norms, Practicability, Standardisation, sample, correlate, sign, nomothetic, idiographic, actuarial prediction, Confirmatory bias, Illusory correlation, Barnum effect, test-user qualification levels, back-translated, item, Essay items, Objective items, supply type, selection type, multiple-choice item, distractors, specific determiners, stencil key, correction for guessing, Item analysis, item difficulty index, ceiling effect, floor effect, item discrimination index, item-total correlation, Distractor analysis, classical test theory, Item response theory, latent trait, item characteristic curve, Rasch model, unidimensionality, local independence, item information function, computerised adaptive testing, differential item functioning, response sets, response style, Acquiescence, Social desirability, deviation hypothesis, behaviourally anchored rating scale, halo effect, true score, reliability coefficient, domain-sampling model, random error, systematic error, test-retest reliability, coefficient of stability, alternate-forms relia-

bility, parallel-forms reliability, coefficient of equivalence, coefficient of stability and equivalence, internal consistency reliability, split-half reliability, Spearman-Brown prophecy formula, Rulon's formula, KR-20, KR-21, coefficient alpha, inter-rater reliability, Cohen's kappa, standard error of measurement, standard error of the difference, difference score, composite score, index of reliability, estimated true score, correction for attenuation, bandwidth-fidelity dilemma, generalizability theory, G study, D study, generalizability coefficient, test information, unified view of validity, Face validity, Content validity, table of specifications, content validity ratio, construct under-representation, construct-irrelevant variance, criterion-related validity, concurrent validity, predictive validity, criterion contamination, Restriction of range, cross-validation, Incremental validity, construct, Construct validity, nomological network, Convergent validity, discriminant validity, multitrait-multimethod matrix, Factorial validity, known-groups method, Consequential validity, ecological validity, validity ceiling, false positive, Sensitivity, Specificity, positive predictive value, base rate, receiver operating characteristic curve, selection ratio, Taylor-Russell tables, raw score, norm-referenced interpretation, norm group, standardisation sample, criterion-referenced interpretation, domain-referenced test, mastery test, ipsative score, national norms, local norms, subgroup norms, age-equivalent norm, mental age, grade-equivalent norm, ratio IQ, deviation IQ, percentile rank, quartiles, deciles, standard score, z score, normal curve, z table, linear transformation, deviation score, normalised standard score, T score, stanine, sten, C scale, scaled score, CEEB score, individual test, group test, verbal test, non-verbal test, performance test, age scale, point scale, routing subtests, basal level, ceiling level, Full Scale IQ, core subtests, index scores, supplemental subtests, Working Memory Index, Processing Speed Index, General Ability Index, Cognitive Proficiency Index, index discrepancies, scatter, PASS theory, culture-free test, culture-fair test, eductive ability, Draw-a-Man test, Bhatia's Battery of Performance Tests of Intelligence, Binet-Kamat Test, Malin's Intelligence Scale for Indian Children, Wechsler Adult Performance Intelligence Scale, developmental scale, developmental age, developmental quotient, social age, social quotient, adaptive behaviour, informant, screening test, diagnostic test, multi-informant assessment, Gesell Developmental Schedules, Bayley Scales of Infant Development, Denver Developmental Screening Test, McCarthy Scales of Children's Abilities, Developmental Screening Test, Developmental Assessment Scales for Indian Infants, Trivandrum Developmental Screening Chart, Vineland Social Maturity Scale, Malin's VSMS, Vineland Adaptive Behavior Scales, Adaptive Behavior Composite, AAMD Adaptive Behavior Scale, K-SADS, intellectual disability, disorders of intellectual development, educable, borderline intellectual functioning, global developmental delay, Rights of Persons with Disabilities Act 2016, benchmark disability, specific learning disability, NIMHANS Index for Specific Learning Disabilities, Childhood Autism Rating Scale, Indian Scale for Assessment of Autism, Autism Diagnostic Interview-Revised, Autism Diagnostic Observation Schedule, Children's Global Assessment Scale, survey test, prognostic test, ability-achievement discrepancy, multiple aptitude battery, Differential Aptitude Tests, General Aptitude Test Battery, cumulative record, convergent thinking, divergent thinking, fluency, flexibility, originality, elaboration, Torrance Tests of Creative Thinking, Remote Associates Test, interest inventory, Strong Vocational Interest Blank, ipsative, RIASEC, Self-Directed Search, method of equal-appearing intervals, scale value, method of summated ratings, Likert scale, cumulative scale, coefficient of reproducibility, social distance scale, semantic differential, law of comparative judgement, scale-discrimination technique, Implicit Association Test, bogus pipeline, objective personality test, L-data, Q-data, T-data, logical-content, Personal Data Sheet, theoretical strategy, Empirical criterion-group keying, factor-analytic strategy, forced-choice, validity scales, Minnesota Multiphasic Personality Inventory, L (Lie) scale, F (Infrequency) scale, K (Correction) scale, neurotic triad, psychotic tetrad, conversion V, demoralisation, Restructured Clinical (RC) scales, Sixteen Personality Factor Questionnaire, stens, Eysenck Personality Questionnaire, NEO Personality Inventory,

Myers-Briggs Type Indicator, Edwards Personal Preference Schedule, California Psychological Inventory, folk concepts, Millon Clinical Multiaxial Inventory, base rate scores, State-Trait Anxiety Inventory, Role Construct Repertory Test, Q-sort, Situational tests, projective technique, projective hypothesis, projection, association techniques, construction techniques, completion techniques, choice or ordering techniques, expression techniques, Rorschach Inkblot Test, inquiry phase, colour shock, location, determinants, form quality, reality testing, popular response, confabulatory response, Comprehensive System, Lambda, Experience Balance, Rorschach Performance Assessment System, Holtzman Inkblot Technique, Thematic Apperception Test, need-press theory, press, thema, Children's Apperception Test, word association test, complex indicators, sentence completion, Rosenzweig Picture-Frustration Study, Draw-A-Person, body-image hypothesis, House-Tree-Person, Neuropsychological assessment, executive functions, fixed battery, flexible battery, process approach, normative comparison, individual comparison, premorbid level, deterioration index, National Adult Reading Test, Halstead-Reitan Neuropsychological Battery, Halstead Impairment Index, Luria-Nebraska Neuropsychological Battery, NIMHANS Neuropsychological Battery, PGI Battery of Brain Dysfunction, Wechsler Memory Scale, Bender Visual-Motor Gestalt Test, visual-motor integration, working memory, Wisconsin Card Sorting Test, perseverative errors, set-shifting, Trail Making Test, response inhibition, thought disorder, Object Sorting Test, Mini-Mental State Examination, Montreal Cognitive Assessment, Glasgow Coma Scale, self-report inventory, clinician-rated scale, Beck Depression Inventory, Yale-Brown Obsessive Compulsive Scale, Positive and Negative Syndrome Scale, Psychopathy Checklist-Revised

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From the past papers

1. The primary difference between a "speed test" and a "power test" in psychological assessment is that a speed test: (Real PYQ 2024)

   (a) Has items of increasing difficulty, while a power test has items of uniform difficulty. (b) Has a strict time limit that prevents most test-takers from completing all items, while a power test has items so difficult that no one can answer them all correctly. (c) Measures reaction time, while a power measures cognitive ability. (d) Is administered individually, while a power test can be administered in groups.

   Answer: (b). A speed test uses easy items under a time limit so strict that few finish; a power test allows generous time, but its items grow so hard that no one passes them all. Option (a) reverses the two, (c) confuses a timed test with reaction time, and (d) confuses timing with individual versus group administration.

2. Which of the following is the most important characteristic of a psychological test? (RML 2024)

   (a) Objectivity (b) Subjectivity (c) Validity (d) Rationality

   Answer (keyed): (a) Objectivity. RML follows Indian texts such as A. K. Singh, which list objectivity first: a score must not depend on who gives or scores the test. Most textbooks and the 2014 *Standards* call validity the most fundamental property, so on other papers choose (c). Subjectivity is a defect, and rationality is not a test characteristic.

3. Which of the following refers to the notion of the confirmatory bias? (RML 2021)

   (a) Clinicians ignore information that does not support their hypothesis (b) Clinicians ignore information that does not support their stereotypes. (c) Clinicians interpret ambiguous information as supporting their hypothesis. (d) All of the above

   Answer: (d) All of the above. Confirmatory bias is the tendency to notice, weigh and interpret information so that it confirms what one already believes; ignoring contrary evidence, ignoring evidence against one's stereotypes and reading ambiguity as support are all forms of it.

4. The improvement in anxiety may sometimes spuriously improve depression. This is called: (RML 2024)

   (a) Halo effect (b) Ceiling effect (c) Hawthorne effect (d) Therapeutic effect

   Answer: (a) Halo effect. One attribute (the improved anxiety) colours the rating of another, so the depression rating improves although the depression has not. A ceiling effect is a test too easy to register further gains, the Hawthorne effect is change because people know they are being studied, and a therapeutic effect is real improvement from treatment.

5. A response style or bias, in which there is tendency to agree with test items regardless of their contents is called: (RML 2024)

   (a) Halo effect (b) Central tendency effect (c) Social desirability effect (d) Acquiescence effect

   Answer: (d) Acquiescence effect. Acquiescence (yea-saying) is agreeing regardless of content and is controlled by balanced keying. Social desirability depends on content (the answer that looks good), central tendency is choosing the middle of the scale, and halo is a rater's error, not a test-taker's.

6. The tendency of the rater/interviewer to judge an individual on the basis of one single attribute is referred to as : (IHBAS 2024)

   (a) Similarity error (b) Central tendency error (c) Contrast error (d) Halo error

   Answer: (d) Halo error. Named by Thorndike (1920), halo lets one attribute or a general impression colour every other rating. Similarity error favours people like the rater, central tendency error rates everyone average, and contrast error rates others as opposite to the rater.

7. Which of the following is assessed by Cronbach's alpha? (RML 2021)

   (a) Concurrent validity (b) Inter-rater reliability (c) Test retest reliability (d) Internal consistency

   Answer: (d) Internal consistency. Alpha (Lee Cronbach, 1951) estimates how well a scale's items hang together within one administration. Concurrent validity needs an outside criterion, inter-rater reliability needs two scorers (kappa or the intraclass correlation), and test-retest reliability needs two occasions.

8. A test developer administers a new anxiety scale to a group of college students and, two weeks later, administers the same scale again to the same group. She then correlates the two sets of scores. This process is used to establish: (Real PYQ 2024)

   (a) Split-half reliability (b) Inter-rater reliability (c) Test-retest reliability (d) Parallel forms reliability

   Answer: (c) Test-retest reliability. Same scale, same people, two occasions, scores correlated: the coefficient of stability. Split-half correlates two halves of one sitting, inter-rater reliability compares scorers, and parallel forms would need a second, equivalent version of the scale.

9. Odd-even method of splitting the test to estimate split-half reliability is MOST appropriate for : (RML 2022)

   (a) Projective tests (b) Neuropsychological tests (c) Power tests (d) Speed tests

   Answer: (c) Power tests. With graded items and generous time, odd and even halves are fair parallel samples. On a speed test the halves match by construction and the coefficient is spuriously high, and projective and many neuropsychological tests lack comparable, objectively scored items to split.

10. To determine whether a newly developed intelligence test truly accurately measures intelligence or not, it is necessary to administer the test with one that has been in the field for some time. Which of the following needs to be looked for? (JEMAS 2025)

    (a) Construct validity (b) Concurrent validity (c) Content validity (d) Predictive validity

    Answer: (b) Concurrent validity. Correlating a new test with an established test given at about the same time is the concurrent type of criterion-related validity. Predictive validity needs a criterion measured later, content validity concerns sampling of the domain, and construct validity rests on a network of evidence rather than one established test.

11. Which of the following statements is TRUE? (RML 2024)

    (a) A test can be valid without being reliable (b) A test can be reliable without being valid (c) If a test measures what it is intended to measure, then by definition it is reliable (d) If a test gives consistent scores when administered repeatedly then it is valid

    Answer: (b) A test can be reliable without being valid. A test can consistently measure the wrong thing. It cannot be valid without being reliable, but option (c) overstates this as true by definition, and consistency alone (d) never proves validity.

12. A high false positive rate in diagnosis means : (IHBAS 2025)

    (a) Diagnosis is too strict (b) Normal people are diagnosed as disordered (c) Diagnosis has low reliability (d) Disordered people go undetected

    Answer: (b) Normal people are diagnosed as disordered. A false positive is a positive call on someone without the condition, the mark of low specificity. Undetected disordered people (d) are false negatives, which an over-strict diagnosis (a) produces; reliability (c) is a different property.

13. Which statistical measure tells us how far individual data points are dispersed from the mean expressed in standard deviation units? (Real PYQ 2025)

    (a) Variance (b) Z-score (Standard score) (c) Coefficient of variation (d) Standard error

    Answer: (b) Z-score (Standard score). $z = (X - M) \div SD$ states how many SDs one score lies from the mean, so scores from different distributions can be compared. Variance (SD squared) describes the spread of a whole distribution, the coefficient of variation compares spread across distributions, and the standard error is the SD of a sampling distribution.

14. IQ if any body has a mental age of 8 and a chronological age of 10 is: (RML 2020)

    (a) 108 (b) 125 (c) 80 (d) 92

    Answer: (c) 80. Ratio IQ = $(MA \div CA) \times 100 = (8 \div 10) \times 100 = 80$. The tempting 125 inverts the ratio ($CA \div MA$); a mental age below the chronological age must give an IQ below 100.

15. Which of the following is a non-linear transformation of raw scores? (Recall set)
    (a) z-score (b) 50 + 10z (c) Stanine (d) Deviation score (X – M)

    Answer: (c) Stanine. Stanines are assigned from percentile ranks in fixed proportions (4, 7, 12, 17, 20, 17, 12, 7, 4 per cent), a normalising, area transformation that forces a normal shape. z, the T score 50 + 10z and the deviation score X – M all apply one multiply-and-add rule to every score, so they are linear and keep the raw distribution's shape.

16. In the WAIS-IV (Wechsler Adult Intelligence Scale - Fourth Edition), how many core subtests are there in the Verbal Comprehension Index and the Perceptual Reasoning Index respectively? (Real PYQ 2025)
    (a) 3 verbal, 4 performance (b) 3 verbal, 3 performance (c) 4 verbal, 3 performance (d) 2 verbal, 4 performance

    Answer: (b) 3 verbal, 3 performance. The VCI core is Similarities, Vocabulary and Information (Comprehension is supplemental); the PRI core is Block Design, Matrix Reasoning and Visual Puzzles (Figure Weights and Picture Completion are supplemental). Counting a supplemental subtest as core produces the tempting 4.

17. A 10-year-old child from a rural area, with no formal schooling, is brought for an intellectual assessment due to academic struggles. The child is illiterate and has had minimal exposure to formal testing materials. Which of the following tests would be the MOST appropriate to use to minimize cultural and educational bias? (Real PYQ 2024)
    (a) Bhatia's Battery of Performance Tests of Intelligence (b) Wechsler Intelligence Scale for Children (WISC-IV) (c) Stanford-Binet Intelligence Scale (5th ed.) (d) Malin's Intelligence Scale for Indian Children (MISIC)

    Answer: (a) Bhatia's Battery. Its five performance subtests need no reading and carry Indian norms. The WISC-IV and SB5 rest on Western verbal content and norms, and MISIC, though Indian, is half verbal and assumes schooling.

18. An IQ test does NOT provide which of the following? (RML 2021)
    (a) High test retest reliability (b) Good predictor of behaviour (c) High internal consistency (d) Good validity

    Answer: (b) Good predictor of behaviour. IQ tests have high test-retest reliability, high internal consistency and good validity for predicting school achievement, but how a person behaves in general depends on personality, motivation and circumstance.

19. Vineland Social Maturity Scale (VSMS) is a measure of : (RML 2022)
    (a) Mental age and Intelligence quotient (b) Social age and social quotient (c) Developmental age and developmental quotient (d) Emotional age and emotional quotient

    Answer: (b) Social age and social quotient. Doll's VSMS converts the items an informant credits into a social age, and SQ = SA / CA × 100. Mental age and IQ come from intelligence tests, developmental age and DQ from developmental scales such as Gesell's or the DST.

20. A key differentiation between global developmental delay and intellectual disability is (JEMAS 2025)
    (a) Age of assessment (under 5 years for GDD, 5 years and above for ID) (b) Presence of social deficits (c) Language delay alone (d) Motor impairment

    Answer: (a) Age of assessment. DSM-5 reserves global developmental delay for children under 5 who are too young for reliable standardised testing and must be reassessed; from 5 years the diagnosis is intellectual disability. The deficits are of the same kind, so social, language or motor problems do not separate them.

21. Name of the Indian tool to assess the severity and disability of children with autism : (IHBAS 2024)

    (a) CGAS (b) Childhood Autism Rating Scale (c) CBCL (d) ISAA

    Answer: (d) ISAA (the paper keys (b) CARS). The Indian Scale for Assessment of Autism (NIMH Secunderabad, 2009; 40 items) is the Indian instrument prescribed for certifying autism and its disability percentage; CARS is the American scale of Schopler and colleagues on which ISAA partly drew. CGAS rates global functioning and the CBCL general behaviour problems.

22. Summated rating scale is also known as : (RML 2022)

    (a) Likert scale (b) Thurstone scale (c) Guttmann scale (d) Equal appearing interval scale

    Answer: (a) Likert scale. Likert (1932) had respondents rate their agreement with each statement and summed the ratings, hence summated ratings. The Thurstone scale and the equal-appearing interval scale are one and the same judge-scaled method, and the Guttman scale is cumulative.

23. The Implicit Association Test is designed to measure: (RML 2022)

    (a) Verbal ability (b) Unconscious stereotyping (c) Hidden mathematical ability (d) Dream content

    Answer: (b) Unconscious stereotyping. Greenwald, McGhee and Schwartz (1998) built the IAT to reveal implicit attitudes and stereotypes from reaction times in a sorting task. It is not an ability test of any kind, and dream content is not its target.

24. What is the primary purpose of achievement tests in psychological assessment? (RML 2025)

    (a) To measure future success potential (b) To assess cognitive abilities related to brain function (c) To measure understanding and knowledge in specific academic areas (d) To evaluate unconscious conflicts and emotions

    Answer: (c) To measure understanding and knowledge in specific academic areas. Achievement tests look back at what has been learnt. Future potential is the aim of aptitude tests, brain-related abilities of neuropsychological tests, and unconscious conflicts of projective techniques.

25. Which of the following statements is TRUE with respect to Minnesota Multiphasic Personality Inventory (MMPI)? (RML 2022)

    (a) It utilizes rational method of test construction (b) It is constructed by assuming specific personality traits and formulating questions to answer them (c) It is derived by finding items that discriminate between criterion and control groups (d) It is true with respect to both a and b

    Answer: (c) It is derived by finding items that discriminate between criterion and control groups. Hathaway and McKinley kept only items that separated patient groups from normal controls, whatever their content (empirical criterion keying). Options (a) and (b) describe the rational, logical-content strategy of Woodworth's Personal Data Sheet.

26. According to Raymond Cattell's classification of personality data sources, which of the following is NOT recognized as a legitimate data type? (Real PYQ 2025)

    (a) Q-data (Questionnaire data) (b) L-data (Life record data) (c) T-data (Objective test data) (d) P-data (Projective data)

    Answer: (d) P-data. Cattell recognised only three sources: L-data (life records such as school and criminal records, and observer ratings), Q-data (self-report questionnaires) and T-data (objective tests in controlled situations whose purpose is hidden).

27. Elevated Lie scale on Multiphasic Questionnaire (MPQ) is indicative of: (IHBAS 2020)
(a) Tendency to fake and manipulate (b) Social desirability (c) Mistrust (d) All the above

Answer: (d) All the above. A high Lie score means the person denies common minor faults: an attempt to fake a good impression, a pull towards social desirability, and a guarded mistrust that admits no flaw to the examiner.

28. How many Rorschach cards are coloured? (RML 2022)
(a) 10 (b) 5 (c) 3 (d) 4

Answer: (b) 5. Two cards (II, III) are black with red and three (VIII, IX, X) are multicoloured, so five carry colour and five (I, IV, V, VI, VII) are achromatic. Three counts only the fully coloured cards; ten is the whole set.

29. Thematic Apperception Test (TAT) is based on which technique of psychological testing? (RML 2022)
(a) Expressive technique (b) Construction technique (c) Association technique (d) Completion technique

Answer: (b) Construction technique. In Lindzey's scheme the person constructs a story about each picture. Association is the Rorschach and word association, completion is sentence completion and the Rosenzweig P-F, and expression is drawing and play.

30. Absence of 'Human' Response on Rorschach Psychodiagnostics Test indicates: (IHBAS 2020)
(a) Apathy (b) Poor interpersonal orientation (c) Lack of empathy (d) All the above

Answer: (b) Poor interpersonal orientation. Human content reflects interest in and understanding of people, so its absence points to poor interpersonal orientation. Apathy and lack of empathy are not the standard reading, so 'All the above' is wrong as well.

31. One of the most widely used tests to assess executive functioning: (Amity 2023)
(a) Wisconsin Card Sorting Test (b) Minnesota Number Ordering Test (c) Manhattan Card Sorting Test (d) None of the above

Answer: (a) Wisconsin Card Sorting Test. Grant and Berg's WCST (1948) measures set-shifting and cognitive flexibility through perseverative errors when the sorting rule changes unannounced, and is most sensitive to dorsolateral prefrontal dysfunction. The Minnesota and Manhattan tests are invented distractors, so 'none of the above' is wrong too.

32. Which of the following is a clinician-rated questionnaire? (RML 2022)
(a) Beck Depression Inventory (b) Beck Anxiety Inventory (c) Hamilton Anxiety Rating Scale (d) Both b and c

Answer: (c) Hamilton Anxiety Rating Scale. Max Hamilton's 14-item anxiety scale (1959) is completed by a trained clinician from interview and observation. Both Beck inventories are self-report: the patient rates his own symptoms, so (a), (b) and (d) are wrong.

33. Impact of psychiatric illness on one's cognitive and intellectual abilities can be estimated on the basis of: (IHBAS 2020)
(a) Deterioration quotient (b) Mediation index (c) Sense of coherence index (d) Theory of Mind index

Answer: (a) Deterioration quotient. Wechsler compared 'hold' subtests (Vocabulary, Information, Picture Completion, Object Assembly) with 'don't hold' subtests (Digit Span, Similarities, Digit Symbol, Block Design) to estimate loss from illness. Mediation is a statistical concept, sense of coherence is Antonovsky's health concept, and theory of mind concerns understanding others' minds.

High-yield sheet

The facts this chapter's questions ask most, most-asked first. Last read before the exam.

1 **Cronbach's alpha** (Lee Cronbach, 1951) measures **internal consistency reliability**: not validity of any kind, not test-retest and not inter-rater reliability. ASKED 10 TIMES

2 The **Spearman-Brown prophecy formula** (Spearman and Brown, 1910) steps a split-half correlation up to full length, 2r / (1 + r), so .60 becomes .75; in general form kr / [1 + (k - 1)r] it shows that adding parallel items raises reliability. ASKED 9 TIMES

3 **Test-retest reliability** correlates the same test given to the same people on two occasions (the coefficient of stability); it suits stable traits such as intelligence and personality, not changing states such as current mood. ASKED 9 TIMES

4 **Criterion-related validity** has two types: **concurrent** (test and criterion measured at about the same time, such as a new intelligence test given with an established one) and **predictive** (criterion measured later, such as entrance scores and later grades). ASKED 9 TIMES

5 The **Thematic Apperception Test** was developed by Christiana Morgan and Henry Murray (1935, Harvard Psychological Clinic), rests on Murray's need-press theory, and is a projective **construction** technique in which the person tells stories about ambiguous pictures of social scenes. ASKED 9 TIMES

6 The three classic MMPI validity scales are **L** (Lie: naive denial of common minor faults), **F** (Infrequency: random responding, faking bad or severe pathology; T above 100 suggests random responding or gross exaggeration) and **K** (Correction: subtle defensiveness; a fraction of K is added to scales 1, 4, 7, 8 and 9). ASKED 8 TIMES

7 **KR-20** (Kuder and Richardson, 1937) is for dichotomously scored items of varying difficulty; KR-21 assumes equal difficulty and gives a lower estimate when difficulties differ; alpha equals KR-20 when every item is scored 0 or 1. ASKED 7 TIMES

8 On the normal curve z = 0 is the 50th percentile, z = +1 the 84th (−1 the 16th), z = +1.5 about the 93rd and z = +2 about the 98th; so IQ 100, 85 and 130 (SD 15) are the 50th, 16th and about 98th percentiles, and IQ 136 on an SD-16 scale (z = 2.25) has about 98.8 per cent below it. ASKED 7 TIMES

9 The **WAIS-IV** (2008; ages 16:0 to 90:11) yields a Full Scale IQ and four index scores, Verbal Comprehension, Perceptual Reasoning, Working Memory and Processing Speed, from 10 core subtests: 3 verbal (Similarities, Vocabulary, Information), 3 perceptual (Block Design, Matrix Reasoning, Visual Puzzles), 2 working memory (Digit Span, Arithmetic) and 2 processing speed (Symbol Search, Coding). ASKED 7 TIMES

10 The **16PF** is Raymond B. Cattell's (1949; fifth edition 1993, 185 items): 16 primary source traits found by factor analysis and five global factors, scored in stens; it measures personality traits, not intelligence. ASKED 7 TIMES

11 **Reliability** means consistency, dependability and freedom from measurement error (all of these): a reliable measure gives stable results on different occasions. ASKED 6 TIMES

12 **SEM = SD × √(1 - r)**, which estimates the range within which a person's true score probably falls: with SD 15 and r = .91 the SEM is 4.5 (±1 SEM 68 percent, ±1.96 SEM 95 percent). ASKED 6 TIMES

13 The **Vineland Adaptive Behavior Scales** (Sparrow, Balla and Cicchetti, 1984; Vineland-3, 2016) measure adaptive functioning in daily life (communication, daily living skills, socialisation, motor skills), not IQ, personality or psychopathology. ASKED 6 TIMES

14 The **MMPI** (Hathaway and McKinley, 1943; 550 items) is an objective personality inventory that assesses personality and psychopathology for psychiatric diagnosis, covering mood, physical concerns and social attitudes; it is not projective, an intelligence test or an aptitude battery. ASKED 6 TIMES

15 Projective tests present ambiguous stimuli to reveal **unconscious processes**; their main advantage is that they avoid self-report biases and are hard to fake. ASKED 6 TIMES

16 John Exner's **Comprehensive System** (1974) is the standard Rorschach scoring system: it codes location, determinants, form quality and content, gives no IQ, and there is no 'Murray system' of Rorschach scoring. ASKED 6 TIMES

17 The **Wisconsin Card Sorting Test** (David Grant and Esta Berg, 1948) is the most widely used test of executive function: it measures set-shifting and cognitive flexibility through perseverative errors and is most sensitive to **dorsolateral prefrontal cortex** dysfunction. ASKED 6 TIMES

18 A test can be **reliable without being valid** but cannot be valid without being reliable: reliability is necessary but not sufficient for validity. ASKED 5 TIMES

19 Wechsler **scaled scores** have a mean of 10 and an SD of 3 (range 1 to 19): 10 is the 50th percentile, 7 is 1 SD below (about the 16th, low average), 13 is 1 SD above and 4 is 2 SD below. ASKED 5 TIMES

20 The **Rorschach** (Hermann Rorschach, *Psychodiagnostik*, 1921) has **10** symmetrical inkblot cards: 5 achromatic (I, IV, V, VI, VII), 2 black and red (II, III) and 3 multicoloured (VIII, IX, X), so five achromatic and five chromatic. ASKED 5 TIMES

21 The **Bender Visual-Motor Gestalt Test** (Lauretta Bender, 1938; nine figures adapted from Wertheimer, 16 in Bender-Gestalt II, 2003) assesses **visual-motor integration** and screens for organic brain damage and developmental immaturity; it is not a personality test. ASKED 5 TIMES

22 The **Halstead-Reitan Neuropsychological Battery** (Halstead 1947; Reitan) is a fixed battery for comprehensive neuropsychological assessment (Category Test, Tactual Performance, Seashore Rhythm, Speech-Sounds Perception, Finger Tapping, Trail Making); an Impairment Index of 0.5 or more suggests brain damage, and the Rorschach is not part of it. ASKED 5 TIMES

23 The **BDI-II** (Aaron Beck, 1996) is a self-report inventory of 21 items rated 0 to 3 for the past two weeks: 0 to 13 minimal, 14 to 19 mild, 20 to 28 moderate, 29 to 63 severe. ASKED 5 TIMES

24 **Sensitivity** = TP / (TP + FN), the proportion of people with the condition whom the test correctly identifies; **specificity** = TN / (TN + FP), the proportion without it whom the test correctly clears. ASKED 4 TIMES

25 **Construct validity** (Cronbach and Meehl, 1955) is the degree to which a test measures the theoretical construct it claims to measure; **convergent validity** is a high correlation with measures of the same construct and **discriminant validity** a low correlation with measures of unrelated constructs. ASKED 4 TIMES

26 **T scores** (McCall, 1922) have a mean of 50 and an SD of 10 ($T = 50 + 10z$) and are used on the MMPI and NEO-PI-R; T 70 is 2 SD above the mean (original MMPI cut-off) and T 65 is 1.5 SD above (MMPI-2 cut-off). ASKED 4 TIMES

27 The **z score**, $z = (X - M) \div SD$, has a mean of 0 and an SD of 1, expresses a score's distance from the mean in SD units and lets results from different distributions be compared. ASKED 4 TIMES

28 **Stanines** run from 1 to 9 (mean 5, SD about 2) in the proportions 4, 7, 12, 17, 20, 17, 12, 7, 4 per cent; stanine 5 holds 20 per cent, the 40th to 60th percentile (mean ± 0.25 SD), and is a normalised, nonlinear transformation. ASKED 4 TIMES

29 The **Vineland Social Maturity Scale** was devised by **Edgar A. Doll** (1935, Vineland Training School): 117 items from birth to adulthood (25 and above), given by informant interview, yielding **social age and social quotient** (SQ = SA / CA × 100). ASKED 4 TIMES

30 A **Likert scale** is a **summated rating scale** (Rensis Likert, 1932): closed-ended statements with a fixed number of ordered alternatives, usually five from strongly agree to strongly disagree, rated by respondents and summed with equal weights. It is a type of attitude scale, not a level of measurement. ASKED 4 TIMES

CHAPTER 3

Social Psychology

IN THIS CHAPTER

3.1 ○ Social Psychology: History, Theories, Concepts and Methods: Define social psychology and set it apart from its neighbours, trace its history from Le Bon, Triplett and Lewin to the replication crisis, use the sociological ideas of anomie, socialisation, status and role, match the main theories to their founders, and outline the field's methods and research ethics.

3.2 ○ Social Cognition and First Impressions: Explain how people think about the social world (cognitive misers, dual processes, schemas, priming, heuristics and the errors of social thought), recognise the self-fulfilling prophecy and the halo effect, and describe how first impressions form, how accurate thin slices are and how people manage the impressions they make.

3.3 ○ Communication, Nonverbal Cues and Attribution: Trace a message through the communication process and its classic models, read nonverbal cues and personal space, explain behaviour with the attribution theories of Heider, Jones and Davis, Kelley and Weiner, and recognise each attributional bias from its description.

3.4 ○ Attitudes, Persuasion and Cognitive Dissonance: Define attitudes, their components, strength and functions, explain how they form and when they predict behaviour (reasoned action, planned behaviour, MODE), analyse persuasion through the Yale factors and the elaboration likelihood model, explain resistance (reactance, forewarning, inoculation), and apply balance and cognitive dissonance theory with its classic paradigms and rivals.

3.5 ○ Stereotypes, Prejudice and Intergroup Relations: Distinguish stereotype, prejudice and discrimination, explain where prejudice comes from (personality, frustration, learning, realistic conflict and social identity), describe the processes that sustain stereotypes and the barriers they create at work, and state when contact reduces prejudice.

3.6 ○ Affiliation, Attraction and Close Relationships: Explain why people affiliate and compare themselves with others, what draws them together, how exchange, equity and investment theories account for staying or leaving, and how self-disclosure, love, break-up and loneliness are described and measured.

3.7 ○ Social Influence: Conformity, Compliance, Obedience and Power: Tell apart conformity, compliance and obedience, explain why people conform or resist, and recall the classic studies of Sherif, Asch, Milgram and Zimbardo, Cialdini's compliance principles and tactics, and French and Raven's bases of power.

3.8 ○ Prosocial Behaviour: Why and When People Help: Explain why and when people help: the empathy-altruism, egoistic, evolutionary and normative accounts, the bystander effect with its five-step model and its two classic explanations, and the situations, personalities and recipient reactions that raise or lower helping.

3.9 ○ Aggression: Its Roots, Causes and Control: Define aggression and its forms, compare the instinct, biological, frustration-aggression, learning and General Aggression Model explanations, link each major cause of aggression to its researchers, and judge which methods of control work and why catharsis fails.

3.10 ●● Groups: Structure, Performance, Decisions and Leadership: Define a group and its structure, map a group with sociometry, order Tuckman's stages, predict when the presence of others raises or lowers effort and performance, diagnose and prevent group polarisation and groupthink, and compare the main theories of leadership.

3.11 ○ Applied Social Psychology: Law, Culture and Social Work: Apply social psychology to health, environment, eyewitnesses, juries and offenders, and name the roots, methods, models, principles and pioneers of social work, from case work and group work to culturally sensitive practice, child welfare and the first Indian school.

Weightage. 136 of the 3180 questions mapped in this book (4.3%) test this chapter, 93 of them from real past papers. 0 of its 11 sections are High yield.

In the early hours of 13 March 1964, Catherine "Kitty" Genovese, a 28-year-old bar manager, parked her car near her flat in Kew Gardens, a quiet neighbourhood of Queens, New York, and started the short walk home. A stranger, Winston Moseley, chased her and stabbed her. A neighbour shouted from a window and Moseley fled, but about ten minutes later he came back, found her lying wounded in a hallway at the back of her building, and attacked her again. She died on the way to hospital.

Two weeks later *The New York Times* put the killing on its front page. For more than half an hour, it reported, 38 respectable, law-abiding citizens had watched a killer stalk and stab a woman, and not one had telephoned the police during the attack. The story went round the world as proof of big-city apathy. Two young psychologists in New York, John Darley and Bibb Latane, suspected another explanation: perhaps each witness, knowing that others could also hear, assumed that someone else would act. They staged an emergency in the laboratory, reported in 1968, in which students talking over an intercom heard another student apparently have a seizure. The more listeners a student believed were present, the less likely that student was to go for help.

The finding became one of the most famous in psychology, yet the newspaper story behind it was badly wrong. Later investigations, among them a 2007 review by the psychologists Rachel Manning, Mark Levine and Alan Collins, found no 38 watching witnesses: few neighbours saw any part of the attacks, most heard only a cry in the night, and one, Sophia Farrar, went down and held Kitty in her arms as she lay dying. In 2016 the *Times* itself conceded that its report had exaggerated the number of witnesses and what they saw. One night in Queens thus holds social psychology in small: how the presence of others changes what each of us does, and how a whole society can come to share a story that is not quite true.

3.1 Social Psychology: History, Theories, Concepts and Methods

LOWER YIELD · 6 QS

Why do you drive more carefully with your father beside you? Social psychology asks how other people, present or imagined, shape what we think, feel and do. This section defines the field, traces its history, and covers its sociological roots, theories and methods.

What social psychology studies

The classic definition is that of **Gordon Willard Allport** (1897 to 1967), from 1954: **social psychology** studies how the thoughts, feelings and behaviour of individuals are influenced by the actual, imagined or implied presence of others: an audience in the hall, a mother whose frown you picture, a "No smoking" sign left by strangers. **Robert A. Baron** and **Nyla R. Branscombe** call it the scientific study of the nature and causes of individual behaviour, feeling and thought in social situations: *scientific* because evidence settles questions, *individual* because the unit is the person, even in a crowd.

Behaviour reflects the **person-situation interaction**: a student silent in class talks freely among friends. **Psychological social psychology** studies the individual's mental processes, mostly by experiment; **sociological social psychology** studies how groups, roles and social

structure shape people, mostly by surveys and observation, drawing on symbolic interactionism (below). Sociology proper studies groups and societies, and personality psychology stable differences between people (Chapter 4); social psychology asks how situations make different people act alike. Common sense has a proverb for every outcome ("birds of a feather flock together", yet "opposites attract"), and hindsight bias (Section 1.8) makes findings look obvious afterwards; only testing shows which holds when.

> **! EXAM TRAP**
> "Actual, imagined or implied presence of others" is **Gordon Allport** (1954), though Baron and Branscombe credit it to his brother Floyd (1924). "Nature and causes of individual behaviour … in social situations" is **Baron and Branscombe**.

Founders and landmarks

Two roots, crowds and experiments, fed the field (Table 3.1). Le Bon's crowd mind, Ringelmann's rope pullers (each pulled less hard as the team grew) and Triplett's children, who wound fishing reels faster in pairs than alone in what is usually called the first social psychology experiment, are taught in Section 3.10. The first two textbooks (1908) pulled opposite ways: the British psychologist **William McDougall** traced social behaviour to inborn instincts, the American sociologist **Edward Alsworth Ross** to imitation and suggestion in crowds. **Floyd Henry Allport** (1924) made the field experimental and individual-centred, and **Muzafer Sherif** (1935, 1936) watched group norms form as people judged the apparent movement of a light in the dark (Section 3.7).

Table 3.1 Landmarks in the history of social psychology.

Year	Who	Landmark
1895	**Gustave Le Bon**	*The Crowd*: contagion, the crowd mind
1898	**Norman Triplett**	First experiment: others speed us up
1908	McDougall; Ross	First textbooks: instinct; imitation
1913	**Max Ringelmann**	Rope pulling: effort falls in groups
1924	Floyd Allport	Experimental, individual-centred text
1935, 1936	Muzafer Sherif	Norm formation (autokinetic effect)
1939	Lewin, **Ronald Lippitt**, **Ralph K. White**	Leadership climates
1945 to 1947	Kurt Lewin	Group dynamics, action research, T-groups
1950	**Theodor W. Adorno** and colleagues	*The Authoritarian Personality*
1951	**Solomon Asch**	Conformity to a wrong majority
1953	**Carl Hovland** and colleagues	Yale persuasion research
1954, 1957	**Leon Festinger**	Social comparison; cognitive dissonance
1958	**Fritz Heider**	*The Psychology of Interpersonal Relations*
1961	Serge Moscovici	Social representations
1963	**Stanley Milgram**	Obedience
1971	**Philip Zimbardo**	Stanford prison study

Year	Who	Landmark
1973	Kenneth Gergen	"Social psychology as history"
2010s	Many teams	Replication crisis

Kurt Lewin ●●

Kurt Lewin (1890 to 1947), a Gestalt-trained refugee from Nazi Germany, is called the father of modern experimental social psychology. His field theory (Section 1.2) states B = f(P, E): behaviour is a function of the person and the environment, together the **life space**, the person plus the *psychological* environment as it exists for that person at that moment, not society, the group or the physical setting.

Lewin wanted research to change real life. In 1945 he founded the Research Center for Group Dynamics at the Massachusetts Institute of Technology (MIT); **group dynamics** is the study of forces within groups. In 1946 he described **action research**, done with the people affected, as a spiral of steps, each a circle of planning, action and fact-finding about the results (Figure 3.1). A 1946 workshop produced the **T-group** (training group), whose members learn from frank feedback on their behaviour in the group, spread by the National Training Laboratories (1947) as sensitivity training. His motto: "There is nothing so practical as a good theory."

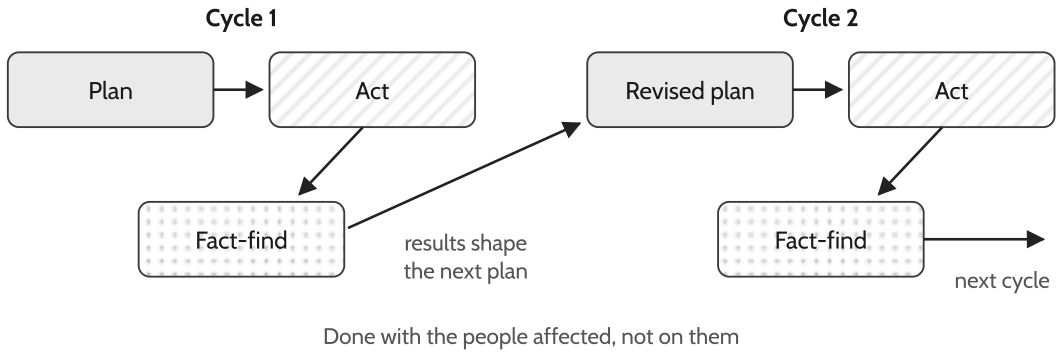


Figure 3.1 Lewin's action research (1946): fact-finding about each action feeds a revised plan.

> **! EXAM TRAP**
>
> **Life space** = person + psychological environment, not society, group or physical environment. Lewin founded **action research**, not psychoanalysis (Freud), psychophysics (Fechner) or functionalism (James).

Crisis and new directions

After 1945 the field grew around the war's questions: persuasion, prejudice, conformity and obedience (Table 3.1). By the late 1960s a crisis had set in: laboratory experiments on students looked artificial, ethically doubtful and culture-bound, and **Kenneth J. Gergen** argued in "Social psychology as history" (1973) that its findings describe one society at one moment, not timeless laws. Europeans made the field more social: **Henri Tajfel** (1919 to 1982) studied group identity (Section 3.5), and Serge Moscovici minorities and shared knowledge. Then came the social cognition era of the 1970s and 1980s (Section 3.2), social neuroscience (the brain activi-

ty behind social thought), implicit processes (automatic attitudes outside awareness, measured by the Implicit Association Test, 1998) and cultural psychology (how culture shapes social thought and behaviour). In the 2010s the **replication crisis** broke: when 100 published psychology studies were repeated in 2015, only about a third gave the original result.

Sociological roots: Durkheim to Mead ○

Emile Durkheim (1858 to 1917), founder of French sociology, treated society as a reality in its own right. **Social facts** are ways of acting, thinking and feeling that exist outside individuals and constrain them, such as laws and customs; **collective representations** are the ideas and symbols a society shares. His most examined idea is **anomie**: normlessness, a state in which social norms are unclear, weak or no longer apply, so people no longer know what is expected of them. He introduced it in *The Division of Labour in Society* (1893); in *Suicide* (1897), anomic suicide rises when sudden change, a crash or a windfall, unsettles the norms. **Robert King Merton** (1910 to 2003) recast anomie as strain (1938), a gap between cultural goals such as wealth and the legitimate means to them, met by conformity, innovation (illegitimate means such as fraud), ritualism (rules kept, goal dropped), retreatism (both rejected) or rebellion (both replaced).

> **! EXAM TRAP**
>
> **Anomie** = normlessness. The progression from traditional to modern society is *modernisation*: Durkheim's mechanical solidarity (bonds of sameness) giving way to organic solidarity (interdependent specialists), or **Ferdinand Tonnies**'s (1887) *Gemeinschaft* (community) giving way to *Gesellschaft* (society). Adapting to a dominant culture's values is *assimilation* or *acculturation*; internalising society's norms is *socialisation*.

A second root is **symbolic interactionism**: people act towards things according to the meanings those things have for them, and meanings, and the self, are built in interaction through symbols, above all language. **George Herbert Mead** (1863 to 1931) founded it; students published his lectures as *Mind, Self and Society* (1934), and **Herbert Blumer** named it in 1937. **Charles Horton Cooley** (1864 to 1929) added the looking-glass self (1902: we imagine how we appear to others, imagine their judgement and feel pride or shame; Chapter 4) and the primary group (1909), such as the family.

Socialisation and resocialisation ○

Socialisation is the lifelong learning of a society's norms, values and roles until they become one's own: *primary* in childhood within the family, *secondary* through school, peers, work and the media, the other agents of socialisation (gender socialisation is in Chapter 6). Merton's **anticipatory socialisation** is adopting a group's norms before joining it, as a medical aspirant already talks like a doctor.

Resocialisation is unlearning old norms, values and roles and learning new ones after a major transition: college, migration, marriage, the army or prison. A student's first weeks at college, before she finds friends and fits in, are a mild case. It is most intense in the **total institutions** described by **Erving Goffman** (1922 to 1982) in *Asylums* (1961), such as prisons, mental hospitals and barracks, which cut people off from society, manage all of daily life and strip away the old identity.

CLINICAL LINK

Long-stay inmates may develop **institutionalisation**: apathy, lost initiative, submissiveness and dependence on routine. **Russell Barton** called it institutional neurosis (1959), a product of the institution rather than the illness (Chapter 7).

! EXAM TRAP

A newcomer adapting to college, a new country or the army shows **resocialisation**, not the *looking-glass self* (a self-image built from imagined judgements), *self-awareness* (attention turned on oneself) or *institutionalisation* (the dependent apathy of long-stay inmates).

Status and roles ○

A **status** is a position in a group: *ascribed* if given at birth (sex, caste), *achieved* if earned (doctor), in **Ralph Linton**'s distinction (1936). A **role** is the behaviour expected of whoever holds a status. Table 3.2 separates the easily confused terms; Zimbardo's prison study shows the power of roles (Section 3.7).

Table 3.2 The vocabulary of roles.

Term	Meaning	Example
Role expectation	What others expect of the person in a position	Patients expect a doctor to listen
Role taking (Mead)	Seeing a situation from another's viewpoint: one person at a time (play stage), all the players at once (game stage), then the community, the *generalised other*	A fielder reading the bowler and wicketkeeper
Playing at a role	A young child's imitative make-believe	Playing "teacher" to dolls
Role playing (role enactment)	The behaviour a person actually engages in while enacting a role	A new teacher taking her first class
Role conflict	Clashing demands: inter-role (between roles) or intra-role (within one)	Mother and employee on a sick child's day
Role strain (**William J. Goode**, 1960)	Felt difficulty in meeting one role's obligations	A student with too many deadlines
Role set (Merton, 1957)	All the role partners of one status	A student's teachers, classmates, parents
Role ambiguity	Unclear expectations	A job with no job description

! EXAM TRAP

"The behaviour an individual engages in while enacting a role" is **role playing**. *Role taking* is mental (seeing through another's eyes), *playing at a role* is a child's make-believe, and *role expectation* lies in other people, not in the performer.

Theoretical perspectives ○

The field works with several theories (Table 3.3), most taught where they are used.

Table 3.3 Theoretical perspectives in social psychology.

Theory	Theorist, year	Core idea
Field theory	Kurt Lewin	B = f(P, E)
Cognitive consistency	Heider (1958); Festinger (1957)	We strive for consistent beliefs
Social exchange	**George C. Homans** (1961); **John Thibaut** and **Harold Kelley** (1959); **Peter Blau** (1964)	Relationships weighed by rewards and costs
Social learning, social cognitive	**Albert Bandura**	Learning by observing models
Psychoanalytic group psychology	**Sigmund Freud** (1921)	Members idealise the leader and identify with one another
Sociobiology, evolutionary	**Edward O. Wilson** (1975)	Natural selection shapes social behaviour; kin selection
Social representations	Serge Moscovici (1961)	Shared common-sense theories
Social identity	Tajfel and **John Turner** (1979)	Self-esteem from group membership
Uncertainty-identity	**Michael A. Hogg**	Group identity reduces self-uncertainty
Need to belong	**Roy F. Baumeister** and **Mark R. Leary** (1995)	A basic drive for lasting bonds
Sociometer	Mark R. Leary	Self-esteem gauges acceptance
Role theory	Mead; Ralph Linton (1936)	Behaviour follows the roles people occupy
Social constructionism	Kenneth Gergen (1985)	Knowledge is built socially, not discovered
Dual-process models	Petty and Cacioppo, others	Fast automatic versus slow deliberate thinking
Liberation psychology	**Ignacio Martin-Baro**	Psychology from the viewpoint of the oppressed

A **social representation** is a shared, changing system of values, ideas and practices, a common-sense theory through which a group makes the unfamiliar familiar: a collective successor to Durkheim's collective representations.

KEY STUDY: MOSCOVICI AND PSYCHOANALYSIS IN FRANCE (1961)

Serge Moscovici (1925 to 2014), a Romanian-born French psychologist, asked in *La Psychanalyse, son image et son public* (1961) how a scientific theory becomes common sense. Surveying Parisians and analysing the Catholic, Communist and popular press, he found each group remaking psychoanalysis by **anchoring**, fitting the unfamiliar into a familiar category (analysis seen as a kind of confession), and **objectification**, turning an abstract idea into a concrete thing (a "complex" became something a person *has*, like a cold). Critics find the concept too broad to test precisely.

> **! EXAM TRAP**
>
> **Social representations** = **Serge Moscovici**. Michael A. Hogg is social identity and uncertainty-identity theory; Baumeister and Leary proposed the need to belong, and Leary the sociometer.

Methods and ethics

Social psychology uses the methods of Chapter 9 (Table 3.4). A theory predicts; results support or revise it, never prove it.

Table 3.4 Research methods in social psychology.

Method	What it does, and its catch
Systematic observation	Records behaviour, often in natural settings (naturalistic); participant observers join the group, and may alter it
Survey	Questions large samples; self-report bias
Correlational	Measures how two variables co-vary; correlation is not causation
Laboratory experiment	Manipulates an independent variable, measures a dependent variable; shows cause, can be artificial
Field experiment	Experiment in a natural setting; realistic, less control
Quasi-experiment	Compares existing groups without random assignment; cause uncertain
Archival	Analyses existing records; no reactivity
Cross-cultural	Compares cultures; tests generality
Meta-analysis	Statistically combines many studies into one overall effect size

Random assignment, giving every participant an equal chance of any condition, lets an experiment show cause. **Elliot Aronson** and **J. Merrill Carlsmith** (1968) separated **experimental realism**, how far a study absorbs participants, from **mundane realism**, how far it resembles everyday life; Milgram's study had much of the first, little of the second. External validity is how far results generalise. A **mediating variable** explains *how* or *why* one variable affects another (a violent film raises aggressive thoughts, which raise aggression); a **moderating variable** changes *when* or *for whom* it does (more in people already angry).

Deception, misleading participants about a study's purpose so they behave naturally, needs two safeguards: informed consent beforehand (what they will do, and that they may leave at any time) and **debriefing** afterwards, by *dehoaxing* (revealing the deception) and *desensitising* (removing distress, such as the belief that one is cruel). The Milgram and Zimbardo controversies made both standard (Section 3.7). Participants also respond to **demand characteristics**, cues revealing what the experimenter hopes to find (**Martin Orne**, 1962), and experimenters bias results through expectancy effects (**Robert Rosenthal**, 1966); double-blind procedures guard against both.

INDIAN CONTEXT

Gardner Murphy led a UNESCO study of social tensions in India, *In the Minds of Men* (1953). **Durganand Sinha** (1922 to 1998) urged indigenising psychology (Section 1.2), **Jai B. P. Sinha** proposed the nurturant-task leader (Section 3.10) and **Udai Pareek** (1925 to 2010) studied role stress. Research centres on caste, communal relations and poverty.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Allport: the actual, imagined or _____ presence of others.
2. Lewin's life space is the person plus the _____ environment.
3. Lewin founded _____ research: planning, action and fact-finding.
4. Durkheim's term for normlessness is _____.
5. Unlearning old norms and learning new ones at college is _____.
6. Behaviour performed while enacting a role is role _____.
7. Social representations theory (1961) is _____'s.
8. Cues revealing the experimenter's hypothesis are _____ characteristics.

Answers: (1) implied, (2) psychological, (3) action, (4) anomie, (5) resocialisation, (6) playing, (7) Serge Moscovici, (8) demand

3.2 Social Cognition and First Impressions LOWER YIELD · 7 QS

A stranger sits down at your hostel mess table, and within seconds you have judged him friendly, clever or arrogant; from then on you notice what fits. This section follows social thought from the frameworks that store it, through the shortcuts and tilts that bend it, to the impressions we form of others and manage in ourselves.

The social thinker ○

Social cognition is how we interpret, analyse, remember and use information about the social world. **Susan Fiske** and **Shelley Taylor** (1984) called people **cognitive misers**: with limited capacity and a flood of social information, we save effort with shortcuts and settle for good-enough answers. In their 1991 revision the thinker became a **motivated tactician**, choosing shortcuts or careful thought according to goals and stakes: you skim a passing stranger but study an interviewer's face.

That choice is **dual-process** thinking. **Automatic processing** is fast, effortless, unintended and often unconscious; **controlled processing** is slow, effortful, deliberate and conscious (System 1 and System 2, Section 1.8). Fatigue and time pressure push us onto the automatic track. The dominant account of how people organise social information in memory and use it to understand behaviour is the **schema approach**.

> **! EXAM TRAP**
>
> "How people organise information in memory and use it to understand behaviour" = the **schema approach**. A *motivational approach* explains social thought by needs and goals; a *perceptual approach* treats knowing people like perceiving objects (Asch's Gestalt view, below); "organisational approach" merely echoes the stem's verb.

Schemas, priming and perseverance

A **schema** is an organised mental framework, built from experience, that holds our knowledge about a person, group, role or event and guides how we take in new information (Bartlett's idea, Section 1.7). Table 3.5 lists the kinds.

Table 3.5 Kinds of social schema.

Schema	What it holds	Example
Person schema	Traits and goals of a person or type	Your cautious uncle
Self-schema	Beliefs about oneself (**Hazel Markus**, 1977; Chapter 4)	"I am a good debater"
Role schema	Expected behaviour in a role	What a professor does
Event schema or **script**	The usual sequence of an event (**Roger Schank** and **Robert Abelson**, 1977)	Eating at a restaurant
Group schema or stereotype	Traits assumed of a group (Section 3.5)	"Engineers are logical"

Schemas steer **attention** towards what fits, shape **encoding** (consistent information is stored readily, though a sharply inconsistent event stands out) and bias **retrieval** towards schema-consistent details. Recently activated schemas dominate. **Priming** is a temporary rise in a schema's availability: after a violent film, a driver who takes your parking spot looks aggressive. **Automatic priming** works below awareness: **John Bargh**, Mark Chen and Lara Burrows (1996) had students unscramble sentences containing words linked to old age (bingo, Florida), and they then walked down the corridor more slowly. It failed to replicate: Stéphane Doyen and colleagues (2012), timing walkers with infrared sensors, found slowing only when experimenters expected it. A primed schema fades by **unpriming** once it is expressed in thought or action (Betsy Sparrow and Daniel Wegner, 2006).

Schemas also resist disconfirmation, the **perseverance effect**. In the classic study of **belief perseverance**, **Lee Ross**, Mark Lepper and Michael Hubbard (1975) gave students false feedback that they were good or poor at telling real suicide notes from fakes; told later that the feedback was invented, they still rated their ability in line with it. Confirmation bias (Section 1.8) keeps schemas alive. Even figures of speech prime schemas: in **reasoning by metaphor** we grasp an abstract idea through a concrete one ("our relationship is on the right track"), and the metaphor shapes the judgement. Worst of all, a schema can make itself come true.

The self-fulfilling prophecy ○

The sociologist **Robert King Merton** (1910 to 2003) named and formalised the **self-fulfilling prophecy** in 1948: "a false definition of the situation evoking a new behaviour which makes the originally false conception come true." A rumour that a sound bank is failing starts a run

on it, and it fails. In social psychology the term covers any expectation, about oneself or another, that leads us to act in ways that make the expected event or behaviour more likely (Figure 3.2).

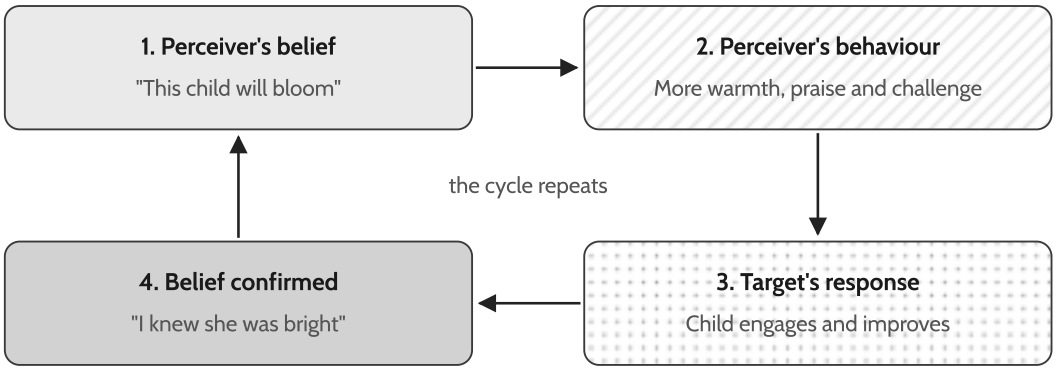


Figure 3.2 The self-fulfilling prophecy: the perceiver's expectation draws matching behaviour from the target, which seems to prove it.

KEY STUDY: PYGMALION IN THE CLASSROOM (1968)

Robert Rosenthal (born 1933) and **Lenore Jacobson** tested every child in a Californian elementary school and told teachers that about 20 percent, in fact chosen at random, would soon "bloom" academically. Eight months later these children had gained more IQ points than their classmates, most clearly in the first two grades; teachers had given them more warmth and challenge. Critics questioned the IQ measures; later reviews find expectancy effects real but usually small.

A high expectation that raises performance is the **Pygmalion effect** (Rosenthal effect), after the sculptor whose statue came to life; a low one that depresses it is the **Golem effect** (Elisha Babad, Jacinto Inbar and Rosenthal, 1982). A perceiver's expectation drawing matching behaviour from the target is **behavioural confirmation**. **Mark Snyder**, Elizabeth Tanke and Ellen Berscheid (1977) gave men a photograph, attractive or not, of the woman they would telephone (it was not her); judges hearing only the women's side rated the "attractive" women warmer and more sociable. Carl Word, Mark Zanna and Joel Cooper (1974) found that white interviewers sat farther away and ended interviews sooner with Black applicants; white applicants treated this coolly then performed worse.

! EXAM TRAP

Acting on an expectation until it comes true = **self-fulfilling prophecy**. **Confirmation bias** only seeks confirming evidence and changes nothing in the world; the **base-rate fallacy** ignores how common something is (Section 1.8); "motivational reaction" is not a term. Merton formalised it, not Erving Goffman (dramaturgy, below; stigma, 1963; total institutions), **Charles Horton Cooley** (1864 to 1929; looking-glass self, 1902; primary group) or **Émile Durkheim** (1858 to 1917; anomie, a state of normlessness).

Heuristics in social judgement

Why shortcuts? Because of **information overload**: demands exceed our capacity, so we use **heuristics**, simple rules giving quick, usually adequate judgements (**Amos Tversky** and **Daniel Kahneman**, 1974; Section 1.8). **Representativeness** judges a quiet, bookish neighbour a librarian, ignoring that managers far outnumber librarians (base-rate neglect). **Availability** judges by ease of recall: people who recalled six of their own assertive acts rated themselves more assertive than those asked for a harder twelve (**Norbert Schwarz** and colleagues, 1991). **Anchoring and adjustment** starts from a value, such as an asking price, and adjusts too little. The **status quo heuristic** ("what is, is good") favours the established: the same chocolate tasted better when said to be on sale since 1937 rather than 2003. The **affect heuristic** (Paul Slovic) lets a feeling of liking or dread stand in for judging risk and benefit.

Errors and tilts in social thought

Other tilts save effort or protect our feelings at the cost of accuracy (Table 3.6).

Table 3.6 Errors and tilts in social thought.

Tilt	Shown or named by	What happens
Negativity bias	Paul Rozin and Edward Royzman (2001)	Bad information outweighs equally strong good information
Optimistic bias (unrealistic optimism)	**Neil Weinstein** (1980)	Good events seem likelier, bad ones less likely, for me than for others
Overconfidence	Section 1.8	Confidence exceeds accuracy
Planning fallacy	Kahneman and Tversky (1979); Roger Buehler, Dale Griffin and Ross (1994)	Underestimating task time: thesis students predicted 33.9 days, took 55.5
Counterfactual thinking	Victoria Medvec, Scott Madey and **Thomas Gilovich** (1995)	Imagining what might have been: upward (better) breeds regret, downward (worse) relief
Magical thinking	**Paul Rozin** (1990)	Law of contagion (contact transfers essence); law of similarity (alike in look, alike in nature)
False consensus effect	Ross, David Greene and Pamela House (1977)	Overestimating how many share our opinions and choices
False uniqueness effect		Underestimating how many share our desirable traits and abilities
Illusion of control	**Ellen Langer** (1975)	Feeling control over chance: a self-chosen lottery ticket is valued more
Ironic rebound after thought suppression	**Daniel Wegner** (1987)	Trying not to think of a white bear makes it return more often

On film of the 1992 Olympics, bronze medallists looked happier than silver medallists: silver thinks upward ("I nearly had gold"), bronze downward ("I nearly got nothing"). Ironic rebound arises because an automatic monitoring process keeps checking for the forbidden thought, and so keeps it primed. **Terror management theory** (Jeff Greenberg, Sheldon Solomon and Tom Pyszczynski, 1986) holds that awareness of certain death creates anxiety we buffer with cultural world-views, self-esteem and supernatural beliefs.

Feeling also steers thought. In **mood-congruent judgement** we notice, recall and judge in line with our mood (Section 1.7). Using **affect as information** we ask "how do I feel about it?":

people telephoned on sunny days reported more life satisfaction than on rainy days, unless first asked about the weather, which let them discount their mood (Schwarz and Gerald Clore, 1983). **Joseph Forgas**'s **affect infusion model** (1995) adds that mood seeps into judgements we must construct, but hardly into ready-made or goal-driven ones.

Forming impressions ●●

Impression formation is the combining of information about a person into an overall judgement. Its founding experiment used only adjectives.

KEY STUDY: ASCH'S WARM AND COLD STRANGER (1946)

Solomon Asch (1907 to 1996), a Gestalt thinker, read participants a stranger's traits: intelligent, skilful, industrious, *warm*, determined, practical, cautious. Others heard *cold* instead. The warm stranger was judged far more generous, happy and sociable; swapping *polite* for *blunt* changed little. Warm-cold were **central traits**, which reorganise the whole impression; polite-blunt were **peripheral traits**. Traits, Asch concluded, form an integrated whole rather than a sum. Critics later showed that centrality depends on context.

Asch also showed a **primacy effect**: a person described as "intelligent, industrious, impulsive, critical, stubborn, envious" was liked more than one given the same words in reverse, because early traits set the meaning of later ones. **Harold Kelley** (1950) told students a guest lecturer was "rather cold" or "very warm"; after the same class, the warm group rated him more considerate and sociable, and more of them joined the discussion.

Such leaps rest on **implicit personality theories**, unspoken beliefs about which traits go together ("warm people are generous"), a term of **Jerome Bruner** and **Renato Tagiuri** (1954). Their best-known product is the **halo effect**: an overall impression, or one salient positive trait, colours ratings of other, unrelated traits. **Edward Lee Thorndike** named it in 1920: army officers who rated a soldier high on physique also rated him high on intelligence, loyalty and leadership. The negative version is the **horns effect**. "What is beautiful is good" is a halo from attractiveness (Section 3.6); in rating scales the halo is a rater error (Chapter 2).

! EXAM TRAP

Halo effect = an overall impression or one positive trait influencing ratings of *other, specific* traits. Not judging by group membership (stereotyping, Section 3.5); not initial impressions shaping later judgements (primacy, or anchoring); not overlooking a liked person's faults (a consequence of a halo, not its definition); not crediting oneself for success and circumstances for failure (self-serving bias, Section 3.3); not overestimating how far others share our opinions (false consensus, Table 3.6).

In **Norman Anderson**'s **information integration theory** (1965) each trait has a value and a weight. An **additive model** sums them, so adding "punctual" to "brilliant, generous" should help; a **weighted averaging model**, which the evidence mostly favours, averages them, so the impression dips. By the **negativity effect**, negative traits weigh more because bad behaviour seems more revealing.

Thin slices and first impressions ○

How fast and how good is a first impression? **Nalini Ambady** and Rosenthal (1992) named **thin slicing**: accurate inferences about people from very brief samples of behaviour. In 1993 they showed students silent clips of college teachers lasting 30 seconds or less (later as little as six); ratings of the clips predicted the teachers' end-of-semester student evaluations. **Malcolm Gladwell** popularised the idea in *Blink* (2005). Janine Willis and Alexander Todorov (2006) found that a 100-millisecond look at a face yields trait judgements, trustworthiness above all, that correlate highly with unhurried ones; longer looks mainly raise confidence. Thin slices beat chance but are not infallible.

> **! EXAM TRAP**
>
> A teacher who watches two students for under a minute and correctly picks the class leader shows **thin slicing**. Not **social facilitation** (better performance of well-learned tasks before others, Section 3.10), not a **self-fulfilling prophecy** (her expectation did not create the leadership), not the **fundamental attribution error** (over-explaining others' behaviour by disposition, Section 3.3).

Managing the impressions we make

We also try to control the impressions others form of us: **impression management**, or **self-presentation**. **Erving Goffman** (1922 to 1982) framed it as theatre in *The Presentation of Self in Everyday Life* (1959). In his **dramaturgy**, on the **front stage** we play a role before an audience (the doctor's calm on the ward); **back stage** we drop it (doctors joking in the staff room). **Face-work** maintains one's own and others' face, the positive social value claimed in an encounter.

Tactics are **self-enhancement**, raising one's own appeal (grooming, recounting achievements), or **other-enhancement**, making the target feel good, chiefly **ingratiation**, winning liking through flattery and agreement, analysed by **Edward E. Jones** (1964). Jones and Thane Pittman (1982) classified five strategies by the impression sought (Table 3.7).

Table 3.7 Jones and Pittman's strategies of self-presentation (1982).

Strategy	Wants to seem	Typical behaviour
Ingratiation	Likeable	Flattery, agreement, favours
Self-promotion	Competent	Displaying achievements
Intimidation	Dangerous	Threats, shows of anger
Exemplification	Morally worthy	Self-sacrifice, staying late
Supplication	Helpless	Pleading need, so others help

Self-monitoring (Snyder, 1974) is the tendency to read situational cues and adjust one's behaviour to fit: high self-monitors are social chameleons, low self-monitors act from inner attitudes. In **self-handicapping** people create an obstacle in advance to excuse possible failure: Steven Berglas and Jones's (1978) students, given unearned success on unsolvable problems, chose a drug said to impair performance before a retest. By the **spotlight effect** we overestimate how much others notice us: students in an embarrassing T-shirt guessed half the room noticed, when about a quarter did (Gilovich, Medvec and Kenneth Savitsky, 2000). The **illu-**

sion of transparency overestimates how well others read our inner states, such as nervousness.

Does it work? Usually: applicants who use it get higher interview ratings. Overdone, it backfires: **Roos Vonk** (1998) described the **slime effect**, strong dislike of people who flatter superiors and scorn subordinates. The self-concept and self-esteem are in Chapter 4.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Fiske and Taylor's (1984) term for people who save mental effort with shortcuts is the _____.
2. Knowledge of the usual sequence of an event, such as eating at a restaurant, is a _____.
3. A false belief that evokes behaviour which makes it come true is a _____, formalised by _____ in 1948.
4. Bronze medallists are happier than silver medallists because of _____ counterfactual thinking.
5. In Asch's study, warm and cold were _____ traits.
6. One positive trait colouring ratings of unrelated traits is the _____ effect, named by _____ in 1920.
7. Accurate judgements of teachers from silent clips of 30 seconds or less illustrate _____.

Answers: (1) cognitive miser, (2) script (event schema), (3) self-fulfilling prophecy; Robert K. Merton, (4) downward, (5) central, (6) halo; Edward Thorndike, (7) thin slicing.

3.3 Communication, Nonverbal Cues and Attribution LOWER YIELD · 12 QS

A friend answers your long message with one word: "ok." Is she busy, annoyed, or simply brief? Two processes are at work: communication, which carries meaning from one head to another through words, faces, gestures and distances, and attribution, by which you decide why someone acted as they did. This section follows the message first, then turns to the rules, and the biases, by which we explain behaviour.

The communication process and its models ○

Communication is the process by which a person or group transmits information, ideas and feelings to another through shared symbols. Every exchange has the same parts (Figure 3.3). The sender, or source, first performs **encoding**, turning an idea into symbols such as words or a gesture. The message travels through a **channel**, the medium that carries it (speech, print, a phone screen, the face). The receiver, or recipient, performs **decoding**, turning the symbols back into meaning, and replies with **feedback**. Anything that distorts the message on the way is **noise**.

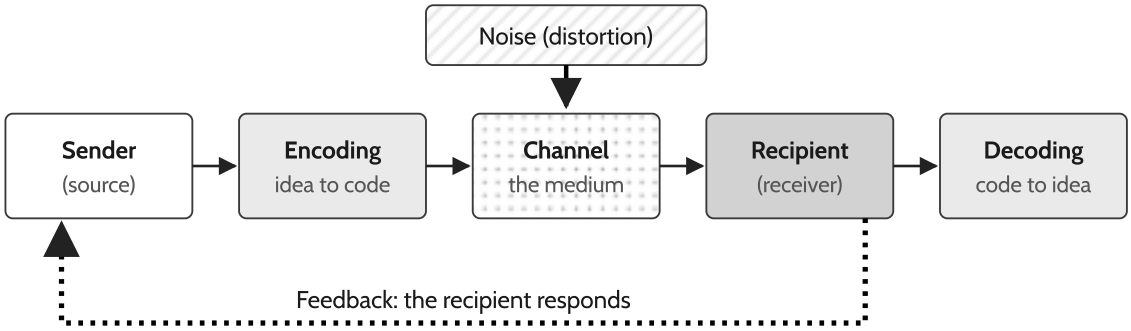


Figure 3.3 The communication process: encoding, channel, recipient, decoding. Noise (dashed arrow) distorts the message in the channel; feedback (dotted line) makes the exchange two-way.

> **! EXAM TRAP**
> The sequence is **encoding, channel, recipient, decoding**: a message must be encoded before it can enter a channel, and received before it can be decoded.

Table 3.8 compares four classic models: those of **Harold Lasswell** (1902 to 1978); **Claude Shannon** (1916 to 2001) and **Warren Weaver** (1894 to 1978); **Wilbur Schramm** (1907 to 1987), for whom a message is understood only where the parties' fields of experience (language, culture, knowledge) overlap; and **David Berlo** (1929 to 1996), whose **SMCR model** adds the source's and receiver's skills, attitudes, knowledge, social system and culture.

Table 3.8 Four classic models of communication.

Model	Year	Elements	Remembered for
Lasswell	1948	Who says what, in which channel, to whom, with what effect	Verbal, linear; no feedback
Shannon and Weaver	1949	Source, transmitter, channel, noise, receiver, destination	Linear; introduced noise
Schramm	1954	Each party encodes, interprets and decodes	Circular; field of experience
Berlo (SMCR)	1960	Source, message, channel, receiver	Skills, attitudes, knowledge, culture

Verbal communication uses words; nonverbal communication uses everything else. In one-way communication (a lecture, a notice) the receiver cannot reply; in two-way communication feedback flows back, which is slower but more accurate. Barriers are **semantic** (words or jargon understood differently), **physical** (noise, distance, a poor network), **psychological** (anxiety, anger, prejudice, inattention) and **cultural** (different norms for gestures, eye contact and directness).

In the **communication networks** studied by **Alex Bavelas** (1913 to 1993) in 1950 and **Harold Leavitt** (1922 to 2007) in 1951, small groups solved problems by passing notes along fixed links. The centralised **wheel**, everyone linked only to a hub, was fastest and most accurate on simple tasks, and the hub became leader, but the others were least satisfied. The **chain** was intermediate. The decentralised **circle** was slower, with no clear leader but the highest morale,

and in the later **all-channel** network everyone talks to everyone, which suits complex problems (Section 3.10).

Nonverbal channels and personal space

Nonverbal communication is the exchange of information through facial expression, gaze, body movement, posture, touch, voice and even scent, apart from the meaning of words. Such cues are hard to suppress, so hidden feelings leak through them. As Section 1.11 showed, **Paul Ekman** (born 1934) and **Wallace Friesen** found six universally recognised expressions (happiness, sadness, fear, anger, surprise, disgust), with contempt added in 1986, while cultures differ in **display rules** about when and how strongly an emotion may be shown. The facial feedback hypothesis is also in Section 1.11.

Eye contact usually signals liking, and avoided eyes suggest dislike or shyness, but **staring**, gaze held regardless of what the other person does, signals hostility and makes people leave. Body language includes posture (an open posture signals ease, frequent self-touching nervousness) and gestures, the most specific being **emblems** with a fixed meaning in one culture, such as the namaste or the Indian head wobble meaning "all right". **Ray Birdwhistell** (1918 to 1994) founded **kinesics**, the study of body movement as communication (*Introduction to Kinesics*, 1952). Touch conveys affection, care or dominance depending on who touches whom and where. **Paralanguage** is the voice apart from the words (pitch, tone, loudness, rate, pauses), which is why "fine" can mean its opposite. Even scent is a channel: subtle changes in body odour, such as those across the menstrual cycle, can affect others who do not consciously notice them.

Edward T. Hall (1914 to 2009) coined **proxemics** (1963), the study of how people use interpersonal distance, and in *The Hidden Dimension* (1966) described four zones (Figure 3.4). He measured them among Americans; contact cultures (Arab, Latin American) stand closer. In *Beyond Culture* (1976) he contrasted **high-context cultures**, where meaning lies largely in the setting, the relationship and nonverbal cues, so speech is indirect (India, Japan, China), with low-context cultures, where explicit words carry it (the United States, Germany). An Indian "I will try to come" may be a polite no.

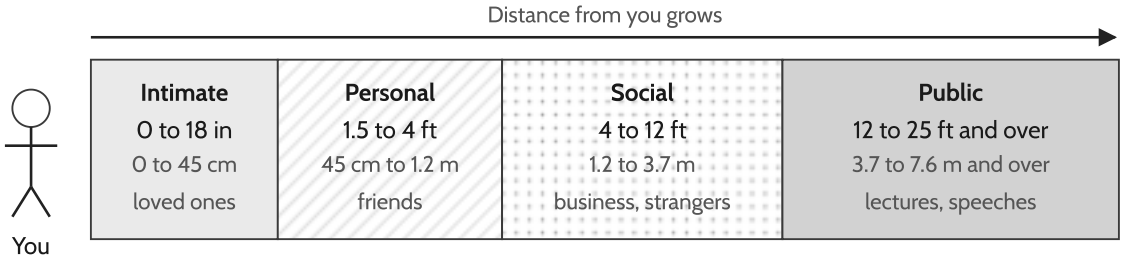


Figure 3.4 Hall's (1966) zones of personal space, not drawn to scale. Learn the boundaries: 18 inches, 4 feet, 12 feet and 25 feet.

Detecting deception

Can nonverbal cues expose a lie? Four help. **Microexpressions** are fleeting facial expressions, lasting a fraction of a second, that leak a concealed emotion before it is masked. **Interchannel discrepancies** are mismatches between channels: a liar may control the face

but not the voice or hands. Eye contact changes: liars blink more, have more dilated pupils, and show too little or unusually much eye contact. Exaggerated expressions, an overdone smile or excessive regret, are the fourth sign, and vocal pitch tends to rise. Yet a meta-analysis of more than 200 studies by **Charles Bond** and **Bella DePaulo** (2006) found accuracy of only about 54 percent, barely above chance, and truths were spotted more often than lies because people assume honesty.

> ! EXAM TRAP
>
> Averted eyes do not reliably mark a lie: liars may stare to look honest. No single cue betrays a lie, and most judges stay near chance.

Heider, Jones and Davis: from acts to causes ●●

Attribution is the process by which people infer the causes of their own and others' behaviour. **Fritz Heider** (1896 to 1988), in *The Psychology of Interpersonal Relations* (1958), described people as amateur scientists; his **naive psychology** (common-sense psychology) concerns the implicit rules they use to infer causes. His key distinction: a **dispositional attribution** (internal) locates the cause in the person (traits, abilities, attitudes, effort), while a **situational attribution** (external) locates it outside (the situation, other people, the environment, luck, roles, culture). "He failed because he is lazy" is dispositional; "the paper was unusually hard" is situational.

> ! EXAM TRAP
>
> Causes resting on the actor's personality traits are **dispositional**; situational, environmental and cultural causes are all external. Matching: Heider = implicit rules of attribution; **Fred Fiedler** (1922 to 2017) = contingency theory of leadership (Section 3.10); **Sigmund Freud** (1856 to 1939) = unconscious conflict; **Albert Bandura** (1925 to 2021) = social learning theory.

Edward E. Jones and **Keith Davis** asked when an act reveals a trait. By their **correspondent inference theory** (1965), we infer that behaviour corresponds to a stable disposition when it is (1) freely chosen; (2) produces **non-common effects**, outcomes that only the chosen option yields, which pinpoint the motive; and (3) low in social desirability, since out-of-role behaviour is more revealing: a guest who calls your mother's biryani too salty reveals more than one who praises it. **Hedonic relevance** (the act affects us) and **personalism** (it seems aimed at us) strengthen the inference.

Kelley's covariation model and causal schemas

Harold Kelley (1921 to 2003) asked whether a cause lies in the person, the stimulus or the circumstances. His **covariation model** (1967) says that, observing behaviour over several occasions, we attribute it to the cause it covaries with, using three kinds of information:

- **Consensus**: do other people react to this stimulus in the same way?
- **Consistency**: does this person react this way to the stimulus on other occasions?
- **Distinctiveness**: does this person react differently to other stimuli?

Ananya laughs at a comedian. If few others laugh, she laughs at him every time and she laughs at every comedian (low, high, low), the cause is Ananya. If everyone laughs, she always does and she laughs at no other comedian (all high), the comedian is funny. If she has never laughed at him before (low consistency), a passing circumstance is responsible. Consistency is high in both main patterns, so only consensus and distinctiveness separate internal from external (Table 3.9).

Table 3.9 Kelley's covariation model.

Consensus	Consistency	Distinctiveness	Attribution
Low	High	Low	Internal: the person
High	High	High	External: the stimulus
Low	High	High	Person and stimulus together
Any	Low	Any	A temporary circumstance

For a single observation, Kelley (1972) proposed causal schemas. By the **discounting principle**, a cause gets less weight when other plausible causes exist (a film star is paid to praise a brand); by the **augmenting principle**, it gets more weight when the effect occurs despite an obstacle (a student tops the class while working nights).

Weiner's three dimensions

Bernard Weiner (born 1935), studying achievement from the 1970s, classified causes by **locus** (internal or external), **stability** and **controllability**. Ability is internal, stable and uncontrollable; effort is internal, unstable and controllable; task difficulty is external and stable; luck is external and unstable. Stability shapes expectations, and the dimensions shape emotion: success put down to internal causes brings pride; failure from low ability brings shame, from low effort guilt. Another person's misfortune seen as controllable ("he drank") provokes anger, and an uncontrollable one ("a lay-off") pity and helping (Section 3.8).

CLINICAL LINK

In the attributional reformulation of learned helplessness, **Lyn Abramson**, **Martin Seligman** (born 1942) and **John Teasdale** (1978) proposed that people who explain bad events by **internal** ("it is me"), **stable** ("it will last") and **global** ("it spoils everything") causes are most vulnerable to depression (Section 1.6; Chapter 8).

Attributional biases ●●●

Our attributions tilt in predictable directions (Table 3.10). The strongest tilt is the **fundamental attribution error** (FAE), named by **Lee Ross** (1942 to 2021) in 1977: overestimating dispositional and underestimating situational causes of other people's behaviour. A man arrives late and spills his coffee; we call him clumsy, not delayed and hurried. The **correspondence bias**, inferring dispositions that correspond to behaviour even when situational causes are present, is usually treated as a synonym (reviewed in 1995 by **Daniel Gilbert**, born 1957, and **Patrick Malone**).

KEY STUDY: JONES AND HARRIS, THE CASTRO ESSAYS (1967)

Method. Jones and **Victor Harris** had participants read essays for or against Fidel Castro, telling some that the writer chose the side and others that it was assigned. **Result.** Even with assigned sides, pro-Castro writers were judged fairly pro-Castro. **Conclusion.** People underweight a situational constraint they know about. **Criticism.** Readers may assume nobody argues well for a view he rejects, but the bias replicates across many tasks.

Why does it happen? The actor is perceptually salient while the situation fades into the background. In Gilbert's two-step model we first make an automatic dispositional inference, then effortfully correct it for the situation; under cognitive load the correction fails.

For our own behaviour the tilt reverses. The **actor-observer effect** (Jones and **Richard Nisbett**, 1971) is attributing our own behaviour to situations and others' to dispositions: he trips because he is clumsy, I trip because the floor was wet. The actor sees the situation; the observer sees the actor. **Michael Storms** (1973) reversed the effect by showing actors and observers videotapes shot from the opposite viewpoint. The **self-serving bias** concerns our own outcomes: successes go to internal causes, failures to external ones (an unfair paper, bad luck); it protects self-esteem and is weaker in collectivist cultures.

The **just-world hypothesis** of **Melvin Lerner** (1965, 1980) holds that people need to believe the world is orderly, predictable and fair, a place where people get what they deserve. An innocent person's suffering threatens the belief, which is restored by deciding the victim deserved it: victim blaming. In Lerner and **Carolyn Simmons**'s experiment (1966), women who watched a fellow student receive apparent electric shocks, and could not end or compensate her suffering, derogated her. **Zick Rubin** and **Letitia Anne Peplau** measured the belief with their Just World Scale (1975).

! EXAM TRAP

Just-world victim blaming comes from the **need to believe that the world is orderly, predictable and fair**, not from sadism, not from the FAE itself (it looks dispositional, but the motive is protecting the belief), and not from downward social comparison with someone worse off (Section 3.6).

Table 3.10 Attributional biases compared.

Bias	Who, when	What happens	Example
Fundamental attribution error (correspondence bias)	Ross 1977	Others' acts put down to disposition	"He snapped at the waiter: he is rude."
Actor-observer effect	Jones and Nisbett 1971	My acts to situations, yours to disposition	"I was late because of traffic; you are careless."
Self-serving bias	**Dale Miller** and **Michael Ross** 1975	My successes internal, my failures external	"I topped through talent; I failed because the paper was unfair."
False consensus effect	Ross 1977 (Section 3.2)	Overestimating how many share my views	"Everyone finds this film boring."
Just-world hypothesis	Lerner 1965, 1980	Victims seen as deserving their fate	"She must have asked for it."

Bias	Who, when	What happens	Example
Ultimate attribution error	**Thomas Pettigrew** 1979 (Section 3.5)	FAE for groups: out-group's bad acts dispositional, good acts luck	"They cheat by nature; he is an exception."
Hostile attribution bias	**Kenneth Dodge** 1980 (Section 3.9)	Ambiguous acts read as hostile; marks aggressive children	"He bumped me on purpose."
Defensive attribution	**Kelly Shaver** 1970	More blame for severe accidents, less if the person resembles us	"He was careless; it could not happen to me."

! EXAM TRAP

Ask whose behaviour and whose outcome. FAE: **other people's** behaviour, disposition overestimated (the reversed wording is a distractor). Actor-observer: my behaviour versus yours. Self-serving: my own successes and failures. Non-attribution distractors: **confirmation bias**, favouring evidence for what we already believe (Section 1.8); **belief bias**, judging an argument valid because its conclusion is believable; the **halo effect**, one good trait colouring the whole impression (Section 3.2); **cognitive dissonance**, discomfort from inconsistent cognitions (Section 3.4); **stereotyping**, a generalised belief about a group (Section 3.5). The optimism bias is in Section 3.2.

Culture and applications

Is the FAE truly fundamental? It is strongest in individualist cultures, which see the person as an independent agent, and weaker in collectivist ones, which see behaviour as embedded in roles and relationships.

INDIAN CONTEXT

Joan Miller (1984) compared Hindu Indians in Mysore with Americans, aged 8, 11, 15 and adult, asking them to explain good and bad acts of people they knew. American adults gave far more dispositional explanations; Hindu adults gave more situational ones (roles, relationships, circumstances). At age 8 the groups barely differed and the gap grew with age: attributional style is learned from culture.

Similarly, **Michael Morris** and **Kaiping Peng** (1994) found that *The New York Times* explained two 1991 American mass murders by the killers' personalities, while a Chinese-language newspaper stressed their situations (isolation, conflicts at work, easy access to guns).

Because attributions shape feelings and effort, changing them helps. In **attribution retraining**, **Timothy Wilson** and **Patricia Linville** (1982) showed worried first-year students evidence that grades usually improve after the first year, inviting an unstable explanation for poor grades; they later earned better grades and dropped out less than controls. Attributional therapy for depression teaches people to credit themselves for success and to stop blaming lasting flaws for failure. Happy couples credit a partner's kindness to disposition and excuse lapses as situational; distressed couples do the reverse (Section 3.6). And computer-mediated communication, stripped of faces and tone, gives the correspondence bias free rein: a terse "ok." reads as rudeness, not haste.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. In the communication process, turning an idea into symbols is _____.
2. Lasswell's model asks who says what, in which _____, to whom, with what effect.
3. Hall's personal distance runs from 1.5 to _____ feet.
4. Fleeting facial expressions that leak a concealed emotion are _____.
5. Low consensus, high consistency and low distinctiveness lead to an _____ attribution.
6. Attributing your own behaviour to the situation but others' to their dispositions is the _____ effect.
7. Victim blaming protects the belief that the world is orderly, predictable and _____.

Answers: (1) encoding, (2) channel, (3) 4, (4) microexpressions, (5) internal (dispositional), (6) actor-observer, (7) fair.

3.4 Attitudes, Persuasion and Cognitive Dissonance LOWER YIELD · 15 QS

Advertisements, election rallies and anti-smoking posters all bet that changing attitudes changes behaviour. This section follows attitudes from their origins to behaviour, persuasion, resistance and dissonance.

What an attitude is

In 1935 **Gordon Willard Allport** (1897 to 1967) called the attitude "the most distinctive and indispensable concept in contemporary American social psychology". An **attitude** is a learned, fairly lasting evaluation of an **attitude object** (a person, group, thing or idea) along a like-dislike dimension. The **tricomponent model** (ABC model) of **Milton J. Rosenberg** and Carl Hovland (1960) gives it three parts: *affect* (feelings: "chai cheers me up"), *behaviour* or conation (action tendencies: "I stop at the stall daily") and *cognition* (beliefs: "tea is good for me").

Attitude strength combines extremity, certainty, accessibility (how fast the attitude comes to mind) and importance. Ambivalence, holding positive and negative evaluations of one object at once, weakens an attitude's hold on behaviour. Explicit attitudes are conscious and self-reported; **implicit attitudes** are automatic evaluations measured from reaction times, as in the Implicit Association Test (Section 3.5), and the two can disagree.

Daniel Katz (1903 to 1998) named four **functions of attitudes** in 1960: *utilitarian* or adjustive (gaining rewards, avoiding punishments), *knowledge* (organising a confusing world), *ego-defensive* (protecting self-esteem from threatening truths, as when scorn for an outgroup props up a shaky self) and *value-expressive* (displaying central values). In 1956 **M. Brewster Smith**, **Jerome Seymour Bruner** and **Robert W. White** had named *object appraisal* (sizing up objects for one's interests), *social adjustment* (easing relations with valued people) and *externalisation* (projecting inner conflicts onto outside objects). Measurement belongs to Section 2.8: Bogardus's social distance scale (1925), Thurstone's equal-appearing intervals (1928), Likert's

summated ratings (1932), Guttman's cumulative scale and Osgood's semantic differential (1957).

How attitudes form ○

Most attitudes are learned (Section 1.6). In classical conditioning a neutral object takes on the feeling of what accompanies it: **Arthur W. Staats** and **Carolyn K. Staats** (1958) paired nationality names such as "Dutch" and "Swedish" with pleasant or unpleasant words, and the pleasantly paired names were later rated more favourably. In **subliminal conditioning** the pairing works without awareness: **Jon A. Krosnick** and colleagues (1992) flashed pleasant or unpleasant pictures too briefly to be noticed before photographs of a stranger, who was then liked more after the pleasant ones.

In instrumental conditioning, views that are praised or agreed with are strengthened, which is why children echo their parents' opinions until adolescence. We also copy the attitudes of models (observational learning) and adopt those of reference groups, the groups whose standards we judge ourselves by, through social comparison, which Leon Festinger (1954) described as evaluating our own opinions and abilities against other people's. Mere exposure is taught in Section 3.6. Identical twins share more attitudes than fraternal twins, and **Abraham Tesser** (1993) found highly heritable attitudes more accessible and harder to change. Online word of mouth (reviews, ratings, posts) now carries the weight of peers.

> **! EXAM TRAP**
> Praise or agreement from a friend **strengthens** an expressed attitude, never weakens it: positive reinforcement raises the likelihood of whatever it follows.

When attitudes predict behaviour ○

KEY STUDY: LAPIERE'S TRAVELS (1934)

In the early 1930s, a time of open anti-Asian prejudice, **Richard Tracy LaPiere** (1899 to 1986) toured the United States with a young Chinese couple. Of about 250 hotels and restaurants, one refused them. Afterwards he wrote to the same places asking whether they would serve Chinese guests, and over 90 percent of those replying said no. Critics note that different staff may have answered the letters.

In 1969 **Allan W. Wicker**'s review found attitude-behaviour correlations rarely above .30. Research then asked when attitudes do predict behaviour.

Strong attitudes predict best. Extremity often reflects **vested interest**: John Sivacek and **William D. Crano** (1982) found that 47 percent of students who would be barred by a higher drinking age agreed to campaign against it, against 12 percent of unaffected students. Certainty (clarity plus a sense of correctness), personal relevance and knowledge about the object also help. So does direct experience. During a Cornell University housing shortage, **Dennis T. Regan** and **Russell H. Fazio** (1977) found that students forced to sleep on lounge cots acted on their attitudes (petitions, committees) far more consistently than students who had only heard about it, and Fazio and **Mark P. Zanna** (1981) showed that direct experience makes attitudes clearer and more accessible. Attitudes formed by indirect experience predict worst.

Specificity must also match. The **principle of compatibility** of **Icek Ajzen** and **Martin Fishbein** (1977) requires attitude and behaviour to correspond in action, target, context and time. In Andrew Davidson and James Jaccard's (1979) study, general attitude towards birth control correlated .08 with women's pill use over the next two years, but attitude towards using the pill in that period correlated .57. Finally, situational constraints and perceived norms intervene: **Deborah A. Prentice** and **Dale T. Miller** (1993) found Princeton students privately less comfortable with heavy drinking than they believed their peers were, and the men's attitudes drifted towards that imagined norm.

> ! EXAM TRAP
>
> Behaviour is hardest to predict from attitudes formed by **indirect experience** (hearsay, media). Direct experience, personal relevance, knowledge of the object, certainty and matching specificity all strengthen the link.

Reasoned action, planned behaviour and MODE ○

For deliberate behaviour, the **theory of reasoned action** (Fishbein and Ajzen, 1975) makes behavioural intention the immediate predictor of behaviour. Intention rests on the *attitude towards the behaviour* (beliefs about its outcomes, weighted by their value) and the **subjective norm** (what important others want, weighted by the motivation to comply). Ajzen (1985, 1991) added **perceived behavioural control**, the belief that one can perform the act, close to Albert Bandura's self-efficacy, making the **theory of planned behaviour**; control affects intention and also behaviour directly. It is the standard model of health behaviours such as condom use and quitting smoking; the health belief model and protection motivation theory are in Section 7.9.

Spontaneous behaviour works differently. In Fazio's **attitude-to-behaviour process model** (1986), an event automatically activates an accessible attitude, which together with knowledge of norms shapes how the situation is defined, and that definition drives behaviour. His **MODE model** (1990: Motivation and Opportunity as DEterminants) unites the accounts: people deliberate only when motivated and given the opportunity (time, capacity); otherwise accessible attitudes act automatically.

> ! EXAM TRAP
>
> Planned behaviour's predictors of intention are attitude, subjective norms and perceived behavioural control, often worded as self-efficacy (RML 2023). Genetics, personality, socioeconomic status, cultural beliefs, social support, health literacy, physical environment, emotional intelligence and coping are not in the model.

Persuasion: who says what to whom ○

Persuasion is attitude change produced by messages. At Yale University **Carl Iver Hovland** (1912 to 1961), **Irving Lester Janis** (1918 to 1990) and **Harold H. Kelley** (1921 to 2003) summarised their programme in *Communication and Persuasion* (1953). The **Yale communication approach** asks who says what to whom (Section 3.3) and varies source, message, audience and channel (Table 3.11). Credibility means expertise plus trustworthiness.

Table 3.11 What makes a message persuasive: the Yale findings.

Variable	What persuades more
Source	Credible, attractive, likeable or similar; fast speech; not seeming to persuade (overheard, arguing against own interest)
One- or two-sided message	One-sided if the audience agrees or is less educated; two-sided (refuting the other side) if it is opposed or better educated (Hovland, Lumsdaine and Sheffield 1949)
Fear appeal	Moderate fear plus clear, doable advice; intense fear invites denial (Janis and Feshbach 1953)
Discrepancy, repetition	Moderate on both; credible sources can push bigger gaps
Order	Primacy if two messages come back to back and judgement is delayed; recency if time separates them and judgement is immediate (Miller and Campbell 1959)
Audience	Moderate self-esteem and intelligence; low need for cognition (for cue-based appeals); impressionable years (late adolescence, early adulthood); good mood; initial view near the message; no forewarning
Channel	Face to face beats mass media; print suits complex messages

Audience factors are **receiver characteristics**, the factors in the audience that influence persuasion. **Need for cognition**, the enjoyment of effortful thinking (**John T. Cacioppo** and **Richard E. Petty**, 1982), makes people weigh arguments rather than cues. **William J. McGuire** (1925 to 2007) explained why moderate self-esteem and intelligence persuade best with his information-processing chain of attention, comprehension, yielding, retention and behaviour: able, confident people comprehend more but yield less.

In the **sleeper effect**, a low-credibility message gains impact over time because the discounting cue (its source) is forgotten while the content is remembered. Hovland and **Walter Weiss** (1951) found that a message credited to the physicist Robert Oppenheimer persuaded more than the same message credited to the Soviet paper *Pravda*, but four weeks later the difference had almost vanished.

> **! EXAM TRAP**
>
> "Receiver characteristics" (printed "receiver-operant characteristics" in IHBAS 2025) are audience factors influencing persuasion, not sensory thresholds or the receiver operating characteristic, the signal-detection curve of hits against false alarms (Section 1.3), not reinforcement controlling recall, and not biases in interpreting messages (biased assimilation, 3.4.7).

Two routes to persuasion ○

Why do these factors work? In the **elaboration likelihood model** (ELM) of Petty and Cacioppo (1981, 1986), elaboration is thinking about a message's arguments, and it depends on *motivation* (personal relevance, need for cognition) and *ability* (time, knowledge, no distraction). With both, people take the **central route**: systematic processing with careful attention to argument quality, giving durable change that resists attack and predicts behaviour. Without them they take the **peripheral route**, using mental shortcuts triggered by cues such as an attractive source, the number of arguments or a good mood, and the change is temporary (Figure 3.5). In a 1981 study by Petty, Cacioppo and Goldman, argument quality decided students' views on a comprehensive-examination proposal due next year, but source expertise decided them when it was a decade away.

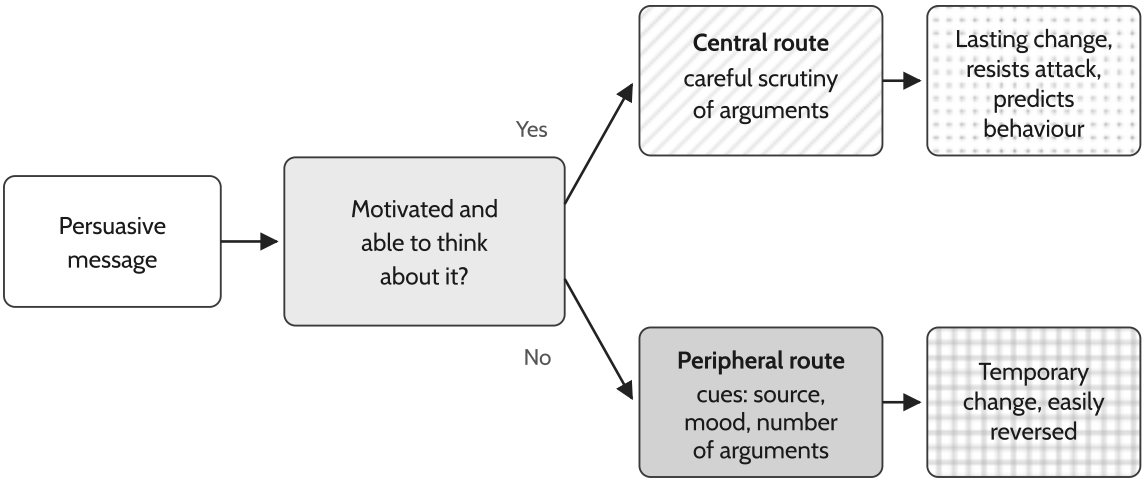


Figure 3.5 The elaboration likelihood model (Petty and Cacioppo, 1986). Motivation and ability decide the route; only central-route change lasts.

Shelly Chaiken's **heuristic-systematic model** (1980) contrasts systematic processing with heuristic processing by rules of thumb ("experts are right"), and lets both run at once.

> **! EXAM TRAP**
>
> Systematic processing is the central route and involves careful attention; heuristic processing is the peripheral route and uses mental shortcuts. The ELM has exactly two routes: there is no "emotional" or "situational" route (IHBAS 2020).

Resisting persuasion ○

Audiences also push back. **Jack Williams Brehm** (1928 to 2009) defined **reactance** in 1966 as the unpleasant state, and motivation, to restore a freedom that is threatened, often by wanting or doing the forbidden thing more; hard selling may even produce a *boomerang effect*, change opposite to the one advocated. Reactance is not compliance (changing behaviour at a direct request), conformity (following group norms) or obedience (following authority), all in Section 3.7, nor social comparison (3.4.2).

Forewarning of persuasive intent aids resistance: **Jonathan L. Freedman** and **David O. Sears** (1965) found teenagers warned ten minutes ahead of a talk against teenage driving were persuaded less than those warned two minutes ahead or not at all, because they had time for counterarguing, generating thoughts that refute the message. People also practise selective avoidance of challenging information and selective exposure to agreeable information. In **inoculation** (McGuire, 1961), a weak attack plus its refutation (refutational defence) protects a belief better than supportive arguments alone, as a vaccine does. **Charles G. Lord**, **Lee Ross** and **Mark R. Lepper** (1979) showed biased assimilation: supporters and opponents of capital punishment read the same mixed evidence, each rated the agreeing study as stronger, and both became more extreme (**attitude polarisation**). Ego depletion, tiredness after a demanding task, weakens resistance.

! EXAM TRAP

Reactance (Brehm, 1966) is the motivation to restore a threatened freedom, so the restricted option grows more attractive. It is not complying with requests or authority, following norms or the majority, the effect of persuasive messages, or comparing one's abilities with others.

CLINICAL LINK

Lecturing a client to stop drinking can provoke reactance; motivational interviewing (Section 7.6) avoids it by letting the client voice the reasons for change.

Consistency theories and cognitive dissonance ●●●

Much resistance reflects a drive for consistency. In the **balance theory** of **Fritz Heider** (1896 to 1988; 1946, 1958), a person (P), another person (O) and an object (X) form a triad of positive or negative relations. It is balanced when the product of the three signs is positive, and imbalanced, motivating change, when it is negative (Figure 3.6). In the **congruity theory** of **Charles E. Osgood** and **Percy H. Tannenbaum** (1955), when a source endorses an object we rate differently, as when an admired cricketer promotes a disliked brand, both evaluations move towards each other.

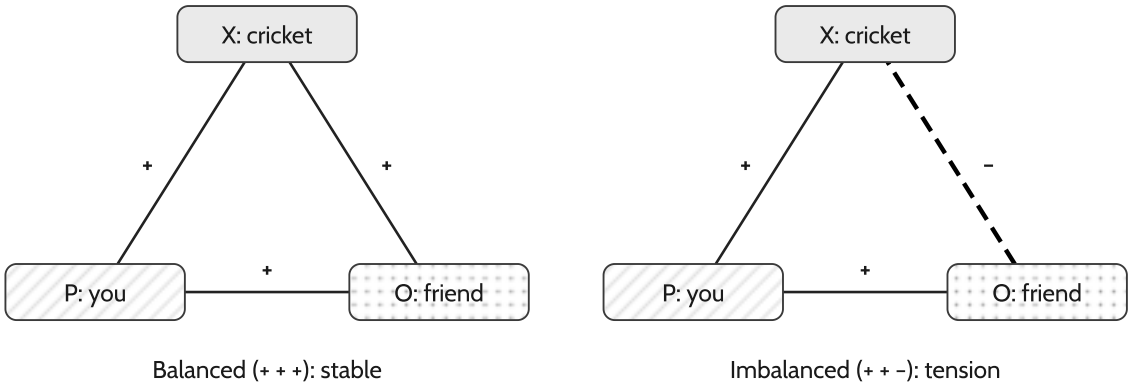


Figure 3.6 Heider's balance theory. If your friend hates the cricket you love (dashed line), you may like cricket less, like the friend less, or win the friend over.

Leon Festinger (1919 to 1989) proposed **cognitive dissonance** theory in 1957. Dissonance is the aversive tension of holding inconsistent cognitions, especially of acting against one's own attitude; it grows with the importance of the cognitions and with the ratio of dissonant to consonant ones. People reduce it by:

- changing the attitude, the usual outcome when the behaviour cannot be undone;
- changing the behaviour;
- adding consonant cognitions and seeking supportive, not conflicting, information;
- **trivialisation**, deciding the inconsistency does not matter (**Linda Simon**, **Jeff Greenberg** and Brehm, 1995);

- **self-affirmation**, bolstering self-worth through other values while the inconsistency stays (**Claude Mason Steele**, 1988);
- denying responsibility.

Take a case. Arvind detests smoking; his grandfather died of lung cancer. At a party his friends smoke, and so does he. "Smoking is deadly" and "I smoked" clash, both matter to him, and he feels dissonance. He cannot unsmoke the cigarette, so he may soften his attitude ("one is harmless"), add a cognition ("I did it for my friends"), blame the pressure, or vow never again. This is not effort justification (valuing what cost effort), an increased drive to smoke, or drive reduction (Hull's need-reduction principle, Section 1.10).

> **! EXAM TRAP**
>
> Cognitive dissonance is Festinger's theory, not Milgram's (obedience), Asch's (conformity) or Zimbardo's (prison simulation), and not operant conditioning, classical conditioning or social learning theory (Section 1.6). Trivialisation and self-affirmation reduce dissonance, not compliance, reactance or self-regulation (control of one's own behaviour towards goals). Dissonance is likeliest when one must decide or act against one's beliefs, not when thoughts and behaviour agree, after praise or during happy memories.

Dissonance paradigms and rival explanations

KEY STUDY: FESTINGER AND CARLSMITH (1959)

Festinger and **James Merrill Carlsmith** had students do an hour of dull tasks (turning pegs, loading spools), then paid them $1 or $20 to tell the next participant the tasks were interesting. The $20 group, like controls who told no lie, later rated the tasks dull; the $1 group rated them more enjoyable. Lacking adequate external justification, they reduced dissonance by changing their attitude. This is **induced compliance** (forced compliance) with insufficient justification, and its result is the *less-leads-to-more effect*.

Table 3.12 sets out the other classic paradigms.

Table 3.12 The classic dissonance paradigms.

Paradigm	Classic study	Finding
Effort justification	Aronson and Judson Mills 1959	A severe, embarrassing initiation made a dull group discussion seem more interesting
Free choice (post-decision)	Brehm 1956	After choosing between similar appliances, the chosen was rated higher, the rejected lower (spreading of alternatives)
Belief disconfirmation	Festinger, Henry Riecken and Stanley Schachter 1956, *When Prophecy Fails*	When a cult's prophecy failed, committed members recruited more
Insufficient deterrence	Aronson and Carlsmith 1963	Children mildly threatened against playing with a toy later liked it less than severely threatened children
Hypocrisy induction	Jeff Stone, Aronson and colleagues 1994	Students who urged condom use, then recalled their own lapses, bought more condoms

What makes inconsistency hurt? **Elliot Aronson** (born 1932) proposed self-consistency: behaviour that threatens a decent, competent self-concept. The new look of **Joel Cooper** and Fazio (1984) required personal responsibility for foreseeable aversive consequences. **Daryl J. Bem**'s **self-perception theory** (1967, 1972) denied any tension: when attitudes are vague we infer them from our own behaviour, as an observer would ("I praised it for a dollar, so I must like it"). Yet when Zanna and Cooper (1974) let people blame their tension on a placebo pill, attitude change vanished, so arousal matters; self-perception fits vague attitudes, dissonance clear ones. Impression management theory (**James T. Tedeschi** and colleagues, 1971) held that people merely present consistent attitudes to others. Culture matters too: **Shinobu Kitayama** and colleagues (2004) found that Japanese participants showed post-decision dissonance when others' views were made salient, Americans when choosing for themselves.

> **! EXAM TRAP**
>
> Dissonance and self-perception both predict that attitudes follow behaviour. Choose Festinger when the stem mentions tension or a clear attitude contradicted; choose Bem when a vague attitude is calmly inferred. Effort justification is one dissonance paradigm, not a synonym for dissonance. The theory dates from 1957; the $1 versus $20 study was published in 1959, though some lists give 1957.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Katz's function that protects self-esteem from threatening truths is the _____ function.
2. Attitudes formed through _____ experience predict behaviour best.
3. Planned behaviour added perceived behavioural _____ to reasoned action.
4. Careful attention to argument quality marks the _____ route.
5. A low-credibility message gaining impact over time shows the _____ effect.
6. Brehm's motivation to restore a threatened freedom is _____.
7. Festinger and Carlsmith's students paid _____ came to like the dull task more.
8. Bem's tension-free alternative to dissonance is _____ theory.

Answers: (1) ego-defensive, (2) direct, (3) control, (4) central, (5) sleeper, (6) reactance, (7) one dollar, (8) self-perception.

3.5 Stereotypes, Prejudice and Intergroup Relations LOWER YIELD · 13 QS

In 1954 two groups of eleven-year-old boys at an Oklahoma camp came to loathe each other within days, though nothing divided them but a name and a tournament. How do ordinary people come to think less of whole groups, and what undoes it?

Stereotype, prejudice and discrimination

Bias has three components, matching the ABC of attitudes (Section 3.4). A **stereotype** is cognitive: a belief about the traits typical of a group's members, what the journalist **Walter Lippmann** called "pictures in our heads" (*Public Opinion*, 1922). **Daniel Katz** and **Kenneth**

Braly (1933) measured them with an adjective checklist: Princeton students agreed strongly on the traits of ten ethnic groups, including groups they had seldom or never met. **Prejudice** is affective: a negative attitude toward people because they belong to a group, in **Gordon Willard Allport**'s words (*The Nature of Prejudice*, 1954) "an antipathy based upon a faulty and inflexible generalization". **Discrimination** is behavioural: unfair negative action toward a group's members.

William Graham Sumner (*Folkways*, 1906) named the **in-group** ("we") and the **out-group** ("they") and defined **ethnocentrism**, judging other groups by the standards of one's own, placed at the centre and above the rest. Allport ranked prejudice in action in five steps of rising harm (**Allport's scale**): **antilocution** (hostile talk), **avoidance**, **discrimination**, **physical attack** and **extermination**. Prejudice is named by its target: **racism**, **sexism** and **ageism** (against the old; coined by **Robert Neil Butler** in 1969). In India, **casteism** is prejudice and discrimination based on caste, a hereditary, endogamous ranking (untouchability is abolished by Article 17 of the Constitution), and **communalism** joins loyalty to one's own religious community with hostility toward others.

> **! EXAM TRAP**
>
> Stereotype = belief, prejudice = feeling, discrimination = action. "Pictures in our heads" is Lippmann, not Allport. Allport's scale starts with antilocution, not avoidance. Ethnocentrism is Sumner (1906).

Where prejudice comes from ○

Early explanations looked inside the prejudiced person. **Theodor Adorno**, **Else Frenkel-Brunswik**, **Daniel Levinson** and **Nevitt Sanford** (*The Authoritarian Personality*, 1950) described an **authoritarian personality**: rigid, conventional, submissive to authority and hostile to minorities, shaped by harsh, punitive parenting, and measured by the **F-scale** (F for fascism). Its modern heirs are **Bob Altemeyer**'s **right-wing authoritarianism** (1981: authoritarian submission, authoritarian aggression, conventionalism) and **Jim Sidanius** and **Felicia Pratto**'s **social dominance orientation**, a preference for a hierarchy in which some groups rank above others.

Scapegoat theory holds that frustration is displaced onto a weak, visible minority (Section 3.9): **Carl Hovland** and **Robert Sears** (1940) found that lynchings in the American South rose as cotton prices fell. In the **social learning view**, children absorb prejudice from parents, peers and media and are rewarded with approval for voicing it.

At the group level, **realistic conflict theory** (named by **Donald Thomas Campbell** in 1965, argued by **Muzafer Sherif** in 1966) holds that competition for scarce resources breeds hostility. Often what hurts is **relative deprivation**, having less than one feels entitled to compared with others: in *The American Soldier* (1949), **Samuel Stouffer** and colleagues found fast-promoted Air Corps men less satisfied with promotion than slow-promoted military police. **Walter Garrison Runciman** (1966) separated **egoistic** (personal) from **fraternal** (group) deprivation; the fraternal kind fuels prejudice and **collective action** such as protest. **Ted Robert Gurr** (*Why Men Rebel*, 1970) traced rebellion to the gap between what people expect and what they get.

KEY STUDY: THE ROBBERS CAVE EXPERIMENT (1954)

Question. Does competition create intergroup hostility, and what removes it? **Method.** Sherif and colleagues took 22 unacquainted boys aged about 11 to 12, in two matched groups, to a summer camp at Robbers Cave State Park, Oklahoma. Phase 1, **in-group formation**: each group lived apart and named itself (the Eagles, the Rattlers). Phase 2, **competition and friction**: a tournament with prizes for the winners. Phase 3, **reduction of conflict**. **Result.** Competition brought name-calling, a burned flag and cabin raids. Mere contact (shared meals, films) failed and even made things worse: meals became food fights. Hostility fell only with **superordinate goals**, which both groups wanted but neither could reach alone: restoring the water supply, pooling money to rent a film, pulling a stalled truck. **Criticism.** The boys were all white, Protestant and middle class, and staff stage-managed events. The report appeared in 1961.

! EXAM TRAP

Superordinate goals requiring cooperation reduced the hostility. Mere contact failed, more competition is what created the hostility, and no dominant leader was assigned. Robbers Cave is not Sherif's autokinetic study of norms (Section 3.7).

Social identity theory ●●●

Sherif's boys traded insults as soon as they learned of each other, before any contest. Is competition even needed? **Henri Tajfel** (1919 to 1982), a Polish Jew who lost most of his family in the Holocaust, looked for the minimal conditions for discrimination. In the **minimal group paradigm** (Tajfel, Billig, Bundy and Flament, 1971), Bristol schoolboys were told they had been grouped by a preference for Paul Klee or Wassily Kandinsky (in truth, at random), then divided points worth money between two anonymous others. Groups with no history, contact or competition still produced in-group favouritism, and many boys chose to **maximise the difference** between groups even at a cost to their own group's absolute gain.

Tajfel and **John Turner** (1947 to 2011) explained this with **social identity theory** (1979, 1986): a large part of our self-concept and self-esteem comes from the groups we belong to. Besides a **personal identity** (one's own traits, achievements and preferences) we have a **social identity** derived from group memberships (Indian, Kannadiga, psychology student). Three processes link membership to bias (Figure 3.7): **social categorisation** into in-groups and out-groups, **social identification** with the in-group, and **social comparison** with out-groups. Since self-esteem rides on the comparison, we seek **positive distinctiveness**: our group must come out ahead. Prejudice thus arises from categorisation itself, with no clash of interests required.

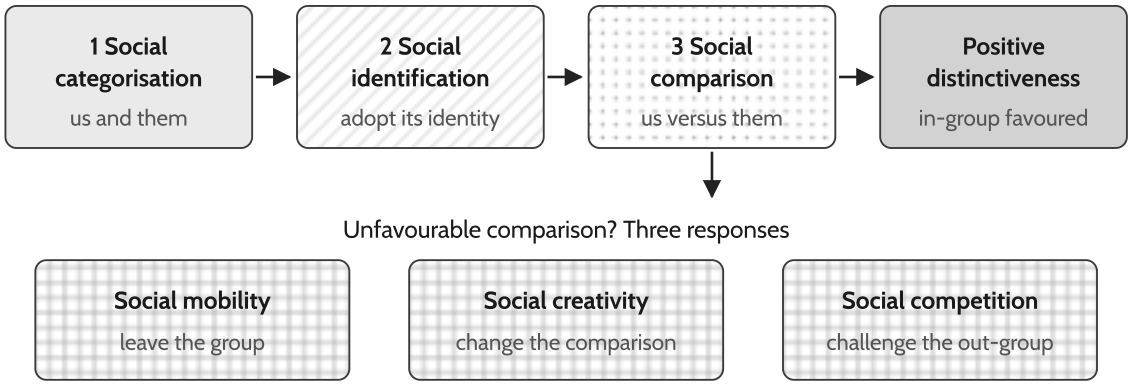


Figure 3.7 Social identity theory (Tajfel and Turner 1979): three processes, and three responses when the in-group compares badly.

When the comparison is unflattering, members may leave for a higher group (**social mobility**, if boundaries are permeable), change the comparison by picking a new dimension or a lower group (**social creativity**: "poorer but kinder"), or challenge the higher group directly (**social competition**). The **self-categorisation theory** of Turner and colleagues (1987) explains when we see ourselves as individuals and when as interchangeable group members: the category that best fits the situation becomes salient. **Marilynn Brewer**'s **optimal distinctiveness theory** (1991) holds that we balance a need to belong against a need to be different, so we identify most with groups that are inclusive yet distinctive.

> ! EXAM TRAP
>
> In social identity theory, identity and prejudice come from **group membership and categorisation into in-groups and out-groups**, driven by the need for a positive social identity. Wrong options: competition for scarce resources (realistic conflict); birthplace, prejudice experienced, economic status, personal achievements, preferences or attributes (personal identity); internal or external attributions (Section 3.3); cognitive dissonance (Section 3.4); social facilitation (Section 3.10); genetic kin selection (Section 3.8); operant reinforcement (Chapter 1).

How stereotypes work in the mind

Categorisation also distorts perception. In the **out-group homogeneity effect** "they are all alike": **George Quattrone** and **Edward Ellsworth Jones** (1980) found Princeton and Rutgers students generalised more from one rival student's choice to his whole university than from an in-group member's; the reverse is **in-group differentiation**. An **illusory correlation** is a perceived group-trait link the data do not support: **David Hamilton** and **Robert Gifford** (1976) gave a large and a small group the same proportion of undesirable acts, yet people paired the two distinctive things, minority and rare bad acts, and judged the small group worse. **Thomas Pettigrew**'s **ultimate attribution error** (1979) extends the fundamental attribution error (Section 3.3) to groups: an out-group's failings are put down to its nature, its successes to luck.

Stereotypes persist because they save effort: **C. Neil Macrae**, **Alan Milne** and **Galen Bodenhausen** (1994) found that people given stereotype labels recalled more traits and did

better on a simultaneous second task, so stereotypes are **energy-saving devices**. They also resist disconfirmation. **Renee Weber** and **Jennifer Crocker** (1983) compared **bookkeeping** (each exception nudges the stereotype), **conversion** (one dramatic case overturns it) and **subtyping** (exceptions are fenced off as atypical, leaving it intact); disconfirmations spread over many typical members changed it most.

The **stereotype content model** of **Susan Fiske**, **Amy Cuddy**, **Peter Glick** and **Jun Xu** (2002) places groups on **warmth** and **competence**, each combination evoking its own emotion (Figure 3.8). Stereotypes can also impair performance through **stereotype threat**, the fear of confirming a negative stereotype about one's group. **Claude Steele** and **Joshua Aronson** (1995) found that Black Stanford students scored below equally able White students on hard verbal items only when these were presented as a test of ability; **Steven Spencer**, Steele and **Diane Quinn** (1999) found the same for women on a hard mathematics test.

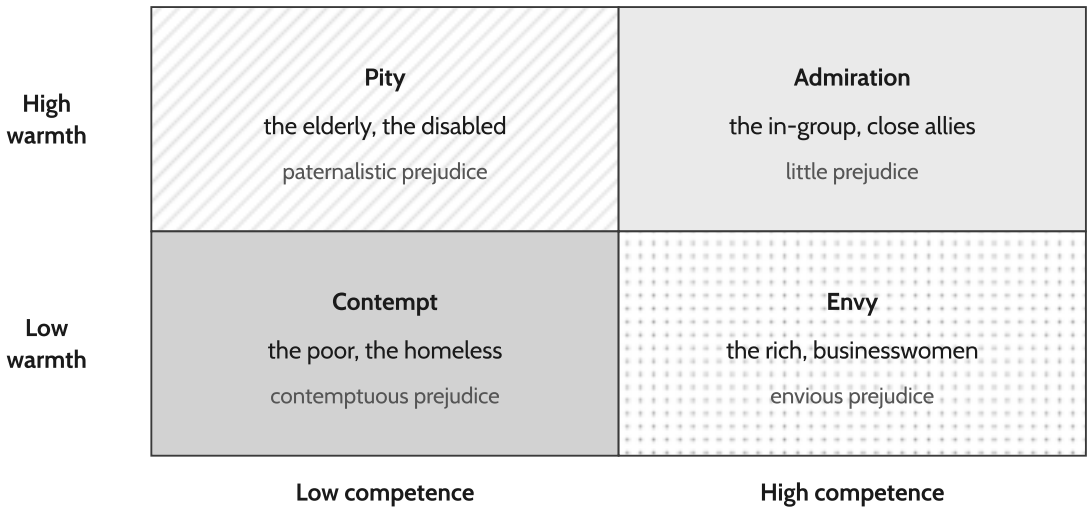


Figure 3.8 The stereotype content model (Fiske and colleagues 2002): warmth and competence decide the emotion a group evokes, so much prejudice is mixed.

Automatic and controlled processes ●●

Nearly everyone knows the cultural stereotypes; is everyone then prejudiced? **Patricia Devine**'s (1989) **dissociation model** says no. High- and low-prejudiced White Americans knew the stereotype of Black Americans equally well, and after stereotype-related words were flashed too fast to see, both judged an ambiguous man as more hostile: activation by **automatic processing** (fast, unintended, outside awareness) happens in everyone. Given time to list their thoughts, however, low-prejudiced people wrote more egalitarian ones. Devine concluded that low-prejudiced people use **controlled processing** (slow, deliberate, effortful) to inhibit the activated stereotype and replace it with egalitarian beliefs, much like breaking a bad habit.

Automatic associations can be measured without asking. The **Implicit Association Test** (IAT) of **Anthony Greenwald**, **Debbie McGhee** and **Jordan Schwartz** (1998) times how fast people sort faces and words with two response keys. Being faster when one group shares a key with "good" words than with "bad" words reveals an implicit, unconscious association (**implicit prejudice** or stereotyping) that people may be unwilling or unable to report. In the

bona fide pipeline of **Russell Fazio** and colleagues (1995), a briefly shown Black or White face primes a good-or-bad judgement of adjectives; faster responses to negative words after Black faces reveal an automatic negative attitude.

> **! EXAM TRAP**
>
> Low-prejudiced people show **controlled processing that refutes the stereotype.** Automatic processing with involuntary stereotyping is the first stage, shared by all and left unchecked by high-prejudiced people; "automatic, voluntary" is self-contradictory; controlled processing prevents involuntary stereotyping rather than causing it. The IAT measures implicit associations by reaction time: not verbal or hidden mathematical ability, and not dream content (psychoanalysis).

Modern and subtle prejudice

Open bigotry has declined in surveys, but prejudice survives in forms that let people keep a fair self-image (Table 3.13).

Table 3.13 Modern and subtle forms of prejudice.

Form	Proposed by	What it is
Modern racism	**John McConahay** (1986)	Denying that discrimination persists; resenting minorities' "special favours"
Symbolic racism	**David Sears**	Hostility voiced as defence of values such as self-reliance
Aversive racism	**Samuel Gaertner** and **John Dovidio** (1986)	Sincere egalitarians with hidden negative feelings discriminate when a non-racial excuse exists
Tokenism	(general term)	A trivial favour or a few minority hires used to deflect charges of bias
Microaggressions	Term: **Chester Pierce** (1970); **Derald Wing Sue** and colleagues (2007)	Brief everyday slights, often unintended
Ambivalent sexism	Glick and Fiske (1996)	**Hostile** sexism (resentment) plus **benevolent** sexism (protective, but casting women as weak)

Gender roles and the workplace ●●

Descriptive gender stereotypes say what women and men are like; **prescriptive** ones say what they should be like, so a forceful woman leader meets backlash. **Alice Eagly**'s **social role theory** (1987; Eagly and **Wendy Wood**, 1999) traces sex differences and gender stereotypes to the division of labour: because men held provider and leader roles and women homemaker and caregiver roles, people infer that each sex has the traits its roles demand, and each sex learns the skills those roles reward. **David Buss**'s evolutionary rival (37 cultures, 1989) found men valuing a mate's youth and looks and women valuing earning prospects, which he traced to reproductive pressures in evolution; Eagly and Wood found these differences smaller in more gender-equal nations.

> **! EXAM TRAP**
>
> Social role theory of gender = **Alice Eagly**. Buss is the evolutionary rival; Clea Mehta and JoNell Strough (2009) reviewed sex segregation in friendships across the life span. Anderson is only a distractor name: no Anderson is linked to social role theory (Craig Anderson, with Brad Bushman, built the General Aggression Model, Section 3.9).

At work, stereotypes become named barriers (Table 3.14). The best known, the **glass ceiling**, is an invisible barrier of attitudes and organisational bias that stops qualified women and minorities from rising to top positions; a 1986 *Wall Street Journal* article popularised the term. The halo effect (one good trait colouring every judgement) and the Pygmalion effect (expectations fulfilling themselves) are impression effects (Section 3.2).

Table 3.14 Named barriers and biases at work.

Term	Source	Meaning
Glass cliff	**Michelle Ryan** and **Alexander Haslam** (2005)	Women more often appointed to lead in a crisis, to precarious posts
Sticky floor	**Catherine White Berheide** (1992)	Women kept in the lowest-paid, least mobile jobs at the bottom
Maternal wall	**Joan Williams**	Bias against mothers, presumed less committed
Glass escalator	**Christine Williams** (1992)	Men in female-dominated jobs such as nursing rise fast
Matthew effect	**Robert King Merton** (1968)	Accumulated advantage: the already recognised get still more credit

> **! EXAM TRAP**
>
> Glass ceiling = blocked at the **top**; sticky floor = stuck at the **bottom**; glass cliff = promoted onto a precarious post in a crisis; maternal wall = blocked by motherhood. The Matthew effect is accumulated advantage, not a barrier.

Reducing prejudice ○

Allport's **contact hypothesis** (1954) holds that contact reduces prejudice only under four conditions: **equal status**, **common goals**, **intergroup cooperation** and **support of authorities, law or custom**. That is why mere contact during conflict failed at Robbers Cave. Pettigrew and **Linda Tropp**'s (2006) meta-analysis of over 500 studies (515, with more than 250,000 participants) confirmed that contact reduces prejudice, most strongly when the conditions hold. In the **extended contact effect** (**Stephen Wright** and colleagues, 1997), merely knowing that an in-group member has a close out-group friend lowers prejudice. **Segregation**, **minimal contact** and **competition** keep prejudice alive.

In **Elliot Aronson**'s **jigsaw classroom** (first used in 1971 in newly desegregated schools in Austin, Texas; book *The Jigsaw Classroom*, 1978), each child in a mixed group learns one piece of the lesson and must teach it, so every child needs every other; liking and self-esteem rose. Other strategies change categories or retrain the mind (Table 3.15).

Table 3.15 Further ways to reduce prejudice.

Strategy	Source	How it works
Recategorisation: common in-group identity model	Gaertner and Dovidio (2000)	"Us" and "them" become one inclusive "we"
Decategorisation	Brewer and **Norman Miller** (1984)	Meet out-group members as individuals
Mutual intergroup differentiation	**Miles Hewstone** and **Rupert Brown** (1986)	Keep both identities salient while cooperating, so liking generalises
Learning not to hate	Social learning view	Parents and teachers model tolerance
Collective guilt	Powell, **Nyla Branscombe** and Schmitt (2005)	Seeing inequality as one's own group's advantage lowers racism
Saying no to stereotypes	**Kerry Kawakami** and colleagues (2000)	Repeatedly negating stereotypic pairings weakens automatic activation
Social influence	**Charles Stangor** and colleagues (2001)	Learning that peers are less prejudiced reduces one's own stereotyping

! EXAM TRAP

Of segregation, minimal contact, competition and intergroup contact with cooperation, only the last reduces prejudice, and only when Allport's conditions hold.

INDIAN CONTEXT

In rural Uttar Pradesh, **Karla Hoff** and **Priyanka Pandey** (2006) found that low-caste boys solved mazes as well as high-caste boys until each boy's caste was announced publicly, an effect resembling stereotype threat. Cooperative contact helps: in **Matt Lowe**'s (2021) field experiment in Uttar Pradesh, young men randomly placed on caste-mixed cricket teams formed more cross-caste friendships, whereas merely playing against other castes did not. **Ashutosh Varshney** (2002) found that cities with strong Hindu-Muslim civic associations stayed peaceful when communal riots struck elsewhere.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. A belief about the traits of a group's members is a _____.
2. Allport's five-step scale begins with _____.
3. At Robbers Cave, hostility fell only with _____ goals.
4. Boys grouped by a supposed preference for Klee or Kandinsky favoured their own group in Tajfel's _____ group paradigm.
5. In social identity theory, the need for positive _____ drives favourable comparison of the in-group.
6. Devine found that low-prejudiced people use _____ processing to inhibit automatically activated stereotypes.
7. Women appointed to lead in a crisis stand on a glass _____; women kept in the lowest jobs face a sticky _____.
8. In Aronson's _____ classroom, each child must teach one piece of the lesson.

Answers: (1) stereotype, (2) antilocution, (3) superordinate, (4) minimal, (5) distinctiveness, (6) controlled, (7) cliff, floor, (8) jigsaw.

3.6 Affiliation, Attraction and Close Relationships

LOWER YIELD · 7 QS

Think back to your first weeks in a hostel: most of the friends you made were probably the people whose rooms were nearest yours. Why do we seek company at all, why these people rather than others, and what keeps a relationship alive or ends it? This section moves from the need to affiliate, through attraction and the theories of close relationships, to love, break-up and loneliness.

The need to affiliate ●●

Roy Baumeister and **Mark Leary** argued in 1995 that the **need to belong**, a drive to form and keep a few lasting, positive relationships, is a fundamental human motive. People differ in its strength; the stable trait is the **need for affiliation** (nAff), listed by **Henry Murray** and measured by **David McClelland** (Section 1.10). The older word is **gregariousness**, the urge to be with others and live in groups. **William McDougall** listed a gregarious instinct in 1908, but gregariousness is now classed as a **social motive**: a secondary, learned or psychosocial motive, like affiliation, achievement and power, that arises and is satisfied in interaction with others. Biological (primary) motives such as hunger and thirst serve bodily survival instead.

! EXAM TRAP

Gregariousness has a **social** motive. "Biological" tempts because McDougall called it an instinct, but instinct theory collapsed (Section 1.10). "Psychological" is not a standard separate class, and personal motives are an individual's own habits, interests and aspirations.

KEY STUDY: SCHACHTER'S ANXIETY AND AFFILIATION EXPERIMENT (1959)

Stanley Schachter (1922 to 1997) had women students met by a fictitious "Dr Gregor Zilstein", who warned of electric shocks that would be painful (high anxiety) or feel like a tickle (low anxiety). Offered a wait alone or with others, about two-thirds of the high-anxiety women chose company, against about one-third of the low-anxiety women, and they wanted only others facing the same shocks: "misery doesn't just love any kind of company, it loves only miserable company". No shocks were given.

Why do frightened people want each other? All three usual answers hold: we affiliate **to obtain information** about what is happening, **to reduce anxiety**, and **to compare ourselves with similar others**. **Craig Hill** (1987) found four affiliation motives: **positive stimulation**, **emotional support**, **attention** and **social comparison**.

The need shows most sharply when thwarted. **Ostracism**, being ignored and excluded, was studied by **Kipling Williams** with **Cyberball** (2000), an online ball-tossing game in which the other "players" soon stop throwing to the participant; minutes of exclusion by strangers lower mood and threaten belonging, self-esteem, control and meaningful existence. Feelings shape liking too: in the **reinforcement-affect model** of **Donn Byrne** and **Gerald Clore** (1970) we like people who make us feel good and, by classical conditioning, even strangers merely present when something pleasant happens.

Social comparison ○

Leon Festinger (1919 to 1989) proposed **social comparison theory** in 1954: people are driven to evaluate their opinions and abilities, and when no objective standard exists they compare themselves with others, preferably similar others. A mark of 85 percent means little until you know how your classmates scored. **Upward social comparison**, with someone better off, can inspire but often brings envy; **downward social comparison**, with someone worse off, was described by **Thomas Wills** (1981) as a route to self-enhancement when self-esteem is threatened. **Abraham Tesser**'s **self-evaluation maintenance model** (1988) adds closeness and relevance: if your cousin tops the medical entrance and medicine matters to you, the **comparison process** stings and you may distance yourself; if she wins a dance contest you do not care about, the **reflection process** lets you bask in her glory.

Social media supply an endless upward stream of other people's best moments. **Andrew Przybylski** and colleagues (2013) defined the **fear of missing out** (FOMO) as a pervasive apprehension that others might be having rewarding experiences from which one is absent; it goes with heavier social media use and lower mood. FOMO is a social comparison phenomenon.

! EXAM TRAP

FOMO is linked to **social comparison**, not classical conditioning (learning by pairing stimuli), operant extinction (a response fading when reinforcement stops) or "cognitive dissonance only" (discomfort from inconsistent cognitions, Section 3.4).

Nearness and familiarity ○

The plainest force of attraction is nearness. In 1950 Festinger, Schachter and **Kurt Back** studied Westgate West, married-student housing at the Massachusetts Institute of Technology where flats were assigned at random. Of next-door neighbours, 41 percent were named as close friends, against 22 percent of those two doors away and 10 percent of those at the far end of the corridor. **Functional distance**, how often architecture brings paths together, mattered too: residents beside the stairs and mailboxes made more friends upstairs. **Mady Segal** (1974) found that police trainees seated alphabetically befriended those with neighbouring surnames. This is the **propinquity effect**: the more we see and interact with a person, the more likely he or she is to become our friend or partner.

The mechanism is largely the **mere exposure effect** of **Robert Zajonc** (1968): repeated exposure to a stimulus, without interaction or reward, increases liking for it, whether nonsense words, Chinese-like characters or faces. **Richard Moreland** and **Scott Beach** (1992) had four women attend a class 0, 5, 10 or 15 times without speaking; students liked them more the more often they came. **Theodore Mita** and colleagues (1977) found that women preferred mirror images of their own faces, the version they see daily, while friends preferred the true image. A stimulus disliked at first, however, can become more disliked.

> **! EXAM TRAP**
>
> "The more we see and interact with someone, the more likely they become our friend or partner" is the **propinquity effect**. The **exposure effect** is narrower (mere exposure to any stimulus raises liking) and is its mechanism; **matching** and **reciprocity** are taught below. For mere exposure, "performing better when observed" is social facilitation, "attitude change after inconsistent behaviour" is cognitive dissonance and "riskier group decisions" is the risky shift (Sections 3.4 and 3.10).

Looks, similarity and reciprocity

Appearance acts first. **Karen Dion**, **Ellen Berscheid** and **Elaine Hatfield** (then Walster) showed in 1972 that attractive people are judged kinder, more sociable and more successful: the **what is beautiful is good** stereotype. At a 1966 computer dance at the University of Minnesota, Walster and colleagues paired students at random; the only predictor of wanting another date was the partner's attractiveness. **Judith Langlois** and **Lori Roggman** (1990) found computer-averaged faces more attractive than most faces composing them (averageness); symmetry also attracts. Yet most people must settle: the **matching hypothesis** (Walster and colleagues 1966; tested by Berscheid and colleagues 1971) holds that we pair with partners of roughly our own attractiveness to avoid rejection, and **Bernard Murstein** (1972) found real couples closer in attractiveness than chance.

Similarity matters more over time. **Theodore Newcomb** measured transfer students' attitudes by post, housed them together, and found that those initially similar became friends (*The Acquaintance Process*, 1961). Byrne's "bogus stranger" experiments gave his **law of attraction** (1971): attraction rises with the *proportion* of similar attitudes. **Milton Rosenbaum**'s **repulsion hypothesis** (1986), that dissimilarity repels while similarity adds nothing, overstates the case, though dissimilarity does weigh slightly more. **Robert Winch**'s (1958) **complementarity of needs**, opposites attracting (dominant with submissive), has found little support.

We also like those who like us (**reciprocity of liking**). In the **gain-loss effect** of **Elliot Aronson** and **Darwyn Linder** (1965), participants who overheard a confederate's evaluations liked her most when they turned from negative to positive, more than when consistently positive, and least when they turned from positive to negative.

Exchange, equity and investment

Why do people stay in some relationships and leave others? **Social exchange theory**, from **John Thibaut** and **Harold Kelley** (1959) and **George Homans** (1961), treats a relationship as a trade: rewards minus costs. The outcome is judged against the **comparison level** (CL), what one feels one deserves from past relationships, which sets satisfaction, and the **comparison level for alternatives** (CLalt), the best outcome available elsewhere, which decides staying. Meera may be dissatisfied with Arjun (below her CL) yet stay because every alternative looks worse (above her CLalt). Table 3.16 sets this beside three relatives.

Table 3.16 Four theories of close relationships.

Theory	Proposed by	Core idea
Social exchange	Thibaut and Kelley 1959; Homans 1961	Rewards minus costs, judged against CL and CLalt
Equity theory	Walster (Hatfield), Berscheid and Walster 1978	Each partner's outcomes in proportion to inputs
Investment model	Caryl Rusbult 1980	Satisfaction, poor alternatives and large investments build commitment
Communal versus exchange	Clark and Mills 1979	Help given to meet needs versus benefits repaid in kind

In **equity theory** (Walster, Berscheid and **G. William Walster** 1978) both the over-benefited, who feel guilty, and the under-benefited, who feel angry, are distressed. **Caryl Rusbult**'s **investment model** explains staying when unhappy: years together, children or a shared home cannot be recovered, so commitment stays high, as in women who return to abusive partners. **Margaret Clark** and **Judson Mills** showed that tit for tat suits **exchange relationships** with strangers; in **communal relationships** with family and close friends, prompt repayment can even reduce liking.

Self-disclosure ○

Relationships deepen through **self-disclosure**, revealing personal information about oneself to another. **Sidney Jourard** (1926 to 1974), in *The Transparent Self* (1964), saw being fully known by someone as a mark of healthy personality; his **Jourard Self-Disclosure Questionnaire** asked how much people had told mother, father, friends and spouse about areas such as tastes, work, money and body. **Irwin Altman** and **Dalmas Taylor**'s **social penetration theory** (1973) pictures the self as an onion: relationships grow in **breadth** (range of topics) and **depth** (intimacy), moving from outer layers to the core. Disclosure invites matching disclosure (**disclosure reciprocity**), and we like those who confide in us.

T. L. Morton (1978), followed by the textbook of **Shelley Taylor**, **Letitia Anne Peplau** and **David Sears**, separated **descriptive self-disclosure**, revealing facts and information about oneself (where you grew up), from **evaluative self-disclosure**, revealing one's personal feel-

ings, opinions and judgements (that you feel lonely). Evaluative disclosure is more intimate and comes later.

> **! EXAM TRAP**
>
> IHBAS 2020 keyed "descriptive self-disclosure" for communicating one's feelings. By the standard definitions that is **evaluative**; descriptive disclosure is about facts. Learn the standard pair, and know that this paper's key reversed it. "Prescriptive" and "non-evaluative" self-disclosure are not standard types.

Love

Zick Rubin (1970) built separate **Liking** (respect, affection) and **Loving Scales** (attachment, caring, intimacy); high scorers on loving gazed at each other longer. Berscheid and Hatfield separated **passionate love**, an intense, aroused longing for union with the other (measured by Hatfield and **Susan Sprecher**'s Passionate Love Scale, 1986), from **companionate love**, the calmer affection we feel for those with whom our lives are deeply entwined. **Robert Sternberg**'s **triangular theory of love** (1986) builds love from **intimacy** (closeness), **passion** (physical attraction and arousal) and **commitment** (the decision to maintain the love); their combinations give eight kinds (Table 3.17).

Table 3.17 Sternberg's eight kinds of love.

Kind	Intimacy	Passion	Commitment	For example
Non-love	no	no	no	casual acquaintances
Liking	yes	no	no	true friendship
Infatuation	no	yes	no	love at first sight
Empty love	no	no	yes	a stale marriage
Romantic love	yes	yes	no	a holiday romance
Companionate love	yes	no	yes	a long, settled marriage
Fatuous love	no	yes	yes	a whirlwind wedding
Consummate love	yes	yes	yes	complete love

The sociologist **John Alan Lee** (1973) described six love styles: eros (passionate, beauty-led), ludus (playful, uncommitted), storge (friendship-based), pragma (practical, a checklist partner), mania (possessive, jealous) and agape (selfless). Berscheid and Walster's **two-factor theory of love** (1974) applies Schachter's two-factor theory of emotion: passionate love is physiological arousal labelled "love" in the presence of a suitable person, so arousal from elsewhere can be misattributed to attraction, as in **Donald Dutton** and **Arthur Aron**'s 1974 bridge study (Section 1.11).

Love also carries the stamp of early attachment (infants in Chapter 6). **Cindy Hazan** and **Phillip Shaver** (1987) classified adults as **secure**, **avoidant** (uneasy with closeness) or **anxious-ambivalent** (craving closeness, fearing abandonment), just over half secure, much as in infants. **Kim Bartholomew** and **Leonard Horowitz** (1991) crossed the model of self with the model of others to get four **adult attachment styles** (Figure 3.9).

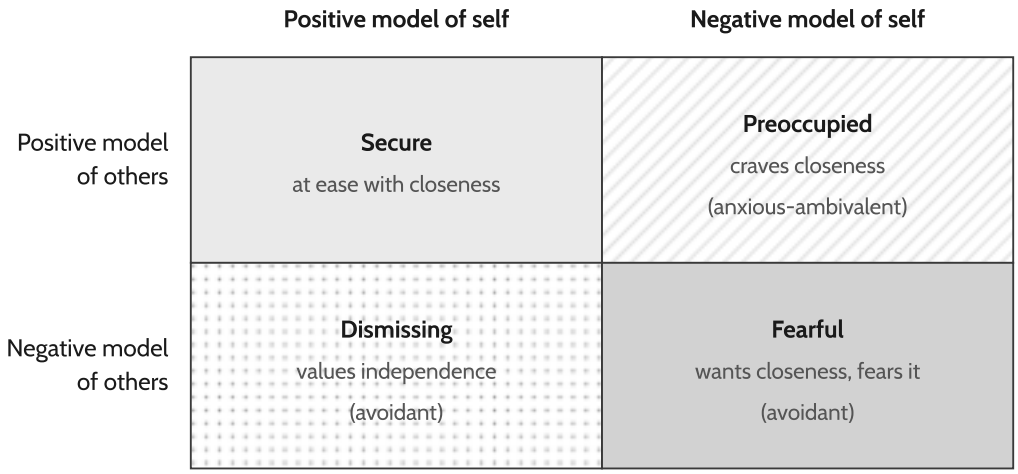


Figure 3.9 Bartholomew and Horowitz's four styles; brackets give Hazan and Shaver's nearest category.

Anxious attachment also feeds **jealousy**, the fear, anger and hurt felt when a rival seems to threaten a valued relationship. **David Buss** and colleagues (1992) reported men more upset by sexual and women by emotional infidelity, an evolutionary reading that critics say depends on question wording.

> **INDIAN CONTEXT**
>
> In Jaipur, **Usha Gupta** and **Pushpa Singh** (1982) found that couples who married for love reported less love after about five years, while those in arranged marriages reported more love once past the newlywed years.

When relationships fail

Rusbult classified responses to dissatisfaction as active or passive and constructive or destructive: **exit** (active, destructive: leaving), **voice** (active, constructive: discussing problems), **loyalty** (passive, constructive: waiting faithfully) and **neglect** (passive, destructive: letting things decay). **Steve Duck** (1982) traced dissolution through four phases: **intrapsychic** (brooding privately on the partner's faults), **dyadic** (confronting the partner), **social** (telling friends and family) and **grave-dressing** (building an acceptable story of the break-up). **John Gottman**'s "four horsemen" are taught with couple therapy in Chapter 7.

Loneliness ○

Peplau and **Daniel Perlman** (1982) defined **loneliness** as the unpleasant experience that occurs when a person's network of social relationships is deficient in quantity or quality. It is subjective: one can be lonely in a crowd. **Robert Weiss** (1973) separated two kinds. **Emotional isolation** (emotional loneliness) is the lack of a close, intimate attachment to one specific person, felt by a widow though friends surround her. **Social isolation** (social loneliness) is the lack of a wider network of friends and associates, felt by a newcomer to a city even in a happy marriage.

> **! EXAM TRAP**
>
> Emotional isolation means no deep attachment to **one specific person**; social isolation means too few **friends or associates**. Incompetence, inadequacy and low self-esteem are correlates of loneliness, not Weiss's types.

The **UCLA Loneliness Scale** of **Daniel Russell**, Peplau and colleagues (1978, revised 1980) has 20 statements such as "I lack companionship". Loneliness goes with depression, poorer health and shorter life (**John Cacioppo** and colleagues), and lonely people often benefit from **social skills training** (starting conversations, disclosing appropriately) and from challenging the negative expectations that keep them withdrawn.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Schachter's anxious participants preferred to wait with others facing the _____ situation.
2. Comparing yourself with someone worse off is _____ social comparison.
3. FOMO is best explained as a _____ phenomenon.
4. Friendship growing from frequent contact with neighbours is the _____ effect.
5. In exchange theory, the standard that decides whether a person stays is the _____.
6. Revealing feelings and opinions is _____ self-disclosure.
7. Passion plus commitment without intimacy is _____ love.
8. Lacking a close attachment to one specific person is Weiss's _____.

Answers: (1) same, (2) downward, (3) social comparison, (4) propinquity, (5) comparison level for alternatives, (6) evaluative, (7) fatuous, (8) emotional isolation.

3.7 Social Influence: Conformity, Compliance, Obedience and Power

LOWER YIELD · 18 QS

Other people change what you do all day, often without a word. You join the queue at a chai stall, lend a friend your notes, fall silent when the invigilator orders it. This section follows influence from the unspoken norm through direct requests to outright commands, then turns to the roles and the bases of power behind them.

Kinds of social influence ●●

Social influence is any effort by one or more people to change the behaviour, attitudes or feelings of others (persuasion, changing attitudes by messages, is in Section 3.4). Three forms differ in how directly pressure is applied (Table 3.18). **Conformity** is changing your behaviour or beliefs to match a group or its norms, although nobody has asked you to. **Compliance** is changing in response to a direct request, which you may accept or refuse. **Obedience** is changing in response to a command from an authority. Textbooks add **symbolic social influence**: absent people still steer us through our mental image of them.

Table 3.18 The forms of social influence.

Form	Source of pressure	Example
Conformity	Group norms; no request	Standing for the national anthem
Compliance	A direct request	A friend asks to borrow your notes
Obedience	A command from an authority	A doctor orders a nurse to give a drug
Symbolic influence	Mental image of absent others	Studying because you picture a parent's disappointment

Herbert Kelman (1958) graded how deep the change goes. In compliance (his sense) a person goes along outwardly to gain reward or avoid punishment, and only while watched. In **identification** he adopts the behaviour to be like an admired person or group. In **internalisation** he adopts it because it fits his own values, and it lasts. The same contrast appears as **public compliance** (going along outwardly while privately disagreeing) versus **private acceptance** (truly coming to believe what the group believes).

> **! EXAM TRAP**
>
> "Changing your behaviour to match others in a group" is **conformity**. *Norming* is a stage of **Bruce Tuckman**'s group development (forming, storming, norming, performing; Section 3.10); *standardisation* means uniform test procedures (Chapter 2). Conformity is change under group pressure with nobody asking; compliance answers a direct request. Conformity is not "always involuntary", and neither term means obedience.

Social norms ○

Social norms are rules indicating how people are expected to behave in a situation. **Explicit norms** are written or stated ("No Parking"); **implicit norms** are unstated yet known to all (do not talk during a concert). **Robert Cialdini**, **Raymond Reno** and **Carl Kallgren** (1990) distinguished **descriptive norms**, what most people actually do, from **injunctive norms**, what is approved or disapproved, what ought to be done. In their littering study, people found a handbill on their car windscreen. More dropped it where the ground was already littered; fewest did where the ground was clean and a confederate conspicuously dropped one, because the lone litterbug highlighted the descriptive norm that nobody litters here. Making the injunctive norm salient (a passer-by picking up litter, in a follow-up) cut littering even in a littered setting. Their **focus theory of normative conduct** holds that a norm guides behaviour only when it is focal, salient at the moment of acting.

Prescriptive norms tell people what they ought to do (greet elders); **proscriptive norms** tell them what not to do (do not cheat). A **perceived norm** is a person's belief about what others do or approve, which may be wrong: students who overestimate their classmates' drinking drink more. Societies whose norms break down suffer anomie (Section 3.1).

> **! EXAM TRAP**
>
> Norms telling people how they *ought* to behave are **prescriptive** (injunctive) norms. "Idealistic norms" is not a standard type; descriptive norms describe what is done; perceived norms are beliefs about norms.

Sherif and Asch: the classic studies ●●

How does a norm form? **Muzafer Sherif** (1935, 1936) used the **autokinetic effect**: in a dark room a stationary point of light seems to move, because there is no frame of reference. Alone, people gave widely different estimates of the movement. In small groups over several sessions their estimates converged on a shared group norm, which each person kept when later tested alone. In an ambiguous situation people used one another as information and privately accepted the answer. **Robert Jacobs** and **Donald Campbell** (1961) had confederates set an exaggerated norm, then replaced members one at a time; the arbitrary norm was handed down for several "generations" after the confederates had gone. **Solomon Asch** (1951, 1956) asked the harder question: will people conform when the answer is obvious?

KEY STUDY: ASCH'S LINE JUDGEMENTS (1951, 1956)

Method: one real participant sat among about seven confederates and said aloud, near the end of the row, which of three comparison lines matched a standard line (Figure 3.10). On 12 of 18 trials, the **critical trials**, every confederate gave the same wrong answer. **Result:** alone, people almost never erred (under 1 percent of judgements); in the group, conformity occurred on about 37 percent of critical trials. About three quarters (76 percent in Baron's account) conformed at least once; about a quarter never did. **Conclusion:** the answer was obvious, so people yielded to avoid standing out: normative influence towards an incorrect majority, mostly public compliance. **Criticism:** only male American students were tested, and such conformity has since declined.

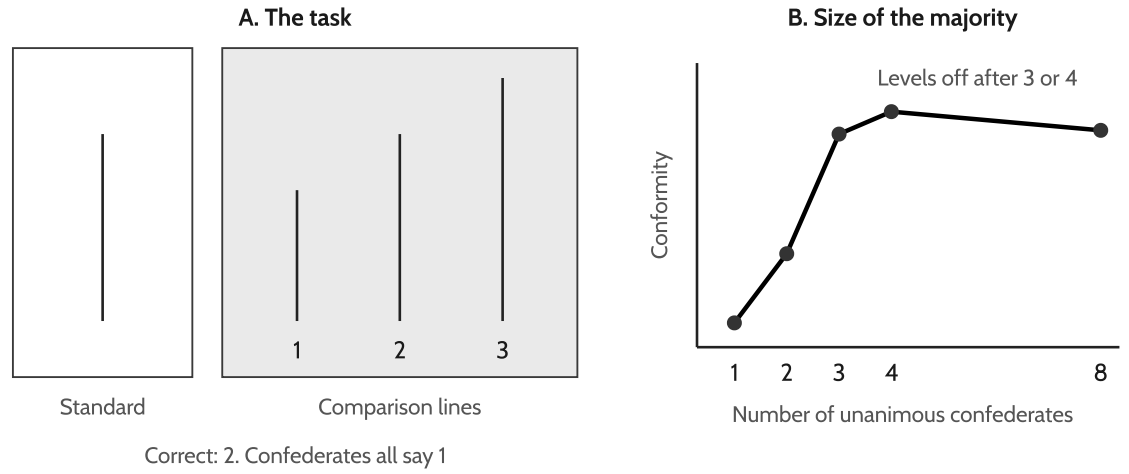


Figure 3.10 Asch's conformity task. (A) The match is obvious, yet on critical trials every confederate named a wrong line. (B) Conformity rose steeply as the unanimous majority grew to three, then levelled off (schematic, after Asch's group-size variation).

Asch's variations show what drives the effect. Conformity rose with the size of the majority up to three or four, then levelled off. A single dissenting partner cut it sharply, even one giving a different wrong answer: unanimity is what matters. Written, private answers cut it steeply too. **Richard Crutchfield** (1955) mechanised the method: people in separate booths saw others' supposed answers as lights on a panel, so no confederates were needed. **Rod Bond** and **Peter Smith**'s (1996) meta-analysis of 133 Asch-type studies in 17 countries found more conformity in collectivist cultures and a decline in the United States since the 1950s.

! EXAM TRAP

Asch tested the **limits of conformity** (not social perception, leadership or coercive power) and showed **normative influence towards an incorrect majority**: not obedience (nobody gave orders), the bystander effect or diffusion of responsibility (Section 3.8), or cognitive dissonance (Section 3.4). Private answers l*owered* conformity. Sherif's dot of light: ambiguity, informational influence, norm formation (not obedience or implicit bias, Section 3.5).

Why people conform ○

Morton Deutsch and **Harold Gerard** (1955) named two motives. **Normative social influence** rests on the desire to be liked and accepted; it produces public compliance and is strongest when a valued group is watching, as in Asch's study. **Informational social influence** rests on the desire to be correct; it produces private acceptance and is strongest when the situation is ambiguous or we lack expertise, as in Sherif's study.

Conformity rises with the group's **cohesiveness** (how much we are attracted to it and want to belong), its size up to a point, its unanimity, its members' status or expertise, public responding, a difficult or ambiguous task, and a collectivist culture. Gender differences are small: **Alice Eagly** and **Linda Carli**'s (1981) review found women slightly more influenceable, mainly under face-to-face group pressure. Conformity keeps traffic and queues orderly, but its downside is that it can lead ordinary people into cruelty (see the power of roles).

! EXAM TRAP

Normative influence is the desire to be **accepted and liked**; informational influence, the desire to be **correct**. "Personal beliefs and values" describe internalisation; "power and authority" describe obedience and the bases of power.

Resisting the majority; minority influence

Conformity pressure can be resisted. **C. R. Snyder** and **Howard Fromkin** (1980) proposed a **need for uniqueness**; later studies found that people told they are "exactly average" resist the majority more. The related desire for **individuation**, to stand out as a distinct person, works the same way. People reminded of a time they held power over others conform less, and pressure that threatens freedom can provoke **reactance** (Section 3.4).

Can a minority sway a majority? **Serge Moscovici**, **Elisabeth Lage** and **Martine Naffrechoux** (1969) showed groups of four participants and two confederates blue slides. When the confederates consistently called every slide "green", about 8 percent of the majority's judgements became "green", and in a later private test participants labelled more blue-green discs green; an inconsistent minority had almost no effect. In his **conversion theory** (1980) Moscovici argued that majorities win quick public compliance, whereas a consistent minority makes people think and produces slower, genuine private conversion. **Edwin Hollander** (1958) explained safe dissent: by conforming early a member earns **idiosyncrasy credits**, goodwill he can later spend on deviation. The most basic influence may be **emotional contagion**, the automatic catching of another's mood, which **Elaine Hatfield**, **John Cacioppo** and **Richard Rapson** (1994) traced to our mimicry of others' faces and postures.

Compliance tactics and cults ●●

Cialdini worked undercover among "compliance professionals" (salespeople, fund-raisers, advertisers) and in *Influence* (1984) traced their tactics to six principles: **reciprocity** (we repay favours and concessions), **commitment and consistency** (we act in line with what we have agreed to), **social proof** or social validation (we copy similar others), **liking** or friendship, **authority** (we defer to real or apparent experts) and **scarcity** (we want what is rare or running out). Table 3.19 matches tactics to principles.

Table 3.19 Compliance tactics and the principle each exploits.

Tactic	What happens	Principle
Ingratiation (**Edward E. Jones**, 1964)	Flattery, agreement or favours make the target like you first	Liking
Foot-in-the-door (1966)	A small request nobody refuses, then the larger real one	Consistency
Low-ball (Cialdini and colleagues, 1978)	A good deal is accepted, then made worse	Commitment
Door-in-the-face (Cialdini and colleagues, 1975)	A huge request is refused, then the smaller real one follows	Reciprocity
That's-not-all (**Jerry Burger**, 1986)	Before any answer, the offer is sweetened	Reciprocity
Playing hard to get	You or your product seem much in demand	Scarcity
Fast-approaching deadline	"Offer ends Sunday"	Scarcity
Pique (Santos, Leve and Pratkanis, 1994)	An odd request ("spare 17 cents?") breaks automatic refusal	Arousing attention
Foot-in-the-mouth (**Daniel Howard**, 1990)	The target first answers "How are you today?"	Commitment to a friendly exchange

KEY STUDY: THE FOOT-IN-THE-DOOR (1966)

Jonathan Freedman and **Scott Fraser** demonstrated the **foot-in-the-door technique** by asking Californian homeowners to put a large, ugly "Drive Carefully" sign on their lawns. About 17 percent of those approached cold agreed, but about 76 percent of those who, two weeks earlier, had agreed to show a small "Be a safe driver" sign. Two explanations: **consistency** (refusing now feels inconsistent) and **self-perception** (watching yourself agree, you infer "I support good causes"; Section 3.4).

The **door-in-the-face technique** reverses the order. Cialdini and his colleagues (1975) asked students to counsel juvenile delinquents two hours a week for two years; all refused. Asked next to take delinquents on a two-hour zoo trip, about 50 percent agreed, against under 17 percent asked only the small favour. The retreat looks like a concession, and reciprocity demands one back (reciprocal concessions). In the **low-ball technique** the initial commitment does the work: a car buyer who agreed to a bargain still buys when the "manager" cancels the discount.

! EXAM TRAP

Foot-in-the-door: **small, then large**. Door-in-the-face: **large, refused, then small**. Low-ball: yes first, then the deal worsens. That's-not-all: the deal improves before any yes. "Reward" is **not** one of Cialdini's six principles, and "conformity technique" is not a compliance tactic: conformity involves no request.

Cults use these principles at full strength. Recruiters begin with **love-bombing**, flooding a newcomer with affection and attention; they escalate commitment step by step (a free dinner, a retreat, a donation, then residence); they isolate recruits from family and former friends, control their information and build an us-versus-them identity. **Robert Jay Lifton** (1961), studying Chinese Communist "thought reform", described such total environments, including milieu control (control of all communication). At Jonestown, Guyana (1978), more than 900 followers of **Jim Jones** died in a mass murder and suicide he ordered. Talking with recruits' families is exactly what cults avoid.

Obedience: Milgram's experiments ○

Would you shock an innocent stranger because a man in a laboratory coat told you to? **Stanley Milgram** (1933 to 1984) asked this partly to understand how ordinary people carried out the Holocaust.

KEY STUDY: MILGRAM'S OBEDIENCE STUDY (1963)

Method: at Yale, 40 men aged 20 to 50 joined a supposed study of punishment and memory. A rigged draw made each the **teacher**; the **learner**, a likeable confederate, sat strapped in next door. For each error the teacher moved one switch up a generator from 15 to 450 volts in 15-volt steps, labelled up to "Danger: severe shock" and "XXX". No real shocks were given. At 300 volts the learner pounded on the wall, then fell silent. When teachers hesitated, the experimenter used four standard prods, from "Please continue" to "You have no other choice, you must go on." **Result:** psychiatrists had predicted that about one in a thousand would go to the end. All 40 reached 300 volts and 26 (65 percent) went to 450 volts, many sweating, trembling and protesting. **Conclusion:** most ordinary people obey a legitimate authority even to the point of apparently harming another person.

His variations (*Obedience to Authority*, 1974) show what raises and lowers obedience (Table 3.20).

Table 3.20 Milgram's variations: percentage who went to 450 volts.

Condition	Full obedience
Remote: learner unseen, pounds on wall (baseline)	65%
Voice feedback: protests heard	62.5%
Proximity: learner in the same room	40%
Touch-proximity: teacher presses learner's hand onto plate	30%
Orders given by telephone	About 20%
Run-down Bridgeport office, no link to Yale	About 48%
An ordinary man gives the orders	20%

Condition	Full obedience
Two peer teachers rebel	10%
Women as teachers	65%

Milgram's explanation was the **agentic state**: within a hierarchy a person stops acting autonomously and becomes the agent of another's wishes, transferring responsibility to the authority. The authority seemed legitimate (Yale, the coat, science); demands escalated gradually, each step barely worse than the last; and events moved too fast for reflection. Obedience falls when people are reminded that they are responsible, see disobedient models, question the authority's expertise and motives, and know about this research.

Diana Baumrind (1964) attacked the study's ethics: participants were deceived and severely stressed. Milgram replied that all were debriefed and 84 percent were glad to have taken part. Burger (2009) ran a safer partial replication, stopping at 150 volts (beyond which nearly all of Milgram's obedient men went to the end): about 70 percent were willing to continue past it. **Hannah Arendt** (1963), reporting the trial of the Nazi official **Adolf Eichmann**, described the **banality of evil**: monstrous crimes done by ordinary people doing their jobs.

CLINICAL LINK

Charles Hofling and colleagues (1966) had an unknown "doctor" telephone hospital nurses and order 20 milligrams of an unfamiliar drug, "Astroten", twice the maximum dose on its label. Of 22 nurses, 21 began to give it before being stopped.

! EXAM TRAP

Milgram's baseline: **65 percent** (26 of 40) to 450 volts, all 40 to 300 volts; not 10, 35 or 100 percent. The lesson is the power of the situation, not personality, and it concerns obedience to authority, not conformity when an answer is ambiguous.

The power of roles ○

KEY STUDY: THE STANFORD PRISON EXPERIMENT (1971)

Method: Philip Zimbardo, with **Craig Haney** and **Curtis Banks**, randomly made 24 psychologically healthy male students guards or prisoners in a mock prison in Stanford's psychology basement. Prisoners were "arrested" at home and known only by numbers; guards wore uniforms and mirrored sunglasses. **Result:** the guards became abusive, and the prisoners, after a brief rebellion, passive and distressed. Planned for two weeks, the study was stopped after six days. **Conclusion:** social roles and situational forces can turn ordinary people cruel. **Criticism:** demand characteristics (guards may have acted as they thought was wanted), Zimbardo's double role as researcher and prison superintendent, and ethical harm.

In the BBC Prison Study (2006), **Stephen Reicher** and **Alexander Haslam** found that guards who did not identify with their role did not turn tyrannical, and prisoners who shared a group identity challenged them: roles shape behaviour when people identify with them. In

The Lucifer Effect (2007) Zimbardo applied his analysis to American soldiers' abuse of Iraqi prisoners at Abu Ghraib.

> **! EXAM TRAP**
>
> "The power of roles" is **Zimbardo's prison experiment**, showing situational forces, not genetics, conditioning or reinforcement schedules (Chapter 1). Not the *Robbers Cave experiment* (Sherif's intergroup conflict study, Section 3.5); not Milgram's *message experiment*, the small-world study (1967), whose letters reached a stranger through about six acquaintances ("six degrees of separation"), nor his lost-letter technique; not "Zilstein's shock research", the fictitious Dr Gregor Zilstein of **Stanley Schachter**'s (1959) affiliation study (Section 3.6).

Bases of social power ○

Social power is the capacity to influence others. **John French** and **Bertram Raven** (1959) described five bases; Raven (1965) added informational power (Table 3.21). Raven later grouped tactics as **harsh** (coercion, impersonal reward, insisting on one's position), which breed resentment, or **soft** (expert, referent and informational), which win lasting acceptance.

Table 3.21 The six bases of social power.

Base	Rests on	Example
Reward	Ability to give rewards	A manager who grants bonuses
Coercive	Ability to punish	A warden who can confine
Legitimate	An accepted right to give orders	A police officer; Milgram's experimenter
Expert	Superior knowledge or skill	A doctor's advice
Referent	Identification with an admired agent	A cricket star's endorsement
Informational	The content of the message itself	A convincing argument

> **! EXAM TRAP**
>
> **Reciprocity** is a Cialdini compliance principle, not one of Raven's six bases of power. Coercion, expertise and information are bases.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Changing in response to a direct request is _____.
2. Norms that say what most people actually do are _____ norms.
3. Sherif used the _____ effect to show how group norms form.
4. Asch's participants conformed on about _____ percent of critical trials.
5. Conforming from a desire to be liked and accepted is _____ social influence.
6. A large request refused, then a smaller one, is the _____ technique.
7. In Milgram's baseline study, _____ percent went to 450 volts.
8. Raven's sixth base of power is _____ power.

Answers: (1) compliance, (2) descriptive, (3) autokinetic, (4) 37, (5) normative, (6) door-in-the-face, (7) 65, (8) informational.

3.8 Prosocial Behaviour: Why and When People Help LOWER YIELD · 15 QS

In March 1964 a young woman was stabbed to death in New York while, the newspapers said, 38 neighbours did nothing. Was this proof that city people no longer cared? Two young psychologists suspected that the number of witnesses was itself the problem. Their answer, the bystander effect, is the heart of this section. First, though: why does anyone help at all?

Prosocial behaviour, helping and altruism

Any action that benefits others, such as sharing, comforting, donating or rescuing, is **prosocial behaviour**. Prosocial behaviour that intentionally benefits a particular person is **helping**. Defined by motive rather than act, **altruism** is helping with no expected reward, often at a cost to oneself, purely for the other's welfare. In **cooperation**, people work together towards a goal that benefits all of them, so nobody need be the giver. A student who lends her notes is helping; if she expects nothing back and loses revision time, she may be acting altruistically; if two friends split the syllabus and swap notes, they are cooperating.

Does true altruism exist? The doctrine of **psychological egoism** says no: every helpful act ultimately serves the helper, by winning praise, avoiding guilt or feeling good. This **egoism versus altruism debate** was fought hardest over empathy.

Empathy and the egoism versus altruism debate ○

The act of perceiving, understanding and responding to another person's emotional state and ideas is **empathy**. Its emotional side, **affective empathy**, is feeling with the person; **cognitive empathy**, or perspective taking, is accurately understanding what they think and feel; **empathic concern** is the warm, other-oriented compassion that follows. **Mark H. Davis** measured these with his Interpersonal Reactivity Index (1980), whose four scales are perspective taking, fantasy, empathic concern and personal distress (one's own anxiety at another's suffering). Feeling concern or pity *for* someone, rather than feeling *with* them, is **sympathy**. Empathy is a foundation of helping, and a core condition a counsellor offers a client (Chapter 7).

! EXAM TRAP

"Perceiving, understanding and responding to the emotional state and ideas of another person" is **empathy**, not sympathy. **Transference** (a client redirecting feelings about past figures onto the therapist) is a therapy term (Chapter 7); **displacement** (shifting an impulse onto a safer target) is a defence mechanism (Chapter 4).

C. Daniel Batson's **empathy-altruism hypothesis** (1981; *The Altruism Question*, 1991) holds that empathic concern produces truly altruistic motivation, a wish to relieve the other's suffering for their sake, whereas personal distress produces only an egoistic wish to escape. To tell the two apart, Batson made escape easy.

KEY STUDY: ELAINE AND THE SHOCKS (BATSON AND COLLEAGUES 1981)

Women students watched "Elaine" react badly to electric shocks and were asked whether they would take her remaining shocks. Some were led to feel high empathy (Elaine shared their values). With **easy escape** they could leave after watching two of ten trials; with **difficult escape** they had to watch all ten. Low-empathy participants helped when escape was hard but mostly left when it was easy, as egoists would. High-empathy participants helped at high rates even when they could walk away. **Criticism:** they might still be avoiding guilt.

Robert Cialdini and colleagues replied with the egoistic **negative-state relief model** (1981, 1987): seeing suffering, or being sad for any reason, makes us feel bad, and we help to lift our own mood. Empathic people helped less when told that a drug had "fixed" their mood, so helping could not make them feel better. The **empathic joy hypothesis** (**K. D. Smith**, **J. P. Keating** and **Ezra Stotland** 1989) says we help for the pleasure of seeing the person relieved: empathic participants helped only if they expected to learn how their help had worked. Batson kept finding helping that these motives cannot explain, and most psychologists now accept that both altruistic and egoistic motives exist.

Genes, status and norms

How could a costly urge to help survive natural selection? **William D. Hamilton**'s **kin selection** theory (1964) answers that helping a relative helps copies of your own genes, so helping is favoured when relatedness multiplied by the benefit exceeds the cost. **Eugene Burnstein**, **Christian Crandall** and **Shinobu Kitayama** (1994) found that people would help close kin more than distant kin or strangers, and young relatives more than old ones, most clearly when life was at stake. For non-relatives, **Robert Trivers** proposed **reciprocal altruism** (1971): helping pays when the favour is likely to be returned.

Two accounts are frankly strategic. According to **competitive altruism** (**Charlie Hardy** and **Mark Van Vugt** 2006), people help to gain status: players gave more when giving was public, and the most generous were preferred as partners. Helping an out-group to reduce its threat to one's in-group is **defensive helping** (**Arie Nadler** and colleagues 2009): students offered more help to a rival school said to outscore them, as if to keep it dependent.

Finally, norms. The **norm of reciprocity** (**Alvin Gouldner** 1960) says we should help those who helped us; the **social responsibility norm** (**Leonard Berkowitz** and L. R. Daniels 1963) says we should help those who depend on us, whether or not they can repay. A **personal**

norm, one's own felt moral obligation to act, predicts helping better than vague social norms.

The bystander effect ●●

In the early hours of 13 March 1964, **Kitty Genovese** (1935 to 1964), aged 28, was stabbed near her apartment building in Kew Gardens, Queens, New York. *The New York Times* reported that 38 neighbours had seen or heard the attack without calling the police. The figure was later shown to be exaggerated (few saw the attack clearly, and some did call or go to her), but the story shocked the country, and commentators blamed big-city apathy.

Bibb Latané (born 1937) and **John Darley** (1938 to 2018) suspected that the witnesses failed to act because there were so many of them. Their experiments (1968) and book *The Unresponsive Bystander* (1970) established the **bystander effect**: the more bystanders present at an emergency, the less likely any one of them is to help, and the slower the help that comes. People are more likely to help when alone.

KEY STUDY: A SEIZURE OVER THE INTERCOM (DARLEY AND LATANÉ 1968)

New York University students talked over an intercom from separate booths. One voice, a tape, began choking and begging for help as if having a seizure. Participants believed that only they, or they and one other, or they and four others could hear. Before the fit ended, 85 percent of those who thought they were alone left the booth to help, 62 percent with one other listener, and 31 percent with four others (Figure 3.11). Non-helpers were not apathetic; many were trembling. **Criticism:** a distressing deception.

In the **smoke-filled room** (Latané and Darley 1968), students filling in questionnaires saw smoke pour through a vent. Alone, 75 percent reported it; beside two passive confederates who kept writing, 10 percent did; in groups of three naive students, someone reported it in 38 percent of groups. In the **lady in distress** study (Latané and **Judith Rodin** 1969), a woman was heard falling next door: 70 percent of people alone went to help, but only 7 percent of those sitting with an unresponsive confederate.

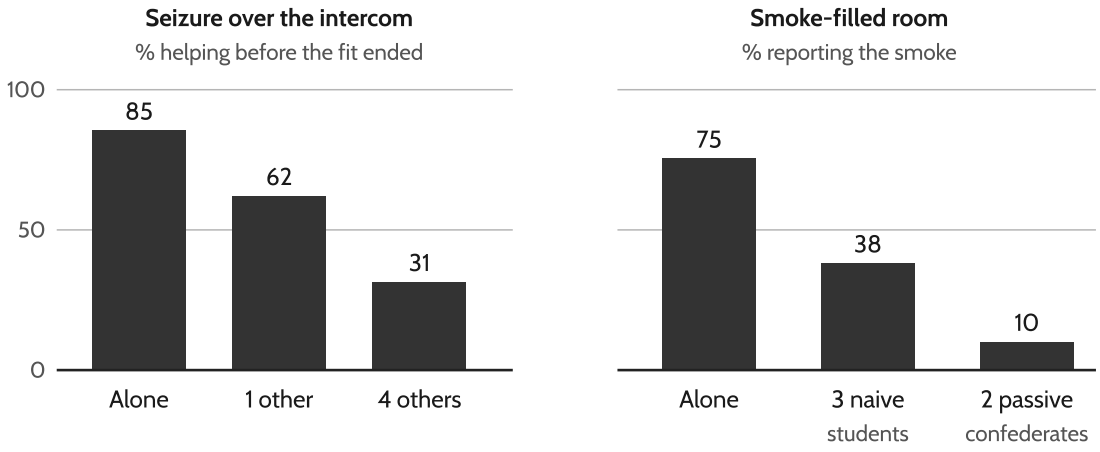


Figure 3.11 The bystander effect in two classic 1968 studies by Darley and Latané. The more people present, the less likely any one person is to act.

! EXAM TRAP

The research followed the murder of **Kitty Genovese**, not Martin Luther King Jr. (assassinated 1968), Emmett Till (a Black teenager lynched in 1955) or Malala Yousafzai (shot by the Taliban in 2012). People help more when **alone**; seeing others not help makes intervention **less** likely. Groups did notice the smoke a little later, but the effect is about helping, not noticing.

Five decisions, and why bystanders fail ●●●

Latané and Darley (1970) traced the failure to five decisions, each of which must be answered yes before help is given (Table 3.22).

Table 3.22 Latané and Darley's decision model of helping.

Step	The bystander must	Obstacle at this step
1	Notice the event	Distraction, hurry, overload
2	Interpret it as an emergency	Ambiguity; pluralistic ignorance
3	Assume responsibility	Diffusion of responsibility
4	Know how to help	Lack of competence
5	Decide to help	Audience inhibition; costs

Two obstacles carry the bystander effect. The first, **diffusion of responsibility** (step 3), is the spreading of responsibility across all the bystanders, so that each feels less personally obliged and assumes someone else will act. In the intercom study nobody could see anyone else, so diffusion alone explains the result. The second, **pluralistic ignorance** (step 2), occurs when bystanders use the behaviour of others to decide whether help is needed and mistakenly conclude that it is not. Among strangers, nobody knows for sure what is happening; each stays calm to avoid looking foolish, reads the others' calm as evidence that all is well, and the whole group hesitates and does nothing, as in the smoke-filled room. A third obstacle, **audience inhibition** (evaluation apprehension), is the fear of being judged for overreacting or helping clumsily.

Push a familiar scene through the model. A woman is beaten in a busy street while a crowd films it on phones and nobody calls for help. Each sees many others who could act (diffusion), sees nobody stepping in and half concludes it is a private quarrel (pluralistic ignorance), and fears looking foolish (audience inhibition). That is the bystander effect, not social loafing or group polarisation.

! EXAM TRAP

The **two classic explanations** are diffusion of responsibility and pluralistic ignorance; asked for one, the key is diffusion of responsibility. Pluralistic ignorance is not ignoring others or eyewitnesses, nor being suggestible about the cause: it is relying on others and wrongly deciding, from their calm, that help is unnecessary. "Someone else will do it" is diffusion; "nobody looks worried, so it must be nothing" is pluralistic ignorance.

Table 3.23 defines the terms most often offered in their place.

Table 3.23 Terms confused with the bystander effect and its causes.

Term	What it actually means	Section
Deindividuation	Loss of self-awareness and restraint in an anonymous crowd	3.10
Social loafing	Less individual effort on a pooled group task	3.10
Social facilitation	Others' presence improves easy, impairs hard tasks	3.10
Group polarisation	Discussion makes a group's view more extreme	3.10
Groupthink	A cohesive group suppresses dissent and decides badly	3.10
Crowd behaviour	Collective action of a large, temporary gathering	3.1, 3.10
Conformity; compliance	Yielding to group pressure; to a direct request	3.7
Normative influence	Conforming to be liked or accepted	3.7
Social comparison	Judging oneself against others	3.6
Mere exposure	Repeated exposure increases liking	3.6
Reactance	Doing the opposite when freedom is threatened	3.4
Cognitive dissonance	Tension from inconsistent beliefs and actions	3.4
Just-world phenomenon	Belief that people get what they deserve	3.3
Halo effect	One good trait colours the whole impression	3.2
Amotivation	No intention to act at all	Ch. 1

The effect weakens when bystanders are friends who can talk (in the lady in distress study, 70 percent of pairs of friends helped against 40 percent of pairs of strangers) and in small towns where people know one another. A meta-analysis by **Peter Fischer** and colleagues (2011) found it much weaker in dangerous emergencies, which are quickly recognised as real and in which other bystanders become physical support.

Weighing costs: the arousal: cost-reward model

Why do bystanders sometimes help at once? In 1969 **Irving Piliavin**, Judith Rodin and **Jane Allyn Piliavin** staged collapses on New York subway trains. A man with a cane who seemed ill was helped in 95 percent of trials, usually within seconds, and larger groups did not help less; a man who smelled of alcohol and carried a bottle was helped in only about half. Their **arousal: cost-reward model** (developed fully in 1981) says that seeing someone in distress causes unpleasant arousal, which bystanders reduce by whichever response costs least. The costs of helping (effort, danger, embarrassment) are weighed against the costs of not helping (guilt, blame, the victim's suffering). The drunk man raised the costs of helping and, seeming responsible for his plight, lowered the guilt of walking away (Figure 3.12).

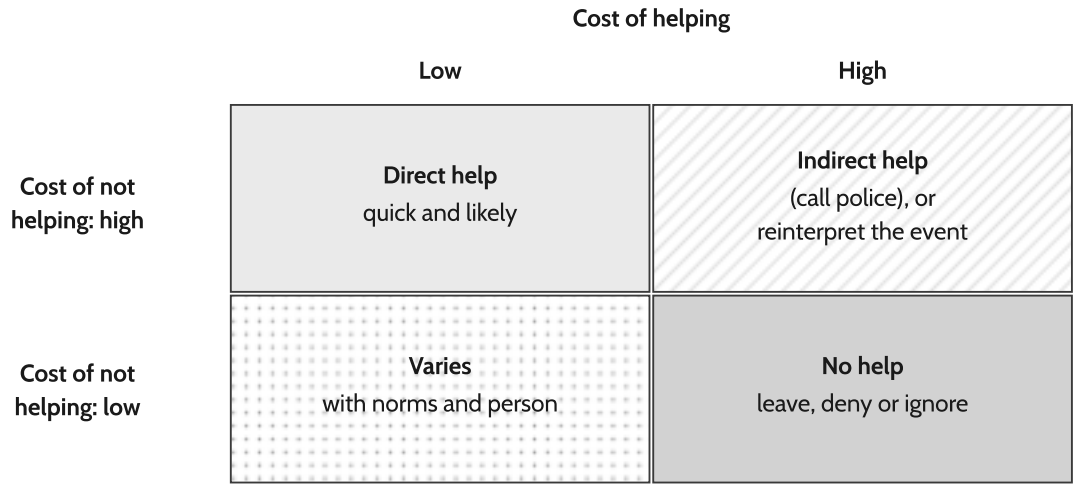


Figure 3.12 Predictions of the arousal: cost-reward model. Help is most likely when helping is cheap and not helping is costly.

Situations that raise or lower helping ○

We help people who are **similar** to us, partly because similarity raises empathic concern. We help less when victims seem **responsible for their plight**: in **Bernard Weiner**'s attribution analysis (Section 3.3), a controllable cause such as drinking invites anger, an uncontrollable one such as illness invites sympathy and help.

KEY STUDY: THE GOOD SAMARITAN (DARLEY AND BATSON 1973)

Princeton Theological Seminary students, some preparing a talk on the parable of the Good Samaritan, were told they had time to spare, were on time, or were already late. On the way they passed a man slumped and groaning in a doorway. Of those in no hurry, 63 percent helped; moderate hurry, 45 percent; great hurry, 10 percent. The topic of the talk made no reliable difference. **Being in a hurry**, not character, decided who stopped.

Stanley Milgram (1970) proposed **urban overload**: city dwellers face so much stimulation that they screen much of it out, people in need included, and helping is lower in big cities than in small towns.

Mood matters. **Alice Isen** and **Paula Levin** (1972) found that students given a cookie were more willing to help, and that people who had just found a dime in a public telephone were far more likely to help a stranger pick up dropped papers (about 88 against 4 percent). This **feel good, do good effect** fades if helping would spoil the mood. **Guilt** also raises helping, as people repair their self-image.

Goodness is contagious. Motorists stopped more often for a woman with a flat tyre if they had just passed another woman being helped (**James Bryan** and **Mary Ann Test** 1967): a **prosocial model** shows what to do and that helping is the norm. After playing a prosocial video game such as Lemmings, 57 percent helped pick up spilt pencils, against 33 percent after a neutral game (**Tobias Greitemeyer** and **Silvia Osswald** 2010). Watching an act of kindness produces **elevation**, an uplifted feeling described by **Jonathan Haidt** (2000) that makes the watcher want to help.

Gratitude raises future helping. **Adam Grant** and **Francesca Gino** (2010) had students comment on a stranger's job application letter; when he thanked them, about twice as many helped again. Being thanked raised the helpers' self-worth (feeling valued by others) and self-efficacy (feeling competent), but only self-worth carried the rise in helping.

> **! EXAM TRAP**
>
> Gratitude raises helping by enhancing helpers' **self-worth and self-efficacy**; pairs built on self-esteem or self-confidence are wrong. If asked which one did the work, it is self-worth.

Helping falls after **social exclusion**, which drains empathy (people told they would end up alone gave less to a student fund; **Jean Twenge** and colleagues 2007); in **darkness** and other conditions of **anonymity**, which remove the approval helping earns (**Chen-Bo Zhong** and colleagues 2010); and when people think of their **time in money terms** (**Sanford DeVoe** and **Jeffrey Pfeffer** 2010).

Who helps, and how help is received ○

The **altruistic personality** is a consistent tendency to help others across situations, rather than helping only when one feels like it, calculating costs and benefits, or being ready only in emergencies. **Hans Werner Bierhoff**, **Renate Klein** and **Peter Kramp** (1991) compared people who had given first aid at road accidents with people who had not, and found five components in the helpers: empathy, belief in a just world, social responsibility, internal locus of control and low egocentrism. **Samuel Oliner** and **Pearl Oliner** (1988) interviewed more than 400 rescuers of Jews in Nazi Europe: compared with non-rescuers, their parents had modelled caring and disciplined by reasoning, and they felt responsible for people beyond their own group. In volunteering, **Allen Omoto** and **Mark Snyder** (1995) found five motives among AIDS volunteers (values, understanding, personal development, community concern, esteem enhancement); the self-oriented ones best predicted who kept serving.

Help is not always welcome. The **threat-to-self-esteem model** (Nadler and **Jeffrey Fisher** 1986) holds that aid carries both support (someone cares) and threat (you are inferior and dependent). It threatens most when the helper is a similar peer, the task is central to one's self-image and the help cannot be repaid; then people resent help or avoid seeking it. Nadler also contrasted **dependency-oriented help**, which hands over the full solution, with **autonomy-oriented help**, which gives the tools to solve the problem oneself. The helper's perceived motive matters too: help that seems freely chosen and caring is welcomed; help that seems forced by a role or given for show is resented.

Are helping and aggression, then, opposites? Not quite. Their motives overlap (people help to gain status or keep rivals dependent), their acts can look alike (a surgeon's knife), and unwanted help can harm. Section 3.9 takes up aggression.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Helping with no expected reward, at a cost to oneself, is _____.
2. Batson's _____ hypothesis says empathic concern produces truly altruistic motivation.
3. Bystander research was prompted by the 1964 murder of _____.
4. In the seizure study, 85 percent of those who thought they were alone helped, against _____ percent with four others.
5. Bystanders who read one another's calm as a sign that help is not needed show _____.
6. Darley and Batson's seminarians helped least when they were in a _____.
7. Gratitude raises helping mainly through the helper's sense of _____.
8. Nadler and Fisher's _____ model explains why help can be resented.

Answers: (1) altruism, (2) empathy-altruism, (3) Kitty Genovese, (4) 31, (5) pluralistic ignorance, (6) hurry, (7) self-worth, (8) threat-to-self-esteem.

3.9 Aggression: Its Roots, Causes and Control LOWER YIELD · 3 QS

A batsman given out wrongly smashes his bat against the boundary board; a schoolgirl spreads a cruel rumour; a hired killer shoots a stranger without anger. All three are aggression, yet their roots differ. This section traces those roots, the triggers, and the methods that actually reduce aggression.

What counts as aggression

Robert A. Baron and **Deborah R. Richardson** (1994) define **aggression** as behaviour directed toward the goal of harming another living being who is motivated to avoid such treatment. So hateful thoughts alone are not aggression (it is behaviour); accidental harm is not, though a punch that misses is (it is intended); and kicking a door or a wanted surgical cut is not (the target must be living and unwilling). Anger is an emotion, aggression a behaviour: you can be furious and do nothing, and the hired killer aggresses without anger.

The main distinction is by goal. In **hostile aggression** (reactive or affective aggression) harm is the end in itself: hot, impulsive, a reply to provocation. In **instrumental aggression** (proactive aggression) harm is a means to another end, such as money or status: cold and planned. Table 3.24 adds the other cuts. The term **relational aggression** comes from **Nicki R. Crick** and **Jennifer K. Grotpeter** (1995), who found that girls used it more than boys; studies that counted only hitting had made girls look unaggressive.

Table 3.24 Further forms of aggression.

Form	What it is	Example
Physical	Harm by bodily force or weapons	A slap
Verbal	Harm by words	Insults, threats
Direct	Face to face; the aggressor is known	A shove

Form	What it is	Example
Indirect	Through others or in secret	An anonymous complaint
Relational	Damages the victim's friendships and status	Exclusion, gossip

Instinct, evolution and biology ○

The oldest answer is that aggression is built in. **Sigmund Freud** (1856 to 1939) proposed in 1920 a **death instinct**, **thanatos**, opposing the life instinct (eros): a drive toward self-destruction that is turned outward as aggression (Chapter 4). The ethologist **Konrad Lorenz** (1903 to 1989), in *On Aggression* (1966), saw an inherited fighting instinct that spaced animals out and let the strongest males breed. His **hydraulic model** pictured aggressive energy building like water in a tank until a releasing stimulus, or eventually none at all, lets it out. Well-armed animals evolved **ritualised fighting** (stags shove with locked antlers rather than gore) and **appeasement gestures** (a beaten wolf bares its throat) that stop the winner's attack; weakly armed humans evolved weak inhibitions, and our weapons now kill beyond the reach of appeasement. Critics note that aggression varies too much across cultures to be one pressure-cooker drive. Both views fed the catharsis idea (3.9.8).

Evolutionary psychology recasts instinct as inherited potential: aggression that won mates and resources was passed on. It predicts that young men commit most violence, mostly against rival men (**male competition**), and that **sexual jealousy** is a leading cause of violence between partners.

The clearest chemical finding is that low **serotonin** activity, shown by low levels of its breakdown product 5-HIAA in cerebrospinal fluid, is linked to impulsive aggression; serotonin helps restrain impulses. **Testosterone** correlates modestly with dominance and aggression. Table 3.25 sets out the usual distractors. The amygdala fuels anger, the hypothalamus organises attack and the prefrontal cortex applies the brakes (Chapter 5). Genes act through environments: in the New Zealand cohort of **Avshalom Caspi** and colleagues (2002), maltreated boys became antisocial adults far more often if they carried the low-activity form of the **MAOA** gene, which codes the enzyme that breaks down serotonin, norepinephrine and dopamine; without maltreatment the variant carried little risk.

Table 3.25 Chemicals and aggression.

Chemical	Main job	Link to aggression
Serotonin	Mood, impulse control	Low: impulsive aggression
Dopamine	Reward, motivation, movement	Not the aggression marker
Epinephrine (adrenaline)	Adrenal fight-or-flight hormone	Rises in anger and fear alike
Norepinephrine	Alertness, arousal	Arousal in any strong emotion
Testosterone	Male sex hormone	High: modestly more aggression

> **! EXAM TRAP**
>
> Low **serotonin**, not low dopamine, epinephrine or norepinephrine, is the impulsive-aggression finding. Testosterone is a hormone, not a neurotransmitter.

Frustration-aggression and its reformulation ○

Social psychologists moved the drive's source outside the person. The **frustration-aggression hypothesis**, put forward at Yale in 1939 by **John Dollard**, **Leonard W. Doob**, **Neal E. Miller**, **O. H. Mowrer** and **Robert R. Sears**, claimed that frustration, the blocking of a goal (Section 1.10), always leads to aggression, and that aggression always comes from frustration. Neither holds: frustration often brings sadness or problem solving, and the hired killer is not frustrated. Miller (1941) soon withdrew the first claim: aggression is only one of several responses to frustration. The group also explained the target. When the frustrator is powerful or absent, aggression shifts to a safer target (**displacement**: scolded by his boss, a clerk shouts at his children); when that target is a weak minority blamed for the trouble, displacement becomes **scapegoating** (Section 3.5).

Leonard Berkowitz (1926 to 2016) reformulated the hypothesis in 1989, in what became his **cognitive neoassociationist model**. Frustration causes aggression only to the extent that it produces **negative affect**, so unexpected or unfair frustrations provoke most. And any aversive event (pain, heat, foul smells, noise) does the same. Negative affect automatically primes both **fight** (rudimentary anger) and **flight** (rudimentary fear); the situation, past learning and later appraisal decide which wins (Figure 3.13). Stimuli associated with violence act as **aggressive cues**, tipping an angry person toward fight.

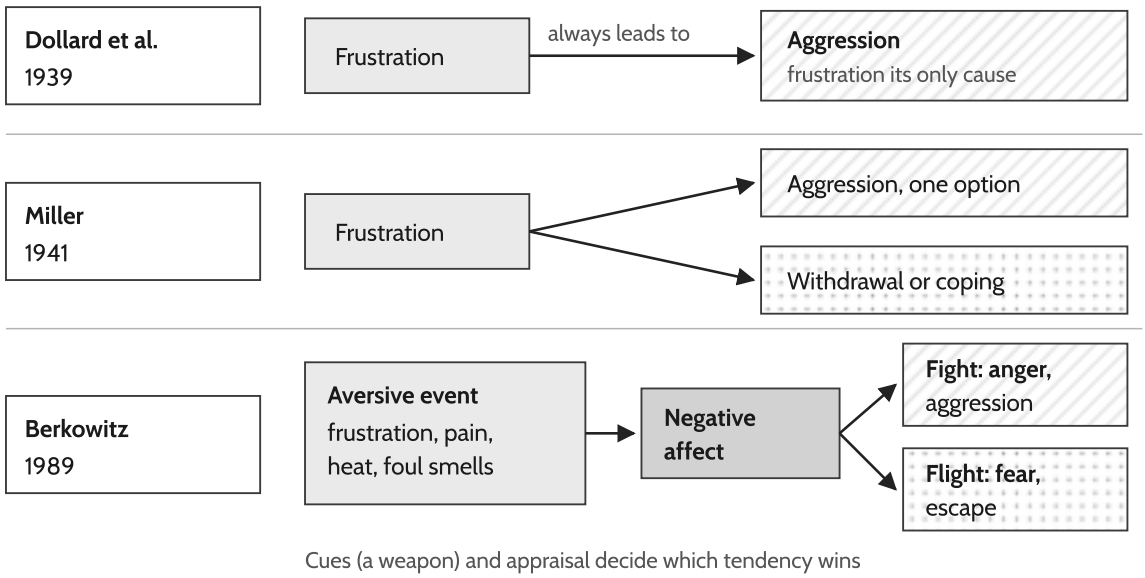


Figure 3.13 From certain cause (1939) to one possible response (1941) to negative affect from any aversive event (1989).

KEY STUDY: THE WEAPONS EFFECT (BERKOWITZ AND LEPAGE, 1967)

In the study by Berkowitz and **Anthony LePage**, male students, angered or not by a partner's electric shocks, could shock him back while a shotgun and a revolver, or badminton racquets, or nothing lay nearby. Angered men beside the guns gave the most shocks. "The finger pulls the trigger, but the trigger may also be pulling the finger." Some replications failed, but meta-analyses support a modest **weapons effect**.

Arousal from elsewhere also feeds aggression. In **Dolf Zillmann**'s **excitation transfer theory** (1971), arousal fades slowly, so arousal left over from one event (a near miss on the Mysuru ring road, a gym session) is misattributed to the next, and a minor annoyance produces rage, especially when the leftover arousal goes unnoticed.

> **! EXAM TRAP**
>
> The **reformulation** is **Berkowitz's** (1989), not Dollard's (the 1939 original), Miller's (the 1941 softening) or Lazarus's (Richard Lazarus: appraisal and stress, Section 1.11). Weapons effect: Berkowitz and LePage; leftover arousal: Zillmann.

Learning, scripts and the General Aggression Model ○

Learning theories ask where the particular ways of aggressing come from. **Albert Bandura** (1925 to 2021) answered: largely from watching others.

KEY STUDY: THE BOBO DOLL EXPERIMENTS (1961 TO 1965)

Bandura, **Dorothea Ross** and **Sheila A. Ross** (1961) let 72 nursery-school children watch an adult punch, kick and shout at an inflatable Bobo doll, or play quietly. Left with the doll, those who had seen aggression copied the model's specific acts and words, unrewarded. Filmed and cartoon models worked almost as well (1963). In 1965, children who saw the model punished (vicarious punishment) imitated less than those who saw it rewarded (vicarious reinforcement), yet a reward for reproducing the acts made all groups equal: consequences to the model affect performance, not learning. **Criticism.** Bobo is built to be hit.

This is **observational learning** through **modelling**, with no direct reinforcement of the learner (Bandura's four processes are in Section 1.6).

> **! EXAM TRAP**
>
> Bobo doll = **observational learning of aggression through modelling**. Not operant conditioning (no child was reinforced for hitting), not classical conditioning of fear (Little Albert, Section 1.6), not sensory deprivation (the McGill isolation studies, Section 1.5).

Repeated observation builds **aggressive scripts** (**L. Rowell Huesmann**): stored programmes for how a conflict unfolds and how to respond ("if disrespected, hit back"), retrieved fast when provoked. The **General Aggression Model** of **Craig A. Anderson** and **Brad J. Bushman** (2002) integrates these theories (Figure 3.14): person and situation inputs shape the present internal state (affect, cognition, arousal); a quick appraisal yields impulsive action, a considered one thoughtful action; and each outcome feeds back, making the next aggressive episode easier.

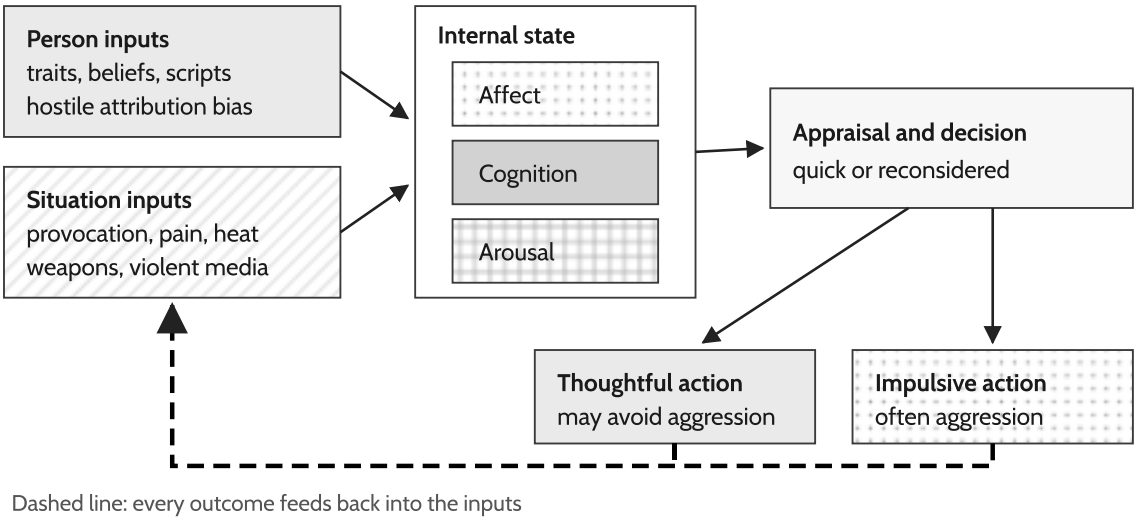


Figure 3.14 The General Aggression Model: inputs act through feelings, thoughts and arousal, and appraisal decides the act.

Social, cultural and personal causes

The strongest trigger is direct **provocation** (insults, unfair criticism, condescension, teasing seen as meant to hurt): people return what they receive, or a little more. Displaced aggression (3.9.3) and leftover arousal add fuel. Table 3.26 gathers the other causes.

Table 3.26 Further causes of aggression.

Cause	Who	Key finding
Social exclusion	**Jean M. Twenge** and colleagues (2001)	Rejected people blasted more noise at others, even at a neutral stranger
Media violence	**Leonard D. Eron** and Huesmann; Anderson (video games)	Boys' TV violence at 8 predicted aggression at 19 and serious crime by 30; violent games raise aggression and lower empathy
The **culture of honour**	**Richard E. Nisbett** and **Dov Cohen** (1996)	Herding regions (the US South) demand violent answers to insults and infidelity; insulted Southerners showed more cortisol, testosterone and aggression
Gender role: **precarious manhood**	**Joseph A. Vandello** and **Jennifer K. Bosson** (2008)	Manhood is hard won and easily lost; in a 2009 follow-up, threatened men punched a bag harder
A **hostile attribution bias**	**Kenneth A. Dodge** (1980)	Aggressive children read ambiguous acts as deliberate, then retaliate
Narcissism: **threatened egotism**	Bushman and **Roy F. Baumeister** (1998)	Insulted narcissists aggressed most; low self-esteem did not predict aggression
Type A pattern	Section 1.11	More hostile, not instrumental, aggression

Media violence works by **modelling**, by **priming** aggressive thoughts and scripts, and by **desensitisation**, the blunting of emotional reactions to real violence. As for sex differences, men are more physically and directly aggressive; women equal or exceed men in relational and indirect aggression; and the gap shrinks under strong provocation.

Heat, alcohol and other aversive conditions

Aversive conditions act as Berkowitz predicted. The **hot temperature effect** is real: across 50 US cities from 1950 to 1995, hotter years brought more murders and assaults but not more property crime (Anderson and colleagues). Yet in Baron's laboratory, very hot, angry participants became *less* aggressive. The **negative affect escape model** of Baron and **Paul A. Bell** explains this: discomfort raises aggression up to a moderate level, beyond which escape takes priority, an inverted U. Alcohol raises aggression in both sexes. In the **alcohol myopia** account of **Claude M. Steele** and **Robert A. Josephs** (1990), alcohol narrows attention to the salient cue (the insult) and pushes out distant ones (consequences, norms). Crowding, uncontrollable noise and pain act likewise, through negative affect.

Bullying, cyberbullying and workplace aggression

Dan Olweus, pioneer of school research in Norway (*Bullying at School*, 1993), defined **bullying** as intentional, repeated aggression against a victim with less power. He distinguished bullies, victims (often anxious and friendless) and bully-victims (provocative victims who also bully, and fare worst); his whole-school programme of supervision, clear rules and consistent action roughly halved bullying. Online, **cyberbullying** follows victims home and reaches vast audiences; anonymity and the unseen victim produce **online disinhibition**, a cousin of deindividuation (Section 3.10). Workplace aggression is mostly verbal, indirect and covert, and perceived injustice often triggers it.

Preventing and controlling aggression

Punishment deters only when it is prompt, certain, strong and seen as justified, conditions courts rarely meet; unfair punishment acts as fresh provocation, and physical punishment models aggression. The **catharsis hypothesis**, rooted in Freud, adopted by Dollard's group (1939) and revived by Lorenz, holds that venting anger safely (punching a pillow, shouting) drains the drive and reduces later aggression. It fails.

KEY STUDY: VENTING FEEDS THE FLAME (BUSHMAN, 2002)

Bushman angered about 600 students with harsh feedback on their essays. One group then hit a punching bag while thinking about the critic, a second while thinking about fitness, and a third sat quietly. When all could blast the critic with noise, the rumination group was angriest and most aggressive; doing nothing produced the least aggression.

What works? **Cognitive interventions**: a sincere **apology**; **pre-attribution**, a warning that someone is ill or stressed, so later rudeness is blamed on circumstances; and a **delay** that lets arousal fade. Exposure to **non-aggressive models** who stay calm under provocation. **Social skills training** in negotiating and complaining without attack. Baron's **incompatible response technique**, which induces empathy or humour, feelings that cannot coexist with anger. Strengthened **self-regulation**, which fatigue, stress and alcohol deplete (Baumeister). And **forgiveness**, which ends the cycle of revenge.

! EXAM TRAP

Which method is **not** supported? **Catharsis**: venting raises aggression. "Severe punishment always deters" is also false.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Aggression harms a living being who is motivated to _____ such treatment.
2. Gossip and exclusion that damage friendships are _____ aggression.
3. Low levels of _____ are linked to impulsive aggression.
4. The frustration-aggression hypothesis was reformulated by _____ in terms of negative affect.
5. Angered men gave more shocks when guns lay nearby: the _____ effect.
6. Leftover arousal intensifying anger later is Zillmann's _____ theory.
7. The Bobo doll studies show observational learning through _____.
8. Bushman's punching-bag study contradicted the _____ hypothesis.

Answers: (1) avoid, (2) relational, (3) serotonin, (4) Leonard Berkowitz, (5) weapons, (6) excitation transfer, (7) modelling, (8) catharsis.

3.10 Groups: Structure, Performance, Decisions and Leadership

MEDIUM YIELD · 25 QS

Your family, your entrance batch and your cricket team all shape how hard you try, what you decide and whom you follow. What turns people into a group, how do groups grow, how do others change our effort, why do groups decide badly, and what makes a leader effective?

What makes a group

A **group** is two or more people who interact, depend on one another for their goals, see themselves as belonging together and share a structure of roles and norms. People at a bus stop share a place, not a group. **Entitativity**, **Donald T. Campbell**'s (1958) term, is how far a collection of people is seen as one coherent unit; frequent interaction, shared goals and similarity raise it, and highly entitative groups such as families matter most to their members. Table 3.27 sorts the kinds of groups.

Table 3.27 Kinds of groups.

Distinction	Who	Meaning
Primary versus secondary	**Charles Horton Cooley** (1909)	Primary: small, intimate, face-to-face (family). Secondary, the later contrast: large, impersonal, goal-based (a company)
Formal versus informal		Set up with assigned roles (a ward team) versus arising on its own (lunch friends)
Membership versus reference	**Herbert Hyman** (1942)	Groups you belong to versus groups whose standards you judge yourself by
In-group versus out-group	**William Graham Sumner** (1906)	"Us" versus "them"; root of ethnocentrism (Section 3.5)
Intimacy groups, task groups, social categories, loose associations	**Brian Lickel** and colleagues (2000)	Most to least entitative: a family; a work team; "women"; a bus queue

A reference group can remake attitudes. In **Theodore Newcomb**'s Bennington College study (late 1930s, published 1943), women from conservative homes who adopted the liberal college as their reference group became more liberal, and most stayed so 25 years later. Formed versus natural groups and social group work are in Section 3.11.

Group structure and membership ○

Every lasting group develops a structure with four parts. **Roles** are behaviours expected of a position. **Kenneth Benne** and **Paul Sheats** (1948) distinguished **task roles** that move the work on (initiator, information seeker, coordinator, evaluator-critic), **maintenance roles** or socio-emotional roles that hold members together (encourager, harmoniser, compromiser, gatekeeper) and **self-oriented roles** that serve the individual at the group's cost (aggressor, blocker, recognition seeker, dominator). **Role differentiation** follows as members specialise, often one leading the task while another keeps the peace. **Status** is rank within the group; high-status members talk more and may deviate more. **Norms** are the group's rules, often unspoken. **Cohesiveness** is the sum of forces holding members in: liking, commitment to the task and pride. Cohesive groups demand more conformity (Section 3.7) and satisfy members more, but perform better only when their norms favour effort, and very high cohesion feeds groupthink.

People join for belonging, goals, self-knowledge, status and identity (Section 3.5), and pay in freedom, time and effort. A high price even raises a group's value: women who passed an embarrassing initiation liked a dull group more (**Elliot Aronson** and **Judson Mills**, 1959), a dissonance effect (Section 3.4).

Richard Moreland and **John Levine** (1982) described **group socialisation** as five phases in which group and person evaluate each other: **investigation** (prospective member), **socialisation** (new member), **maintenance** (full member), **resocialisation** (marginal member, as commitment falls) and **remembrance** (ex-member). Entry, acceptance, divergence and exit mark the transitions. Sometimes the whole group breaks. A **schism** is the splintering of a group into distinct factions. **Fabio Sani** (2005) found that clergy who left the Church of England after it first ordained women in 1994 felt the change had subverted its identity, and that their protests would go unheard.

> **! EXAM TRAP**
>
> Splintering into factions is a **schism**. A *chapter* is a local branch, *division* an everyday word and *deviation* one member breaking a norm.

Sociometry: mapping who chooses whom ●●

A hostel's room list shows its official structure; how would you find the real one? **Jacob Levy Moreno** (1889 to 1974) answered with **sociometry**, a method of measuring an individual's social behaviour and relations in a group from the attractions and rejections among its members. He set it out in *Who Shall Survive?* (1934), with **Helen Hall Jennings**, after regrouping a girls' training school by the residents' own choices.

In the **sociometric test** each member privately names whom he or she would choose, and sometimes reject, for working, living or playing together. Answers fill a **sociometric matrix**, choosers down the side and the chosen across the top; a column total divided by the number

of other members (N minus 1) is a **sociometric index** of choice status. Drawn out, the matrix becomes a **sociogram**, a diagram of choices and rejections that shows attraction, social distance and patterns of choice within a group (Figure 3.15). A **star** receives many choices, an **isolate** none; a **neglectee** receives few but is not rejected; a **rejectee** draws rejections. A mutual pair choose each other, a chain runs A to B to C, and a **clique** chooses within itself. The sociogram thus captures implicit and explicit communication and interaction patterns; teachers use it to find isolated children. Moreno also founded psychodrama (Chapter 7).

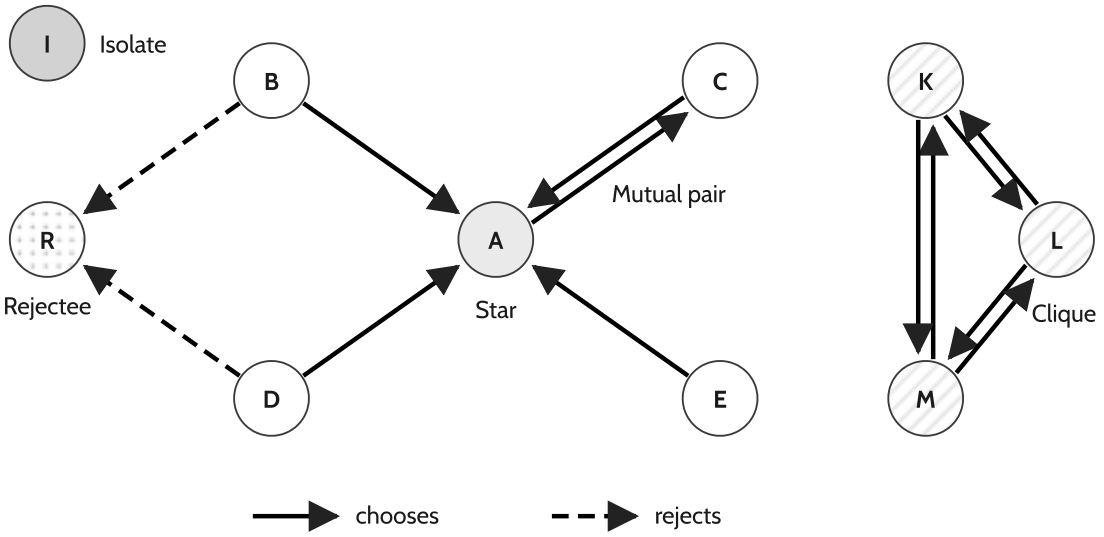


Figure 3.15 A sociogram. A is the star, A and C a mutual pair, K, L and M a clique; R is rejected and nobody chooses I.

> **! EXAM TRAP**
>
> Sociometry, not the sample method (Chapter 9), the interview method or the *Verstehen* method (interpretive understanding of social action, **Wilhelm Dilthey**, **Max Weber**), measures social behaviour in a group. A sociogram does not measure IQ, a leader's personality or economic status. A **genogram** is a three-generation family tree (Chapter 7), an **ecomap** a family's links with its environment (Section 3.11), a **cognitive map** Tolman's mental map (Section 1.6).

How groups develop •••

In 1965 **Bruce Tuckman** reviewed studies of therapy, training and laboratory groups and proposed four stages; with **Mary Ann Jensen** in 1977 he added a fifth (Figure 3.16). In **forming**, members orient themselves, are polite and unsure, and depend on the leader. In **storming**, conflict and resistance surface as members struggle over leadership, roles and the group's definition. In **norming**, cohesion grows, norms and roles are agreed, and the group clarifies its purpose and strategy for reaching its goals. In **performing**, the work gets done. In **adjourning**, the group disbands, often with a sense of loss.

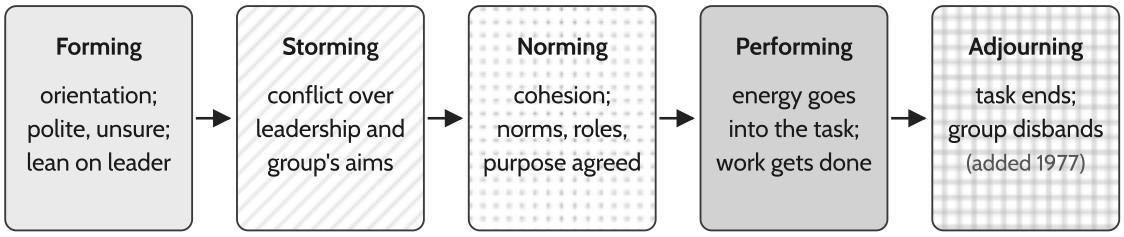


Figure 3.16 Tuckman's stages of group development, in their fixed order.

Real groups do not always march in step. **Connie Gersick** (1988) found that project teams with deadlines show **punctuated equilibrium**: inertia, a burst of change near the midpoint of their time, then focused work. The Boston model used in social group work is in Section 3.11.

> **! EXAM TRAP**
>
> **Proximity** is not a stage of group formation; it is a cause of attraction (Section 3.6). "Clarified its purpose and strategy" is **norming**; "struggle over leadership and group definition" is **storming**. The order is forming, storming, norming, performing, adjourning.

Performing in front of others ●●

Does an audience help you or hurt you? The question is as old as experimental social psychology.

> **KEY STUDY: TRIPLETT'S FISHING REELS (1898)**
>
> **Norman Triplett** noticed in cycling records that riders were faster when paced or racing a rival than when riding alone against the clock. He then had children wind a fishing reel as fast as possible, alone or beside another child doing the same; many wound faster in company, though some did worse. Published in 1898, it is often called the first social psychology experiment. Criticism: the gains were small and uneven.

Floyd Henry Allport named the effect **social facilitation** (1920, 1924), better performance when others are present, and called working side by side on the same task coaction. Yet others sometimes made people worse, called social inhibition or impairment. **Robert Zajonc** resolved the puzzle in 1965 with **drive theory**: the mere presence of others raises arousal, and arousal strengthens the **dominant response**, the one most likely in that situation. On a simple or well-learned task the dominant response is correct, so performance improves; on a complex or new task it is likely wrong, so performance worsens (Figure 3.17).

Push a case through it. Tim has trained his voice for years but has hardly danced, and his school musical needs both. The audience raises his arousal: his singing improves and his dancing deteriorates. An audience helps most on easy, well-learned tasks, not novel, difficult, creative or first-time ones. Zajonc, **Alexander Heingartner** and **Edward Herman** (1969) found it even in cockroaches: watched by other roaches, they ran a straight runway faster but a maze more slowly.

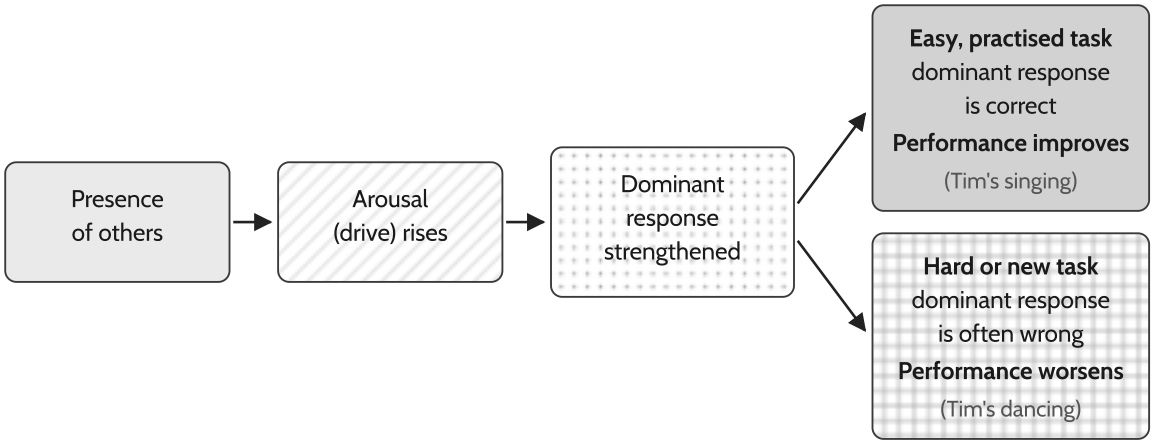


Figure 3.17 Zajonc's drive theory (1965): one rise in arousal, two opposite outcomes.

Two rivals refine the account. In **evaluation apprehension** (**Nickolas Cottrell** and colleagues, 1968), concern about being judged does the arousing: a blindfolded audience produced no facilitation. In **distraction-conflict theory** (**Robert S. Baron**, 1986), attending to both task and onlookers creates conflict that arouses and narrows attention. The arousal-performance curve is the Yerkes-Dodson law (Section 1.10).

> **! EXAM TRAP**
>
> Triplett used **fishing reels**. Rope pulling is Ringelmann, the shock generator Milgram (Section 3.7), a maze with food reward animal learning (Section 1.6). Diffusion of responsibility explains bystanders not helping (Section 3.8).

Social loafing and deindividuation ●●●

Now pull a rope with seven others: nobody can tell how hard you pull. The French engineer **Max Ringelmann** found that the bigger the team, the less each member pulled (published 1913). Part of this **Ringelmann effect** is coordination loss, but **Alan Ingham** and colleagues (1974) showed that blindfolded people who merely believed others were pulling also pulled less: motivation falls too.

That loss is **social loafing**: less effort when individual efforts are pooled than when working alone. **Bibb Latané**, **Kipling Williams** and **Stephen Harkins** (1979) had blindfolded students in headphones clap and shout alone or in supposed groups of two, four or six: each person's output fell as the group grew. **Steven Karau** and Williams's collective effort model (1993) explains why: people work hard only when they expect their effort to matter for an outcome they value. Loafing therefore falls when each contribution is identifiable and evaluated, the task is challenging, appealing or involving, members are friends or committed, effort is believed to bring valued rewards, and the group is small. A teacher such as Miss Pooja should grade both the group and each member; group grades alone, random assignment or optional presentations leave individual effort hidden.

> **! EXAM TRAP**
>
> "Groups loaf less when evaluation apprehension is absent" is **false**: removing the fear of judgement lets people loaf. Social loafing's other name is the **Ringelmann effect**; the free-rider effect is a related but separate term (Table 3.28). The Triplett effect is social facilitation, the McGurk effect is lips changing the speech sound heard (Section 1.3), and the sleeper effect is delayed persuasion (Section 3.4).

Loafing is not universal. **P. Christopher Earley** (1989, 1993) found that Chinese management trainees worked harder in a group than alone, where Americans loafed: **social striving**, typical of collectivist cultures. Table 3.28 separates the look-alike terms.

Table 3.28 Effort and restraint in groups: look-alike terms.

Term	Who	What happens
Social loafing (Ringelmann effect)	Ringelmann 1913; Latané and colleagues 1979	Less effort when contributions are pooled
Free-rider effect	**Norbert Kerr** and **Steven Bruun** 1983	Letting others carry the load because one's input seems dispensable
Sucker effect	Kerr 1983	Cutting effort to avoid being exploited by loafers
Social compensation	Williams and Karau 1991	Working harder to cover weak teammates on an important task
Social striving	Earley 1989, 1993	Working harder in a group than alone
Deindividuation	Festinger and colleagues 1952	Lost self-awareness and restraint under anonymity
Bystander effect	Darley and Latané 1968	Less helping when others are present (Section 3.8)

Gustave Le Bon, in *The Crowd* (1895), argued that people in a mob lose their minds to a contagious group mind. **Leon Festinger**, **Albert Pepitone** and Newcomb (1952) named the state **deindividuation**: anonymity and shared accountability lower self-awareness, so restraints weaken. **Philip Zimbardo** (1969) found that hooded women gave longer shocks than name-tagged women, and **Edward Diener** and colleagues (1976) saw anonymous Halloween trick-or-treaters in groups take extra sweets far more often than named children alone. The **SIDE model** (social identity model of deindividuation effects; **Stephen Reicher**, **Russell Spears** and **Tom Postmes**, 1995) adds that anonymity strengthens a salient group identity, so people follow group norms more, violent or peaceful. Cyberbullying and online disinhibition, cruelty behind a screen name, are deindividuation, not social facilitation, polarisation or the bystander effect (bullying is in Section 3.9).

Group decision making ●●

Do several heads beat one? **James H. Davis** (1973) showed that group decisions follow **social decision schemes**: **majority wins** on opinions, **truth wins** when one member can demonstrate a correct answer, and **first shift**, where the group follows the first member to change position, as in deadlocked juries.

Groups are often more extreme, not more moderate. In 1961 **James Stoner** found that groups recommended riskier courses than their members had alone, the **risky shift**, but later studies found that groups that began cautious became more cautious. **Serge Moscovici** and

Marisa Zavalloni (1969) showed the general rule: French students' liking for de Gaulle and dislike of Americans both grew after discussion. This is **group polarisation**: discussion pushes members towards a more extreme version of their initial leaning, through **persuasive arguments** (more arguments heard for the favoured side), **social comparison** (wanting to look better than average on the valued side) and **social identity** (converging on the group's prototype).

> ! EXAM TRAP
>
> More extreme group decisions are **group polarisation**. "Extremity" is not a technical term, groupthink is cohesive suppression of dissent, and conformity is yielding to a norm (Section 3.7).

Irving Janis, in *Victims of Groupthink* (1972; revised 1982), analysed fiascoes such as Kennedy's Bay of Pigs invasion (1961) and the ignored warnings before Pearl Harbor (1941); later analysts added the 1986 *Challenger* launch. **Groupthink** is a mode of thinking in highly cohesive groups in which the drive for unanimity overrides realistic appraisal of alternatives: collective decision making that discourages dissent and leads to poor decisions. Antecedents are high cohesion, insulation, a directive leader, high stress and no methodical procedures. The eight symptoms are an illusion of invulnerability (excess optimism and risk), collective rationalisation (warnings explained away), belief in the group's morality, stereotyped views of out-groups, direct pressure on dissenters, self-censorship of doubts, an illusion of unanimity (silence read as agreement) and self-appointed mindguards who screen out contrary information. Remedies: the leader stays impartial and withholds his or her view early, appoints a devil's advocate or a subgroup to critique the plan, invites outside experts, uses subgroups and holds a second-chance meeting.

> ! EXAM TRAP
>
> To curb groupthink, the leader should **ask a subset of the group to critique** the plan. Team-building raises cohesion, stating his view first steers the group, and forbidding argument silences dissent: all make it worse. Groupthink is not individual thinking, brainstorming or mere peer pressure, and it yields poor decisions.

Groups also waste what members know. **Garold Stasser** and **William Titus** (1985) spread facts so that the best student-body candidate stood out only if members pooled unshared facts, a **hidden profile**; groups discussed mainly shared facts (biased information sampling) and chose worse. In **brainstorming**, popularised by **Alex Osborn**, members produce many ideas with criticism postponed, yet as many people working alone produce more, because of **production blocking**: one person speaks at a time and ideas fade while waiting. The **nominal group technique** has members write ideas silently, share them in turn and vote privately; the **Delphi technique**, from the RAND Corporation, polls experts anonymously over rounds with feedback until opinion converges. **Escalation of commitment** is continuing to invest in a failing course to justify past costs.

Cooperation, conflict and fairness

In a **social dilemma**, each person gains by acting selfishly but all lose if everyone does. In the **prisoner's dilemma**, two suspects questioned apart each do best by betraying the other, yet both do worse if both betray than if both stay silent. **Garrett Hardin**'s **tragedy of the commons** (1968) is the many-person version: each herder adds cows to the shared pasture until it is ruined, as with groundwater or fish stocks today. Communication, trust, small groups and sanctions raise cooperation.

Conflict yields to **bargaining**, best ending in an **integrative agreement** that gives both sides more than a split by trading on issues each values differently; to **superordinate goals** that neither side can reach alone (Sherif's Robbers Cave, Section 3.5); and to **Charles Osgood**'s **GRIT** (graduated and reciprocated initiatives in tension reduction, 1962), small announced concessions that invite reciprocation. Conflict is often about fairness: **distributive justice** is fair outcomes, judged in **J. Stacy Adams**'s equity theory (1965) by comparing our outcome-to-input ratio with others'; **procedural justice** is fair procedures; **transactional** or interactional **justice** is courteous treatment and honest explanation.

Leadership ○

Leadership is influencing group members towards shared goals. The **great-person theory** held that leaders are born with special traits, but **Ralph Stogdill**'s 1948 review found traits only weakly linked to leadership: a leader in one situation may not lead in another. Attention turned to what leaders do.

KEY STUDY: LEWIN, LIPPITT AND WHITE (1939)

Kurt Lewin, **Ronald Lippitt** and **Ralph White** had adults lead clubs of ten-year-old boys in three styles. Under **autocratic** leaders, who made every decision, boys worked while watched but stopped when the leader left, and grew hostile or apathetic. Under **democratic** leaders, who guided group decisions, work was more original, continued without the leader, and the boys liked this style best. Under **laissez-faire** leaders, who left the boys alone, little was done. Criticism: small American samples.

The Ohio State studies, using the Leader Behaviour Description Questionnaire (LBDQ), found two independent dimensions: initiating structure (organising work and roles) and consideration (warmth and concern for subordinates); one leader can be high on both. The Michigan studies contrasted job-centred with employee-centred supervisors. **Robert Blake** and **Jane Mouton**'s managerial grid (1964) crosses concern for production with concern for people, from impoverished (1,1), country club (1,9), authority-compliance (9,1) and middle of the road (5,5) to the ideal, team management (9,9).

Contingency theories match leader to situation. In **Fred Fiedler**'s **contingency model** (1967), leaders rate their **least preferred co-worker** (LPC): rating even that person kindly (high LPC) marks a relationship-oriented leader, rating harshly (low LPC) a task-oriented one. Situational favourableness comes from three dimensions: **leader-member relations**, **task structure** and **position power** (the leader's formal power and authority to reward and punish); good relations count most. Task-oriented leaders do best in very favourable or very unfavourable situations, relationship-oriented leaders in between; since style is fixed, Fiedler advised changing the situation. Table 3.29 lists later theories.

! EXAM TRAP

Employee motivation is not one of Fiedler's dimensions. The three are leader-member relations, task structure and position power ("intrinsic power and authority" in some papers).

Table 3.29 Later theories of leadership.

Theory	Who	Core idea
Situational leadership	**Paul Hersey** and **Kenneth Blanchard** (1969)	Fit style to followers' readiness: telling, selling, participating, delegating
Path-goal theory	**Robert House** (1971)	Clear the path to goals: directive, supportive, participative or achievement-oriented
Normative decision model	**Victor Vroom** and **Philip Yetton** (1973)	A decision tree sets how far to involve subordinates
Idiosyncrasy credits	**Edwin Hollander** (1958)	Conformity and competence earn room to innovate (Section 3.7)
Leader-member exchange	**George Graen** (1975)	Close ties with an in-group of followers, formal ones with an out-group
Transactional leadership	**James MacGregor Burns** (1978)	Rewards exchanged for performance
Transformational leadership	Burns (1978); **Bernard Bass** (1985)	Idealised influence, inspirational motivation, intellectual stimulation, individualised consideration
Charismatic leadership	Weber; House (1977)	Devotion to a leader seen as extraordinary
Servant leadership	**Robert Greenleaf** (1970)	The leader serves followers' growth first

INDIAN CONTEXT

J. B. P. Sinha's **nurturant-task leader** (1980) is firm about work but caring like a parent, with nurturance conditional on the task being done; it suits hierarchical Indian workplaces and should grow into participative leadership. Women and leadership are in Section 3.5; Herzberg's work motivation is in Section 1.10.

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. A diagram of choices and rejections within a group is a _____.
2. Tuckman's stage of struggle over leadership is _____.
3. Zajonc: others' presence strengthens the _____ response.
4. Social loafing is also called the _____ effect.
5. Working harder in a group, seen in collectivist cultures, is social _____.
6. Discussion making views more extreme is group _____.
7. Fiedler's dimensions: leader-member relations, task structure and _____.
8. Splintering of a group into factions is a _____.

Answers: (1) sociogram, (2) storming, (3) dominant, (4) Ringelmann, (5) striving, (6) polarisation, (7) position power, (8) schism.

3.11 Applied Social Psychology: Law, Culture and Social Work

LOWER YIELD · 15 QS

Kurt Lewin wanted social psychology to change lives, through action research that studies a problem while trying to solve it (Section 3.1). This closing section follows that wish into hospitals and crowded cities, police stations and courtrooms, and then into social work, the profession that helps individuals, groups and communities.

Health, environment, work and happiness

Why does the same stress make one person ill and leave another well? Partly because of other people. **Social support** is the perception or experience of being loved, cared for and part of a network of mutual help. **Sheldon Cohen** and **Thomas Ashby Wills** (1985) contrasted two models. In the main-effect model, social integration benefits health at any level of stress. In the **stress-buffering hypothesis**, support protects chiefly under high stress, by making the stressor seem less threatening and by softening the reaction to it. Loneliness is in Section 3.6, and health behaviour models in Chapter 7.

Places matter too (personal space is in Section 3.3). **Daniel Stokols** (1972) separated density, a physical count of people per unit of space, from **crowding**, the unpleasant feeling that there are too many people, with lost control and privacy: a festival is dense yet enjoyable, while a lift with three strangers feels crowded. In 1962 **John B. Calhoun** let rats breed with ample food but limited space. Their society collapsed into a **behavioural sink** of neglected young, soaring infant mortality, aggression and withdrawal. In people, density harms mainly when it removes control. **Irwin Altman** (1975) described **territoriality**, the marking and defending of places: primary territories are owned and central to life (a home), secondary territories are used regularly but shared (a canteen table) and public territories are open to anyone for a while (a park bench). Noise, likewise, harms most when it is unpredictable and uncontrollable.

At work, social psychology informs motivation (Herzberg, Section 1.10), the Hawthorne effect, in which people change their behaviour because they know they are being studied (Chapter 9), and leadership (Section 3.10). **Ed Diener** (1946 to 2021) defined **subjective well-being** as a person's own evaluation of life: high life satisfaction, frequent positive feelings and infrequent negative ones.

Eyewitness testimony ○

Why do defence lawyers so often protest the use of eyewitness testimony? Witnesses are not especially forgetful, and poor eyesight is rarely the issue; the trouble is that a memory can be vivid, confident and false. **Source monitoring** is the process of judging where a memory came from: seen, heard about or imagined (**Marcia Johnson** and colleagues, 1993). A source monitoring error attaches a genuine memory to the wrong origin, so a face seen in a mugshot album, on the news or as an innocent bystander is taken for the culprit's. In one celebrated case a rape victim identified the psychologist **Donald Thomson**, who had been on the live television programme she was watching when she was attacked: his alibi was the source of her memory. Misleading questions act the same way. In the 'smashed' versus 'hit' study of **Elizabeth Loftus** and **John Palmer** (1974; Section 1.7), the wording of a question changed what witnesses remembered, down to broken glass that was never there.

A confident witness is not usually right: across studies, confidence predicts accuracy only weakly, and it grows after the event. In the **post-identification feedback effect** (**Gary Wells** and colleagues, 1998), witnesses told 'Good, you identified the suspect' later claimed a better view and more certainty, even when they had picked an innocent person. Some sources of error the legal system cannot control. In **weapon focus** a visible weapon draws attention away from the face; in **own-race bias** people recognise faces of their own race more accurately than faces of other races; and high stress during the crime usually impairs identification. Other sources it can control, chiefly the lineup (Table 3.30).

Table 3.30 Lineup reforms that reduce false identifications.

Reform	What is done	Why it helps
Double-blind	The officer running it does not know the suspect	No unintended hints or feedback
Sequential lineup	Faces shown one at a time, each judged yes or no	Stops choosing the closest match
Unbiased instructions	Witness told the culprit may not be present	Removes pressure to choose someone
Fair fillers	Others match the witness's description	The suspect does not stand out
Confidence recorded at once	Certainty noted in the witness's words	Before feedback inflates it

The **cognitive interview** of **Ronald Fisher** and **Edward Geiselman** (first tested in 1984) gives four instructions: **reinstate the context** (mentally return to the scene and your feelings), **report everything**, however trivial, **change the order** (recall backwards as well as forwards) and **change the perspective** (describe what another person present would have seen). Because a memory has several retrieval routes, it draws out much more correct detail than a standard police interview, with few added errors.

> **! EXAM TRAP**
>
> Defence lawyers protest eyewitness testimony because **source errors can cause a false identification**, not because witnesses forget more, see poorly or are usually right when confident. The Ronald Fisher of the cognitive interview is not Ronald Aylmer Fisher, the statistician.

Juries, profiling and rehabilitation ○

In the **story model** of **Nancy Pennington** and **Reid Hastie** (1986), jurors faced with conflicting evidence build a narrative of what happened from the evidence and their own knowledge of how such events go, then choose the verdict that fits the most coherent story; evidence presented in story order persuades best. Pretrial publicity against the accused raises guilty verdicts, and instructions to ignore it help little. Attractive defendants are more often acquitted and given lighter sentences, unless their looks aided the crime, as in a swindle. Stereotypes about race, sex and class bias verdicts (Section 3.5), though careful deliberation weakens them.

Offender profiling infers the characteristics of an unknown offender from the crime. FBI agents such as **Robert Ressler** and **John Douglas** interviewed convicted murderers and built a top-down typology. **Organised** offenders plan, control the scene, bring and remove a weapon and are socially competent; **disorganised** offenders act on impulse, leave a chaotic

scene and a weapon of opportunity, and are socially inadequate. **David Canter**'s **investigative psychology** works bottom-up, from statistical patterns across many solved cases; his 1986 profile helped catch England's 'railway rapist'. Support for the organised-disorganised split is weak.

Profiling and the cognitive interview help to catch offenders; rehabilitation tries to change them. After **Robert Martinson**'s 'nothing works' (Key study), meta-analyses from the late 1980s built the 'what works' evidence. Successful programmes are **community-based** rather than institutional, keeping offenders tied to work, family and prosocial peers; they are **cognitive-behavioural**, structured and multimodal. They follow the **risk-need-responsivity** (RNR) principles of **Don Andrews** and **James Bonta** (1990): match treatment intensity to the risk of reoffending (**risk**); target criminogenic needs, the changeable causes of crime such as antisocial attitudes, antisocial peers and substance misuse (**need**); and teach in the way the offender learns best (**responsivity**). Punishment alone does not reduce reoffending.

KEY STUDY: MARTINSON AND 'NOTHING WORKS' (1974)

Question: does rehabilitation reduce reoffending? **Method:** Martinson and colleagues reviewed 231 controlled evaluations of correctional programmes. **Result:** 'with few and isolated exceptions', rehabilitation had had no appreciable effect on recidivism; the press shortened this to 'nothing works', and policy turned to punishment. **Criticism:** many programmes were weak or badly run, and Martinson himself withdrew the strong claim in 1979.

! EXAM TRAP

A quality of successful offender rehabilitation is **being community-based**. Cognitive interviews and offender profiling are investigative tools, not treatment, and a focus on the relationship with parents is not a defining quality.

Social work: definition, roots and methods ●●

The 2014 global definition of the International Federation of Social Workers (IFSW) and the International Association of Schools of Social Work (IASSW) calls social work a practice-based profession and academic discipline that promotes social change and development, social cohesion, and the empowerment and liberation of people. Its lens is the **person-in-environment**: problems arise between people and their families, neighbourhoods and institutions, so help may target either side.

It has two roots. The **Charity Organisation Society** (London, 1869) sent 'friendly visitors' to investigate applicants and coordinate relief, the seed of case work. The **settlement houses** placed educated volunteers among the poor to press for reform: **Toynbee Hall** (London, 1884, under **Samuel Barnett**) and **Hull House** (Chicago, 1889, founded by **Jane Addams** and **Ellen Gates Starr**), the seed of group work and community organisation.

The **primary methods**, or direct methods, are social case work, social group work and community organisation; the secondary or auxiliary methods are social welfare administration, social action and social work research. **Direct practice**, face-to-face work with clients, has historically been conceptualised as case work and case management, as individual psy-

chotherapy (clinical social work) and as generalist practice. Policy practice, shaping laws and agency rules, is indirect or macro practice.

India's first school of social work, and Asia's, was the **Sir Dorabji Tata Graduate School of Social Work**, opened in Bombay in 1936 under the American missionary **Clifford Manshardt**; in 1944 it was renamed the **Tata Institute of Social Sciences**. The Delhi School of Social Work followed in 1946, and psychiatric social work training grew up at Ranchi and at NIMHANS, Bengaluru.

> **! EXAM TRAP**
>
> 'Tata Institute of Social Profession' and 'Tata Institute of Social Service' are invented, and the Tata Institute of Social Sciences is the 1944 name; the first school was the **Sir Dorabji Tata Graduate School of Social Work**. Policy practice is not a form of direct practice.

Social case work ●●

Social case work, the oldest method, helps one person or family at a time with problems of social functioning, through a professional relationship and community resources. Its founder, **Mary Ellen Richmond** (1861 to 1928), gave it a method in *Social Diagnosis* (1917), which taught workers to gather evidence systematically before judging, and defined it in *What Is Social Case Work?* (1922) as the processes that develop personality through adjustments consciously effected, individual by individual, between people and their social environment.

The process runs in phases (Figure 3.18): **intake** (can this agency help?), **study** (gathering the facts), **assessment** or **social diagnosis** (defining the problem and its causes), **intervention** or treatment, **evaluation**, and **termination** with **follow-up**. Content analysis, the coding of themes in texts, is a research technique, not a phase. The worker's techniques are **interviewing** (the central one), **observation**, **counselling**, **supportive techniques** (reassurance, encouragement), **home visits**, **collateral contacts** (speaking, with consent, to relatives, teachers or employers), **environmental modification** (changing the client's surroundings, such as arranging a pension) and **referral**. Lobbying legislators belongs to social action, not case work.

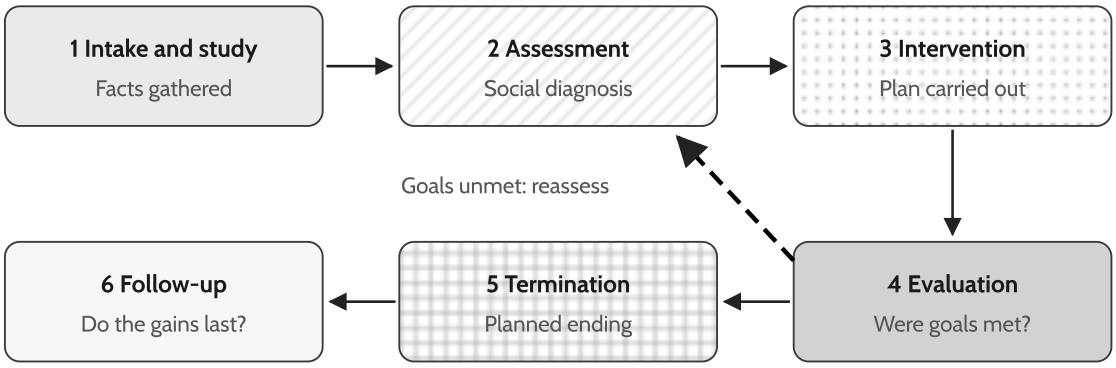


Figure 3.18 The social case work process. When evaluation finds goals unmet, the worker returns to assessment (dashed arrow).

By the 1930s case work had split into the **diagnostic school**, built on Freud, and the **functional school**, built on Otto Rank's will therapy (Table 3.31). **Florence Hollis** developed the diagnostic school into the psychosocial approach, and **Charlotte Towle**'s *Common Human Needs* (1945) taught that every welfare applicant has the same basic needs as anyone else.

Table 3.31 Schools and models of social case work.

Approach	Key names	Central idea
Diagnostic (psychosocial)	**Gordon Hamilton** (1940); Hollis (1964)	Understand the personality, diagnose, then treat; the client as person-in-situation
Functional	**Jessie Taft**, **Virginia Robinson** (Pennsylvania)	The client's will drives change, within the agency's function and time limits
Problem-solving	**Helen Harris Perlman** (1957)	Four Ps: a person with a problem comes to a place for a process of help
Task-centred	**William Reid** and **Laura Epstein** (1972)	Brief, time-limited work on agreed problems through concrete tasks

Whatever the school, the relationship carries the help. **Felix Biestek** (1957) set out seven principles of the casework relationship: **individualisation** (each client is unique), **purposeful expression of feelings**, **controlled emotional involvement** (responding with feeling without being swept away), **acceptance**, a **non-judgemental attitude**, **client self-determination** (the right to make one's own choices) and **confidentiality**.

KEY STUDY: FISCHER, 'IS CASEWORK EFFECTIVE?' (1973)

Question: does professional case work help clients? **Method: Joel Fischer** reviewed the controlled studies comparing clients who received professional case work with control groups who did not. **Result:** there was no good evidence that case work worked, and in some studies treated clients did worse. **Impact:** the review shook the profession and pushed it toward empirical, evidence-based practice. **Criticism:** the studies often had vague goals and crude outcome measures.

! EXAM TRAP

Richmond founded case work; **Joel Fischer** studied its effectiveness. Irwin Epstein wrote on social work research, John Goyder on survey nonresponse, and Jacob Cohen was the statistician of kappa, effect size and power (Chapter 9). Not part of the case work process: **content analysis**. Not a case work technique: **lobbying**.

Social group work ●●●

Social group work uses the group itself as a helper: members gain from one another as well as from the worker. **Grace Longwell Coyle** (1892 to 1962), whose *Social Process in Organized Groups* appeared in 1930, established group work as a method within social work. **Gisela Konopka** (1910 to 2003) later carried it into treatment settings such as institutions and child guidance clinics. **Harleigh B. Trecker** listed ten principles: planned group formation, specific objectives, a purposeful worker-group relationship, continuous individualisation, guided group interaction, democratic group self-determination, flexible functional organisation, progressive programme experiences, resource utilisation and evaluation.

Three models dominate. The social goals model, from the settlements, builds social consciousness and responsible citizenship. The remedial model of **Robert Vinter** uses the group to treat individual members, with the worker as change agent. The reciprocal, or mediating, model of **William Schwartz** makes the group a mutual-aid system, with the worker mediating between members and between the group and society.

Ronald Toseland and **Robert Rivas** distinguish **formed groups**, deliberately convened by a worker or agency for an identified purpose, usually with a professionally trained leader and an agency auspice (a support group for caregivers), from **natural groups**, which come together spontaneously out of friendship, shared interest or circumstance (a circle of friends, a street gang). All groups share certain properties: a purpose, stages of development and criteria for their composition, along with structure, norms, roles and cohesion (Section 3.10). No group needs an 'analytic base'.

Some members serve the group in hidden ways. **Lawrence Shulman** described the scapegoat, blamed for what the group dislikes in itself; the deviant member, whose odd behaviour often voices what others feel; the **gatekeeper**, who blocks or diverts the group, by joking or changing the subject, whenever talk nears a painful topic; the internal leader, who takes initiative from within; the defensive member, who denies having a problem; the quiet member; and the monopoliser, who dominates the talk. Groups also change over time. The **Boston model** of **James Garland**, **Hubert Jones** and **Ralph Kolodny** (1965) has five stages: pre-affiliation (approach and avoidance), power and control, intimacy, differentiation and separation. Tuckman's forming to adjourning sequence is in Section 3.10.

> **! EXAM TRAP**
>
> Group work as a method: **Grace Coyle**, not Mary Richmond (case work), Charlotte Towle (*Common Human Needs*) or Gisela Konopka (group work in treatment settings). 'Focusing on change and possibilities' is not one of Trecker's principles. Formed groups do not come together spontaneously.

Community organisation

Community organisation works with whole communities. **Murray G. Ross** (1955) defined it as a process by which a community identifies and ranks its needs, develops the confidence and will to work on them, finds resources, takes action, and in doing so develops cooperative attitudes and practices. **Jack Rothman** (1968) distinguished three models (Table 3.32). Social action is linked above all to **Saul Alinsky** (1909 to 1972), who organised Chicago's Back of the Yards neighbourhood and wrote *Rules for Radicals* (1971).

Table 3.32 Rothman's three models of community organisation.

Model	Aim	Worker's role
Locality development	Self-help and consensus through broad local participation	Enabler
Social planning	Expert, data-based solving of problems such as housing or health	Planner, fact-gatherer
Social action	Shifting power and resources to a disadvantaged group, through conflict if needed	Activist, advocate

Culturally sensitive practice and child welfare ●●

Cultures differ in measurable ways. Surveying IBM employees in 40 countries, **Geert Hofstede** (1980) found four dimensions: power distance (acceptance of unequal power), **individualism-collectivism** (a self defined as an independent 'I' or by the 'we' of family and group), masculinity-femininity (competition and achievement versus care) and uncertainty avoidance (discomfort with ambiguity, eased by rules). Long-term orientation and indulgence-restraint were added later. Collectivism brings more conformity (Section 3.7), a weaker fundamental attribution error (Section 3.3) and social striving (Section 3.10).

Ethnic-sensitive practice (**Wynetta Devore** and **Elfriede Schlesinger**, 1981) uses descriptive knowledge specific to different racial, cultural and ethnic groups in every phase of practice, so that the worker grasps the client's 'ethnic reality', the outlook shaped jointly by ethnic group and social class. Three neighbouring terms differ: ethno-psychotherapy (**Georges Devereux**) joins psychoanalysis and anthropology to treat clients from other cultures; sensitivity training is the T-group method (Section 3.1); and matching assigns a worker of the client's own ethnicity. **Cultural competence** (**Derald Wing Sue** and colleagues) combines awareness of one's own biases, knowledge of the client's worldview and culturally appropriate skills.

How does a worker learn a world not his own? The **ethnographic interview** of the anthropologist **James Spradley** (1979), adapted for social work by **James Leigh** (1998), treats the client as a cultural guide who explains what things mean in their world: 'Can you tell me why this band is so popular among your age group?' A psychodynamic inquiry ('What does that remind you of from childhood?') seeks unconscious conflict; a task-focused inquiry ('What could you do before next week?') seeks concrete steps; a strengths-based inquiry ('How have you coped so far?'), from the **strengths perspective** of **Dennis Saleebey** (1992), seeks resources rather than deficits. The **ecomap** of **Ann Hartman** (1978) draws the family in a circle linked to school, work, services and friends by lines marking strong, weak or stressful ties; the **genogram** charts three or more generations of a family.

> **! EXAM TRAP**
> Descriptive knowledge specific to racial, cultural and ethnic groups is **ethnic-sensitive practice**, not ethno-psychotherapy, sensitivity training or matching. Asking a client to explain what something means in their world is **ethnographic** inquiry.

Child maltreatment takes four forms: physical abuse, sexual abuse, emotional abuse and **neglect**, the failure to provide for a child's basic needs. Neglect has several categories: **physical neglect** (inadequate food, housing, clothing or hygiene); **health or medical neglect** (failing to immunise, to seek medical care or to follow medical directives); **inadequate supervision** (children left unattended); **educational neglect**; and **emotional neglect**. Using licensed day care is responsible substitute care, not neglect. Sexual offences against children in India fall under the POCSO Act 2012 (Chapter 7).

SCAN YOUR BRAIN

Fill in the blanks. The answers are below; cover them first.

1. Support that protects mainly under high stress fits the _____ hypothesis.
2. Attaching a genuine memory to the wrong origin is a _____ error.
3. The cognitive interview's fourth instruction is to change the _____.
4. Andrews and Bonta's principles are risk, need and _____.
5. The founder of social case work, author of *Social Diagnosis* (1917), was _____.
6. Group work was established as a method within social work by _____.
7. In Shulman's terms, the member who blocks painful topics is the _____.
8. India's first school of social work (Bombay, 1936) was the _____.

Answers: (1) stress-buffering, (2) source monitoring, (3) perspective, (4) responsivity, (5) Mary Ellen Richmond, (6) Grace Coyle, (7) gatekeeper, (8) Sir Dorabji Tata Graduate School of Social Work.

The Murder of Kitty Genovese, Revisited

Return to Kew Gardens with the whole chapter behind you. Allport's definition (Section 3.1) fits that night exactly: the presence of other neighbours, actual and imagined, changed what each one did. The public's verdict of apathy blamed character and ignored the situation, the fundamental attribution error (Section 3.3); diffusion of responsibility and the bystander research it launched are Section 3.8. Neighbours who took the cries for a drunken quarrel show how we read ambiguous scenes (Section 3.2); the persuasive power of a front-page story is Section 3.4; Moseley's violence belongs to Section 3.9. And the 38-witnesses legend, repeated in textbooks for some forty years before psychologists checked it, carries the lesson of the replication crisis: in social psychology even the stories need testing.

Chapter summary

3.1 Social Psychology: History, Theories, Concepts and Methods. Social psychology studies how individuals' thoughts, feelings and behaviour are influenced by the actual, imagined or implied presence of others (**Gordon Allport**, 1954). It grew from Le Bon's crowd (1895), Triplett's first experiment (1898) and the 1908 texts of McDougall and Ross; **Kurt Lewin**, its modern father, added the life space (person plus psychological environment), group dynamics and **action research**. A crisis around 1970, European work and the social cognition era followed. From sociology came Durkheim's **anomie** (normlessness), symbolic interactionism, socialisation and **resocialisation**, status and role. Its theories include Moscovici's **social representations** (1961); its methods run from observation to experiment, with debriefing after deception.

3.2 Social Cognition and First Impressions. Social cognition is how we interpret, remember and use social information. As **cognitive misers** (Fiske and Taylor, 1984) we lean on automatic processing, **schemas** (scripts, self-schemas, stereotypes), priming and heuristics. Schemas persist (belief perseverance) and can fulfil themselves: Merton's **self-fulfilling prophecy** (1948), the Pygmalion effect (Rosenthal and Jacobson, 1968) and the Golem effect. Social thought tilts towards negativity, optimism, the planning fallacy, counterfactuals, magical thinking, false consensus and

the illusion of control. Asch's **central traits** (warm-cold) and the **halo effect** (Thorndike, 1920) shape impressions; **thin slices** (Ambady and Rosenthal) are surprisingly accurate; impression management follows Goffman's dramaturgy and Jones and Pittman's strategies.

3.3 Communication, Nonverbal Cues and Attribution. Communication runs **encoding, channel, recipient, decoding**, with feedback and noise; the classic models are Lasswell (1948), Shannon and Weaver (1949), Schramm (1954) and Berlo's SMCR (1960). Nonverbal channels include face, gaze, body language, touch and paralanguage; Hall's proxemics sets intimate, personal, social and public zones, and lies are detected with only about 54 percent accuracy. **Attribution** theory began with Heider (1958): dispositional versus situational causes. Jones and Davis explained correspondent inferences, Kelley used consensus, consistency and distinctiveness, and Weiner added stability and controllability. The **fundamental attribution error** (Ross 1977), actor-observer effect, self-serving bias and just-world hypothesis tilt our explanations, more so in individualist cultures.

3.4 Attitudes, Persuasion and Cognitive Dissonance. An **attitude** is a learned evaluation of an object with affective, behavioural and cognitive parts (Rosenberg and Hovland, 1960), serving Katz's utilitarian, knowledge, ego-defensive and value-expressive functions. LaPiere (1934) exposed the attitude-behaviour gap; strong, specific attitudes formed by **direct experience** predict best, and the **theory of planned behaviour** (Ajzen) adds perceived control to attitude and subjective norm. Persuasion depends on source, message and audience (the Yale approach) and runs by the **central** or **peripheral route** (Petty and Cacioppo). **Reactance** (Brehm, 1966), forewarning and inoculation aid resistance. Festinger's **cognitive dissonance** (1957) explains why $1 beat $20; Bem's self-perception theory is its main rival.

3.5 Stereotypes, Prejudice and Intergroup Relations. A **stereotype** is a belief about a group, **prejudice** a negative attitude (Allport, 1954) and **discrimination** unfair action. Prejudice has been traced to the authoritarian personality, displaced frustration, social learning, **realistic conflict** (at Robbers Cave only **superordinate goals** ended hostility) and relative deprivation. Tajfel's minimal groups showed that categorisation alone breeds in-group favouritism; **social identity theory** (Tajfel and Turner, 1979) says much of our self-esteem comes from our groups. Low-prejudiced people inhibit automatically activated stereotypes by controlled processing (Devine, 1989); the IAT measures implicit bias. Social roles (Eagly) shape gender stereotypes, which build glass ceilings and sticky floors. Contact reduces prejudice under Allport's conditions.

3.6 Affiliation, Attraction and Close Relationships. The **need to belong** (Baumeister and Leary 1995) is fundamental; gregariousness is a **social** motive. Schachter (1959) showed that anxiety raises affiliation with others in the same plight, and we affiliate for information, anxiety reduction and **social comparison** (Festinger 1954), which also explains FOMO. Attraction grows from **propinquity** and **mere exposure** (Zajonc 1968), looks, matching, similarity and reciprocity. Exchange, equity and investment theories explain staying or leaving. Self-disclosure deepens relationships (descriptive facts, evaluative feelings). Love is passionate or companionate, triangular (Sternberg 1986) and shaped by attachment style. Weiss separated **emotional** from **social** loneliness.

3.7 Social Influence: Conformity, Compliance, Obedience and Power. Social influence takes three forms: **conformity** (matching a group with no request), **compliance** (yielding to a direct request) and **obedience** (following an authority's command). Norms may be descriptive or injunctive, prescriptive or proscriptive. Sherif's autokinetic study showed norms forming through infor-

mational influence; Asch's line studies showed normative conformity to an obviously wrong majority (about 37 percent of critical trials). Consistent minorities can convert majorities (Moscovici). Cialdini traced compliance tactics such as the foot-in-the-door and door-in-the-face to six principles. In Milgram's study 65 percent obeyed to 450 volts; Zimbardo's prison study showed the power of roles. French and Raven named six bases of power.

3.8 Prosocial Behaviour: Why and When People Help. **Altruism** is helping with no expected reward, at a cost to oneself. Batson's **empathy-altruism hypothesis** (the Elaine study) claims true altruism; Cialdini's **negative-state relief model** and the **empathic joy hypothesis** are egoistic, while kin selection (Hamilton), reciprocal altruism (Trivers), competitive altruism, defensive helping and norms add other motives. After the 1964 murder of **Kitty Genovese**, Latané and Darley showed the **bystander effect**: people help more when alone (seizure study, 85 versus 31 percent). Their five-step model locates its two classic causes, **diffusion of responsibility** and **pluralistic ignorance.** Costs (Piliavin), hurry, mood, models, gratitude, the altruistic personality and the threat-to-self-esteem model complete the picture.

3.9 Aggression: Its Roots, Causes and Control. **Aggression** is behaviour aimed at harming a living being who wants to avoid it (Baron and Richardson, 1994): **hostile** or **instrumental**, physical or verbal, direct, indirect or **relational.** Freud's thanatos and Lorenz's hydraulic instinct gave way to biology (low **serotonin**, MAOA with maltreatment) and to the **frustration-aggression hypothesis** (Dollard and colleagues, 1939), softened by Miller (1941) and reformulated by **Berkowitz** (1989) as negative affect plus aggressive cues (the weapons effect). Bandura's Bobo doll studies showed observational learning; the General Aggression Model integrates these. Provocation, exclusion, media violence, honour cultures, heat and alcohol raise aggression; **catharsis** fails, while apology, incompatible responses and self-regulation help.

3.10 Groups: Structure, Performance, Decisions and Leadership. A **group** is two or more interacting, interdependent people with a shared identity and a structure of roles, status, norms and cohesiveness. Moreno's **sociometry** (1934) maps choices in a **sociogram** of stars, isolates and cliques. Groups pass through Tuckman's **forming, storming, norming, performing** and adjourning. Others' presence brings **social facilitation** on easy tasks and impairment on hard ones (Triplett 1898; Zajonc 1965); pooled effort brings **social loafing** (Ringelmann), reversed as social striving in collectivist cultures; anonymity brings **deindividuation.** Discussion causes **group polarisation**, and cohesion with insulation causes Janis's **groupthink.** Leadership theory runs from traits and styles to Fiedler's contingency model and transformational leadership.

3.11 Applied Social Psychology: Law, Culture and Social Work. Social support protects health, chiefly as a **stress buffer** (Cohen and Wills, 1985); crowding is felt, density is counted (Stokols). Defence lawyers challenge eyewitnesses because **source monitoring errors** cause false identifications and confidence predicts accuracy weakly; double-blind sequential lineups and the **cognitive interview** help. Jurors build stories; effective rehabilitation is community-based, cognitive-behavioural and follows **risk-need-responsivity**. Social work grew from the Charity Organisation Society and settlements. Mary Richmond founded **case work** (intake to follow-up); Fischer (1973) doubted its effectiveness. Grace Coyle established **group work**; Trecker gave its principles and Shulman its member roles. India's first school was the **Sir Dorabji Tata Graduate School of Social Work** (1936).

Who's who

Table 3.33 The people in this chapter.

Name	When	Examined for
Gordon Willard Allport	1897 to 1967	Classic definition of social psychology (1954): the actual, imagined or implied presence of others
Robert A. Baron	Modern textbook	With Branscombe: the scientific study of the nature and causes of individual behaviour, feeling and thought in social situations
Nyla R. Branscombe	Modern textbook	With Baron: definition of social psychology
Gustave Le Bon	1895	The Crowd: the crowd mind, contagion and suggestion
Norman Triplett	1898	Usually called the first social psychology experiment (cyclists, fishing reels; social facilitation)
Max Ringelmann	Published 1913	Rope pulling: effort per person falls as the group grows (root of social loafing)
William McDougall	1908	One of the first two social psychology textbooks; instinct based
Edward Alsworth Ross	1908	Social Psychology; sociological, imitation and suggestion in crowds
Floyd Henry Allport	1924	Social Psychology: experimental and individual centred
Muzafer Sherif	1935, 1936	Group norm formation with the autokinetic effect; The Psychology of Social Norms
Kurt Lewin	1890 to 1947	Father of modern experimental social psychology; B = f(P, E) and the life space; action research (1946); group dynamics; Research Center for Group Dynamics at MIT (1945); T-groups
Ronald Lippitt	1939	With Lewin and White: autocratic, democratic and laissez-faire leadership climates
Ralph K. White	1939	With Lewin and Lippitt: leadership climates
Theodor W. Adorno	1950	The Authoritarian Personality (with colleagues)
Solomon Asch	1951	Conformity to a wrong majority
Carl Hovland	1953	Yale persuasion research
Leon Festinger	1954, 1957	Social comparison theory; cognitive dissonance theory
Fritz Heider	1958	The Psychology of Interpersonal Relations; cognitive consistency
Stanley Milgram	1963	Obedience studies; ethics controversy; high experimental, low mundane realism
Philip Zimbardo	1971	Stanford prison study; the power of roles; ethics controversy
Kenneth J. Gergen	1973, 1985	"Social psychology as history" (1973); social constructionism
Henri Tajfel	1919 to 1982	European social psychology; social identity theory (with Turner, 1979)
John Turner	1979	With Tajfel: social identity theory
Serge Moscovici	1925 to 2014	Social representations theory (1961): anchoring and objectification; minority influence
Emile Durkheim	1858 to 1917	Social facts, collective representations, anomie (The Division of Labour in Society 1893; Suicide 1897)
Robert King Merton	1910 to 2003	Strain theory of anomie (1938); anticipatory socialisation; role set (1957)
Ferdinand Tonnies	1887	Gemeinschaft (community) and Gesellschaft (society)
George Herbert Mead	1863 to 1931	Symbolic interactionism; Mind, Self and Society (1934); role taking, play and game stages, generalised other
Herbert Blumer	1937	Named symbolic interactionism
Charles Horton Cooley	1864 to 1929	Looking-glass self (1902); primary group (1909)
Erving Goffman	1922 to 1982	Total institutions, Asylums (1961)
Russell Barton	1959	Institutional neurosis
Ralph Linton	1936	Ascribed versus achieved status
William J. Goode	1960	Role strain
George C. Homans	1961	Social exchange theory
John Thibaut	1959	With Kelley: social exchange, The Social Psychology of Groups
Harold Kelley	1959	With Thibaut: social exchange
Peter Blau	1964	Social exchange: Exchange and Power in Social Life

Name	When	Examined for
Albert Bandura	1977, 1986	Social learning and social cognitive theory
Sigmund Freud	1921	Group Psychology and the Analysis of the Ego; not the founder of action research
Edward O. Wilson	1975	Sociobiology
Michael A. Hogg	2000s	Uncertainty-identity theory; social identity
Roy F. Baumeister	1995	With Leary: the need to belong
Mark R. Leary	1995	Need to belong (with Baumeister); sociometer theory of self-esteem
Richard E. Petty	1986	With Cacioppo: dual-process (elaboration likelihood) model
John T. Cacioppo	1986	With Petty: dual-process model
Ignacio Martin-Baro	1942 to 1989	Liberation psychology
Gustav Theodor Fechner	1860	Psychophysics (distractor for Lewin; Chapter 1)
William James	1890	Functionalism (distractor for Lewin; Chapter 1)
Elliot Aronson	1968	With Carlsmith: experimental versus mundane realism
J. Merrill Carlsmith	1968	With Aronson: experimental versus mundane realism
Martin Orne	1962	Demand characteristics
Robert Rosenthal	1966	Experimenter expectancy effects
Gardner Murphy	1953	UNESCO study of social tensions in India, In the Minds of Men
Durganand Sinha	1922 to 1998	Indigenisation of psychology in India
Jai B. P. Sinha	1980	Nurturant-task leader
Udai Pareek	1925 to 2010	Organisational role stress
Susan Fiske	1984	The cognitive miser (with Shelley Taylor); later the motivated tactician (1991)
Shelley Taylor	1984	The cognitive miser, with Susan Fiske
Frederic Charles Bartlett	1886 to 1969	Origin of the schema concept in reconstructive memory (1932)
Hazel Markus	1977	Self-schemas
Roger Schank	1977	Scripts (event schemas), with Robert Abelson
Robert Abelson	1977	Scripts (event schemas), with Roger Schank
John Bargh	1996	Automatic priming: the elderly prime slowed walking (with Mark Chen and Lara Burrows)
Mark Chen	1996	Co-author of the elderly-priming study
Lara Burrows	1996	Co-author of the elderly-priming study
Stéphane Doyen	2012	Failed replication of the elderly-priming study (slowing only when experimenters expected it)
Betsy Sparrow	2006	Unpriming, with Daniel Wegner
Daniel Wegner	1948 to 2013	Thought suppression and ironic rebound, the white bear (1987); unpriming (2006)
Lee Ross	1942 to 2021	Belief perseverance (1975); false consensus effect (1977); planning fallacy study (1994)
Mark Lepper	1975	Belief perseverance, with Ross and Hubbard
Michael Hubbard	1975	Belief perseverance, with Ross and Lepper
Lenore Jacobson	1968	Pygmalion in the Classroom, with Robert Rosenthal
Elisha Babad	1982	Golem effect, with Jacinto Inbar and Robert Rosenthal
Jacinto Inbar	1982	Golem effect, with Babad and Rosenthal
Mark Snyder	1974, 1977	Self-monitoring (1974); behavioural confirmation telephone study (1977)
Elizabeth Tanke	1977	Behavioural confirmation telephone study, with Snyder and Berscheid
Ellen Berscheid	1977	Behavioural confirmation telephone study, with Snyder and Tanke
Carl Word	1974	Interviewer study of the self-fulfilling prophecy, with Zanna and Cooper
Mark Zanna	1974	Interviewer study, with Word and Cooper
Joel Cooper	1974	Interviewer study, with Word and Zanna
Émile Durkheim	1858 to 1917	Anomie
Amos Tversky	1937 to 1996	Heuristics of judgement (1974); planning fallacy (1979)
Daniel Kahneman	1934 to 2024	Heuristics of judgement (1974); planning fallacy (1979)

Name	When	Examined for
Norbert Schwarz	1983, 1991	Ease of retrieval in availability (1991); affect as information, with Gerald Clore (1983)
Paul Slovic	2000s	Affect heuristic
Paul Rozin	1990, 2001	Magical thinking: laws of contagion and similarity; negativity bias (with Edward Royzman, 2001)
Edward Royzman	2001	Negativity bias, with Paul Rozin
Neil Weinstein	1980	Optimistic bias (unrealistic optimism)
Roger Buehler	1994	Planning fallacy study, with Dale Griffin and Lee Ross
Dale Griffin	1994	Planning fallacy study
Victoria Medvec	1995, 2000	Olympic medallists and counterfactual thinking (1995); spotlight effect (2000)
Scott Madey	1995	Olympic medallists and counterfactual thinking
Thomas Gilovich	1995, 2000	Bronze happier than silver medallists (1995); spotlight effect (2000)
David Greene	1977	False consensus effect, with Ross and House
Pamela House	1977	False consensus effect, with Ross and Greene
Ellen Langer	born 1947	Illusion of control, the lottery-ticket study (1975)
Jeff Greenberg	1986	Terror management theory, with Solomon and Pyszczynski
Sheldon Solomon	1986	Terror management theory
Tom Pyszczynski	1986	Terror management theory
Gerald Clore	1983	Affect as information, the sunny and rainy day study, with Norbert Schwarz
Joseph Forgas	1995	Affect infusion model
Jerome Bruner	1915 to 2016	Implicit personality theory, with Renato Tagiuri (1954)
Renato Tagiuri	1954	Implicit personality theory, with Jerome Bruner
Edward Lee Thorndike	1874 to 1949	Named the halo effect (1920) from officers' ratings of soldiers
Norman Anderson	1965	Information integration theory: weighted averaging versus additive models of impression formation
Nalini Ambady	1959 to 2013	Thin slicing, with Robert Rosenthal (1992, 1993)
Malcolm Gladwell	2005	Popularised thin slicing in Blink
Janine Willis	2006	100-millisecond face judgements, with Alexander Todorov
Alexander Todorov	2006	100-millisecond face judgements, with Janine Willis
Edward E. Jones	1926 to 1993	Ingratiation (1964); self-presentation strategies with Pittman (1982); self-handicapping with Berglas (1978)
Thane Pittman	1982	Five strategies of self-presentation, with Edward E. Jones
Steven Berglas	1978	Self-handicapping, with Edward E. Jones
Kenneth Savitsky	2000	Spotlight effect, with Gilovich and Medvec
Roos Vonk	1998	Slime effect
Harold Lasswell	1902 to 1978	Communication model (1948): who says what, in which channel, to whom, with what effect
Claude Shannon	1916 to 2001	Shannon and Weaver model of communication (1949); introduced noise
Warren Weaver	1894 to 1978	Shannon and Weaver model (1949): source, transmitter, channel, noise, receiver, destination
Wilbur Schramm	1907 to 1987	Interactive, circular model of communication (1954); field of experience
David Berlo	1929 to 1996	SMCR model of communication (1960): source, message, channel, receiver
Alex Bavelas	1913 to 1993	Communication networks in small groups (1950)
Harold Leavitt	1922 to 2007	Communication networks (1951): wheel fastest and most accurate on simple tasks, circle highest morale
Paul Ekman	born 1934	Six universal facial expressions plus contempt; display rules; microexpressions
Wallace Friesen	1969 onwards	With Ekman: universal facial expressions and display rules
Ray Birdwhistell	1918 to 1994	Founded kinesics, the study of body movement as communication (Introduction to Kinesics, 1952)

Name	When	Examined for
Edward T. Hall	1914 to 2009	Proxemics (1963); four zones of personal space in The Hidden Dimension (1966); high- and low-context cultures (1976)
Charles Bond	2006	Meta-analysis with DePaulo: lie detection accuracy about 54 percent
Bella DePaulo	2006	Meta-analysis with Bond: lie detection accuracy about 54 percent
Fred Fiedler	1922 to 2017	Contingency theory of leadership
Keith Davis	1965	Correspondent inference theory with Jones: free choice, non-common effects, low social desirability
Bernard Weiner	born 1935	Attribution theory of achievement and emotion: locus, stability, controllability
Lyn Abramson	1978	Attributional reformulation of learned helplessness: internal, stable, global causes
Martin Seligman	born 1942	Learned helplessness; its attributional reformulation (1978)
John Teasdale	1978	Attributional reformulation of learned helplessness
Daniel Gilbert	born 1957	Two-step model of attribution; review of the correspondence bias with Malone (1995)
Patrick Malone	1995	Review of the correspondence bias with Gilbert
Victor Harris	1967	Castro essay study of the fundamental attribution error, with Jones
Richard Nisbett	born 1941	Actor-observer effect with Jones (1971)
Michael Storms	1973	Videotape study that reversed the actor-observer effect
Dale Miller	1975	Review of the self-serving bias with Michael Ross
Michael Ross	1975	Review of the self-serving bias with Dale Miller
Melvin Lerner	1965, 1980	Just-world hypothesis; victim blaming
Carolyn Simmons	1966	Lerner and Simmons experiment: observers derogated an innocent victim they could not help
Zick Rubin	1975	Just World Scale, with Peplau
Letitia Anne Peplau	1975	Just World Scale, with Rubin
Thomas Pettigrew	1979	Ultimate attribution error
Kenneth Dodge	1980	Hostile attribution bias in aggressive children
Kelly Shaver	1970	Defensive attribution
Joan Miller	1984	Hindu Indians in Mysore gave more situational and fewer dispositional explanations than Americans; the gap grew with age
Michael Morris	1994	With Peng: English-language newspapers explained murders dispositionally, a Chinese-language newspaper situationally
Kaiping Peng	1994	With Morris: culture and attribution in newspaper reports of murders
Timothy Wilson	1982	Attribution retraining of worried first-year students, with Linville
Patricia Linville	1982	Attribution retraining of worried first-year students, with Wilson
Milton J. Rosenberg	1960	Tricomponent (ABC) model of attitudes, with Carl Hovland
Carl Iver Hovland	1912 to 1961	Yale communication approach, Communication and Persuasion (1953); sleeper effect with Weiss (1951); one- versus two-sided messages (1949); ABC model (1960)
Irving Lester Janis	1918 to 1990	Yale communication approach (1953); fear appeals with Feshbach (1953)
Harold H. Kelley	1921 to 2003	Co-author of Communication and Persuasion (1953), the Yale approach
Daniel Katz	1903 to 1998	Four functions of attitudes: utilitarian, knowledge, ego-defensive, value-expressive (1960)
M. Brewster Smith	1956	Functions of attitudes: object appraisal, social adjustment, externalisation (with Bruner and White)
Jerome Seymour Bruner	1956	Functions of attitudes with Smith and White
Robert W. White	1956	Functions of attitudes with Smith and Bruner
Emory S. Bogardus	1925	Social distance scale (Section 2.8)
Louis Leon Thurstone	1928	Equal-appearing intervals attitude scale (Section 2.8)
Rensis Likert	1932	Summated ratings attitude scale (Section 2.8)
Louis Guttman	1944	Cumulative (scalogram) attitude scale (Section 2.8)
Charles E. Osgood	1916 to 1991	Congruity theory with Tannenbaum (1955); semantic differential (1957)
Percy H. Tannenbaum	1955	Congruity theory with Osgood

Name	When	Examined for
Arthur W. Staats	1958	Classical conditioning of attitudes: nationality names paired with pleasant or unpleasant words (with Carolyn Staats)
Carolyn K. Staats	1958	Classical conditioning of attitudes (with Arthur Staats)
Jon A. Krosnick	1992	Subliminal conditioning of attitudes towards a stranger
Abraham Tesser	1993	Heritable attitudes are more accessible and resistant to change
Richard Tracy LaPiere	1899 to 1986	Attitude-behaviour gap: Chinese couple refused once in about 250 places, yet over 90 percent of letters refused (1934)
Allan W. Wicker	1969	Review: attitude-behaviour correlations rarely above .30
John Sivacek	1982	Vested interest and the drinking-age campaign (with Crano)
William D. Crano	1982	Vested interest strengthens the attitude-behaviour link
Dennis T. Regan	1977	Cornell housing shortage: direct experience strengthens attitude-behaviour consistency (with Fazio)
Russell H. Fazio	1977, 1986, 1990	Direct experience (1977, 1981); attitude-to-behaviour process model (1986); MODE model (1990); new look of dissonance with Cooper (1984)
Mark P. Zanna	1974, 1981	Direct experience and attitude accessibility (with Fazio, 1981); placebo misattribution study of dissonance arousal (with Cooper, 1974)
Icek Ajzen	born 1942	Theory of reasoned action (1975) and planned behaviour (1985, 1991); principle of compatibility (1977)
Martin Fishbein	1936 to 2009	Theory of reasoned action with Ajzen (1975); principle of compatibility (1977)
Andrew R. Davidson	1979	Birth-control pill study of attitude-behaviour compatibility (with Jaccard)
James J. Jaccard	1979	Compatibility study with Davidson: specific attitudes predict specific acts
Deborah A. Prentice	1993	Perceived campus drinking norms and pluralistic ignorance (with Miller)
Dale T. Miller	1993	Perceived campus drinking norms (with Prentice)
Arthur A. Lumsdaine	1949	One-sided versus two-sided messages (with Hovland and Sheffield)
Fred D. Sheffield	1949	One-sided versus two-sided messages (with Hovland and Lumsdaine)
Seymour Feshbach	1953	Fear appeals with Janis: mild fear changed tooth-brushing most
Norman Miller	1959	Primacy versus recency in persuasion (with Campbell)
Donald T. Campbell	1959	Primacy versus recency in persuasion (with Miller)
R. Goldman	1981	Involvement study of the elaboration likelihood model (with Petty and Cacioppo)
William J. McGuire	1925 to 2007	Information-processing chain of persuasion; inoculation theory (1961)
Walter Weiss	1951	Source credibility and the sleeper effect (with Hovland)
Robert Oppenheimer	1904 to 1967	Physicist used as the credible source in Hovland and Weiss (1951)
Shelly Chaiken	1980	Heuristic-systematic model of persuasion
Jack Williams Brehm	1928 to 2009	Reactance theory (1966); free-choice dissonance and spreading of alternatives (1956); trivialisation (1995)
Jonathan L. Freedman	1965	Forewarning increases resistance to persuasion (with Sears)
David O. Sears	1965	Forewarning study with Freedman
Charles G. Lord	1979	Biased assimilation and attitude polarisation (with Ross and Lepper)
Mark R. Lepper	1979	Biased assimilation (with Lord and Ross)
Linda Simon	1995	Trivialisation as dissonance reduction (with Greenberg and Brehm)
Claude Mason Steele	1988	Self-affirmation reduces dissonance
Clark Leonard Hull	1884 to 1952	Drive-reduction theory (Section 1.10)
James Merrill Carlsmith	1936 to 1984	$1 versus $20 induced-compliance study with Festinger (1959); forbidden toy with Aronson (1963)
Judson Mills	1959	Severe initiation study of effort justification (with Aronson)
Henry W. Riecken	1956	When Prophecy Fails, belief disconfirmation (with Festinger and Schachter)
Stanley Schachter	1922 to 1997	When Prophecy Fails (1956) with Festinger and Riecken
Jeff Stone	1994	Hypocrisy induction and condom buying (with Aronson and colleagues)
Daryl J. Bem	born 1938	Self-perception theory (1967, 1972), rival to dissonance
James T. Tedeschi	1971	Impression management theory of dissonance effects

Name	When	Examined for
Shinobu Kitayama	2004	Culture and dissonance: Japanese show post-decision dissonance when others' views are salient
Walter Lippmann	1889 to 1974	Public Opinion (1922): stereotypes as pictures in our heads
Kenneth Braly	1933	Katz and Braly checklist study of Princeton students
William Graham Sumner	1840 to 1910	Folkways (1906): ethnocentrism, in-group and out-group
Robert Neil Butler	1927 to 2010	Coined ageism (1969)
Theodor Adorno	1903 to 1969	The Authoritarian Personality (1950); F-scale
Else Frenkel-Brunswik	1908 to 1958	Co-author of The Authoritarian Personality (1950)
Daniel Levinson	1920 to 1994	Co-author of The Authoritarian Personality (1950)
Nevitt Sanford	1909 to 1995	Co-author of The Authoritarian Personality (1950)
Bob Altemeyer	1981	Right-wing authoritarianism: submission, aggression, conventionalism
Jim Sidanius	1945 to 2021	Social dominance orientation (with Felicia Pratto)
Felicia Pratto	1990s	Social dominance orientation (with Jim Sidanius)
Robert Sears	1908 to 1989	Hovland and Sears (1940) cotton prices and lynchings
Donald Thomas Campbell	1916 to 1996	Named realistic conflict theory (1965)
Samuel Stouffer	1900 to 1960	The American Soldier (1949): relative deprivation
Walter Garrison Runciman	1934 to 2020	Egoistic versus fraternal relative deprivation (1966)
Ted Robert Gurr	1936 to 2017	Why Men Rebel (1970): expectations versus attainments
Michael Billig, R. P. Bundy and Claude Flament	1971	Co-authors of the minimal group experiments with Tajfel
Paul Klee and Wassily Kandinsky	Painters	Their paintings were the basis of the minimal group assignment
Marilynn Brewer	1991	Optimal distinctiveness theory; decategorisation (1984, with Miller)
George Quattrone	1980	Out-group homogeneity effect (with Jones)
Edward Ellsworth Jones	1926 to 1993	Out-group homogeneity effect (Quattrone and Jones, 1980)
David Hamilton	1976	Distinctiveness-based illusory correlation (with Gifford)
Robert Gifford	1976	Illusory correlation (Hamilton and Gifford)
C. Neil Macrae	1994	Stereotypes as energy-saving devices (with Milne and Bodenhausen)
Alan Milne	1994	Stereotypes as energy-saving devices
Galen Bodenhausen	1994	Stereotypes as energy-saving devices
Renee Weber	1983	Bookkeeping, conversion and subtyping models (with Crocker)
Jennifer Crocker	1983	Models of stereotype change (with Weber)
Amy Cuddy	2002	Stereotype content model
Peter Glick	1996	Ambivalent sexism: hostile and benevolent (with Fiske); stereotype content model (2002)
Jun Xu	2002	Stereotype content model
Claude Steele	1995	Stereotype threat (with Joshua Aronson)
Joshua Aronson	1995	Stereotype threat (with Steele)
Steven Spencer	1999	Stereotype threat in women's mathematics performance (with Steele and Quinn)
Diane Quinn	1999	Stereotype threat and women's mathematics performance
Patricia Devine	1989	Dissociation model: automatic activation, controlled inhibition by low-prejudiced people
Anthony Greenwald	1998	Implicit Association Test (with McGhee and Schwartz)
Debbie McGhee	1998	Implicit Association Test
Jordan Schwartz	1998	Implicit Association Test
Russell Fazio	1995	Bona fide pipeline
John McConahay	1986	Modern racism and the Modern Racism Scale
David Sears	1980s	Symbolic racism
Samuel Gaertner	1986	Aversive racism (with Dovidio); common in-group identity model (2000)
John Dovidio	1986	Aversive racism; common in-group identity model (2000, with Gaertner)
Derald Wing Sue	2007	Microaggressions: taxonomy of everyday slights (with colleagues)
Chester Pierce	1970	Coined the term microaggression

Name	When	Examined for
Alice Eagly	1987	Social role theory of gender
Wendy Wood	1999	Social role theory with Eagly; reanalysis of Buss's mate preference data
David Buss	1989	Evolutionary account of sex differences in mate preferences (37 cultures)
Clea Mehta and JoNell Strough	2009	Sex segregation in friendships across the life span (a distractor for social role theory)
Craig Anderson and Brad Bushman	2002	General Aggression Model (Section 3.9); not linked to social role theory
Michelle Ryan	2005	Glass cliff (with Haslam)
Alexander Haslam	2005	Glass cliff (with Ryan)
Catherine White Berheide	1992	Sticky floor
Joan Williams	2000s	Maternal wall: bias against mothers
Christine Williams	1992	Glass escalator
Linda Tropp	2006	Meta-analysis of over 500 contact studies (with Pettigrew)
Stephen Wright	1997	Extended contact effect
Miles Hewstone	1986	Mutual intergroup differentiation (with Brown)
Rupert Brown	1986	Mutual intergroup differentiation (with Hewstone)
Adam Powell, Nyla Branscombe and Michael Schmitt	2005	Collective guilt framed as in-group advantage lowers racism
Kerry Kawakami	2000	Training to say no to stereotypes
Charles Stangor	2001	Social influence: learning peers are less prejudiced reduces stereotyping
Karla Hoff and Priyanka Pandey	2006	Caste identity made public lowered low-caste boys' performance (Uttar Pradesh)
Matt Lowe	2021	Caste-mixed cricket teams increased cross-caste friendships (Uttar Pradesh)
Ashutosh Varshney	2002	Hindu-Muslim civic associations and communal peace in Indian cities
Roy Baumeister	born 1953	The need to belong as a fundamental motive (with Mark Leary, 1995)
Mark Leary	1995	The need to belong (with Roy Baumeister)
Henry Murray	1893 to 1988	Listed the need for affiliation among human needs
David McClelland	1917 to 1998	Measured achievement, power and affiliation needs from TAT stories
Craig Hill	1987	Four affiliation motives: positive stimulation, emotional support, attention, social comparison
Kipling Williams	2000	Ostracism research; the Cyberball game
Donn Byrne	1970, 1971	Reinforcement-affect model (with Gerald Clore); law of attraction (proportion of similar attitudes)
Thomas Wills	1981	Downward social comparison as self-enhancement
Andrew Przybylski	2013	Defined and measured the fear of missing out (FOMO)
Kurt Back	1950	Westgate West housing study (with Festinger and Schachter)
Mady Segal	1974	Police trainees seated alphabetically befriended alphabetical neighbours
Robert Zajonc	1923 to 2008	Mere exposure effect (1968)
Richard Moreland	1992	Classroom exposure study (with Scott Beach)
Scott Beach	1992	Classroom exposure study (with Richard Moreland)
Theodore Mita	1977	Mirror-image face preference (with Dermer and Knight)
Karen Dion	1972	What is beautiful is good (with Berscheid and Walster)
Elaine Hatfield (Walster)	born 1937	Computer dance and matching hypothesis (1966); equity theory (1978); passionate and companionate love; Passionate Love Scale (1986)
Judith Langlois	1990	Averaged faces are attractive (with Lori Roggman)
Lori Roggman	1990	Averaged faces are attractive (with Judith Langlois)
Bernard Murstein	1972	Couples matched in physical attractiveness
Theodore Newcomb	1903 to 1984	Acquaintance study of transfer students; similarity breeds friendship (1961)
Milton Rosenbaum	1986	Repulsion hypothesis
Robert Winch	1958	Complementarity of needs in mate selection
Darwyn Linder	1965	Gain-loss effect (with Elliot Aronson)
George Homans	1910 to 1989	Social exchange theory (1961)

Name	When	Examined for
G. William Walster	1978	Equity theory (with Elaine Walster and Ellen Berscheid)
Caryl Rusbult	1952 to 2010	Investment model of commitment (1980); exit, voice, loyalty, neglect
Margaret Clark	1979	Communal versus exchange relationships (with Judson Mills)
Sidney Jourard	1926 to 1974	Self-disclosure; The Transparent Self (1964); Jourard Self-Disclosure Questionnaire
Irwin Altman	1973	Social penetration theory: breadth and depth (with Dalmas Taylor)
Dalmas Taylor	1973	Social penetration theory (with Irwin Altman)
T. L. Morton	1978	Descriptive versus evaluative self-disclosure
Susan Sprecher	1986	Passionate Love Scale (with Elaine Hatfield)
Robert Sternberg	born 1949	Triangular theory of love (1986): intimacy, passion, commitment
John Alan Lee	1973	Six love styles: eros, ludus, storge, pragma, mania, agape
Donald Dutton	1974	Suspension-bridge study of misattributed arousal (with Arthur Aron)
Arthur Aron	1974	Suspension-bridge study (with Donald Dutton)
Cindy Hazan	1987	Adult attachment styles: secure, avoidant, anxious-ambivalent (with Phillip Shaver)
Phillip Shaver	1987	Adult attachment styles (with Cindy Hazan)
Kim Bartholomew	1991	Four-category model: secure, preoccupied, dismissing, fearful (with Leonard Horowitz)
Leonard Horowitz	1991	Four-category attachment model (with Kim Bartholomew)
Usha Gupta	1982	Love in love versus arranged marriages in Jaipur (with Pushpa Singh)
Pushpa Singh	1982	Love in love versus arranged marriages in Jaipur (with Usha Gupta)
Steve Duck	1982	Phases of relationship dissolution: intrapsychic, dyadic, social, grave-dressing
John Gottman	born 1942	Four horsemen that predict divorce (Chapter 7)
Daniel Perlman	1982	Definition of loneliness (with Letitia Anne Peplau)
Robert Weiss	1973	Emotional versus social isolation (loneliness)
Daniel Russell	1978, 1980	UCLA Loneliness Scale
John Cacioppo	1951 to 2018	Loneliness and health
Herbert Kelman	1958	Three processes of influence: compliance, identification and internalisation
Bruce Tuckman	1965	Stages of group development (forming, storming, norming, performing); norming is a distractor for conformity
Robert Cialdini	1975 to 1990	Six principles of compliance in Influence (1984); door-in-the-face (1975) and low-ball (1978); descriptive and injunctive norms and the littering study (1990)
Raymond Reno	1990	Descriptive versus injunctive norms and focus theory, with Cialdini and Kallgren
Carl Kallgren	1990	Descriptive versus injunctive norms and focus theory, with Cialdini and Reno
Robert Jacobs	1961	Arbitrary autokinetic norm passed across generations, with Donald Campbell
Donald Campbell	1916 to 1996	Arbitrary autokinetic norm passed across generations, with Robert Jacobs (1961)
Richard Crutchfield	1955	Booth and light-panel apparatus for testing conformity without confederates
Rod Bond	1996	Meta-analysis of Asch-type studies with Peter Smith: more conformity in collectivist cultures, decline over time
Peter Smith	1996	Meta-analysis of Asch-type studies with Rod Bond
Morton Deutsch	1920 to 2017	Normative versus informational social influence, with Harold Gerard (1955)
Harold Gerard	1955	Normative versus informational social influence, with Morton Deutsch
Linda Carli	1981	Review with Alice Eagly on gender and influenceability
C. R. Snyder	1980	Need for uniqueness, with Howard Fromkin
Howard Fromkin	1980	Need for uniqueness, with C. R. Snyder

Name	When	Examined for
Elisabeth Lage	1969	Blue-green slide study of minority influence with Moscovici
Martine Naffrechoux	1969	Blue-green slide study of minority influence with Moscovici
Edwin Hollander	1958	Idiosyncrasy credits
Elaine Hatfield	1994	Emotional contagion, with Cacioppo and Rapson
Richard Rapson	1994	Emotional contagion, with Hatfield and Cacioppo
Jerry Burger	1986; 2009	That's-not-all technique (1986); partial replication of Milgram (2009)
Daniel Howard	1990	Foot-in-the-mouth technique
Michael Santos, Craig Leve and Anthony Pratkanis	1994	Pique technique
Jonathan Freedman	1966	Foot-in-the-door technique, with Scott Fraser ('Be a safe driver' sign)
Scott Fraser	1966	Foot-in-the-door technique, with Jonathan Freedman
Robert Jay Lifton	1961	Thought reform and the psychology of totalism; milieu control
Jim Jones	1931 to 1978	Cult leader of the Jonestown mass murder and suicide (1978)
Diana Baumrind	1964	Ethical critique of Milgram's obedience study
Hannah Arendt	1906 to 1975	Banality of evil (1963), from the Eichmann trial
Adolf Eichmann	1906 to 1962	Nazi official whose trial Arendt reported
Charles Hofling	1966	Hospital nurses study: 21 of 22 began to give an overdose ordered by phone
Craig Haney	1971	Co-researcher of the Stanford Prison Experiment
Curtis Banks	1971	Co-researcher of the Stanford Prison Experiment
Stephen Reicher	2006	BBC Prison Study with Alexander Haslam
John French	1959	Five bases of social power, with Bertram Raven
Bertram Raven	1959; 1965	Bases of social power with John French; added informational power (1965)
Mark H. Davis	1980	Interpersonal Reactivity Index: perspective taking, fantasy, empathic concern, personal distress
C. Daniel Batson	1981, 1991	Empathy-altruism hypothesis; the Elaine shock study with easy versus difficult escape; Good Samaritan study with Darley (1973)
K. D. Smith, J. P. Keating and Ezra Stotland	1989	Empathic joy hypothesis: helping for the pleasure of the victim's relief, which needs feedback
William D. Hamilton	1936 to 2000	Kin selection theory (1964)
Eugene Burnstein, Christian Crandall and Shinobu Kitayama	1994	More help to closer and younger kin, most clearly when life is at stake
Robert Trivers	born 1943	Reciprocal altruism (1971)
Charlie Hardy and Mark Van Vugt	2006	Competitive altruism: helping to gain status
Arie Nadler	1986, 2009	Defensive helping; threat-to-self-esteem model (with Fisher); dependency- versus autonomy-oriented help
Jeffrey Fisher	1986	Threat-to-self-esteem model of receiving help (with Nadler)
Alvin Gouldner	1920 to 1980	Norm of reciprocity (1960)
Leonard Berkowitz	1926 to 2016	Social responsibility norm (with Daniels, 1963)
L. R. Daniels	1963	Social responsibility norm (with Berkowitz)
Kitty Genovese	1935 to 1964	Murdered in Kew Gardens, Queens, on 13 March 1964; her case prompted bystander research
Bibb Latané	born 1937	Bystander effect with Darley: seizure and smoke-filled room studies (1968), The Unresponsive Bystander (1970), five-step decision model
John Darley	1938 to 2018	Bystander effect with Latané (1968, 1970); Good Samaritan study with Batson (1973)
Judith Rodin	born 1944	Lady in distress study with Latané (1969); New York subway study with the Piliavins (1969)
Martin Luther King Jr.	1929 to 1968	Wrong option: his assassination did not prompt bystander research
Emmett Till	1941 to 1955	Wrong option: his lynching did not prompt bystander research
Malala Yousafzai	born 1997	Wrong option: her shooting (2012) did not prompt bystander research
Peter Fischer	2011	Meta-analysis: the bystander effect is weaker in dangerous emergencies
Irving Piliavin	1969	New York subway study; arousal: cost-reward model

Name	When	Examined for
Jane Allyn Piliavin	1969, 1981	New York subway study; arousal: cost-reward model
Alice Isen	1942 to 2012	Feel good, do good: cookie and dime studies with Levin (1972)
Paula Levin	1972	Cookie and dime studies with Isen
James Bryan and Mary Ann Test	1967	Prosocial models: the flat tyre study
Tobias Greitemeyer and Silvia Osswald	2010	Prosocial video games increase helping
Jonathan Haidt	born 1963	Elevation: the uplift felt on witnessing kindness (2000)
Adam Grant and Francesca Gino	2010	Gratitude raises helpers' self-worth and self-efficacy; only self-worth carries later helping
Jean Twenge	2007	Social exclusion reduces helping
Chen-Bo Zhong	2010	Darkness and anonymity reduce prosocial behaviour
Sanford DeVoe and Jeffrey Pfeffer	2010	Thinking of time in money terms reduces volunteering
Hans Werner Bierhoff, Renate Klein and Peter Kramp	1991	Components of the altruistic personality: empathy, belief in a just world, social responsibility, internal locus of control, low egocentrism
Samuel Oliner and Pearl Oliner	1988	Rescuers of Jews in Nazi Europe; The Altruistic Personality
Allen Omoto and Mark Snyder	1995	Volunteer motives among AIDS volunteers
Deborah R. Richardson	1994	Definition of aggression with Baron
Nicki R. Crick	1995	Relational aggression (with Grotpeter); more used by girls
Jennifer K. Grotpeter	1995	Relational aggression with Crick
Konrad Lorenz	1903 to 1989	On Aggression (1966): fighting instinct, hydraulic model, ritualised fighting, appeasement gestures
Avshalom Caspi	2002	Low-activity MAOA gene plus childhood maltreatment predicts antisocial behaviour
John Dollard	1900 to 1980	Frustration-aggression hypothesis (1939), with Doob, Miller, Mowrer and Sears
Leonard W. Doob	1909 to 2000	Co-author of the frustration-aggression hypothesis (1939)
Neal E. Miller	1909 to 2002	Co-author of the frustration-aggression hypothesis (1939); revision (1941): aggression is only one response to frustration
O. H. Mowrer	1907 to 1982	Co-author of the frustration-aggression hypothesis (1939)
Robert R. Sears	1908 to 1989	Co-author of the frustration-aggression hypothesis (1939)
Anthony LePage	1967	Weapons effect study with Berkowitz
Dolf Zillmann	1971	Excitation transfer theory: leftover arousal intensifies later anger
Richard Lazarus	1922 to 2002	Appraisal and stress (Section 1.11); wrong option for the reformulation of the frustration-aggression hypothesis
Dorothea Ross	1961	Bobo doll study with Bandura and Sheila Ross
Sheila A. Ross	1961	Bobo doll study with Bandura and Dorothea Ross
L. Rowell Huesmann	1984 and 1986	Aggressive scripts; longitudinal media violence research with Eron
Craig A. Anderson	2002	General Aggression Model (with Bushman); violent video games; heat and violent crime
Brad J. Bushman	2002	General Aggression Model (with Anderson); venting study showing catharsis fails (2002); threatened egotism (1998)
Jean M. Twenge	2001	Social exclusion increases aggression
Leonard D. Eron	1972	Longitudinal study: TV violence viewed at 8 predicted aggression at 19 and serious crime by 30
Richard E. Nisbett	1996	Culture of honour (with Cohen)
Dov Cohen	1996	Culture of honour (with Nisbett)
Joseph A. Vandello	2008	Precarious manhood (with Bosson)
Jennifer K. Bosson	2008	Precarious manhood (with Vandello); threatened men punched harder (2009)
Kenneth A. Dodge	1980	Hostile attribution bias in aggressive children
Paul A. Bell	1992	Negative affect escape model of heat and aggression (with Baron)
Claude M. Steele	1990	Alcohol myopia (with Josephs)
Robert A. Josephs	1990	Alcohol myopia (with Steele)

Name	When	Examined for
Dan Olweus	1993	Bullying (Bullying at School): repeated aggression against a weaker victim; bullies, victims, bully-victims
Herbert Hyman	1942	Coined the reference group
Brian Lickel	2000	Intimacy groups, task groups, social categories and loose associations, ranked by entitativity
Kenneth Benne	1948	Task, maintenance and self-oriented roles in groups, with Paul Sheats
Paul Sheats	1948	Task, maintenance and self-oriented roles in groups, with Kenneth Benne
John Levine	1982	Group socialisation model with Richard Moreland
Fabio Sani	2005	Schism: groups splinter when members see a change as subverting the group's identity
Jacob Levy Moreno	1889 to 1974	Founded sociometry and the sociogram (Who Shall Survive? 1934); founded psychodrama
Helen Hall Jennings	1934	Moreno's collaborator on Who Shall Survive? and sociometry
Wilhelm Dilthey	1833 to 1911	Verstehen, interpretive understanding (a distractor for sociometry)
Max Weber	1864 to 1920	Verstehen method; charismatic authority
Mary Ann Jensen	1977	Added adjourning to Tuckman's stages
Connie Gersick	1988	Punctuated equilibrium model of group development
Alexander Heingartner	1969	Cockroach social facilitation study with Zajonc and Herman
Edward Herman	1969	Cockroach social facilitation study with Zajonc and Heingartner
Nickolas Cottrell	1968	Evaluation apprehension explanation of social facilitation
Robert S. Baron	1986	Distraction-conflict theory of social facilitation
Alan Ingham	1974	Pseudo-group rope pulling showed motivation loss, not only coordination loss
Stephen Harkins	1979	Social loafing clapping and shouting study
Steven Karau	1993	Collective effort model with Kipling Williams; social compensation (1991)
P. Christopher Earley	1989, 1993	Social striving: Chinese trainees worked harder in groups
Norbert Kerr	1983	Sucker effect; free-rider effect with Steven Bruun
Steven Bruun	1983	Free-rider effect with Norbert Kerr
Albert Pepitone	1952	Named deindividuation with Festinger and Newcomb
Edward Diener	1946 to 2021	Halloween trick-or-treater study of deindividuation (1976)
Russell Spears	1995	SIDE model of deindividuation effects
Tom Postmes	1995	SIDE model of deindividuation effects
James H. Davis	1973	Social decision schemes: majority wins, truth wins, first shift
James Stoner	1961	Discovered the risky shift
Marisa Zavalloni	1969	Group polarisation with Serge Moscovici
Irving Janis	1918 to 1990	Groupthink (Victims of Groupthink, 1972; revised 1982): eight symptoms and remedies
Garold Stasser	1985	Hidden profiles and biased information sampling, with William Titus
William Titus	1985	Hidden profile study with Garold Stasser
Alex Osborn	1888 to 1966	Popularised brainstorming
Garrett Hardin	1915 to 2003	Tragedy of the commons (1968)
Charles Osgood	1916 to 1991	GRIT, graduated and reciprocated initiatives in tension reduction (1962)
J. Stacy Adams	1965	Equity theory of distributive justice
Ralph Stogdill	1948	Review showing traits weakly predict leadership
Ralph White	1939	Leadership styles study with Lewin and Lippitt
Robert Blake	1964	Managerial grid with Jane Mouton
Jane Mouton	1964	Managerial grid with Robert Blake
Paul Hersey	1969	Situational leadership with Kenneth Blanchard: telling, selling, participating, delegating
Kenneth Blanchard	1969	Situational leadership with Paul Hersey
Robert House	1971, 1977	Path-goal theory (1971); charismatic leadership theory (1977)
Victor Vroom	1973	Normative decision model with Philip Yetton

Name	When	Examined for
Philip Yetton	1973	Normative decision model with Victor Vroom
George Graen	1975	Leader-member exchange (LMX)
James MacGregor Burns	1918 to 2014	Transactional versus transforming leadership (Leadership, 1978)
Bernard Bass	1925 to 2007	Transformational leadership and its four components (1985)
Robert Greenleaf	1904 to 1990	Servant leadership (1970)
J. B. P. Sinha	1980	Nurturant-task leader, an Indian leadership model
Edward Chace Tolman	1886 to 1959	Cognitive map (distractor for the sociogram; Section 1.6)
Frederick Herzberg	1923 to 2000	Work motivation (Section 1.10)
Sheldon Cohen	1985	Stress-buffering hypothesis of social support (with Wills)
Thomas Ashby Wills	1985	Stress-buffering versus main-effect models of social support (with Cohen)
Daniel Stokols	1972	Density (physical) versus crowding (psychological)
John B. Calhoun	1917 to 1995	Crowded rat colonies and the behavioural sink (1962)
Ed Diener	1946 to 2021	Subjective well-being: life satisfaction, frequent positive and infrequent negative feelings
Marcia Johnson	1993	Source monitoring framework (with colleagues)
Donald Thomson	Twentieth century	Australian psychologist wrongly identified by a victim who had seen him on live television: a source error
Elizabeth Loftus	born 1944	Misinformation effect ('smashed' versus 'hit', with Palmer, 1974); weapon focus
John Palmer	1974	Loftus and Palmer study of leading questions and memory
Gary Wells	1998	Post-identification feedback effect; lineup reform
Ronald Fisher	1984	Cognitive interview (with Geiselman); not the statistician
Edward Geiselman	1984	Cognitive interview (with Fisher)
Ronald Aylmer Fisher	1890 to 1962	Statistician; a name trap for the cognitive interview
Nancy Pennington	1986	Story model of juror decision making (with Hastie)
Reid Hastie	1986	Story model of juror decision making (with Pennington)
Robert Ressler	1980s	FBI organised versus disorganised offender typology
John Douglas	1980s	FBI offender profiling (with Ressler)
David Canter	1986	Investigative psychology; bottom-up, statistical offender profiling
Robert Martinson	1927 to 1979	'What works?' (1974), read as 'nothing works' in offender rehabilitation
Don Andrews	1990	Risk-need-responsivity principles of offender rehabilitation (with Bonta)
James Bonta	1990	Risk-need-responsivity principles (with Andrews)
Samuel Barnett	1844 to 1913	Toynbee Hall, London, the first settlement house (1884)
Jane Addams	1860 to 1935	Hull House, Chicago (1889, with Ellen Gates Starr)
Ellen Gates Starr	1859 to 1940	Co-founder of Hull House (1889)
Clifford Manshardt	1936	First director of the Sir Dorabji Tata Graduate School of Social Work, Bombay
Mary Ellen Richmond	1861 to 1928	Founder of social case work: Social Diagnosis (1917), What Is Social Case Work? (1922)
Otto Rank	1884 to 1939	Will therapy, the basis of the functional school of case work
Gordon Hamilton	1940	Diagnostic school of case work (Theory and Practice of Social Case Work)
Florence Hollis	1964	Psychosocial approach; the person-in-situation
Charlotte Towle	1945	Common Human Needs
Jessie Taft	1930s	Functional (Rankian) school of case work, Pennsylvania
Virginia Robinson	1930s	Functional (Rankian) school of case work, Pennsylvania
Helen Harris Perlman	1957	Problem-solving model of case work; the four Ps
William Reid	1972	Task-centred casework (with Laura Epstein)
Laura Epstein	1972	Task-centred casework (with William Reid)
Felix Biestek	1957	Seven principles of the casework relationship
Joel Fischer	1973	'Is casework effective?': no good evidence that case work worked
Irwin Epstein	Contemporary	Social work research (a distractor for Fischer)

Name	When	Examined for
John Goyder	Contemporary	Survey nonresponse research (a distractor for Fischer)
Jacob Cohen	1923 to 1998	Statistician of kappa, effect size and power (a distractor for Fischer)
Grace Longwell Coyle	1892 to 1962	Established group work as a method in social work; Social Process in Organized Groups (1930)
Gisela Konopka	1910 to 2003	Group work in treatment settings
Harleigh B. Trecker	Mid-twentieth century	Ten principles of social group work
Robert Vinter	1960s	Remedial model of group work
William Schwartz	1960s	Reciprocal (mediating) model of group work
Ronald Toseland	Contemporary	Formed versus natural groups (with Rivas)
Robert Rivas	Contemporary	Formed versus natural groups (with Toseland)
Lawrence Shulman	Contemporary	Member roles in groups: scapegoat, deviant, gatekeeper, internal leader, defensive, quiet, monopoliser
James Garland	1965	Boston model of group development (with Jones and Kolodny)
Hubert Jones	1965	Boston model of group development
Ralph Kolodny	1965	Boston model of group development
Murray G. Ross	1955	Classic definition of community organisation
Jack Rothman	1968	Three models of community organisation: locality development, social planning, social action
Saul Alinsky	1909 to 1972	Social action organising; Rules for Radicals (1971)
Geert Hofstede	1928 to 2020	Cultural dimensions (1980): power distance, individualism-collectivism, masculinity-femininity, uncertainty avoidance
Wynetta Devore	1981	Ethnic-sensitive practice (with Schlesinger)
Elfriede Schlesinger	1981	Ethnic-sensitive practice (with Devore)
Georges Devereux	1908 to 1985	Ethno-psychotherapy (ethnopsychiatry)
James Spradley	1979	The ethnographic interview
James Leigh	1998	Ethnographic interviewing in social work: the client as cultural guide
Dennis Saleebey	1992	The strengths perspective in social work
Ann Hartman	1978	The ecomap

Key terms

social psychology, person-situation interaction, Psychological social psychology, sociological social psychology, life space, group dynamics, action research, T-group, replication crisis, Social facts, collective representations, anomie, symbolic interactionism, Socialisation, anticipatory socialisation, Resocialisation, total institutions, institutionalisation, status, role, Role taking, Role playing, Role conflict, Role strain, Role set, social representation, anchoring, objectification, experimental realism, mundane realism, mediating variable, moderating variable, debriefing, demand characteristics, Social cognition, cognitive misers, motivated tactician, Automatic processing, controlled processing, schema approach, schema, script, Priming, unpriming, belief perseverance, self-fulfilling prophecy, Pygmalion effect, Golem effect, behavioural confirmation, status quo heuristic, Negativity bias, Optimistic bias, Planning fallacy, Counterfactual thinking, Magical thinking, False consensus effect, Illusion of control, Terror management theory, affect infusion model, central traits, implicit personality theories, halo effect, thin slicing, impression management, dramaturgy, Self-monitoring, self-handicapping, spotlight effect, slime effect, Communication, encoding, decoding, feedback, noise, SMCR model, communication networks, Nonverbal communication, display rules, kinesics, Paralanguage, proxemics, high-context cultures, Microexpressions, Interchannel discrepancies, Attribution, naive psychology, dispositional attribution, situational attribution, correspondent inference theory, non-common effects, covariation model, Consensus,

Consistency, Distinctiveness, discounting principle, augmenting principle, fundamental attribution error, correspondence bias, actor-observer effect, self-serving bias, just-world hypothesis, Ultimate attribution error, Hostile attribution bias, attribution retraining, attitude, attitude object, tricomponent model, Attitude strength, implicit attitudes, functions of attitudes, subliminal conditioning, vested interest, principle of compatibility, theory of reasoned action, subjective norm, perceived behavioural control, theory of planned behaviour, attitude-to-behaviour process model, MODE model, Persuasion, Yale communication approach, receiver characteristics, Need for cognition, sleeper effect, elaboration likelihood model, central route, peripheral route, heuristic-systematic model, reactance, Forewarning, inoculation, attitude polarisation, balance theory, congruity theory, cognitive dissonance, trivialisation, self-affirmation, induced compliance, self-perception theory, stereotype, Prejudice, Discrimination, in-group, ethnocentrism, Allport's scale, casteism, authoritarian personality, Scapegoat theory, realistic conflict theory, relative deprivation, superordinate goals, minimal group paradigm, social identity theory, positive distinctiveness, self-categorisation theory, optimal distinctiveness theory, out-group homogeneity effect, illusory correlation, subtyping, stereotype content model, stereotype threat, Implicit Association Test, bona fide pipeline, Modern racism, Aversive racism, Ambivalent sexism, social role theory, glass ceiling, Glass cliff, Sticky floor, contact hypothesis, jigsaw classroom, need to belong, need for affiliation, gregariousness, Ostracism, reinforcement-affect model, social comparison theory, Upward social comparison, downward social comparison, self-evaluation maintenance model, fear of missing out, propinquity effect, mere exposure effect, what is beautiful is good, matching hypothesis, law of attraction, complementarity of needs, gain-loss effect, Social exchange theory, comparison level, comparison level for alternatives, equity theory, investment model, communal relationships, self-disclosure, social penetration theory, descriptive self-disclosure, evaluative self-disclosure, passionate love, companionate love, triangular theory of love, two-factor theory of love, loneliness, Emotional isolation, Social isolation, UCLA Loneliness Scale, Social influence, Conformity, Compliance, Obedience, symbolic social influence, identification, internalisation, public compliance, private acceptance, Social norms, descriptive norms, injunctive norms, focus theory of normative conduct, Prescriptive norms, proscriptive norms, perceived norm, autokinetic effect, Normative social influence, Informational social influence, cohesiveness, need for uniqueness, conversion theory, idiosyncrasy credits, emotional contagion, reciprocity, foot-in-the-door technique, door-in-the-face technique, low-ball technique, love-bombing, agentic state, banality of evil, Social power, prosocial behaviour, helping, altruism, cooperation, psychological egoism, empathy, affective empathy, cognitive empathy, empathic concern, sympathy, empathy-altruism hypothesis, negative-state relief model, empathic joy hypothesis, kin selection, reciprocal altruism, competitive altruism, defensive helping, norm of reciprocity, social responsibility norm, personal norm, bystander effect, diffusion of responsibility, pluralistic ignorance, audience inhibition, arousal: cost-reward model, urban overload, feel good, do good effect, prosocial model, elevation, social exclusion, altruistic personality, threat-to-self-esteem model, dependency-oriented help, autonomy-oriented help, aggression, hostile aggression, instrumental aggression, relational aggression, death instinct, hydraulic model, ritualised fighting, appeasement gestures, serotonin, MAOA, frustration-aggression hypothesis, displacement, scapegoating, cognitive neoassociationist model, aggressive cues, weapons effect, excitation transfer theory, observational learning, modelling, aggressive scripts, General Aggression Model, provocation, culture of honour, precarious manhood, threatened egotism, desensitisation, hot temperature effect, negative affect escape model, alcohol myopia, bullying, cyberbullying, catharsis hypothesis, incompatible response technique, self-regulation, group, Entitativity, group socialisation, schism, sociometry, sociometric test, sociogram, forming, storming, norming, performing, adjourning, punctuated equilibrium, social facilitation, drive theory, evaluation apprehension, Ringelmann effect, social loafing, social

striving, deindividuation, SIDE model, risky shift, group polarisation, Groupthink, hidden profile, production blocking, Delphi technique, tragedy of the commons, GRIT, great-person theory, contingency model, least preferred co-worker, Transformational leadership, nurturant-task leader, Social support, stress-buffering hypothesis, crowding, behavioural sink, territoriality, subjective well-being, Source monitoring, post-identification feedback effect, weapon focus, own-race bias, cognitive interview, story model, Offender profiling, investigative psychology, risk-need-responsivity, person-in-environment, Charity Organisation Society, Direct practice, Social case work, social diagnosis, diagnostic school, functional school, formed groups, natural groups, gatekeeper, Boston model, Community organisation, individualism-collectivism, Ethnic-sensitive practice, Cultural competence, ethnographic interview, strengths perspective, ecomap, genogram, neglect

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From the past papers

1. Who is credited with elaborating the "social representation theory"? (RML 2022)
   (a) Serge Moscovici (b) Michael A. Hogg (c) Roy F. Baumeister (d) Mark E. Leary

   Answer: (a) Serge Moscovici. Moscovici launched social representations theory in 1961 with his study of how psychoanalysis became French common sense, through anchoring and objectification. Hogg is known for social identity and uncertainty-identity theory; Baumeister and Mark R. Leary for the need to belong (1995), and Leary for the sociometer theory of self-esteem.

2. Which one refers to the behaviour that an individual engages in while enacting a role? (RML 2020)
   (a) Role taking (b) Role playing (c) Playing at a role (d) Role expectation

   Answer: (b) Role playing. Role playing (role enactment) is the actual behaviour of the person performing a role. Role taking is Mead's mental act of seeing a situation from another's viewpoint, playing at a role is a young child's make-believe, and role expectation is what other people expect of the occupant.

3. Some students experience an intense period of adaptation when they leave their old school environment and start college, especially during those initial days before they find companions, start to fit in and feel comfortable. This process is best described as: (RML 2021)
   (a) Looking-glass self (b) Self-awareness (c) Institutionalisation (d) Re-socialisation

   Answer: (d) Re-socialisation. Resocialisation is unlearning old norms and roles and learning new ones after a major transition such as starting college. The looking-glass self is Cooley's self-image built from others' imagined judgements, self-awareness is attention turned on oneself, and institutionalisation is the apathetic dependence of long-stay inmates.

4. A tendency to act in a way that is consistent with one's own belief, expectation, or cognition about an event or behavior, thereby increasing the likelihood that the event or behavior will occur, is termed as: (IHBAS 2020)
   (a) Self-fulfilling prophecy (b) Confirmation bias (c) Motivational reaction (d) Base-rate fallacy

   Answer: (a) Self-fulfilling prophecy. Acting on an expectation in ways that make it come true is Merton's self-fulfilling prophecy (1948). Confirmation bias only seeks evidence that fits a belief and changes nothing in the world, the base-rate fallacy ignores how common something is, and 'motivational reaction' is not a term.

5. What is the halo effect? (IHBAS 2024)

   (a) The tendency to judge others based on their group membership (b) The influence of a positive trait on the perception of other traits (c) The tendency to overlook negative behaviors in individuals (d) The effect of initial impressions on subsequent judgments

   Answer: (b) The influence of a positive trait on the perception of other traits. Thorndike (1920) named the halo effect: one salient positive trait or an overall impression colours ratings of unrelated traits. Judging by group membership is stereotyping, the effect of initial impressions is the primacy effect (or anchoring), and overlooking faults is a consequence of a halo, not its definition.

6. A teacher watches two students interact for less than a minute and correctly infers who is the class leader. This example best illustrates: (IHBAS 2025)

   (a) Social facilitation (b) Self-fulfilling prophecy (c) Thin slicing (d) Fundamental attribution error

   Answer: (c) Thin slicing. Ambady and Rosenthal showed that very brief samples of behaviour (silent clips of 30 seconds or less) yield accurate judgements. Social facilitation is better performance of well-learned tasks before others, the teacher's expectation did not create the leadership, and the fundamental attribution error over-explains behaviour by disposition.

7. Arrange the process of communication in sequence (RML 2020)

   (a) coding, decoding, recipient, channel (b) channel, encoding, recipient, decoding (c) channel, decoding, recipient, encoding (d) encoding, channel, recipient, decoding

   Answer: (d) encoding, channel, recipient, decoding. The sender first encodes the idea into symbols, the message travels through a channel to the recipient, who then decodes it. The other options send a message through the channel before it is encoded or decode it before it is received.

8. The tendency of people to attribute their own behavior to external causes and other's behavior to internal causes is referred to as : (IHBAS 2024)

   (a) Self-serving bias (b) Fundamental attribution error (c) Actor-Observer effect (d) Stereotyping

   Answer: (c) Actor-Observer effect. Jones and Nisbett (1971): as actors we explain our own behaviour by the situation, as observers we explain others' by their dispositions. The fundamental attribution error concerns only other people's behaviour, the self-serving bias concerns our own successes and failures, and stereotyping is a generalised belief about a group.

9. The "just-world hypothesis" explains victim-blaming as a result of: (Real PYQ 2024)

   (a) People's inherent sadism and desire to see others suffer. (b) The need to believe that the world is orderly, predictable, and fair. (c) A fundamental error in attributing behavior to dispositional factors. (d) The process of social comparison with a less fortunate other.

   Answer: (b) The need to believe that the world is orderly, predictable, and fair. For Melvin Lerner, an innocent person's suffering threatens this belief, so observers restore it by deciding the victim deserved it. Observers are distressed rather than sadistic, the dispositional blame is the result rather than the cause, and downward social comparison is a different process.

10. Arvind is a college student who has strong negative feelings about smoking cigarettes. His grandfather died of lung cancer and he believes smoking is disgusting. One night at a party, Arvind's friends all go outside for a cigarette and he decides to smoke as well. What will he most likely experience as a result? (RML 2021)

    (a) Justification of efforts (b) Increased drive to smoke (c) Cognitive Dissonance (d) Drive Reduction

    Answer: (c) Cognitive dissonance. Acting against a strong, important attitude creates Festinger's aversive tension. Effort justification is a different paradigm (valuing what cost effort), nothing suggests a stronger urge to smoke, and drive reduction is Hull's motivation theory.

11. In which of the following scenarios would it be most difficult to predict behaviour from attitudes? (RML 2021)

    (a) When the attitude has a great deal of personal relevance for the individual (b) When the attitude is formed by indirect experience (c) When the individual has a lot of knowledge about the attitude object (d) When the attitude is formed by the individual's direct experience

    Answer: (b) When the attitude is formed by indirect experience. Hearsay-based attitudes are weak and less accessible. Personal relevance, knowledge of the object and direct experience (Regan and Fazio, 1977) all strengthen the attitude-behaviour link.

12. In Social Psychology, reactance is : (IHBAS 2024)

    (a) The tendency to comply with direct requests (b) The motivation to restore freedom when it is threatened (c) The effect of persuasive messages on attitude change (d) The tendency to follow social norms without question

    Answer: (b) The motivation to restore freedom when it is threatened. Brehm (1966) defined reactance this way; it often makes the forbidden option more attractive. Complying with requests is compliance, following norms is conformity, and message-driven attitude change is persuasion.

13. According to social identity theory, prejudice arises primarily due to : (IHBAS 2025)

    (a) Competition for scarce resources (b) Cognitive dissonance (c) Categorization into in-groups and out-groups (d) Social facilitation

    Answer: (c) Categorization into in-groups and out-groups. Tajfel's minimal groups showed favouritism with no resources at stake, so categorisation plus the need for positive distinctiveness is enough. Competition for scarce resources is Sherif's realistic conflict theory; cognitive dissonance (Festinger) and social facilitation (performance in the presence of others) are unrelated.

14. Low-prejudiced people engage in: (IHBAS 2020)

    (a) Automatic information processing resulting in involuntary stereotyping (b) Controlled information processing resulting in refuting stereotype (c) Controlled information processing resulting in involuntary stereotyping (d) Automatic information processing resulting in voluntary stereotyping

    Answer: (b) Controlled information processing resulting in refuting stereotype. Devine (1989) found stereotypes are activated automatically in high- and low-prejudiced people alike; low-prejudiced people then use controlled processing to inhibit them. Option (a) is the first stage common to all; (c) and (d) pair the processes wrongly.

15. The phenomenon where a qualified individual is prevented from advancing to higher positions in their career due to an invisible barrier based on gender or minority status is called: (Real PYQ 2024)

    (a) The Matthew Effect (b) The Glass Ceiling Effect (c) The Halo Effect (d) The Pygmalion Effect

    Answer: (b) The Glass Ceiling Effect. The glass ceiling is an invisible barrier of attitudes and organisational bias near the top, popularised by a 1986 Wall Street Journal article. The Matthew effect is accumulated advantage (Merton, 1968); halo and Pygmalion effects concern impressions and expectations, not career barriers.

16. 'The more we see and interact with a person, the more likely he or she is to become our friend or sexual partner.' This denotes: (IHBAS 2020)

    (a) Propinquity effect (b) Exposure effect (c) Matching hypothesis (d) Reciprocity

    Answer: (a) Propinquity effect. Nearness and frequent interaction breed friendship (Festinger, Schachter and Back 1950). The exposure effect is narrower (mere repeated exposure to any stimulus raises liking) and is its mechanism; matching is choosing partners of similar attractiveness; reciprocity is liking those who like us.

17. Emotional isolation means: (IHBAS 2020)

    (a) Feeling a lack of deep emotional attachment to one specific person (b) Suffering from a lack of friends/associates (c) Suffering from feelings of incompetence (d) Feeling a sense of inadequacy and low self-esteem

    Answer: (a) Lack of deep attachment to one specific person. Robert Weiss (1973) separated emotional isolation from social isolation, the lack of a wider network of friends and associates (option b). Incompetence and low self-esteem are correlates of loneliness, not its types.

18. Which concept is closely related to FOMO (Fear of Missing Out) in social media research? (JEMAS 2025)

    (a) Social comparison (b) Classical conditioning (c) Operant extinction (d) Cognitive dissonance only

    Answer: (a) Social comparison. FOMO (Przybylski and colleagues 2013) arises from constant upward comparison with others' curated experiences, in the tradition of Festinger's social comparison theory (1954). Conditioning, extinction and dissonance are learning and consistency processes, not comparison.

19. Which of the following is an example of the power of roles? (IHBAS 2024)

    (a) Robber's Cave Experiment (b) Milgram's message experiment (c) Zilstein's shock research (d) Zimbardo's prison experiment

    Answer: (d) Zimbardo's prison experiment. Students randomly made guards or prisoners took on their roles so fully that the study was stopped after six days. Robbers Cave studied intergroup conflict, Milgram's message experiment was the small-world study, and Dr Zilstein was the fictitious experimenter of Schachter's affiliation study.

20. What is normative social influence? (IHBAS 2024)

    (a) Influence based on the desire to be correct and accurate (b) Influence based on the desire to be accepted and liked by others (c) Influence based on personal beliefs and values (d) Influence based on perceived power and authority

    Answer: (b) The desire to be accepted and liked. Deutsch and Gerard (1955) contrasted it with informational influence, the desire to be correct (option a). Personal values describe internalisation, and power and authority describe obedience.

21. According to Cialdini, techniques for gaining compliance rest to a large extent on six basic principles. These do NOT include: (RML 2022)

    (a) Scarcity (b) Social validation (c) Friendship/liking (d) Reward

    Answer: (d) Reward. The six principles are reciprocity, commitment and consistency, social validation (social proof), friendship or liking, authority and scarcity.

22. The bystander effect, where the presence of others inhibits helping, is primarily explained by two factors: (Real PYQ 2024)

    (a) Social loafing and deindividuation (b) Diffusion of responsibility and pluralistic ignorance (c) Social comparison and normative influence (d) Group polarization and groupthink

    Answer: (b) Diffusion of responsibility and pluralistic ignorance. These are Latané and Darley's two classic mechanisms: each bystander assumes someone else will act, and each reads the others' calm as a sign that nothing is wrong. The other pairs are group phenomena of other kinds (Sections 3.6, 3.7 and 3.10).

23. A state that occurs when bystanders in an emergency situation use the behaviour of others to determine whether help is actually required is called: (IHBAS 2020)

    (a) Pluralistic ignorance (b) Group conformity (c) Group compliance (d) Crowd behavior

    Answer: (a) Pluralistic ignorance. Bystanders look to one another to decide whether help is needed and, seeing calm faces, wrongly conclude it is not. Conformity and compliance are yielding to group pressure or to a direct request, and crowd behaviour is the collective action of a large gathering.

24. Altruistic Personality is characterized by: (IHBAS 2020)

    (a) Offering help only when one feels like (b) Consistently helping others (c) Calculating the cost and benefits of helping (d) Readiness to help in an emergency

    Answer: (b) Consistently helping others. The altruistic personality is a stable tendency to help across situations; Bierhoff, Klein and Kramp (1991) found its components to be empathy, belief in a just world, social responsibility, internal locus of control and low egocentrism. Helping only when in the mood, cost-benefit calculation and readiness only in emergencies describe situational helping.

25. Who reformulated the "frustration-aggression hypothesis"? (RML 2022)

    (a) Dollard (b) Miller (c) Berkowitz (d) Lazarus

    Answer: (c) Berkowitz. Leonard Berkowitz (1989) recast it: frustration leads to aggression only to the extent that it produces negative affect, with aggressive cues tipping the balance. Dollard and colleagues proposed the original (1939), Miller only softened it (1941), and Richard Lazarus is known for appraisal and stress.

26. Albert Bandura's Bobo doll experiment is a classic demonstration of: (RML Practice 1)

    (a) Operant conditioning through direct reinforcement (b) Observational learning of aggression through modeling (c) Classical conditioning of fear (d) The effects of sensory deprivation

    Answer: (b) Observational learning of aggression through modelling. Children copied an aggressive adult's specific acts without ever being reinforced, so it is not operant conditioning; classical conditioning of fear is Little Albert, and sensory deprivation is the McGill isolation research.

27. Low levels of which neurotransmitter leads to aggression (Entrance Mock 1)

    (a) Serotonin (b) Dopamine (c) Epinephrine (d) Norepinephrine

    Answer: (a) Serotonin. Low serotonin activity (low 5-HIAA in cerebrospinal fluid) is linked to impulsive aggression. Dopamine concerns reward and movement, while epinephrine and norepinephrine mark arousal in any strong emotion.

28. Which of the following is not a stage of group formation? (JEMAS 2025)

    (a) Storming (b) Proximity (c) Norming (d) Performing

    Answer: (b) Proximity. Tuckman's stages are forming, storming, norming, performing and adjourning. Proximity is a cause of interpersonal attraction, not a stage of group development.

29. Tim would like to participate in his high school's musical. He has been taking vocal lessons for years and is an excellent singer, but does not have much experience as a dancer. The director gives him a part where he has to sing and dance. Based on the theory of social facilitation, how will the presence of an audience affect Tim's performance? (RML 2021)

    (a) His dancing will improve, but his singing will be impaired (b) Both his singing and dancing will improve (c) His singing will improve, but his dancing will be impaired (d) Both his singing and dancing will be impaired

    Answer: (c) His singing will improve, but his dancing will be impaired. By Zajonc's drive theory, an audience raises arousal, which strengthens the dominant response: correct on a well-learned task (singing), often wrong on an unpractised one (dancing). Options (b) and (d) ignore that the effect depends on how well learned the task is.

30. A group of scientists is deciding on the use of live animals in research; most of the group has used live animals. How might the leader reduce groupthink? (RML 2021)

    (a) Make a cohesive team-building exercise a mandatory part of the meeting (b) Relate his opinion at the beginning of the discussion (c) Ask a subset of individuals in the group to critique animal research (d) Ask the group members not to argue with one another's opinions

    Answer: (c) Ask a subset to critique animal research. Janis's remedies include a devil's advocate or critical subgroup. Team-building raises cohesion, a leader stating views first steers the group, and forbidding argument suppresses dissent, so (a), (b) and (d) all worsen groupthink.

31. Defense attorneys often protest the prosecutor's use of eyewitness testimony because: (Amity 2023)

    (a) Eyewitnesses are more prone to forgetting (b) A confident witness is usually right (c) Source errors may cause an eyewitness to make a false identification (d) Poor eyesight of eyewitnesses tends to invalidate their testimony

    Answer: (c) Source errors may cause an eyewitness to make a false identification. A face met elsewhere (a mugshot, the news, a bystander) is misattributed to the crime. Witnesses are not unusually forgetful, confidence predicts accuracy only weakly and grows after feedback, and eyesight is rarely the issue.

32. Which of the following is not a part of the social case work process? (RML 2020)

    (a) Content analysis (b) Treatment (c) Assessment (d) Evaluation

    Answer: (a) Content analysis. Case work runs intake and study, assessment (social diagnosis), treatment or intervention, evaluation, termination and follow-up. Content analysis, coding themes in texts, is a research technique.

33. Who among the following established group work as a method within the profession of social work? (RML 2020)

    (a) Mary Richmond (b) Charlotte Towle (c) Grace Coyle (d) Geisla Konopka

    Answer: (c) Grace Coyle. Her *Social Process in Organized Groups* (1930) founded group work as a method. Richmond founded case work, Towle wrote *Common Human Needs*, and Konopka carried group work into treatment settings later.

High-yield sheet

The facts this chapter's questions ask most, most-asked first. Last read before the exam.

1 The **fundamental attribution error** (named by Lee Ross, 1977), also called the **correspondence bias**, is the tendency to overestimate dispositional and underestimate situational causes of *other people's* behaviour. ASKED 6 TIMES

2 **Leon Festinger** proposed **cognitive dissonance** theory (1957): holding inconsistent cognitions, or acting against one's attitude, produces aversive tension that people reduce, usually by changing the attitude when the behaviour cannot be undone. ASKED 6 TIMES

3 **Diffusion of responsibility** is the main explanation of the bystander effect: responsibility spreads across all the bystanders, so each feels less personally obliged and assumes someone else will act (Latané and Darley). ASKED 6 TIMES

4 **Social loafing**, also called the Ringelmann effect, is reduced effort when individual efforts are pooled (Latané, Williams and Harkins 1979); it falls when each contribution is identifiable and evaluated, so a teacher should grade both the group and each member. Its collectivist reversal is **social striving** (Earley). ASKED 6 TIMES

5 **Social identity theory** (Henri Tajfel and John Turner, 1979): a large part of our self-concept and self-esteem comes from the groups we belong to, and prejudice arises from categorisation into in-groups and out-groups and the need for positive distinctiveness, not from competition for resources. ASKED 5 TIMES

6 **Solomon Asch** (1951, 1956): among about seven confederates unanimously giving an obviously wrong line judgement, participants conformed on about 37 percent of critical trials and about three quarters conformed at least once; this is normative influence towards an incorrect majority, not obedience. ASKED 5 TIMES

7 **Pluralistic ignorance**: in an ambiguous emergency each bystander uses the calm behaviour of the others to decide whether help is needed, and mistakenly concludes that it is not (the smoke-filled room). ASKED 5 TIMES

8 **Bruce Tuckman** (1965) ordered group development as forming, storming (conflict over leadership and the group's definition), norming (cohesion; purpose and strategy clarified) and performing; adjourning was added with Mary Ann Jensen in 1977. Proximity is not a stage. ASKED 5 TIMES

9 **Social facilitation** began with Norman Triplett's 1898 fishing-reel study; Zajonc's drive theory (1965) says the presence of others raises arousal and strengthens the dominant response, improving easy, well-learned tasks and impairing new or difficult ones. ASKED 4 TIMES

10 The **halo effect** (Edward Thorndike, 1920): an overall impression or one positive trait influences ratings of a person's other, unrelated traits; the negative version is the horns effect. It is not primacy, stereotyping, self-serving bias or false consensus. ASKED 3 TIMES

11 **Normative social influence** rests on the desire to be liked and accepted; **informational social influence** on the desire to be correct (Deutsch and Gerard, 1955). Sherif's autokinetic study shows informational influence and the formation of group norms. ASKED 3 TIMES

12 The **bystander effect**: the more bystanders present, the less likely any one is to help and the slower the help; people help more when alone (seizure study: 85 percent alone versus 31 percent with four others). ASKED 3 TIMES

13 A **sociogram**, the tool of Jacob Moreno's sociometry (1934), maps choices and rejections to show social distance and patterns of choice in a group (stars, isolates, cliques); it captures implicit and explicit interaction patterns and does not measure IQ, personality or economic status. ASKED 3 TIMES

14 **Groupthink** (Irving Janis 1972) is a mode of thinking in highly cohesive groups where the drive for unanimity overrides realistic appraisal, discouraging dissent and producing poor decisions; a devil's advocate or critical subgroup and an impartial leader reduce it. ASKED 3 TIMES

15 **Kurt Lewin** (1890 to 1947), father of modern social psychology: in B = f(P, E) the **life space** is the person plus the psychological environment, and he founded **action research** (1946) and group dynamics. ASKED 2 TIMES

16 The **self-fulfilling prophecy** was named and formalised by the sociologist **Robert K. Merton** (1948): a false definition of a situation evokes behaviour that makes it come true. It is not confirmation bias or the base-rate fallacy. ASKED 2 TIMES

17 Melvin Lerner's **just-world hypothesis** explains victim blaming by the need to believe that the world is orderly, predictable and fair. ASKED 2 TIMES

18 In the **elaboration likelihood model** (Petty and Cacioppo), systematic processing with careful attention to argument quality is the **central route**; reliance on mental shortcuts and cues is the **peripheral route**. There is no emotional or situational route. ASKED 2 TIMES

19 **Reactance** (Jack Brehm, 1966) is the motivation to restore a threatened freedom, often by wanting the forbidden behaviour more. ASKED 2 TIMES

20 The **glass ceiling** is an invisible barrier that stops qualified women and minorities from rising to top positions; the term was popularised by a 1986 Wall Street Journal article. ASKED 2 TIMES

21 Intergroup hostility falls with cooperation: at Robbers Cave (Sherif, 1954) only **superordinate goals** ended it, and Allport's **contact hypothesis** (1954) requires equal status, common goals, cooperation and support of authorities, law or custom. ASKED 2 TIMES

22 The **propinquity effect**: the more we see and interact with a person, the more likely he or she becomes our friend or partner (Festinger, Schachter and Back 1950); its mechanism is Robert Zajonc's **mere exposure effect** (1968), in which repeated exposure to a stimulus increases liking for it. ASKED 2 TIMES

23 **Conformity** is changing to match a group with no direct request; **compliance** is changing in response to a direct request; **obedience** is following a command from an authority. ASKED 2 TIMES

24 In **Milgram**'s 1963 baseline study, 26 of 40 men (65 percent) obeyed to the maximum 450 volts, and all went to 300 volts. ASKED 2 TIMES

25 **Zimbardo**'s Stanford Prison Experiment (1971) demonstrates the power of social roles and situational forces; planned for two weeks, it was stopped after six days. ASKED 2 TIMES

26 **Cialdini**'s six compliance principles are reciprocity, commitment and consistency, social proof, liking, authority and scarcity (not reward); the foot-in-the-door (Freedman and Fraser, 1966) starts with a small request, then a larger one. ASKED 2 TIMES

27 The **two classic explanations** of the bystander effect are diffusion of responsibility and pluralistic ignorance, not social loafing, deindividuation, group polarisation or groupthink. ASKED 2 TIMES

28 **Social case work** was founded by Mary Ellen Richmond (*Social Diagnosis*, 1917); its process is intake and study, assessment (social diagnosis), intervention, evaluation, termination and follow-up, and its techniques include interviewing, observation, counselling, home visits, collateral contacts and referral, but not content analysis or lobbying. ASKED 2 TIMES

29 **Grace Coyle** established group work as a method within social work; **formed groups** are convened for an identified purpose with a trained leader and an agency auspice, whereas natural groups come together spontaneously. ASKED 2 TIMES

30 In Lawrence Shulman's terms the **gatekeeper** blocks or diverts the group from painful topics; Trecker's principles include democratic group self-determination, flexible functional organisation and progressive programme experiences. ASKED 2 TIMES